Solutions Manual

Engineering Optimization

Theory and Practice

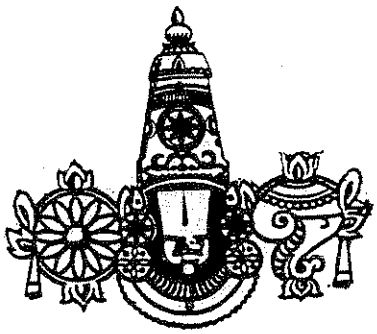
Fourth Edition

Singiresu S. Rao

Professor and Chairman

Department of Mechanical and Aerospace Engineering

University of Miami, Coral Gables, Florida



Contents

Chapter 1	Introduction to Optimization	1
Chapter 2	Classical Optimization Techniques	52
Chapter 3	Linear Programming I: Simplex Method	108
Chapter 4	Linear Programming II: Additional Topics and Extensions	247
Chapter 5	Nonlinear Programming I: One-Dimensional Minimization Methods	292
Chapter 6	Nonlinear Programming II: Unconstrained Optimization Techniques	334
Chapter 7	Nonlinear Programming III: Constrained Optimization Techniques	413
Chapter 8	Geometric Programming	492
Chapter 9	Dynamic Programming	517
Chapter 10	Integer Programming	549
Chapter 11	Stochastic Programming	576
Chapter 12	Optimal Control and Optimality Criteria Methods	594
Chapter 13	Modern Methods of Optimization	607
Chapter 14	Practical Aspects of Optimization	620

© Singiresu S. Rao, March 2010 (*srao@miami.edu*)

Chapter 1
Introduction to Optimization

1.1 Let x_A, x_B, x_C, x_D denote the quantity of fertilizer A, B, C, D produced per week (tons).

Let P_i = profit per ton of fertilizer i $(i = A, B, C, D)$

$$P_i = SP_i - PC_i - \left(C_N x_{Ni} + C_{Ph} x_{Phi} + C_{Po} x_{Poi} + C_{ch} x_{chi}\right)$$

where
- SP_i = selling price of fertilizer i per ton
- PC_i = production cost of fertilizer i per ton
- C_N = cost of nitrates per ton = \$ 1500
- C_{Ph} = cost of phosphates per ton = \$ 500
- C_{Po} = cost of potash per ton = \$ 1000
- C_{ch} = cost of inert chalk base = \$ 100
- x_{Phi} = tons of nitrate required for 1 ton of fertilizer i
- x_{Phi} = tons of phosphates required for 1 ton of fertilizer i
- x_{Poi} = tons of potash required for 1 ton of fertilizer i
- x_{chi} = tons of inert chalk base required for 1 ton of fertilizer i

$$P_A = 350 - 100 - \left(1500 * 0.05 + 500 * 0.1 + 1000 * 0.05 + 100 * 0.8\right) = -5$$

$$P_B = 550 - 150 - \left(1500 * 0.05 + 500 * 0.15 + 1000 * 0.1 + 100 * 0.7\right) = 80$$

$P_C = 450 - 200 - (1500 * 0.1 + 500 * 0.2 + 1000 * 0.1 + 100 * 0.6) = -160$

$P_D = 700 - 250 - (1500 * 0.15 + 500 * 0.05 + 1000 * 0.15 + 100 * 0.65) = -15$

Minimize

$f = -\text{profit} = -P_A x_A - P_B x_B - P_C x_C - P_D x_D$

$\therefore f = 5 x_A - 80 x_B + 160 x_C + 15 x_D \quad (1)$

Availability constraints:

$0.05 x_A + 0.05 x_B + 0.1 x_C + 0.15 x_D \leq 1000$ (nitrates) (2)

$0.1 x_A + 0.15 x_B + 0.2 x_C + 0.05 x_D \leq 2000$ (phosphates) (3)

$0.05 x_A + 0.1 x_B + 0.1 x_C + 0.15 x_D \leq 1500$ (potash) (4)

Production requirements:

$x_A \geq 5000, \quad x_B \geq 0, \quad x_C \geq 0, \quad x_D \geq 4000 \quad (5)$

$\therefore$ Problem is defined by Eqs. (1) to (5).

1.2 $\vec{x} = \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} A/A_{ref} \\ x/h \end{Bmatrix}$

Objective functions:

$$f_1(\vec{x}) = 2\rho h x_2 \sqrt{1+x_1^2}\, A_{ref} \quad \text{(weight)}$$

$$f_2(\vec{x}) = \frac{P h (1+x_1^2)^{1.5} \sqrt{1+x_1^4}}{2\sqrt{2}\, E\, x_1^2 x_2 A_{ref}} \quad \text{(deflection)}$$

Behavior constraints:

$$\sigma_1(\vec{x}) \le \sigma_0$$

$$\sigma_2(\vec{x}) \le \sigma_0$$

with

$$\sigma_1 = \frac{P(1+x_1)\sqrt{1+x_1^2}}{2\sqrt{2}\, x_1 x_2 A_{ref}}, \quad \sigma_2 = \frac{P(x_1-1)\sqrt{1+x_1^2}}{2\sqrt{2}\, x_1 x_2 A_{ref}}$$

side constraints:

$$x_i^{min} \le x_i \le x_i^{max}, \quad i = 1, 2$$

Data: $E = 30 \times 10^6$ psi, $\rho = 0.283$ lb/in^3, $P = 10000$ lb,
$\sigma_0 = 20000$ psi, $A_{ref} = 1$ in^2, $h = 100$ in,
$x_1^{min} = x_2^{min} = 0.1$, $x_1^{max} = 2.0$, $x_2^{max} = 2.5$

(a) For given data,

$$f_1(\vec{x}) = 56.6\, x_2 \sqrt{1+x_1^2} \tag{1}$$

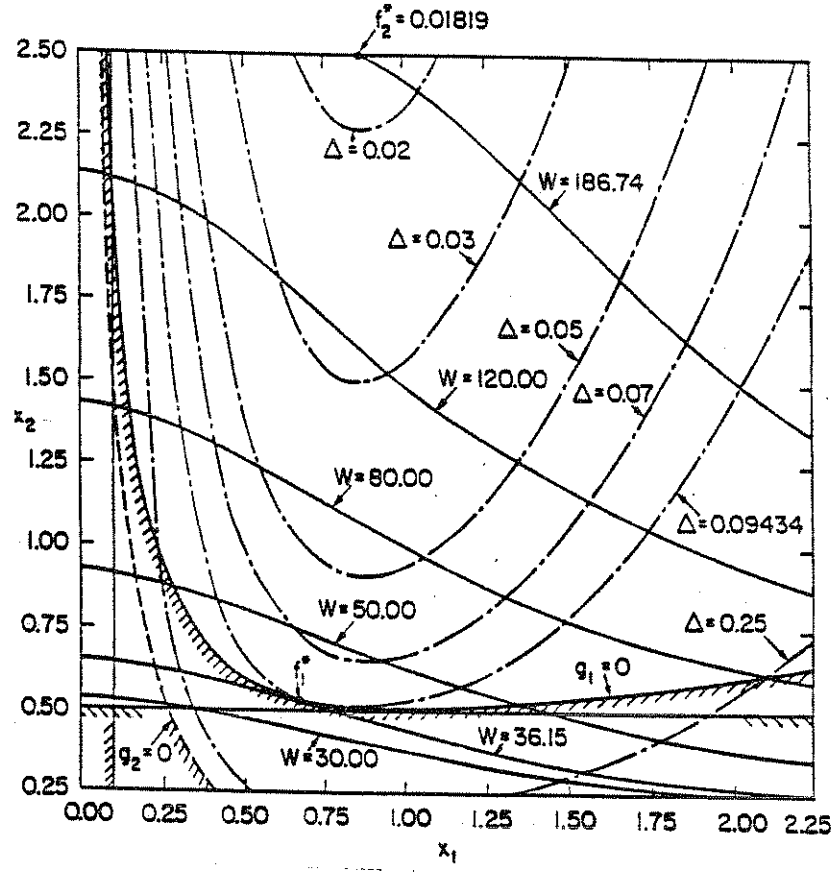
$$g_1(\vec{x}) \Rightarrow \frac{(1+x_1)\sqrt{1+x_1^2}}{x_1 x_2} \le 5.656854 \tag{2}$$

$$g_2(\vec{x}) \Rightarrow \frac{(x_1-1)\sqrt{1+x_1^2}}{x_1 x_2} \le 5.656854 \tag{3}$$

$$0.1 \le x_1 \le 2, \quad 0.1 \le x_2 \le 2.5 \tag{4}$$

Graphical optimization shown in the figure.

Optimum solution: $\vec{x}^* = \begin{Bmatrix} 0.65 \\ 0.53521 \end{Bmatrix}$, $f_1^* = 36.12969$



$f_1 = W$ = weight

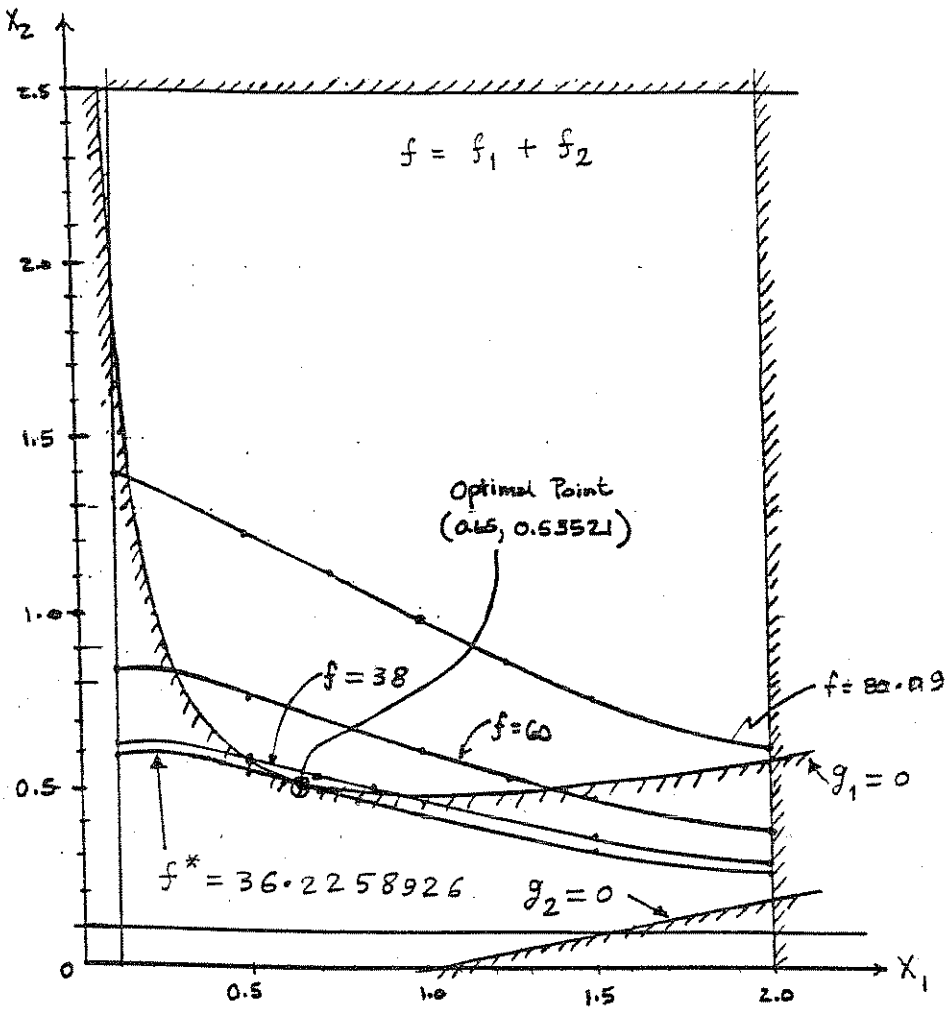
$f_2 = \Delta$ = deflection

(b) $$f_2(\vec{x}) = \frac{0.0117851\,1\,\left(1+x_1^2\right)^{1.5}\sqrt{1+x_1^4}}{x_1^2\,x_2} \qquad (5)$$

constraints are given by Eqs. (2), (3) and (4).

Optimum solution: $\vec{x}^* = \begin{Bmatrix} 0.9 \\ 2.5 \end{Bmatrix}$, $f_2^* = 0.01823$

(Graphical solution is indicated in the above figure).



(c) $f_1(\vec{x}) + f_2(\vec{x})$

$$= 56.6\, x_2 \sqrt{1 + x_1^2} + \frac{0.0117851 \left(1 + x_1^2\right)^{1.5} \sqrt{1 + x_1^4}}{x_1^2\, x_2} \qquad (6)$$

Constraints are given by Eqs. (2), (3) and (4).

Optimum solution:

$$\vec{x}^* = \begin{Bmatrix} 0.65 \\ 0.53521 \end{Bmatrix}, \quad f_3^* = 36.2258926$$

(Graphical solution is indicated in the above figure).

1.3

Jobs: $i = 1, 2, \ldots, 10$

Stations: $j = 1, 2, \ldots, n$

t_i = time required for job i = $t_{fi} - t_{si}$, $i = 1, 2, \ldots, 10$

where t_{fi} = time at which job i is completed,

t_{si} = time at which job i is started

Let t_E = time at which all jobs are completed.

For station j, idle time = $I_j = t_E - \sum_{i=1}^{10} x_{ij}\, t_i$

where $x_{ij} = \begin{cases} 1 & \text{if job } i \text{ is assigned to station } j \\ 0 & \text{otherwise} \end{cases}$

Total idle time = $f = \sum_{j=1}^{n} I_j = \sum_{j=1}^{n} \sum_{i=1}^{10} \left(t_E - x_{ij}\, t_i \right)$ (1)

At station j,

$$x_{ij}\, t_{si} = \sum_{k=1}^{i-1} x_{kj}\, t_k \quad ; \quad j = 1, 2, \ldots, n \qquad (2)$$

For job i,

$$t_{fi} = t_i + t_{si} \quad ; \quad i = 1, 2, \ldots, 10 \qquad (3)$$

Find $n, x_{11}, x_{12}, \ldots, x_{10,n}$

to minimize f given by Eq. (1)

subject to

$t_{s5} \geq t_{f1}$, $t_{s5} \geq t_{f3}$,

$t_{s6} \geq t_{f2}$, $t_{s6} \geq t_{f3}$, $t_{s6} \geq t_{f4}$,

$t_{s7} \geq t_{f5}$, $t_{s7} \geq t_{f6}$,

$t_{s8} \geq t_{f6}$, $t_{s9} \geq t_{f7}$, $t_{s9} \geq t_{f8}$, $t_{s10} \geq t_{f9}$

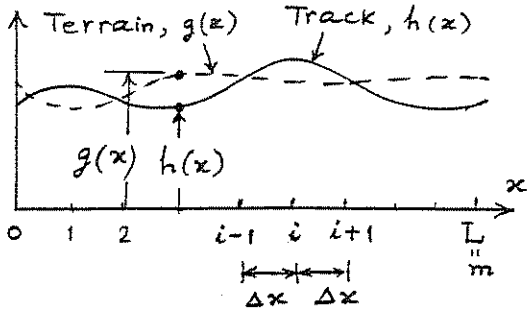
and Eqs. (2) and (3)

with $t_1 = 4$, $t_2 = 8$, $t_3 = 7$, $t_4 = 6$, $t_5 = 3$,

$t_6 = 5$, $t_7 = 1$, $t_8 = 9$, $t_9 = 2$, $t_{10} = 8$

1.4

Divide the track length into m intervals of equal length as shown in the figure:



Design variables:

$$\{h_0, h_1, h_2, \ldots, h_m\} \qquad (1)$$

with $h_i = h(x = x_i)$

and $x_i = i.(\Delta x)$; $\Delta x = (L/m)$.

objective function: (to be minimized)

$$f = c \int_0^L |g(x) - h(x)|\, dx$$

$$\approx c \sum_{i=1}^{m} |g(x_i) - h_i| \cdot \Delta x \qquad (2)$$

where c is a proportionality constant.

Constraints:

slope:

$$\left| \frac{h_{i+1} - h_i}{\Delta x} \right| \leq r_1 \quad ; \quad i = 0, 1, \ldots, m-1 \qquad (3)$$

Rate of change of slope:

$$\left| \frac{\frac{h_{i+2} - h_{i+1}}{\Delta x} - \frac{h_{i+1} - h_i}{\Delta x}}{\Delta x} \right| = \left| \frac{h_{i+2} - 2h_{i+1} + h_i}{(\Delta x)^2} \right| \leq r_2 \; ;$$

$$i = 0, 1, \ldots, m-2 \qquad (4)$$

second derivative of slope:

$$\left| \frac{\frac{h_{i+3} - 2h_{i+2} + h_{i+1}}{(\Delta x)^2} - \frac{h_{i+2} - 2h_{i+1} + h_i}{(\Delta x)^2}}{\Delta x} \right|$$

$$= \left| \frac{h_{i+3} - 3h_{i+2} + 3h_{i+1} - h_i}{(\Delta x)^3} \right| \leq r_3 \; ; \; i = 0, 1, \ldots, m-3 \qquad (5)$$

End conditions: (6)

$$h_0 = a \; , \quad h_m = b$$

Problem defined by Eqs. (1) to (6) is an O.C. problem. (It has serial structure).

1.5

Let x_i = number of units produced in week i $(i = 1, 2)$

Assume: (1) Every thing produced is sold by end of second week.

(2) More units can be produced in 1st week, if economical.

(3) Total units in the two weeks is 600.

Minimize $f = x_1(4x_1^2) + x_2(4x_2^2) + 10(x_1 - 200)$

subject to

$$200 \le x_1 \le 600$$
$$0 \le x_2 \le 600$$
$$x_1 + x_2 = 600$$

Graphical solution:

Contours of f are drawn as follows:

$$f = 4x_1^3 + 4x_2^3 + 10x_1 - 2000 = c = \text{constant}$$

$c = 6 \times 10^8$:

x_2	100	200	300	400	500	600
x_1	530.1	521.7	497	441	292	—

$c = 4 \times 10^8$:

x_2	100	200	300	400	500
x_1	463	451	417	330	−292

$c = 3 \times 10^8$:

x_2	100	200	300	400	500
x_1	420	406	363.4	222.4	−368

$c = 2.75 \times 10^8$:

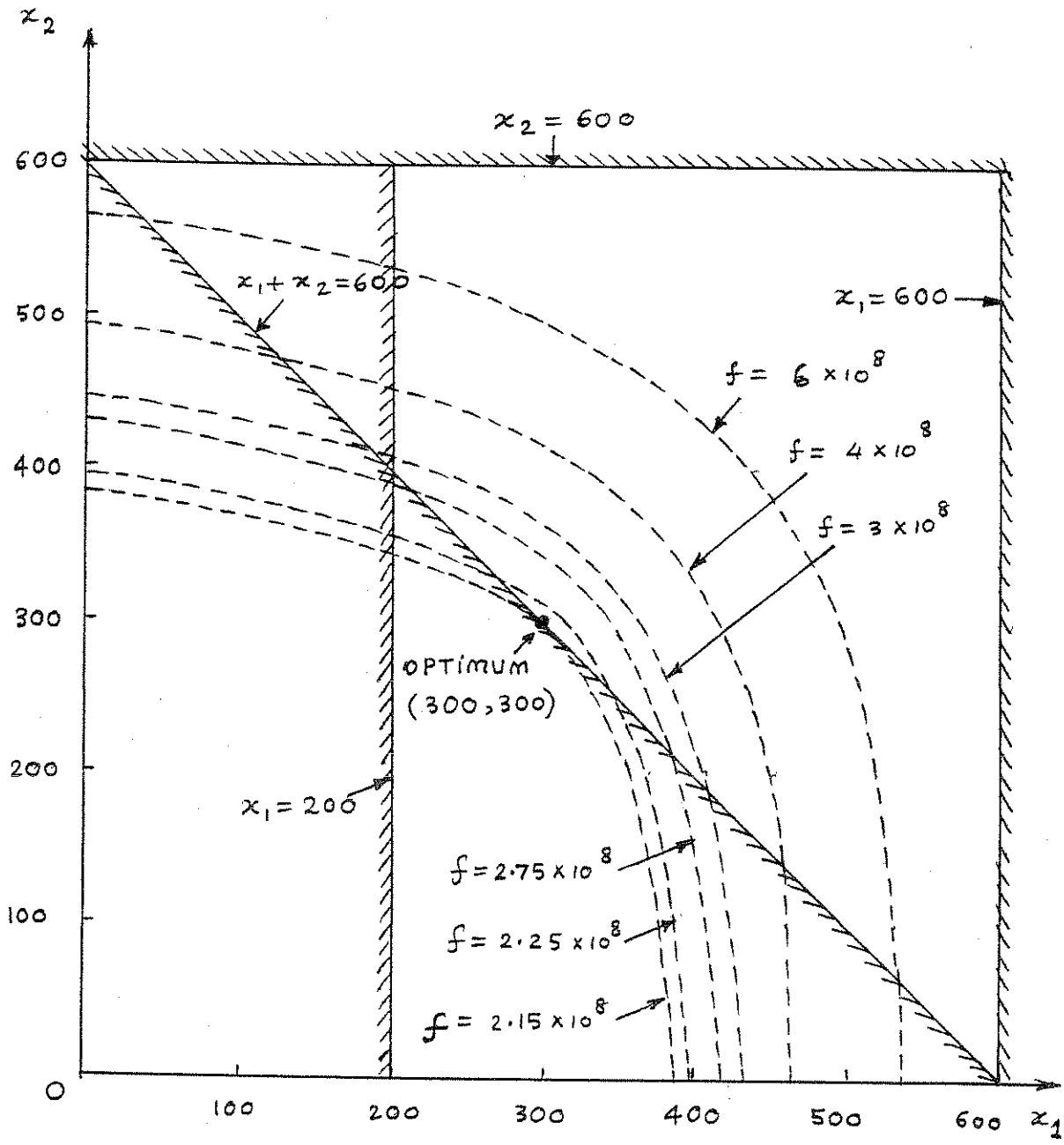
x_2	100	200	300	400	500
x_1	407	393	346	168	−383

$c = 2.25 \times 10^8$:

x_2	100	200	300	400
x_1	380	364	308	−197

$c = 2.15 \times 10^8$:

x_2	100	200	300	400
x_1	375	358	299	−217


x_2
$x_2 = 600$
600
500
400
300
200
100
0
$x_1 + x_2 = 600$
$x_1 = 600$
$f = 6 \times 10^8$
$f = 4 \times 10^8$
$f = 3 \times 10^8$
OPTIMUM
(300, 300)
$x_1 = 200$
$f = 2.75 \times 10^8$
$f = 2.25 \times 10^8$
$f = 2.15 \times 10^8$
100
200
300
400
500
600
x_1

1.6

$$\vec{x} = \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} r \\ l \end{Bmatrix}$$

$$x = x_1 \cos\theta + x_2 \sqrt{1 - \left(\frac{x_1 \sin\theta}{x_2}\right)^2} \qquad (1)$$

$$\dot{x} = \frac{dx}{dt} = -x_1 \omega \left\{ \sin\theta + \frac{x_1 \sin 2\theta}{2x_2 \sqrt{1 - \left(\frac{x_1 \sin\theta}{x_2}\right)^2}} \right\} \qquad (2)$$

where $\omega = \frac{d\theta}{dt} = \dot{\theta}$

$$\ddot{x} \simeq -x_1 \omega^2 \left(\cos\theta + \frac{x_1}{x_2} \cos 2\theta\right) \qquad (3)$$

To plot the constraint on x, we rearrange Eq. (1) as:

$$(x - x_1 \cos\theta)^2 = x_2^2 \left(1 - \frac{x_1^2}{x_2^2} \sin^2\theta\right) = x_2^2 - x_1^2 \sin^2\theta$$

or $$x_2 = \left\{ (x_1 \sin\theta)^2 + (x - x_1 \cos\theta)^2 \right\}^{\frac{1}{2}} \qquad (4)$$

To plot the contours of $\dot{x}$, Eq. (2) is rewritten as:

$$(\dot{x} + x_1 \omega \sin\theta)^2 = -\frac{x_1^2 \omega \sin 2\theta}{2x_2 \sqrt{1 - \frac{x_1^2 \sin^2\theta}{x_2^2}}}$$

or

$$(\dot{x} + x_1 . \omega . \sin\theta)^2 = \frac{(x_1^2 \omega \sin 2\theta)^2}{4(x_2^2 - x_1^2 \sin^2\theta)}$$

or $$x_2 = \sqrt{\frac{1}{4} \left(\frac{x_1^2 \omega \sin 2\theta}{\dot{x} + x_1 \omega \sin\theta}\right)^2 + (x_1 \sin\theta)^2} \qquad (5)$$

To plot the contours of $\ddot{x}$, Eq. (3) is rewritten as:

$$x_2 (\ddot{x} + x_1 \omega^2 \cos\theta) = -x_1^2 \omega^2 \cos 2\theta$$

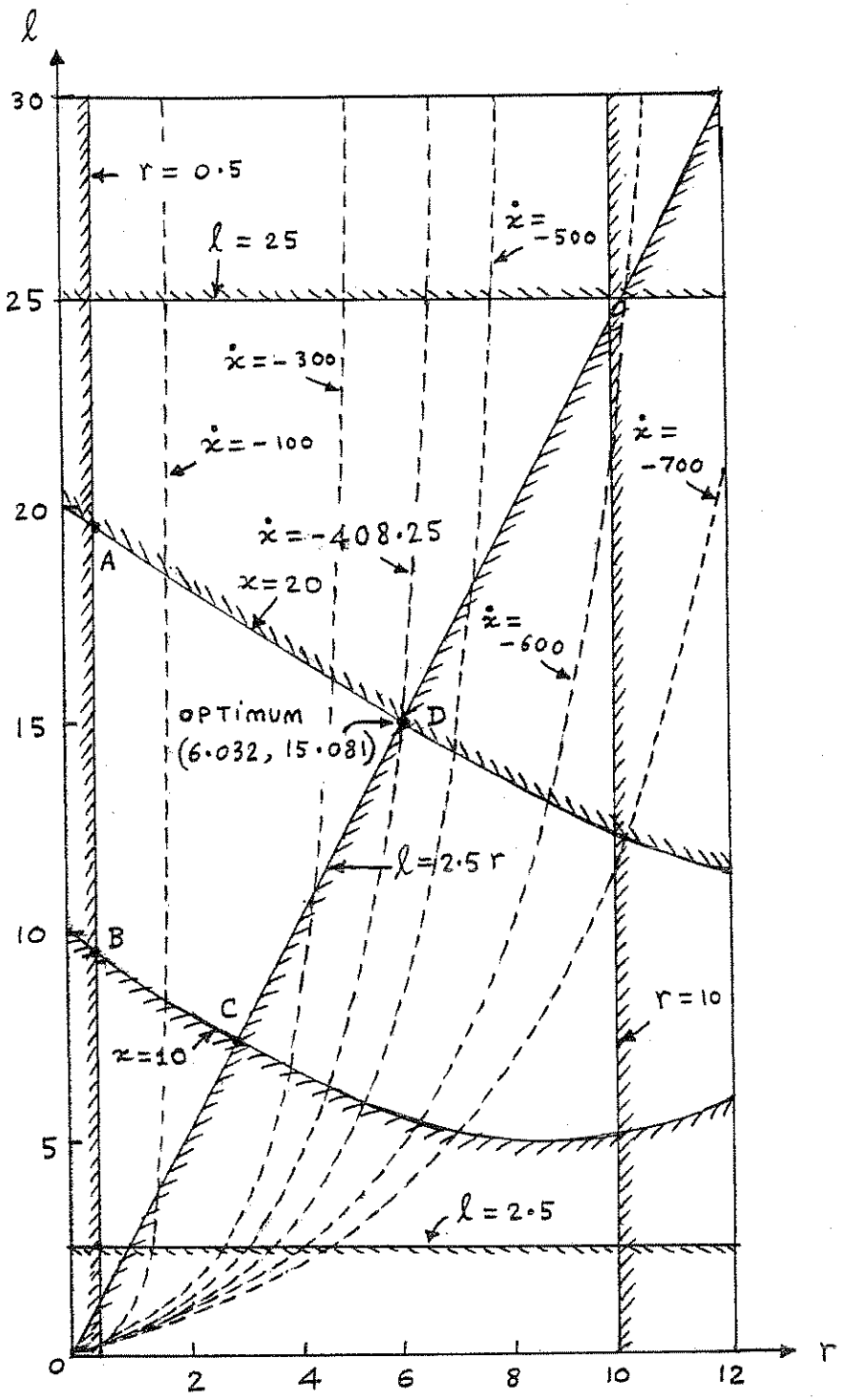
or $$x_2 = \frac{-x_1^2 \omega^2 \cos 2\theta}{\ddot{x} + x_1 \omega^2 \cos\theta} \qquad (6)$$

Data: $\theta = 30^\circ$, $\omega = 100$ rad/s

Constraints: $l \geq 2.5\, r \Rightarrow \frac{x_2}{x_1} \geq 2.5$

$0.5 \leq x_1 \leq 10$, $2.5 \leq x_2 \leq 25$

$10 \leq x \leq 20$

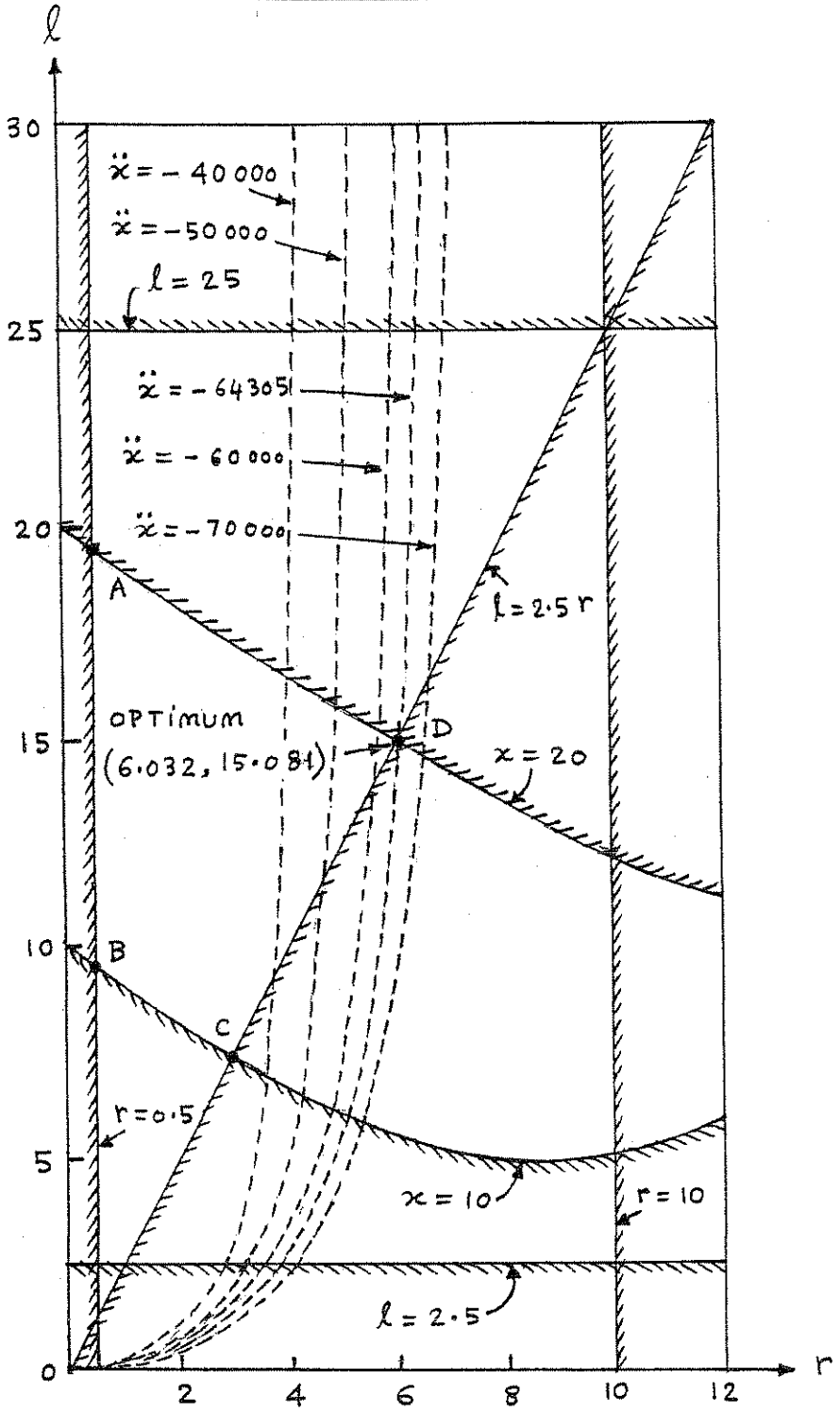


Feasible space = ABCDA

Optimum solution :

$r^* = 6.032$, $l^* = 15.081$, $\dot{x}_{opt} = -408.25$

1.7



Feasible space = ABCDA

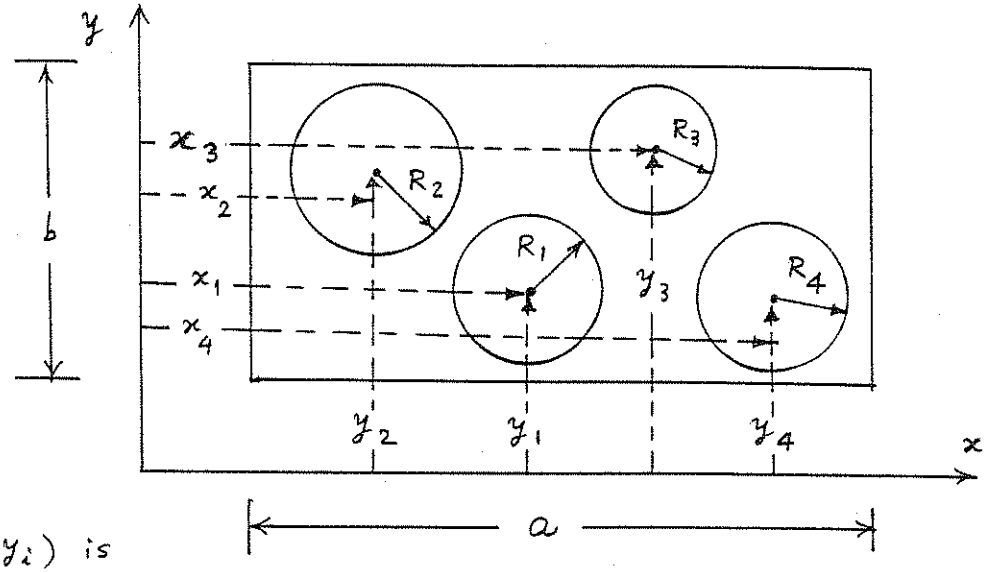
optimum solution: $r^* = 6.032$, $l^* = 15.081$

$\ddot{x}_{opt} = -64305$

(Formulation: given in solution of problem 1.7)

1.8

$$\vec{X} = \begin{Bmatrix} a \\ b \\ x_1 \\ y_1 \\ x_2 \\ y_2 \\ x_3 \\ y_3 \\ x_4 \\ y_4 \end{Bmatrix}$$



where (x_i, y_i) is the center of disk i.

Minimize $f = ab - \pi (R_1^2 + R_2^2 + R_3^2 + R_4^2)$

subject to the constraints

$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \geq R_1 + R_2$$

$$\sqrt{(x_1 - x_3)^2 + (y_1 - y_3)^2} \geq R_1 + R_3$$

$$\sqrt{(x_1 - x_4)^2 + (y_1 - y_4)^2} \geq R_1 + R_4$$

$$\sqrt{(x_2 - x_3)^2 + (y_2 - y_3)^2} \geq R_2 + R_3$$

$$\sqrt{(x_2 - x_4)^2 + (y_2 - y_4)^2} \geq R_2 + R_4$$

$$\sqrt{(x_3 - x_4)^2 + (y_3 - y_4)^2} \geq R_3 + R_4$$

$$a \geq x_i + R_i \; ; \; i = 1, 2, 3, 4$$

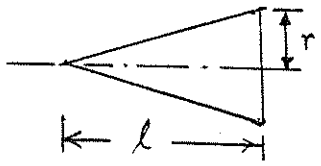
$$b \geq y_i + R_i \; ; \; i = 1, 2, 3, 4$$

$$x_i \geq R_i \; ; \; i = 1, 2, 3, 4$$

$$y_i \geq R_i \; ; \; i = 1, 2, 3, 4$$

1.9 $\vec{X} = \begin{Bmatrix} R_1 \\ R_2 \end{Bmatrix}$

(a) Volume of a cone $= \frac{1}{3}\pi r^2 \ell$



Volume of cone clutch:

$$f(\vec{X}) = \frac{1}{3}\pi\left(R_1^2 \ell_1 - R_2^2 \ell_2\right)$$

$$= \frac{1}{3}\frac{\pi}{\tan\alpha}\left(R_1^3 - R_2^3\right) \quad (1)$$

$$T = \frac{2\pi f p}{3\sin\alpha}\left(R_1^3 - R_2^3\right) \quad (2)$$

Using $p = \frac{F}{A}$, $A = \pi(R_1^2 - R_2^2)$, Eq. (2) becomes

$$T = \frac{2\pi(0.5)\ 30}{3(0.5)\ \pi(R_1^2 - R_2^2)}\left(R_1^3 - R_2^3\right) = 20\left(\frac{R_1^3 - R_2^3}{R_1^2 - R_2^2}\right) \quad (3)$$

constraints:

$$g_1(\vec{X}) = \frac{R_1^3 - R_2^3}{R_1^2 - R_2^2} - 5 \geq 0 \quad (4)$$

$$g_2(\vec{X}) = R_1 - 2R_2 \geq 0 \quad (5)$$

$$g_3(\vec{X}) = R_1 \geq 0 \quad (6)$$

$$g_4(\vec{X}) = R_1 - 15 \leq 0 \quad (7)$$

$$g_5(\vec{X}) = R_2 \geq 0 \quad (8)$$

$$g_6(\vec{X}) = R_2 - 10 \leq 0 \quad (9)$$

Plotting $g_1 = 0$:

R_1	0.0	1.0	2.0	3.0	3.5	4.0	4.5	5.0
R_2	5.0	4.83	4.37	3.65	3.16	2.56	1.77	0.0

Contours of f (Eq. 1):

For $f = 50.0$:

R_1	3.02	3.5	4.0	5.0	6.0	7.0	10.0
R_2	0.0	2.48	3.32	4.6	5.73	6.8	9.9

for $f = 100.0$:

R_1	3.81	4.0	4.5	5.0	6.0	7.0	10.0
R_2	0.0	2.07	3.3	4.1	5.4	6.6	9.8

for $f = 141.9$:

R_1	4.27	4.5	5.0	5.5	6.0	8.0	10.0
R_2	0.0	2.34	3.6	4.45	5.16	7.6	9.73

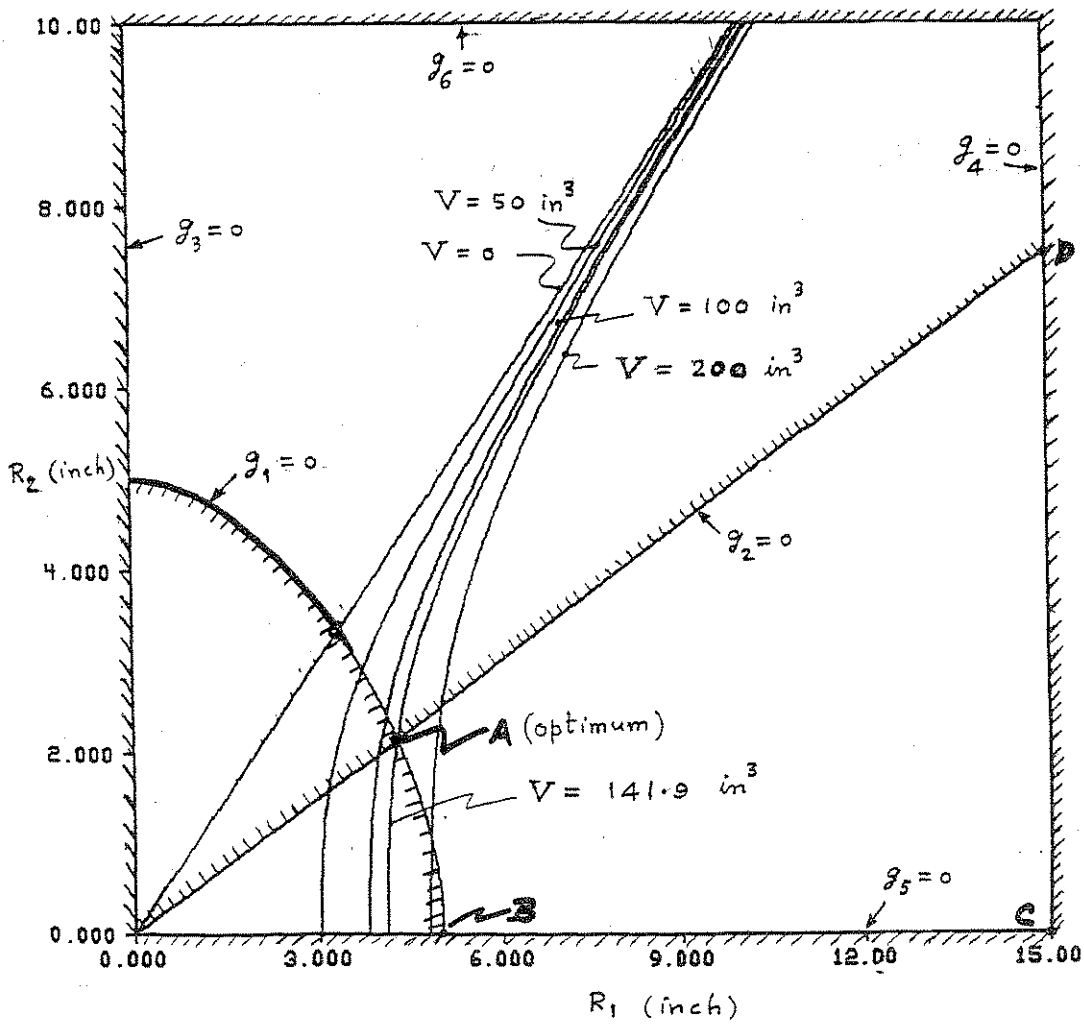
for $f = 0.0$:

R_1	1.0	2.0	3.0	5.0	10.0
R_2	1.0	2.0	3.0	5.0	10.0

Graphical solution : indicated in the figure.

Optimum point : $\vec{X}^* = \begin{Bmatrix} R_1^* \\ R_2^* \end{Bmatrix} = \begin{Bmatrix} 4.472 \\ 2.236 \end{Bmatrix}$, $f^* = 141.9\ in^3$

(Feasible space: A B C D A) $\Rightarrow$ Point A



(b) Constraints = same as in part (a) except that g_2 is changed to $R_1 - 2R_2 \le 0$.

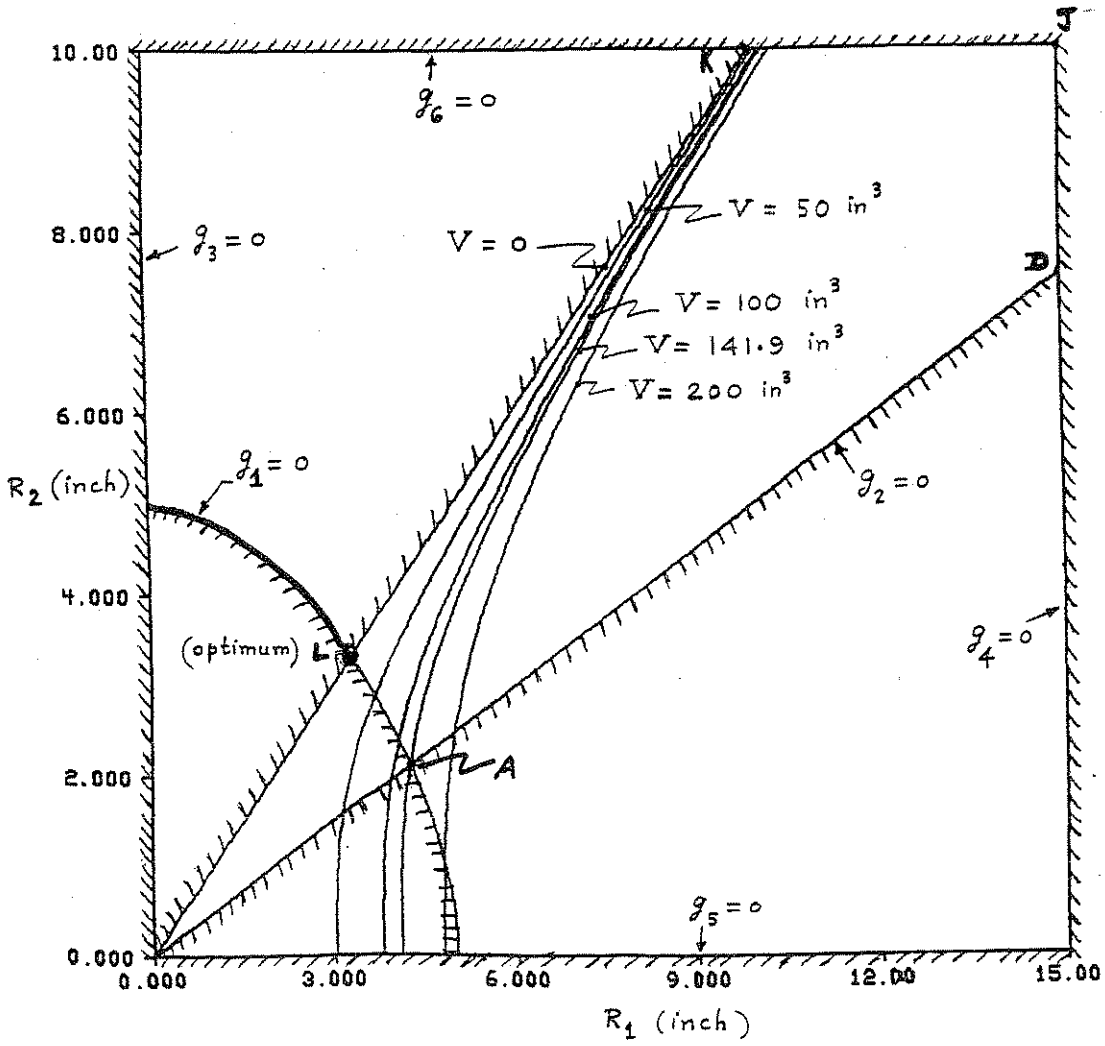
Objective function = same as in part (a).

Feasible space = A D J K L A

optimum point = L

$$\vec{X}^* = \begin{Bmatrix} R_1^* \\ R_2^* \end{Bmatrix} = \begin{Bmatrix} 3.536 \\ 3.536 \end{Bmatrix}, \quad f^* = 0 \text{ in}^3$$

Details of graphical solution: shown in the figure.



(c) Constraint equation for g_1 changes.

$$T = \frac{\pi f p R_2}{\sin\alpha}(R_1^2 - R_2^2)$$

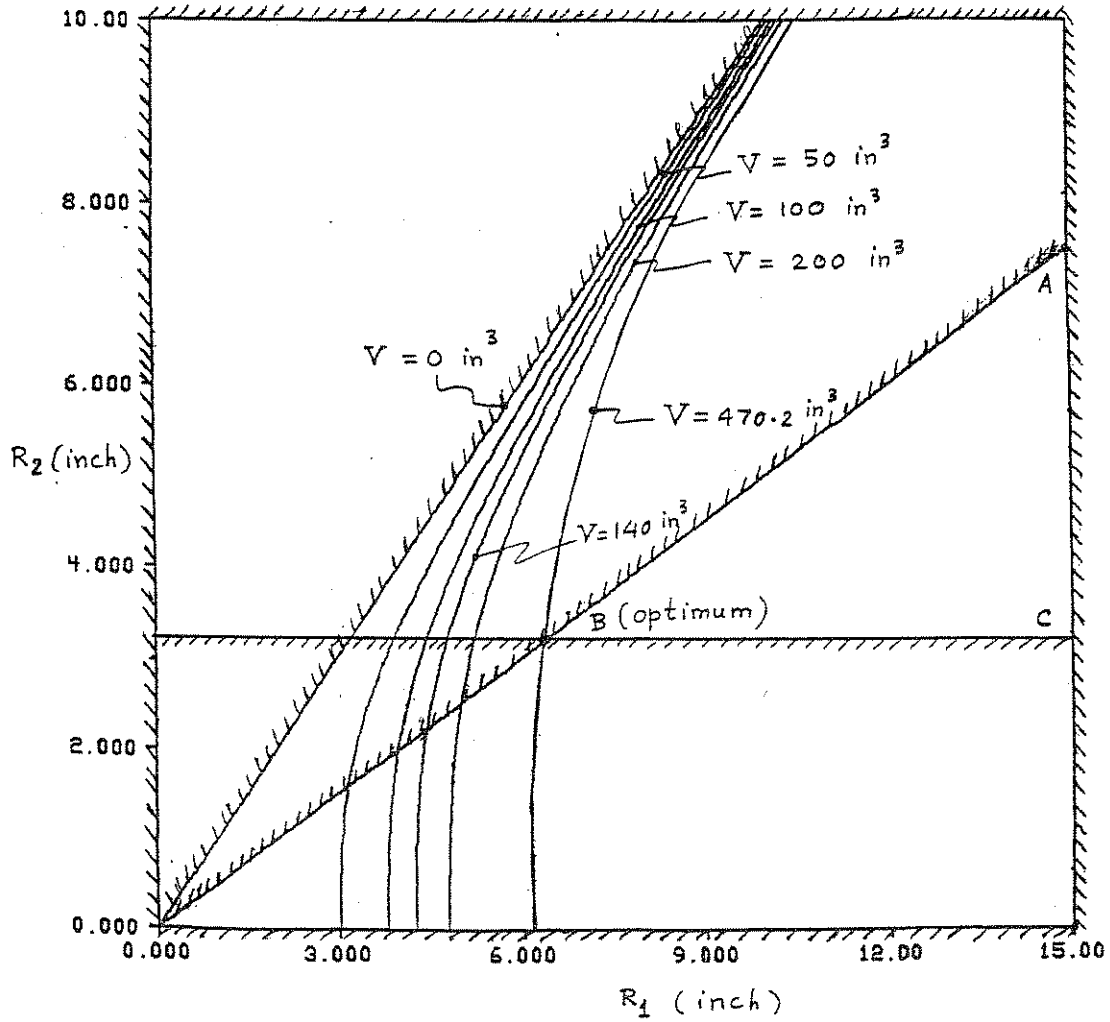
with $$p = \frac{F}{\pi(R_1^2 - R_2^2)}$$

$\therefore$ $T \geq 100$ becomes:

$$g_1(\vec{x}) = 30 R_2 - 100 \geq 0 \qquad (10)$$

Feasible space = ABCA

Optimum point = B (6.67, 3.33), $f^* = 470.2\ in^3$



1.10

$$\vec{X} = \begin{Bmatrix} \bar{r}_o \\ \bar{r}_i \end{Bmatrix}$$

$$\text{weight} = w = \pi \rho \ell (r_o^2 - r_i^2) \qquad (1)$$

$$\text{induced stress} = \tau = \frac{T r_o}{J} = \frac{4 T r_o}{\pi (r_o^4 - r_i^4)} \qquad (2)$$

Random variables: $\rho, \ell, r_o, r_i, T, \tau_o$

$$\bar{\rho} = 0.3, \quad \bar{\ell} = 50, \quad \bar{r}_o, \quad \bar{r}_i, \quad \bar{T} = 10^6, \quad \bar{\tau}_o = 5 * 10^4$$

$$\sigma_\rho = 0.01, \quad \sigma_\ell = \tfrac{1}{3}, \quad \sigma_{r_o} = \sigma_{r_i} = 0.02, \quad \sigma_T = 10^4, \quad \sigma_{\tau_o} = 5000$$

Partial derivatives:

$$\frac{\partial w}{\partial \rho} = \pi \ell (r_o^2 - r_i^2), \qquad \frac{\partial w}{\partial \ell} = \pi \rho (r_o^2 - r_i^2)$$

$$\frac{\partial w}{\partial r_o} = \pi \rho \ell (2 r_o), \qquad \frac{\partial w}{\partial r_i} = \pi \rho \ell (-2 r_i)$$

$$\frac{\partial \tau}{\partial T} = 4 r_o \Big/ \{\pi (r_o^4 - r_i^4)\}$$

$$\frac{\partial \tau}{\partial r_o} = \frac{4T}{\pi} \frac{3 r_o^4 + r_i^4}{(r_o^4 - r_i^4)^2}, \qquad \frac{\partial \tau}{\partial r_i} = -\frac{16 T r_o r_i^3}{\pi (r_o^4 - r_i^4)^2}$$

Mean and standard deviations:

$$\bar{w} = \pi \bar{\rho} \bar{\ell} (\bar{r}_o^2 - \bar{r}_i^2) = 47.1240 (\bar{r}_o^2 - \bar{r}_i^2) \qquad (3)$$

$$\sigma_w = \left\{ \left(\frac{\partial w}{\partial \rho}\Big|_m\right)^2 \sigma_\rho^2 + \left(\frac{\partial w}{\partial \ell}\Big|_m\right)^2 \sigma_\ell^2 + \left(\frac{\partial w}{\partial r_o}\Big|_m\right)^2 \sigma_{r_o}^2 + \left(\frac{\partial w}{\partial r_i}\Big|_m\right)^2 \sigma_{r_i}^2 \right\}^{\frac{1}{2}}$$

$$= \left\{ 2.4674 (\bar{r}_o^2 - \bar{r}_i^2)^2 + 986.9651 * 10^{-4} (\bar{r}_o^2 - \bar{r}_i^2)^2 + 3.5531 \bar{r}_o^2 + 3.5531 \bar{r}_i^2 \right\}^{\frac{1}{2}} \qquad (4)$$

$$\bar{\tau} = \frac{4 \bar{T} \bar{r}_o}{\pi (\bar{r}_o^4 - \bar{r}_i^4)} = \frac{1.2732 * 10^6 \bar{r}_o}{(\bar{r}_o^4 - \bar{r}_i^4)} \qquad (5)$$

$$\sigma_{\bar{\tau}} = \left\{ \left(\frac{\partial \tau}{\partial T}\Big|_m \right)^2 \sigma_T^2 + \left(\frac{\partial \tau}{\partial r_o}\Big|_m \right)^2 \sigma_{r_o}^2 + \left(\frac{\partial \tau}{\partial r_i}\Big|_m \right)^2 \sigma_{r_i}^2 \right\}^{\frac{1}{2}}$$

$$= \frac{1}{\left(\bar{r}_o^4 - \bar{r}_i^4 \right)^2} \left[1.6211 * 10^8 \; \bar{r}_o^2 \left(\bar{r}_o^4 - \bar{r}_i^4 \right)^2 \right.$$

$$\left. + 648.453 * 10^6 \left(3\bar{r}_o^4 + \bar{r}_i^4 \right)^2 + 103.7524 * 10^8 \; \bar{r}_o^2 \bar{r}_i^6 \right]^{\frac{1}{2}} \qquad (6)$$

Optimization problem:

Find $\vec{x} = \{ \bar{r}_o \quad \bar{r}_i \}^T$

which minimizes $f(\vec{x}) = \bar{\omega} + \sigma_{\omega}$

subject to

$$2.326 \sqrt{\sigma_{\tau}^2 + \sigma_{\tau_o}^2} \leq \bar{\tau} - \bar{\tau}_o$$

$$\bar{r}_o \geq \bar{r}_i$$

$$\bar{r}_i \geq 0, \quad \bar{r}_o \geq 0$$

where $\bar{\omega}$, σ_{ω}, σ_{τ} and $\bar{\tau}$ are given by Eqs. (3), (4), (6) and (5), respectively.

Graphical solution:

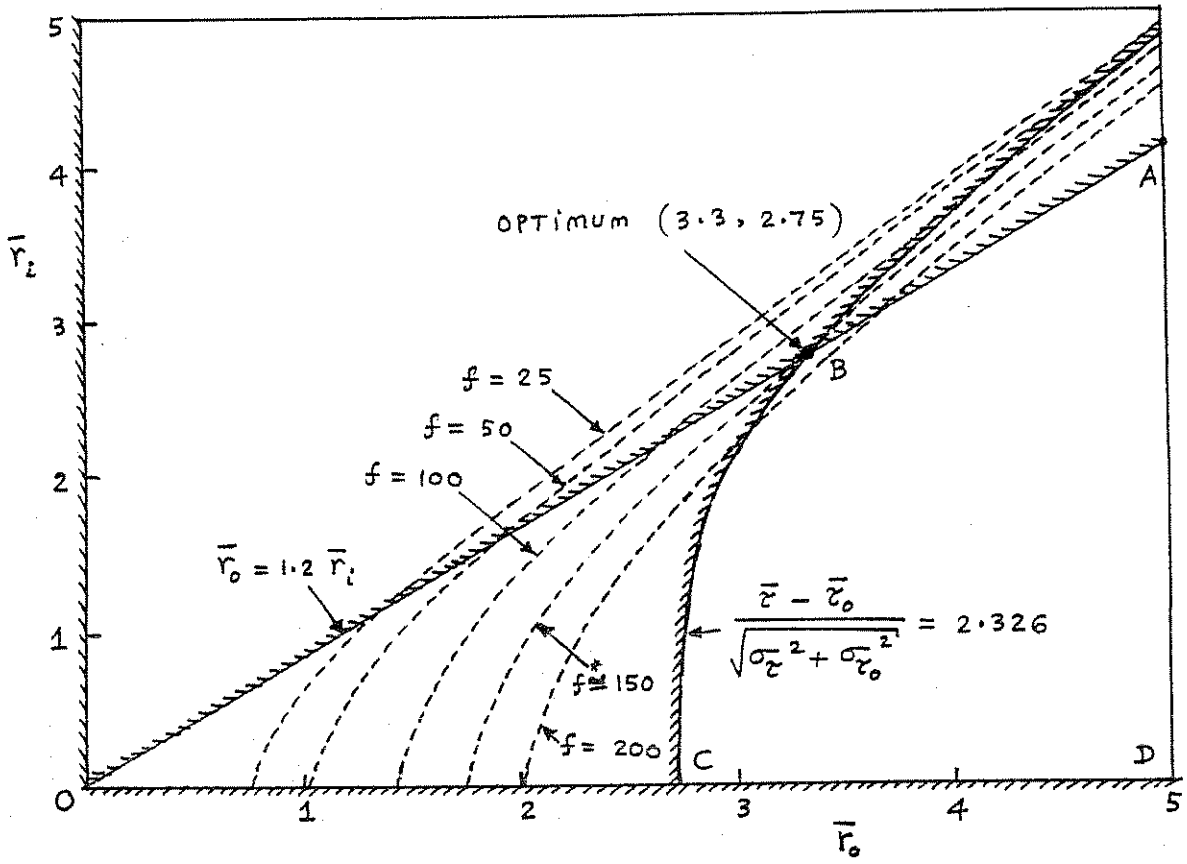
See figure on next page.

To avoid $\bar{r}_o = \bar{r}_i$, the constraint is modified to $\bar{r}_o = 1.2\,\bar{r}_i$ in the figure.

Feasible space: ABCD

optimum solution: $\bar{r}_i^* = 2.75''$, $\bar{r}_o = 3.30''$

$f \simeq 150.$



1.11

(a) $f = x_1^2 x_2^3 \quad ; \quad x_1 > 0, \; x_2 > 0$

$\ln f = 2 \ln x_1 + 3 \ln x_2$

or $\bar{f} = 2 y_1 + 3 y_2$

where $\bar{f} = \ln f$, $y_1 = \ln x_1$ and $y_2 = \ln x_2$

(b) $f = x_1^{x_2} \quad ; \quad x_1 > 0$

let $y_1 = x_1^{x_2} \equiv 10^{y_2 x_2}$ (1)

so that $x_1 = 10^{y_2}$ (2)

Eq. (1) can be separated now:

$y_2 x_2 = \log_{10} y_1$

$$\ln y_2 x_2 = \ln (\log_{10} y_1) = \ln y_2 + \ln x_2 \quad (3)$$

Eq. (2) $\Rightarrow$ $$y_2 = \log_{10} x_1 \quad (4)$$

$$\therefore \ln (\log_{10} y_1) = \ln (\log_{10} x_1) + \ln x_2 \quad (5)$$

1.12

The design variables are D, L, d and N and hence $X = \begin{Bmatrix} D \\ L \\ d \\ N \end{Bmatrix}$

The objective function is given by

Overall cost = cost of the tube + cost of the shell + cost of the floor space + cost of pumping cold fluid + cost of maintenance

$$\therefore f(X) = \alpha_2 NL + \alpha_3 D^{2.5} L + \alpha_4 DL + \alpha_{10} \left(\frac{\alpha_5 L}{d^5 N^2} + \alpha_6 NdL \right) \quad (E_1)$$

The constraint on the total length of the tube is

$$NL \geq \alpha_1 \quad (E_2)$$

and the constraint on the thermal energy transferred can be expressed as

$$\frac{\alpha_7}{N^{1.2} dL^{1.4}} + \frac{\alpha_8}{d^{0.2} L} \geq \alpha_9 \quad (E_3)$$

The restriction on the cross sectional area is given by

$$\frac{\pi D^2}{4} \geq Nd^2 \quad (E_4)$$

and the lower bounds on the design variables can be taken as

$$D \geq 0 \quad (E_5)$$
$$d \geq 0 \quad (E_6)$$
$$L \geq 0 \quad (E_7)$$
$$N \geq 0 \quad (E_8)$$

As the objective function, (E_1), and the constraints, (E_2) to (E_8), are posynomials in D, d, L and N, the problem is a geometric programming problem.

1.13 Find $\vec{x} = \{R_1, R_2, R_3, R_4, R_5\}^T$

to minimize $$f(\vec{x}) = \sum_{i=1}^{5} I_i^2 R_i = \sum_{i=1}^{5} \frac{c_i^2}{R_i}$$

subject to $$(I_i)_{min} \le \left(I_i = \frac{c_i}{R_i}\right) \le (I_i)_{max} \; ; \; i = 1, 2, \ldots, 5$$

$$R_i \ge 0 \; , \quad i = 1, 2, \ldots, 5$$

where c_i is the voltage drop, $c_i = R_i I_i$.

1.14 Define variables x_{ij} such that $x_{ij} = 1$ if she visits city j immediately after city i, and $= 0$ otherwise.

Design vector: $\vec{x} = \{x_{ij}\}$

Minimize $f(\vec{x})$ = total distance $= \sum_{i=1}^{n} \sum_{j=1}^{n} d_{ij} x_{ij}$

Constraints:

Each city must be visited once:

$$\sum_{i=1}^{n} x_{ij} = 1 \quad \text{for} \quad j = 1, 2, \ldots, n \quad \text{and} \quad i \neq j$$

She must leave every city:

$$\sum_{j=1}^{n} x_{ij} = 1 \quad \text{for} \quad i = 1, 2, \ldots, n \quad \text{and} \quad i \neq j$$

1.15

Design variables:

x_1 = acres of barley, x_2 = acres of oats

x_3 = acres of rice, x_4 = acres of wheat

x_5 = acres leased.

Costs per acre:

Barley: labor + water + fertilizer costs

$\$(300 + 10000 * 0.02 + 100 * 2) = \$700$

oats: $\$(200 + 7000 * 0.02 + 120 * 2) = \$580$

Rice: $\$(250 + 6000 * 0.02 + 160 * 2) = \$690$

Wheat: $\$(360 + 8000 * 0.02 + 200 * 2) = \$920$

Total farming cost = $\$(700 x_1 + 580 x_2 + 690 x_3 + 920 x_4)$

Using interest of 4%, total cost is:

$$1.04\,(700 x_1 + 580 x_2 + 690 x_3 + 920 x_4)$$

Profits per acre:

Barley: \$750, Oats: \$600, Rice: \$750, Wheat: \$800

Net return = (profits − costs) from crops + income from lease

$$= \left[(750 - 1.04 * 700) x_1 + (600 - 1.04 * 580) x_2 + (750 - 1.04 * 690) x_3 + (800 - 1.04 * 920) x_4\right] + 200 x_5$$

$$\therefore \quad f = 22.0 x_1 - 3.2 x_2 + 32.4 x_3 - 156.0 x_4 + 200.0 x_5 \qquad (1)$$

(Maximize)

Constraints:

Total acres: $x_1 + x_2 + x_3 + x_4 + x_5 = 200$ (2)

Available loan: $700 x_1 + 580 x_2 + 690 x_3 + 920 x_4 \le 50{,}000$ (3)

Available water:

$$10000 x_1 + 7000 x_2 + 6000 x_3 + 8000 x_4 \le 4 \times 10^5 \qquad (4)$$

Nonnegativity requirements:

$$x_i \ge 0 \;;\; i = 1, 2, 3, 4, 5 \qquad (5)$$

Problem is defined by Eqs. (1) to (5).

1.16 Design variables : $\{x_{ij}\}$

$$x_{ij} = \begin{cases} 1 & \text{if site } i \text{ is selected for target } j \\ 0 & \text{otherwise} \end{cases}$$

constraints :

$$\sum_{i=1}^{2} x_{ij} = 1 \quad , \quad j = 1, 2, 3, 4$$

Minimize total cost :

$$f = (4\,x_{11} + 7\,x_{21}) + (x_{12} + 9\,x_{22}) + (9\,x_{13} + 5\,x_{23}) + (7\,x_{14} + 2\,x_{24}) + 11\,(x_{11} + x_{12} + x_{13} + x_{14}) + 13\,(x_{21} + x_{22} + x_{23} + x_{24})$$

1.17 $\vec{x} = \begin{Bmatrix} x_1 \\ x_2 \\ \vdots \\ x_5 \end{Bmatrix}$

Minimize: $f(\vec{x}) = \text{power} = k_4\, N B\, x_1\, x_3\, x_5$

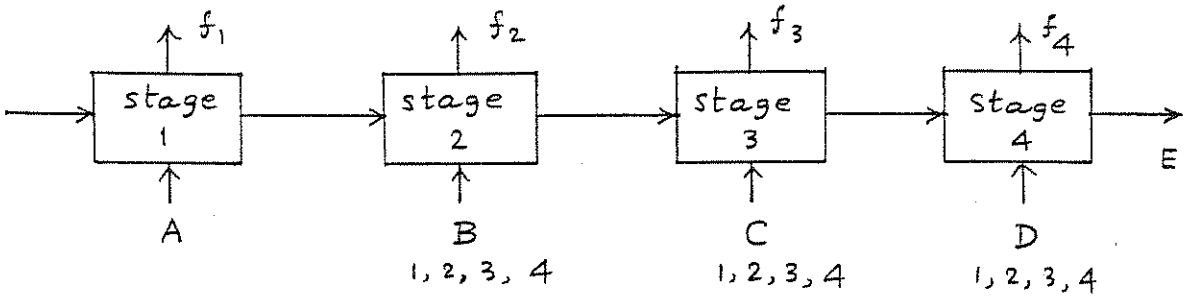
Constraints:

Air gap: $\frac{x_2}{2} - 2x_3 - 2x_4 \le k_1 \sqrt{x_2 + 7.5}$

Temperature: $k_2\, x_1\, x_2^{-1}\, x_4^{-1}\, x_5^{2} \le k_3\, x_1\, x_2\, \Delta T$

Lower limits: $x_i > 0 \; ; \; i = 1, 2, \ldots, 5$

1.18



Problem can be considered as a 4-stage discrete variable o.c. problem.

Decision (control) variables:

A (one), B (four choices), C (four choices), D (four choices)

state variables:

paths (lines) AB_i, B_iC_j, C_jD_k, D_kE

Cost functions:

f_1 = cost of pipeline AB_i

f_2 = cost of pipeline B_iC_j

f_3 = cost of pipeline C_jD_k

f_4 = cost of pipeline D_kE

O.C. problem:

Find (B_i, C_j, D_k)

to minimize $f = \sum_{i=1}^{4} f_i$

subject to the constraints:

AB_i is to be followed by B_iC_j

B_iC_j is to be followed by C_jD_k

C_jD_k is to be followed by D_kE

1.19

weight $= \rho l b d = (0.3)(50)\, b d = 15\, b d$

stress induced $= \dfrac{P_y}{bd} + \dfrac{6 P_x l}{b d^2} = \dfrac{25}{bd} + \dfrac{3000}{b d^2}$

Buckling stress = buckling load / area

$$= \frac{\pi^2 E\, b d^3}{48 l^2 (bd)} = \frac{\pi^2 (30 \times 10^6)\, b d^3}{48 (50^2)(bd)}$$

$$= 2467.4126\, d^2$$

Optimization problem:

Find $\vec{x} = \{ b \quad d \}^T$

to minimize $f(\vec{x}) = 15\, b d$

subject to

$$g_1(\vec{x}) = \frac{25}{bd} + \frac{3000}{b d^2} - 30000 \le 0 \qquad \text{(yielding)}$$

$$g_2(\vec{x}) = \frac{25}{bd} + \frac{3000}{b d^2} - 2467.4126\, d^2 \le 0 \qquad \text{(buckling)}$$

$$g_3(\vec{x}) = b - 2d \le 0$$

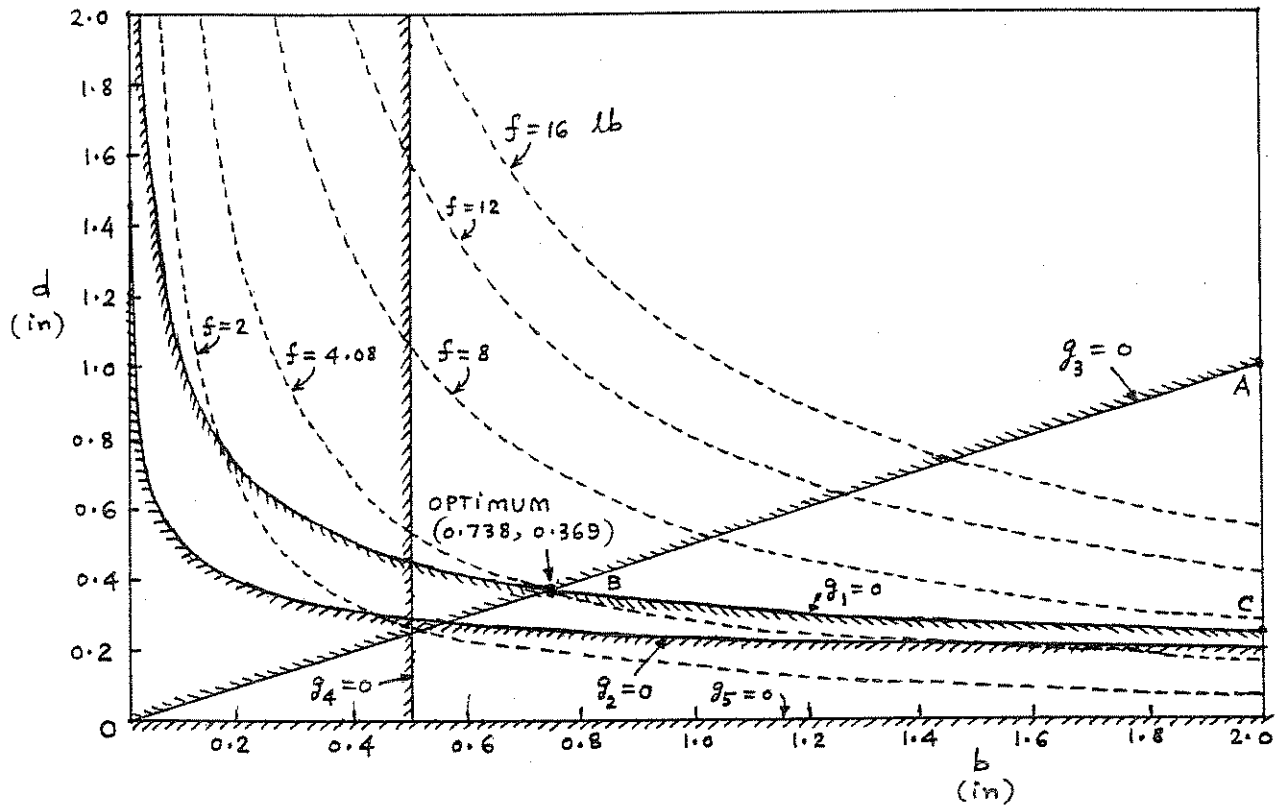
$$g_4(\vec{x}) = 0.5 - b \le 0$$

$$g_5(\vec{x}) = -d \le 0$$

Feasible space: ABCA (see figure)

optimum point: $b^* = 0.738''$, $d^* = 0.369''$

$f^* = 4.08$ lb.


2·0
1·8
1·6
1·4
1·2
1·0
0·8
0·6
0·4
0·2
0
d
(in)
f = 16 lb
f = 12
f = 8
f = 4·08
f = 2
g3 = 0
A
OPTIMUM
(0·738, 0·369)
B
g1 = 0
C
g4 = 0
g2 = 0
g5 = 0
0·2
0·4
0·6
0·8
1·0
1·2
1·4
1·6
1·8
2·0
b
(in)

1.20 $\vec{x} = \begin{Bmatrix} d \\ t \end{Bmatrix}$

Axial load in each bar = P

$P \sin\theta = W$ or $P = \dfrac{W\sqrt{b^2+h^2}}{h}$

Minimize: $f(\vec{x}) = 2\rho(\pi dt)\sqrt{b^2+h^2}$

Constraints:

yield stress: $g_1(\vec{x})$: $\dfrac{P}{\pi dt} = \dfrac{W\sqrt{b^2+h^2}}{h(\pi dt)} \le \sigma_y$

$P_{cri} = \dfrac{\pi^2 E I}{l^2}$: where $I = \dfrac{\pi}{64}(d_o^4 - d_i^4) = \dfrac{\pi dt(d^2+t^2)}{8}$

Buckling: $g_2(\vec{x})$: $P \le P_{cri}$

or $\dfrac{W\sqrt{b^2+h^2}}{h} \le \dfrac{\pi^3 E}{8} \cdot \dfrac{(d^2+t^2)}{(b^2+h^2)} \cdot d \cdot t$

side constraints:

$t \le d$, $d \ge 0$, $t \ge 0$

1.21 Optimization problem:

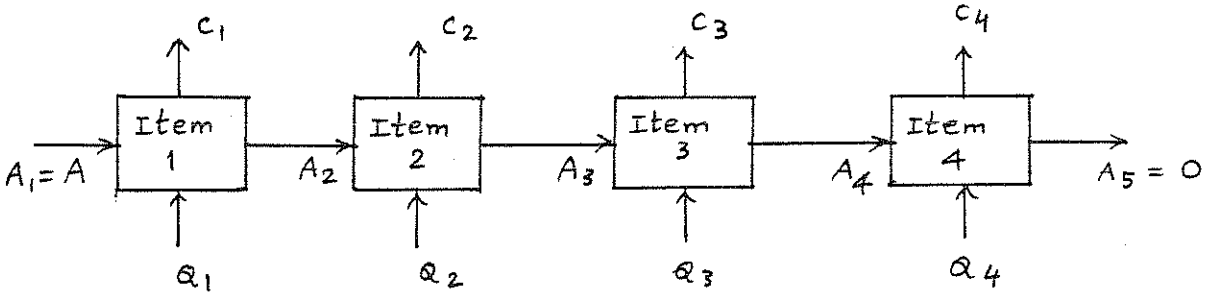
$\vec{x} = \begin{Bmatrix} Q_1 \\ Q_2 \\ Q_3 \\ Q_4 \end{Bmatrix}$,

Minimize $f = C = \sum_{i=1}^{4}\left(\dfrac{a_i}{Q_i} + b_i Q_i\right)$

subject to

$\sum_{i=1}^{4} Q_i d_i \le A$

Dynamic programming model:



Four-state discrete dynamic programming problem.

Q_i = lot size of item i ; $i = 1, 2, 3, 4$

(control or decision variables)

c_i = cost of ordering item i = $\frac{a_i}{Q_i} + b_i Q_i$

A_i = storage space (area) available for storing items $i, i+1, \ldots, 4$

1.22 Let location of H be defined by: (x_H, y_H)

$$\vec{x} = \begin{Bmatrix} x_H \\ y_H \end{Bmatrix}$$

Minimize $f(\vec{x}) = |x_H + 25| + |x_H - 20| + |x_H - 40| + |x_H + 30|$

$$+ |y_H + 35| + |y_H + 15| + |y_H - 30| + |y_H - 20|$$

subject to:

$$-15 \le x_H \le 15$$

$$-10 \le y_H \le 10$$

1.23 Let x_1, x_2 = amounts of A, B mixed to produce a unit weight of C

Minimize: $f(x_1, x_2) = x_1 + 2x_2$

subject to:

copper content: $0.8 x_1 + 0.6 x_2 \ge 0.75$

Zinc content: $0.1 x_1 + 0.2 x_2 \ge 0.15$

Lead content: $0.06 x_1 + 0.18 x_2 \ge 0.16$

Tin content: $0.04 x_1 + 0.02 x_2 \ge 0.03$

side constraints: $x_1 \ge 0, \quad x_2 \ge 0$

1.24

Design variables:

x_{ij} = fraction of the capacity of the ith process plant used to produce j^{th} grade oil

$(i = 1, 2, 3 \; ; \; j = 1, 2, 3, 4)$

Assume: 1. Production cost is proportional to the fraction of the capacity used for the specific grade of oil.

2. Production does not exceed the demand (since there is no incentive to produce more)

Objective function (for minimization):

$$\begin{aligned} f(\vec{x}) = {} & 750\, x_{11} + 800\, x_{21} + 900\, x_{31} + 10\{50 - 100\, x_{11} \\ & - 150\, x_{21} - 200\, x_{31}\} \\ & + 900\, x_{12} + 950\, x_{22} + 1000\, x_{32} + 12\{150 - 100\, x_{12} \\ & - 150\, x_{22} - 200\, x_{32}\} \\ & + 1000\, x_{13} + 1100\, x_{23} + 1200\, x_{33} + 16\{100 - 100\, x_{13} \\ & - 150\, x_{23} - 200\, x_{33}\} \\ & + 1200\, x_{14} + 1400\, x_{24} + 1600\, x_{34} + 20\{75 - 100\, x_{14} \\ & - 150\, x_{24} - 200\, x_{34}\} \end{aligned}$$

= cost of production + penalty for not meeting the demand

Constraints:

To meet the demand;

$$100\, x_{11} + 150\, x_{21} + 200\, x_{31} \leq 50$$
$$100\, x_{12} + 150\, x_{22} + 200\, x_{32} \leq 150$$
$$100\, x_{13} + 150\, x_{23} + 200\, x_{33} \leq 100$$
$$100\, x_{14} + 150\, x_{24} + 200\, x_{34} \leq 75$$

To utilize the capacity of each process plant;

$$x_{11} + x_{12} + x_{13} + x_{14} = 1.0$$
$$x_{21} + x_{22} + x_{23} + x_{24} = 1.0$$
$$x_{31} + x_{32} + x_{33} + x_{34} = 1.0$$

Side constraints:

$$0 \le x_{ij} \le 1 \quad ; \quad i = 1, 2, 3 \; ; \; j = 1, 2, 3, 4$$

1.25 Design vector: $\vec{x} = \begin{Bmatrix} t_m \\ t_d \end{Bmatrix}$

t_m, t_d = time spent on math, design

Maximize: $f(\vec{x}) = 4\, g_m + 3\, g_d = 4\left(\frac{t_m}{6}\right) + 3\left(\frac{t_d}{5}\right)$

Constraints:

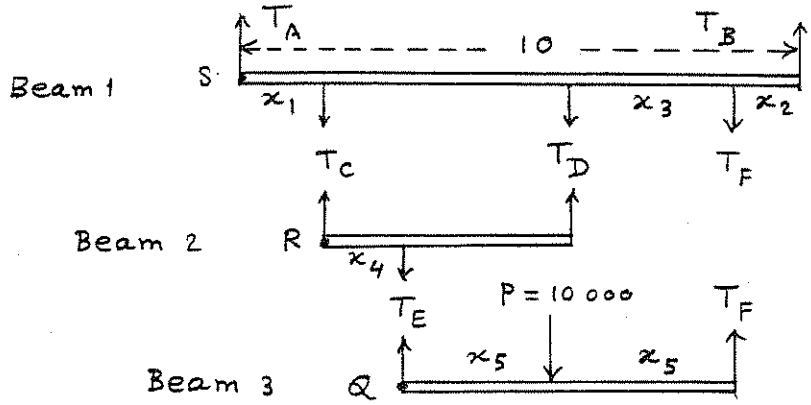
$t_m + t_d \le 40$ (maximum time available)

$t_d \ge \frac{75}{60}\, t_m$ (at least 75 minutes for design for every 60 minutes for math)

$0 \le t_m \le 24$ (24 hrs to get "A" in math)

$0 \le t_d \le 20$ (20 hrs to get "A" in design)

1.26



$$2x_5 = 10 - x_1 - x_4 - x_2 \quad \text{or} \quad x_5 = 5 - \tfrac{1}{2}x_1 - \tfrac{1}{2}x_2 - \tfrac{1}{2}x_4 \qquad (1)$$

$$\vec{x} = \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix}$$

Objective functions: $f_1(\vec{x}) = T_A(\vec{x})$ (2)

$f_2(\vec{x}) = T_B(\vec{x})$ (3)

Constraints:

$T_C \geq 0$, $T_D \geq 0$, $T_E \geq 0$, $T_F \geq 0$ (4)

$x_i \geq 0$, $i = 1, 2, 3, 4$ (5)

Analysis of system:

Beam 3: Force equilibrium: $T_E + T_F = P = 10000$ (6)

Moment equilibrium: $P \cdot x_5 - 2x_5 T_F = 0$ (7)

Beam 2: Force equilibrium: $T_C + T_D = T_E$ (8)

Moment equilibrium: $T_E x_4 - T_D\{10 - (x_2 + x_3 + x_1)\} = 0$ (9)

Beam 1: Force equilibrium: $T_A + T_B = T_C + T_D + T_F$ (10)

Moment equilibrium: $T_C x_1 + T_D\{10 - (x_2 + x_3)\}$

$+ T_F(x_1 + x_4 + 2x_5) - 10 T_B = 0$ (11)

Solution of equilibrium equations:

Eq. (7): $P = 10\,000 = 2T_F$, $T_F = 5000$ (12)

Eqs. (6) and (12): $T_E = T_F = 5000$ (13)

Eq. (9): $$T_D = \frac{T_E\, x_4}{10 - x_1 - x_2 - x_3} = \frac{5000\, x_4}{10 - x_1 - x_2 - x_3} \qquad (14)$$

Eqs. (8) and (14):

$$T_C = T_E - T_D = 5000 - \frac{5000\, x_4}{10 - x_1 - x_2 - x_3} \qquad (15)$$

Eq. (11):

$$10\, T_B = T_C\, x_1 + T_D\,(10 - x_2 - x_3) + T_F\,(x_1 + x_4 + 2x_5)$$

or

$$T_B = \left\{500 - \left(\frac{500\, x_4}{10 - x_1 - x_2 - x_3}\right)\right\} x_1 + \left(\frac{500\, x_4}{10 - x_1 - x_2 - x_3}\right)(10 - x_2 - x_3) + 500\,(10 - x_2) \qquad (16)$$

Eqs. (10) and (16):

$$T_A = T_C + T_D + T_F - T_B$$

$$= 10000 - \left\{500 - \left(\frac{500\, x_4}{10 - x_1 - x_2 - x_3}\right)\right\} x_1 - \left(\frac{500\, x_4}{10 - x_1 - x_2 - x_3}\right)(10 - x_2 - x_3) - 500\,(10 - x_2) \qquad (17)$$

Optimization problem:

Find $\vec{x}$ to minimize the objective functions $f_1(\vec{x})$ and $f_2(\vec{x})$ given by Eqs. (2) and (3) subject to the constraints of Eqs. (4) and (5).

1.27

$$\vec{x} = \begin{Bmatrix} p \\ d \\ h \\ s \end{Bmatrix} = \begin{Bmatrix} \text{pitch} \\ \text{major diameter} \\ \text{nut height} \\ \text{length of screw} \end{Bmatrix}$$

Minimize: $f = \frac{\pi s}{8}(2d^2 - 2dp + p^2) = \text{volume of screw}$

$$\equiv \frac{\pi}{4}\left\{\frac{d^2 + (d-p)^2}{2}\right\} s \qquad (1)$$

constraints:

screw should be self-locking:

$$\pi \mu d_m > l \qquad (2)$$

where $d_m = d - \frac{p}{2}$ = mean diameter

and $l = np$ $\left(\begin{matrix} n=1 \text{ for single threads,} \\ n=2 \text{ for double threads} \end{matrix}\right)$

shear stress in screw thread (τ):

$$\tau = \frac{2F}{\pi d h} \leq \frac{0.577\,\sigma_y}{F.S.} \qquad (3)$$

where F = load applied on screw

$F.S.$ = factor of safety

Bearing stress in threads (σ):

$$\frac{-4pF}{\pi h (d^2 - d_r^2)} \leq \frac{\sigma_y}{F.S.} \qquad (4)$$

where $d_r = d - p$ = minor diameter.

Critical buckling load:

$$\frac{\pi^3 E (d-p)^4}{16 s^2} - F \leq 0 \qquad (5)$$

where E = Young's modulus.

$\therefore$ Optimization problem is defined by Eqs. (1) to (5).

1.28 (a)

$$\vec{X} = \{ x_1 \quad x_2 \quad x_3 \quad x_4 \}^T \tag{1}$$

Minimize $f(\vec{x}) = \rho l (x_1 x_2 - x_3 x_4) = 34.08 (x_1 x_2 - x_3 x_4)$ (2)

Constraints:

Deflection due to self weight and $F_y \leq 0.5$ inch

$$\frac{5 w_0 l^4}{384 E I_{xx}} + \frac{F_y l^3}{48 E I_{xx}} \leq 0.5 \tag{3}$$

where

w_0 = self weight per unit length of beam

$$= \rho (x_1 x_2 - x_3 x_4) = 0.284 (x_1 x_2 - x_3 x_4) \tag{4}$$

$$I_{xx} = \frac{1}{12} (x_1 x_2^3 - x_3 x_4^3) \tag{5}$$

Using $F_y = 300$, $l = 120$, $E = 30 \times 10^6$, and Eqs. (4) and (5) in Eq. (3), we obtain

$$0.610262 (x_1 x_2 - x_3 x_4) + 8.64 \leq (x_1 x_2^3 - x_3 x_4^3) \tag{6}$$

Axial load, P, should not cause buckling.

Assume: Interaction of self weight or F_y with P (during buckling) is negligible.

$$P \leq \frac{\pi^2 E I_{xx}}{l^2} \quad \text{(To avoid buckling in } yz\text{-plane)} \tag{7}$$

$$P \leq \frac{\pi^2 E I_{yy}}{l^2} \quad \text{(To avoid buckling in } xz\text{-plane)} \tag{8}$$

where

$$I_{xx} = \frac{1}{12} (x_1 x_2^3 - x_3 x_4^3) \tag{9}$$

$$I_{yy} = \frac{1}{12} (x_2 x_1^3 - x_4 x_3^3) \tag{10}$$

Eqs. (7) to (10) lead to:

$$40000 - 1713.481 (x_1 x_2^3 - x_3 x_4^3) \leq 0 \tag{11}$$

$$40000 - 1713.481 (x_2 x_1^3 - x_4 x_3^3) \leq 0 \tag{12}$$

side constraints:

$$x_i \geq 0, \quad i = 1, 2, 3, 4 \tag{13}$$

$$x_1 - x_3 \geq 0 \tag{14}$$

$$x_2 - x_4 \geq 0 \tag{15}$$

Optimization problem:

Find $\vec{x}$ to minimize $f(\vec{x})$ given by Eq. (2) subject to the constraints of Eqs. (6), (11), (12), and (13) to (15).

(b) For a solid section, $x_3 = x_4 = 0$.

Problem becomes:

Find $\vec{x} = \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix}$

to minimize $f(\vec{x}) = 34.08\, x_1 x_2$

subject to

$$g_1 = \frac{0.610262}{x_2^2} + \frac{8.64}{x_1 x_2^3} \leq 1$$

$$g_2 = 40000 - 1713.481\, x_1 x_2^3 \leq 0$$

$$g_3 = 40000 - 1713.481\, x_2 x_1^3 \leq 0$$

$g_4, g_5 = x_i \geq 0 \; ; \; i = 1, 2$

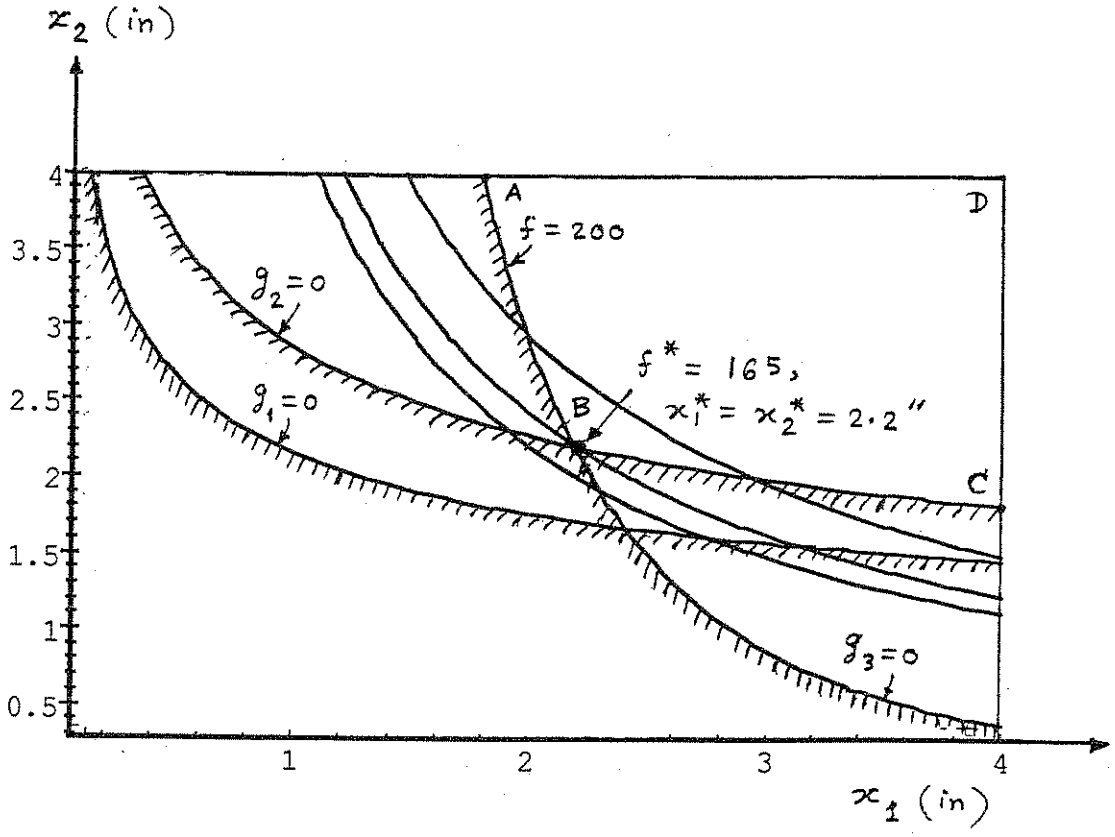
Graphical solution:

shown in the figure on next page.

Feasible space: ABCDA

Optimum solution: $x_1^* = 2.2''$, $x_2^* = 2.2''$

$f^* \simeq 165$ lb.


x_2 (in)
4
3.5
3
2.5
2
1.5
1
0.5
1
2
3
4
x_1 (in)
A
B
C
D
$f = 200$
$f^* = 165,$
$x_1^* = x_2^* = 2.2''$
$g_1 = 0$
$g_2 = 0$
$g_3 = 0$

1.29 Design vector:

$$\vec{x} = \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} \text{outer radius of shell and head} \\ \text{wall thickness of vessel} \\ \text{length of shell} \\ \text{wall thickness of head} \end{Bmatrix}$$

Minimize $f(\vec{x})$ = volume of material

$$= \pi x_3 \left\{ x_1^2 - (x_1 - x_2)^2 \right\} + \frac{4\pi}{3} \left\{ x_1^3 - (x_1 - x_4)^3 \right\}$$

subject to:

volume: $g_1(\vec{x}) = \pi x_3 (x_1 - x_2)^2 + \frac{4\pi}{3} (x_1 - x_4)^3 - 4{,}619{,}606 \leq 0$

(1 gallon = 230.9803 in^3)

thickness constraints:

$$g_2(\vec{x}) = x_2 - \frac{p R}{S \cdot e + 0.4\, p} \leq 0$$

$$g_3(\vec{x}) = x_4 - \frac{p \cdot R}{S \cdot e + 0.8\, p} \leq 0$$

side constraints:

$$x_i \geq 0 \;, \quad i = 1, 2, 3, 4$$

1.30 Design vector: $\vec{x} = \{ r_i \;, r_o \;, b \;, h \}^T \geq \vec{0}$

Minimize: $f(\vec{x})$ = volume of material $= \frac{3\pi}{2} \left(\frac{r_i + r_o}{2} \right) bh$

subject to:

$$\sigma_B = \frac{M c_i}{A e r_i} \leq \sigma_y$$

$$\sigma_A = \frac{M c_o}{A e r_o} \leq \sigma_y$$

where $r_o = r_i + h$; $M = F \cdot R$; $R = \frac{r_o + r_i}{2}$; $r_n = \frac{h}{\ln \frac{r_o}{r_i}}$;

$e = R - r_n$; $c_i = r_n - r_i$; $c_o = r_o - r_n$; $A = b \cdot h$.

1.31

$\vec{x} = \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix}$

Minimize $f(\vec{x}) = 0.1\, x_1 + 0.05773\, x_2$

subject to

$$\frac{0.6}{x_1} + \frac{0.3464}{x_2} - 0.1 \le 0$$

$$x_1 \ge 6$$

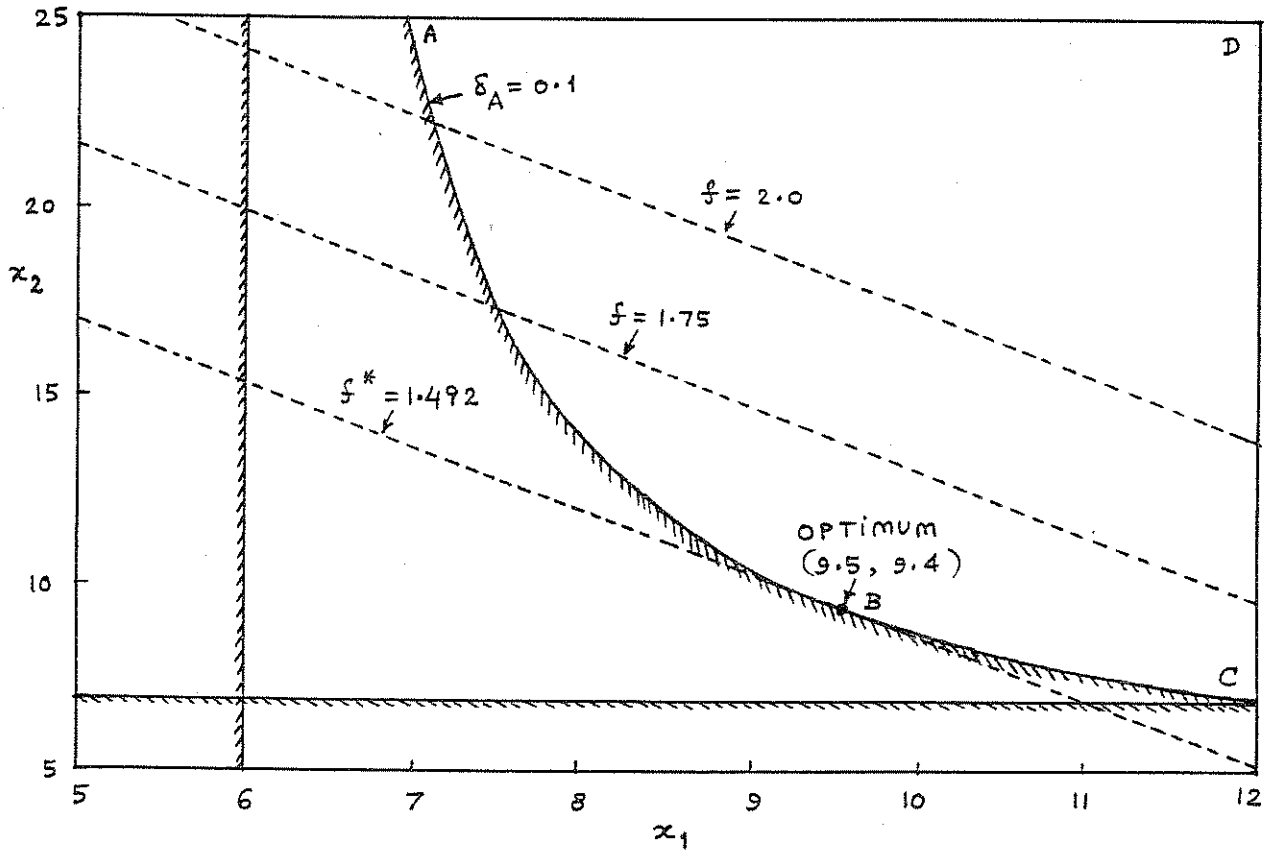
$$x_2 \ge 7$$

Graphical solution:

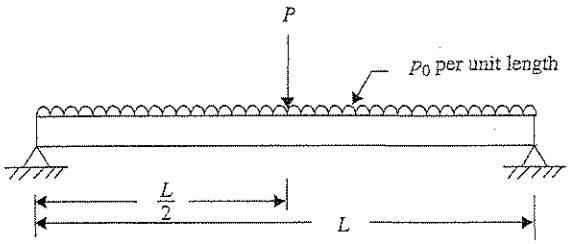
shown in the figure below.

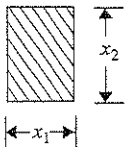
Feasible space: ABCDA

optimum solution: $x_1^* = 9.5$, $x_2^* = 9.4$, $f^* = 1.492$.



1.32





Cross-section

(a) Stress due to distributed load:

Max. bending moment $= M_{A1} = \frac{p_0 L}{2} \cdot \frac{L}{2} - \frac{p_0 L}{2} \cdot \frac{L}{4} = \frac{p_0 L^2}{8}$

$$\sigma_{A1} = \frac{M_{A1} \cdot y}{I} = \frac{p_0 L^2}{8} \cdot \frac{\left(\frac{x_2}{2}\right)}{\frac{1}{12} x_1 x_2^3} = c_1 x_1^{-1} x_2^{-2}$$

where $c_1 = \frac{3}{4} p_0 L^2 = \frac{3}{4} (10^6)(1^2) = 0.75 \times 10^6$

Stress due to concentrated load:

Max. bending moment $= M_{A2} = \frac{P}{2} \cdot \frac{L}{2} = \frac{PL}{4}$

$$\sigma_{A2} = \frac{M_{A2} \cdot y}{I} = \frac{PL}{4} \frac{\left(\frac{x_2}{2}\right)}{\frac{1}{12} x_1 x_2^3} = c_2 x_1^{-1} x_2^{-2}$$

where $c_2 = \frac{3PL}{2} = 10^5 (1)(1.5) = 0.15 \times 10^6$

Total stress $= \sigma_t = \sigma_{A1} + \sigma_{A2} = 0.90 \, x_1^{-1} x_2^{-2} (10^6)$

stress constraint: $\sigma_t - \sigma_0 \leq 0$

or $g_1(\vec{x}) = 0.90 \, x_1^{-1} x_2^{-2} (10^6) - 220 \times 10^6 \leq 0$

or $0.90 \, x_1^{-1} x_2^{-2} - 220 \leq 0$

Deflection due to distributed load:

$$\delta_{A1} = \frac{5}{384} \frac{p_0 L^4}{EI} = \frac{5}{384} \frac{10^6 (1^4)}{(207 \times 10^9)\left(\frac{1}{12} x_1 x_2^3\right)}$$

$$= \frac{754.831 \times 10^{-9}}{x_1 x_2^3}$$

Deflection due to concentrated load:

$$\delta_{A2} = \frac{PL^3}{48 EI} = \frac{10^5 (1^3)}{48 (207 \times 10^9)\left(\frac{1}{12} x_1 x_2^3\right)} = \frac{120.773 \times 10^{-9}}{x_1 x_2^3}$$

Total deflection $= \delta_t = \delta_{A1} + \delta_{A2} = \dfrac{875.604 \times 10^{-9}}{x_1 x_2^3}$

Deflection constraint : $\delta_t - \delta_0 \le 0$

or $g_2(\vec{x}) = 875.604 \times 10^{-9}\, x_1^{-1} x_2^{-3} - 0.02 \le 0$

objective function for minimization:

$f(\vec{x}) = \text{weight} = \rho_w L x_1 x_2 = 76.5 \times 10^3 (1) x_1 x_2$

$= 76500\, x_1 x_2$

other constraints:

$g_3(\vec{x}) = x_1 - x_2 \le 0$; $g_4(\vec{x}) = 0.04 - x_1 \le 0$;

$g_5(\vec{x}) = x_1 - 0.12 \le 0$; $g_6(\vec{x}) = 0.06 - x_2 \le 0$;

$g_7(\vec{x}) = x_2 - 0.20 \le 0$

(b)

```
function problem 132

lb=[0.04 0.06];ub=[0.12 0.20];

x0=[0.05 0.15];
f0=obj(x0);g0=gcons(x0);

[xv,fval]=fmincon(@obj,x0,[],[],[],[],lb,ub,@cons)
gx=gcons(xv);

fprintf('*******Results Problem 4*********\n');
fprintf('*********************************\n');
fprintf(' x0(1)          x0(2)\n');
fprintf(' %6.4f          %6.4f\n', x0(1),x0(2));
fprintf('*********************************\n')
fprintf(' f(x0)\n');
fprintf(' %6.4f\n',f0);
fprintf('*********************************\n')
fprintf(' g1(x0)      g2(x0)      g3(x0)\n');
fprintf(' %6.4f     %6.4f     %6.4f\n',g0(1),g0(2),g0(3));
fprintf('*********************************\n')
fprintf(' x*(1)          x*(2)\n');
fprintf(' %6.4f          %6.4f\n', xv(1),xv(2));
fprintf('*********************************\n')
fprintf(' f(x*)\n');
fprintf(' %6.4f\n',fval);
```

```
fprintf('*********************************\n')
fprintf(' g1(x*)      g2(x*)      g3(x*)\n');
fprintf(' %6.4f     %6.4f     %6.4f\n',gx(1),gx(2),gx(3));
end
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
function f=obj(x)
f=76500*x(1)*x(2);
end
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
function [c,ceq]=cons(x)
c(1)=0.00403409/(x(1)*(x(2)^2))-1;
c(2)=43.7809375*(10^-6)/(x(1)*(x(2)^3))-1;
c(3)=x(1)/x(2)-1;
ceq=0;
end
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
function g=gcons(v);
g(1)=0.00403409/(v(1)*(v(2)^2))-1;
g(2)=43.7809375*(10^-6)/(v(1)*(v(2)^3))-1;
g(3)=v(1)/v(2)-1;
end
```

Results

```
*********************************

 x0(1)          x0(2)

 0.0500         0.1500

*********************************

 f(x0)

 573.7500

*********************************

 g1(x0)     g2(x0)     g3(x0)

 2.5859     -0.7406    -0.6667

*********************************
```

x*(1) x*(2)

0.1009 0.2000

****

f(x*)

1543.0394 %%%%% large than f(x0)

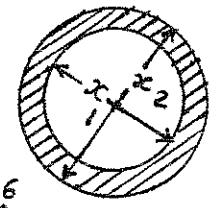
****

g1(x*) g2(x*) g3(x*)

0.0000 -0.9457 -0.4957

1.33

$I = \frac{\pi}{64}(x_2^4 - x_1^4)$



(a) $\sigma_{A1} = \frac{p_0 L^2}{8} \cdot \frac{x_2}{2} \cdot \frac{64}{\pi} \cdot \frac{1}{(x_2^4 - x_1^4)}$

$= c_1 \left(\frac{x_2}{x_2^4 - x_1^4} \right)$ where $c_1 = \frac{4}{\pi} p_0 L^2 = 1.273236 \times 10^6$

$\sigma_{A2} = \frac{PL}{4} \cdot \frac{x_2}{2} \cdot \frac{64}{\pi} \cdot \frac{1}{(x_2^4 - x_1^4)} = c_2 \frac{x_2}{(x_2^4 - x_1^4)}$

where $c_2 = \frac{8}{\pi} PL = 0.254647 \times 10^6$

Stress constraint: $g_1(\vec{x}) = \sigma_t - \sigma_0 \leq 0$

where $\sigma_t = \sigma_{A1} + \sigma_{A2} = 1.527883 \times 10^6 \frac{x_2}{(x_2^4 - x_1^4)}$

and $\sigma_0 = 220 \times 10^6$.

$\delta_{A1} = \frac{5}{384} \frac{p_0 L^4}{EI} = \frac{5}{384} \cdot \frac{10^6 (1^4)}{207 \times 10^9} \cdot \frac{64}{\pi} \cdot \frac{1}{(x_2^4 - x_1^4)}$

$= 1281 \times 10^{-9} / (x_2^4 - x_1^4)$

$\delta_{A2} = \frac{PL^3}{48EI} = \frac{1}{48} \cdot \frac{10^5 (1^3)}{207 \times 10^9} \cdot \frac{64}{\pi} \cdot \frac{1}{(x_2^4 - x_1^4)}$

$= 205.03 \times 10^{-9} / (x_2^4 - x_1^4)$

Deflection constraint: $g_2(\vec{x}) = \delta_t - \delta_0 \leq 0$

where $\delta_t = \delta_{A1} + \delta_{A2} = [1486.03/(x_2^4 - x_1^4)](10^{-9})$

and $\delta_0 = 0.02$

objective function: $f(\vec{x}) = \rho_w L \frac{\pi}{4}(x_2^2 - x_1^2)$

$= 60083.1 (x_2^2 - x_1^2)$

other constraints:

$g_3(\vec{x}) = x_1 - x_2 \leq 0$; $g_4(\vec{x}) = 0.04 - x_1 \leq 0$;

$g_5(\vec{x}) = x_1 - 0.12 \leq 0$; $g_6(\vec{x}) = 0.06 - x_2 \leq 0$;

$g_7(\vec{x}) = x_2 - 0.20 \leq 0$; $g_8(\vec{x}) = x_1 - 0.75 x_2 \leq 0$

(b) function problem 133

```
lb=[0.04 0.06];ub=[0.12 0.20];
x0=[0.10 0.15];
f0=obj(x0);g0=gcons(x0);

[xv,fval]=fmincon(@obj,x0,[],[],[],[],lb,ub,@cons)
gx=gcons(xv);

fprintf('***************Results Problem 5***************\n');
fprintf('***************************************************\n');
fprintf('  x0(1)                x0(2)\n');
fprintf(' %8.4f              %8.4f\n', x0(1),x0(2));
fprintf('***************************************************\n');
fprintf('  f(x0)\n');
fprintf('  %8.4f\n',f0);
fprintf('***************************************************\n');
fprintf('  g1(x0)         g2(x0)         g3(x0)\n');
fprintf('  %8.4f       %8.4f       %8.4f\n',g0(1),g0(2),g0(3));
fprintf('***************************************************\n');
fprintf('  x*(1)              x*(2)\n');
fprintf(' %8.4f          %8.4f\n', xv(1),xv(2));
fprintf('***************************************************\n');
```

```
fprintf('  f(x*)\n');
fprintf(' %8.4f\n',fval);
fprintf('*************************************************\n');
fprintf('  g1(x*)              g2(x*)          g3(x*)\n');
fprintf(' %8.4f  %8.4f  %8.4f\n',gx(1),gx(2),gx(3));
end
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
function f=obj(x)
f=60083.1*(x(2)^2-x(1)^2);
end
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
function [c,ceq]=cons(x)
c(1)=(9424.0*(10^(-9)))*x(2)/(x(2)^4-x(1)^4)-1;
c(2)=(74.30158*(10^(-6)))/(x(2)^4-x(1)^4)-1;
c(3)=x(1)/x(2)-0.75;
ceq=0;
end
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
function g=gcons(xv);
g(1)=(9424.0*(10^-9))*xv(2)/(xv(2)^4-xv(1)^4)-1;
g(2)=(74.30158*(10^-6))/(xv(2)^4-xv(1)^4)-1;
g(3)=xv(1)/xv(2)-1;
end
```

Results

```
************************************************

  x0(1)              x0(2)

  0.1000             0.1500

************************************************

  f(x0)

  751.0387

************************************************

  g1(x0)        g2(x0)        g3(x0)

  -0.9965       -0.8171       -0.3333

************************************************
```

```
x*(1)          x*(2)

0.1188         0.1188

**********************************************

f(x*)

-0.0000

**********************************************

g1(x*)                  g2(x*)                  g3(x*)

-1423800248259.4729  -94525671071191.0620   0.0000
```

1.34

```
function problem 134
clear;clc;

x0=[12 12];
f0=obj(x0);g0=gcons(x0);

[xv,fval]=fmincon(@obj,x0,[],[],[],[],[],[],@cons);
gx=gcons(xv);
fprintf('******Results Problem1*******\n');
fprintf('*****************************\n')
fprintf(' x0(1)     x0(2)\n');
fprintf(' %6.4f   %6.4f\n', x0(1),x0(2));
fprintf('*****************************\n')
fprintf(' g(x0) \n');
fprintf(' %6.4f   %6.4f   %6.4f\n',g0(1),g0(2),g0(3));
fprintf('*****************************\n')
fprintf(' f(x0)\n');
fprintf('%8.4f\n',f0);
fprintf('*****************************\n')
fprintf('  x*(1)        x*(2)\n');
fprintf('  %6.4f       %6.4f\n', xv(1),xv(2));
fprintf('*****************************\n')
fprintf('   f(x*)\n');
fprintf(' %8.4f\n',fval);
fprintf('*****************************\n')
fprintf('    g(x*) \n');
fprintf('   %6.4f   %6.4f   %6.4f\n',gx(1),gx(2),gx(3));
end
```

```
%%%%%%%%%objective%%%%%%%%%
function f=obj(x);
f=0.1*x(1)+0.05773*x(2);
end
%%%%%%%% constraint%%%%%%%%
function [c,ceq]=cons(x);
c(1)=0.6/x(1)+0.3464/x(2)-0.1;
c(2)=6-x(1);
c(3)=7-x(2);
ceq=0;
end
%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%% value of constraint%%%%%%%%
function g=gcons(xv);
g(1)=0.6/xv(1)+0.3464/xv(2)-0.1;
g(2)=6-xv(1);
g(3)=7-xv(2);
end
```

Results

```
*****************************

 x0(1)    x0(2)

 12.0000   12.0000

*****************************

 g(x0)

 -0.0211   -6.0000   -5.0000

*****************************

 f(x0)

  1.8928

*****************************

 x*(1)       x*(2)

 9.4639      9.4642

*****************************
```

```
f(x*)

1.4928

**************************

g(x*)

0.0000  -3.4639  -2.4642
```

1.35

```
function problem 135  %%%GA
lb=[0.1 0.1];ub=[2 2.5];
x0=[0.5 1.0];
f0=obj(x0);g0=gcons(x0);

[xv,fval]=fmincon(@obj,x0,[],[],[],[],lb,ub,@cons);
gx=gcons(xv);

fprintf('****** Results Problem 2 ********\n');
fprintf('*********************************\n')
fprintf(' x0(1)       x0(2)\n');
fprintf(' %6.4f      %6.4f\n', x0(1),x0(2));
fprintf('*********************************\n')
fprintf(' f(x0)\n');
fprintf('%8.4f\n',f0);
fprintf('*********************************\n')
fprintf(' g1(x0)       g2(x0) \n');
fprintf(' %6.4f      %6.4f\n',g0(1),g0(2));
fprintf('*********************************\n')
fprintf('  x*(1)       x*(2)\n');
fprintf('  %6.4f      %6.4f\n', xv(1),xv(2));
fprintf('*********************************\n')
fprintf('  f(x*)\n');
fprintf(' %8.4f\n',fval);
fprintf('*********************************\n')
fprintf('  g1(x*)       g2(x*) \n');
fprintf('  %6.4f      %6.4f\n',gx(1),gx(2));
end

%%%%%%%%%%%%%%%%%%%%%
function f=obj(x);
f=56.6*x(2)*sqrt(1+x(1)^2);
end
%%%%%%%%%%%%%%%%%%%%%
```

```
function [c,ceq]=cons(x);
c(1)=(x(1)+1)*sqrt(1+x(1)^2)/(x(1)*x(2))-5.656854;
c(2)=(x(1)-1)*sqrt(1+x(1)^2)/(x(1)*x(2))-5.656854;
ceq=0;
end
%%%%%%%%%%%%%%%%
function g=gcons(v);
g(1)=(v(1)+1)*sqrt(1+v(1)^2)/(v(1)*v(2))-5.656854;
g(2)=(v(1)-1)*sqrt(1+v(1)^2)/(v(1)*v(2))-5.656854;
end
```

Results

```
********************************

 x0(1)      x0(2)

 0.5000     1.0000

*****************************

 f(x0)

 63.2807

*****************************

 g1(x0)      g2(x0)

 -2.3028     -6.7749

*****************************

 x*(1)      x*(2)

 0.6573     0.5334

*****************************

 f(x*)

 36.1273

*****************************
```

g1(x*)	g2(x*)
0.0	-6.8266

1.36

```
function problem 136

lb=[0.1 0.1];ub=[2 2.5];

x0=[0.5 1.5];
f0=obj(x0);g0=gcons(x0);

[xv,fval]=fmincon(@obj,x0,[],[],[],[],lb,ub,@cons)
gx=gcons(xv);

fprintf('*******results problem 3*********\n');
fprintf('*********************************\n')
fprintf('  x0(1)            x0(2)\n');
fprintf(' %6.4f           %6.4f\n', x0(1),x0(2));
fprintf('*********************************\n')
fprintf(' f(x0)\n');
fprintf(' %6.4f\n',f0);
fprintf('*********************************\n')
fprintf(' g(x0) \n');
fprintf(' %6.4f           %6.4f\n',g0(1),g0(2));
fprintf('*********************************\n')
fprintf('  x*(1)            x*(2)\n');
fprintf(' %6.4f           %6.4f\n', xv(1),xv(2));
fprintf('*********************************\n')
fprintf(' f(x*)\n');
fprintf(' %6.4f\n',fval);
fprintf('*********************************\n')
fprintf(' g(x*) \n');
fprintf(' %6.4f           %6.4f\n',gx(1),gx(2));

end

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
function f=obj(x);
f=0.01178511*((1+x(1)^2)^1.5)*sqrt(1+x(1)^4)/(x(1)*x(1)*x(2));
end
%%%%%%%%%%%%%%%%%%%%
function [c,ceq]=cons(x)
c(1)=(x(1)+1)*sqrt(1+x(1)^2)/(x(1)*x(2))-5.656854;
c(2)=(x(1)-1)*sqrt(1+x(1)^2)/(x(1)*x(2))-5.656854;
ceq=0;
end
%%%%%%%%%%%%%%%%%%%%%%%%%%
```

```
function g=gcons(v);
g(1)=(v(1)+1)*sqrt(1+v(1)^2)/(v(1)*v(2))-5.656854;
g(2)=(v(1)-1)*sqrt(1+v(1)^2)/(v(1)*v(2))-5.656854;
end
```

results

****

x0(1)	x0(2)
0.5000	1.5000

****

f(x0)

0.0453

****

g(x0)

-3.4208	-6.4022

****

x*(1)	x*(2)
0.8656	2.5000

****

f(x*)

0.0182

****

g(x*)

-4.5166	-5.7390

Chapter 2
Classical Optimization Techniques

2.1 The current flowing through the load resistance is given by

$$\text{Current (i)} = \frac{\text{Voltage } (V)}{\text{Total resistance } (R + r)}$$

and the power delivered to the load (P) is

$$P = i^2 r = \left(\frac{V}{R + r}\right)^2 r = \frac{V^2 r}{R^2 + r^2 + 2rR}$$

The necessary condition for maximum of P is

$$\frac{dP}{dr} = \frac{(R^2 + r^2 + 2rR)\,(V^2) - (V^2 r)\,(2r + 2R)}{(R^2 + r^2 + 2rR)^2}$$

$$= \frac{V^2 (R^2 - r^2)}{(R^2 + r^2 + 2rR)^2} = 0$$

i.e. $$r = \pm R.$$

Since the resistance cannot be negative, one obtains $r = R$ as a necessary condition for the maximum of P. The second derivative of P gives

$$\frac{d^2 P}{dr^2} = \frac{(R^2 + r^2 + 2rR)^2\,(-2rV^2) - V^2 (R^2 - r^2)\,[2\,(R^2 + r^2 + 2rR)\,(2R + 2r)]}{(R^2 + r^2 + 2rR)^4}$$

and

$$\left(\frac{d^2 P}{dr^2}\right)_{r=R} = -\frac{V^2}{8R^3} < 0$$

Since (d^2P/dr^2) at $r = R$ is negative, the sufficiency condition is satisfied and maximum power delivered is given by

$$P_{\max} = \left(\frac{V^2 r}{R^2 + r^2 + 2rR}\right)_{r=R} = \frac{V^2}{4R} \text{ watts}$$

2.2

$$\frac{df}{dx} = \frac{(x-1)(x-3)^3(4x^3) - x^4\{(x-3)^3 + (x-1)3(x-3)^2\}}{(x-1)^2(x-3)^6}$$

$$= \frac{(x-1)(x-3)^3 x^3\left[4 - \frac{x}{x-1} - \frac{3x}{x-3}\right]}{(x-1)^2(x-3)^6}$$

$$= \frac{12x^3 - 10x^4}{(x-1)^2(x-3)^4} = 0 \quad \text{at} \quad x = 0,\ 1.2$$

$$\frac{d^2f}{dx^2} = \frac{\left[(x-1)^2(x-3)^4\{36x^2 - 40x^3\} - (12x^3 - 10x^4)\left[2(x-1)(x-3)^4 + (x-1)^2 4(x-3)^3\right]\right]}{(x-1)^4(x-3)^8}$$

$= 0$ at $x=0$ and -41.1523 at $x = 1.2$

$\therefore$ f is a relative maximum at $x = 1.2$

2.3

$f = 4x^3 - 18x^2 + 27x$, $\quad \frac{df}{dx} = 3(4x^2 - 12x + 9)$

$= 0$ at $x = 1.5$

$\frac{d^2f}{dx^2} = 24x - 36$

$= 0$ at $x = 1.5$

$\frac{d^3f}{dx^3} = 24 > 0$ $\quad \therefore$ $x^* = 1.5$ is an inflection point

---

2.4

$\eta = \tan\alpha \,/\, \tan(\alpha+\phi)$, $\quad \phi$ = constant

$$\frac{d\eta}{d\alpha} = \{\sec^2\alpha \tan(\alpha+\phi) - \sec^2(\alpha+\phi)\tan\alpha\} \,/\, \tan^2(\alpha+\phi)$$

If $\tan(\alpha+\phi) \neq 0$, $\frac{d\eta}{d\alpha} = 0$ gives

$$\{\sin(\alpha+\phi)\cdot\cos(\alpha+\phi) - \sin\alpha\cdot\cos\alpha\} \,/\, \{\cos^2\alpha \cdot \cos^2(\alpha+\phi)\} = 0$$

If $\cos\alpha\cdot\cos(\alpha+\phi) \neq 0$, $d\eta/d\alpha = 0$ gives

$$\sin 2(\alpha+\phi) - \sin 2\alpha = 0$$

i.e., $\sin(2\alpha+2\phi) = \sin(2\alpha)$

i.e., $2\alpha + 2\phi = \pi \pm 2\alpha$

i.e., $\alpha^* = \frac{\pi}{4} - \frac{\phi}{2}$ $\quad$ (since $\phi \neq \frac{\pi}{2}$)

$$\frac{d^2\eta}{d\alpha^2} = \frac{1}{\tan^4(\alpha+\phi)}\Big\{\tan^2(\alpha+\phi)\big[2\sec^2\alpha\tan\alpha\tan(\alpha+\phi) + \sec^2\alpha\sec^2(\alpha+\phi) - 2\sec^2(\alpha+\phi)\tan(\alpha+\phi)\tan\alpha - \sec^2(\alpha+\phi)\sec^2\alpha\big] - \big[\sec^2\alpha\tan(\alpha+\phi) - \sec^2(\alpha+\phi)\tan\alpha\big]\,2\tan(\alpha+\phi)\sec^2(\alpha+\phi)\Big\}$$

$$= \frac{1}{\tan^4(\alpha+\phi)}\Big\{2\tan^2(\alpha+\phi)\tan\alpha\left(\sec^2\alpha - \sec^2(\alpha+\phi)\right)\Big\}$$

at $\alpha = \alpha^*$

$$\left.\frac{d^2\eta}{d\alpha^2}\right|_{\alpha^*} = \frac{2\tan\left(\frac{\pi}{4}-\frac{\phi}{2}\right)}{\tan\left(\frac{\pi}{4}+\frac{\phi}{2}\right)}\left\{\frac{1}{\cos^2\left(\frac{\pi}{4}-\frac{\phi}{2}\right)} - \frac{1}{\cos^2\left(\frac{\pi}{4}+\frac{\phi}{2}\right)}\right\}$$

Noting that $\cos^2\left(\frac{\pi}{4}-\frac{\phi}{2}\right) = \frac{1}{2}(1+\sin\phi)$

and $\cos^2\left(\frac{\pi}{4}+\frac{\phi}{2}\right) = \frac{1}{2}(1-\sin\phi)$

$\left.\frac{d^2\eta}{d\alpha^2}\right|_{\alpha^*}$ can be seen to be negative when $\phi \in (0, \frac{\pi}{2})$

$\therefore \alpha^*$ is a relative maximum.

$$\eta(\alpha^*) = \frac{\tan\left(\frac{\pi}{4}-\frac{\phi}{2}\right)}{\tan\left(\frac{\pi}{4}+\frac{\phi}{2}\right)} = \frac{\sin\left(\frac{\pi}{4}-\frac{\phi}{2}\right)\cdot\cos\left(\frac{\pi}{4}+\frac{\phi}{2}\right)}{\cos\left(\frac{\pi}{4}-\frac{\phi}{2}\right)\cdot\sin\left(\frac{\pi}{4}+\frac{\phi}{2}\right)}$$

$$= \frac{\left(\cos\frac{\phi}{2}-\sin\frac{\phi}{2}\right)^2}{\left(\cos\frac{\phi}{2}+\sin\frac{\phi}{2}\right)^2} = \frac{1-2\sin\frac{\phi}{2}\cos\frac{\phi}{2}}{1+2\sin\frac{\phi}{2}\cos\frac{\phi}{2}}$$

$$\therefore \eta_{max} = \eta(\alpha^*) = \frac{1-\sin\phi}{1+\sin\phi}.$$

2.5

$$f^{(1)}(x) = 60x^5 - 240x^4 + 60x^3 + 600x^2 - 240x - 480$$
$$= 60(x+1)^2(x-2)^3 = 0 \quad \text{at} \quad x = -1, 2$$

$$f^{(2)}(x) = 300x^4 - 960x^3 + 180x^2 + 1200x - 240$$

$$f^{(3)}(x) = 1200x^3 - 2880x^2 + 360x + 1200$$

$$f^{(4)}(x) = 3600x^2 - 5760x + 360$$

$f^{(2)}(-1) = 0$, $f^{(3)}(-1) = +2,520$

$f^{(2)}(2) = 0$, $f^{(3)}(2) = 0$, $f^{(4)}(2) = +3,240$

At $x = -1$, $f^{(1)}(-1) = f^{(2)}(-1) = 0$ and $f^{(3)}(-1) = +$

Neither minimum, nor maximum.

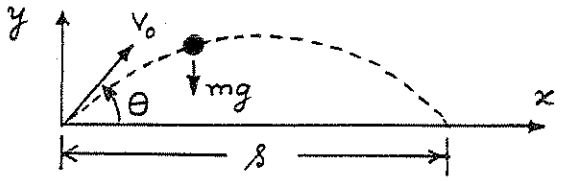
At $x = 2$, $f^{(1)}(2) = f^{(2)}(2) = f^{(3)}(2) = 0$, $f^{(4)}(2) = +$

Relative minimum.

2.6

Let initial velocity $= V_0$

angle $= \theta$



vertical component of $V_0 = V_0 \sin\theta$

Horizontal component of $V_0 = V_0 \cos\theta$

Under gravitational acceleration of g,

height of projectile at time $t = h = V_0 \sin\theta \cdot t - \frac{1}{2} g t^2$ (1)

At time t, horizontal distance travelled $= s = V_0 \cos\theta \cdot t$ (2)

When projectile falls on ground after time t_0, $h = 0$.

Eq. (1) gives $t_0 = \dfrac{2 V_0 \sin\theta}{g}$ (3)

Horizontal distance in time t_0 is given by Eq. (2):

$$s = V_0 \cos\theta \left(\frac{2 V_0 \sin\theta}{g}\right) = \frac{V_0^2}{g} \sin 2\theta \qquad (4)$$

For maximum range, $ds/d\theta = 0$ or $\theta = \frac{\pi}{4}$

$\left(\left.\frac{d^2 s}{d\theta^2}\right|_{\theta = \frac{\pi}{4}} = -\frac{4 V_0^2}{g} = \text{negative}\right)$

$\therefore$ Angular orientation for maximum range $= \frac{\pi}{4}$.

2.7

speed of signalling $= v = x^2 \log\frac{1}{x}$

$$\frac{dv}{dx} = 2x \log\frac{1}{x} + x^2 \cdot (x)(-1/x^2) = 0 \Rightarrow \log\frac{1}{x} = \frac{1}{2}$$

$\therefore x^* = 1/\sqrt{e}$

$$\frac{d^2 v}{dx^2} = \frac{d}{dx}\left(2x \log\frac{1}{x} - x\right) = 2 \log\frac{1}{x} - 3 = -2 \text{ at } x^* = 1/\sqrt{e}$$

$\therefore$ speed is greatest when $x = x^* = 1/\sqrt{e}$.

2.8 Horsepower $= P = k\,u(V-u)$, k = constant

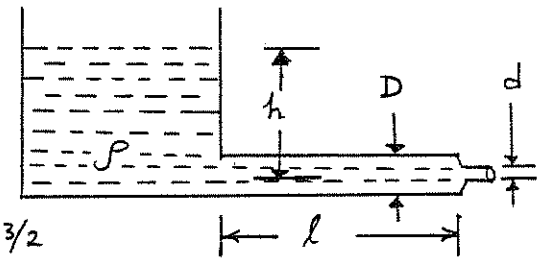
$$\frac{dP}{du} = k(V-2u) = 0 \quad \text{at} \quad u^* = V/2$$

$$\frac{d^2P}{du^2} = -2k < 0$$

$\therefore$ Efficiency is maximum when $u = u^* = V/2$.

2.9 T = kinetic energy

$$= \frac{\pi}{4}\rho d^2 \left(\frac{2g D^5 h}{D^5 + 4 f l d^4}\right)^{3/2}$$



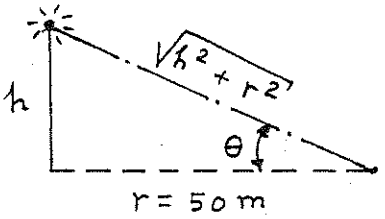
Let $a = \frac{\pi}{4}\rho$ and $b = (2g D^5 h)^{3/2}$

$$\frac{dT}{dd} = 2abd\left\{\frac{(D^5+4fl d^4)^{3/2} - 12 fl d^4 (D^5+4fl d^4)^{1/2}}{(D^5+4fl d^4)^3}\right\} = 0$$

$$\Rightarrow \quad d^* = \left(\frac{D^5}{8fl}\right)^{1/4}$$

$$\left.\frac{d^2T}{dd^2}\right|_{d^*} < 0 \text{ (can be shown).}$$

2.10 f = intensity $= \dfrac{\sin\theta}{(h^2+r^2)} = \dfrac{h}{(h^2+r^2)^{3/2}}$



$$\frac{df}{dh} = \frac{1}{(h^2+r^2)^{3/2}} - \frac{3h^2}{(h^2+r^2)^{5/2}} = 0$$

$$\Rightarrow \quad h^* = \pm r/\sqrt{2} \quad \Rightarrow \quad h^* = 50/\sqrt{2} = 35.36 \text{ m}$$

$$\frac{d^2f}{dh^2} = \frac{-4h}{(h^2+r^2)^{5/2}} < 0 \quad \text{at} \quad h^*$$

$\therefore$ optimum height $= h^* = 35.36$ m

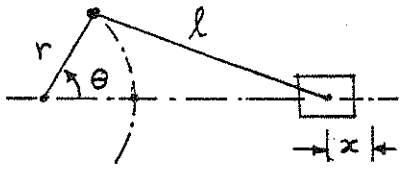
2.11 $x = r(1-\cos\theta) + \dfrac{r^2}{4l}(1-\cos 2\theta)$; $\theta = \omega t$

(a) For $r = 1$ and $l = 5$, $x = 1 - \cos\theta + \frac{1}{20}(1-\cos 2\theta)$

$$x = \frac{21}{20} - \cos\theta - \frac{1}{20}\cos 2\theta$$

$$v = \frac{dx}{dt} = -(-\sin\theta)\,\dot{\theta} - \frac{1}{20}(-\sin 2\theta)(2\dot{\theta})$$

$$= \dot{\theta}\sin\theta + \frac{1}{10}\dot{\theta}\sin 2\theta \;;\; \dot{\theta} = \omega$$



$$\frac{dv}{d\theta} = \omega\cos\theta + \frac{\omega}{5}\cos 2\theta = 0$$

$$\Rightarrow -\cos\theta = \frac{1}{5}\cos 2\theta = \frac{1}{5}(2\cos^2\theta - 1)$$

$$\Rightarrow 2\cos^2\theta + 5\cos\theta - 1 = 0 \quad \Rightarrow \cos\theta = -\frac{5}{4} \pm \frac{\sqrt{33}}{4}$$

$\cos\theta^* = +0.186$, $\quad \theta^* = 79.28^\circ$

$$\frac{d^2v}{d\theta^2} = -\omega\sin\theta - \frac{2\omega}{5}\sin 2\theta = -1.129\,\omega \quad \text{at } \theta^* = 79.28^\circ$$

$\therefore$ Piston velocity is maximum at $\theta^* = 79.28^\circ$

(b) At $\theta = 79.28^\circ$,

$$x = \frac{21}{20} - \cos 79.28^\circ - \frac{1}{20}\cos 2(79.28^\circ) = 0.911$$

from end of stroke

2.12

$$\left|[A] - \lambda[I]\right| = \begin{vmatrix} 3-\lambda & 1 & -1 \\ 1 & 3-\lambda & -1 \\ -1 & -1 & 5-\lambda \end{vmatrix} = 0$$

i.e., $(3-\lambda)\{(3-\lambda)(5-\lambda) - 1\} - 1\{(5-\lambda) - 1\} - 1\{-1 + (3-\lambda)\} = 0$

i.e., $\lambda^3 - 11\lambda^2 + 36\lambda - 36 = (\lambda-2)(\lambda-3)(\lambda-6) = 0$

i.e., $\lambda_1 = 2,\ \lambda_2 = 3,\ \lambda_3 = 6$

$\therefore$ [A] is positive definite

2.13

$$\left|[B] - \lambda[I]\right| = \begin{vmatrix} 4-\lambda & 2 & -4 \\ 2 & 4-\lambda & -2 \\ -4 & -2 & 4-\lambda \end{vmatrix} = 0$$

i.e., $(4-\lambda)\{12 - 8\lambda + \lambda^2\} - 2\{8 - 2\lambda - 8\} - 4\{-4 + 16 - 4\lambda\} = 0$

i.e., $\lambda^3 - 12\lambda^2 + 24\lambda = 0$

This gives $\lambda_1 = 0$, $\lambda_{2,3} = 6 \pm \sqrt{12} = 2.5359,\ 9.4641$

$\therefore$ [B] is positive semi-definite.

2.14 $\left|[C]-\lambda[I]\right| = \begin{vmatrix} -1-\lambda & -1 & -1 \\ -1 & -2-\lambda & -2 \\ -1 & -2 & -3-\lambda \end{vmatrix} = 0$

i.e., $(-1-\lambda)\{6+5\lambda+\lambda^2-4\}+1\{3+\lambda-2\}-1\{2-(2+\lambda)\}=0$

i.e., $\lambda^3+6\lambda^2+5\lambda+1=0$

Letting $f=\lambda^3+6\lambda^2+5\lambda+1$, the roots of $f=0$ can be found by trial and error:

$\lambda_1 \approx -0.305$, $\lambda_2 \approx -0.643$, $\lambda_3 \simeq -5.049$

$\therefore$ $[C]$ is negative definite

2.15 $\left|[A_1]\right| = +3$, $\left|[A_2]\right| = \begin{vmatrix} 3 & 1 \\ 1 & 3 \end{vmatrix} = +8$,

$\left|[A_3]\right| = \begin{vmatrix} 3 & 1 & -1 \\ 1 & 3 & -1 \\ -1 & -1 & 5 \end{vmatrix} = 3(15-1)-1(5-1)-1(-1+3) = +36$

$\therefore$ $[A]$ = positive definite

2.16 $\left|[B_1]\right| = +4$, $\left|[B_2]\right| = \begin{vmatrix} 4 & 2 \\ 2 & 4 \end{vmatrix} = +12$,

$\left|[B_3]\right| = \begin{vmatrix} 4 & 2 & -4 \\ 2 & 4 & -2 \\ -4 & -2 & 4 \end{vmatrix} = 4(16-4)-2(8-8)-4(-4+16) = 0$

$\therefore$ $[B]$ = positive semi-definite

2.17 $\left|[C_1]\right| = -1$, $\left|[C_2]\right| = \begin{vmatrix} -1 & -1 \\ -1 & -2 \end{vmatrix} = +1$,

$\left|[C_3]\right| = \begin{vmatrix} -1 & -1 & -1 \\ -1 & -2 & -2 \\ -1 & -2 & -3 \end{vmatrix} = -1(6-4)+1(3-2)-1(2-2) = -1$

$\therefore$ $[C]$ = negative definite

2.18 Let $f = c + \{b_1 \;\; b_2 \;\; b_3\}\begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} + \frac{1}{2}\{x_1 \;\; x_2 \;\; x_3\}\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}\begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix}$

comparing this with given function, we find:

$c=2$, $b_1=4$, $b_2=0$, $b_3=-5$, $a_{11}=-2$, $a_{22}=-2$, $a_{33}=-2$,

$a_{12} = 2,\ a_{13} = 6,\ a_{23} = 0,\ a_{21} = 2,\ a_{31} = 6,\ a_{32} = 0.$

$$[A] = \begin{bmatrix} -2 & 2 & 6 \\ 2 & -2 & 0 \\ 6 & 0 & -2 \end{bmatrix}$$

$$|[A_1]| = -2, \quad |[A_2]| = \begin{vmatrix} -2 & 2 \\ 2 & -2 \end{vmatrix} = 4 - 4 = 0,$$

$$|[A_3]| = -2(4-0) - 2(-4-0) + 6(0+12) = +72$$

$\therefore [A]$ = negative semi-definite

2.19

$$|[A_1]| = +4, \quad |[A_2]| = \begin{vmatrix} 4 & -3 \\ -3 & 0 \end{vmatrix} = 0 - 9 = -9,$$

$$|[A_3]| = \begin{vmatrix} 4 & -3 & 0 \\ -3 & 0 & 4 \\ 0 & 4 & 2 \end{vmatrix} = 4(0-16) + 3(-6-0) + 0 = -82$$

$[A]$ = indefinite matrix

2.20

$$|[A_1]| = -14, \quad |[A_2]| = \begin{vmatrix} -14 & 3 \\ 3 & -1 \end{vmatrix} = 14 - 9 = +5,$$

$$|[A_3]| = \begin{vmatrix} -14 & 3 & 0 \\ 3 & -1 & 4 \\ 0 & 4 & 2 \end{vmatrix} = -14(-2-16) - 3(6-0) + 0 = +234$$

$\therefore [A]$ = negative definite matrix

2.21

For given data, $\frac{EA}{s} = \frac{(207 \times 10^9)(10^{-5})}{(0.75^2 + 4^2)^{\frac{1}{2}}} = 50.8637 \times 10^4$

$$\left(\frac{l}{2s}\right)^2 = \left(\frac{1.5}{2 * 4.0697}\right)^2 = 0.033962, \quad \left(\frac{h}{s}\right)^2 = \left(\frac{4}{4.0697}\right)^2 = 0.96604$$

$$f = 17.274 \times 10^3 x_1^2 + 49.1364 \times 10^4 x_2^2 - 8.66 \times 10^3 x_1 - 5.00 \times 10^3 x_2$$

$$\frac{\partial f}{\partial x_1} = 34.544 \times 10^3 x_1 - 8.66 \times 10^3 = 0 \Rightarrow x_1^* = 0.2507 \text{ m}$$

$$\frac{\partial f}{\partial x_2} = 98.2728 \times 10^4 x_2 - 5.00 \times 10^3 = 0 \Rightarrow x_2^* = 5.0879 \times 10^{-3} \text{ m}$$

$$[J]|_{(x_1^*, x_2^*)} = \left[\frac{\partial^2 f}{\partial x_i \partial x_j}\right](x_1^*, x_2^*) = \begin{bmatrix} 34.544 \times 10^3 & 0 \\ 0 & 98.2728 \times 10^4 \end{bmatrix}$$

= positive definite

$\therefore$ f is minimized at (x_1^*, x_2^*)

2.22

$$f = 20x_1 + 26x_2 + 4x_1x_2 - 4x_1^2 - 3x_2^2$$

$$\partial f/\partial x_1 = 20 + 4x_2 - 8x_1 = 0 \quad , \quad \partial f/\partial x_2 = 26 + 4x_1 - 6x_2 = 0$$

Solution of these equations: $x_1^* = 7, \quad x_2^* = 9$

$$[J]_{\vec{x}^*} = \left[\frac{\partial^2 f}{\partial x_i \partial x_j}\right]_{\vec{x}^*} = \begin{bmatrix} -8 & 4 \\ 4 & -6 \end{bmatrix}$$

$$|[J_1]| = -8, \quad |[J_2]| = 48 - 16 = +32$$

$[J]$ = negative definite

$\therefore$ f = maximum at $(x_1^* = 7, \; x_2^* = 9)$

2.23

f = sum of squares of differences between graph and tabulated values

$$= \{a + b(0) - 380\}^2 + \{a + 0.25b - 200\}^2 + \{a + 0.5b - 100\}^2 + \{a + 0.75b - 20\}^2 + \{a + 1.00b - 0\}^2$$

$$\frac{\partial f}{\partial a} = 2(a - 380) + 2(a + 0.25b - 200) + 2(a + 0.5b - 100) + 2(a + 0.75b - 20) + 2(a + b) = 10a + 5b - 1400 = 0 \quad (1)$$

$$\frac{\partial f}{\partial b} = 2(a + 0.25b - 200)(0.25) + 2(a + 0.5b - 100)(0.5) + 2(a + 0.75b - 20)(0.75) + 2(a + b) = 5a + 3.75b - 230 = 0 \quad (2)$$

Solution of Eqs. (1) and (2) gives

$$a = 328, \quad b = -376$$

$$[J] = \begin{bmatrix} \frac{\partial^2 f}{\partial a^2} & \frac{\partial^2 f}{\partial a \partial b} \\ \frac{\partial^2 f}{\partial b \partial a} & \frac{\partial^2 f}{\partial b^2} \end{bmatrix} = \begin{bmatrix} 10 & 5 \\ 5 & 3.75 \end{bmatrix} = \text{positive definite}$$

$\therefore$ $t = 328 - 376\,d$ minimizes f.

2.24

$$f = (x_1 - 1)^2 e^{x_2} + x_1 \approx f(\vec{x}_0) + df(\vec{x}_0) + \frac{1}{2!} d^2 f(\vec{x}_0)$$

with $$df(\vec{x}_0) = h_1 \frac{\partial f}{\partial x_1} + h_2 \frac{\partial f}{\partial x_2}$$

and $$d^2 f(\vec{x}_0) = h_1^2 \frac{\partial^2 f}{\partial x_1^2} + h_2^2 \frac{\partial^2 f}{\partial x_2^2} + 2h_1 h_2 \frac{\partial^2 f}{\partial x_1 \partial x_2}$$

Here $$\frac{\partial f}{\partial x_1} = 2(x_1 - 1)e^{x_2} + 1, \quad \frac{\partial f}{\partial x_2} = (x_1 - 1)^2 e^{x_2},$$

$$\frac{\partial^2 f}{\partial x_1^2} = 2\,e^{x_2}, \quad \frac{\partial^2 f}{\partial x_2^2} = (x_1-1)^2\,e^{x_2}, \quad \frac{\partial^2 f}{\partial x_1 \partial x_2} = 2(x_1-1)\,e^{x_2}$$

(a) At $\vec{X}_0 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$, $\frac{\partial f}{\partial x_1} = -2e^0 + 1 = -1$, $\frac{\partial f}{\partial x_2} = e^0 = 1$,

$\frac{\partial^2 f}{\partial x_1^2} = 2$, $\frac{\partial^2 f}{\partial x_2^2} = 1$, $\frac{\partial^2 f}{\partial x_1 \partial x_2} = -2$, $f = 1$

$$\therefore f \simeq 1 + h_1(-1) + h_2(1) + \frac{1}{2!}\{h_1^2(2) + h_2^2(1) + 2h_1h_2(-2)\}$$

$$\simeq 1 - h_1 + h_2 + h_1^2 + \tfrac{1}{2}h_2^2 - 2h_1h_2$$

with $h_1 = x_1 - 0 = x_1$ and $h_2 = x_2 - 0 = x_2$.

(b) At $\vec{X}_0 = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$, $\frac{\partial f}{\partial x_1} = 1$, $\frac{\partial f}{\partial x_2} = 0$, $\frac{\partial^2 f}{\partial x_1^2} = 2e$, $\frac{\partial^2 f}{\partial x_2^2} = 0$,

$\frac{\partial^2 f}{\partial x_1 \partial x_2} = 0$, $f = 1$

$$\therefore f \simeq 1 + h_1(1) + h_2(0) + \frac{1}{2!}\{h_1^2(2e) + h_2^2(0) + 2h_1h_2(0)\}$$

$$\simeq 1 + h_1 + e\,h_1^2$$

with $h_1 = x_1 - 1$ and $h_2 = x_2 - 1$.

2.25 **The third order Taylor's series approximation of the function f about the point $\mathbf{X}^*$ is given by**

$$f(\mathbf{X}) = f\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} + df\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} + \frac{1}{2!}d^2f\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} + \frac{1}{3!}d^3f\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$$

where

$$f\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} = e^{-2},$$

$$df\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} = h_1\frac{\partial f}{\partial x_1}\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} + h_2\frac{\partial f}{\partial x_2}\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} + h_3\frac{\partial f}{\partial x_3}\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$$

$$= (h_1e^{x_3} + h_2 2x_2x_3 + h_3x_2^2 + h_3x_1e^{x_3})\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$$

$$= h_1e^{-2} + h_3e^{-2}$$

$$d^2f\begin{pmatrix}1\\0\\-2\end{pmatrix}=\sum_{i=1}^{3}\sum_{j=1}^{3}h_ih_j\frac{\partial^2 f}{\partial x_i\partial x_j}\begin{pmatrix}1\\0\\-2\end{pmatrix}$$

$$=\left(h_1^2\frac{\partial^2 f}{\partial x_1^2}+h_2^2\frac{\partial^2 f}{\partial x_2^2}+h_3^2\frac{\partial^2 f}{\partial x_3^2}+2h_1h_2\frac{\partial^2 f}{\partial x_1\partial x_2}\right.$$

$$\left.+2h_2h_3\frac{\partial^2 f}{\partial x_2\partial x_3}+2h_1h_3\frac{\partial^2 f}{\partial x_1\partial x_3}\right)\begin{pmatrix}1\\0\\-2\end{pmatrix}$$

$$=[h_1^2(0)+h_2^2(2x_3)+h_3^2(x_1e^{x_3})+2h_1h_2(0)$$

$$+2h_2h_3(2x_2)+2h_1h_3(e^{x_3})]\begin{pmatrix}1\\0\\-2\end{pmatrix}$$

$$=-4h_2^2+e^{-2}h_3^2+2h_1h_3e^{-2},\quad\text{and}$$

$$d^3f\begin{pmatrix}1\\0\\-2\end{pmatrix}=\sum_{i=1}^{3}\sum_{j=1}^{3}\sum_{k=1}^{3}h_ih_jh_k\frac{\partial^3 f}{\partial x_i\partial x_j\partial x_k}\begin{pmatrix}1\\0\\-2\end{pmatrix}$$

$$=\left[h_1^3\frac{\partial^3 f}{\partial x_1^3}+h_2^3\frac{\partial^3 f}{\partial x_2^3}+h_3^3\frac{\partial^3 f}{\partial x_3^3}+3h_1^2h_2\frac{\partial^3 f}{\partial x_1^2\partial x_2}\right.$$

$$+3h_1^2h_3\frac{\partial^3 f}{\partial x_1^2\partial x_3}+3h_1h_2^2\frac{\partial^3 f}{\partial x_1\partial x_2^2}+3h_1h_3^2\frac{\partial^3 f}{\partial x_1\partial x_3^2}$$

$$+3h_2^2h_3\frac{\partial^3 f}{\partial x_2^2\partial x_3}+3h_2h_3^2\frac{\partial^3 f}{\partial x_2\partial x_3^2}$$

$$\left.+6h_1h_2h_3\frac{\partial^3 f}{\partial x_1\partial x_2\partial x_3}\right]\begin{pmatrix}1\\0\\-2\end{pmatrix}$$

$$=[h_1^3(0)+h_2^3(0)+h_3^3(x_1e^{x_3})+3h_1^2h_2(0)+3h_1^2h_3(0)$$

$$+3h_1h_2^2(0)+3h_1h_3^2(e^{x_3})+3h_2^2h_3(2)+3h_2h_3^2(0)$$

$$+6h_1h_2h_3(0)]\begin{pmatrix}1\\0\\-2\end{pmatrix}=h_3^3e^{-2}+3h_1h_3^2e^{-2}+6h_2^2h_3$$

Thus the Taylor's series approximation is given by

$$f(X)\simeq e^{-2}+e^{-2}(h_1+h_3)+\frac{1}{2!}(-4h_2^2+e^{-2}h_3^2+2h_1h_3e^{-2})$$

$$+\frac{1}{3!}e^{-2}(h_3^3+3h_1h_3^2+6h_2^2h_3e^2)$$

where $h_1=x_1-1$, $h_2=x_2$ and $h_3=x_3+2$.

2.26

f = sales $= 12xy - x^2 - 3y^2$

cost constraint: $1000(x+y) = 48{,}000$ or $x+y = 48$

Solution by elimination: $y = 48 - x$

$$f = 12x(48-x) - x^2 - 3(48-x)^2 = -16x^2 + 864x - 6912$$

$$\frac{df}{dx} = -32x + 864 = 0 \Rightarrow x^* = 27,\ y^* = 48 - x^* = 21$$

$$\frac{d^2f}{dx^2} = -32 \text{ , } \therefore f = \text{maximum at } (x^* = 27,\ y^* = 21).$$

$$f_{max} = 12(27)(21) - 27^2 - 3(21^2) = 4{,}752$$

2.27

$$f(x) = \frac{1}{10\sqrt{2\pi}}\, e^{-\frac{1}{2}\left(\frac{x-100}{10}\right)^2}$$

$$\frac{df}{dx} = \frac{1}{10\sqrt{2\pi}}\left(-\tfrac{1}{2}\right)(2)\left(\frac{x-100}{10}\right)\left(\tfrac{1}{10}\right) e^{-\frac{1}{2}\left(\frac{x-100}{10}\right)^2} = 0 \text{ at } x^* = 100$$

$$\frac{d^2f}{dx^2} = -\frac{1}{100\sqrt{2\pi}}\left(\tfrac{1}{10}\right) e^{-\frac{1}{2}\left(\frac{x-100}{10}\right)^2}$$

$$- \frac{1}{100\sqrt{2\pi}}\left(\frac{x-100}{10}\right)\left(-\tfrac{1}{2}\right)(2)\left(\frac{x-100}{10}\right)\left(\tfrac{1}{10}\right) \cdot e^{-\frac{1}{2}\left(\frac{x-100}{10}\right)^2}$$

$$= -\frac{1}{1000\sqrt{2\pi}}\, e^{-\frac{1}{2}\left(\frac{x-100}{10}\right)^2}\left\{1 - \left(\frac{x-100}{10}\right)^2\right\} = \text{negative}$$

at $x^* = 100$

$\therefore$ $x^* = 100$ maximizes $f(x)$.

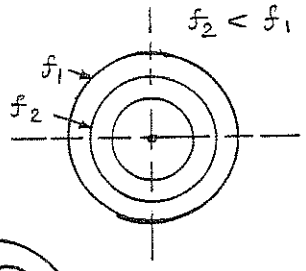
$$f_{max} = \frac{1}{10\sqrt{2\pi}\, e}.$$

2.28

(a) Eigenvalues of the Hessian matrix, β_1, β_2 are positive with $\beta_1 = \beta_2$:

Contours of f: circles

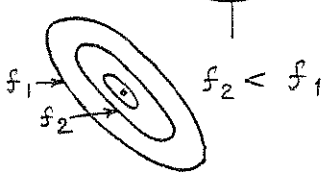
Type of stationary point: minimum.



(b) β_1, β_2 are positive; $\beta_1 > \beta_2$:

Contours of f: ellipses

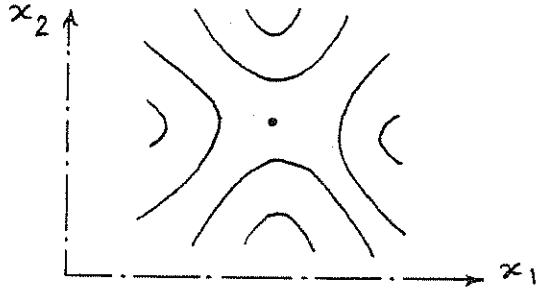
Type of stationary point: minimum.



(c) $|\beta_1| = |\beta_2|$ and β_1, β_2 have opposite signs:

Contours:
hyperbolas (symmetric)

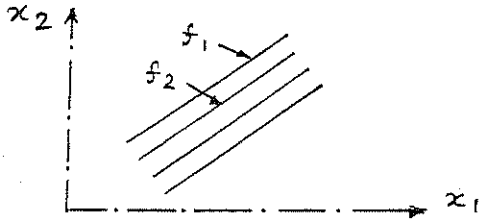
Type of stationary point:
saddle point.



(d) $\beta_1 > 0$, $\beta_2 = 0$:

Contours:
straight lines

Type of stationary point:
None.



2.29 $f = 2 - x^2 - y^2 + 4xy$

Contours are plotted in the range
$-5 \le x \le 5$ and
$-5 \le y \le 5$

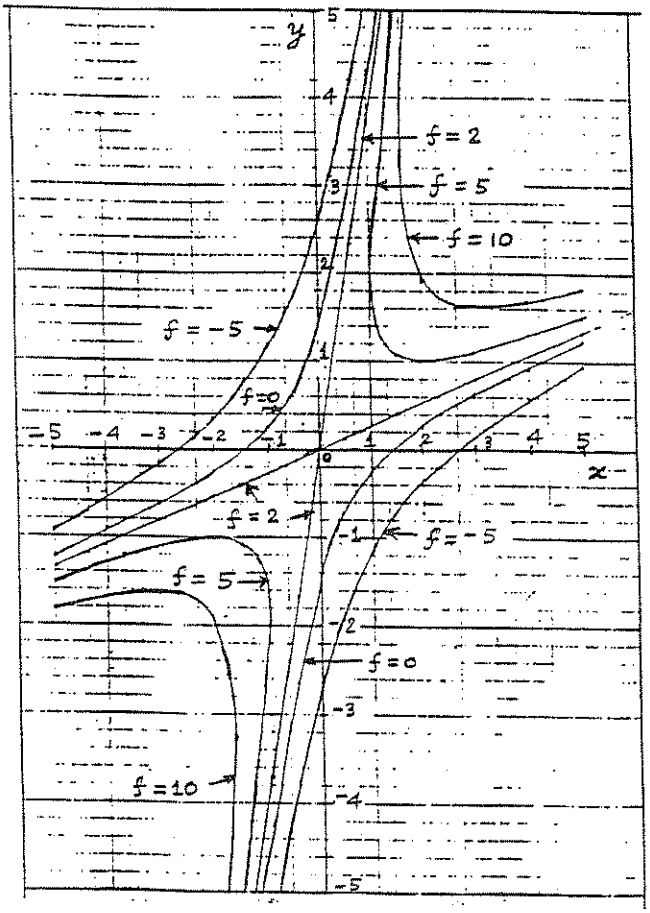
When $f = c$ = constant, we have

$$y^2 - 4yx - 2 - c = 0$$

$$y = \frac{4x \pm \sqrt{16x^2 + 4(2+c)}}{2}$$

By giving a series of values to x, the corresponding values of y are found, and contours are plotted (as shown).

Nature of stationary point: saddle point at (0,0).

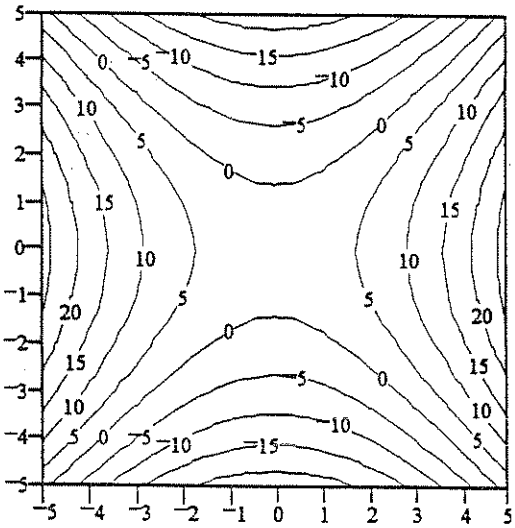


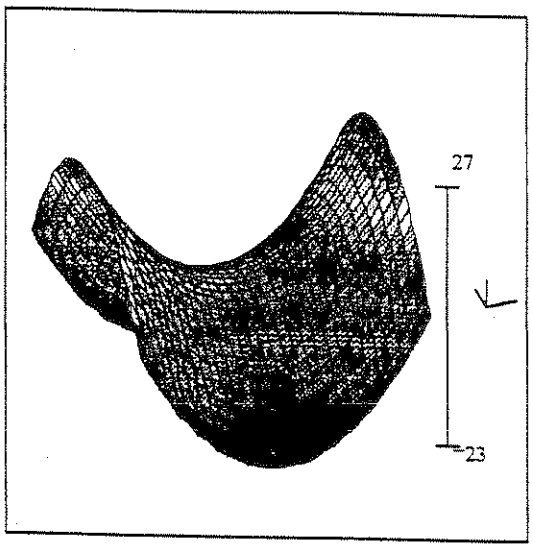
2.30 $f = x^2 - y^2 + 2 \quad \Rightarrow \quad y^2 = x^2 + 2 - c$ where $f = c$ = constant

$$y = \pm\sqrt{x^2 + 2 - c}$$

By giving a series of values to x in the range, $-5 \le x \le 5$, the corresponding values of y are found, and the contours are plotted as shown below.

Nature of stationary point: saddle point at (0, 0).





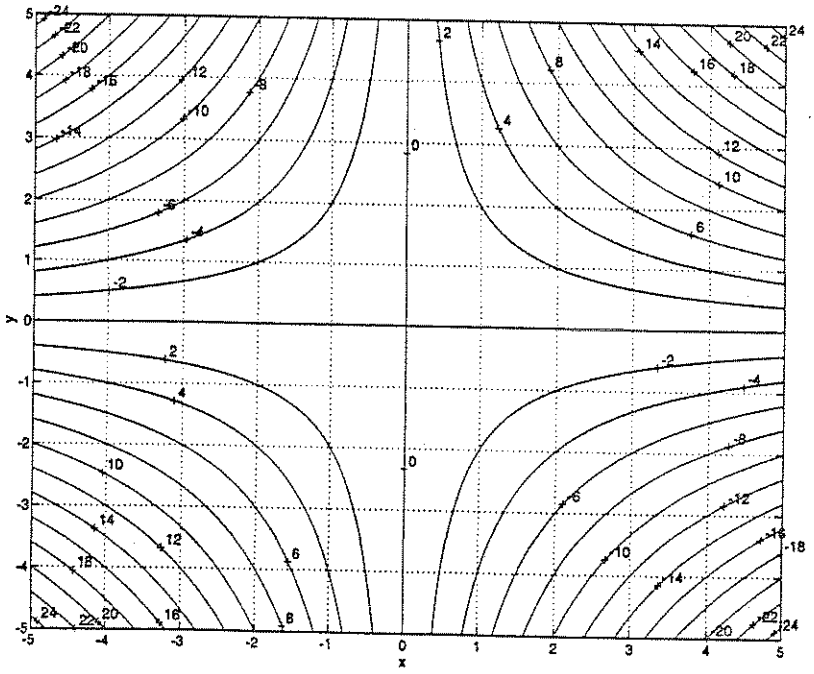
2.31 $f = xy$

If $f = c$ = constant,

$$y = \frac{c}{x}$$

Contours: plotted as shown in figure.

Nature of stationary point: Saddle point at (0,0).



2.32

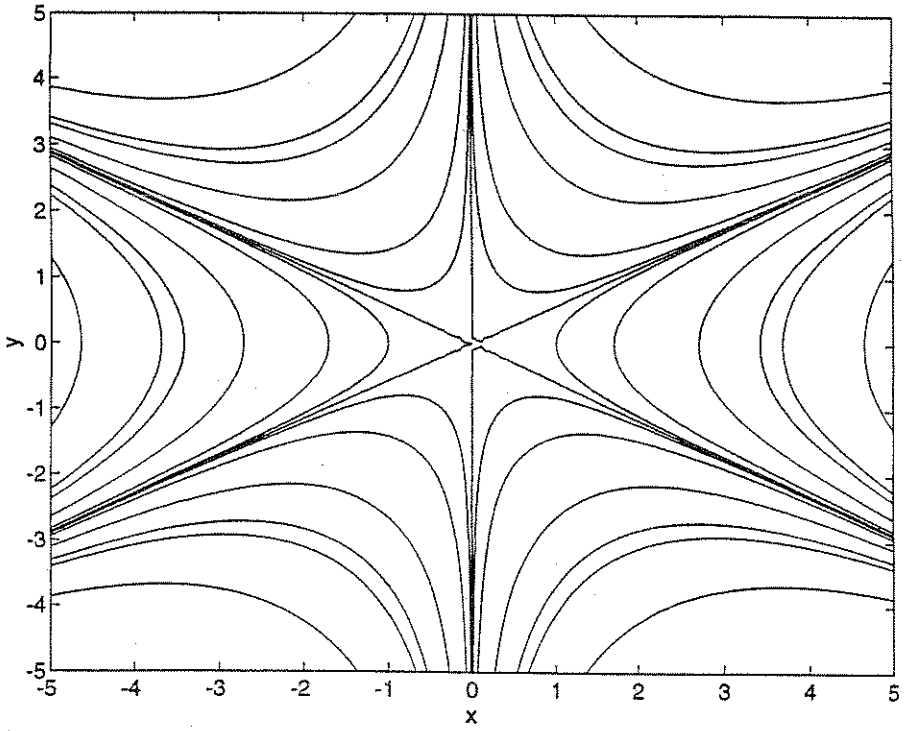
$f = x^3 - 3xy^2$

When $f = c =$ constant, $\quad 3xy^2 = x^3 - c$

or $\quad y = \pm\sqrt{\dfrac{x^3 - c}{3x}} \qquad (1)$

By giving different values to x in the range, $-5 \le x \le 5$, the values of y are found from Eq. (1). Contours of f are plotted by changing the value of c (shown below).

Nature of stationary point: Multiple saddle point at $(0, 0)$.



2.33 $f = x_1^2 + (x_2-1)^2$, $df = \frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2$

$= (2x_1)\, dx_1 + 2(x_2-1)\, dx_2$

$g = -2x_1^2 + x_2 - 4 = 0$, $dg = \frac{\partial g}{\partial x_1} dx_1 + \frac{\partial g}{\partial x_2} dx_2$

$= (-4x_1)\, dx_1 + (1)\, dx_2$

$= 0$ gives $dx_2 = (4x_1)\, dx_1$

At $\vec{X} = \begin{Bmatrix} 0 \\ 4 \end{Bmatrix}$, dx_1 = arbitrary and $dx_2 = 0$

2.34 f = volume $= \frac{\pi D^2}{4} H$; surface area $= \frac{\pi D^2}{4} + \pi D H = S$

$$H = \left\{ \frac{S - \frac{\pi D^2}{4}}{\pi D} \right\} \qquad (1)$$

$$f = \frac{\pi D^2}{4} \left\{ \frac{S}{\pi D} - \frac{D}{4} \right\} = \frac{D \cdot S}{4} - \frac{\pi}{16} D^3$$

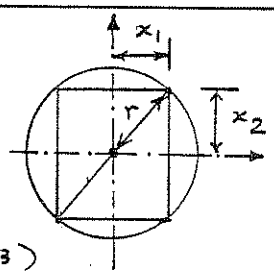
$$\frac{df}{dD} = \frac{S}{4} - \frac{3\pi}{16} D^2 = 0 \Rightarrow D^* = \sqrt{\frac{4S}{3\pi}} \qquad (2)$$

Eqs. (1) and (2) give: $H = \frac{S}{\pi D} - \frac{D}{4} = \frac{S}{\pi}\sqrt{\frac{3\pi}{4S}} - \frac{1}{4}\sqrt{\frac{4S}{3\pi}} = \sqrt{\frac{S}{3\pi}}$

$\frac{d^2 f}{dD^2} = -\frac{3\pi}{8} D$ = negative at D^*

$\therefore$ volume is maximum when $D^* = \sqrt{\frac{4S}{3\pi}}$ and $H^* = \sqrt{\frac{S}{3\pi}}$.

2.35 (a)



Max. $f = 4 x_1 x_2$ (1)

subject to: $g_1 = x_1^2 + x_2^2 - r^2 = 0$ (2)

$L(x_1, x_2, \lambda) = 4 x_1 x_2 + \lambda (x_1^2 + x_2^2 - r^2)$

$\partial L/\partial x_1 = 4 x_2 + 2\lambda x_1 = 0 \Rightarrow x_2 = -\frac{1}{2}\lambda x_1$ (3)

$\partial L/\partial x_2 = 4 x_1 + 2\lambda x_2 = 0 \Rightarrow x_1 = -\frac{1}{2}\lambda x_2$ (4)

$\partial L/\partial \lambda = x_1^2 + x_2^2 - r^2 = 0$ (5)

Eqs. (3) and (4): $x_2 = -\frac{1}{2}\lambda\left(-\frac{1}{2}\lambda x_2\right)$ or $\lambda^2 = 4$

$\lambda = -2$ ($+2$ gives negative values of x_1 and x_2)

With $\lambda = -2$, Eqs. (3), (4) and (5) give:

$2 x_1^2 = r^2$ $\therefore$ $x_1^* = x_2^* = r/\sqrt{2}$, $\lambda^* = -2$

$$L_{11} = \frac{\partial^2 L}{\partial x_1^2} = 2\lambda = -4 \;, \qquad L_{12} = \frac{\partial^2 L}{\partial x_1 \partial x_2} = 4$$

$$L_{21} = \frac{\partial^2 L}{\partial x_2 \partial x_1} = 4 \;, \quad \frac{\partial^2 L}{\partial x_2^2} = 2\lambda = -4$$

$$g_{11} = \frac{\partial g_1}{\partial x_1} = 2x_1 = \sqrt{2}\, r \;, \qquad g_{12} = \frac{\partial g_1}{\partial x_2} = 2x_2 = \sqrt{2}\, r$$

Eq. (2.44) gives:

$$\begin{vmatrix} (-4-z) & 4 & \sqrt{2}\, r \\ 4 & (-4-z) & \sqrt{2}\, r \\ \sqrt{2}\, r & \sqrt{2}\, r & 0 \end{vmatrix} = (-4-z)\left[0 - 2r^2\right] - 4\left[0 - 2r^2\right] + \sqrt{2}\, r\left[4\sqrt{2}\, r + \sqrt{2}\, r\,(4+z)\right] = 0$$

i.e., $\quad (4+z)\, 2r^2 + 8r^2 + 8r^2 + 2r^2 (4+z) = 0$

i.e., $\quad 4r^2 z + 32 r^2 = 0$

i.e., $\quad z = -8 \quad \therefore$ f is maximized when each side $= \sqrt{2}\, r$

(b) Max. $f = 4(x_1 + x_2)$ (6)

subject to $g_1 = x_1^2 + x_2^2 - r^2 = 0$ (7)

$$L = 4(x_1 + x_2) + \lambda (x_1^2 + x_2^2 - r^2) \qquad (8)$$

$\partial L/\partial x_1 = 4 + 2\lambda x_1 = 0$ (9) $\Rightarrow x_1 = -2/\lambda$

$\partial L/\partial x_2 = 4 + 2\lambda x_2 = 0$ (10) $\Rightarrow x_2 = -2/\lambda$

$\partial L/\partial \lambda = x_1^2 + x_2^2 - r^2 = 0$ (11)

$= 4/\lambda^2 + 4/\lambda^2 - r^2 = 0$ or $\lambda^2 = 8/r^2$ or $\lambda = -\frac{2\sqrt{2}}{r}$ (12)

$\therefore \; x_1^* = x_2^* = -\frac{2}{\lambda^*} = \frac{r}{\sqrt{2}}$

$$L_{11} = \frac{\partial^2 L}{\partial x_1^2} = 2\lambda = -\frac{4\sqrt{2}}{r} \;, \qquad L_{12} = \frac{\partial^2 L}{\partial x_1 \partial x_2} = 0 = L_{21} \;,$$

$$L_{22} = \frac{\partial^2 L}{\partial x_2^2} = 2\lambda = -\frac{4\sqrt{2}}{r}$$

$$g_{11} = \frac{\partial g_1}{\partial x_1} = 2x_1 = \sqrt{2}\, r \;, \qquad g_{12} = \frac{\partial g_1}{\partial x_2} = 2x_2 = \sqrt{2}\, r$$

Eq. (2.44) gives:

$$\begin{vmatrix} \left(-\frac{4\sqrt{2}}{r}-\lambda\right) & 0 & \sqrt{2}\,r \\ 0 & \left(-\frac{4\sqrt{2}}{r}-\lambda\right) & \sqrt{2}\,r \\ \sqrt{2}\,r & \sqrt{2}\,r & 0 \end{vmatrix} = \left(-\frac{4\sqrt{2}}{r}-\lambda\right)\{0-2r^2\} - 0 + \sqrt{2}\,r\left\{0-\sqrt{2}\,r\left(-\frac{4\sqrt{2}}{r}-\lambda\right)\right\}$$

$$= 0$$

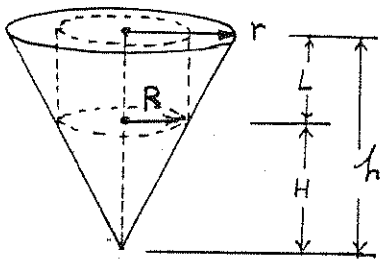
i.e., $\left(\frac{4\sqrt{2}}{r}+\lambda\right)2r^2 + 2r^2\left(\frac{4\sqrt{2}}{r}+\lambda\right) = 0$

i.e., $16\sqrt{2}\,r + 4r^2\lambda = 0$ or $\lambda = -\frac{4\sqrt{2}}{r}$ = negative.

$\therefore$ f is maximized when each side $= \sqrt{2}\,r$.

2.36

Let R = radius of beam
L = length of beam



f = volume of beam $= \pi R^2 L$ (2)

Eqs. (1) and (2):

$$f = \pi L \frac{r^2}{h^2}(h-L)^2 \qquad (3)$$

$$\frac{h}{r} = \frac{H}{R} = \frac{h-L}{R}$$

$$R = \frac{(h-L)r}{h} \qquad (1)$$

$$\frac{\partial f}{\partial L} = \frac{\pi r^2}{h^2}\{(h-L)^2 + L(2)(h-L)(-1)\}$$

$$= 0 \text{ at } L^* = \frac{h}{3} \qquad (4)$$

From (1) and (4), $R^* = \frac{\left(h-\frac{h}{3}\right)r}{h} = \frac{2r}{3}$ (5)

$$\frac{\partial f}{\partial L} = \frac{\pi r^2}{h^2}\{(h-L)^2 - 2L(h-L)\}$$

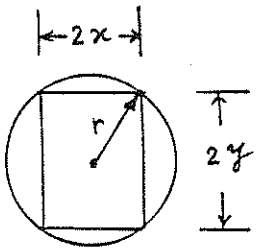
$$\frac{\partial^2 f}{\partial L^2} = \frac{\pi r^2}{h^2}\{2(h-L)(-1) - 2(h-L) - 2L(-1)\}$$

$$= \frac{\pi r^2}{h^2}\{-4(h-L)+2L\} = \frac{\pi r^2}{h^2}\left\{-\frac{8h}{3}+\frac{2h}{3}\right\} = -2\pi r^2/h$$

= negative at L^*

$\therefore$ f is maximum when $L^* = \frac{h}{3}$ and $R^* = \frac{2r}{3}$.

2.37



$$f = \frac{k}{x y^3}, \qquad g_1 = x^2 + y^2 - r^2 = 0$$

$$L = \frac{k}{x y^3} + \lambda (x^2 + y^2 - r^2)$$

$$\partial L/\partial x = -\frac{k}{x^2 y^3} + 2\lambda x = 0 \qquad (1)$$

$$\partial L/\partial y = -\frac{3k}{x y^4} + 2\lambda y = 0 \qquad (2)$$

$$\partial L/\partial \lambda = x^2 + y^2 - r^2 = 0 \qquad (3)$$

Eqs. (1) and (2): $\lambda = \dfrac{k}{2x^3 y^3} = \dfrac{3k}{2 x y^5} \Rightarrow y = \sqrt{3}\, x \quad (4)$

Eqs. (3) and (4): $x^2 + 3x^2 - r^2 = 0$ or $x = \dfrac{r}{2}$

$$\therefore \; x^* = \frac{r}{2}, \quad y^* = \frac{\sqrt{3}}{2} r$$

$$\lambda^* = \frac{k}{2 x^{*3} y^{*3}} = \frac{32k}{3\sqrt{3}\, r^6}$$

$$L_{11} = \frac{\partial^2 L}{\partial x^2} = \frac{2k}{x^3 y^3} + 2\lambda = \frac{2k(8)(8)}{r^3 (3\sqrt{3} r^3)} + \frac{64k}{3\sqrt{3} r^6} = \frac{192k}{3\sqrt{3} r^6}$$

$$L_{12} = L_{21} = \frac{\partial^2 L}{\partial x \partial y} = \frac{3k}{x^2 y^4} = \frac{3k(4)(16)}{r^2 (9) r^4} = \frac{64k}{3 r^6}$$

$$L_{22} = \frac{\partial^2 L}{\partial y 2} = \frac{12k}{x y^5} + 2\lambda = \frac{12k(2)(32)}{9\sqrt{3}\, r^6} + \frac{64k}{3\sqrt{3} r^6} = \frac{320k}{3\sqrt{3}\, r^6}$$

$$g_{11} = \frac{\partial g_1}{\partial x} = 2x = r, \qquad g_{12} = \frac{\partial g_1}{\partial y} = 2y = \sqrt{3}\, r$$

Eq. (2.44):

Let $c = \dfrac{64k}{3\sqrt{3} r^6}$

$$\begin{vmatrix} 3c - z & \sqrt{3}\, c & r \\ \sqrt{3}\, c & 5c - z & \sqrt{3}\, r \\ r & \sqrt{3}\, r & 0 \end{vmatrix} = (3c - z)(0 - 3r^2) - \sqrt{3}\, c (0 - \sqrt{3}\, r^2) + r(3cr - 5cr + rz) = 0$$

i.e., $-9cr^2 + 3r^2 z + 3cr^2 + 3cr^2 - 5cr^2 + r^2 z = 0$

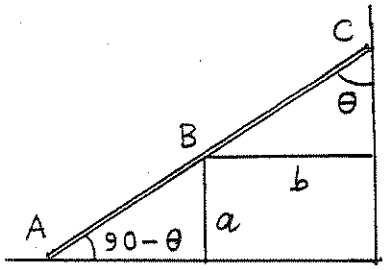
i.e. $z = 8c$ = positive

$\therefore$ f is minimized for $x^* = r/2$, $y^* = \sqrt{3}\, r/2$.

2.38

l = length of ladder = AB + BC

$$= \frac{a}{\cos\theta} + \frac{b}{\sin\theta} = a\sec\theta + b\,\mathrm{cosec}\,\theta$$



$$\frac{dl}{d\theta} = a\sec\theta\tan\theta - b\,\mathrm{cosec}\,\theta\cot\theta = 0$$

$$\theta^* = \tan^{-1}\left[\left(\frac{b}{a}\right)^{\frac{1}{3}}\right] = \tan^{-1} z \quad \text{where} \quad z = \left(\frac{b}{a}\right)^{1/3}$$

$$\sin\theta^* = z/\sqrt{1+z^2}\,, \quad \cos\theta^* = 1/\sqrt{1+z^2}$$

$$\therefore\ l^* = a/\cos\theta^* + b/\sin\theta^* = \left(a^{2/3} + b^{2/3}\right)^{3/2}$$

Sufficiency condition:

$$\frac{d^2l}{d\theta^2} = \frac{d}{d\theta}\left(a\sin\theta\sec^2\theta - b\cos\theta\,\mathrm{cosec}^2\theta\right)$$

$$= \frac{a\cos\theta}{\cos^2\theta} + \frac{2a\sin\theta}{\cos^2\theta}\,\frac{\sin\theta}{\cos\theta} + \frac{b\sin\theta}{\sin^2\theta} + \frac{2b\cos\theta}{\sin^2\theta}\,\frac{\cos\theta}{\sin\theta}$$

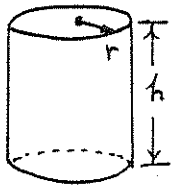
$$= \frac{a}{\cos\theta}\left(1 + 2\tan^2\theta\right) + \frac{b}{\sin\theta}\left(1 + 2\cot^2\theta\right) = \text{positive}$$

$\therefore\ l^*$ is minimum at θ^*

2.39

Maximize $f = \pi r^2 h$ with $2\pi r^2 + 2\pi r h = S$

$$h = \left(S - 2\pi r^2\right)/2\pi r$$



Using substitution,

$$f = \pi r^2\left(\frac{S - 2\pi r^2}{2\pi r}\right) = \frac{Sr}{2} - \pi r^3$$

$$\frac{df}{dr} = \frac{S}{2} - 3\pi r^2 = 0 \qquad \therefore\ r^* = \sqrt{\frac{S}{6\pi}}$$

$$h^* = \frac{S - 2\pi\left(\frac{S}{6\pi}\right)}{2\pi\sqrt{S/6\pi}} = 2\sqrt{\frac{S}{6\pi}} = 2r^* = \text{diameter}$$

$$\frac{d^2f}{dr^2} = -6\pi r = -6\pi\sqrt{\frac{S}{6\pi}} = -\sqrt{6\pi S} = \text{negative}$$

$\therefore\ f$ is maximum when diameter = height

with $r^* = \sqrt{\frac{S}{6\pi}}$.

2.40

Let r = radius, h = height

Minimize $f = 2\pi r^2 + 2\pi r h$ (1)

subject to $\pi r^2 h = V$ (2)

Eq. (2): $h = V/\pi r^2$

Substitution gives Eq. (1): $f = 2\pi r^2 + \frac{2V}{r}$ (3)

$$\frac{df}{dr} = 4\pi r - 2\frac{V}{r^2} = 0 \Rightarrow r^* = \left(V/2\pi\right)^{1/3} \quad (4)$$

Eqs. (2) and (4): $h^* = V/\pi r^{*2} = 2\left(\frac{V}{2\pi}\right)^{1/3} = 2r^*$

$$\frac{d^2f}{dr^2} = 4\pi + \frac{4V}{r^3} = 4\pi + \frac{4V(2\pi)}{V} = 12\pi \text{ at } r^*$$

$$= \text{positive}$$

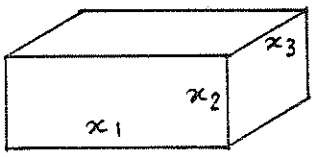
$\therefore$ f is minimum at r^* and h^*.

2.41

Maximize f = volume $= x_1 x_2 x_3$ (1)

subject to

$$g = 2x_1x_2 + 2x_2x_3 + x_1x_3 - S = 0 \quad (2)$$



$$L = x_1x_2x_3 + \lambda(2x_1x_2 + 2x_2x_3 + x_1x_3 - S)$$

$$\partial L/\partial x_1 = x_2x_3 + \lambda(2x_2 + x_3) = 0 \quad (3)$$

$$\partial L/\partial x_2 = x_1x_3 + \lambda(2x_1 + 2x_3) = 0 \quad (4)$$

$$\partial L/\partial x_3 = x_1x_2 + \lambda(2x_2 + x_1) = 0 \quad (5)$$

$$\partial L/\partial \lambda = 2x_1x_2 + 2x_2x_3 + x_1x_3 - S = 0 \quad (6)$$

Eqs. (1) and (2): $\lambda = -\frac{x_2x_3}{2x_2 + x_3} = -\frac{x_1x_3}{2x_1 + 2x_3} \Rightarrow x_1 = 2x_2$

Eqs. (2) and (3): $\lambda = -\frac{x_1x_3}{2x_1 + 2x_3} = -\frac{x_1x_2}{2x_2 + x_1} \Rightarrow x_3 = 2x_2$

Eq. (6): $2(2x_2)x_2 + 2x_2(2x_2) + (2x_2)(2x_2) - S = 0$

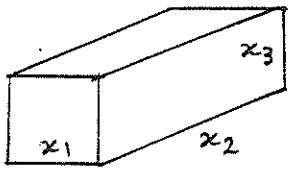
$$\therefore x_1^* = x_3^* = \sqrt{\frac{S}{3}}, \quad x_2^* = \sqrt{\frac{S}{12}}$$

$$f_{max} = x_1^* x_2^* x_3^* = \left(\frac{S^3}{108}\right)^{1/2}$$

2.42

Minimize $f = 2x_1x_3 + x_1x_2 + 2x_2x_3$ (1)

subject to $x_1x_2x_3 = V$ (2)



Solution by substitution:

Eq. (2): $x_1 = V/x_2x_3$ (3)

Eqs. (1) and (3): $f = \frac{2V}{x_2} + \frac{V}{x_3} + 2x_2x_3$ (4)

$$\frac{\partial f}{\partial x_2} = -\frac{2V}{x_2^2} + 2x_3 = 0 \quad \Rightarrow \quad x_3 = V/x_2^2 \qquad (5)$$

$$\frac{\partial f}{\partial x_3} = -\frac{V}{x_3^2} + 2x_2 = 0 \quad \Rightarrow \quad x_2 = V/(2x_3^2) \qquad (6)$$

Eqs. (5) and (6): $x_2 = \frac{V}{2x_3^2} = \frac{V}{2}\cdot\frac{x_2^4}{V^2} = \frac{x_2^4}{2V}$

$$\therefore \quad x_2^* = (2V)^{1/3}, \quad x_3^* = \left(\frac{V}{4}\right)^{1/3}, \quad x_1^* = (2V)^{1/3}$$

Sufficiency conditions:

$$\frac{\partial^2 f}{\partial x_2^2} = \frac{4V}{x_2^3} = 2, \quad \frac{\partial^2 f}{\partial x_3^2} = \frac{2V}{x_3^3} = 8, \quad \frac{\partial^2 f}{\partial x_2 \partial x_3} = 2$$

$$[J]_{\vec{x}^*} = \begin{bmatrix} 2 & 2 \\ 2 & 8 \end{bmatrix}, \quad J_1 = +2, \quad J_2 = +12$$

$[J]$ = positive definite at $\vec{x}^*$

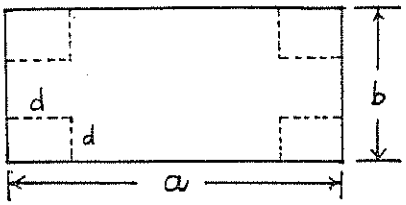
$\therefore$ f is minimum at $\vec{x}^*$

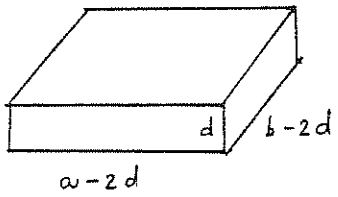
2.43

f = volume

$= (a-2d)(b-2d)\,d$

$= 4d^3 - 2d^2(a+b) + dab$





$$\frac{df}{dd} = 12d^2 - 4d(a+b) + ab = 0$$

$$d^* = \frac{4(a+b) \pm \sqrt{16(a+b)^2 - 48ab}}{24}$$

$$= \frac{1}{6}(a+b) \pm \frac{1}{6}\sqrt{a^2+b^2-ab}$$

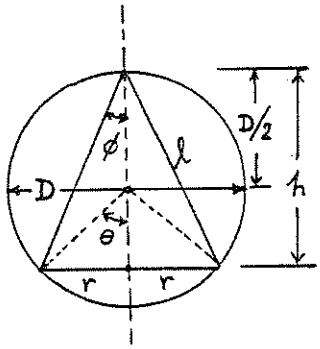
$$\frac{d^2f}{dd^2} = 24d - 4(a+b) = 4(a+b) \pm 4\sqrt{a^2+b^2-ab} - 4(a+b)$$

$$= \pm 4\sqrt{a^2+b^2-ab}$$

$\therefore$ f is maximum at $d^* = \frac{1}{6}(a+b) - \frac{1}{6}\sqrt{a^2+b^2-ab}$.

2.44 Maximize $f = \frac{1}{3}\pi r^2 h$

with $r = \frac{D}{2}\sin\theta$, $h = \frac{D}{2}(1+\cos\theta)$



(a) $f = \frac{\pi}{24} D^3 \sin^2\theta\,(1+\cos\theta)$

$$\frac{df}{d\theta} = \frac{\pi D^3}{24}\left[2\sin\theta\cos\theta\,(1+\cos\theta) + \sin^2\theta(-\sin\theta)\right]$$

$$= \frac{\pi D^3}{24}\sin\theta\left[2\cos\theta + 2\cos^2\theta - (1-\cos^2\theta)\right] = 0$$

at $\sin\theta = 0$ or $3\cos^2\theta + 2\cos\theta - 1 = 0$

$$\cos\theta = \frac{-2\pm\sqrt{4+12}}{6} = -1 \text{ or } \frac{1}{3}$$

$\therefore$ Acceptable solution is $\cos\theta = \frac{1}{3}$, $\sin\theta = \sqrt{1-\left(\frac{1}{3}\right)^2} = \frac{2\sqrt{2}}{3}$

$$h^* = \frac{D}{2}\left(1+\frac{1}{3}\right) = \frac{2}{3}D, \quad r^* = \frac{D}{2}\,\frac{2\sqrt{2}}{3} = \frac{\sqrt{2}\,D}{3}.$$

(b) Maximize f = curved surface $= \pi r \ell$

with $\ell = D\cos\phi$, $r = \ell\sin\phi = D\sin\phi\cos\phi$, $h = \ell\cos\phi = D\cos^2\phi$

$$f = \pi D^2 \sin\phi\cos^2\phi$$

$$\frac{df}{d\phi} = \pi D^2\left\{\sin\phi\,(-2\cos\phi\sin\phi) + \cos\phi\cdot\cos^2\phi\right\} = 0$$

at $\cos\phi\left\{-2\sin^2\phi + \cos^2\phi\right\} = 0$

i.e., $\cos\phi = 0$ or $3\cos^2\phi = 2$

Acceptable solution is $\cos\phi = \sqrt{\frac{2}{3}}$, $\sin\phi = \frac{1}{\sqrt{3}}$

$$\therefore\ h^* = D\cos^2\phi = \frac{2D}{3}$$

Sufficiency condition:

$$\frac{d^2f}{d\phi^2} = \pi D^2\left\{2\sin^3\phi - 2\cos\phi\,(2\sin\phi\cos\phi) + 3\cos^2\phi\,(-\sin\phi)\right\}$$

$$= \pi D^2\left\{2\left(\frac{1}{\sqrt{3}}\right)^3 - 4\left(\frac{2}{3}\right)\left(\frac{1}{\sqrt{3}}\right) - 3\left(\frac{2}{3}\right)\left(\frac{1}{\sqrt{3}}\right)\right\}$$

$$= \pi D^2\left(-\frac{4}{\sqrt{3}}\right) = \text{negative}$$

$\therefore$ f is maximum at $h^* = \frac{2D}{3}$.

2.45

THEOREM 2.6 *A sufficient condition for $f(\mathbf{X})$ to have a relative minimum at $\mathbf{X}^*$ is that the quadratic, Q, defined by*

$$Q = \sum_{i=1}^{n} \sum_{j=1}^{n} \frac{\partial^2 L}{\partial x_i \partial x_j} dx_i \, dx_j \tag{1}$$

evaluated at $\mathbf{X} = \mathbf{X}^*$ must be positive definite for all values of $d\mathbf{X}$ for which the constraints are satisfied.

Proof: The Lagrange function is defined as

$$L(\mathbf{X}, \boldsymbol{\lambda}) = f(\mathbf{X}) + \sum_{j=1}^{m} \lambda_j g_j(\mathbf{X}) \tag{2}$$

If $\mathbf{X}^*$ is an extreme point of $f(\mathbf{X})$, it should be feasible and hence it must satisfy the constraints

$$g_j(\mathbf{X}^*) = 0, \; j = 1, 2, \ldots, m \tag{3}$$

If $d\mathbf{X}$ is a vector of admissible variations about the point $\mathbf{X}^*$, the point $\mathbf{X}^* + d\mathbf{X}$ must also satisfy the given constraints. Thus

$$g_j(\mathbf{X}^* + d\mathbf{X}) = 0, \; j = 1, 2, \ldots, m \tag{4}$$

Equations (2) to (4) lead to

$$L(\mathbf{X}^* + d\mathbf{X}, \boldsymbol{\lambda}) - L(\mathbf{X}^*, \boldsymbol{\lambda}) = f(\mathbf{X}^* + d\mathbf{X}) - f(\mathbf{X}^*) \tag{5}$$

Since $\mathbf{X}^*$ is a constrained relative minimum of $f(\mathbf{X})$, we have

$$f(\mathbf{X}^* + d\mathbf{X}) - f(\mathbf{X}^*) > 0 \tag{6}$$

Equations (5) and (6) give

$$L(\mathbf{X}^* + d\mathbf{X}, \boldsymbol{\lambda}) - L(\mathbf{X}^*) > 0 \tag{7}$$

The Taylor's series expansion of $L(\mathbf{X}^* + d\mathbf{X}, \boldsymbol{\lambda})$ about the point $\mathbf{X}^*$ (keeping $\boldsymbol{\lambda}$ fixed) gives

$$L(\mathbf{X}^* + d\mathbf{X}, \boldsymbol{\lambda}) = L(\mathbf{X}^*, \boldsymbol{\lambda}) + \sum_{i=1}^{n} \frac{\partial L}{\partial x_i}(\mathbf{X}, \boldsymbol{\lambda})\Big|_{\mathbf{X} = \mathbf{X}^*} dx_i + \sum_{i=1}^{n} \sum_{j=1}^{n} \frac{\partial^2 L(\mathbf{X}, \boldsymbol{\lambda})}{\partial x_i \partial x_j}\Big|_{\mathbf{X} = \mathbf{X}^* + \theta d\mathbf{X}} dx_i \, dx_j, \; 0 < \theta < 1 \tag{8}$$

Since the necessary conditions state that $(\partial L/\partial x_i)\ (X^*, \lambda) = 0, i = 1, 2, \ldots, n$, Eq. (8) becomes

$$L(X^* + dX, \lambda) - L(X^*, \lambda) = \sum_{i=1}^{n} \sum_{j=1}^{n} \frac{\partial^2 L(X, \lambda)}{\partial x_i \partial x_j}\bigg|_{X^* + \theta\, dX} dx_i\, dx_j \qquad (9)$$

If it is assumed that the second partial derivatives, $\frac{\partial^2 L}{\partial x_i \partial x_j}(X, \lambda)$, are continuous in the neighbourhood of X^*, the sign of

$$\sum_{i=1}^{n} \sum_{j=1}^{n} \frac{\partial^2 L}{\partial x_i \partial x_j}(X, \lambda)\bigg|_{X^* + \theta\, dX} dx_i\, dx_j \qquad (10)$$

will be same as that of

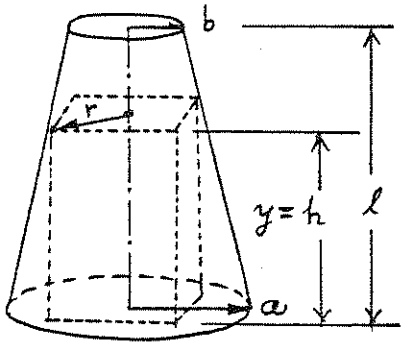
$$\sum_{i=1}^{n} \sum_{j=1}^{n} \frac{\partial^2 L(X^*, \lambda)}{\partial x_i \partial x_j}\, dx_i\, dx_j \qquad (11)$$

for all sufficiently small admissible variations dX. Thus in order to satisfy Eq. (7), we need to have

$$Q = \sum_{i=1}^{n} \sum_{j=1}^{n} \frac{\partial^2 L}{\partial x_i \partial x_j}(X^*, \lambda)\, dx_i\, dx_j > 0 \qquad (12)$$

for all admissible dx_i, $i = 1, 2, \ldots, n$. Hence the theorem is proved.

2.46 Radius of log at y = $r(y) = a - \left(\frac{a-b}{\ell}\right) y$



Area of section at height y

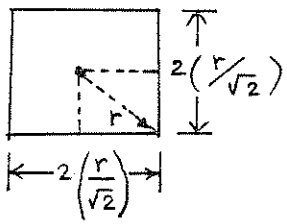
$$= \left(2\,\frac{r}{\sqrt{2}}\right)^2 = 2r^2$$

Volume of beam = $f = (2r^2)\,y$

$$f = 2y\left(a - \frac{a-b}{\ell}\,y\right)^2$$

$$= 2y(a - cy)^2 \quad \text{with} \quad c = \frac{a-b}{\ell}$$

$$= 2a^2 y - 4ac\,y^2 + 2c^2 y^3$$



square section at height y

$$\frac{df}{dy} = 2a^2 - 4ac(2y) + 2c^2(3y^2) = 0$$

$$= 2(a - cy)(a - 3cy) = 0$$

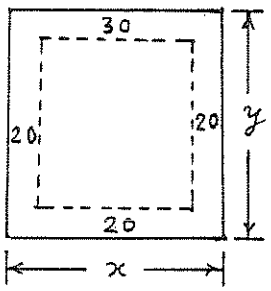
at $y = \frac{a}{c}, \frac{a}{3c}$

$$\frac{d^2 f}{dy^2} = -8ac + 12c^2 y = \begin{cases} 4ac & \left(\text{at } y = \frac{a}{c}\right) \\ -4ac & \left(\text{at } y = \frac{a}{3c}\right) \end{cases}$$

$\therefore$ f is maximum at $y^* = h^* = \frac{a}{3c} = \frac{a\ell}{3(a-b)}$.

2.47 Maximize $f = (x-40)(y-50) = xy - 50x - 40y + 2000$

subject to $g_1 = xy - 50{,}000 = 0$



$$L = xy - 40y - 50x + 2000 + \lambda(xy - 50\,000)$$

$$\partial L/\partial x = y - 50 + \lambda y = 0 \qquad (1)$$

$$\partial L/\partial y = x - 40 + \lambda x = 0 \qquad (2)$$

$$\partial L/\partial \lambda = xy - 50\,000 = 0 \qquad (3)$$

Eqs. (1) and (2): $\lambda = \frac{50-y}{y} = \frac{40-x}{x} \Rightarrow 50x - xy = 40y - xy$

$$x = \frac{4}{5}y \qquad (4)$$

Eqs. (3) and (4): $x^* = 200$ mm, $y^* = 250$ mm, $\lambda^* = -0.8$

Sufficiency conditions:

$$\frac{\partial^2 L}{\partial x^2} = L_{11} = 0, \quad \frac{\partial^2 L}{\partial x \partial y} = L_{12} = L_{21} = 1 + \lambda = 0.2$$

$$\frac{\partial^2 L}{\partial y^2} = L_{22} = 0, \quad g_{11} = \frac{\partial g_1}{\partial x} = y, \quad g_{12} = \frac{\partial g_1}{\partial y} = x$$

Eq. (2.44):

$$\begin{vmatrix} 0 - z & 0.2 & 250 \\ 0.2 & 0 - z & 200 \\ 250 & 200 & 0 \end{vmatrix} = 0$$

i.e., $-z(-40000) - 0.2(-50000) + 250(40 + 250z) = 0$

i.e., $40000z + 10000 + 10000 + 62500z = 0$

i.e. $z = -20000/102500 = -0.1951 = \text{negative}$

$\therefore$ Printed area is maximum at $x^* = 200$ mm, $y^* = 250$ mm.

2.48 (a) Direct substitution:

From constraint, $x_1 = -x_2 - 2x_3 + 3$

$$f(x_2, x_3) = 9 - 8(-x_2 - 2x_3 + 3) - 6x_2 - 4x_3 + 2(-x_2 - 2x_3 + 3)^2 + 2x_2^2 + x_3^2 + 2(-x_2 - 2x_3 + 3)(x_2 + x_3)$$

$$= -15 + 8x_2 + 18x_3 - 3x_3^2 - 6x_2x_3 + 2(-x_2 - 2x_3 + 3)^2$$

$$\frac{\partial f}{\partial x_2} = 8 - 6x_3 + 4(-x_2 - 2x_3 + 3)(-1) = 0 \qquad (1)$$

$$\frac{\partial f}{\partial x_3} = 18 - 6x_3 - 6x_2 + 4(-x_2 - 2x_3 + 3)(-2) = 0 \qquad (2)$$

Eqs. (1) and (2):

$$\left.\begin{aligned} 4x_2 + 2x_3 &= 4 \\ 2x_2 + 10x_3 &= 6 \end{aligned}\right\} \quad x_2 = \frac{7}{9}, \; x_3 = \frac{4}{9}$$

$$\therefore \vec{X}^* = \left\{ \frac{4}{3}, \frac{7}{9}, \frac{4}{9} \right\}$$

(b) constrained variation:

Let independent variable be x_1. Here $m = 1$.

Eq. (2.26) with $x_k = x_2$ and x_3 gives:

$$\begin{vmatrix} \frac{\partial f}{\partial x_2} & \frac{\partial f}{\partial x_1} \\ \frac{\partial g}{\partial x_2} & \frac{\partial g}{\partial x_1} \end{vmatrix} = \begin{vmatrix} (-6+4x_2+2x_1) & (-8+4x_1+2x_2+2x_3) \\ 1 & 1 \end{vmatrix} = 0$$

$$\Rightarrow \quad x_1 - x_2 + x_3 = 1 \qquad (1)$$

$$\begin{vmatrix} \frac{\partial f}{\partial x_3} & \frac{\partial f}{\partial x_1} \\ \frac{\partial g}{\partial x_3} & \frac{\partial g}{\partial x_1} \end{vmatrix} = \begin{vmatrix} (-4+2x_1+2x_3) & (-8+4x_1+2x_2+2x_3) \\ 2 & 1 \end{vmatrix} = 0$$

$$\Rightarrow \quad 3x_1 + 2x_2 + x_3 = 6 \qquad (2)$$

constraint: $\quad g = x_1 + x_2 + 2x_3 = 3 \qquad (3)$

Eqs. (2) − (1) $\Rightarrow \quad 2x_1 + 3x_2 = 5 \qquad (4)$

Eqs. (3) + (1) $\Rightarrow \quad 2x_1 + 3x_3 = 4 \qquad (5)$

Eqs. (2) − 2*(3) $\Rightarrow \quad x_1 - 3x_3 = 0 \qquad (6)$

Eqs. (6), (5) and (4) $\Rightarrow \quad x_1^* = \frac{4}{3}, \quad x_2^* = \frac{7}{9}, \quad x_3^* = \frac{4}{9}$

(c) Lagrange multiplier method:

$$L = 9 - 8x_1 - 6x_2 - 4x_3 + 2x_1^2 + 2x_2^2 + x_3^2 + 2x_1x_2 + 2x_1x_3 + \lambda(x_1 + x_2 + 2x_3 - 3)$$

$\partial L/\partial x_1 = -8 + 4x_1 + 2x_2 + 2x_3 + \lambda = 0 \qquad (1)$

$\partial L/\partial x_2 = -6 + 4x_2 + 2x_1 + \lambda = 0 \qquad (2)$

$\partial L/\partial x_3 = -4 + 2x_3 + 2x_1 + 2\lambda = 0 \qquad (3)$

$\partial L/\partial \lambda = x_1 + x_2 + 2x_3 - 3 = 0 \qquad (4)$

Eqs. (2) and (3): $\quad x_2 = \frac{3}{2} - \frac{1}{2}x_1 - \frac{1}{4}\lambda \qquad (5)$

$\quad x_3 = 2 - x_1 - \lambda \qquad (6)$

Eqs. (5), (6) and (1): $\quad x_1 = 1 + \frac{3}{2}\lambda \qquad (7)$

Eqs. (5), (6) and (3): $\quad \frac{5}{2} - \frac{3}{2}x_1 - \frac{9}{4}\lambda = 0 \qquad (8)$

Eqs. (7) and (8) $\Rightarrow \quad \lambda^* = 2/9 \quad \therefore \quad x_1^* = \frac{4}{3}, \quad x_2^* = 7/9, \quad x_3^* = 4/9.$

2.49 (a) Direct substitution:

constraints give: $x_1 = x_2$ and $x_3 = 1 - x_1 - x_2 = 1 - 2x_1$

$$f = \frac{1}{2}(x_1^2 + x_2^2 + x_3^2) = \frac{1}{2}\left(x_1^2 + x_1^2 + \{1-2x_1\}^2\right)$$

$$= \frac{1}{2}(6x_1^2 - 4x_1 + 1)$$

$$\frac{df}{dx_1} = 6x_1 - 2 = 0 \Rightarrow x_1^* = \frac{1}{3}, \quad x_2^* = \frac{1}{3}, \quad x_3^* = \frac{1}{3}$$

(b) Constrained variation:

Let x_3 be independent.

$$f = \frac{1}{2}(x_1^2 + x_2^2 + x_3^2)$$

$$g_1 = x_1 - x_2 = 0 \qquad (1)$$

$$g_2 = x_1 + x_2 + x_3 - 1 = 0 \qquad (2)$$

$$J\left(\frac{g_1, g_2}{x_1, x_2}\right) = \begin{vmatrix} \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} \\ \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix} = 2 \neq 0 \Rightarrow \text{Eq. (2.26) can be applied.}$$

$$\begin{vmatrix} \frac{\partial f}{\partial x_3} & \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} \\ \frac{\partial g_1}{\partial x_3} & \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} \\ \frac{\partial g_2}{\partial x_3} & \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} \end{vmatrix} = \begin{vmatrix} x_3 & x_1 & x_2 \\ 0 & 1 & -1 \\ 1 & 1 & 1 \end{vmatrix} = 2x_3 - x_1 - x_2 = 0 \qquad (3)$$

Eqs. (1) and (2): $x_1 = x_2$ (4)

$x_1 = 1 - x_2 - x_3$ (5)

Eq. (3): $x_1 = 2x_3 - x_2$ (6)

Eqs. (5) and (6): $x_3^* = \frac{1}{3}$, $x_2^* = \frac{1}{3}$, $x_1^* = \frac{1}{3}$.

(c) Lagrange multipliers:

$$L = \frac{1}{2}(x_1^2 + x_2^2 + x_3^2) + \lambda_1(x_1 - x_2) + \lambda_2(x_1 + x_2 + x_3 - 1)$$

$$\partial L/\partial x_1 = x_1 + \lambda_1 + \lambda_2 = 0 \qquad (1)$$

$$\partial L/\partial x_2 = x_2 - \lambda_1 + \lambda_2 = 0 \qquad (2)$$

$$\partial L/\partial x_3 = x_3 + \lambda_2 = 0 \qquad (3)$$

$$\partial L/\partial \lambda_1 = x_1 - x_2 = 0 \qquad (4)$$

$$\partial L/\partial \lambda_2 = x_1 + x_2 + x_3 - 1 = 0 \qquad (5)$$

Eqs. (1) – (5): $\lambda_1^* = 0, \quad \lambda_2^* = -\frac{1}{3}$

$x_1^* = x_2^* = x_3^* = \frac{1}{3} \quad ; \quad f^* = \frac{1}{6}.$

2.50 Maximize $f = \dfrac{6xyz}{x + 2y + 2z}$ subject to $xyz = 16$

constraint gives: $x = 16/yz$ (1)

$$f = \frac{6\left(\frac{16}{yz}\right)yz}{\left(\frac{16}{yz}\right) + 2y + 2z} = \frac{96yz}{16 + 2y^2z + 2yz^2}$$

$$\frac{\partial f}{\partial y} = \frac{(16 + 2y^2z + 2yz^2)(96z) - 96yz(4yz + 2z^2)}{(16 + 2y^2z + 2yz^2)^2} = 0$$

i.e., $1536z + 192y^2z^2 + 192yz^3 - 384y^2z^2 - 192yz^3 = 0$

i.e., $192y^2z^2 = 1536z \Rightarrow y^2z = 8$ (2)

$$\frac{\partial f}{\partial z} = (16 + 2y^2z + 2yz^2)(96y) - 96yz(2y^2 + 4yz) = 0$$

i.e., $1536y + 192y^3z + 192y^2z^2 - 192y^3z - 384y^2z^2 = 0$

i.e., $192y^2z^2 = 1536y \Rightarrow z^2y = 8$ (3)

Eqs. (2) and (3): $yz = \frac{8}{y} = \frac{8}{z} = \frac{16}{x}$ from Eq. (1)

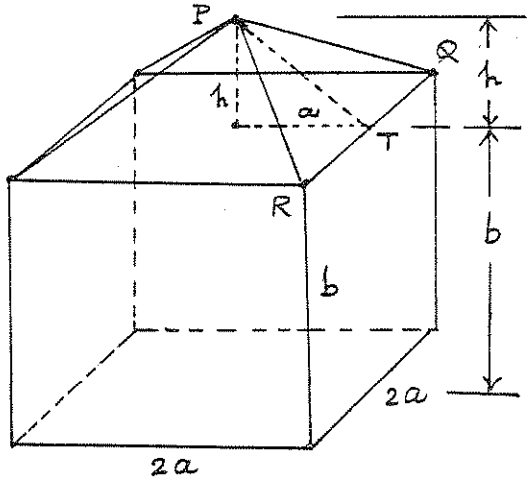
or $y = z, \quad x = 2z$

$\therefore \quad x^* = 4, \quad y^* = 2, \quad z^* = 2 \quad ; \quad f^* = 8.$

2.51 V = volume enclosed

$= (2a)(2a)(b) + \frac{1}{3}(\text{base area})\,h$

$= 4a^2b + \frac{1}{3}(4a^2)h$ (1)



Surface area = S:

$S = 4(2a \cdot b) + 4(\text{area of } \Delta PQR)$

area of $PQR = \frac{1}{2}(QR)(PT)$

$= \frac{1}{2}(2a)\sqrt{h^2 + a^2}$

$S = 8ab + 4a\sqrt{a^2 + h^2}$ (2)

Eq. (1): $V = 4a^2\left(b + \frac{h}{3}\right)$

Eq. (2): $S = 8ab + \frac{8ah}{3} - \frac{8ah}{3} + 4a\sqrt{h^2+a^2}$

$$\therefore \quad S = \frac{2V}{a} - \frac{8ah}{3} + 4a\sqrt{h^2+a^2} \qquad (3)$$

$$\frac{\partial S}{\partial h} = -\frac{8a}{3} + 4a\left(\tfrac{1}{2}\right)\left(h^2+a^2\right)^{-\frac{1}{2}}(2h) = 0$$

i.e., $$\frac{4ah}{\sqrt{a^2+h^2}} = \frac{8a}{3} \Rightarrow \frac{9}{4}h^2 = a^2 + h^2$$

$$\Rightarrow \quad a = \frac{\sqrt{5}}{2}h \qquad (4)$$

$$\frac{\partial S}{\partial a} = -\frac{2V}{a^2} - \frac{8h}{3} + 4\sqrt{h^2+a^2} + 4a\left(\tfrac{1}{2}\right)\left(h^2+a^2\right)^{-\frac{1}{2}}(2a) = 0$$

i.e., $$-\frac{2}{a^2}\left(4a^2b + \frac{4a^2h}{3}\right) - \frac{8h}{3} + 4\sqrt{h^2+a^2} + \frac{4a^2}{\sqrt{h^2+a^2}} = 0 \qquad (5)$$

Using Eq. (4), Eq. (5) becomes:

$$-\frac{2(4)}{5h^2}\left(4b * \frac{5h^2}{4} + \frac{4h}{3} * \frac{5h^2}{4}\right) - \frac{8h}{3} + 4\sqrt{h^2+\frac{5h^2}{4}} + \frac{5h^2}{\sqrt{h^2+\frac{5h^2}{4}}} = 0$$

i.e., $$-\frac{8}{5h^2}\left(5bh^2 + \frac{5h^3}{3}\right) - \frac{8h}{3} + 4\left(\frac{3h}{2}\right) + 5h^2\left(\frac{2}{3h}\right) = 0$$

i.e., $$-8b - \frac{8h}{3} - \frac{8h}{3} + 6h + \frac{10}{3}h = 0$$

i.e., $$-8b + 4h = 0 \Rightarrow h = 2b \qquad (6)$$

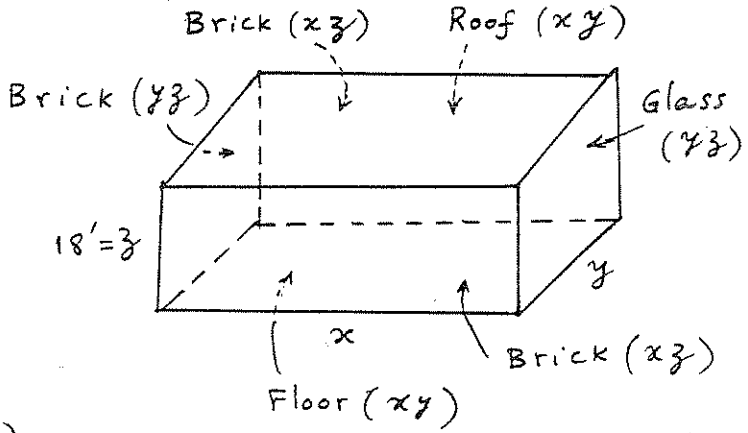
2.52 f = construction cost

$$= c(yz + 2xz + 2yz + 3xy)$$

$$= c(18y + 36x + 36y + 3xy) \qquad (1)$$

constraint:

$$xy = 22500 \qquad (2)$$



Let c = cost of brick wall per ft^2.

Eq.(2): $y = 22500/x$ (3)

Substitution of Eq.(3) in Eq.(1) gives:

$$f = c\left(18 * \frac{22500}{x} + 36x + 36 * \frac{22500}{x} + 3x * \frac{22500}{x}\right)$$

$$= c\left(67500 + 36x + \frac{1,215,000}{x}\right)$$

$$\frac{df}{dx} = c\left(36 - \frac{1,215,000}{x^2}\right) = 0 \Rightarrow x^2 = \frac{1,215,000}{36} = 33750$$

$$\therefore x^* = 183.7117 \text{ ft} \; ; \; y^* = \frac{22500}{x^*} = 122.4745 \text{ ft.}$$

Sufficiency condition:

$$\left.\frac{d^2 f}{dx^2}\right|_{x^*} = \left.\frac{c(2)(1,215,000)}{x^3}\right|_{x^*} = \frac{c(2,430,000)}{6,200,267.8} = 0.3919c$$

= positive

$\therefore (x^*, y^*)$ minimize f.

2.53 f = heat loss $= c[4xy + 2(xz + xz + yz) + 5yz + 1 \cdot xy]$

$= c(5xy + 72x + 126y)$ (1)

where c is a constant

constraint: $g = xy - 22500 = 0$ (2)

$L(x, y, \lambda) = c(5xy + 72x + 126y) + \lambda(xy - 22500)$

$\partial L/\partial x = 5cy + 72c + \lambda y = 0$ (3)

$\partial L/\partial y = 5cx + 126c + \lambda x = 0$ (4)

$\partial L/\partial \lambda = xy - 22500 = 0$ (5)

Eqs.(3) and (4): $\lambda = \frac{-c(72 + 5y)}{y} = \frac{-c(5x + 126)}{x}$

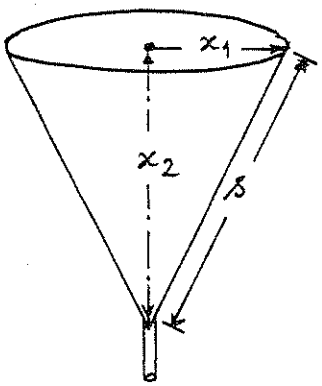
i.e., $-72x - 5xy = -5xy - 126y$

i.e., $x = \frac{7}{4}y$ (6)

Eqs.(5) and (6): $y\left(\frac{7}{4}y\right) = 22500$ or $y^* = 113.3893$ ft

and $x^* = \frac{7}{4}y^* = 198.4313$ ft.

2.54



f = lateral surface area $= \pi x_1 s$

$$= \pi x_1 \sqrt{x_1^2 + x_2^2} \qquad (1)$$

constraint:

Volume $= 200\ \text{in}^3 \Rightarrow$

$$g = \frac{1}{3}\pi x_1^2 x_2 - 200 = 0 \qquad (2)$$

From Eq. (2), $\quad x_2 = \dfrac{600}{\pi x_1^2} \qquad (3)$

Eqs. (1) and (3) give:

$$f = \pi x_1 \sqrt{x_1^2 + \frac{360{,}000}{\pi^2 x_1^4}} \qquad (4)$$

$$\frac{df}{dx_1} = \pi \sqrt{x_1^2 + \frac{360{,}000}{\pi^2 x_1^4}} + \pi x_1 \left(\frac{1}{2}\right) \frac{\left(2x_1 - \dfrac{360{,}000\,(4)}{\pi^2 x_1^5}\right)}{\sqrt{x_1^2 + \dfrac{360{,}000}{\pi^2 x_1^4}}} = 0$$

i.e.,

$$\pi\left(x_1^2 + \frac{360{,}000}{\pi^2 x_1^4}\right) + \frac{\pi x_1}{2}\left(2x_1 - \frac{1{,}440{,}000}{\pi^2 x_1^5}\right) = 0$$

i.e., $\quad 2x_1^2 - \dfrac{360{,}000}{\pi^2 x_1^4} = 0 \quad \Rightarrow \quad x_1^6 = \dfrac{180{,}000}{\pi^2}$

$$\therefore\ x_1^* = \left(\frac{180{,}000}{\pi^2}\right)^{1/6} = 5.1305 \text{ in.}$$

$$x_2^* = \left(600 \Big/ \pi x_1^{*2}\right) = \frac{600}{\pi}\left(\frac{\pi^2}{180{,}000}\right)^{1/3} = \left(\frac{1200}{\pi}\right)^{\frac{1}{3}} = 7.2556 \text{ in.}$$

2.55

$df^* = \lambda^* db$

From Example 2.10, we have $\quad \lambda^* = -1$ and $f^* = 16\pi$

(a) When $\quad A_0 = 25\pi, \quad db = 25\pi - 24\pi = \pi$

$$df^* = (-1)\,\pi = -\pi$$

$$f^*_{new} = f^*_{old} + df^* = 16\pi - \pi = 15\pi$$

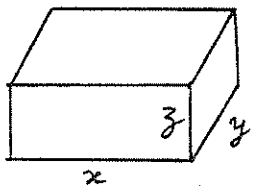
(b) When $\quad A_0 = 22\pi, \quad db = 22\pi - 24\pi = -2\pi$

$$df^* = (-1)(-2\pi) = 2\pi$$

$$f^*_{new} = f^*_{old} + df^* = 16\pi + 2\pi = 18\pi$$

2.56 Minimize $f = 4(x+y+z)$

Subject to: $g = xyz - V = 0$



$V = 1000 \text{ in}^3$

(a) $L = 4(x+y+z) + \lambda(xyz - V)$

$\partial L/\partial x = 4 + \lambda yz = 0 \quad (1)$

$\partial L/\partial y = 4 + \lambda xz = 0 \quad (2)$

$\partial L/\partial z = 4 + \lambda xy = 0 \quad (3)$

$\partial L/\partial \lambda = xyz - V = 0 \quad (4)$

Eqs. (1) to (3): $\lambda = -\frac{4}{yz} = -\frac{4}{xz} = -\frac{4}{xy}$

$\Rightarrow x = y = z \quad (5)$

Eqs. (4) and (5): $x^* = y^* = z^* = V^{\frac{1}{3}} = 10 \text{ inch}$

$\lambda^* = -\frac{4}{100} = -0.04$

$f^* = 4(x^* + y^* + z^*) = 120 \text{ inch}$

(b) $df^* = \lambda^* db$

Here $db = 200$ and $\lambda^* = -0.04$

$df^* = (-0.04)(200) = -8$

$f^*_{new} = f^*_{old} + df^* = 120 - 8 = 112 \text{ inch}$

(c) Exact solution (by solving the new problem):

Min. $f = 4(x+y+z)$

subject to $g = xyz - 1200 = 0$

$x^* = y^* = z^* = (1200)^{1/3} = 10.6266 \text{ inch}$

$f^*_{new} = 4(x^* + y^* + z^*) = 127.5192 \text{ inch}$

2.57 From Problem 2.48,

$x_1^* = \frac{4}{3}, \; x_2^* = \frac{7}{9}, \; x_3^* = \frac{4}{9}, \; \lambda^* = \frac{2}{9}, \; f^* = \frac{1}{9}$

$b = 3$

(a) constraint is: $x_1 + x_2 + 2x_3 - 4 = 0$

$db = 1$, $df^* = \lambda^* db = \frac{2}{9}(1) = \frac{2}{9}$

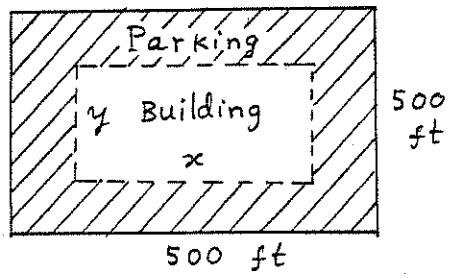
$f^*_{new} = f^*_{old} + df^* = \frac{1}{9} + \frac{2}{9} = \frac{1}{3}$ (f^* increased)

(b) constraint is: $x_1 + x_2 + 2x_3 - 2 = 0$

$db = -1$, $df^* = \lambda^* db = \frac{2}{9}(-1) = -\frac{2}{9}$

$f^*_{new} = f^*_{old} + df^* = -\frac{1}{9}$ (f^* decreased)

2.58 Design variables: x, y, n (n = number of storys)



constraints:

$n\,xy = 8 * 10^5$ (1)

$12n \le 75$; $n = 1, 2, \ldots, 6$ (2)

$500 * 500 - xy \ge 0.1(8 * 10^5)$ (3)

Minimize $f = 5 * 10^5 h + 2000 F + 500 P$

$= 5 * 10^5 (12n) + 2000 (n\,xy) + 500(25 * 10^4 - xy)$

$= 60 * 10^5 n + 2000\, n\,xy + 125 * 10^6 - 500\, xy$ (4)

Eq. (3): $xy \le 25 * 10^4 - 0.8 * 10^5 = 17 * 10^4$ (5)

Using $xy = 8 * 10^5 / n$ from Eq. (1), f becomes:

Min. $f = 6 * 10^6 n + 16 * 10^8 - \frac{4 * 10^8}{n} + 126 * 10^6$ (6)

Since n is discrete, we can use $\frac{df}{dn} = 0$.

With $n = 1$, $f = 13.2 * 10^8$ from Eq. (6); $xy = 8 * 10^5$ from (1)

$n = 2$, $f = 15.14 * 10^8$ from Eq. (6); $xy = 4 * 10^5$ "

$n = 3$, $f = 16.11 * 10^8$ "; $xy = 2.67 * 10^5$ "

$n = 4$, $f = 16.50 * 10^8$ "; $xy = 2 * 10^5$ "

$n = 5$, $f = 16.76 * 10^8$ "; $xy = 1.6 * 10^5$ "

$n = 6$, $f = 16.95 * 10^8$ "; $xy = 1.33 * 10^5$ "

$n = 5$ satisfies the constraint Eq. (5) and minimizes f.

$\therefore f_{min} = \$\, 16.76 * 10^8$; $n^* = 5$; $(xy)^* = 16 * 10^4\ ft^2$.

2.61 Minimize $f = (x_1-2)^2 + (x_2-1)^2$

subject to: $g_1: \; x_1 + x_2 - 2 \le 0$

$g_2: \; x_1^2 - x_2 \le 0$

At optimum, $-\frac{\partial f}{\partial x_i} = \sum_{j \in J_1} \lambda_j \frac{\partial g_j}{\partial x_i}$; $i = 1,2$

$\frac{\partial f}{\partial x_1} = 2(x_1-2)$, $\frac{\partial f}{\partial x_2} = 2(x_2-1)$

$\frac{\partial g_1}{\partial x_1} = 1$, $\frac{\partial g_1}{\partial x_2} = 1$; $\frac{\partial g_2}{\partial x_1} = 2x_1$, $\frac{\partial g_2}{\partial x_2} = -1$

At $\vec{X}_1 = \begin{Bmatrix} 1.5 \\ 0.5 \end{Bmatrix}$, $g_1 = 0$, and $g_2 = 2.25 - 0.5 = 1.75 > 0$ (violated)

$\therefore \vec{X}_1$ is not optimum.

At $\vec{X}_2 = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$, $g_1 = 0$ and $g_2 = 0$ (both active).

$-2(x_1-2) = \lambda_1 + \lambda_2(2x_1) \Rightarrow \lambda_1 + 2\lambda_2 = 2$

$-2(x_2-1) = \lambda_1 - \lambda_2 \Rightarrow \lambda_1 - \lambda_2 = 0$

$\therefore \lambda_1 = \frac{2}{3}$, $\lambda_2 = \frac{2}{3}$ (Both positive)

$\therefore \vec{X}_2$ is a local minimum.

At $\vec{X}_3 = \begin{Bmatrix} 2 \\ 0 \end{Bmatrix}$, $g_1 = 0$, and $g_2 = 4 - 0 = 4$ (violated)

$\therefore \vec{X}_3$ is not optimum.

2.62 Max. $f = 2x_1 + \beta x_2$

subject to: $g_1: \; x_1^2 + x_2^2 - 5 \le 0$

$g_2: \; x_1 - x_2 - 2 \le 0$

$\nabla f = \begin{Bmatrix} 2 \\ \beta \end{Bmatrix}$, $\nabla g_1 = \begin{Bmatrix} 2x_1 \\ 2x_2 \end{Bmatrix}$, $\nabla g_2 = \begin{Bmatrix} 1 \\ -1 \end{Bmatrix}$

At $\vec{X}^* = \begin{Bmatrix} 1 \\ 2 \end{Bmatrix}$, $g_1 = 0$ (active) and $g_2 = -3$ (satisfied)

K-T conditions: $\frac{\partial f}{\partial x_i} + \lambda_1 \frac{\partial g_1}{\partial x_i} = 0$; $i = 1,2$

$\lambda_1 < 0$

Since $\nabla g_1 = \begin{Bmatrix} 2 \\ 4 \end{Bmatrix}$ at $\vec{X}^*$, K-T conditions give

$i = 1$: $\quad 2 + 2\lambda_1 = 0 \Rightarrow \lambda_1 = -1$

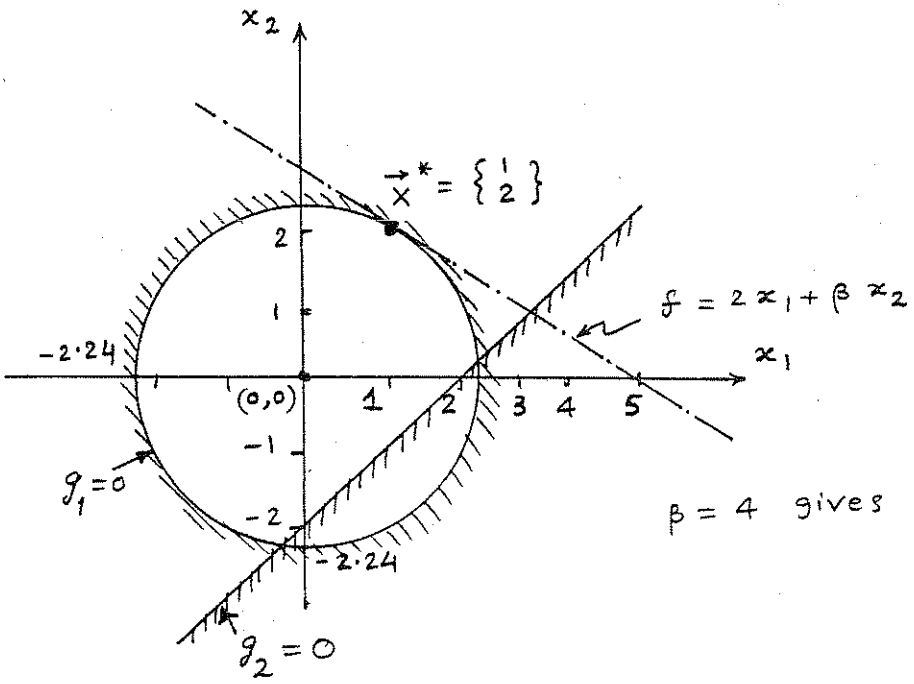
$i = 2$: $\quad \beta + 4\lambda_1 = 0 \Rightarrow \beta + 4(-1) = 0 \Rightarrow \beta = 4$

Graphical procedure:

Plot the constraints: $g_1 = 0$ is a circle with radius $= \sqrt{5}$

$g_2 = 0$ is a straight line

At optimum point, objective function contour should be tangential to $g_1 = 0$.



$\beta = 4$ gives $f = 10$.

2.63

Min. $f = x_1 + x_2$

s.t. $g_1 = -x_1^2 - x_2 + 2 \leq 0$

$g_2 = 4 - x_1 - 3x_2 \leq 0$

$g_3 = x_1 + x_2^4 - 30 \leq 0$

(a) At $\vec{x} = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$, $g_1 = 0$, $g_2 = 0$, $g_3 = -28$.

(g_1 and g_2 are active)

$\frac{\partial f}{\partial x_1} = 1$, $\frac{\partial f}{\partial x_2} = 1$; $\frac{\partial g_1}{\partial x_1} = -2x_1 = -2$, $\frac{\partial g_1}{\partial x_2} = -1$;

$\frac{\partial g_2}{\partial x_1} = -1$, $\frac{\partial g_2}{\partial x_2} = -3$.

K-T conditions:

$$\frac{\partial f}{\partial x_i} + \lambda_1 \frac{\partial g_1}{\partial x_i} + \lambda_2 \frac{\partial g_2}{\partial x_i} = 0 \; ; \; i = 1, 2$$

$\lambda_1 > 0$, $\lambda_2 > 0$

Here for $i = 1$, $1 - 2\lambda_1 - \lambda_2 = 0$

for $i = 2$, $1 - \lambda_1 - 3\lambda_2 = 0$

Solution gives: $\lambda_1 = 0.4$, $\lambda_2 = 0.2$ (Both > 0)

$\therefore$ K-T conditions are satisfied;

$\vec{x} = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$ is optimum.

(b) $\lambda_1 = 0.4$, $\lambda_2 = 0.2$

2.64

Min. $f = x_1^2 + x_2^2 + x_3^2$

subject to:

g_1: $-x_1 - x_2 - x_3 + 5 \leq 0$

g_2: $2 - x_2 x_3 \leq 0$

g_3: $-x_1 \leq 0$

g_4: $-x_2 \leq 0$

g_5: $-x_3 + 2 \leq 0$

(a) At $\vec{x}_1 = \begin{Bmatrix} 3/2 \\ 3/2 \\ 2 \end{Bmatrix}$, $g_1 = -\frac{3}{2} - \frac{3}{2} - 2 + 5 = 0$,

$g_2 = 2 - \left(\frac{3}{2}\right)(2) = -1$, $g_3 = -\frac{3}{2}$,

$g_4 = -\frac{3}{2}$, $g_5 = -2 + 2 = 0$

(g_1 and g_5 are active)

Using K-T conditions,

$$\nabla f + \lambda_1 \nabla g_1 + \lambda_5 \nabla g_5 = \vec{0} \qquad (1)$$

$$\nabla f\Big|_{\vec{x}_1} = \begin{Bmatrix} 2x_1 \\ 2x_2 \\ 2x_3 \end{Bmatrix}_{\vec{x}_1} = \begin{Bmatrix} 3 \\ 3 \\ 4 \end{Bmatrix}, \quad \nabla g_1\Big|_{\vec{x}_1} = \begin{Bmatrix} -1 \\ -1 \\ -1 \end{Bmatrix},$$

$$\nabla g_5\Big|_{\vec{x}_1} = \begin{Bmatrix} 0 \\ 0 \\ -1 \end{Bmatrix}$$

$$\text{Eq. (1)} \Rightarrow \quad \begin{Bmatrix} 3 \\ 3 \\ 4 \end{Bmatrix} + \lambda_1 \begin{Bmatrix} -1 \\ -1 \\ -1 \end{Bmatrix} + \lambda_5 \begin{Bmatrix} 0 \\ 0 \\ -1 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}$$

i.e.,

$3 - \lambda_1 = 0 \quad \Rightarrow \lambda_1 = 3$

$3 - \lambda_1 = 0 \quad \Rightarrow \lambda_1 = 3$

$4 - \lambda_1 - \lambda_5 = 0 \quad \Rightarrow \lambda_5 = 4 - 3 = 1$

Since $\lambda_1 > 0$ and $\lambda_5 > 0$, $\vec{x}_1$ is optimum.

(b) $\vec{x}_2 = \begin{Bmatrix} 4/3 \\ 2/3 \\ 3 \end{Bmatrix}$, $g_1 = -\frac{4}{3} - \frac{2}{3} - 3 + 5 = 0$,

$g_2 = 2 - \frac{2}{3}(3) = 0$, $g_3 = -\frac{4}{3}$, $g_4 = -\frac{2}{3}$,

$g_5 = -3 + 2 = -1$

(g_1 and g_2 are active)

Using K-T conditions,

$$\nabla f + \lambda_1 \nabla g_1 + \lambda_2 \nabla g_2 = \vec{0} \qquad (2)$$

Here $\nabla f\Big|_{\vec{x}_2} = \begin{Bmatrix} 8/3 \\ 4/3 \\ 6 \end{Bmatrix}$, $\nabla g_1\Big|_{\vec{x}_2} = \begin{Bmatrix} -1 \\ -1 \\ -1 \end{Bmatrix}$, $\nabla g_2\Big|_{\vec{x}_2} = \begin{Bmatrix} 0 \\ -3 \\ -2/3 \end{Bmatrix}$

Eq. (2) $\Rightarrow$ $\begin{Bmatrix} 8/3 \\ 4/3 \\ 6 \end{Bmatrix} + \lambda_1 \begin{Bmatrix} -1 \\ -1 \\ -1 \end{Bmatrix} + \lambda_2 \begin{Bmatrix} 0 \\ -3 \\ -2/3 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}$

i.e., $\frac{8}{3} - \lambda_1 = 0 \quad \Rightarrow \lambda_1 = \frac{8}{3}$

$\frac{4}{3} - \lambda_1 - 3\lambda_2 = 0 \quad \Rightarrow \lambda_2 = \frac{1}{3}\left(\frac{4}{3} - \lambda_1\right) = -\frac{4}{9}$

$6 - \lambda_1 - \frac{2}{3}\lambda_2 = 0$

since $\lambda_2 < 0$, K-T conditions are not satisfied.

(c) $\vec{X}_3 = \begin{Bmatrix} 2 \\ 1 \\ 2 \end{Bmatrix}$, $g_1 = -2-1-2+5 = 0$, $g_2 = 2-(1)(2) = 0$,

$g_3 = -2$, $g_4 = -1$, $g_5 = -2+2 = 0$

(g_1, g_2 and g_5 are active)

$\nabla f\big|_{\vec{X}_3} = \begin{Bmatrix} 4 \\ 2 \\ 4 \end{Bmatrix}$, $\nabla g_1\big|_{\vec{X}_3} = \begin{Bmatrix} -1 \\ -1 \\ -1 \end{Bmatrix}$, $\nabla g_2\big|_{\vec{X}_3} = \begin{Bmatrix} 0 \\ -4 \\ -2 \end{Bmatrix}$,

$\nabla g_5\big|_{\vec{X}_3} = \begin{Bmatrix} 0 \\ 0 \\ -1 \end{Bmatrix}$

K-T conditions yield:

$\nabla f + \lambda_1 \nabla g_1 + \lambda_2 \nabla g_2 + \lambda_5 \nabla g_5 = \vec{0}$

$\begin{Bmatrix} 4 \\ 2 \\ 4 \end{Bmatrix} + \lambda_1 \begin{Bmatrix} -1 \\ -1 \\ -1 \end{Bmatrix} + \lambda_2 \begin{Bmatrix} 0 \\ -4 \\ -2 \end{Bmatrix} + \lambda_5 \begin{Bmatrix} 0 \\ 0 \\ -1 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}$

i.e., $4 - \lambda_1 = 0 \quad \Rightarrow \lambda_1 = 4$

$2 - \lambda_1 - 4\lambda_2 = 0 \quad \Rightarrow \lambda_2 = \frac{1}{4}(2-4) = -\frac{1}{2}$

$4 - \lambda_1 - 2\lambda_2 - \lambda_5 = 0 \Rightarrow \lambda_5 = 4 - 4 + 1 = +1$

since $\lambda_2 < 0$, K-T conditions are not satisfied.

$\therefore \vec{X}_3$ is not optimum.

2.65

Min. $f = (x_1-1)^2 + (x_2-5)^2$

subject to: $g_1: \; -x_1^2 + x_2 - 4 \le 0$

$g_2: \; -(x_1-2)^2 + x_2 - 3 \le 0$

$$\nabla f = \begin{Bmatrix} 2(x_1-1) \\ 2(x_2-5) \end{Bmatrix}, \quad \nabla g_1 = \begin{Bmatrix} -2x_1 \\ 1 \end{Bmatrix}, \quad \nabla g_2 = \begin{Bmatrix} -2(x_1-2) \\ 1 \end{Bmatrix}$$

(a) At $\vec{X}_1 = \begin{Bmatrix} -1 \\ 5 \end{Bmatrix}$, $g_1 = 0$ and $g_2 = -7$.

(g_1 is active)

Usable feasible direction is given by:

$$\nabla f^T \vec{S} < 0 \Rightarrow (-4 \quad 0) \begin{Bmatrix} s_1 \\ s_2 \end{Bmatrix} < 0 \Rightarrow -4s_1 < 0$$

$$\nabla g_1^T \vec{S} \le 0 \Rightarrow (2 \quad 1) \begin{Bmatrix} s_1 \\ s_2 \end{Bmatrix} \le 0 \Rightarrow 2s_1 + s_2 \le 0$$

since $\nabla f = \begin{Bmatrix} -4 \\ 0 \end{Bmatrix}$ and $\nabla g_1 = \begin{Bmatrix} 2 \\ 1 \end{Bmatrix}$

$\therefore \vec{S} = \begin{Bmatrix} 1 \\ -3 \end{Bmatrix}$ is a usable feasible direction.

(b) At $\vec{X}_2 = \begin{Bmatrix} 2 \\ 3 \end{Bmatrix}$, $g_1 = -5$ and $g_2 = 0$

(g_2 is active).

$$\nabla f = \begin{Bmatrix} 2 \\ -4 \end{Bmatrix} \text{ and } \nabla g_2 = \begin{Bmatrix} 0 \\ 1 \end{Bmatrix}.$$

Usable feasible direction is given by:

$$\nabla f^T \vec{S} < 0 \Rightarrow (2 \quad -4) \begin{Bmatrix} s_1 \\ s_2 \end{Bmatrix} < 0 \Rightarrow 2s_1 - 4s_2 < 0$$

$$\nabla g_2^T \vec{S} \le 0 \Rightarrow (0 \quad 1) \begin{Bmatrix} s_1 \\ s_2 \end{Bmatrix} \le 0 \Rightarrow s_2 \le 0$$

$\therefore \vec{S} = \begin{Bmatrix} -3 \\ -1 \end{Bmatrix}$ is a usable feasible direction.

2.66

Min. $f = x_1^2 - x_2$

subject to: $g_1: -x_1 \le 0$

$g_2: -26 + x_1^2 + x_2^2 \le 0$

$g_3: 6 - x_1 - x_2 \le 0$

At $\vec{x} = \begin{Bmatrix} 5 \\ 1 \end{Bmatrix}$, $g_1 < 0$ and $g_2 = g_3 = 0$ (active).

$$\nabla f = \begin{Bmatrix} 2x_1 \\ -1 \end{Bmatrix} = \begin{Bmatrix} 10 \\ -1 \end{Bmatrix}, \quad \nabla g_2 = \begin{Bmatrix} 2x_1 \\ 2x_2 \end{Bmatrix} = \begin{Bmatrix} 10 \\ 2 \end{Bmatrix}, \quad \nabla g_3 = \begin{Bmatrix} -1 \\ -1 \end{Bmatrix}$$

(a) $\vec{s}^T \nabla f = (0 \quad 1) \begin{Bmatrix} 10 \\ -1 \end{Bmatrix} = -1 < 0$ (usable)

$\vec{s}^T \nabla g_2 = (0 \quad 1) \begin{Bmatrix} 10 \\ 2 \end{Bmatrix} = 2 > 0$ (not feasible)

$\vec{s}^T \nabla g_3 = (0 \quad 1) \begin{Bmatrix} -1 \\ -1 \end{Bmatrix} = -1 < 0$ (feasible)

$\therefore \vec{s} = \begin{Bmatrix} 0 \\ 1 \end{Bmatrix}$ is usable; but not feasible.

(b) $\vec{s}^T \nabla f = (-1 \quad 1) \begin{Bmatrix} 10 \\ -1 \end{Bmatrix} = -11 < 0$ (usable)

$\left.\begin{array}{l} \vec{s}^T \nabla g_2 = (-1 \quad 1) \begin{Bmatrix} 10 \\ 2 \end{Bmatrix} = -8 < 0 \\ \vec{s}^T \nabla g_3 = (-1 \quad 1) \begin{Bmatrix} -1 \\ -1 \end{Bmatrix} = 0 \end{array}\right\}$ (feasible)

$\therefore \vec{s} = \begin{Bmatrix} -1 \\ 1 \end{Bmatrix}$ is usable and feasible.

(c) $\vec{s}^T \nabla f = (1 \quad 0) \begin{Bmatrix} 10 \\ -1 \end{Bmatrix} = 10 > 0$ (not usable)

$\left.\begin{array}{l} \vec{s}^T \nabla g_2 = (1 \quad 0) \begin{Bmatrix} 10 \\ 2 \end{Bmatrix} = 10 > 0 \\ \vec{s}^T \nabla g_3 = (1 \quad 0) \begin{Bmatrix} -1 \\ -1 \end{Bmatrix} = -1 < 0 \end{array}\right\}$ (not feasible)

$\therefore \vec{s} = \begin{Bmatrix} 1 \\ 0 \end{Bmatrix}$ is not usable; not feasible.

(d) $\vec{s}^T \nabla f = (-1 \quad 2) \begin{Bmatrix} 10 \\ -1 \end{Bmatrix} = -12$ (usable)

$$\vec{S}^T \nabla g_2 = (-1 \quad 2) \begin{Bmatrix} 10 \\ 2 \end{Bmatrix} = -6 < 0$$

$$\vec{S}^T \nabla g_3 = (-1 \quad 2) \begin{Bmatrix} -1 \\ -1 \end{Bmatrix} = -1 < 0 \quad \Big\} \text{ (feasible)}$$

$\therefore \vec{S} = \begin{Bmatrix} -1 \\ 2 \end{Bmatrix}$ is usable ; feasible.

2.67

Min. $f = x_1^3 - 6x_1^2 + 11x_1 + x_3$

subject to: $g_1: \; x_1^2 + x_2^2 - x_3^2 \le 0$

$g_2: \; 4 - x_1^2 - x_2^2 - x_3^2 \le 0$

$g_3: -x_1 \le 0, \quad g_4: -x_2 \le 0, \quad g_5 = -x_3 \le 0$

$g_6: \; x_3 - 5 \le 0$

At $\vec{X} = \begin{Bmatrix} 0 \\ \sqrt{2} \\ \sqrt{2} \end{Bmatrix}$, g_1, g_2 and g_3 are active.

$$\frac{\partial f}{\partial x_1} = 3x_1^2 - 12x_1 + 11 = 11, \quad \frac{\partial f}{\partial x_2} = 0, \quad \frac{\partial f}{\partial x_3} = 1$$

$$\frac{\partial g_1}{\partial x_1} = 2x_1 = 0, \quad \frac{\partial g_1}{\partial x_2} = 2x_2 = 2\sqrt{2}, \quad \frac{\partial g_1}{\partial x_3} = -2x_3 = -2\sqrt{2}$$

$$\frac{\partial g_2}{\partial x_1} = -2x_1 = 0, \quad \frac{\partial g_2}{\partial x_2} = -2x_2 = -2\sqrt{2}, \quad \frac{\partial g_2}{\partial x_3} = -2x_3 = -2\sqrt{2}$$

$$\frac{\partial g_3}{\partial x_1} = -1, \quad \frac{\partial g_3}{\partial x_2} = 0, \quad \frac{\partial g_3}{\partial x_3} = 0$$

K-T conditions are:

$$\frac{\partial f}{\partial x_i} + \lambda_1 \frac{\partial g_1}{\partial x_i} + \lambda_2 \frac{\partial g_2}{\partial x_i} + \lambda_3 \frac{\partial g_3}{\partial x_i} = 0 \; ; \; i = 1, 2, 3$$

$$\lambda_i > 0 \; ; \; i = 1, 2, 3$$

These equations, in the present case, yield:

$$11 + \lambda_1(0) + \lambda_2(0) + \lambda_3(-1) = 0 \Rightarrow \lambda_3 = 11$$

$$0 + \lambda_1(2\sqrt{2}) + \lambda_2(-2\sqrt{2}) + \lambda_3 = 0 \Rightarrow \lambda_1 = \lambda_2$$

$$1 + \lambda_1(-2\sqrt{2}) + \lambda_2(-2\sqrt{2}) + \lambda_3 = 0 \Rightarrow \lambda_1 = (1/4\sqrt{2}) = \lambda_2$$

Since $\lambda_1 > 0$, $\lambda_2 > 0$, $\lambda_3 > 0$, given vector $\vec{X}$ is optimum.

2.68

Min. $f = x_1^2 + 2x_2^2 + 3x_3^2$ (1)

subject to: $g_1: x_1 - x_2 - 2x_3 - 12 \le 0$ (2)

$g_2: x_1 + 2x_2 - 3x_3 - 8 \le 0$ (3)

Kuhn-Tucker conditions:

$$\frac{\partial f}{\partial x_i} + \lambda_1 \frac{\partial g_1}{\partial x_i} + \lambda_2 \frac{\partial g_2}{\partial x_i} = 0 \; ; \quad i = 1,2,3 \qquad (4)$$

$\lambda_1 g_1 = 0 \; ; \quad g_1 \le 0$

$\lambda_2 g_2 = 0 \; ; \quad g_2 \le 0$

$\lambda_1 \ge 0 \; ; \quad \lambda_2 \ge 0$

$\frac{\partial f}{\partial x_1} = 2x_1, \quad \frac{\partial f}{\partial x_2} = 4x_2, \quad \frac{\partial f}{\partial x_3} = 6x_3$

$\frac{\partial g_1}{\partial x_1} = 1, \quad \frac{\partial g_1}{\partial x_2} = -1, \quad \frac{\partial g_1}{\partial x_3} = -2$

$\frac{\partial g_2}{\partial x_1} = 1, \quad \frac{\partial g_2}{\partial x_2} = 2, \quad \frac{\partial g_2}{\partial x_3} = -3$

Equations (4):

$2x_1 + \lambda_1 + \lambda_2 = 0$ (5)

$4x_2 - \lambda_1 + 2\lambda_2 = 0$ (6)

$6x_3 - 2\lambda_1 - 3\lambda_2 = 0$ (7)

(i) Assume $g_1 = 0$, and $g_2 = 0$ (both active):

Eqs. (5) to (7): $x_1 = -\frac{1}{2}\lambda_1 - \frac{1}{2}\lambda_2$ (8)

$x_2 = \frac{\lambda_1}{4} - \frac{1}{2}\lambda_2$ (9)

$x_3 = \frac{1}{3}\lambda_1 + \frac{1}{2}\lambda_2$ (10)

Using Eqs. (8) to (10) in $g_1 = 0$ and $g_2 = 0$; we get

$$x_1 - x_2 - 2x_3 - 12 = -\frac{1}{2}\lambda_1 - \frac{1}{2}\lambda_2 - \frac{\lambda_1}{4} + \frac{1}{2}\lambda_2 - \frac{2}{3}\lambda_1 - \lambda_2 - 12 = 0$$

$$\Rightarrow -\frac{17}{12}\lambda_1 - \lambda_2 = 12 \qquad (11)$$

and

$$x_1 + 2x_2 - 3x_3 - 8 = -\frac{\lambda_1}{2} - \frac{\lambda_2}{2} + \frac{\lambda_1}{2} - \lambda_2 - \lambda_1 - \frac{3}{2}\lambda_2 - 8 = 0$$

$$\Rightarrow \quad -\lambda_1 - 3\lambda_2 = 8 \qquad (12)$$

Solution of Eqs. (11) and (12): $\lambda_1 = -8.6154$, $\lambda_2 = 0.2051$

K-T conditions are not satisfied.

(ii) Assume $g_1 = 0$ (active):

$\lambda_2 = 0$ (since g_2 is not active)

Eqs. (5) to (7) $\Rightarrow$ $x_1 = -\frac{1}{2}\lambda_1$, $x_2 = \frac{1}{4}\lambda_1$, $x_3 = \frac{1}{3}\lambda_1$ (13)

Eqs. (13) and $g_1 = 0$ yield:

$$-\frac{1}{2}\lambda_1 - \frac{1}{4}\lambda_1 - \frac{2}{3}\lambda_1 - 12 = 0 \quad \Rightarrow \quad \lambda_1 = -\frac{12}{17}(12) = -8.4706$$

K-T conditions not satisfied.

(iii) Assume $g_2 = 0$ (active):

$\lambda_1 = 0$ (since g_1 is not active)

Eqs. (5) to (7) $\Rightarrow$ $x_1 = -\frac{1}{2}\lambda_2$, $x_2 = -\frac{1}{2}\lambda_2$, $x_3 = \frac{1}{2}\lambda_2$ (14)

Eqs. (14) and $g_2 = 0$ give:

$$-\frac{1}{2}\lambda_2 - \lambda_2 - \frac{3}{2}\lambda_2 - 8 = 0 \quad \Rightarrow \quad \lambda_2 = -\frac{8}{3}$$

K-T conditions are not satisfied.

(iv) Assume no constraint to be active:

$\lambda_1 = \lambda_2 = 0$

Eqs. (5) to (7) $\Rightarrow$ $x_1 = 0$, $x_2 = 0$, $x_3 = 0$ (15)

At this vector, $g_1 = -12 < 0$

$g_2 = -8 < 0$

$\therefore$ optimum solution is $\vec{X} = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}$ and $f^* = 0$.

2.69 (a) Graphical Method:

$$\text{Min.} \quad f = (x_1 - 1)^2 + (x_2 - 5)^2 \qquad (1)$$

subject to

$$g_1: \quad -x_1^2 + x_2 - 4 \le 0 \qquad (2)$$

$$g_2: \quad -(x_1 - 2)^2 + x_2 - 3 \le 0 \qquad (3)$$

Equation (1) is a circle of radius $\sqrt{f}$ with center at (1, 5).

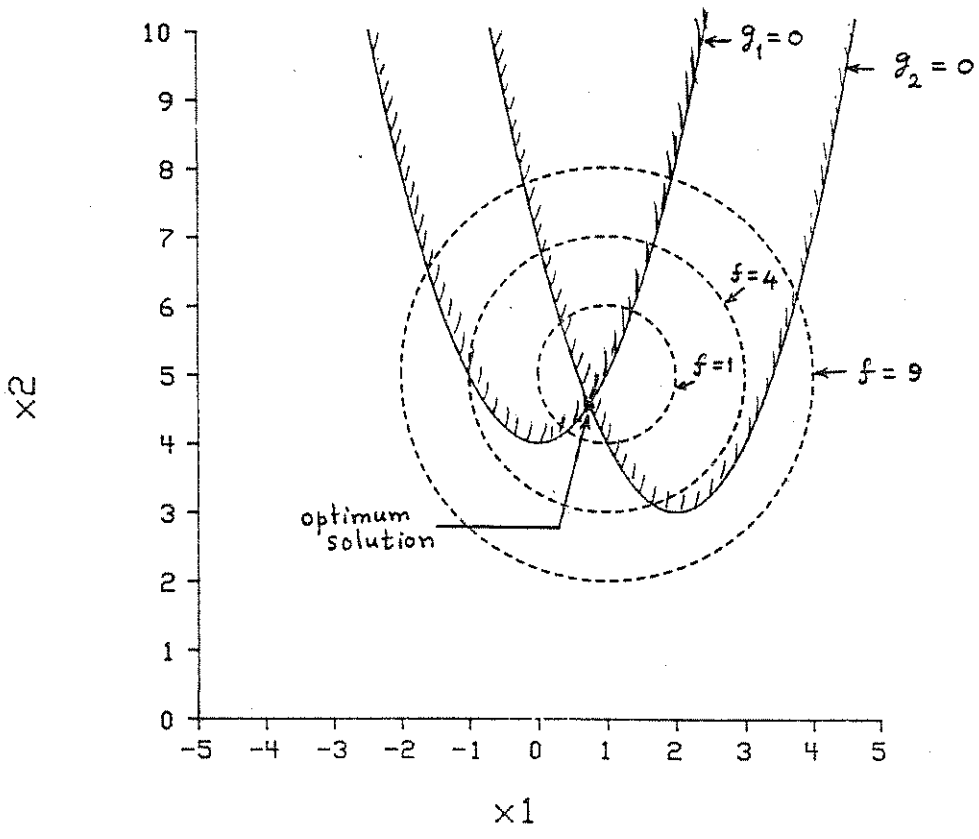
Eq. (2): $x_2 = 4 + x_1^2$:

x_1	−3	−2	−1	0	1	2	3
x_2	13	8	5	4	5	8	13

Eq. (3): $x_2 = 3 + (x_1 - 2)^2$:

x_1	−2	−1	0	1	2	3	4	5
x_2	19	12	7	4	3	4	7	12

Graphical optimization is indicated below. optimum solution is given by $\vec{x}^* = \begin{Bmatrix} 0.75 \\ 4.56 \end{Bmatrix}$ with $f_{min} = 0.256$.



(b) Kuhn-Tucker Conditions:

K-T conditions: $\frac{\partial f}{\partial x_i} + \lambda_1 \frac{\partial g_1}{\partial x_i} + \lambda_2 \frac{\partial g_2}{\partial x_i} = 0 \quad , \; i = 1, 2 \qquad (1)$

$\lambda_j \, g_j = 0 \quad , \; j = 1, 2 \qquad (2)$

$g_j \leq 0 \quad , \; j = 1, 2 \qquad (3)$

$\lambda_j \geq 0 \quad , \; j = 1, 2 \qquad (4)$

Eqs. (1): $2(x_1 - 1) - \lambda_1 \, 2x_1 - \lambda_2 \, 2(x_1 - 2) = 0 \qquad (5)$

$2(x_2 - 5) + \lambda_1 + \lambda_2 = 0 \qquad (6)$

Eqs. (2): $\lambda_1 \left(- x_1^2 + x_2 - 4 \right) = 0 \qquad (7)$

$\lambda_2 \left(- x_1^2 + 4x_1 + x_2 - 7 \right) = 0 \qquad (8)$

Eqs. (3): $- x_1^2 + x_2 - 4 \leq 0 \qquad (9)$

$- x_1^2 + 4x_1 + x_2 - 7 \leq 0 \qquad (10)$

Eqs. (4): $\lambda_1 \geq 0 \qquad (11)$

$\lambda_2 \geq 0 \qquad (12)$

From Eq. (8), $\lambda_2 = 0$ or $- x_1^2 + 4x_1 + x_2 - 7 = 0$.

Let $\lambda_2 = 0$: From Eq. (6), $2x_2 - 10 + \lambda_1 = 0$ or $\lambda_1 = 10 - 2x_2 \qquad (13)$

Using Eqs. (13) and (5), we find

$2x_1 - 2 - 2x_1 (10 - 2x_2) = 0 \quad$ or $\quad x_1 = \frac{1}{2x_2 - 9} \qquad (14)$

Eqs. (13), (14) and (7):

$$\left(10 - 2x_2\right)\left(- \frac{1}{(2x_2 - 9)^2} + x_2 - 4 \right) = 0$$

$$x_2 = 5 \quad \text{or} \quad (x_2 - 4)(2x_2 - 9)^2 - 1 = 0$$

If $x_2 = 5$, $\lambda_1 = 0$, $\lambda_2 = 0$, $x_1 = 1 \Rightarrow$ Eq. (10) is violated.

If $(x_2 - 4)(2x_2 - 9)^2 - 1 = 0$; $4x_2^3 - 52x_2^2 + 225x_2 - 325 = 0$.

$x_2 = 5 \; , \; 4 + i\,0.5 \; , \; 4 - i\,0.5$

Imaginary values need not be considered.

If $x_2 = 5$, $x_1 = 1$ from Eq. (14), $\lambda_2 = 0$, and $\lambda_1 = 1$ from Eq. (13). $\Rightarrow$ Eq. (10) is violated.

Let $-x_1^2 + 4x_1 + x_2 - 7 = 0$:

$$x_2 = x_1^2 - 4x_1 + 7 \qquad (15)$$

Eqs. (7) and (15): $\lambda_1(-x_1^2 + x_1^2 - 4x_1 + 7 - 4) = 0$

i.e. $\lambda_1(-4x_1 + 3) = 0$

Let $\lambda_1 = 0$.

Eqs. (5) and (6): $\lambda_2 = -2(x_1^2 - 4x_1 + 2)$ (16)

$$2x_1 - 2 - 4(x_1^2 - 4x_1 + 2)(2 - x_1) = 0$$

or $2x_1^3 - 12x_1^2 + 21x_1 - 9 = 0$

or $x_1 = 0.634,\ 2.366,\ 3.0$

If $x_1 = 0.634$, $x_2 = 4.866$ from Eq. (15),

$\lambda_2 = 0.2681$ from Eq. (16), $\lambda_1 = 0 \Rightarrow$ Eq. (9) violated.

If $x_1 = 2.366$, $x_2 = 3.134$ from Eq. (15),

$\lambda_2 = 3.7321$ from Eq. (16), $\lambda_1 = 0 \Rightarrow f = 5.3479$

and all equations satisfied.

If $x_1 = 3.0$, $x_2 = 4.0$ from Eq. (15),

$\lambda_2 = 2$ from Eq. (16), $\lambda_1 = 0 \Rightarrow f = 5$

and all equations satisfied.

Let $-4x_1 + 3 = 0$.

$x_1 = 0.75$, $x_2 = 4.5625$ from Eq. (15).

Eqs. (5) and (6): $-0.5 - 1.5\lambda_1 + 2.5\lambda_2 = 0$

$-0.875 + \lambda_1 + \lambda_2 = 0$

Solution gives: $\lambda_1 = 0.4219$, $\lambda_2 = 0.4531$

$f = 0.2539$ (all constraints satisfied).

$\therefore \vec{X}^* = \begin{Bmatrix} 0.75 \\ 4.5625 \end{Bmatrix}$, $f^* = 0.2539$.

Note: Some cases were not considered.

2.70

Max. $f = 8x_1 + 4x_2 + x_1x_2 - x_1^2 - x_2^2$

subject to:
$$2x_1 + 3x_2 - 24 \le 0$$
$$-5x_1 + 12x_2 - 24 \le 0$$
$$x_2 - 5 \le 0$$

Kuhn-Tucker conditions:

$$\frac{\partial f}{\partial x_i} + \lambda_1 \frac{\partial g_1}{\partial x_i} + \lambda_2 \frac{\partial g_2}{\partial x_i} + \lambda_3 \frac{\partial g_3}{\partial x_i} = 0 \quad ; \; i = 1, 2$$

$$8 + x_2 - 2x_1 + 2\lambda_1 - 5\lambda_2 = 0 \qquad (1)$$

$$4 + x_1 - 2x_2 + 3\lambda_1 + 12\lambda_2 + \lambda_3 = 0 \qquad (2)$$

$\lambda_j g_j = 0$, $j = 1, 2, 3$

$$\lambda_1 (2x_1 + 3x_2 - 24) = 0 \qquad (3)$$

$$\lambda_2 (-5x_1 + 12x_2 - 24) = 0 \qquad (4)$$

$$\lambda_3 (x_2 - 5) = 0 \qquad (5)$$

$g_j \le 0$, $j = 1, 2, 3$

$$2x_1 + 3x_2 - 24 \le 0 \qquad (6)$$

$$-5x_1 + 12x_2 - 24 \le 0 \qquad (7)$$

$$x_2 - 5 \le 0 \qquad (8)$$

$\lambda_j \le 0$, $j = 1, 2, 3$

$$\lambda_1 \le 0, \quad \lambda_2 \le 0, \quad \lambda_3 \le 0 \qquad (9)$$

Eq. (5) shows: $\lambda_3 = 0$ or $x_2 = 5$.

Let $\lambda_3 = 0$:

Eqs. (1) and (2):
$$8 + x_2 - 2x_1 + 2\lambda_1 - 5\lambda_2 = 0$$
$$4 + x_1 - 2x_2 + 3\lambda_1 + 12\lambda_2 = 0$$

solution of these equations is

$$x_2 = \frac{8}{3}\lambda_1 + \frac{19}{3}\lambda_2 + \frac{16}{3} \qquad (10)$$

$$x_1 = \frac{7}{3}\lambda_1 + \frac{2}{3}\lambda_2 + \frac{20}{3} \qquad (11)$$

Substitute Eqs. (10) and (11) into Eqs. (3) and (4):

$$\lambda_1\left(\frac{14}{3}\lambda_1+\frac{4}{3}\lambda_2+\frac{40}{3}+8\lambda_1+19\lambda_2+16-24\right)=0$$

$$\lambda_2\left(-\frac{35}{3}\lambda_1-\frac{10}{3}\lambda_2-\frac{100}{3}+32\lambda_1+76\lambda_2+64-24\right)=0$$

i.e.,

$$\lambda_1\left(\frac{38}{3}\lambda_1+\frac{61}{3}\lambda_2+\frac{16}{3}\right)=0 \qquad (12)$$

$$\lambda_2\left(\frac{61}{3}\lambda_1+\frac{218}{3}\lambda_2+\frac{20}{3}\right)=0 \qquad (13)$$

Four possible solutions to Eqs. (12) and (13):

(i) $\lambda_1=0$, $\frac{61}{3}\lambda_1+\frac{218}{3}\lambda_2+\frac{20}{3}=0$:

$$\lambda_1=0, \quad \lambda_2=-\frac{20}{218} \qquad (14)$$

Eqs. (10), (11) and (14) give: $x_1=6.6055$, $x_2=4.7523$

Eq. (6) is violated.

(ii) $\lambda_1=0$, $\lambda_2=0$:

Eqs. (10) and (11) give: $x_1=\frac{20}{3}$, $x_2=\frac{16}{3}$

Eq. (8) is violated.

(iii) $\frac{38}{3}\lambda_1+\frac{61}{3}\lambda_2+\frac{16}{3}=0$, $\frac{61}{3}\lambda_1+\frac{218}{3}\lambda_2+\frac{20}{3}=0$:

Simultaneous solution of these equations:

$$\lambda_2=\frac{216}{4563}=0.04733; \quad \lambda_1=-0.4970$$

Eq. (9) violated.

(iv) $\lambda_2=0$, $\frac{38}{3}\lambda_1+\frac{61}{3}\lambda_2+\frac{16}{3}=0$:

$$\lambda_2=0, \quad \lambda_1=-\frac{16}{38}=-0.4210 \qquad (15)$$

Eqs. (15), (10) and (11) give: $x_2=4.21$, $x_1=5.684$

All equations are satisfied.

Let $x_2=5$:

Eq. (1) gives: $x_1=\lambda_1-2.5\lambda_2+6.5$ (16)

Eqs. (16), (3) and (4):

$$\lambda_1\left(2\lambda_1-5\lambda_2+13+15-24\right)=0$$

$$\lambda_2\left(-5\lambda_1+12.5\lambda_2-32.5-24+60\right)=0$$

i.e., $\lambda_1 (2\lambda_1 - 5\lambda_2 + 4) = 0$ (17)

$\lambda_2 (-5\lambda_1 + 12.5\,\lambda_2 + 3.5) = 0$ (18)

Four possible solutions to Eqs. (17) and (18):

(i) $\lambda_2 = 0$, $2\lambda_1 - 5\lambda_2 + 4 = 0$:

$\lambda_2 = 0$, $\lambda_1 = -2$ (19)

Eqs. (19) and (1) give: $x_1 = 4.5$, $x_2 = 5.0$

Eq. (7) is violated.

(ii) $\lambda_1 = 0$, $\lambda_2 = 0$:

Eq. (1) gives: $x_1 = 6.5$, $x_2 = 5.0$

Eq. (6) is violated.

(iii) $\lambda_1 = 0$, $-5\lambda_1 + 12.5\,\lambda_2 + 3.5 = 0$:

$\lambda_1 = 0$, $\lambda_2 = -0.28$ (20)

Eqs. (20) and (1) give: $x_1 = 7.2$, $x_2 = 5.0$

Eq. (6) is violated.

(iv) $2\lambda_1 - 5\lambda_2 + 4 = 0$, $-5\lambda_1 + 12.5\,\lambda_2 - 3.5 = 0$:

solution of these equations is:

$$\left.\begin{aligned} 5\lambda_1 - 12.5\,\lambda_2 &= -10 \\ 5\lambda_1 - 12.5\,\lambda_2 &= -3.5 \end{aligned}\right\} \text{ no solution!}$$

∴ Only one solution satisfies all equations.

$x_1^* = 4.21$, $x_2^* = 5.684$

$\lambda_1^* = 0$, $\lambda_2^* = -0.4210$

$f^* = 36.2105$

2.71 Maximize $f = (x-1)^2$

subject to $g_1: \; -2-x \le 0$, $g_2: \; x-4 \le 0$

optimum points: $x^* = -2 \; , \; 4$

At $x^* = -2$, g_1 is active and at $x^* = 4$, g_2 is active.

Since the constraints are linear, the constraint qualification is satisfied at the optimum points.

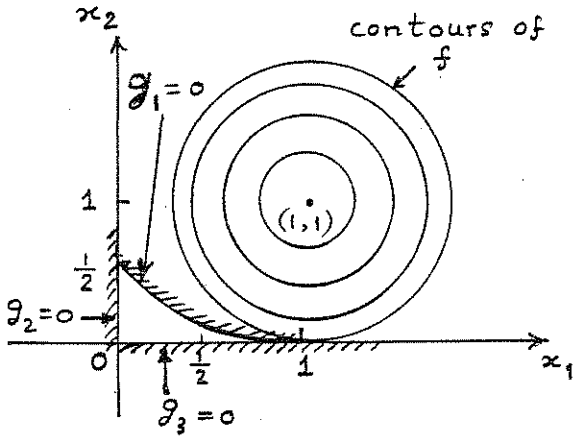
2.72 Maximize $f = (x_1-1)^2 + (x_2-1)^2$

subject to

$2x_2 - (1-x_1)^3 \le 0$ or $g_1 \le 0$

$x_1 \ge 0$ or $-x_1 \le 0$ or $g_2 \le 0$

$x_2 \ge 0$ or $-x_2 \le 0$ or $g_3 \le 0$



Graphical optimization shows the optimum point as $\vec{x}^* = \begin{Bmatrix} 1 \\ 0 \end{Bmatrix}$

At $\vec{x}^*$, g_1 and g_3 are active.

$$\nabla g_1 \Big|_{\vec{x}^*} = \begin{Bmatrix} 3(1-x_1)^2 \\ 2 \end{Bmatrix}_{\vec{x}^*} = \begin{Bmatrix} 0 \\ 2 \end{Bmatrix}$$

$$\nabla g_3 \Big|_{\vec{x}^*} = \begin{Bmatrix} 0 \\ -1 \end{Bmatrix}$$

Since the gradients $\nabla g_1 \Big|_{\vec{x}^*}$ and $\nabla g_3 \Big|_{\vec{x}^*}$ are not linearly independent, the constraint qualification is not satisfied.

Kuhn-Tucker conditions:

$$\frac{\partial f}{\partial x_i} + \lambda_1 \frac{\partial g_1}{\partial x_i} + \lambda_3 \frac{\partial g_3}{\partial x_i} = 0 \; ; \; i = 1,2$$

$$\lambda_i > 0 \; ; \; i = 1,3$$

Here $\nabla f \Big|_{\vec{x}^*} = \begin{Bmatrix} 2(x_1-1) \\ 2(x_2-1) \end{Bmatrix}_{\vec{x}^*} = \begin{Bmatrix} 0 \\ -2 \end{Bmatrix}$

Kuhn-Tucker conditions become: $0 + \lambda_1(0) + \lambda_3(0) = 0$

$$-2 + \lambda_1(2) + \lambda_3(-1) = 0$$

or $\lambda_3 = 2\lambda_1 - 2$

Since λ_1 can have any value, the K-T conditions are not satisfied at $\vec{x}^*$.

2.73

$$f = -4x_1 + x_1^2 - 2x_1x_2 + 2x_2^2$$

$$\frac{\partial f}{\partial x_1} = -4 + 2x_1 - 2x_2 \quad , \quad \frac{\partial f}{\partial x_2} = -2x_1 + 4x_2$$

$$\frac{\partial^2 f}{\partial x_1^2} = 2 \quad , \quad \frac{\partial^2 f}{\partial x_1 \partial x_2} = -2 \quad , \quad \frac{\partial^2 f}{\partial x_2^2} = 4$$

$$[J]_f = \begin{bmatrix} 2 & -2 \\ -2 & 4 \end{bmatrix} \Rightarrow J_1 = +2, \quad J_2 = 8 - 4 = +4$$

$[J]_f$ = positive definite, f = convex

$$\left.\begin{aligned} g_1 &= 2x_1 + x_2 - 6 \le 0 \\ g_2 &= x_1 - 4x_2 \le 0 \end{aligned}\right\}$$ Linear constraints; can be considered convex

$\therefore$ Problem is convex.

2.74

(a) $$f = 2x_1 + 3x_2 - x_1^3 - 2x_2^2$$

$$\frac{\partial f}{\partial x_1} = 2 - 3x_1^2 \quad , \quad \frac{\partial f}{\partial x_2} = 3 - 4x_2$$

$$\frac{\partial^2 f}{\partial x_1^2} = -6x_1 \quad , \quad \frac{\partial^2 f}{\partial x_1 \partial x_2} = 0 \quad , \quad \frac{\partial^2 f}{\partial x_2^2} = -4$$

$$[J]_f = \begin{bmatrix} -6x_1 & 0 \\ 0 & -4 \end{bmatrix} \Rightarrow J_1 = -6x_1 \quad , \quad J_2 = 24x_1$$

For $x_1 \ge 0$, $J_1 \le 0$ and $J_2 = \ge 0$

If $x_1 = 0$ is excluded, $[J]_f$ = negative definite and hence f = concave (strictly)

otherwise, f = concave.

constraints are linear; can be considered convex.

$\therefore$ Problem is not convex.

(b) $f = 9x_1^2 - 18x_1x_2 + 13x_1 - 4$

$$\frac{\partial f}{\partial x_1} = 18x_1 - 18x_2 + 13 \quad , \quad \frac{\partial f}{\partial x_2} = -18x_1$$

$$\frac{\partial^2 f}{\partial x_1^2} = 18 \quad , \quad \frac{\partial^2 f}{\partial x_1 \partial x_2} = -18 \quad , \quad \frac{\partial^2 f}{\partial x_2^2} = 0$$

$$[J]_f = \begin{bmatrix} 18 & -18 \\ -18 & 0 \end{bmatrix} \Rightarrow \quad J_1 = +18 \quad , \quad J_2 = -324$$

$[J]_f$ is indefinite.

$\therefore$ Problem is not convex.

2.75

$$\frac{\partial f}{\partial x_1} = 200(x_2 - x_1^2)(-2x_1) + 2(1-x_1)(-1)$$
$$= -400\,x_1(x_2 - x_1^2) - 2(1-x_1)$$

$$\frac{\partial f}{\partial x_2} = 200(x_2 - x_1^2)$$

$$\frac{\partial g_1}{\partial x_1} = 1, \quad \frac{\partial g_1}{\partial x_2} = -2x_2$$

$$\frac{\partial g_2}{\partial x_1} = -2x_1, \quad \frac{\partial g_2}{\partial x_2} = 1$$

$$\frac{\partial g_3}{\partial x_1} = -1, \quad \frac{\partial g_3}{\partial x_2} = 0$$

$$\frac{\partial g_4}{\partial x_1} = 1, \quad \frac{\partial g_4}{\partial x_2} = 0$$

$$\frac{\partial g_5}{\partial x_1} = 0, \quad \frac{\partial g_5}{\partial x_2} = 1$$

g_1: $x_1 - x_2^2 \le 0$

g_2: $x_2 - x_1^2 \le 0$

g_3: $-\frac{1}{2} - x_1 \le 0$

g_4: $x_1 - \frac{1}{2} \le 0$

g_5: $x_2 - 1 \le 0$

(a) $\vec{X}_1 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$; g_1 and g_2 are active.

$$\frac{\partial f}{\partial x_i} + \lambda_1 \frac{\partial g_1}{\partial x_i} + \lambda_2 \frac{\partial g_2}{\partial x_i} = 0 \quad ; i = 1, 2$$

Equations to be satisfied

$\lambda_1 > 0$, $\lambda_2 > 0$

Since $\frac{\partial f}{\partial x_1} = -2$, $\frac{\partial f}{\partial x_2} = 0$, $\frac{\partial g_1}{\partial x_1} = \frac{\partial g_2}{\partial x_2} = 1$ and $\frac{\partial g_1}{\partial x_2} = \frac{\partial g_2}{\partial x_1} = 0$,

above equations yield:

$$-2 + \lambda_1 = 0 \Rightarrow \lambda_1 = 2$$
$$\lambda_2 = 0 \Rightarrow \lambda_2 = 0 \text{ (not positive)}$$

K-T conditions not satisfied ; $\vec{X}_1$ is not optimum.

(b) $\vec{X}_2 = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$; no constraint is active.

$$\frac{\partial f}{\partial x_i} = 0 \quad ; i = 1, 2$$

Equations to be satisfied.

Here $\left.\frac{\partial f}{\partial x_1}\right|_{\vec{X}_2} = -2$ and $\left.\frac{\partial f}{\partial x_2}\right|_{\vec{X}_2} = -200$

conditions not satisfied ; $\vec{X}_2$ is not optimum.

(c) $\vec{X}_3 = \begin{Bmatrix} -1/2 \\ 1/4 \end{Bmatrix}$; g_2 and g_3 are active.

$$\frac{\partial f}{\partial x_i} + \lambda_2 \frac{\partial g_2}{\partial x_i} + \lambda_3 \frac{\partial g_3}{\partial x_i} = 0 \; ; \quad i = 1, 2$$

$\lambda_2 > 0, \quad \lambda_3 > 0$

Equations to be satisfied

At $\vec{X}_3$, $\frac{\partial f}{\partial x_1} = -400\left(-\frac{1}{2}\right)\left(\frac{1}{4} - \frac{1}{4}\right) - 2\left(1 + \frac{1}{2}\right) = -3$

$\frac{\partial f}{\partial x_2} = 0, \quad \frac{\partial g_2}{\partial x_1} = -2\left(-\frac{1}{2}\right) = 1, \quad \frac{\partial g_2}{\partial x_2} = 1,$

$\frac{\partial g_3}{\partial x_1} = -1$ and $\frac{\partial g_3}{\partial x_2} = 0.$

Above equations become: $-3 + 1(\lambda_2) - 1(\lambda_3) = 0$

$+1(\lambda_2) = 0$

$\therefore \lambda_2 = 0, \quad \lambda_3 = -3$

K-T conditions not satisfied; $\vec{X}_3$ is not optimum.

2.76 Minimize $f = x_1^2 + x_2^2 - x_1 x_2 - 7x_1 - 4x_2$

subject to: $g_1 = 2x_1 + 3x_2 - 24 \le 0$

$g_2 = -5x_1 + 12x_2 - 24 \le 0$

$g_3 = x_2 - 4 \le 0, \quad g_4 = -x_1 \le 0, \quad g_5 = -x_2 \le 0$

$$\nabla f = \begin{Bmatrix} \partial f/\partial x_1 \\ \partial f/\partial x_2 \end{Bmatrix} = \begin{Bmatrix} 2x_1 - x_2 - 7 \\ 2x_2 - x_1 - 4 \end{Bmatrix}$$

$$\nabla g_1 = \begin{Bmatrix} 2 \\ 3 \end{Bmatrix}, \quad \nabla g_2 = \begin{Bmatrix} -5 \\ 12 \end{Bmatrix}, \quad \nabla g_3 = \begin{Bmatrix} 0 \\ 1 \end{Bmatrix}$$

$\vec{X}_1 = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$; All $g_j(\vec{X}_1) < 0$; it is an interior feasible point.

$$\nabla f\big|_{\vec{X}_1} = \begin{Bmatrix} -6 \\ -3 \end{Bmatrix}; \quad \vec{S} = -\nabla f = \begin{Bmatrix} 6 \\ 3 \end{Bmatrix} \text{ is usable feasible.}$$

$\vec{X}_2 = \begin{Bmatrix} 6 \\ 4 \end{Bmatrix}$; $g_1 = 0, \quad g_3 = 0$; $\nabla f\big|_{\vec{X}_2} = \begin{Bmatrix} 1 \\ -2 \end{Bmatrix}$

$\vec{S}^T \nabla f < 0 \Rightarrow (s_1 \quad s_2)\begin{Bmatrix} 1 \\ -2 \end{Bmatrix} < 0$

$\vec{S}^T \nabla g_1 \le 0 \Rightarrow (s_1 \quad s_2)\begin{Bmatrix} 2 \\ 3 \end{Bmatrix} \le 0$

$\vec{S} \nabla g_3 \le 0 \Rightarrow (s_1 \quad s_2)\begin{Bmatrix} 0 \\ 1 \end{Bmatrix} \le 0$

$\therefore \vec{S} = \begin{Bmatrix} s_1 \\ s_2 \end{Bmatrix} = \begin{Bmatrix} -4 \\ -1 \end{Bmatrix}$ can be taken as a usable feasible direction.

Chapter 3
Linear Programming I:
Simplex Method

3.1 Minimize $F = -f = 2x_1 + x_2 - 5x_3$

subject to
$$x_1 - 2x_2 + x_3 + x_4 = 8$$
$$-3x_1 + 2x_2 + x_5 = 18$$
$$-2x_1 - x_2 + 2x_3 - x_6 = 4$$

x_1, x_2, x_3 unrestricted; $x_4 \geq 0$, $x_5 \geq 0$, $x_6 \geq 0$

Define $x_i = x_i^+ - x_i^-$ with $x_i^+ \geq 0$, $x_i^- \geq 0$; $i = 1, 2, 3$

Final problem:

Minimize $f = 2x_1^+ - 2x_1^- + x_2^+ - x_2^- - 5x_3^+ + 5x_3^-$

subject to

$$x_1^+ - x_1^- - 2x_2^+ + 2x_2^- + x_3^+ - x_3^- + x_4 = 8$$
$$-3x_1^+ + 3x_1^- + 2x_2^+ - 2x_2^- + x_5 = 18$$
$$-2x_1^+ + 2x_1^- - x_2^+ + x_2^- + 2x_3^+ - 2x_3^- - x_6 = 4$$
$$x_i^+ \geq 0, \quad i = 1, 2, 3; \quad x_i^- \geq 0, \quad i = 1, 2, 3$$
$$x_i \geq 0, \quad i = 4, 5, 6$$

3.2 Minimize $F = -f = -x_1 + 8x_2$

subject to
$$3x_1 + 2x_2 - x_3 = 6$$
$$9x_1 + 7x_2 + x_4 = 108$$
$$-2x_1 + 5x_2 + x_5 = 35$$

Let $x_1 = x_1^+ - x_1^-$, $x_2 = x_2^+ - x_2^-$

$x_i^+ \geq 0$, $x_i^- \geq 0$, $i = 1, 2$

$x_3 \geq 0$, $x_4 \geq 0$, $x_5 \geq 0$

Final problem:

Minimize $F = -x_1^+ + x_1^- + 8x_2^+ - 8x_2^-$

subject to

$3x_1^+ - 3x_1^- + 2x_2^+ - 2x_2^- - x_3 = 6$

$9x_1^+ - 9x_1^- + 7x_2^+ - 7x_2^- + x_4 = 108$

$-2x_1^+ + 2x_1^- + 5x_2^+ - 5x_2^- + x_5 = 35$

$x_i^+ \geq 0$, $i = 1, 2$; $x_i^- \geq 0$, $i = 1, 2$

$x_i \geq 0$, $i = 3, 4, 5$

3.3

$6x_1 - 2x_2 + 3x_3 = 11$ (E_1)

$4x_1 + 7x_2 + x_3 = 21$ (E_2)

$5x_1 + 8x_2 + 9x_3 = 48$ (E_3)

$(E_1) \div 6 \Rightarrow \quad x_1 - \frac{1}{3}x_2 + \frac{1}{2}x_3 = \frac{11}{6}$ (E_4)

$(E_2) - 4(E_4) \Rightarrow \quad \frac{25}{3}x_2 - x_3 = \frac{41}{3}$ (E_5)

$(E_2) \div 4 \Rightarrow \quad x_1 + \frac{7}{2}x_2 + \frac{1}{4}x_3 = \frac{21}{4}$ (E_6)

$(E_3) - 5(E_4) \Rightarrow \quad \frac{29}{3}x_2 + \frac{13}{2}x_3 = \frac{233}{6}$ (E_7)

$\frac{3}{25}(E_5) \Rightarrow \quad x_2 - \frac{3}{25}x_3 = \frac{41}{25}$ (E_8)

$(E_4) + \frac{1}{3}(E_8) \Rightarrow \quad x_1 + \frac{23}{50}x_3 = \frac{119}{50}$ (E_9)

$(E_7) - \frac{29}{3}(E_8) \Rightarrow \quad x_3 = 3$ (E_{10})

$(E_9) - \frac{23}{50}(E_{10}) \Rightarrow \quad x_1 = 1$ (E_{11})

$(E_8) + \frac{3}{25}(E_{10}) \Rightarrow \quad x_2 = 2$ (E_{12})

$\therefore \quad x_1 = 1, \quad x_2 = 2, \quad x_3 = 3$

3.4 Let reservoir capacity $= x_1$

and water released to irrigation district $= x_2$

Assume : Reservoir is filled by end of wet season and emptied by end of dry season.

Minimize $f = 5.4\,x_2 + 8.4\,x_2 + 25\,x_1 = 25\,x_1 + 13.8\,x_2$

subject to

$$4.5*10^6 + 1.2*10^6 - 0.5*10^6 - x_1 \geq 0.3\,x_2 \quad \text{(wet season)}$$

$$\text{or} \quad 5.2*10^6 - x_1 - 0.3\,x_2 \geq 0$$

$$1.1*10^6 + 0.3*10^6 - 0.2*10^6 + x_1 \geq 0.7\,x_2 \quad \text{(summer season)}$$

$$\text{or} \quad 1.2*10^6 + x_1 - 0.7\,x_2 \geq 0$$

$$x_1 \geq 0, \quad x_2 \geq 0, \quad 4.5*10^6 \leq x_1$$

3.5

$$4x_1 - 7x_2 + 2x_3 = -8 \qquad (I_0)$$

$$3x_1 + 4x_2 - 5x_3 = -8 \qquad (II_0)$$

$$5x_1 + x_2 - 8x_3 = -34 \qquad (III_0)$$

$$x_1 - \frac{7}{4}x_2 + \frac{1}{2}x_3 = -2 \qquad (I_1) = \frac{1}{4}(I_0)$$

$$\frac{37}{4}x_2 - \frac{13}{2}x_3 = -2 \qquad (II_1) = (II_0) - 3(I_1)$$

$$\frac{39}{4}x_2 - \frac{21}{2}x_3 = -24 \qquad (III_1) = (III_0) - 5(I_1)$$

$$x_1 + 0\cdot x_2 - \frac{27}{37}x_3 = -\frac{88}{37} \qquad (I_2) = (I_1) + \frac{7}{4}(II_2)$$

$$+ x_2 - \frac{26}{37}x_3 = -\frac{8}{37} \qquad (II_2) = \frac{4}{37}(II_1)$$

$$-\frac{135}{37}x_3 = -\frac{810}{37} \qquad (III_2) = (III_1) - \frac{39}{4}(II_2)$$

$$x_3 = 6 \qquad (III_3) = -\frac{37}{135}(III_2)$$

$$x_2 = 4 \qquad (II_3) = (II_2) + \frac{26}{37}(III_3)$$

$$x_1 = 2 \qquad (I_3) = (I_2) + \frac{27}{37}(III_3)$$

3.6

$2x_1 + x_2 + x_3 = 9$ $\quad (I_0)$

$x_1 + x_2 + x_3 = 6$ $\quad (II_0)$

$2x_1 + 3x_2 + x_3 = 13$ $\quad (III_0)$

$x_1 = 3$ $\quad (I_1) = (I_0) - (II_0)$

$x_2 = 2$ $\quad (II_1) = \frac{1}{2}(III_0 - I_0)$

$x_1 + 3x_2 + x_3 = 10$ $\quad (III_1) = (III_0) + (II_0) - (I_0)$

$x_3 = 1$ $\quad (III_2) = (III_1) - (I_1) - 3(II_1)$

$\therefore \quad x_1 = 3, \quad x_2 = 2, \quad x_1 = 1$

3.7

Maximize $f = 2x_1 + 6x_2$

subject to $g_1 = -x_1 + x_2 - 1 \leq 0$

$g_2 = 2x_1 + x_2 - 2 \leq 0$

$x_i \geq 0, \quad i = 1, 2$

$g_1 = 0$ at $(-1, 0), (0, 1), (1, 2), (2, 3)$

$g_2 = 0$ at $(-1, 4), (0, 2), (1, 0), (2, -2)$

Contours of $f = 2x_1 + 6x_2$:

$f = -2$ at $(-1, 0), (1, -\frac{2}{3}), (3, -\frac{4}{3})$

$f = 2$ at $(1, 0)$

$f = 0$ at $(0, 0), (3, -1)$

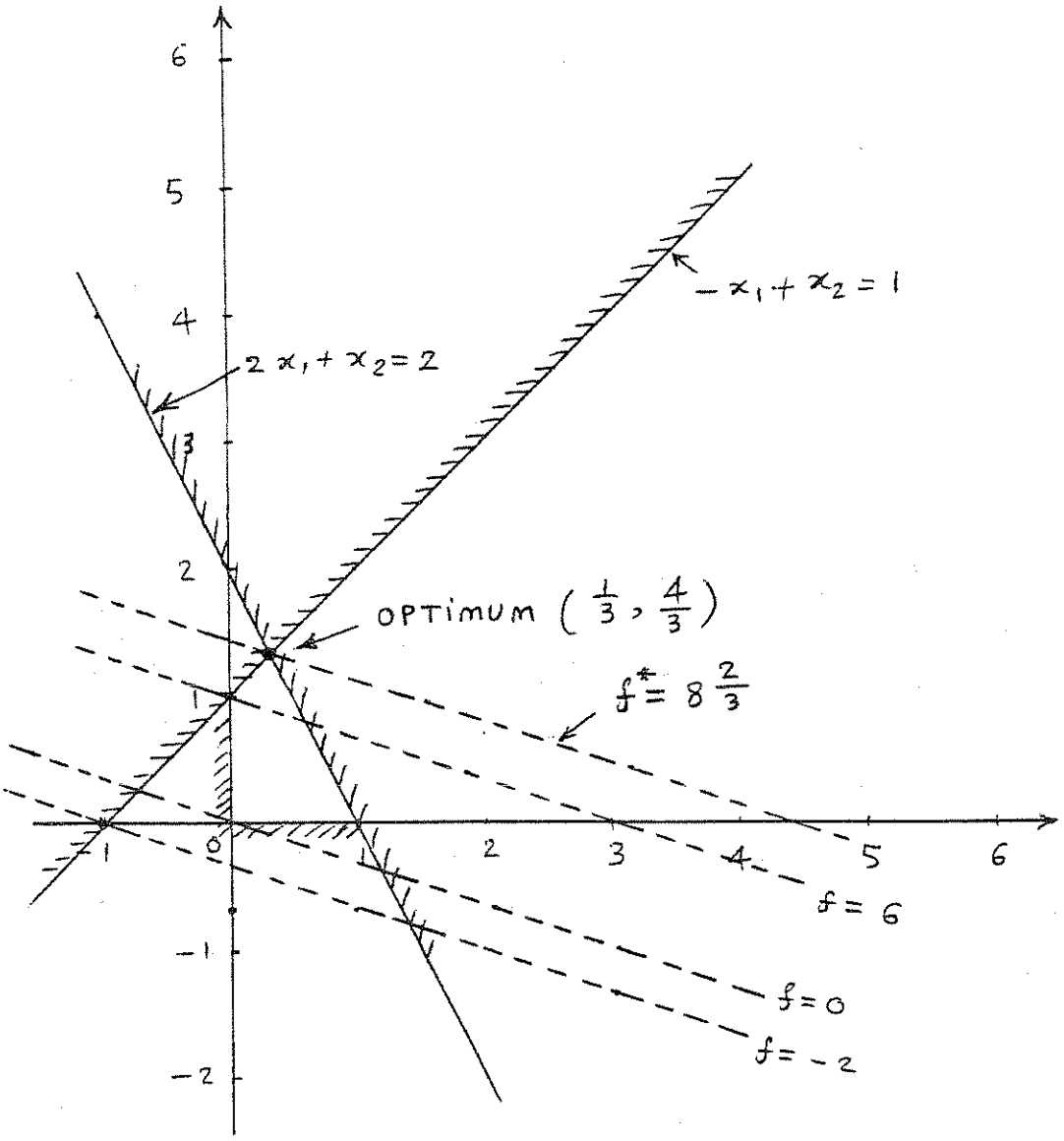
$f = 6$ at $(0, 1), (3, 0)$

$f = \frac{26}{3}$ at $(\frac{1}{3}, \frac{4}{3}), (2, \frac{7}{9})$

Graphical optimization gives:

$\vec{x}^* = (\frac{1}{3}, \frac{4}{3})$

$f^* = 8\frac{2}{3}$


6
5
4
3
2
1
0
-1
-2
1
2
3
4
5
6
$-x_1 + x_2 = 1$
$2x_1 + x_2 = 2$
OPTIMUM $(\frac{1}{3}, \frac{4}{3})$
$f^* = 8\frac{2}{3}$
$f = 6$
$f = 0$
$f = -2$

3.8

Minimize $f = -3x_1 + 2x_2$

subject to $g_1 = x_1 + x_2 - 5 \leq 0$

$0 \leq x_1 \leq 4$

$1 \leq x_2 \leq 6$

$g_1 = 0$ at $(0, 5), (5, 0)$

Contours of $f = -3x_1 + 2x_2$:

$f = 0$ at $(2, 3), (0, 0)$

$f = -12$ at $(2, -3), (5, 1\frac{1}{2}), (4, 0)$

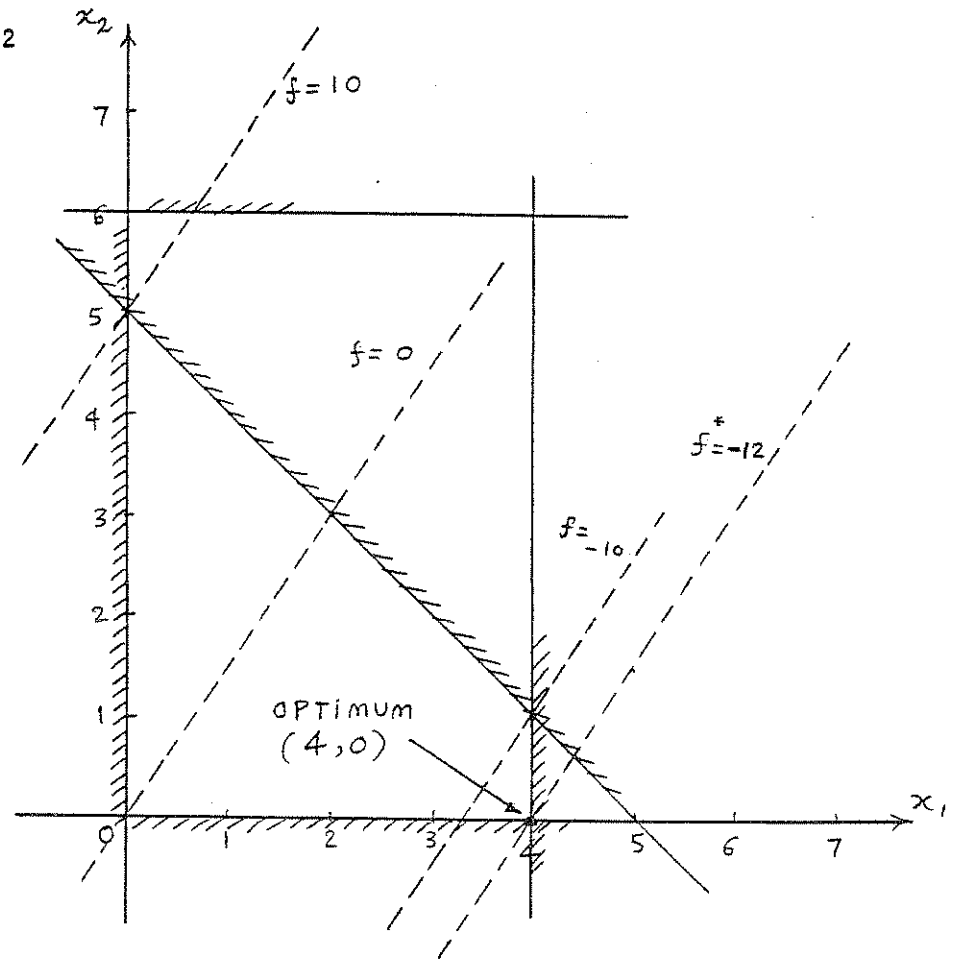
$f = 10$ at $(0, 5), (2, 8)$

$f = -10$ at $(4, 1), (2, -2)$

Graphical optimization gives:

$\vec{X}^* = \begin{pmatrix} 4 \\ 0 \end{pmatrix}$

with $f^* = -12$



3.9

Minimize $f = 3x_1 + 2x_2$ subject to $g_1 = 8x_1 + x_2 - 8 \geq 0$

$g_2 = 2x_1 + x_2 - 6 \geq 0$

$g_3 = x_1 + 3x_2 - 6 \geq 0$

$g_4 = x_1 + 6x_2 - 8 \geq 0$

$x_1 \geq 0, \quad x_2 \geq 0$

$g_1 = 0$ at $(0, 8), (1, 0)$

$g_2 = 0$ at $(0, 6), (3, 0)$

$g_3 = 0$ at $(0, 2), (6, 0)$

$g_4 = 0$ at $(0, 4/3), (8, 0)$

contours of $f = 3x_1 + 2x_2$:

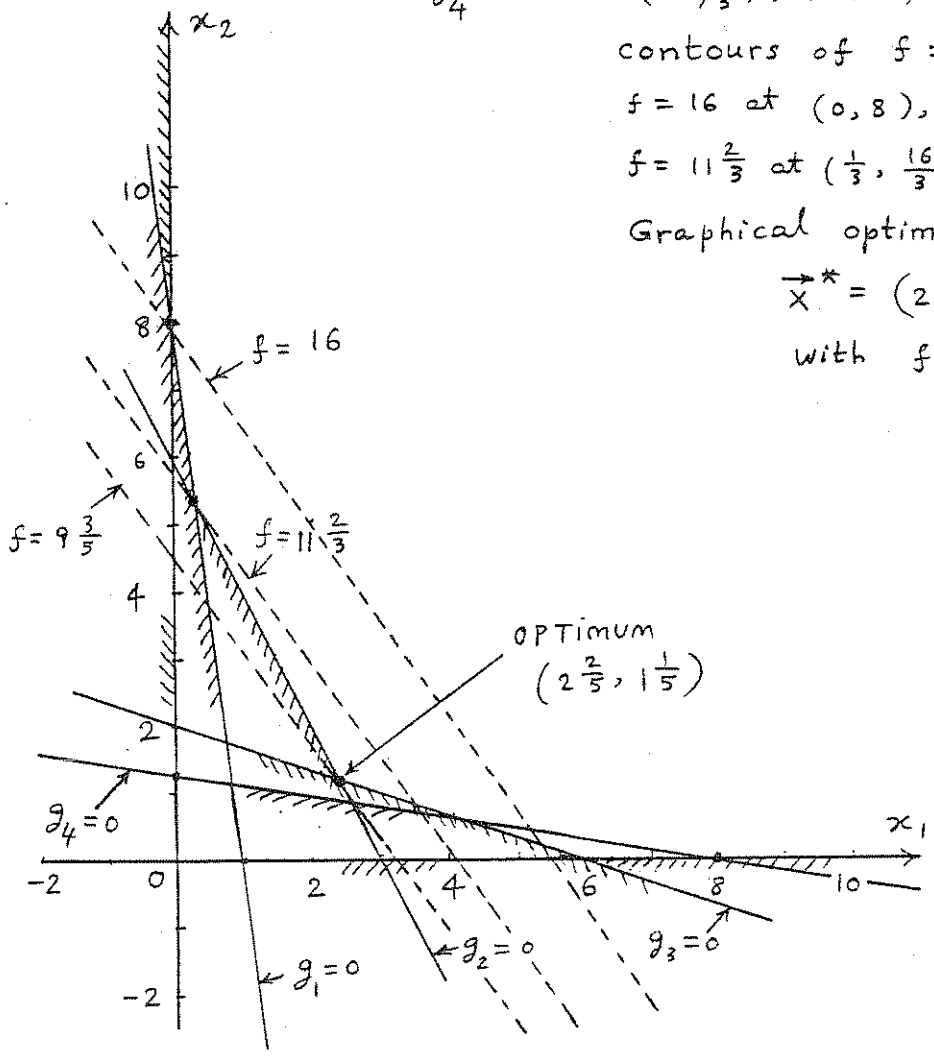
$f = 16$ at $(0, 8), (2, 5)$

$f = 11\frac{2}{3}$ at $(\frac{1}{3}, \frac{16}{3}), (4, -\frac{1}{6})$

Graphical optimization gives:

$\vec{x}^* = (2\frac{2}{5}, 1\frac{1}{5})$

with $f^* = 9\frac{3}{5}$.



3.10

Minimize $f = x_1^2 x_2^2$ (1)

subject to $x_1^3 x_2^2 \geq e^3$ (2)

$x_1 x_2^4 \geq e^4$ (3)

$x_1^2 x_2^3 \leq e$ (4)

$x_1 > 0, \; x_2 > 0$ (5)

Taking logarithms, Eqs. (1) to (5) become:

Minimize $F = 2x_3 + 2x_4$ (6)

subject to

$3x_3 + 2x_4 \geq 3$ (7) or $g_1 = 3x_3 + 2x_4 - 3 \geq 0$

$x_3 + 4x_4 \geq 4$ (8) or $g_2 = x_3 + 4x_4 - 4 \geq 0$

$2x_3 + 3x_4 \leq 1$ (9) or $g_3 = 2x_3 + 3x_4 - 1 \leq 0$

$x_3 \geq -\infty$ (10)

$x_4 \geq -\infty$ (11)

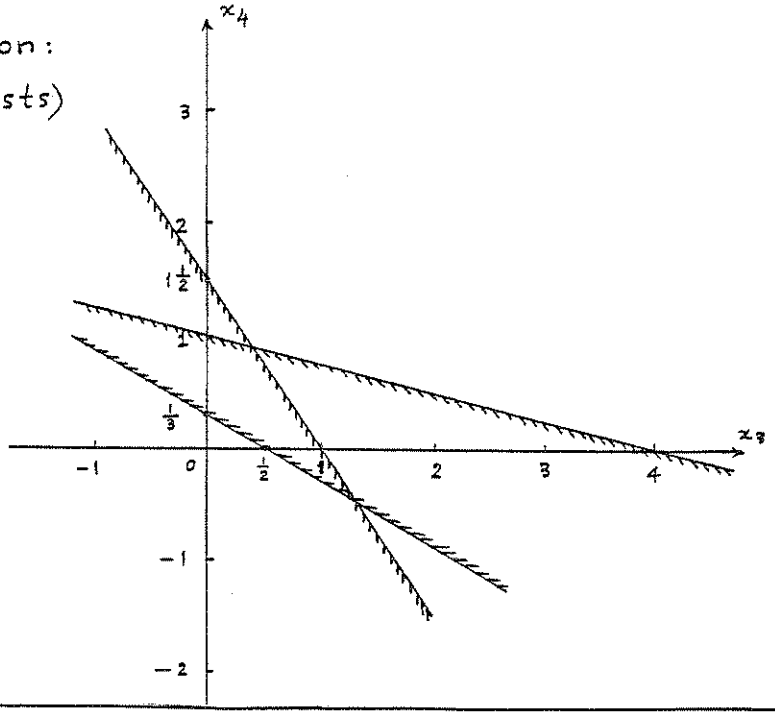
where $x_3 = \ln x_1, \quad x_4 = \ln x_2$ (12)

$g_1 = 0$ at $(x_3 = 0, x_4 = 1.5), (1, 0), (2, -1.5), (3, -3)$

$g_2 = 0$ at $(x_3 = 0, x_4 = 1), (4, 0)$

$g_3 = 0$ at $(x_3 = 0, x_4 = 1/3), (0.5, 0), (5, -3)$

Graphical optimization:
(No feasible point exists)
∴ No solution.



3.11

THEOREM 3.6 *Let S be a closed convex polyhedron. Then the minimum of a linear function over S is attained at an extreme point of S.*

Proof: Suppose $\mathbf{X}^*$ minimizes the objective function $\mathbf{c}^T\mathbf{X}$ over S and an equivalent minimum does not occur at an extreme point.

Then $$\mathbf{c}^T\mathbf{X} < \mathbf{c}^T\mathbf{X}_i^e, \quad i = 1, 2, \ldots, p$$

where X_i^e are the finite number of extreme points of the polyhedral set.

Thus, for $$0 < \lambda_i < 1,$$

$$\lambda_i \mathbf{c}^T\mathbf{X}^* < \lambda_i \mathbf{c}^T\mathbf{X}_i^e$$

and

$$\sum_{i=1}^{p} \lambda_i \mathbf{c}^T\mathbf{X}^* < \sum_{i=1}^{p} \lambda_i \mathbf{c}^T\mathbf{X}_i^e \qquad (E_1)$$

Now choose $\lambda_i = \lambda_i^*$ so that

$$\mathbf{X}^* = \sum_{i=1}^{p} \lambda_i^* \mathbf{X}_i^e, \ \lambda_i^* \geqslant 0 \text{ and } \sum_{i=1}^{p} \lambda_i^* = 1$$

This is possible according to the Theorem 3.5. Thus the inequality (E_1) becomes

$$\sum_{i=1}^{p} \lambda_i \mathbf{c}^T\mathbf{X}^* = \mathbf{c}^T\mathbf{X}^* < \sum_{i=1}^{p} \lambda_i^* \mathbf{c}^T\mathbf{X}_i^e = \mathbf{c}^T \sum_{i=1}^{p} \lambda_i^* \mathbf{X}_i^e = \mathbf{c}^T\mathbf{X}^*$$

which is a contradiction.

3.12

Maximize $f = 6x + 7y$

subject to $g_1 = 7x + 6y - 42 \leq 0$

$g_2 = 5x + 9y - 45 \leq 0$

$g_3 = x - y - 4 \leq 0$

$g_4 = -x \leq 0$

$g_5 = -y \leq 0$

$g_1 = 0$ at $(0, 7)$, $(6, 0)$

$g_2 = 0$ at $(0, 5)$, $(9, 0)$

$g_3 = 0$ at $(0, -4)$, $(4, 0)$, $(7, 3)$

Contours of $f = 6x + 7y$

$f = 24$ at $(4, 0)$ and $(0, 24/7)$

$f = 36$ at $(6, 0)$ and $(0, 36/7)$

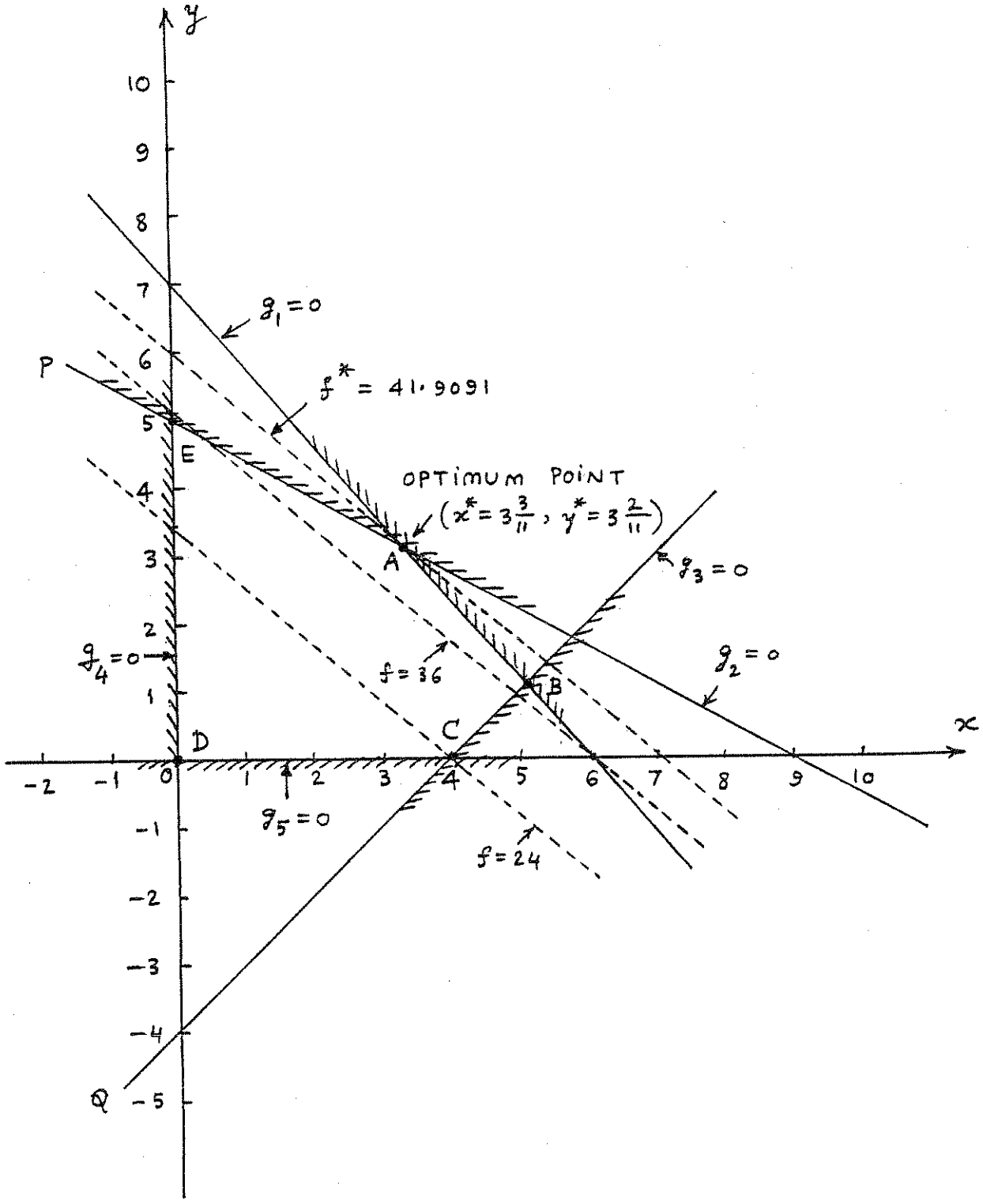
Feasible space = ABCDEA

optimum solution:

$\vec{X}^*$ = Point A $\left(3\frac{3}{11}, 3\frac{2}{11}\right)$

$f^* = 6\left(\frac{36}{11}\right) + 7\left(\frac{35}{11}\right)$

$= 41.9091$


y
x
$g_1 = 0$
$g_2 = 0$
$g_3 = 0$
$g_4 = 0$
$g_5 = 0$
$f^* = 41.9091$
OPTIMUM POINT
$(x^* = 3\frac{3}{11}, y^* = 3\frac{2}{11})$
f = 36
f = 24
A
B
C
D
E
P
Q

3.13 This problem is same as Problem 3.12 except that g_4 and g_5 are not present here.

Feasible space = PEABCQ

Optimum solution : $\vec{x}^* =$ Point A $= \begin{Bmatrix} 3\frac{3}{11} \\ 3\frac{2}{11} \end{Bmatrix}$, $f^* = 41.9091$

3.14 Maximize $f = 19x + 7y$

subject to $g_j \leq 0$, $j = 1,2,3,4,5$ (as in Problem 3.12).

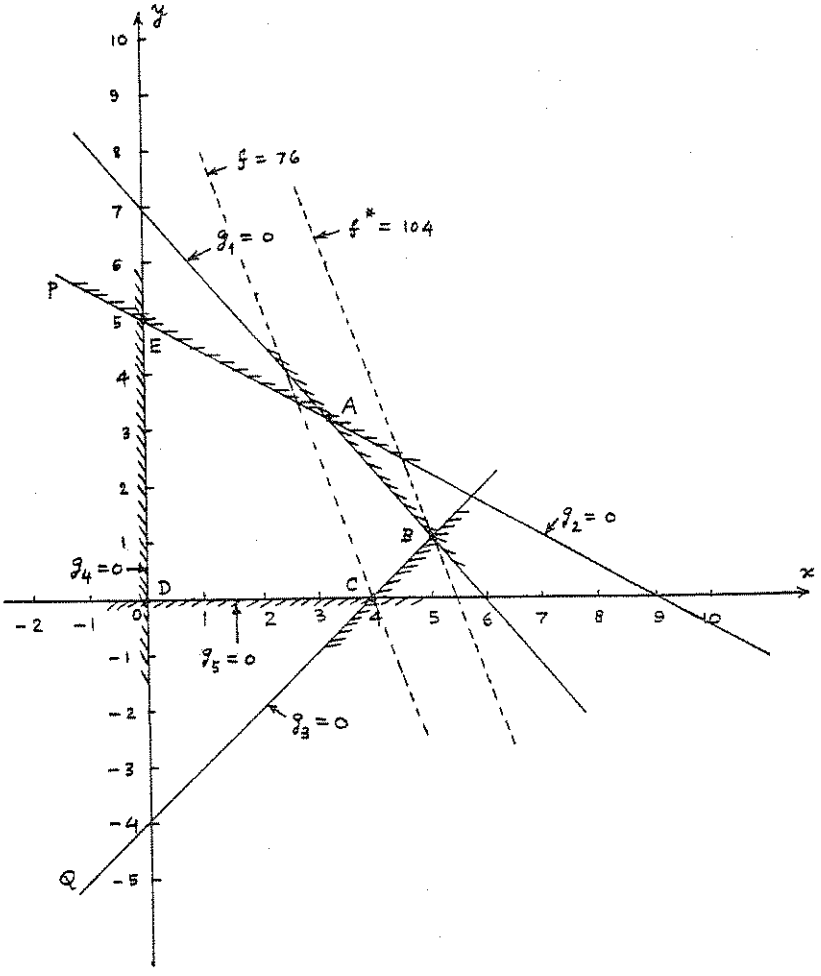
contours of $f = 19x + 7y$:

$f = 76$ at $(4,0)$ and $(2, \frac{38}{7})$

$f = 104$ at $(5\frac{1}{13}, 1\frac{1}{13})$ and $(4,4)$

Feasible space = ABCDEA

Optimum point = B $= (5\frac{1}{13}, 1\frac{1}{13})$, $f^* = 104$.



3.15 This problem is same as Problem 3.14 except that g_4 and g_5 are not present here.

Feasible space = P E A B C Q

Optimum point = point B = $\left(5\frac{1}{13}, 1\frac{1}{13}\right)$, $f^* = 104$.

3.16 Maximize $f = x + 2y$

subject to:

$$g_1 = x - y + 8 \geq 0$$
$$g_2 = 5x - y \geq 0$$
$$g_3 = x + y - 8 \geq 0$$
$$g_4 = -x + 6y - 12 \geq 0$$
$$g_5 = -5x - 2y + 68 \geq 0$$
$$g_6 = -x + 10 \geq 0$$
$$g_7 = x \geq 0$$
$$g_8 = y \geq 0$$

$g_1 = 0$ at $(0, 8)$, $(-8, 0)$, $(4, 12)$

$g_2 = 0$ at $(0, 0)$, $(1, 5)$

$g_3 = 0$ at $(0, 8)$, $(8, 0)$

$g_4 = 0$ at $(0, 2)$, $(-12, 0)$, $(6, 3)$

$g_5 = 0$ at $(0, 34)$, $\left(\frac{68}{5}, 0\right)$, $\left(2, \frac{58}{2}\right)$, $(10, 9)$

$g_6 = 0$ at $(10, y)$

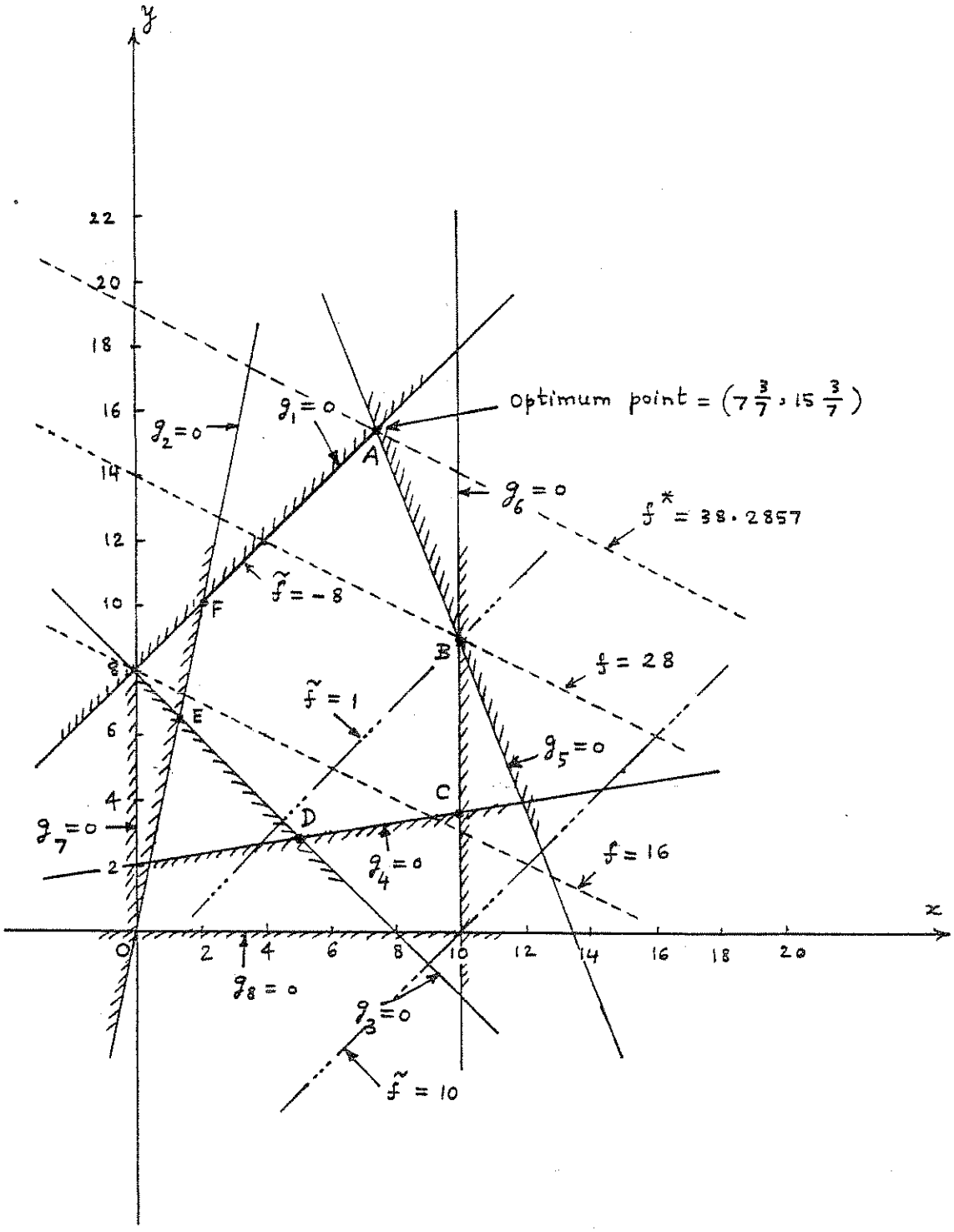
Contours of $f = x + 2y$:

$f = 28$ at $(0, 14)$, $(4, 12)$

$f = 16$ at $(0, 8)$, $(4, 6)$

Feasible space: A B C D E F A

Optimum solution: $\vec{x}^* = \left(7\frac{3}{7}, 15\frac{3}{7}\right)$ with $f^* = 38.2857$


y
x
22
20
18
16
14
12
10
8
6
4
2
0
2
4
6
8
10
12
14
16
18
20
A
B
C
D
E
F
optimum point = $(7\frac{3}{7}, 15\frac{3}{7})$
$g_1 = 0$
$g_2 = 0$
$g_3 = 0$
$g_4 = 0$
$g_5 = 0$
$g_6 = 0$
$g_7 = 0$
$g_8 = 0$
$f^* = 38.2857$
$f = 28$
$f = 16$
$\tilde{f} = -8$
$\tilde{f} = 1$
$\tilde{f} = 10$

3.17 Minimize $\tilde{f} = x - y$

subject to $g_j \geq 0$, $j = 1, 2, \ldots, 8$ (as in Problem 3.16)

Contours of $\tilde{f} = x - y$:

$\tilde{f} = 10$ at $(10, 0)$, $(15, 5)$

$\tilde{f} = 1$ at $(10, 9)$, $(12, 11)$

$\tilde{f} = -8$ at $(4, 12)$, $(8, 16)$

Feasible space = ABCDEFA in figure of Problem 3.16

Optimum solution: All points on "FA"

$\tilde{f}^* = -8$.

3.18 Maximize $f = x + 2y$

subject to

$$g_1 = x - y + 8 \geq 0$$

$$g_2 = 5x - y \geq 0$$

$$g_3 = x + y - 8 \geq 0$$

$$g_4 = -x + 6y - 12 \geq 0$$

$$g_5 = 5x + 2y - 68 \geq 0$$

$$g_6 = -x + 10 \geq 0$$

$$g_7 = x \geq 0$$

$$g_8 = y \geq 0$$

$g_1 = 0$ at $(0, 8)$, $(4, 12)$

$g_2 = 0$ at $(0, 0)$, $(1, 5)$

$g_3 = 0$ at $(0, 8)$, $(8, 0)$

$g_4 = 0$ at $(0, 2)$, $(6, 3)$

$g_5 = 0$ at $(6, 19)$, $(10, 9)$, $(12, 4)$

$g_6 = 0$ at $(10, y)$

Contours of f:

$f = 46$ at $(10, 18)$, $(12, 17)$

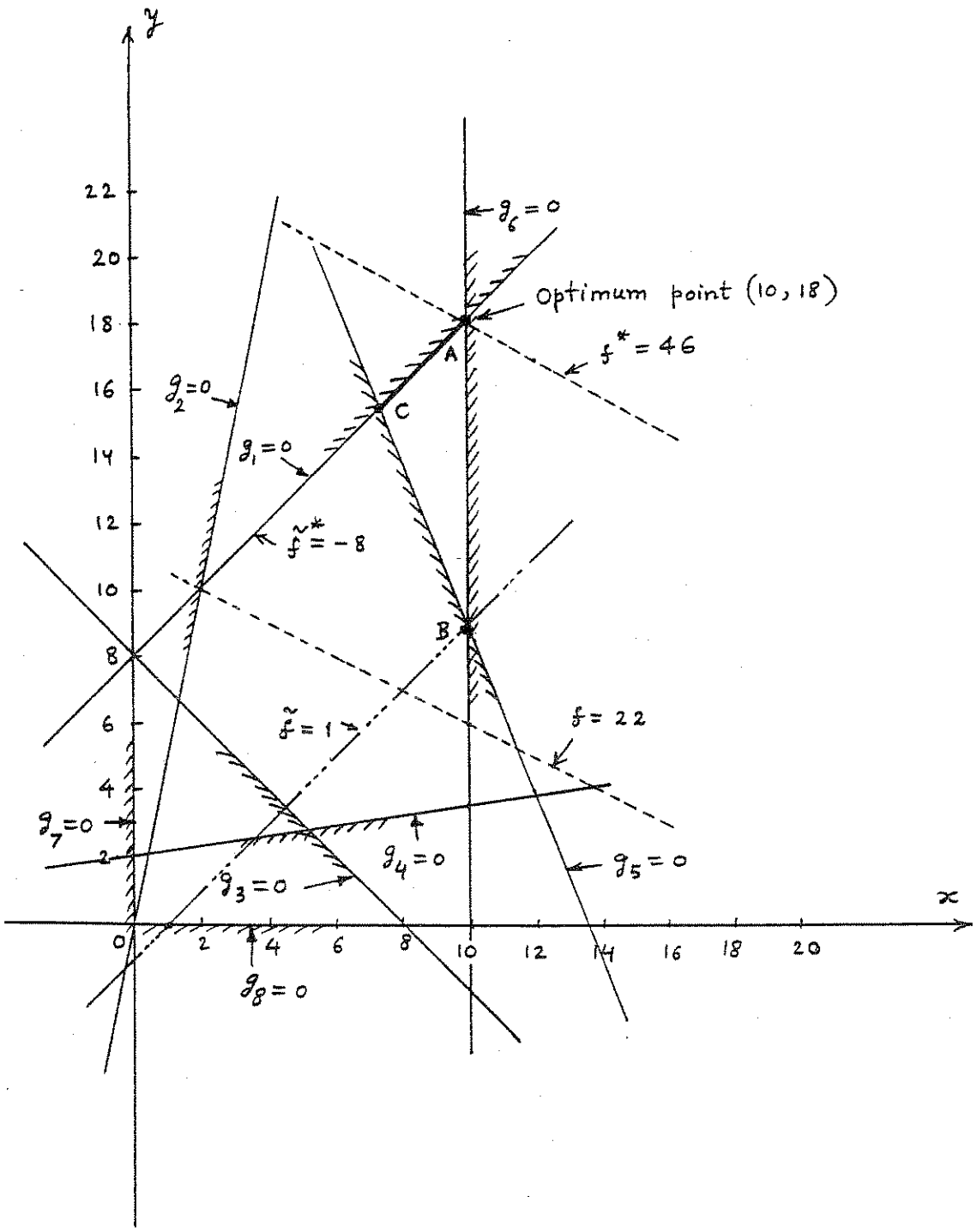
$f = 22$ at $(2, 10)$, $(10, 6)$

Feasible space = ABCA

optimum point = A

$= (10, 18)$

$f^* = 46$


y
x
22
20
18
16
14
12
10
8
6
4
2
0
2
4
6
8
10
12
14
16
18
20
$g_6 = 0$
Optimum point (10, 18)
$f^* = 46$
A
$g_2 = 0$
C
$g_1 = 0$
$\tilde{f}^* = -8$
B
$\tilde{f} = 1$
$f = 22$
$g_7 = 0$
$g_4 = 0$
$g_5 = 0$
$g_3 = 0$
$g_8 = 0$

3.19 Minimize $\tilde{f} = x - y$

subject to $g_j \geq 0$, $j = 1$ to 8 (same as in Problem 3.18)

Contours of $\tilde{f} = x - y$:

$\tilde{f} = -8$ at $(10, 18)$, $(20, 28)$, $(0, 8)$

$\tilde{f} = 1$ at $(10, 9)$, $(1, 0)$

Feasible space = ABCA of figure in Problem 3.18

Optimum point = every point on "CA"

$f^* = -8$

3.20 Maximize $f = x + 3y$

subject to $g_1 = -4x + 3y - 12 \leq 0$

$g_2 = x + y - 7 \leq 0$

$g_3 = x - 4y - 2 \leq 0$

$g_4 = -x \leq 0$

$g_5 = -y \leq 0$

$g_1 = 0$ at $(0, 4)$, $(-3, 0)$

$g_2 = 0$ at $(0, 7)$, $(7, 0)$

$g_3 = 0$ at $(0, -\frac{1}{2})$, $(2, 0)$

contours of $f = x + 3y$:

$f = 12$ at $(0, 4)$, $(3, 3)$

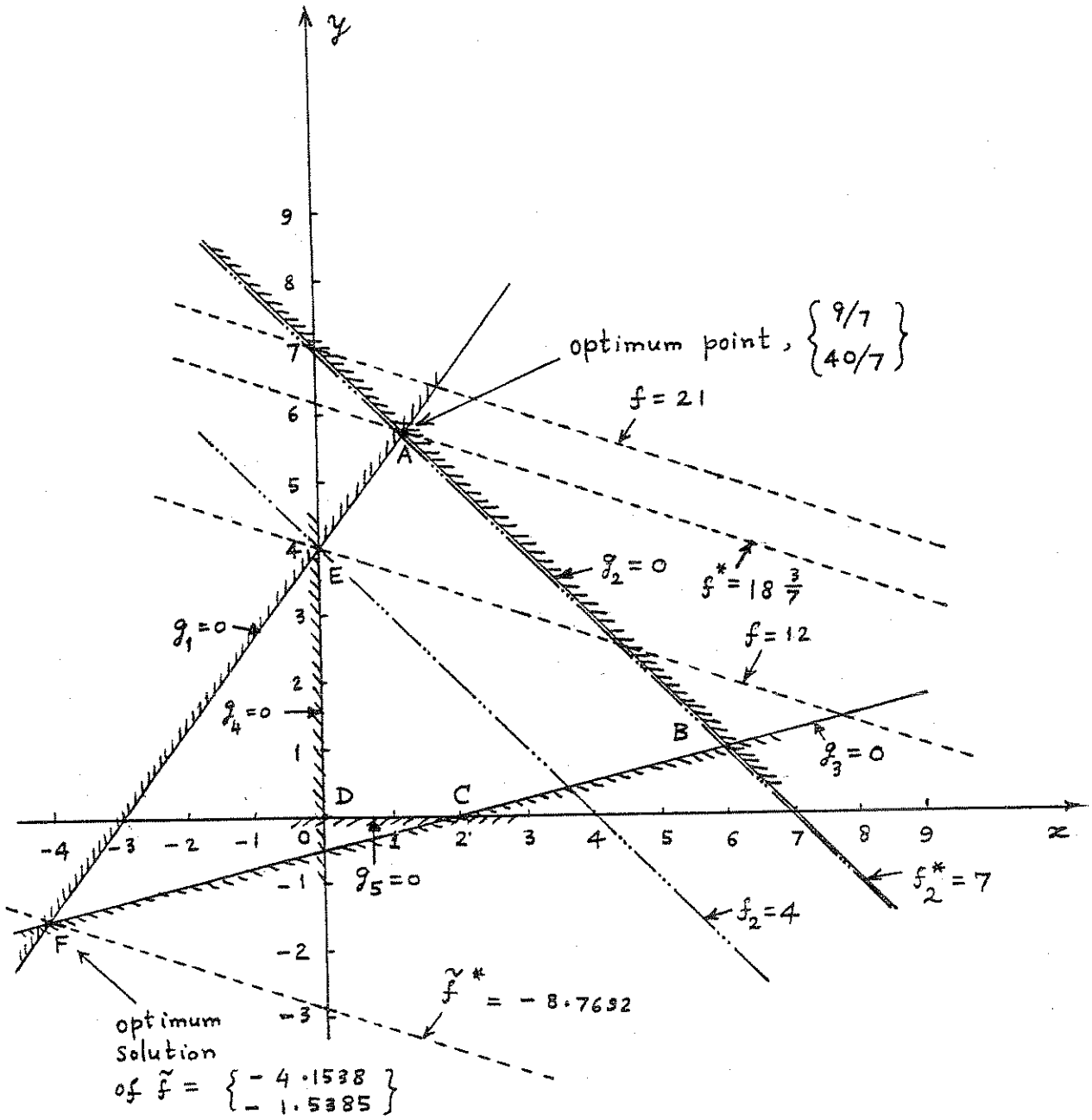
$f = 21$ at $(0, 7)$, $(6, 5)$

Feasible region = A B C D E A

Optimum solution:

Point A = $\vec{X}^* = \begin{Bmatrix} 9/7 \\ 40/7 \end{Bmatrix}$

with $f^* = 18\frac{3}{7}$


y
x
optimum point, $\begin{Bmatrix} 9/7 \\ 40/7 \end{Bmatrix}$
$f = 21$
$f^* = 18\frac{3}{7}$
$f = 12$
$f_2^* = 7$
$f_2 = 4$
$\tilde{f}^* = -8.7692$
$g_1 = 0$
$g_2 = 0$
$g_3 = 0$
$g_4 = 0$
$g_5 = 0$
A
B
C
D
E
F
optimum Solution of $\tilde{f} = \begin{Bmatrix} -4.1538 \\ -1.5385 \end{Bmatrix}$

3.21 Minimize $\tilde{f} = x + 3y$

subject to $g_1 = -4x + 3y - 12 \le 0$

$g_2 = x + y - 7 \le 0$

$g_3 = x - 4y - 2 \le 0$

x, y unrestricted in sign.

Feasible space = ABCFEA in the figure of solution of problem 3.20.

Contours of $\tilde{f} = x + 3y$:

$\tilde{f} = 12$ at $(0,4)$, $(3,3)$

$\tilde{f}^* = -\frac{54}{13} - \frac{60}{13} = -8.7692$

$\tilde{f} = -8.7692$ at $(x = 0, y = -2.9231)$

Optimum point = F:

$\vec{x}^* = \left(-\frac{54}{13}, -\frac{20}{13}\right)$

Point F: $-4x + 3y = 12$

$4x - 16y = 8$

$-13y = 20$

$y = -20/13$

$x = 2 + 4y = \frac{26 - 80}{13}$

$= -\frac{54}{13}$

3.22 Maximize $f_2 = x + y$

subject to $g_j \le 0$, $j = 1$ to 5 (same as in Problem 3.20)

Feasible space = ABCDEA in the figure of solution of Problem 3.20

Contours of $f_2 = x + y$:

$f_2 = 4$ at $(0,4)$, $(4,0)$

$f_2 = 7$ at $(0,7)$, $(7,0)$

Optimum solution: All points between A and B with $f_2^* = 7$, $A = \begin{Bmatrix} 9/7 \\ 40/7 \end{Bmatrix}$, and $B = \begin{Bmatrix} 6 \\ 1 \end{Bmatrix}$.

3.23 Maximize $f = x + 3y$ subject to $g_1 = -4x + 3y - 12 \le 0$

$g_2 = x + y - 7 \le 0$

$g_3 = -x + 4y + 2 \le 0$

$g_4 = -x \le 0$

$g_5 = -y \le 0$

$g_1 = 0$ at $(0, 4)$, $(-3, 0)$

$g_2 = 0$ at $(0, 7)$, $(7, 0)$

$g_3 = 0$ at $(0, -\frac{1}{2})$, $(2, 0)$

Contours of $f = x + 3y$:

$f = 12$ at $(0, 4)$, $(3, 3)$

$f = 9$ at $(6, 1)$, $(3, 2)$

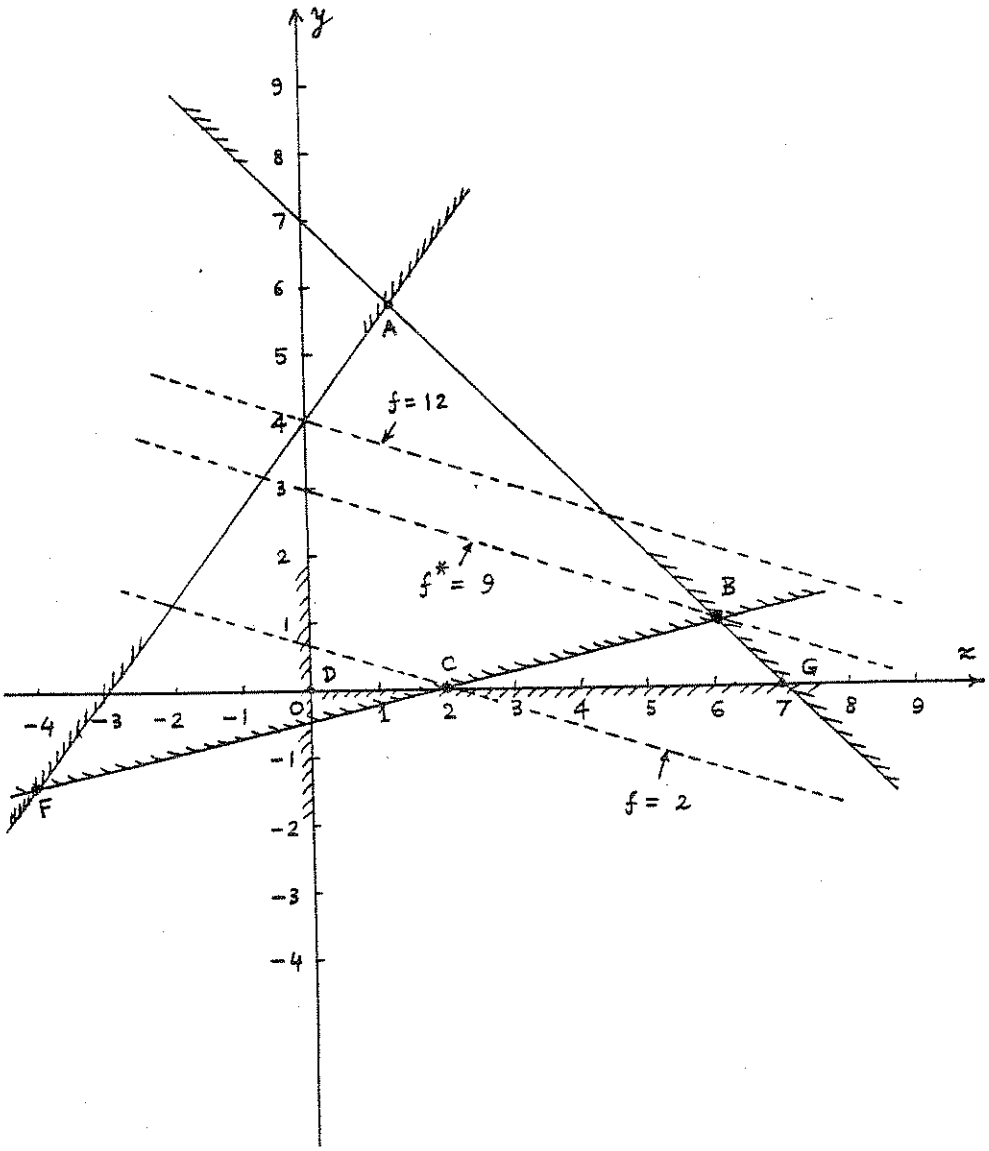
$f = 2$ at $(2, 0)$, $(0, 2/3)$

Feasible space = B G C B.

Optimum point $(\vec{X}^*)$:

$B = \begin{Bmatrix} 6 \\ 1 \end{Bmatrix}$

$f^* = 9$



3.24

Minimize $f = x - 8y$

subject to

$$g_1 = 3x + 2y - 6 \geq 0$$
$$g_2 = -x + y + 6 \geq 0$$
$$g_3 = -9x - 7y + 108 \geq 0$$
$$g_4 = -3x - 7y + 70 \geq 0$$
$$g_5 = 2x - 5y + 35 \geq 0$$
$$g_6 = +x \geq 0$$
$$g_7 = +y \geq 0$$

$g_1 = 0$ at $(0, 3), (2, 0)$

$g_2 = 0$ at $(0, -6), (6, 0)$

$g_3 = 0$ at $(0, \frac{108}{7}), (12, 0), (8.1, 5)$

$g_4 = 0$ at $(0, 10), (\frac{70}{3}, 0), (7, 7)$

$g_5 = 0$ at $(0, 7), (-\frac{35}{2}, 0), (5, 9)$

Contours of $f = x - 8y$:

$f = -24$ at $(0, 3), (8, 4)$

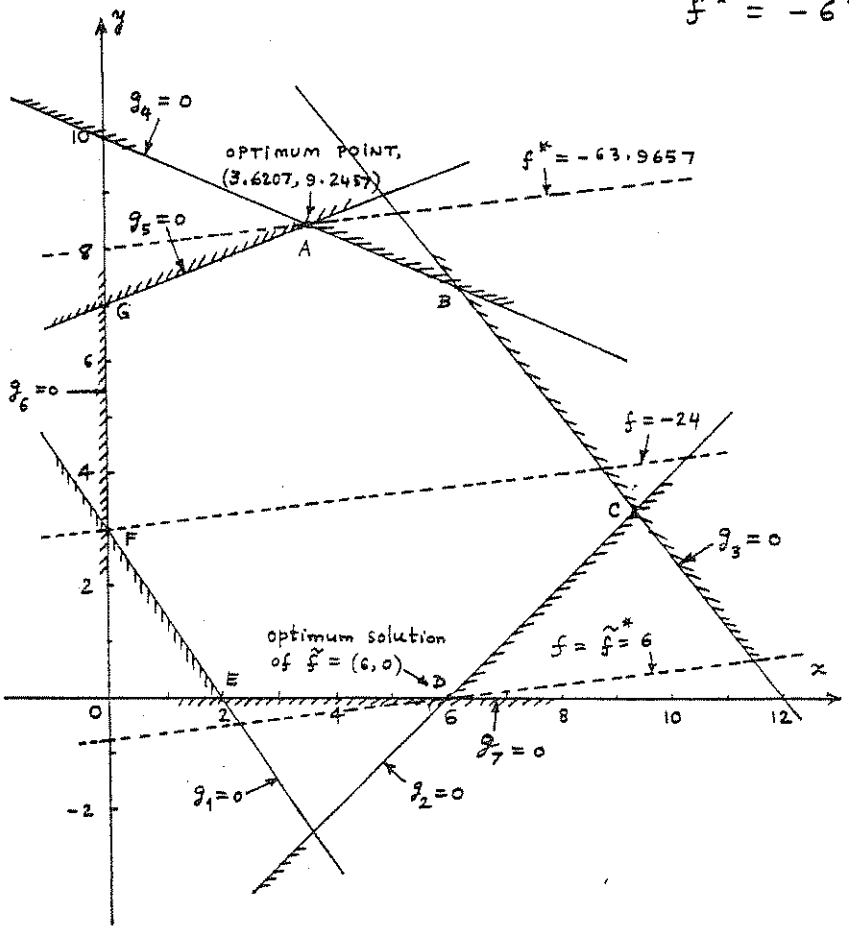
$f = 6$ at $(6, 0), (-2, -1)$

Feasible region: ABCDEFGA

optimum point = A:

$(x^* = 3.6207,\ y^* = 9.2457)$

$f^* = -63.9657$



3.25

Maximize $\tilde{f} = x - 8y$

subject to $g_j \geq 0$, $j = 1$ to 7 (same as in Problem 3.24)

Contours of $\tilde{f} = x - 8y$:

$\tilde{f} = -24$ at $(0,3)$, $(8,4)$

$\tilde{f} = -80$ at $(0,10)$, $(8,11)$

Feasible region: ABCDEFGA in figure of solution of Problem 3.24

optimum point: D = $(6,0)$ with $\tilde{f}^* = 6$

3.26

Maximize $f = x - 8y$

subject to

$g_1 = 3x + 2y - 6 \geq 0$

$g_2 = -x + y + 6 \geq 0$

$g_3 = -9x - 7y + 108 \geq 0$

$g_4 = -3x - 7y + 70 \geq 0$

$g_5 = 2x - 5y + 35 \geq 0$

$g_6 = x \geq 0$; y is unrestricted in sign

$g_1 = 0$ to $g_6 = 0$ can be plotted as in Problem 3.24.

Feasible space: ABCDEFGHA

Contours of $f = x - 8y$:

$f = -24$ at $(0,3)$, $(8,4)$

$f = 6$ at $(6,0)$, $(-2,-1)$

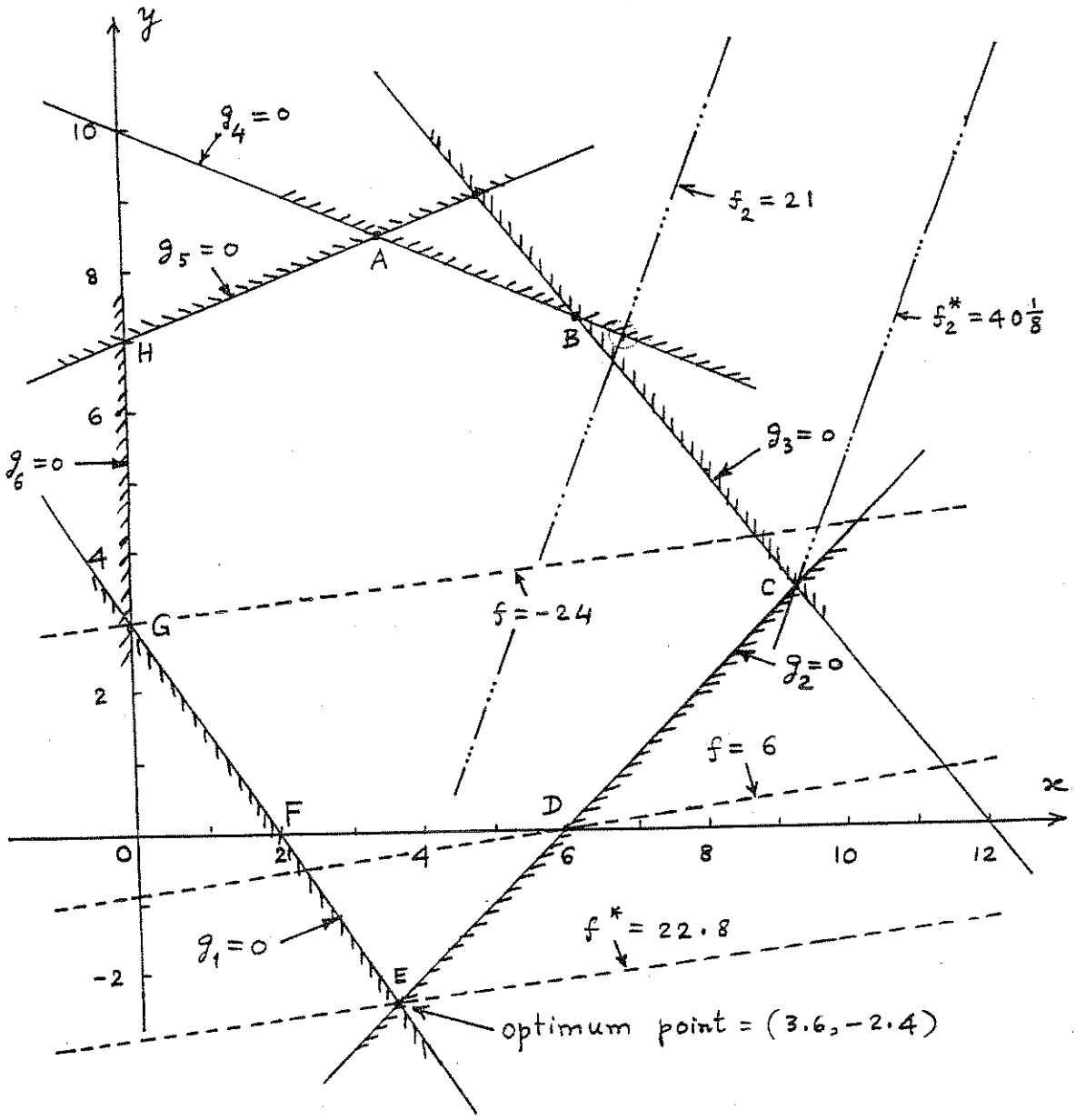
Point E = intersection of $g_1 = 0$ and $g_2 = 0$:

$$+3x + 2y = 6$$
$$-2x + 2y = -12$$
$$5x = 18$$
$$x = +18/5$$
$$y = -6 + x = -12/5$$

optimum point = E

$= \vec{x}^* = (3.6, -2.4)$

$f^* = \frac{18}{5} + \frac{96}{5} = \frac{114}{5} = 22\frac{4}{5}$


y
x
$g_4 = 0$
$g_5 = 0$
$g_6 = 0$
$g_3 = 0$
$g_2 = 0$
$g_1 = 0$
$f_2 = 21$
$f_2^* = 40\frac{1}{8}$
$f = -24$
$f = 6$
$f^* = 22.8$
A
B
C
D
E
F
G
H
optimum point = (3.6, −2.4)
10
8
6
4
2
0
-2
-4
2
6
8
10
12

3.27 Maximize $f_2 = 5x - 2y$

subject to $g_j \geq 0$; $j = 1, 2, \ldots, 7$ (same as in Problem 3.24)

Feasible region = A B C D F G H A (in the figure of solution of Problem 3.26)

Contours of $f_2 = 5x - 2y$:

$f_2 = 21$ at $(7, 7)$, $(9, 12)$

$f_2 = 5 * \frac{75}{8} - 2 * \frac{27}{8}$

$= (375 - 54)/8 = 321/8$

$= 40\frac{1}{8}$ at $\left(\frac{75}{8}, \frac{27}{8}\right)$

and at $\left(12, 9\frac{15}{16}\right)$

Point C:

$9x + 7y = 108$

$7x - 7y = 42$

$16x = 150$

$x = 75/8$

$y = x - 6 = \frac{75 - 48}{8}$

$= 27/8$

Optimum solution = Point C = $\left(x^* = \frac{75}{8}, y^* = \frac{27}{8}\right)$ with $f_2^* = 40\frac{1}{8}$.

3.28 Minimize $f = x - 4y$

subject to $g_1 = -x + y - 4 \leq 0$

$g_2 = 4x + 5y - 45 \leq 0$

$g_3 = 5x - 2y - 20 \leq 0$

$g_4 = 5x + 2y - 10 \leq 0$

$g_5 = -x \leq 0$

$g_6 = -y \leq 0$

$g_1 = 0$ at $(0, 4)$, $(-4, 0)$, $(2, 6)$

$g_2 = 0$ at $(0, 9)$, $(11\frac{1}{4}, 0)$

$g_3 = 0$ at $(0, -10)$, $(4, 0)$, $(8, 10)$

$g_4 = 0$ at $(0, 5)$, $(2, 0)$

Contours of $f = x - 4y$:

$f = -20$ at $(0, 5)$, $(4, 6)$

$f = 2$ at $(2, 0)$, $(6, 1)$

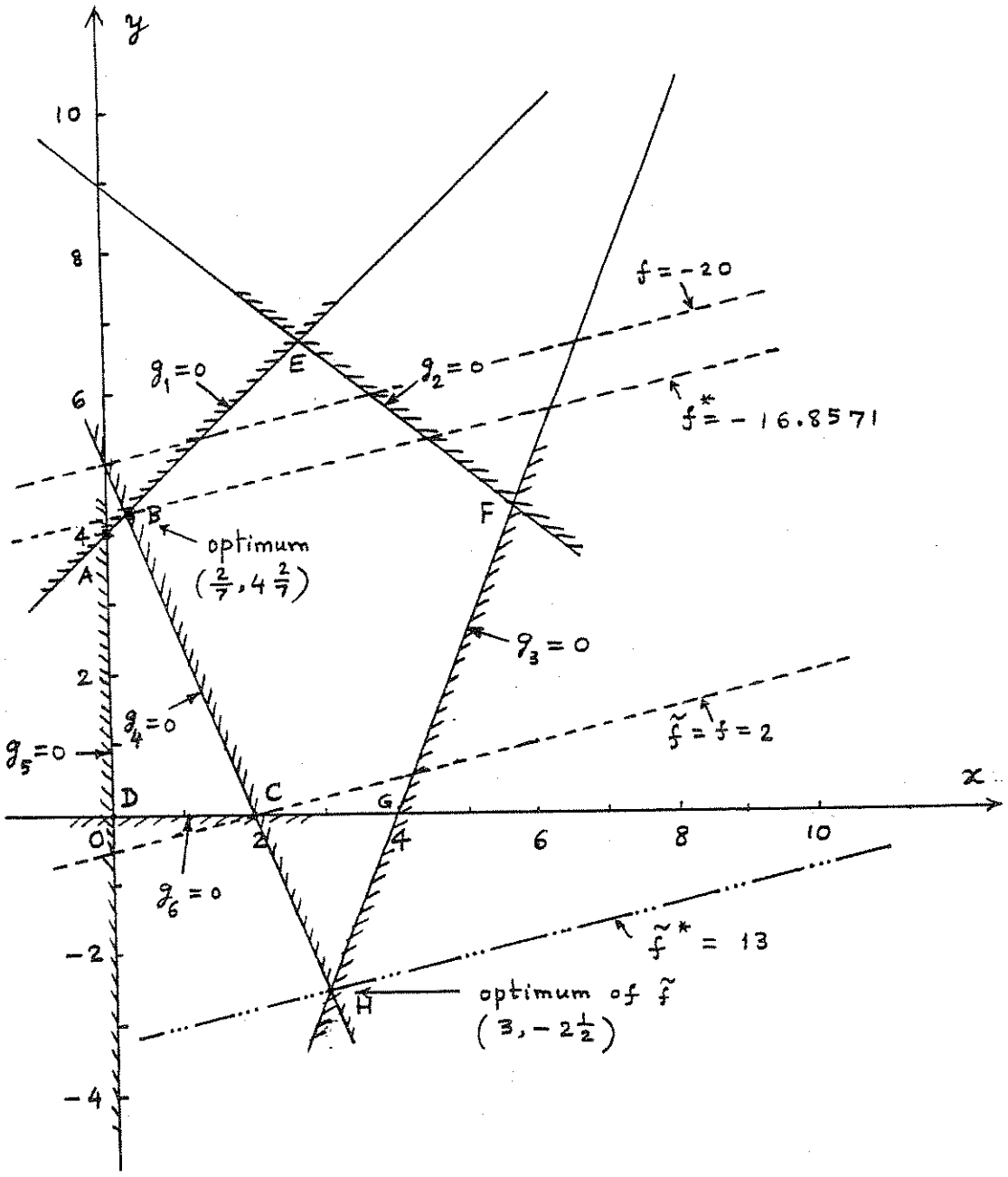
$f = -16.8571$ at $\left(\frac{2}{7}, \frac{30}{7}\right)$, $(8, 6.2143)$

Feasible space = A B C D A in figure

Optimum point:

$B = \left(\frac{2}{7}, 4\frac{2}{7}\right)$

with $f^* = -16.8571$


y
x
10
8
6
4
2
-2
-4
0
2
4
6
8
10
A
B
C
D
E
F
G
H
$g_1=0$
$g_2=0$
$g_3=0$
$g_4=0$
$g_5=0$
$g_6=0$
$f=-20$
$f^*=-16.8571$
optimum
$(\frac{2}{7}, 4\frac{2}{7})$
$\tilde{f}=f=2$
$\tilde{f}^*=13$
optimum of $\tilde{f}$
$(3, -2\frac{1}{2})$

3.29 Maximize $\tilde{f} = x - 4y$

subject to $g_1 = -x + y - 4 \le 0$

$g_2 = 4x + 5y - 45 \le 0$

$g_3 = 5x - 2y - 20 \le 0$

$g_4 = -5x - 2y + 10 \le 0$

$g_5 = -x \le 0$

y is unrestricted in sign

Feasible space = BEFGHCB in Figure of Problem 3.28.

Point H: intersection of $g_3 = 0$ and $g_4 = 0$:

$$\begin{array}{r} 5x - 2y = 20 \\ -5x - 2y = -10 \\ \hline 10x = 30,\ x = 3 \\ 2y = 5x - 20 \\ y = -2\tfrac{1}{2} \end{array}$$

$\vec{X}^* =$ Point H $= \left(3, -2\tfrac{1}{2}\right)$

$f^* = 3 + 10 = 13$

3.30 Minimize $f = x - 4y$

subject to $g_1 = -x + y - 4 \le 0$

$g_2 = 4x + 5y - 45 \le 0$

$g_3 = 5x - 2y - 20 \le 0$

$g_4 = -5x - 2y + 10 \le 0$

$g_5 = -x \le 0$

$g_6 = -y \le 0$

$g_1 = 0$ at $(0,4), (2,6)$

$g_2 = 0$ at $(0,9), (11\tfrac{1}{4}, 0)$

$g_3 = 0$ at $(4,0), (8,10)$

$g_4 = 0$ at $(0,5), (2,0)$

Contours of $f = x - 4y$:

$f = -20$ at $(0,5), (4,6)$

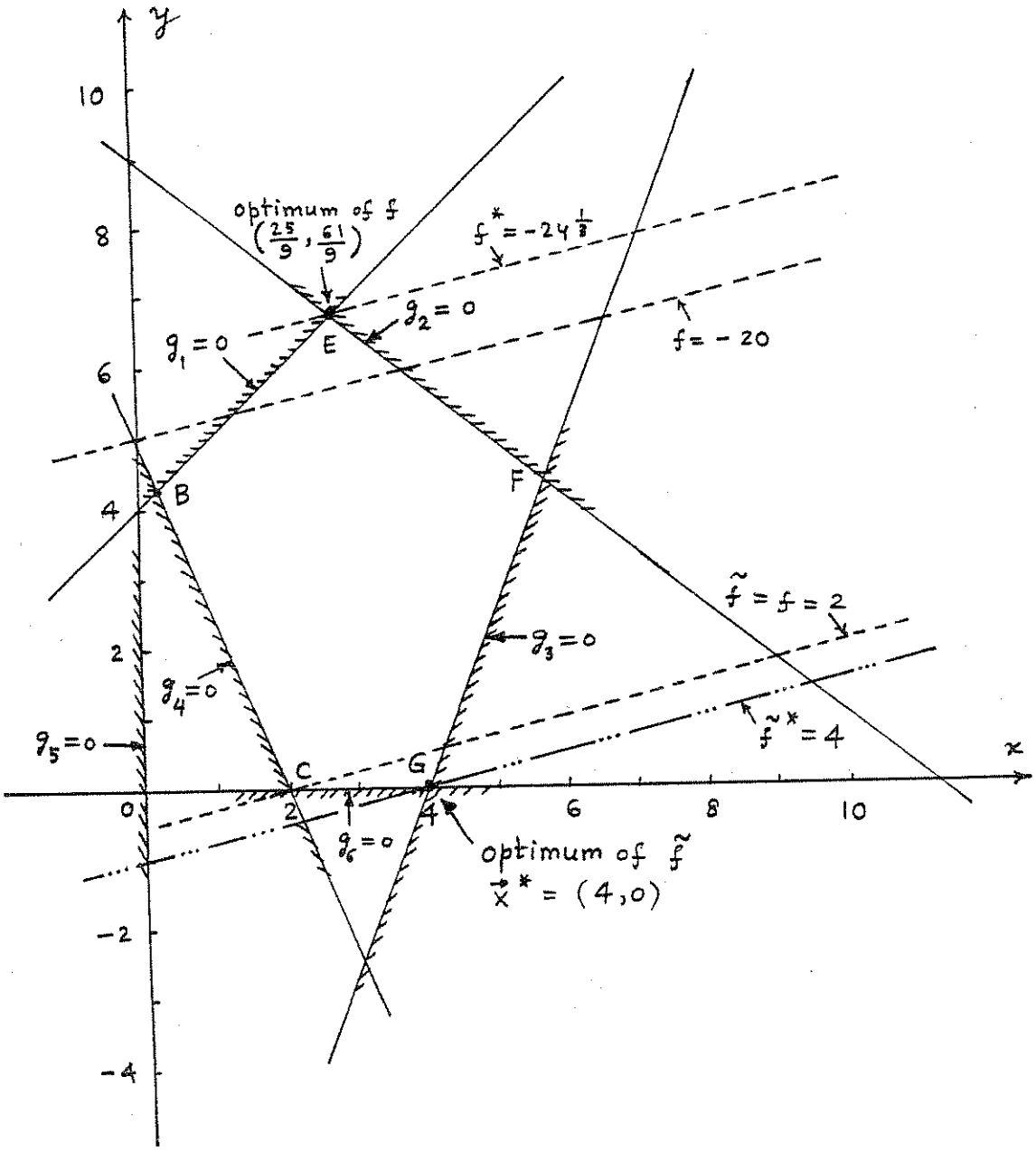
$f = 2$ at $(2,0), (6,1)$

Feasible space = BEFGCB in the figure

Point E $= \left(\frac{25}{9}, \frac{61}{9}\right) = \vec{X}^*$ = optimum point

$f^* = \frac{25}{9} - 4\left(\frac{61}{9}\right) = -\frac{219}{9}$

$= -24\tfrac{1}{3}$


y
x
10
8
6
4
2
0
-2
-4
2
4
6
8
10
optimum of f
$(\frac{25}{9}, \frac{61}{9})$
$f^* = -24\frac{1}{9}$
$g_1 = 0$
$g_2 = 0$
$f = -20$
E
B
F
C
G
$\tilde{f} = f = 2$
$g_3 = 0$
$g_4 = 0$
$g_5 = 0$
$\tilde{f}^* = 4$
$g_6 = 0$
optimum of $\tilde{f}$
$\vec{x}^* = (4,0)$

3.31 Maximize $\tilde{f} = x - 4y$

subject to $g_j \le 0$, $j = 1, 2, \ldots, 6$ (same as in Problem 3.30)

Feasible space: BEFGCB of figure of problem 3.30

$\tilde{f} = 4$ at $(4, 0)$, $(0, -1)$

optimum point = G = $(4, 0)$ with $f^* = 4$

3.32 Minimize $f = 4x + 5y$

subject to

$g_1 = 10x + y - 10 \ge 0$

$g_2 = 5x + 4y - 20 \ge 0$

$g_3 = 3x + 7y - 21 \ge 0$

$g_4 = x + 12y - 12 \ge 0$

$g_5 = x \ge 0$

$g_6 = y \ge 0$

$g_1 = 0$ at $(0, 10)$, $(1, 0)$

$g_2 = 0$ at $(0, 5)$, $(4, 0)$

$g_3 = 0$ at $(0, 3)$, $(7, 0)$

$g_4 = 0$ at $(0, 1)$, $(12, 0)$

Feasible space: space above ABCDEFG in figure

$f = 50$ at $(0, 10)$, $(12\frac{1}{2}, 0)$

$f = 25$ at $(0, 5)$, $(6\frac{1}{4}, 0)$

Point D = intersection of $g_2 = 0$ and $g_3 = 0$:

$$= \left(\frac{56}{23}, \frac{45}{23}\right)$$

$15x + 12y = 60$

$15x + 35y = 105$

$-23y = -45$

$y = \frac{45}{23}$;

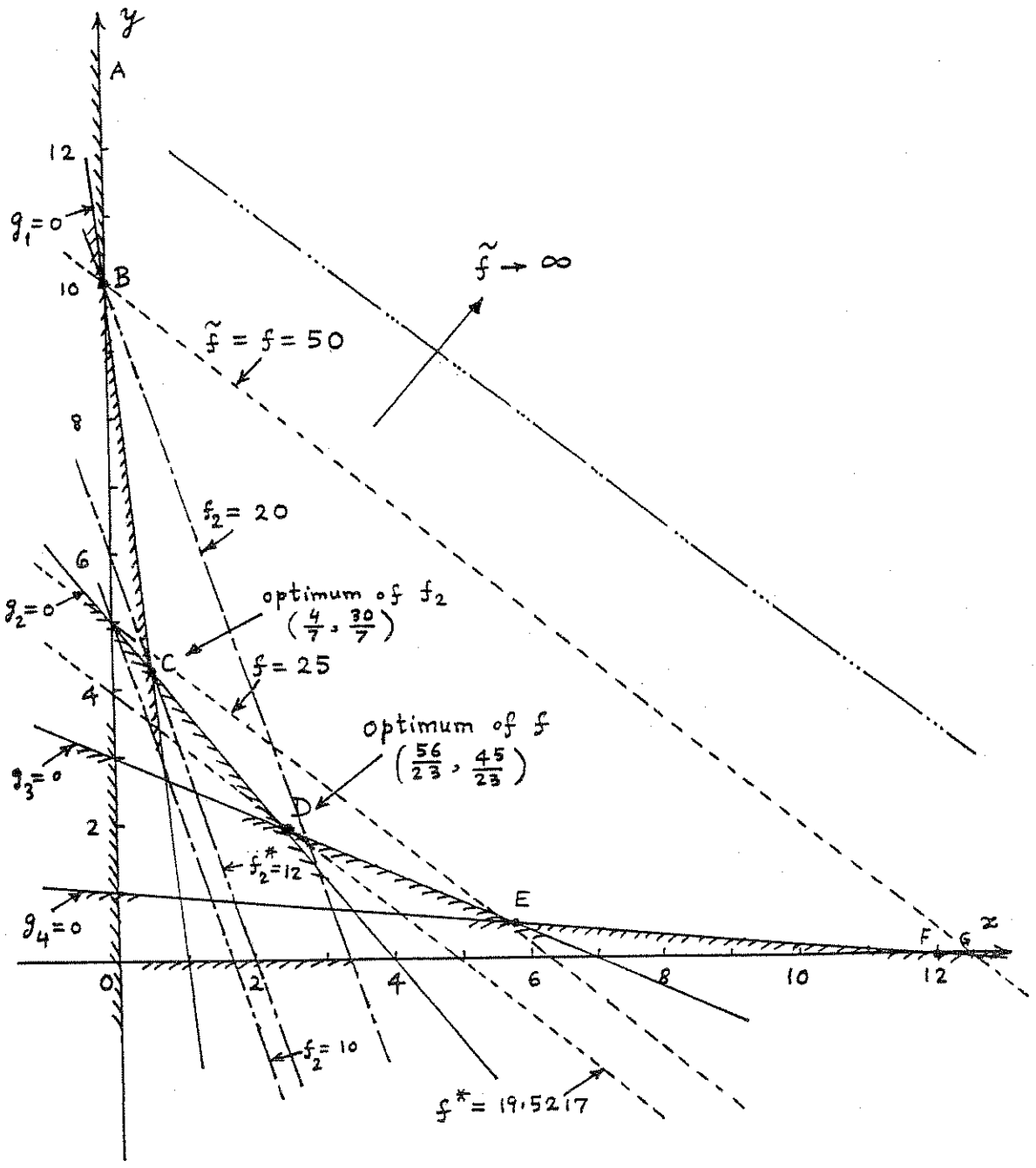
$5x = 20 - \frac{180}{23} = \frac{280}{23}$

$x = \frac{56}{23}$

optimum solution = $\vec{x}^* =$ point D

$$= \left(\frac{56}{23}, \frac{45}{23}\right)$$

$f^* = \frac{449}{23} = 19.5217$


y
A
12
$g_1=0$
10
B
$\tilde{f} \to \infty$
$\tilde{f} = f = 50$
8
$f_2 = 20$
6
$g_2=0$
optimum of f_2
$(\frac{4}{7}, \frac{30}{7})$
C
$f = 25$
4
optimum of f
$(\frac{56}{23}, \frac{45}{23})$
$g_3=0$
D
2
$f_2^* = 12$
E
$g_4=0$
F
G
x
0
2
4
6
8
10
12
$f_2 = 10$
$f^* = 19.5217$

3.33 Maximize $\tilde{f} = 4x + 5y$

subject to $g_j \geq 0$, $j = 1, 2, \ldots, 6$ (same as in Problem 3.32)

Optimum solution = unbounded

Feasible space = space above ABCDEFG in figure of Problem 3.32

3.34 Minimize $f_2 = 6x + 2y$

subject to $g_j \geq 0$, $j = 1, 2, \ldots, 6$ (same as in Problem 3.32)

Feasible space = space above ABCDEFG in figure of Problem 3.32

Contours of $f_2 = 6x + 2y$:

$f_2 = 20$ at $(0, 10)$, $(2, 4)$

$f_2 = 10$ at $(0, 5)$, $(2, -1)$

Point C = Optimum of f_2 = intersection of $g_1 = 0$ and $g_2 = 0$:

$= \vec{x}^* = \left(\frac{4}{7}, \frac{30}{7}\right)$

with $f_2^* = 12$

$10x + y = 10$

$10x + 8y = 40$

$-7y = -30, \quad y = \frac{30}{7}$

$5x = 20 - \frac{120}{7} = \frac{20}{7}$

$x = \frac{4}{7}$

3.35 Minimize $f = 6x + 2y$

subject to $g_1 = 10x + y - 10 \geq 0$

$g_2 = 5x + 4y - 20 \geq 0$

$g_3 = 3x + 7y - 21 \geq 0$

$g_4 = x + 12y - 12 \geq 0$

x, y unrestricted in sign

$g_1 = 0$ at $(0, 10)$, $(1, 0)$

$g_2 = 0$ at $(0, 5)$, $(4, 0)$

$g_3 = 0$ at $(0, 3)$, $(7, 0)$

$g_4 = 0$ at $(0, 1)$, $(12, 0)$

Feasible space: space above ABCDEFG in figure below.

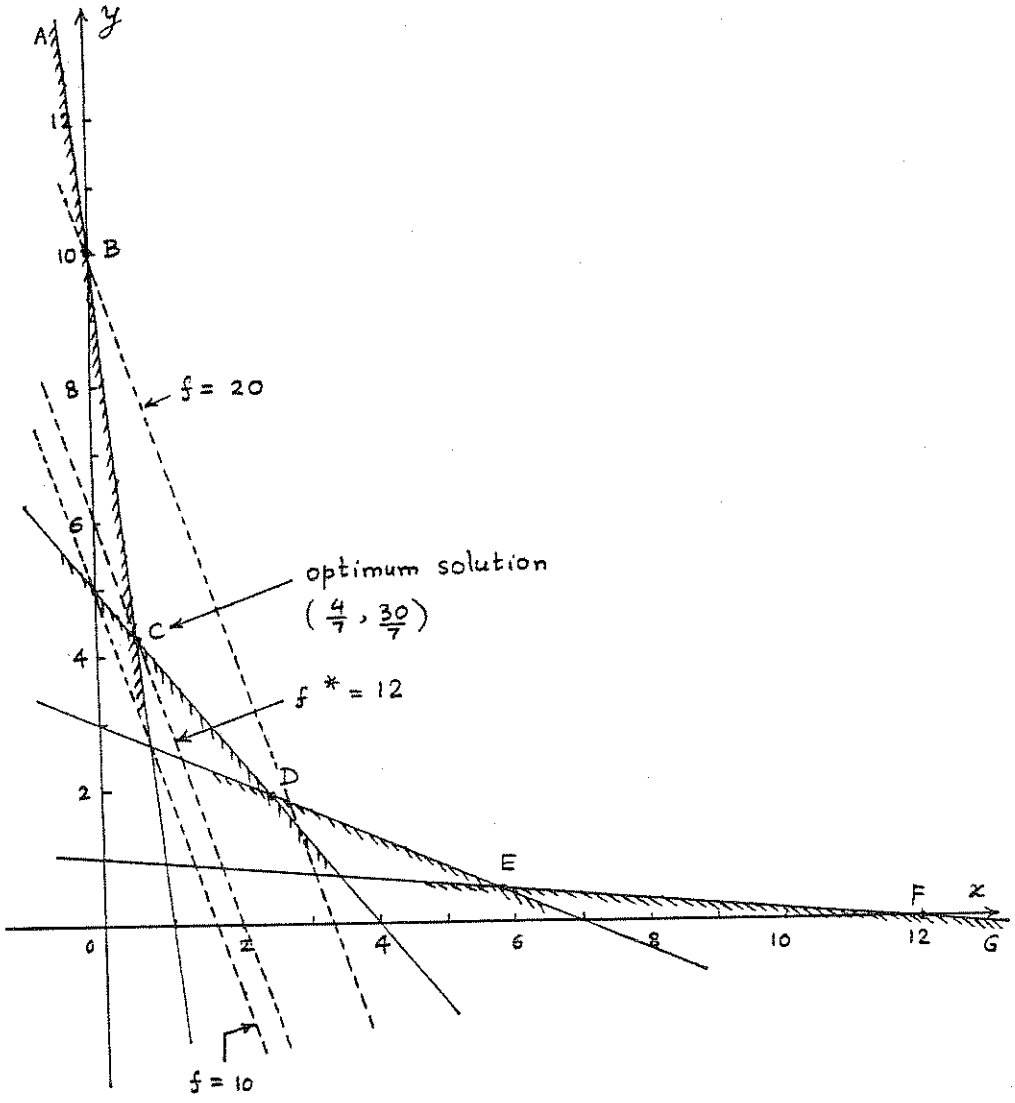
Contours of $f = 6x + 2y$:

$f = 20$ at $(0,10)$, $(2,4)$

$f = 10$ at $(0,5)$, $(2,-1)$

$f = 12$ at $\left(\frac{4}{7}, \frac{30}{7}\right)$, $(1,3)$

Optimum point = point $C = \left(\frac{4}{7}, \frac{30}{7}\right)$ with $f^* = 12$.



3.36 Minimize $f = 5x + 2y$

subject to: $g_1 = -3x - 4y + 24 \geq 0$

$g_2 = -x + y + 3 \geq 0$

$g_3 = x + 4y - 4 \geq 0$

$g_4 = 3x + y - 3 \geq 0$

$g_5 = +x \geq 0$

$g_6 = +y \geq 0$

$g_1 = 0$ at $(0, 6), (8, 0)$

$g_2 = 0$ at $(3, 0), (0, -3)$

$g_3 = 0$ at $(0, 1), (4, 0)$

$g_4 = 0$ at $(0, 3), (1, 0)$

$f = 12$ at $(0, 6), (2, 1)$

$f = 15$ at $(3, 0), (1, 5)$

Point D $\Rightarrow$

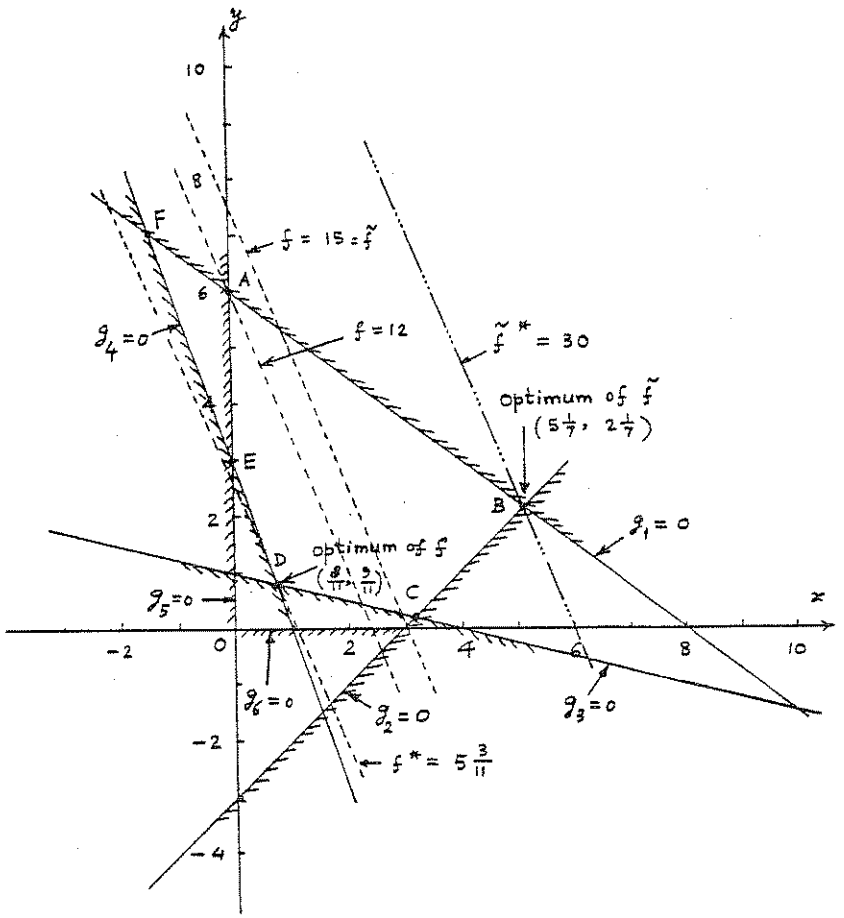
$$3x + 12y = 12$$
$$3x + y = 3$$
$$11y = 9$$
$$y = 9/11$$
$$x = 4 - 4\left(\frac{9}{11}\right) = \frac{8}{11}$$

Feasible space = ABCDEA

optimum point = point D

$$= \left(\frac{8}{11}, \frac{9}{11}\right) = \vec{x}^*$$

$$f^* = \frac{58}{11} = 5\frac{3}{11}$$



3.37 Maximize $\tilde{f} = 5x + 2y$

subject to $g_j \geq 0$, $j = 1, 2, \ldots, 6$ (as in Problem 3.36)

Feasible space = ABCDEA

Point B = optimum point = $(5\frac{1}{7}, 2\frac{1}{7})$

$\tilde{f}^* = 5\left(\frac{36}{7}\right) + 2\left(\frac{15}{7}\right) = \frac{210}{7} = 30$

$\tilde{f} = 30$ at $\left(\frac{36}{7}, \frac{15}{7}\right)$, $(6, 0)$

3.38 Minimize $f = 5x + 2y$

subject to $g_1 = -3x - 4y + 24 \geq 0$

$g_2 = -x + y + 3 \geq 0$

$g_3 = x + 4y - 4 \geq 0$

$g_4 = 3x + y - 3 \geq 0$

$g_5 = y \geq 0$

x = unrestricted in sign

Feasible space = ABCDEFA in Figure of Problem 3.36.

Optimum point = point D = $(8/11, 9/11)$

$f^* = 5\frac{3}{11}$

3.39 Maximize $f = 5x + 2y$

subject to $g_1 = -3x - 4y + 24 \geq 0$

$g_2 = -x + y + 3 \geq 0$

$g_3 = -x - 4y + 4 \geq 0$

$g_4 = 3x + y - 3 \geq 0$

$g_5 = +x \geq 0$

$g_6 = +y \geq 0$

Constraint boundaries can be identified as:

$g_1 = 0 \Rightarrow (0, 6), (8, 0)$

$g_2 = 0 \Rightarrow (3, 0), (0, -3)$

$g_3 = 0 \Rightarrow (0, 1), (4, 0)$

$g_4 = 0 \Rightarrow (0, 3), (1, 0)$

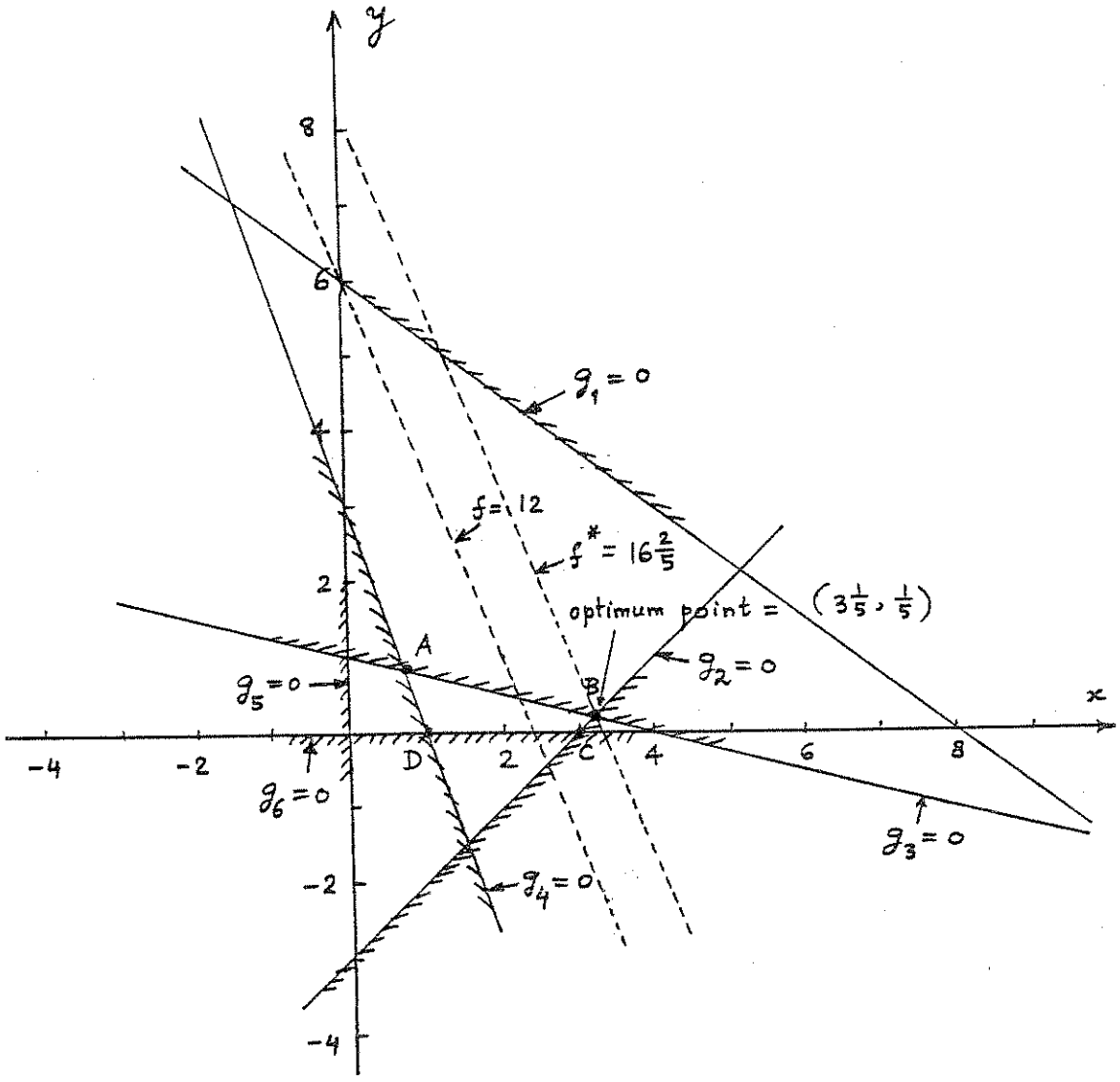
Feasible space = ABCDA in figure below.

$f = 12$ at $(0,6)$, $(2,1)$

$f = 15$ at $(3,0)$, $(1,5)$

optimum point = Point $B = \left(\frac{16}{5}, \frac{1}{5}\right) = \vec{x}^*$

$$f^* = 5\left(\frac{16}{5}\right) + 2\left(\frac{1}{5}\right) = 16\frac{2}{5}$$



3.40 Maximize $f = 3x + 2y$

subject to

$$g_1 = 9x + 10y - 330 \le 0$$
$$g_2 = -21x + 4y - 36 \le 0$$
$$g_3 = -x - 2y + 6 \le 0$$
$$g_4 = 6x - y - 72 \le 0$$
$$g_5 = 3x + y - 54 \le 0$$
$$g_6 = -x \le 0$$
$$g_7 = -y \le 0$$

$g_1 = 0$ at $(0, 33)$, $(10, 24)$

$g_2 = 0$ at $(0, 9)$, $(-2, -1.5)$

$g_3 = 0$ at $(0, 3)$, $(6, 0)$

$g_4 = 0$ at $(12, 0)$, $(15, 18)$

$g_5 = 0$ at $(18, 0)$, $(15, 9)$

contours of f:

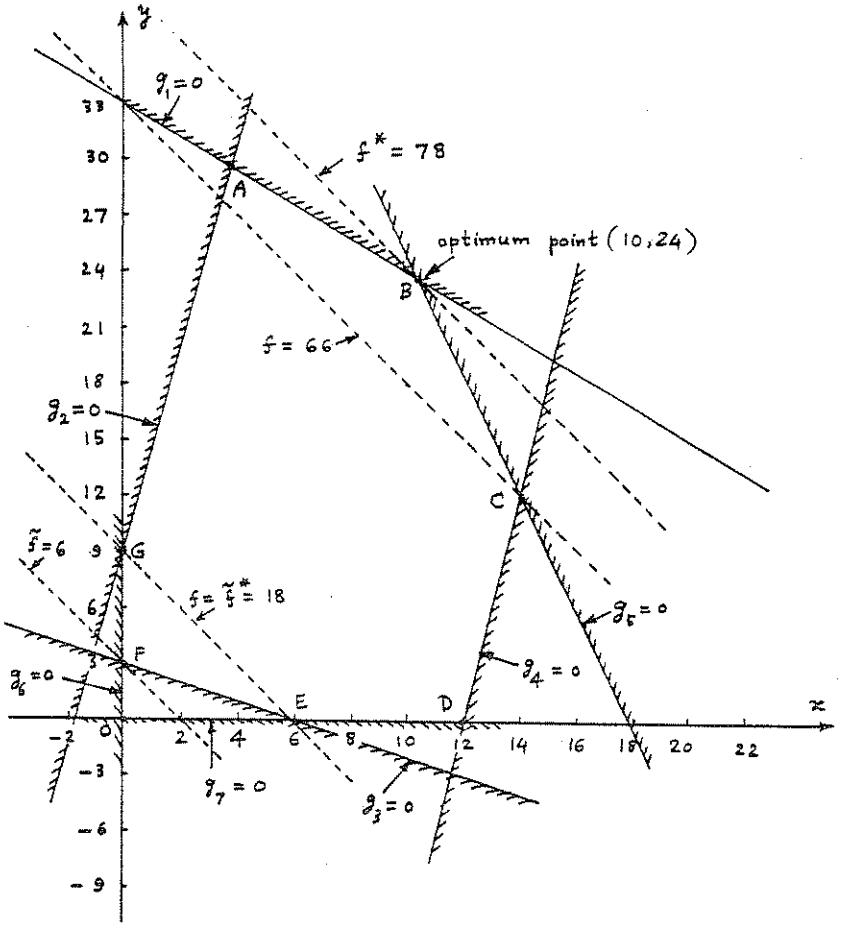
$f = 18$ at $(0, 9)$, $(2, 6)$

$f = 66$ at $(0, 33)$, $(10, 18)$

Feasible space = ABCDEFGA

optimum point = B = $(10, 24)$

with $f^* = 78$



3.41 Maximize $\tilde{f} = 3x + 2y$

subject to $g_1 = 9x + 10y - 330 \leq 0$

$g_2 = -21x + 4y - 36 \leq 0$

$g_3 = x + 2y - 6 \leq 0$

$g_4 = 6x - y - 72 \leq 0$

$g_5 = 3x + y - 54 \leq 0$

$g_6 = -x \leq 0$

$g_7 = -y \leq 0$

Here g_3 is reversed compared to that of Problem 3.41.

Feasible space = EFOE in figure of Problem 3.41.

Optimum point = point E = (6, 0)

with $\tilde{f}^* = 18$

3.42 Maximize $f = 3x + 2y$

subject to $g_1 = 9x + 10y - 330 \leq 0$

$g_2 = -21x + 4y - 36 \leq 0$

$g_3 = x + 2y - 6 \leq 0$

$g_4 = 6x - y - 72 \leq 0$

$g_5 = -3x - y + 54 \leq 0$

$g_6 = -x \leq 0$

$g_7 = -y \leq 0$

$g_1 = 0$ at (0, 33), (10, 24)

$g_2 = 0$ at (0, 9), (−2, −1.5)

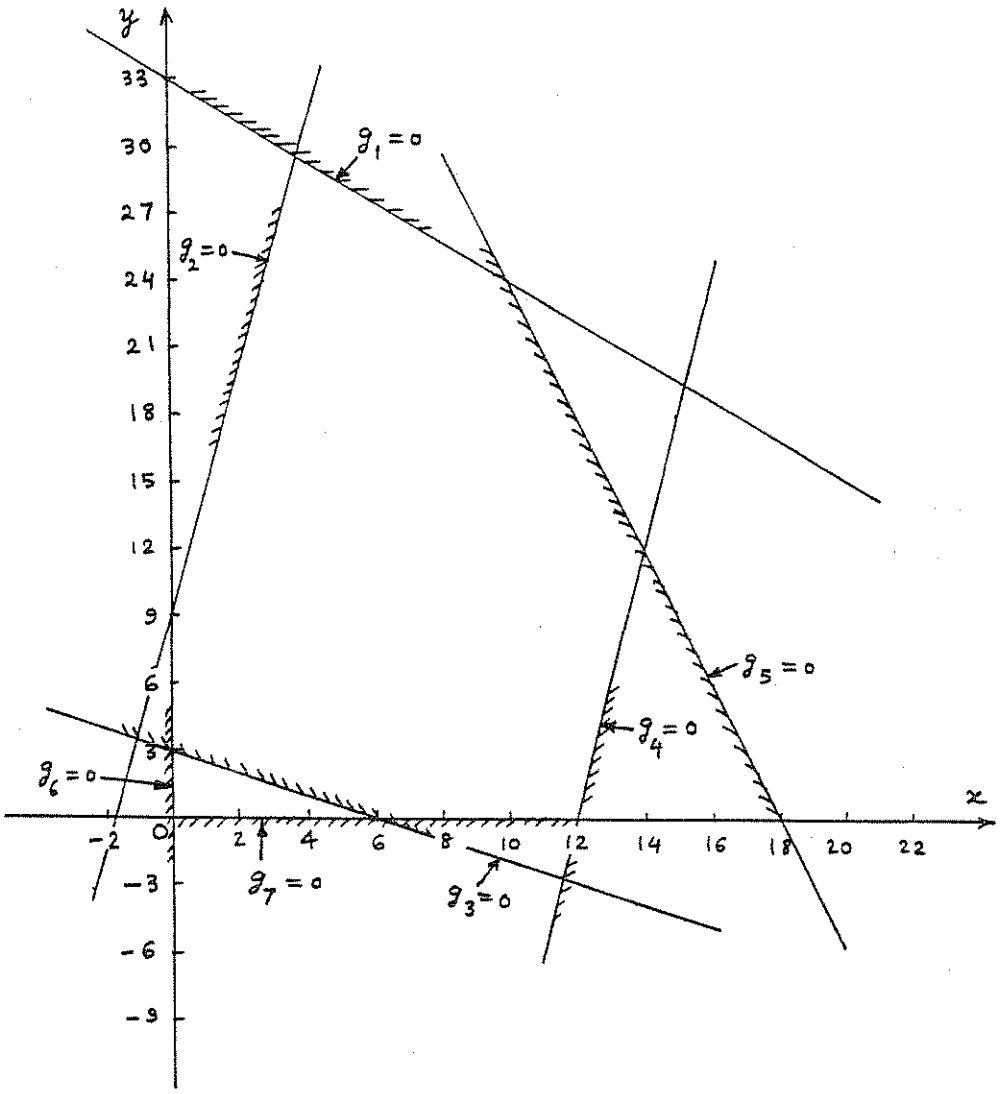
$g_3 = 0$ at (0, 3), (6, 0)

$g_4 = 0$ at (12, 0), (15, 18)

$g_5 = 0$ at (18, 0), (15, 9)

No feasible point (see figure below).

∴ No optimum point exists.



3.43

Maximize $f = 3x + 2y$

subject to

$$g_1 = -21x + 4y - 36 \leq 0$$

$$g_2 = -x - 2y + 6 \leq 0$$

$$g_3 = 6x - y - 72 \leq 0$$

$$g_4 = -x \leq 0$$

$$g_5 = -y \leq 0$$

$g_1 = 0$ at $(0, 9)$, $(-2, -1.5)$

$g_2 = 0$ at $(0, 3)$, $(6, 0)$

$g_3 = 0$ at $(12, 0)$, $(15, 18)$

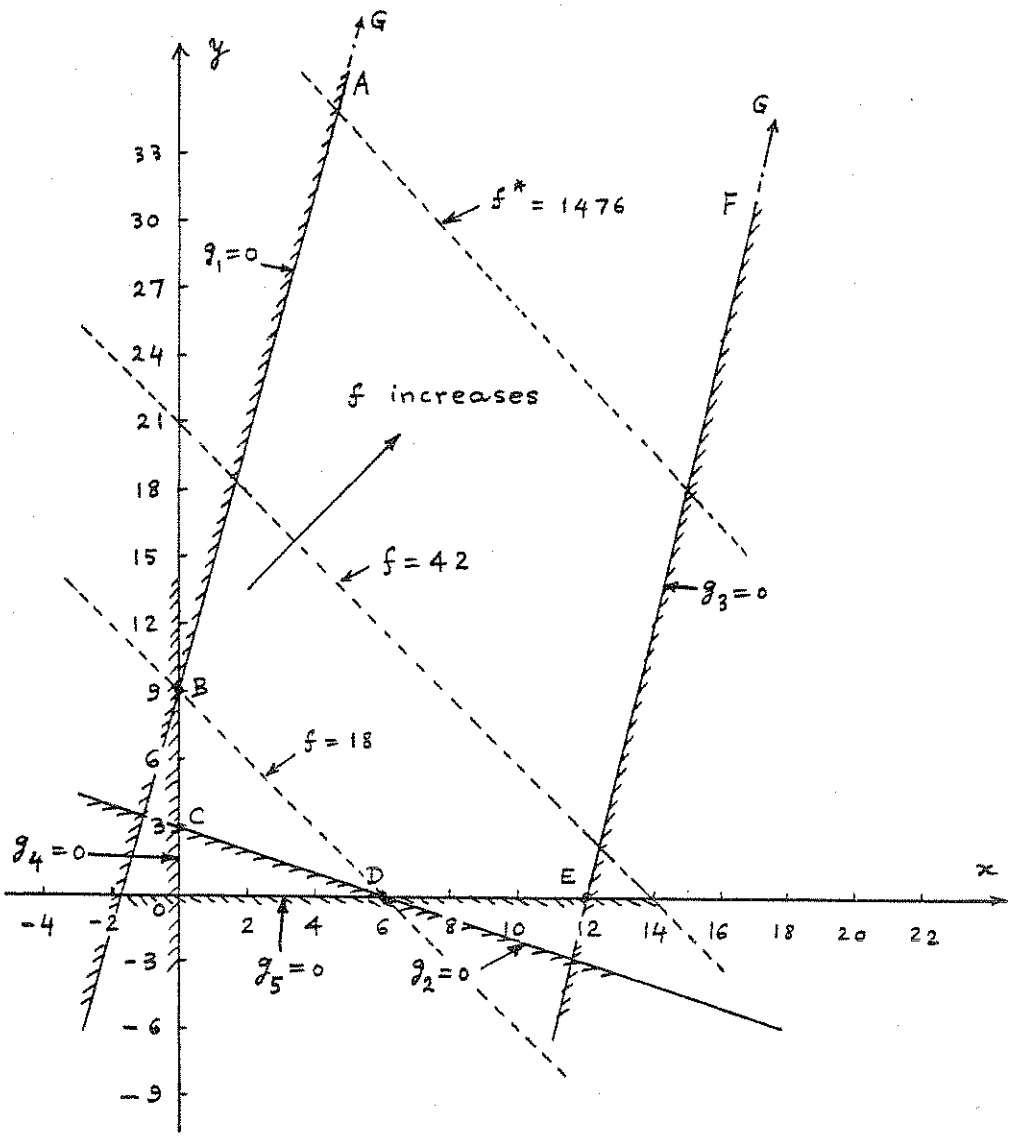
$f = 18$ at $(0, 9), (2, 6)$

$f = 42$ at $(0, 21), (2, 18)$

Feasible space = A B C D E F

Optimum solution = G = intersection point of $g_1 = 0$ and $g_3 = 0$

$= (108, 576)$ with $f^* = 1476$



3.44 Given system of equations:

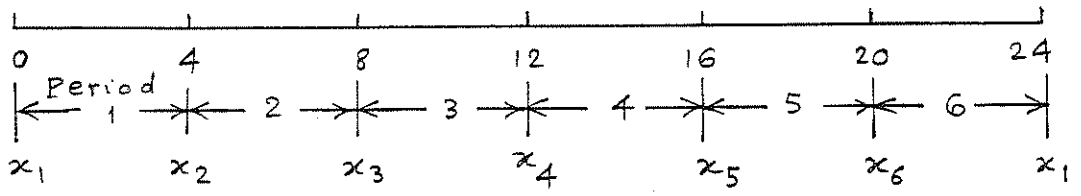
x_1	x_2	x_3	x_4	b_i	
(2)	3	−2	−7	2	
1	1	−1	3	12	
1	−1	1	5	8	
Pivot on a_{11}:					
1	3/2	−1	−7/2	1	
0	(−1/2)	0	13/2	11	
0	−5/2	2	17/2	7	
Pivot on a_{22}:					
1	0	−1	16	34	
0	1	0	−13	−22	
0	0	(2)	−24	−48	
Pivot on a_{33}:					
1	0	0	4	10	canonical form with x_1, x_2, x_3 as basic variables
0	1	0	−13	−22	
0	0	1	(−12)	−24	
Pivot on a_{34}:					
1	0	1/3	0	2	Canonical form with x_1, x_2, x_4 as basic variables
0	1	(−13/12)	0	4	
0	0	−1/12	1	2	
Pivot on a_{23}:					
1	(4/13)	0	0	42/13	Canonical form with x_1, x_3, x_4 as basic variables
0	−12/13	1	0	−48/13	
0	−1/13	0	1	22/13	
Pivot on a_{12}:					
13/4	1	0	0	21/2	Canonical form with x_2, x_3, x_4 as basic variables
3	0	1	0	6	
1/4	0	0	1	5/2	

3.45

x_1	x_2	x_3	x_4	x_5	x_6	b_i	Basic variables	Nonbasic variables	Feasible ?	f
20	9	6	(1)	1	0	20	$x_5 = 20$,	$x_1 = x_2 = x_3 = x_4$	Yes	0
10	4	2	1	0	1	10	$x_6 = 10$	$= 0$		
20	9	6	1	1	0	20	$x_4 = 20$,	$x_1 = x_2 = x_3 =$	No	380
−10	−5	−4	0	(−1)	1	−10	$x_6 = -10$	$x_5 = 0$		
10	4	2	1	0	1	10	$x_4 = 10$,	$x_1 = x_2 = x_3 =$	Yes	190
10	5	(4)	0	1	−1	10	$x_5 = 10$	$x_6 = 0$		
5	3/2	0	1	(−1/2)	3/2	5	$x_3 = 5/2$,	$x_1 = x_2 = x_5$	Yes	245
5/2	5/4	1	0	1/4	−1/4	5/2	$x_4 = 5$	$= x_6 = 0$		
−10	−3	0	−2	1	(−3)	−10	$x_3 = 5$,	$x_1 = x_2 = x_4$	No	300
5	2	1	1/2	0	1/2	5	$x_5 = -10$	$= x_6 = 0$		
10/3	1	0	2/3	−1/3	1	10/3	$x_3 = 10/3$,	$x_1 = x_2 = x_4$	Yes	200
10/3	(3/2)	1	1/6	1/6	0	10/3	$x_6 = 10/3$	$= x_5 = 0$		
10/9	0	−2/3	5/9	(−4/9)	1	10/9	$x_2 = 20/9$,	$x_1 = x_3 = x_4$	Yes	$231\frac{1}{9}$
20/9	1	2/3	1/9	1/9	0	20/9	$x_6 = 10/9$	$= x_5 = 0$		
−5/2	0	3/2	(−5/4)	1	−9/4	−5/2	$x_2 = 5/2$,	$x_1 = x_3 = x_4$	No	260
5/2	1	1/2	1/4	0	1/4	5/2	$x_5 = -5/2$	$= x_6 = 0$		
(2)	0	−6/5	1	−4/5	9/5	2	$x_2 = 2$,	$x_1 = x_3 = x_5$	Yes	246
2	1	4/5	0	1/5	−1/5	2	$x_4 = 2$	$= x_6 = 0$		optimum
1	0	(−3/5)	1/2	−2/5	9/10	1	$x_1 = 1$,	$x_3 = x_4 = x_5$	Yes	240
0	1	2	−1	1	−2	0	$x_2 = 0$	$= x_6 = 0$		
−5/3	0	1	−5/6	2/3	−3/2	−5/3	$x_2 = 10/3$,	$x_1 = x_4 = x_5$	No	$246\frac{2}{3}$
(10/3)	1	0	2/3	−1/3	1	10/3	$x_3 = -5/3$	$= x_6 = 0$		

x_1	x_2	x_3	x_4	x_5	x_6	b_i	Basic variables	Nonbasic variables	Feasible ?	f
0	$1/2$	1	$(-1/2)$	$1/2$	-1	0	$x_1 = 1$,	$x_2 = x_4 = x_5$	Yes	240
1	$3/10$	0	$1/5$	$-1/10$	$3/10$	1	$x_3 = 0$	$= x_6 = 0$		
0	-1	-2	1	(-1)	2	0	$x_1 = 1$,	$x_2 = x_3 = x_5$	Yes	240
1	$1/2$	$2/5$	0	$1/10$	$-1/10$	1	$x_4 = 0$	$= x_6 = 0$		
0	1	2	-1	1	(-2)	0	$x_1 = 1$,	$x_2 = x_3 = x_4$	Yes	240
1	$2/5$	$1/5$	$1/10$	0	$1/10$	1	$x_5 = 0$	$= x_6 = 0$		
0	$-1/2$	-1	$1/2$	$-1/2$	1	0	$x_1 = 1$,	$x_2 = x_3 = x_4$	Yes	240
1	$9/20$	$3/10$	$1/20$	$1/20$	0	1	$x_6 = 0$	$= x_5 = 0$		

3.46



Let x_i = number of attendants starting at the beginning of period i $(i = 1, 2, \ldots, 6)$

Problem: Minimize $f = x_1 + x_2 + x_3 + x_4 + x_5 + x_6$

subject to

$x_6 + x_1 \geq 4$ (0 – 4 hours)

$x_1 + x_2 \geq 7$ (4 – 8 hours)

$x_2 + x_3 \geq 8$ (8 – 12 hours)

$x_3 + x_4 \geq 9$ (12 – 16 hours)

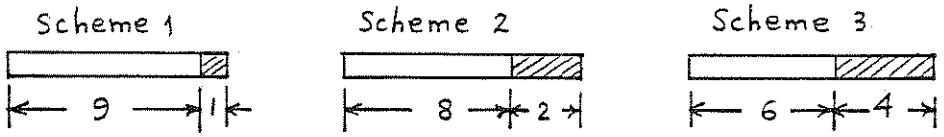
$x_4 + x_5 \geq 14$ (16 – 20 hours)

$x_5 + x_6 \geq 3$ (20 – 24 hours)

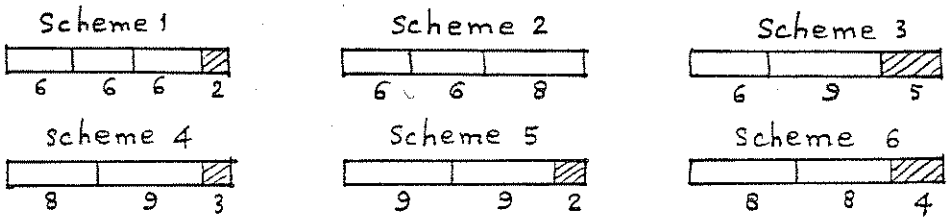
$x_i \geq 0$, $i = 1, 2, \ldots, 6$

3.47

Width 10:



Width 20:



Let x_i = number of rolls of width 10 m to be cut according to scheme i $(i = 1, 2, 3)$

y_i = number of rolls of width 20 m to be cut according to scheme i $(i = 1, 2, \ldots, 6)$

Number of 6 m – width rolls produced = $x_3 + 3y_1 + 2y_2 + y_3$

Number of 8 m – width rolls produced = $x_2 + y_2 + y_4 + 2y_6$

Number of 9 m – width rolls produced = $x_1 + y_3 + y_4 + 2y_5$

Let z_1 = number of surplus rolls of width 6 m

z_2 = number of surplus rolls of width 8 m

z_3 = number of surplus rolls of width 9 m

$$z_1 = x_3 + 3y_1 + 2y_2 + y_3 - 100$$

$$z_2 = x_2 + y_2 + y_4 + 2y_6 - 300$$

$$z_3 = x_1 + y_3 + y_4 + 2y_5 - 200$$

Problem: Minimize $f = x_1 + 2x_2 + 4x_3 + 2y_1 + 5y_3 + 3y_4 + 2y_5 + 4y_6 + 6z_1 + 8z_2 + 9z_3$

subject to

$$x_3 + 3y_1 + 2y_2 + y_3 - 100 = z_1$$

$$x_2 + y_2 + y_4 + 2y_6 - 300 = z_2$$

$$x_1 + y_3 + y_4 + 2y_5 - 200 = z_3$$

$x_i \geq 0$ $(i = 1,2,3)$, $y_j \geq 0$ $(j = 1, \ldots, 6)$, $z_k \geq 0$ $(k = 1,2,3)$

$$\vec{x} = \begin{Bmatrix} \{x_i\} \\ \{y_j\} \\ \{z_k\} \end{Bmatrix}$$

3.48

Maximize $f = x_1 + x_2 + x_3$

subject to

$$6x_1 + 6x_2 + 3x_3 + 4.5w_1 + 6w_2 + 4w_3 \le 9W_1$$

$$\text{or} \quad 6x_1 + 6x_2 + 3x_3 \le 24{,}100 \tag{1}$$

$$3x_1 + 3x_2 + 4x_3 + 4.5w_1 + 3w_2 + 5w_3 \le 9W_2$$

$$\text{or} \quad 3x_1 + 3x_2 + 4x_3 \le 15{,}500 \tag{2}$$

$$4x_2 + x_3 + 4w_2 + 2w_3 \le 8W_2$$

$$\text{or} \quad 4x_2 + x_3 \le 14{,}800 \tag{3}$$

$$2x_3 + 2w_3 \le 4W_3$$

$$\text{or} \quad 2x_3 \le 3{,}600 \tag{4}$$

$$x_3 + 2w_3 \le 4W_3$$

$$\text{or} \quad x_3 \le 3{,}800 \tag{5}$$

$$x_i \ge 0, \quad i = 1, 2, 3$$

given system x_4 to x_8 = slack variables in Eqs. (1) to (5).

x4:	6.000	6.000	3.000	1.000	0.000	0.000	0.000	0.000	24100.000
x5:	3.000	3.000	4.000	0.000	1.000	0.000	0.000	0.000	15500.000
x6:	0.000	4.000	1.000	0.000	0.000	1.000	0.000	0.000	14800.000
x7:	0.000	0.000	2.000	0.000	0.000	0.000	1.000	0.000	3600.000
x8:	0.000	0.000	1.000	0.000	0.000	0.000	0.000	1.000	3800.000
-f:	-1.000	-1.000	-1.000	0.000	0.000	0.000	0.000	0.000	0.000

minimum value of c1:-0.10000000E+01 variable out of basis: 1 result of pivoting on element: 1 1

x1:	1.000	1.000	0.500	0.167	0.000	0.000	0.000	0.000	4016.667
x5:	0.000	0.000	2.500	-0.500	1.000	0.000	0.000	0.000	3450.000
x6:	0.000	4.000	1.000	0.000	0.000	1.000	0.000	0.000	14800.000
x7:	0.000	0.000	2.000	0.000	0.000	0.000	1.000	0.000	3600.000
x8:	0.000	0.000	1.000	0.000	0.000	0.000	0.000	1.000	3800.000
-f:	0.000	0.000	-0.500	0.167	0.000	0.000	0.000	0.000	4016.667

minimum value of c1:-0.50000000E+00 variable out of basis: 3 result of pivoting on element: 2 3

x1:	1.000	1.000	0.000	0.267	-0.200	0.000	0.000	0.000	3326.667
x3:	0.000	0.000	1.000	-0.200	0.400	0.000	0.000	0.000	1380.000
x6:	0.000	4.000	0.000	0.200	-0.400	1.000	0.000	0.000	13420.000
x7:	0.000	0.000	0.000	0.400	-0.800	0.000	1.000	0.000	840.000
x8:	0.000	0.000	0.000	0.200	-0.400	0.000	0.000	1.000	2420.000
-f:	0.000	0.000	0.000	0.067	0.200	0.000	0.000	0.000	4706.667

3.49 Minimize $f = 5x_1 - 80x_2 + 160x_3 + 15x_4$

subject to

$$0.05x_1 + 0.05x_2 + 0.10x_3 + 0.15x_4 \leq 1000 \quad (1)$$

$$0.1x_1 + 0.15x_2 + 0.2x_3 + 0.05x_4 \leq 2000 \quad (2)$$

$$0.05x_1 + 0.1x_2 + 0.1x_3 + 0.15x_4 \leq 1500 \quad (3)$$

$$0 \leq x_1$$

$$0 \leq x_2$$

$$0 \leq x_3$$

$$0 \leq x_4$$

$$x_1 \leq 5000 \quad (4)$$

$$x_4 \leq 4000 \quad (5)$$

x_5 to x_9 = slack variables in Eqs. (1) to (5).

given system

x5:	0.5000E-01	0.5000E-01	0.1000E+00	0.1500E+00	0.1000E+01
	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.1000E+04
x6:	0.1000E+00	0.1500E+00	0.2000E+00	0.5000E-01	0.0000E+00
	0.1000E+01	0.0000E+00	0.0000E+00	0.0000E+00	0.2000E+04
x7:	0.5000E-01	0.1000E+00	0.1000E+00	0.1500E+00	0.0000E+00
	0.0000E+00	0.1000E+01	0.0000E+00	0.0000E+00	0.1500E+04
x8:	0.1000E+01	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
	0.0000E+00	0.0000E+00	0.1000E+01	0.0000E+00	0.5000E+04
x9:	0.0000E+00	0.0000E+00	0.0000E+00	0.1000E+01	0.0000E+00
	0.0000E+00	0.0000E+00	0.0000E+00	0.1000E+01	0.4000E+04
-f:	0.5000E+01	-0.8000E+02	0.1600E+03	0.1500E+02	0.0000E+00
	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00

minimum value of c1:-0.80000000E+02 variable out of basis: 2
result of pivoting on element: 2 2

x5:	0.1667E-01	0.0000E+00	0.3333E-01	0.1333E+00	0.1000E+01
	-0.3333E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.3333E+03
x2:	0.6667E+00	0.1000E+01	0.1333E+01	0.3333E+00	0.0000E+00
	0.6667E+01	0.0000E+00	0.0000E+00	0.0000E+00	0.1333E+05
x7:	-0.1667E-01	0.0000E+00	-0.3333E-01	0.1167E+00	0.0000E+00
	-0.6667E+00	0.1000E+01	0.0000E+00	0.0000E+00	0.1667E+03
x8:	0.1000E+01	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
	0.0000E+00	0.0000E+00	0.1000E+01	0.0000E+00	0.5000E+04
x9:	0.0000E+00	0.0000E+00	0.0000E+00	0.1000E+01	0.0000E+00
	0.0000E+00	0.0000E+00	0.0000E+00	0.1000E+01	0.4000E+04
-f:	0.5833E+02	0.0000E+00	0.2667E+03	0.4167E+02	0.0000E+00
	0.5333E+03	0.0000E+00	0.0000E+00	0.0000E+00	0.1067E+07

Solution: $\vec{x}^* = \{ x_1^* \quad x_2^* \quad x_3^* \quad x_4^* \}^T = \{ 0 \quad 0.1333 * 10^5 \quad 0 \quad 0 \}^T$

$f^* = -0.1067 * 10^7$

3.50

Maximize $f = 22x_1 - 3.2x_2 + 32.4x_3 - 156x_4 + 200x_5 - 50000 * 0.04$

subject to

$$700x_1 + 580x_2 + 690x_3 + 920x_4 \leq 50000$$

$$x_1 + x_2 + x_3 + x_4 + x_5 = 200$$

$$10000x_1 + 7000x_2 + 6000x_3 + 8000x_4 \leq 4 * 10^5$$

$$x_i \geq 0 \;, \quad i = 1, 2, \ldots, 5$$

Solution:

$$\vec{x}^* = \begin{Bmatrix} x_1^* \\ x_2^* \\ x_3^* \\ x_4^* \\ x_5^* \end{Bmatrix} = \begin{Bmatrix} 40.0 \\ 0.0 \\ 0.0 \\ 0.0 \\ 160.0 \end{Bmatrix}$$

$$f_{max} = 0.3288 * 10^5$$

given system

x6:	0.7000E+01	0.5800E+01	0.6900E+01	0.9200E+01	0.0000E+00
	0.1000E+01	0.0000E+00	0.5000E+03		
x5:	0.1000E+01	0.1000E+01	0.1000E+01	0.1000E+01	0.1000E+01
	0.0000E+00	0.0000E+00	0.2000E+03		
x7:	0.1000E+02	0.7000E+01	0.6000E+01	0.8000E+01	0.0000E+00
	0.0000E+00	0.1000E+01	0.4000E+03		
-f:	-0.2200E+02	0.3200E+01	-0.3240E+02	0.1560E+03	-0.2000E+03
	0.0000E+00	0.0000E+00	0.0000E+00		

minimum value of c1:-0.20000000E+03 variable out of basis: 5
result of pivoting on element: 2 5

x6:	0.7000E+01	0.5800E+01	0.6900E+01	0.9200E+01	0.0000E+00
	0.1000E+01	0.0000E+00	0.5000E+03		
x5:	0.1000E+01	0.1000E+01	0.1000E+01	0.1000E+01	0.1000E+01
	0.0000E+00	0.0000E+00	0.2000E+03		
x7:	0.1000E+02	0.7000E+01	0.6000E+01	0.8000E+01	0.0000E+00
	0.0000E+00	0.1000E+01	0.4000E+03		
-f:	0.1780E+03	0.2032E+03	0.1676E+03	0.3560E+03	0.0000E+00
	0.0000E+00	0.0000E+00	0.4000E+05		

3.51 $\vec{X} = \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} M_b \\ M_c \end{Bmatrix}$

Minimize $f = 20\,x_1 + 16\,x_2$

subject to

$$2x_1 + x_2 \geq 17$$
$$x_1 + x_2 \geq 12$$
$$2.5 \leq x_1$$
$$6 \leq x_2$$

constraints can be expressed as:

$$2x_1 + x_2 - x_3 + x_7 = 17 \quad (1)$$
$$x_1 + x_2 - x_4 + x_8 = 12 \quad (2)$$
$$x_1 - x_5 + x_9 = 2.5 \quad (3)$$
$$x_2 - x_6 + x_{10} = 6 \quad (4)$$

where x_3, x_4, x_5, x_6 are surplus variables and x_7, x_8, x_9, x_{10} are artificial variables.

Infeasibility form = $w = x_7 + x_8 + x_9 + x_{10}$
(objective for Phase I)

Using Eqs. (1) to (4), w can be expressed as

$$w = -4x_1 - 3x_2 + x_3 + x_4 + x_5 + x_6 + 37.5$$

Solution: $x_1^* = 5$, $x_2^* = 7$, $f^* = 212.0$

```
PHASE-I OF SIMPLEX METHOD
given system

  x 7:  0.2000E+01  0.1000E+01 -0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
        0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.1700E+02
  x 8:  0.1000E+01  0.1000E+01  0.0000E+00 -0.1000E+01  0.0000E+00  0.0000E+00
        0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.1200E+02
  x 9:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.1000E+01  0.0000E+00
        0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.2500E+01
 x10:  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.1000E+01
        0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.6000E+01
   -f:  0.2000E+02  0.1600E+02  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
   -w: -0.4000E+01 -0.3000E+01  0.1000E+01  0.1000E+01  0.1000E+01  0.1000E+01
        0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.3750E+02

minimum value of ds:-0.40000000E+01  variable out of basis:     1
result of pivoting on element:     3     1


  x 7:  0.0000E+00  0.1000E+01 -0.1000E+01  0.0000E+00  0.2000E+01  0.0000E+00
        0.1000E+01  0.0000E+00 -0.2000E+01  0.0000E+00  0.1200E+02
  x 8:  0.0000E+00  0.1000E+01  0.0000E+00 -0.1000E+01  0.1000E+01  0.0000E+00
        0.0000E+00  0.1000E+01 -0.1000E+01  0.0000E+00  0.9500E+01
  x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.1000E+01  0.0000E+00
        0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.2500E+01
 x10:  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.1000E+01
        0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.6000E+01
   -f:  0.0000E+00  0.1600E+02  0.0000E+00  0.0000E+00  0.2000E+02  0.0000E+00
        0.0000E+00  0.0000E+00 -0.2000E+02  0.0000E+00 -0.5000E+02
   -w:  0.0000E+00 -0.3000E+01  0.1000E+01  0.1000E+01 -0.3000E+01  0.1000E+01
        0.0000E+00  0.0000E+00  0.4000E+01  0.0000E+00 -0.2750E+02

minimum value of ds:-0.30000000E+01  variable out of basis:     2
result of pivoting on element:     4     2


  x 7:  0.0000E+00  0.0000E+00 -0.1000E+01  0.0000E+00  0.2000E+01  0.1000E+01
        0.1000E+01  0.0000E+00 -0.2000E+01 -0.1000E+01  0.6000E+01
  x 8:  0.0000E+00  0.0000E+00  0.0000E+00 -0.1000E+01  0.1000E+01  0.1000E+01
        0.0000E+00  0.1000E+01 -0.1000E+01 -0.1000E+01  0.3500E+01
  x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.1000E+01  0.0000E+00
        0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.2500E+01
  x 2:  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.1000E+01
        0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.6000E+01
   -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.2000E+02  0.1600E+02
        0.0000E+00  0.0000E+00 -0.2000E+02 -0.1600E+02 -0.1460E+03
   -w:  0.0000E+00  0.0000E+00  0.1000E+01  0.1000E+01 -0.3000E+01 -0.2000E+01
        0.0000E+00  0.0000E+00  0.4000E+01  0.3000E+01 -0.9500E+01

minimum value of ds:-0.30000000E+01  variable out of basis:     5
result of pivoting on element:     1     5


  x 5:  0.0000E+00  0.0000E+00 -0.5000E+00  0.0000E+00  0.1000E+01  0.5000E+00
        0.5000E+00  0.0000E+00 -0.1000E+01 -0.5000E+00  0.3000E+01
  x 8:  0.0000E+00  0.0000E+00  0.5000E+00 -0.1000E+01  0.0000E+00  0.5000E+00
       -0.5000E+00  0.1000E+01  0.0000E+00 -0.5000E+00  0.5000E+00
  x 1:  0.1000E+01  0.0000E+00 -0.5000E+00  0.0000E+00  0.0000E+00  0.5000E+00
        0.5000E+00  0.0000E+00  0.0000E+00 -0.5000E+00  0.5500E+01
  x 2:  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.1000E+01
        0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.6000E+01
   -f:  0.0000E+00  0.0000E+00  0.1000E+02  0.0000E+00  0.0000E+00  0.6000E+01
       -0.1000E+02  0.0000E+00  0.0000E+00 -0.6000E+01 -0.2060E+03
   -w:  0.0000E+00  0.0000E+00 -0.5000E+00  0.1000E+01  0.0000E+00 -0.5000E+00
```

```
        0.1500E+01  0.0000E+00  0.1000E+01  0.1500E+01 -0.5000E+00

minimum value of ds:-0.50000000E+00  variable out of basis:     3
result of pivoting on element:     2     3


  x 5:  0.0000E+00  0.0000E+00  0.0000E+00 -0.1000E+01  0.1000E+01  0.1000E+01
        0.0000E+00  0.1000E+01 -0.1000E+01 -0.1000E+01  0.3500E+01
  x 3:  0.0000E+00  0.0000E+00  0.1000E+01 -0.2000E+01  0.0000E+00  0.1000E+01
       -0.1000E+01  0.2000E+01  0.0000E+00 -0.1000E+01  0.1000E+01
  x 1:  0.1000E+01  0.0000E+00  0.0000E+00 -0.1000E+01  0.0000E+00  0.1000E+01
        0.0000E+00  0.1000E+01  0.0000E+00 -0.1000E+01  0.6000E+01
  x 2:  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.1000E+01
        0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.6000E+01
   -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.2000E+02  0.0000E+00 -0.4000E+01
        0.0000E+00 -0.2000E+02  0.0000E+00  0.4000E+01 -0.2160E+03
   -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.1000E+01  0.1000E+01  0.1000E+01  0.1000E+01  0.0000E+00

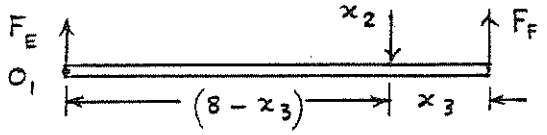
PHASE-II OF SIMPLEX METHOD
system at start of phase-II

  x 5:  0.0000E+00  0.0000E+00  0.0000E+00 -0.1000E+01  0.1000E+01  0.1000E+01
        0.3500E+01
  x 3:  0.0000E+00  0.0000E+00  0.1000E+01 -0.2000E+01  0.0000E+00  0.1000E+01
        0.1000E+01
  x 1:  0.1000E+01  0.0000E+00  0.0000E+00 -0.1000E+01  0.0000E+00  0.1000E+01
        0.6000E+01
  x 2:  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.1000E+01
        0.6000E+01
   -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.2000E+02  0.0000E+00 -0.4000E+01
       -0.2160E+03

minimum value of cs:-0.40000000E+01  variable out of basis:     6
result of pivoting on element:     2     6


  x 5:  0.0000E+00  0.0000E+00 -0.1000E+01  0.1000E+01  0.1000E+01  0.0000E+00
        0.2500E+01
  x 6:  0.0000E+00  0.0000E+00  0.1000E+01 -0.2000E+01  0.0000E+00  0.1000E+01
        0.1000E+01
  x 1:  0.1000E+01  0.0000E+00 -0.1000E+01  0.1000E+01  0.0000E+00  0.0000E+00
        0.5000E+01
  x 2:  0.0000E+00  0.1000E+01  0.1000E+01 -0.2000E+01  0.0000E+00  0.0000E+00
        0.7000E+01
   -f:  0.0000E+00  0.0000E+00  0.4000E+01  0.1200E+02  0.0000E+00  0.0000E+00
       -0.2120E+03
```

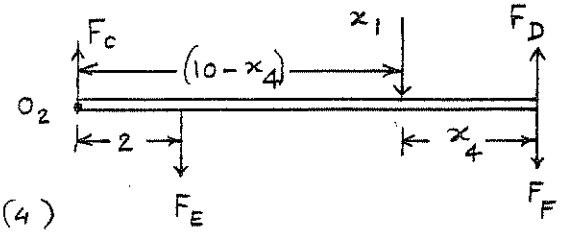
3.52 Equilibrium equations:



$$F_E + F_F = x_2 \qquad (1)$$

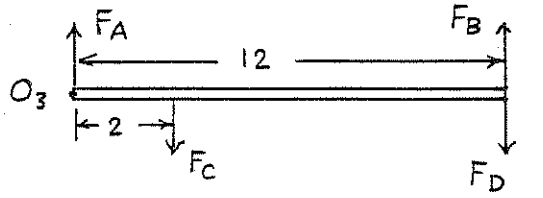
$$x_2(8-x_3) = F_F(8)$$

or $$8x_2 - x_2 x_3 = 8F_F \qquad (2)$$



$$F_C + F_D = x_1 + F_E + F_F \qquad (3)$$

$$x_1(10-x_4) + 2F_E + 10F_F = 10F_D \qquad (4)$$



$$F_A + F_B = F_C + F_D \qquad (5)$$

$$2F_C + 12F_D - 12F_B = 0 \qquad (6)$$

Eqs. (1) and (2) give:

$$F_E = \frac{1}{8}x_2 x_3 \qquad (7)$$

$$F_F = \frac{1}{8}(8x_2 - x_2 x_3) \qquad (8)$$

Eqs. (3) and (4), along with Eqs. (7) and (8), yield:

$$F_C = \frac{1}{10}(x_1 x_4 + x_2 x_3) \qquad (9)$$

$$F_D = x_1 + x_2 - \frac{1}{10}x_1 x_4 - \frac{1}{10}x_2 x_3 \qquad (10)$$

Eqs. (5) and (6), in conjunction with Eqs. (9) and (10), give:

$$F_A = \frac{1}{12}(x_1 x_4 + x_2 x_3) \qquad (11)$$

$$F_B = x_1 + x_2 - \frac{1}{12}x_1 x_4 - \frac{1}{12}x_2 x_3 \qquad (12)$$

Problem:

Maximize $f = x_1 + x_2$

subject to

$$F_A = \frac{1}{12}(x_1 x_4 + x_2 x_3) \le 300$$

$$F_B = x_1 + x_2 - \frac{1}{12}(x_1 x_4 + x_2 x_3) \le 300$$

$$F_C = \frac{1}{10}(x_1 x_4 + x_2 x_3) \le 200$$

$$F_D = x_1 + x_2 - \frac{1}{10}(x_1 x_4 + x_2 x_3) \le 200$$

$$F_E = \frac{1}{8}x_2 x_3 \le 100$$

$$F_F = x_2 - \frac{1}{8}x_2 x_3 \le 100$$

$$x_i \ge 0, \quad i = 1, 2, 3, 4$$

3.53

Let (x_1, x_2, x_3) = number of parts (A, B, C) produced per day

Profit = income − expenses

$$= (50x_1 + 75x_2 + 100x_3) - \{(5x_1 + 10x_2 + 15x_3)$$

$+250$ if $x_1 > 0$ or $x_3 > 0$

$+200$ if $x_2 > 0$ or $x_3 > 0$

$+300$ if $x_1 > 0$ or $x_2 > 0$ $\}$

Constraints:

$$0 \le x_1 \le 15$$

$$0 \le x_2 \le 20$$

$$0 \le x_3 \le 10$$

Problem:

Find $\vec{x}$ to maximize the profit subject to the stated constraints.

3.54

Minimize $f = \sum_{i=1}^{8} (y_i + z_i)$

subject to

$$x_1 + 25 = y_1 - z_1$$
$$x_1 - 20 = y_2 - z_2$$
$$x_1 - 40 = y_3 - z_3$$
$$x_1 + 30 = y_4 - z_4$$
$$x_2 + 35 = y_5 - z_5$$
$$x_2 + 15 = y_6 - z_6$$
$$x_2 - 30 = y_7 - z_7$$
$$x_2 - 20 = y_8 - z_8$$
$$-15 \le x_1 \le 15$$
$$-10 \le x_2 \le 10$$
$$y_i \ge 0, \quad i = 1, 2, \ldots, 8$$
$$z_i \ge 0, \quad i = 1, 2, \ldots, 8$$

with $\vec{x} = \{ x_1, x_2; y_1, y_2, \ldots, y_8; z_1, z_2, \ldots, z_8 \}^T$.

Solution:

$$\vec{x}^* = \{ 15, 10; 40, 0, 0, 45, 45, 25, 0, 0; 0, 5, 25, 0, 0, 0, 20, 10 \}^T$$

$$f^* = 215.0$$

3.55 Minimize $f = x_1 + 2x_2$

subject to

$$0.8\,x_1 + 0.6\,x_2 \geq 0.75$$

$$0.1\,x_1 + 0.2\,x_2 \geq 0.15$$

$$0.6\,x_1 + 0.18\,x_2 \geq 0.16$$

$$0.04\,x_1 + 0.02\,x_2 \geq 0.03$$

$$x_1 \geq 0, \quad x_2 \geq 0$$

Constraints can be expressed as:

$$0.8\,x_1 + 0.6\,x_2 - x_3 + x_7 = 0.75$$

$$0.1\,x_1 + 0.2\,x_2 - x_4 + x_8 = 0.15$$

$$0.6\,x_1 + 0.18\,x_2 - x_5 + x_9 = 0.16$$

$$0.04\,x_1 + 0.02\,x_2 - x_6 + x_{10} = 0.03$$

$x_3, \ldots, x_6$ = surplus variables

$x_7, \ldots, x_{10}$ = artificial variables

Infeasibility form:

$$w = x_7 + x_8 + x_9 + x_{10}$$

or $$w = -1.54\,x_1 - x_2 + x_3 + x_4 + x_5 + x_6 + 1.09$$

Solution: $x_1^* = 1.5, \quad x_2^* = 0, \quad f^* = 1.5$

```
PHASE-I OF SIMPLEX METHOD
given system

 x 7:  0.8000E+00  0.6000E+00 -0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
       0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.7500E+00
 x 8:  0.1000E+00  0.2000E+00  0.0000E+00 -0.1000E+01  0.0000E+00  0.0000E+00
       0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.1500E+00
 x 9:  0.6000E+00  0.1800E+00  0.0000E+00  0.0000E+00 -0.1000E+01  0.0000E+00
       0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.1600E+00
 x10:  0.4000E-01  0.2000E-01  0.0000E+00  0.0000E+00  0.0000E+00 -0.1000E+01
       0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.3000E-01
  -f:  0.1000E+01  0.2000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
       0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
  -w: -0.1540E+01 -0.1000E+01  0.1000E+01  0.1000E+01  0.1000E+01  0.1000E+01
       0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.1090E+01

minimum value of ds:-0.15400000E+01  variable out of basis:    1
result of pivoting on element:    3    1


 x 7:  0.0000E+00  0.3600E+00 -0.1000E+01  0.0000E+00  0.1333E+01  0.0000E+00
       0.1000E+01  0.0000E+00 -0.1333E+01  0.0000E+00  0.5367E+00
 x 8:  0.0000E+00  0.1700E+00  0.0000E+00 -0.1000E+01  0.1667E+00  0.0000E+00
       0.0000E+00  0.1000E+01 -0.1667E+00  0.0000E+00  0.1233E+00
 x 1:  0.1000E+01  0.3000E+00  0.0000E+00  0.0000E+00 -0.1667E+01  0.0000E+00
       0.0000E+00  0.0000E+00  0.1667E+01  0.0000E+00  0.2667E+00
 x10:  0.0000E+00  0.8000E-02  0.0000E+00  0.0000E+00  0.6667E-01 -0.1000E+01
       0.0000E+00  0.0000E+00 -0.6667E-01  0.1000E+01  0.1933E-01
  -f:  0.0000E+00  0.1700E+01  0.0000E+00  0.0000E+00  0.1667E+01  0.0000E+00
       0.0000E+00  0.0000E+00 -0.1667E+01  0.0000E+00 -0.2667E+00
  -w:  0.0000E+00 -0.5380E+00  0.1000E+01  0.1000E+01 -0.1567E+01  0.1000E+01
       0.0000E+00  0.0000E+00  0.2567E+01  0.0000E+00 -0.6793E+00

minimum value of ds:-0.15666666E+01  variable out of basis:    5
result of pivoting on element:    4    5


 x 7:  0.0000E+00  0.2000E+00 -0.1000E+01  0.0000E+00  0.0000E+00  0.2000E+02
       0.1000E+01  0.0000E+00  0.0000E+00 -0.2000E+02  0.1500E+00
 x 8:  0.0000E+00  0.1500E+00  0.0000E+00 -0.1000E+01  0.0000E+00  0.2500E+01
       0.0000E+00  0.1000E+01  0.0000E+00 -0.2500E+01  0.7500E-01
 x 1:  0.1000E+01  0.5000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.2500E+02
       0.0000E+00  0.0000E+00  0.0000E+00  0.2500E+02  0.7500E+00
 x 5:  0.0000E+00  0.1200E+00  0.0000E+00  0.0000E+00  0.1000E+01 -0.1500E+02
       0.0000E+00  0.0000E+00 -0.1000E+01  0.1500E+02  0.2900E+00
  -f:  0.0000E+00  0.1500E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.2500E+02
       0.0000E+00  0.0000E+00  0.0000E+00 -0.2500E+02 -0.7500E+00
  -w:  0.0000E+00 -0.3500E+00  0.1000E+01  0.1000E+01  0.0000E+00 -0.2250E+02
       0.0000E+00  0.0000E+00  0.1000E+01  0.2350E+02 -0.2250E+00

minimum value of ds:-0.22500000E+02  variable out of basis:    6
result of pivoting on element:    1    6


 x 6:  0.0000E+00  0.1000E-01 -0.5000E-01  0.0000E+00  0.0000E+00  0.1000E+01
       0.5000E-01  0.0000E+00  0.0000E+00 -0.1000E+01  0.7500E-02
 x 8:  0.0000E+00  0.1250E+00  0.1250E+00 -0.1000E+01  0.0000E+00  0.0000E+00
      -0.1250E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.5625E-01
 x 1:  0.1000E+01  0.7500E+00 -0.1250E+01  0.0000E+00  0.0000E+00  0.0000E+00
       0.1250E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.9375E+00
 x 5:  0.0000E+00  0.2700E+00 -0.7500E+00  0.0000E+00  0.1000E+01  0.0000E+00
       0.7500E+00  0.0000E+00 -0.1000E+01  0.0000E+00  0.4025E+00
  -f:  0.0000E+00  0.1250E+01  0.1250E+01  0.0000E+00  0.0000E+00  0.0000E+00
      -0.1250E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.9375E+00
  -w:  0.0000E+00 -0.1250E+00 -0.1250E+00  0.1000E+01  0.0000E+00  0.0000E+00
```

```
        0.1125E+01  0.0000E+00  0.1000E+01  0.1000E+01 -0.5625E-01

minimum value of ds:-0.12500000E+00  variable out of basis:     2
result of pivoting on element:     2     2


  x 6:  0.0000E+00  0.0000E+00 -0.6000E-01  0.8000E-01  0.0000E+00  0.1000E+01
        0.6000E-01 -0.8000E-01  0.0000E+00 -0.1000E+01  0.3000E-02
  x 2:  0.0000E+00  0.1000E+01  0.1000E+01 -0.8000E+01  0.0000E+00  0.0000E+00
       -0.1000E+01  0.8000E+01  0.0000E+00  0.0000E+00  0.4500E+00
  x 1:  0.1000E+01  0.0000E+00 -0.2000E+01  0.6000E+01  0.0000E+00  0.0000E+00
        0.2000E+01 -0.6000E+01  0.0000E+00  0.0000E+00  0.6000E+00
  x 5:  0.0000E+00  0.0000E+00 -0.1020E+01  0.2160E+01  0.1000E+01  0.0000E+00
        0.1020E+01 -0.2160E+01 -0.1000E+01  0.0000E+00  0.2810E+00
   -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+02  0.0000E+00  0.0000E+00
        0.0000E+00 -0.1000E+02  0.0000E+00  0.0000E+00 -0.1500E+01
   -w:  0.0000E+00  0.0000E+00  0.1192E-06  0.0000E+00  0.0000E+00  0.0000E+00
        0.1000E+01  0.1000E+01  0.1000E+01  0.1000E+01 -0.8941E-07

PHASE-II OF SIMPLEX METHOD
system at start of phase-II

  x 6:  0.0000E+00  0.0000E+00 -0.6000E-01  0.8000E-01  0.0000E+00  0.1000E+01
        0.3000E-02
  x 2:  0.0000E+00  0.1000E+01  0.1000E+01 -0.8000E+01  0.0000E+00  0.0000E+00
        0.4500E+00
  x 1:  0.1000E+01  0.0000E+00 -0.2000E+01  0.6000E+01  0.0000E+00  0.0000E+00
        0.6000E+00
  x 5:  0.0000E+00  0.0000E+00 -0.1020E+01  0.2160E+01  0.1000E+01  0.0000E+00
        0.2810E+00
   -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+02  0.0000E+00  0.0000E+00
       -0.1500E+01
```

3.56 From the formulation of the problem 1.24, the LP problem can be stated as follows:

Minimize $f = 500 - 250\,y_1 - 700\,y_2 - 1100\,y_3$
$+ 1800 - 300\,y_4 - 850\,y_5 - 1400\,y_6$
$+ 1600 - 600\,y_7 - 1300\,y_8 - 2000\,y_9$
$+ 1500 - 800\,y_{10} - 1600\,y_{11} - 2400\,y_{12}$

subject to

$$100\,y_1 + 150\,y_2 + 200\,y_3 \le 50$$
$$100\,y_4 + 150\,y_5 + 200\,y_6 \le 150$$
$$100\,y_7 + 150\,y_8 + 200\,y_9 \le 100$$
$$100\,y_{10} + 150\,y_{11} + 200\,y_{12} \le 75$$
$$y_1 + y_4 + y_7 + y_{10} = 1$$
$$y_2 + y_5 + y_8 + y_{11} = 1$$
$$y_3 + y_6 + y_9 + y_{12} = 1$$

$0 \le y_i \le 1$, $i = 1, 2, \ldots, 12$

Solution:

$f_{min} = 5400 - 3575 = \$\ 1825$

$\vec{Y}^{*T} = \{ y_1, y_2, \ldots, y_{12} \}^*$

$= \{0.5, 0.0, 0.0, 0.5,$
$0.666667, 0.0, 0.0, 0.333333,$
$0.25, 0.0, 0.0, 0.75 \}$

3.57

$$x_1 + x_2 + x_3 = 40$$
$$5x_1 - 4x_2 + x_4 = 0$$
$$x_1 + x_5 = 24$$
$$x_2 + x_6 = 20$$
$$-\frac{2}{3}x_1 - 0.6x_2 = f$$

given system

x3:	1.000	1.000	1.000	0.000	0.000	0.000	40.000
x4:	5.000	-4.000	0.000	1.000	0.000	0.000	0.000
x5:	1.000	0.000	0.000	0.000	1.000	0.000	24.000
x6:	0.000	1.000	0.000	0.000	0.000	1.000	20.000
-f:	-0.667	-0.600	0.000	0.000	0.000	0.000	0.000

minimum value of c1:-0.66666698E+00 variable out of basis: 1
result of pivoting on element: 2 1

x3:	0.000	1.800	1.000	-0.200	0.000	0.000	40.000
x1:	1.000	-0.800	0.000	0.200	0.000	0.000	0.000
x5:	0.000	0.800	0.000	-0.200	1.000	0.000	24.000
x6:	0.000	1.000	0.000	0.000	0.000	1.000	20.000
-f:	0.000	-1.133	0.000	0.133	0.000	0.000	0.000

minimum value of c1:-0.11333337E+01 variable out of basis: 2
result of pivoting on element: 4 2

x3:	0.000	0.000	1.000	-0.200	0.000	-1.800	4.000
x1:	1.000	0.000	0.000	0.200	0.000	0.800	16.000
x5:	0.000	0.000	0.000	-0.200	1.000	-0.800	8.000
x2:	0.000	1.000	0.000	0.000	0.000	1.000	20.000
-f:	0.000	0.000	0.000	0.133	0.000	1.133	22.667

3.58

$$-x_1 + x_2 + x_3 = 1$$
$$2x_1 + x_2 + x_4 = 2$$
$$-2x_1 - 6x_2 = f$$

given system

x3:	-1.000	1.000	1.000	0.000	1.000
x4:	2.000	1.000	0.000	1.000	2.000
-f:	-2.000	-6.000	0.000	0.000	0.000

minimum value of c1:-0.60000000E+01 variable out of basis: 2
result of pivoting on element: 1 2

x2:	-1.000	1.000	1.000	0.000	1.000
x4:	3.000	0.000	-1.000	1.000	1.000
-f:	-8.000	0.000	6.000	0.000	6.000

minimum value of c1:-0.80000000E+01 variable out of basis: 1
result of pivoting on element: 2 1

x2:	0.000	1.000	0.667	0.333	1.333
x1:	1.000	0.000	-0.333	0.333	0.333
-f:	0.000	0.000	3.333	2.667	8.667

3.59 Minimize $f = -6x - 7y$

subject to

$$7x + 6y + z_1 = 42$$
$$5x + 9y + z_2 = 45$$
$$x - y + z_3 = 4$$
$$x \geq 0,\ y \geq 0,\ z_i \geq 0,\ i = 1, 2, 3$$

(z_i are slack variables)

Since the equations are in canonical form, we can take z_1, z_2 and z_3 as basic variables and use Phase II of simplex method to solve the problem.

Basic variables	x	y	z_1	z_2	z_3	b_i''	(b_i''/a_{is}'') for $a_{is}'' > 0$
z_1	7	6	1	0	0	42	7
z_2	5	[9]	0	1	0	45	5 ←
z_3	1	−1	0	0	1	4	
$-f$	−6	−7	0	0	0	0	

↑ most negative c_i''

Result of pivoting:

z_1	[11/3]	0	1	−6/9	0	12	36/11 ←
y	5/9	1	0	1/9	0	5	9
z_3	14/9	0	0	1/9	1	9	81/14
$-f$	−19/9	0	0	7/9	0	35	

↑ most negative c_i''

Result of pivoting:

x	1	0	3/11	−2/11	0	36/11
y	0	1	−5/33	7/33	0	35/11
z_3	0	0	−14/33	13/33	1	43/11
$-f$	0	0	19/33	13/33	1	$41\frac{10}{11}$

All $c_i'' \geq 0$; optimum solution: $x^* = \frac{36}{11}$, $y^* = \frac{35}{11}$, $f^* = -41\frac{10}{11}$, $z_3^* = \frac{43}{11}$.

3.60 Maximize $\tilde{f} = 6x + 7y$

subject to
$$7x + 6y \leq 42$$
$$5x + 9y \leq 45$$
$$x - y \leq 4$$
x and y unrestricted in sign

Let $x = x_1 = x_1' - x_1''$, $y = x_2 = x_2' - x_2''$

and y_1, y_2, y_3 = slack variables.

Problem becomes:

Minimize $f = -6x_1' + 6x_1'' - 7x_2' + 7x_2''$

subject to
$$7x_1' - 7x_1'' + 6x_2' - 6x_2'' + y_1 = 42$$
$$5x_1' - 5x_1'' + 9x_2' - 9x_2'' + y_2 = 45$$
$$x_1' - x_1'' - x_2' + x_2'' + y_3 = 4$$
with all x_i' , x_i'' , $y_i \geq 0$

Simplex method:

Basic variables	variables							b_i''	$\left(\frac{b_i''}{a_{is}''}\right)$ for $a_{is}'' > 0$	
	x_1'	x_1''	x_2'	x_2''	y_1	y_2	y_3			
y_1	7	−7	6	−6	1	0	0	42	7	
y_2	5	−5	(9)	−9	0	1	0	45	5	←
y_3	1	−1	−1	1	0	0	1	4		
$-f$	−6	6	−7	7	0	0	0	0		
			↑							
y_1	(11/3)	−11/3	0	0	1	−2/9	0	12	36/11	←
x_2'	5/9	−5/9	1	−1	0	1/9	0	5	9	
y_3	14/9	−14/9	0	0	0	1/9	1	9	81/14	
$-f$	−19/9	19/9	0	0	0	7/9	0	35		
	↑									

x_1'	1	−1	0	0	$\frac{3}{11}$	$-\frac{2}{11}$	0	$\frac{36}{11}$
x_2'	0	0	1	−1	$-\frac{5}{33}$	$\frac{7}{33}$	0	$\frac{35}{11}$
y_3	0	0	0	0	$-\frac{14}{33}$	$\frac{13}{33}$	1	$\frac{43}{11}$
$-f$	0	0	0	0	$\frac{19}{33}$	$\frac{13}{33}$	0	$\frac{461}{11}$

optimum solution: $x_1' = \frac{36}{11}$, $x_2' = \frac{35}{11}$, $y_3 = \frac{43}{11}$

$x^* = \frac{36}{11}$, $y^* = \frac{35}{11}$, $f^* = f_{min} = -\frac{461}{11}$

3.61 Minimize $f = -19x - 7y$

subject to

$$\begin{aligned} 7x + 6y + z_1 &= 42 \\ 5x + 9y + z_2 &= 45 \\ x - y + z_3 &= 4 \\ -19x - 7y - f &= 0 \end{aligned}$$

(z_i are slack variables)

Basic variables	variables x	y	z_1	z_2	z_3	b_i''	(b_i''/a_{is}'') for $a_{is}'' > 0$
z_1	7	6	1	0	0	42	6
z_2	5	9	0	1	0	45	9
z_3	[1]	−1	0	0	1	4	4 ←
$-f$	−19	−7	0	0	0	0	
Result of pivoting:							
z_1	0	[13]	1	0	−7	14	14/13 ←
z_2	0	14	0	1	−5	25	25/14
x	1	−1	0	0	1	4	
$-f$	0	−26	0	0	19	76	
Result of pivoting:							
y	0	1	1/13	0	−7/13	14/13	
z_2	0	0	−14/13	1	33/13	129/13	
x	1	0	1/13	0	6/13	66/13	
$-f$	0	0	2	0	5	104	

Optimum solution:

$$x^* = 66/13, \quad y^* = 14/13, \quad f^* = -104, \quad z_2^* = 129/13.$$

3.62

Minimize $f = -19x - 7y$

subject to
$$7x + 6y \le 42$$
$$5x + 9y \le 45$$
$$x - y \le 4$$
x and y unrestricted in sign

Defining $x = x_1 - x_2$ and $y = y_1 - y_2$, the problem becomes:

Minimize $f = -19x_1 + 19x_2 - 7y_1 + 7y_2$

subject to
$$7x_1 - 7x_2 + 6y_1 - 6y_2 + z_1 = 42$$
$$5x_1 - 5x_2 + 9y_1 - 9y_2 + z_2 = 45$$
$$x_1 - x_2 - y_1 + y_2 + z_3 = 4$$
all $x_i \ge 0$, $y_i \ge 0$, $z_i \ge 0$

Basic Variables	x_1	x_2	y_1	y_2	z_1	z_2	z_3	b_i''	$\left(\frac{b_i''}{a_{is}''}\right)$ for $a_{is}'' > 0$	
z_1	7	−7	6	−6	1	0	0	42	6	
z_2	5	−5	9	−9	0	1	0	45	9	
z_3	[1]	−1	−1	1	0	0	1	4	4	←
$-f$	−19	19	−7	7	0	0	0	0		
	↑									
z_1	0	0	13	−13	1	0	−7	14	14/13	←
z_2	0	0	14	−14	0	1	−5	25	25/14	
x_1	1	−1	−1	1	0	0	1	4		
$-f$	0	0	−28	28	0	0	19	76		
			↑							
y_1	0	0	1	−1	1/13	0	−7/13	14/13		
z_2	0	0	0	0	−14/13	1	2.5385	9.9231		
x_1	1	−1	0	0	1/13	0	6/13	5.0769		
$-f$	0	0	0	0	2	0	5	104.0		

optimum solution: $x^* = 5.0769$, $y^* = 1.0769$, $f^* = -104.0$

3.63

$$
\begin{aligned}
-x_1 + x_2 + x_3 &= 8 \\
5x_1 - x_2 - x_6 + x_9 &= 0 \\
x_1 + x_2 - x_7 + x_{10} &= 8 \\
-x_1 + 6x_2 - x_8 + x_{11} &= 12 \\
5x_1 + 2x_2 + x_4 &= 68 \\
x_1 + x_5 &= 10
\end{aligned}
$$

objective function:

$$f = -x_1 - 2x_2$$

Infeasibility form:

$$w = x_9 + x_{10} + x_{11}$$

or $$w = -5x_1 - 6x_2 + x_6 + x_7 + x_8 + 20$$

PHASE-I OF SIMPLEX METHOD
given system

x 3:	-0.1000E+01	0.1000E+01	0.1000E+01	0.0000E+00	0.0000E+00	0.0000E+00
	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.8000E+01
x 9:	0.5000E+01	-0.1000E+01	0.0000E+00	0.0000E+00	0.0000E+00	-0.1000E+01
	0.0000E+00	0.0000E+00	0.1000E+01	0.0000E+00	0.0000E+00	0.0000E+00
x10:	0.1000E+01	0.1000E+01	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
	-0.1000E+01	0.0000E+00	0.0000E+00	0.1000E+01	0.0000E+00	0.8000E+01
x11:	-0.1000E+01	0.6000E+01	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
	0.0000E+00	-0.1000E+01	0.0000E+00	0.0000E+00	0.1000E+01	0.1200E+02
x 4:	0.5000E+01	0.2000E+01	0.0000E+00	0.1000E+01	0.0000E+00	0.0000E+00
	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.6800E+02
x 5:	0.1000E+01	0.0000E+00	0.0000E+00	0.0000E+00	0.1000E+01	0.0000E+00
	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.1000E+02
-f:	-0.1000E+01	-0.2000E+01	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
-w:	-0.5000E+01	-0.6000E+01	0.0000E+00	0.0000E+00	0.0000E+00	0.1000E+01
	0.1000E+01	0.1000E+01	0.0000E+00	0.0000E+00	0.0000E+00	-0.2000E+02

```
minimum value of ds:-0.60000000E+01  variable out of basis:     2
result of pivoting on element:     4     2

 x 3: -0.8333E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
       0.0000E+00  0.1667E+00  0.0000E+00  0.0000E+00 -0.1667E+00  0.6000E+01

 x 9:  0.4833E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.1000E+01
       0.0000E+00 -0.1667E+00  0.1000E+01  0.0000E+00  0.1667E+00  0.2000E+01

 x10:  0.1167E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
      -0.1000E+01  0.1667E+00  0.0000E+00  0.1000E+01 -0.1667E+00  0.6000E+01

 x 2: -0.1667E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
       0.0000E+00 -0.1667E+00  0.0000E+00  0.0000E+00  0.1667E+00  0.2000E+01

 x 4:  0.5333E+01  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00
       0.0000E+00  0.3333E+00  0.0000E+00  0.0000E+00 -0.3333E+00  0.6400E+02

 x 5:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00
       0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+02

  -f: -0.1333E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
       0.0000E+00 -0.3333E+00  0.0000E+00  0.0000E+00  0.3333E+00  0.4000E+01

  -w: -0.6000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01
       0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01 -0.8000E+01

minimum value of ds:-0.60000000E+01  variable out of basis:     1
result of pivoting on element:     2     1

 x 3:  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00 -0.1724E+00
       0.0000E+00  0.1379E+00  0.1724E+00  0.0000E+00 -0.1379E+00  0.6345E+01

 x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.2069E+00
       0.0000E+00 -0.3448E-01  0.2069E+00  0.0000E+00  0.3448E-01  0.4138E+00

 x10:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.2414E+00
      -0.1000E+01  0.2069E+00 -0.2414E+00  0.1000E+01 -0.2069E+00  0.5517E+01

 x 2:  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.3448E-01
       0.0000E+00 -0.1724E+00  0.3448E-01  0.0000E+00  0.1724E+00  0.2069E+01

 x 4:  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.1103E+01
       0.0000E+00  0.5172E+00 -0.1103E+01  0.0000E+00 -0.5172E+00  0.6179E+02

 x 5:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.2069E+00
       0.0000E+00  0.3448E-01 -0.2069E+00  0.0000E+00 -0.3448E-01  0.9586E+01

  -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.2759E+00
       0.0000E+00 -0.3793E+00  0.2759E+00  0.0000E+00  0.3793E+00  0.4552E+01

  -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.2414E+00
       0.1000E+01 -0.2069E+00  0.1241E+01  0.0000E+00  0.1207E+01 -0.5517E+01
```

```
minimum value of ds:-0.24137926E+00  variable out of basis:     6
result of pivoting on element:     3     6

 x 3:  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
      -0.7143E+00  0.2857E+00  0.0000E+00  0.7143E+00 -0.2857E+00  0.1029E+02

 x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
      -0.8571E+00  0.1429E+00  0.0000E+00  0.8571E+00 -0.1429E+00  0.5143E+01

 x 6:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01
      -0.4143E+01  0.8571E+00 -0.1000E+01  0.4143E+01 -0.8571E+00  0.2286E+02

 x 2:  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
      -0.1429E+00 -0.1429E+00  0.0000E+00  0.1429E+00  0.1429E+00  0.2857E+01

 x 4:  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00
       0.4571E+01 -0.4286E+00  0.0000E+00 -0.4571E+01  0.4286E+00  0.3657E+02

 x 5:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00
       0.8571E+00 -0.1429E+00  0.0000E+00 -0.8571E+00  0.1429E+00  0.4857E+01

  -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
      -0.1143E+01 -0.1429E+00  0.0000E+00  0.1143E+01  0.1429E+00  0.1086E+02

  -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
       0.1192E-06 -0.1490E-07  0.1000E+01  0.1000E+01  0.1000E+01 -0.4768E-06

minimum value of ds:-0.14901161E-07  variable out of basis:     8
result of pivoting on element:     3     8

 x 3:  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00 -0.3333E+00
       0.6667E+00  0.0000E+00  0.3333E+00 -0.6667E+00  0.0000E+00  0.2667E+01

 x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.1667E+00
      -0.1667E+00  0.0000E+00  0.1667E+00  0.1667E+00  0.0000E+00  0.1333E+01

 x 8:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1167E+01
      -0.4833E+01  0.1000E+01 -0.1167E+01  0.4833E+01 -0.1000E+01  0.2667E+02

 x 2:  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.1667E+00
      -0.8333E+00  0.0000E+00 -0.1667E+00  0.8333E+00  0.0000E+00  0.6667E+01

 x 4:  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.5000E+00
       0.2500E+01  0.0000E+00 -0.5000E+00 -0.2500E+01  0.0000E+00  0.4800E+02

 x 5:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.1667E+00
       0.1667E+00  0.0000E+00 -0.1667E+00 -0.1667E+00  0.0000E+00  0.8667E+01

  -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1667E+00
      -0.1833E+01  0.0000E+00 -0.1667E+00  0.1833E+01  0.0000E+00  0.1467E+02

  -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1738E-07
       0.4719E-07  0.0000E+00  0.1000E+01  0.1000E+01  0.1000E+01 -0.7947E-07
```

```
PHASE-II OF SIMPLEX METHOD
system at start of phase-II

 x 3:  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00 -0.3333E+00
       0.6667E+00  0.0000E+00  0.2667E+01
 x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.1667E+00
      -0.1667E+00  0.0000E+00  0.1333E+01
 x 8:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1167E+01
      -0.4833E+01  0.1000E+01  0.2667E+02
 x 2:  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.1667E+00
      -0.8333E+00  0.0000E+00  0.6667E+01
 x 4:  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.5000E+00
       0.2500E+01  0.0000E+00  0.4800E+02
 x 5:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.1667E+00
       0.1667E+00  0.0000E+00  0.8667E+01
  -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1667E+00
      -0.1833E+01  0.0000E+00  0.1467E+02

minimum value of cs:-0.18333335E+01  variable out of basis:     7
result of pivoting on element:     1     7


 x 7:  0.0000E+00  0.0000E+00  0.1500E+01  0.0000E+00  0.0000E+00 -0.5000E+00
       0.1000E+01  0.0000E+00  0.4000E+01
 x 1:  0.1000E+01  0.0000E+00  0.2500E+00  0.0000E+00  0.0000E+00 -0.2500E+00
       0.0000E+00  0.0000E+00  0.2000E+01
 x 8:  0.0000E+00  0.0000E+00  0.7250E+01  0.0000E+00  0.0000E+00 -0.1250E+01
       0.0000E+00  0.1000E+01  0.4600E+02
 x 2:  0.0000E+00  0.1000E+01  0.1250E+01  0.0000E+00  0.0000E+00 -0.2500E+00
       0.0000E+00  0.0000E+00  0.1000E+02
 x 4:  0.0000E+00  0.0000E+00 -0.3750E+01  0.1000E+01  0.0000E+00  0.1750E+01
       0.0000E+00  0.0000E+00  0.3800E+02
 x 5:  0.0000E+00  0.0000E+00 -0.2500E+00  0.0000E+00  0.1000E+01  0.2500E+00
       0.0000E+00  0.0000E+00  0.8000E+01
  -f:  0.0000E+00  0.0000E+00  0.2750E+01  0.0000E+00  0.0000E+00 -0.7500E+00
       0.0000E+00  0.0000E+00  0.2200E+02

minimum value of cs:-0.75000006E+00  variable out of basis:     6
result of pivoting on element:     5     6


 x 7:  0.0000E+00  0.0000E+00  0.4286E+00  0.2857E+00  0.0000E+00  0.0000E+00
       0.1000E+01  0.0000E+00  0.1486E+02
 x 1:  0.1000E+01  0.0000E+00 -0.2857E+00  0.1429E+00  0.0000E+00  0.0000E+00
       0.0000E+00  0.0000E+00  0.7429E+01
 x 8:  0.0000E+00  0.0000E+00  0.4571E+01  0.7143E+00  0.0000E+00  0.0000E+00
       0.0000E+00  0.1000E+01  0.7314E+02
 x 2:  0.0000E+00  0.1000E+01  0.7143E+00  0.1429E+00  0.0000E+00  0.0000E+00
       0.0000E+00  0.0000E+00  0.1543E+02
 x 6:  0.0000E+00  0.0000E+00 -0.2143E+01  0.5714E+00  0.0000E+00  0.1000E+01
       0.0000E+00  0.0000E+00  0.2171E+02
 x 5:  0.0000E+00  0.0000E+00  0.2857E+00 -0.1429E+00  0.1000E+01  0.0000E+00
       0.0000E+00  0.0000E+00  0.2571E+01
  -f:  0.0000E+00  0.0000E+00  0.1143E+01  0.4286E+00  0.0000E+00  0.0000E+00
       0.0000E+00  0.0000E+00  0.3829E+02
```

3.64 Constraints Infeasibility form } same as in Problem 3.63

objective function: $f = x_1 - x_2$

```
PHASE-I OF SIMPLEX METHOD
given system

 x 3: -0.1000E+01  0.1000E+01  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
       0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.8000E+01

 x 9:  0.5000E+01 -0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.1000E+01
       0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00

x10:  0.1000E+01  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
     -0.1000E+01  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.8000E+01

x11: -0.1000E+01  0.6000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
       0.0000E+00 -0.1000E+01  0.0000E+00  0.0000E+00  0.1000E+01  0.1200E+02

 x 4:  0.5000E+01  0.2000E+01  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00
       0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.6800E+02

 x 5:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00
       0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+02

  -f:  0.1000E+01 -0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
       0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00

  -w: -0.5000E+01 -0.6000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01
       0.1000E+01  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.2000E+02

minimum value of ds:-0.60000000E+01  variable out of basis:     2
result of pivoting on element:     4     2

 x 3: -0.8333E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
       0.0000E+00  0.1667E+00  0.0000E+00  0.0000E+00 -0.1667E+00  0.6000E+01

 x 9:  0.4833E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.1000E+01
       0.0000E+00 -0.1667E+00  0.1000E+01  0.0000E+00  0.1667E+00  0.2000E+01

x10:  0.1167E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
     -0.1000E+01  0.1667E+00  0.0000E+00  0.1000E+01 -0.1667E+00  0.6000E+01

 x 2: -0.1667E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
       0.0000E+00 -0.1667E+00  0.0000E+00  0.0000E+00  0.1667E+00  0.2000E+01

 x 4:  0.5333E+01  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00
       0.0000E+00  0.3333E+00  0.0000E+00  0.0000E+00 -0.3333E+00  0.6400E+02

 x 5:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00
       0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+02

  -f:  0.8333E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
       0.0000E+00 -0.1667E+00  0.0000E+00  0.0000E+00  0.1667E+00  0.2000E+01

  -w: -0.6000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01
       0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01 -0.8000E+01
```

```
minimum value of ds:-0.60000000E+01  variable out of basis:     1
result of pivoting on element:     2     1

  x 3:  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00 -0.1724E+00
        0.0000E+00  0.1379E+00  0.1724E+00  0.0000E+00 -0.1379E+00  0.6345E+01

  x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.2069E+00
        0.0000E+00 -0.3448E-01  0.2069E+00  0.0000E+00  0.3448E-01  0.4138E+00

  x10:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.2414E+00
       -0.1000E+01  0.2069E+00 -0.2414E+00  0.1000E+01 -0.2069E+00  0.5517E+01

  x 2:  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.3448E-01
        0.0000E+00 -0.1724E+00  0.3448E-01  0.0000E+00  0.1724E+00  0.2069E+01

  x 4:  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.1103E+01
        0.0000E+00  0.5172E+00 -0.1103E+01  0.0000E+00 -0.5172E+00  0.6179E+02

  x 5:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.2069E+00
        0.0000E+00  0.3448E-01 -0.2069E+00  0.0000E+00 -0.3448E-01  0.9586E+01

   -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1724E+00
        0.0000E+00 -0.1379E+00 -0.1724E+00  0.0000E+00  0.1379E+00  0.1655E+01

   -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.2414E+00
        0.1000E+01 -0.2069E+00  0.1241E+01  0.0000E+00  0.1207E+01 -0.5517E+01

minimum value of ds:-0.24137926E+00  variable out of basis:     6
result of pivoting on element:     3     6

  x 3:  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
       -0.7143E+00  0.2857E+00  0.0000E+00  0.7143E+00 -0.2857E+00  0.1029E+02

  x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
       -0.8571E+00  0.1429E+00  0.0000E+00  0.8571E+00 -0.1429E+00  0.5143E+01

  x 6:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01
       -0.4143E+01  0.8571E+00 -0.1000E+01  0.4143E+01 -0.8571E+00  0.2286E+02

  x 2:  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
       -0.1429E+00 -0.1429E+00  0.0000E+00  0.1429E+00  0.1429E+00  0.2857E+01

  x 4:  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00
        0.4571E+01 -0.4286E+00  0.0000E+00 -0.4571E+01  0.4286E+00  0.3657E+02

  x 5:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00
        0.8571E+00 -0.1429E+00  0.0000E+00 -0.8571E+00  0.1429E+00  0.4857E+01

   -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.7143E+00 -0.2857E+00  0.0000E+00 -0.7143E+00  0.2857E+00 -0.2286E+01

   -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.1192E-06 -0.1490E-07  0.1000E+01  0.1000E+01  0.1000E+01 -0.4768E-06
```

```
minimum value of ds:-0.14901161E-07  variable out of basis:     8
result of pivoting on element:     3     8

  x 3:  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00 -0.3333E+00
        0.6667E+00  0.0000E+00  0.3333E+00 -0.6667E+00  0.0000E+00  0.2667E+01

  x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.1667E+00
       -0.1667E+00  0.0000E+00  0.1667E+00  0.1667E+00  0.0000E+00  0.1333E+01

  x 8:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1167E+01
       -0.4833E+01  0.1000E+01 -0.1167E+01  0.4833E+01 -0.1000E+01  0.2667E+02

  x 2:  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.1667E+00
       -0.8333E+00  0.0000E+00 -0.1667E+00  0.8333E+00  0.0000E+00  0.6667E+01

  x 4:  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.5000E+00
        0.2500E+01  0.0000E+00 -0.5000E+00 -0.2500E+01  0.0000E+00  0.4800E+02

  x 5:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.1667E+00
        0.1667E+00  0.0000E+00 -0.1667E+00 -0.1667E+00  0.0000E+00  0.8667E+01

   -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.3333E+00
       -0.6667E+00  0.0000E+00 -0.3333E+00  0.6667E+00  0.0000E+00  0.5333E+01

   -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1738E-07
        0.4719E-07  0.0000E+00  0.1000E+01  0.1000E+01  0.1000E+01 -0.7947E-07


PHASE-II OF SIMPLEX METHOD
system at start of phase-II

  x 3:  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00 -0.3333E+00
        0.6667E+00  0.0000E+00  0.2667E+01
  x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.1667E+00
       -0.1667E+00  0.0000E+00  0.1333E+01
  x 8:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1167E+01
       -0.4833E+01  0.1000E+01  0.2667E+02
  x 2:  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.1667E+00
       -0.8333E+00  0.0000E+00  0.6667E+01
  x 4:  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.5000E+00
        0.2500E+01  0.0000E+00  0.4800E+02
  x 5:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.1667E+00
        0.1667E+00  0.0000E+00  0.8667E+01
   -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.3333E+00
       -0.6667E+00  0.0000E+00  0.5333E+01

minimum value of cs:-0.66666663E+00  variable out of basis:     7
result of pivoting on element:     1     7

  x 7:  0.0000E+00  0.0000E+00  0.1500E+01  0.0000E+00  0.0000E+00 -0.5000E+00
        0.1000E+01  0.0000E+00  0.4000E+01
  x 1:  0.1000E+01  0.0000E+00  0.2500E+00  0.0000E+00  0.0000E+00 -0.2500E+00
        0.0000E+00  0.0000E+00  0.2000E+01
  x 8:  0.0000E+00  0.0000E+00  0.7250E+01  0.0000E+00  0.0000E+00 -0.1250E+01
        0.0000E+00  0.1000E+01  0.4600E+02
  x 2:  0.0000E+00  0.1000E+01  0.1250E+01  0.0000E+00  0.0000E+00 -0.2500E+00
        0.0000E+00  0.0000E+00  0.1000E+02
  x 4:  0.0000E+00  0.0000E+00 -0.3750E+01  0.1000E+01  0.0000E+00  0.1750E+01
        0.0000E+00  0.0000E+00  0.3800E+02
  x 5:  0.0000E+00  0.0000E+00 -0.2500E+00  0.0000E+00  0.1000E+01  0.2500E+00
        0.0000E+00  0.0000E+00  0.8000E+01
   -f:  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00  0.8000E+01
```

3.65

$$
\begin{aligned}
-x_1 + x_2 + x_3 &= 8 \\
-5x_1 + x_2 + x_4 &= 0 \\
x_1 + x_2 - x_6 + x_9 &= 8 \\
-x_1 + 6x_2 - x_7 + x_{10} &= 12 \\
5x_1 + 2x_2 - x_8 + x_{11} &= 68 \\
x_1 + x_5 &= 10 \\
-x_1 - 2x_2 &= f \\
x_9 + x_{10} + x_{11} &= w \\
\text{or } -5x_1 - 9x_2 + x_6 + x_7 + x_8 + 88 &= w
\end{aligned}
$$

PHASE-I OF SIMPLEX METHOD
given system

x 3:	-0.1000E+01	0.1000E+01	0.1000E+01	0.0000E+00	0.0000E+00	0.0000E+00
	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.8000E+01
x 4:	-0.5000E+01	0.1000E+01	0.0000E+00	0.1000E+01	0.0000E+00	0.0000E+00
	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
x 9:	0.1000E+01	0.1000E+01	0.0000E+00	0.0000E+00	0.0000E+00	-0.1000E+01
	0.0000E+00	0.0000E+00	0.1000E+01	0.0000E+00	0.0000E+00	0.8000E+01
x10:	-0.1000E+01	0.6000E+01	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
	-0.1000E+01	0.0000E+00	0.0000E+00	0.1000E+01	0.0000E+00	0.1200E+02
x11:	0.5000E+01	0.2000E+01	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
	0.0000E+00	-0.1000E+01	0.0000E+00	0.0000E+00	0.1000E+01	0.6800E+02
x 5:	0.1000E+01	0.0000E+00	0.0000E+00	0.0000E+00	0.1000E+01	0.0000E+00
	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.1000E+02
-f:	-0.1000E+01	-0.2000E+01	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
-w:	-0.5000E+01	-0.9000E+01	0.0000E+00	0.0000E+00	0.0000E+00	0.1000E+01
	0.1000E+01	0.1000E+01	0.0000E+00	0.0000E+00	0.0000E+00	-0.8800E+02

minimum value of ds:-0.90000000E+01 variable out of basis: 2
result of pivoting on element: 2 2

x 3:	0.4000E+01	0.0000E+00	0.1000E+01	-0.1000E+01	0.0000E+00	0.0000E+00
	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.8000E+01
x 2:	-0.5000E+01	0.1000E+01	0.0000E+00	0.1000E+01	0.0000E+00	0.0000E+00
	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
x 9:	0.6000E+01	0.0000E+00	0.0000E+00	-0.1000E+01	0.0000E+00	-0.1000E+01
	0.0000E+00	0.0000E+00	0.1000E+01	0.0000E+00	0.0000E+00	0.8000E+01

```
x10:  0.2900E+02  0.0000E+00  0.0000E+00 -0.6000E+01  0.0000E+00  0.0000E+00
     -0.1000E+01  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.1200E+02

x11:  0.1500E+02  0.0000E+00  0.0000E+00 -0.2000E+01  0.0000E+00  0.0000E+00
      0.0000E+00 -0.1000E+01  0.0000E+00  0.0000E+00  0.1000E+01  0.6800E+02

x 5:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00
      0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+02

 -f: -0.1100E+02  0.0000E+00  0.0000E+00  0.2000E+01  0.0000E+00  0.0000E+00
      0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00

 -w: -0.5000E+02  0.0000E+00  0.0000E+00  0.9000E+01  0.0000E+00  0.1000E+01
      0.1000E+01  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.8800E+02

minimum value of ds:-0.50000000E+02  variable out of basis:     1
result of pivoting on element:     4     1

x 3:  0.0000E+00  0.0000E+00  0.1000E+01 -0.1724E+00  0.0000E+00  0.0000E+00
      0.1379E+00  0.0000E+00  0.0000E+00 -0.1379E+00  0.0000E+00  0.6345E+01

x 2:  0.0000E+00  0.1000E+01  0.0000E+00 -0.3448E-01  0.0000E+00  0.0000E+00
     -0.1724E+00  0.0000E+00  0.0000E+00  0.1724E+00  0.0000E+00  0.2069E+01

x 9:  0.0000E+00  0.0000E+00  0.0000E+00  0.2414E+00  0.0000E+00 -0.1000E+01
      0.2069E+00  0.0000E+00  0.1000E+01 -0.2069E+00  0.0000E+00  0.5517E+01

x 1:  0.1000E+01  0.0000E+00  0.0000E+00 -0.2069E+00  0.0000E+00  0.0000E+00
     -0.3448E-01  0.0000E+00  0.0000E+00  0.3448E-01  0.0000E+00  0.4138E+00

x11:  0.0000E+00  0.0000E+00  0.0000E+00  0.1103E+01  0.0000E+00  0.0000E+00
      0.5172E+00 -0.1000E+01  0.0000E+00 -0.5172E+00  0.1000E+01  0.6179E+02

x 5:  0.0000E+00  0.0000E+00  0.0000E+00  0.2069E+00  0.1000E+01  0.0000E+00
      0.3448E-01  0.0000E+00  0.0000E+00 -0.3448E-01  0.0000E+00  0.9586E+01

 -f:  0.0000E+00  0.0000E+00  0.0000E+00 -0.2759E+00  0.0000E+00  0.0000E+00
     -0.3793E+00  0.0000E+00  0.0000E+00  0.3793E+00  0.0000E+00  0.4552E+01

 -w:  0.0000E+00  0.0000E+00  0.0000E+00 -0.1345E+01  0.0000E+00  0.1000E+01
     -0.7241E+00  0.1000E+01  0.0000E+00  0.1724E+01  0.0000E+00 -0.6731E+02

minimum value of ds:-0.13448277E+01  variable out of basis:     4
result of pivoting on element:     3     4

x 3:  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00 -0.7143E+00
      0.2857E+00  0.0000E+00  0.7143E+00 -0.2857E+00  0.0000E+00  0.1029E+02

x 2:  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.1429E+00
     -0.1429E+00  0.0000E+00  0.1429E+00  0.1429E+00  0.0000E+00  0.2857E+01

x 4:  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00 -0.4143E+01
      0.8571E+00  0.0000E+00  0.4143E+01 -0.8571E+00  0.0000E+00  0.2286E+02

x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.8571E+00
      0.1429E+00  0.0000E+00  0.8571E+00 -0.1429E+00  0.0000E+00  0.5143E+01

x11:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.4571E+01
     -0.4286E+00 -0.1000E+01 -0.4571E+01  0.4286E+00  0.1000E+01  0.3657E+02
```

```
 x 5:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.8571E+00
      -0.1429E+00  0.0000E+00 -0.8571E+00  0.1429E+00  0.0000E+00  0.4857E+01

  -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.1143E+01
      -0.1429E+00  0.0000E+00  0.1143E+01  0.1429E+00  0.0000E+00  0.1086E+02

  -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.4571E+01
       0.4286E+00  0.1000E+01  0.5571E+01  0.5714E+00  0.0000E+00 -0.3657E+02

minimum value of ds:-0.45714273E+01  variable out of basis:     6
result of pivoting on element:     6     6

 x 3:  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.8333E+00  0.0000E+00
       0.1667E+00  0.0000E+00  0.0000E+00 -0.1667E+00  0.0000E+00  0.1433E+02

 x 2:  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.1667E+00  0.0000E+00
      -0.1667E+00  0.0000E+00  0.0000E+00  0.1667E+00  0.0000E+00  0.3667E+01

 x 4:  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.4833E+01  0.0000E+00
       0.1667E+00  0.0000E+00  0.0000E+00 -0.1667E+00  0.0000E+00  0.4633E+02

 x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00
       0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+02

x11:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.5333E+01  0.0000E+00
       0.3333E+00 -0.1000E+01  0.0000E+00 -0.3333E+00  0.1000E+01  0.1067E+02

 x 6:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1167E+01  0.1000E+01
      -0.1667E+00  0.0000E+00 -0.1000E+01  0.1667E+00  0.0000E+00  0.5667E+01

  -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1333E+01  0.0000E+00
      -0.3333E+00  0.0000E+00  0.0000E+00  0.3333E+00  0.0000E+00  0.1733E+02

  -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.5333E+01  0.0000E+00
      -0.3333E+00  0.1000E+01  0.1000E+01  0.1333E+01  0.0000E+00 -0.1067E+02

minimum value of ds:-0.33333331E+00  variable out of basis:     7
result of pivoting on element:     5     7

 x 3:  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.3500E+01  0.0000E+00
       0.0000E+00  0.5000E+00  0.0000E+00  0.0000E+00 -0.5000E+00  0.9000E+01

 x 2:  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00 -0.2500E+01  0.0000E+00
       0.0000E+00 -0.5000E+00  0.0000E+00  0.0000E+00  0.5000E+00  0.9000E+01

 x 4:  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.7500E+01  0.0000E+00
       0.0000E+00  0.5000E+00  0.0000E+00  0.0000E+00 -0.5000E+00  0.4100E+02

 x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00
       0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+02

 x 7:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.1600E+02  0.0000E+00
       0.1000E+01 -0.3000E+01  0.0000E+00 -0.1000E+01  0.3000E+01  0.3200E+02

 x 6:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.1500E+01  0.1000E+01
       0.0000E+00 -0.5000E+00 -0.1000E+01  0.0000E+00  0.5000E+00  0.1100E+02

  -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.4000E+01  0.0000E+00
       0.0000E+00 -0.1000E+01  0.0000E+00  0.0000E+00  0.1000E+01  0.2800E+02

  -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.4768E-06  0.0000E+00
       0.0000E+00  0.0000E+00  0.1000E+01  0.1000E+01  0.1000E+01 -0.1144E-04
```

```
minimum value of ds:-0.47683716E-06  variable out of basis:     5
result of pivoting on element:     1     5

 x 5:  0.0000E+00  0.0000E+00  0.2857E+00  0.0000E+00  0.1000E+01  0.0000E+00
       0.0000E+00  0.1429E+00  0.0000E+00  0.0000E+00 -0.1429E+00  0.2571E+01

 x 2:  0.0000E+00  0.1000E+01  0.7143E+00  0.0000E+00  0.0000E+00  0.0000E+00
       0.0000E+00 -0.1429E+00  0.0000E+00  0.0000E+00  0.1429E+00  0.1543E+02

 x 4:  0.0000E+00  0.0000E+00 -0.2143E+01  0.1000E+01  0.0000E+00  0.0000E+00
       0.0000E+00 -0.5714E+00  0.0000E+00  0.0000E+00  0.5714E+00  0.2171E+02

 x 1:  0.1000E+01  0.0000E+00 -0.2857E+00  0.0000E+00  0.0000E+00  0.0000E+00
       0.0000E+00 -0.1429E+00  0.0000E+00  0.0000E+00  0.1429E+00  0.7429E+01

 x 7:  0.0000E+00  0.0000E+00  0.4571E+01  0.0000E+00  0.0000E+00  0.0000E+00
       0.1000E+01 -0.7143E+00  0.0000E+00 -0.1000E+01  0.7143E+00  0.7314E+02

 x 6:  0.0000E+00  0.0000E+00  0.4286E+00  0.0000E+00  0.0000E+00  0.1000E+01
       0.0000E+00 -0.2857E+00 -0.1000E+01  0.0000E+00  0.2857E+00  0.1486E+02

  -f:  0.0000E+00  0.0000E+00  0.1143E+01  0.0000E+00  0.0000E+00  0.0000E+00
       0.0000E+00 -0.4286E+00  0.0000E+00  0.0000E+00  0.4286E+00  0.3829E+02

  -w:  0.0000E+00  0.0000E+00  0.1362E-06  0.0000E+00  0.0000E+00  0.0000E+00
       0.0000E+00  0.6812E-07  0.1000E+01  0.1000E+01  0.1000E+01 -0.1022E-04


PHASE-II OF SIMPLEX METHOD
system at start of phase-II

 x 5:  0.0000E+00  0.0000E+00  0.2857E+00  0.0000E+00  0.1000E+01  0.0000E+00
       0.0000E+00  0.1429E+00  0.2571E+01
 x 2:  0.0000E+00  0.1000E+01  0.7143E+00  0.0000E+00  0.0000E+00  0.0000E+00
       0.0000E+00 -0.1429E+00  0.1543E+02
 x 4:  0.0000E+00  0.0000E+00 -0.2143E+01  0.1000E+01  0.0000E+00  0.0000E+00
       0.0000E+00 -0.5714E+00  0.2171E+02
 x 1:  0.1000E+01  0.0000E+00 -0.2857E+00  0.0000E+00  0.0000E+00  0.0000E+00
       0.0000E+00 -0.1429E+00  0.7429E+01
 x 7:  0.0000E+00  0.0000E+00  0.4571E+01  0.0000E+00  0.0000E+00  0.0000E+00
       0.1000E+01 -0.7143E+00  0.7314E+02
 x 6:  0.0000E+00  0.0000E+00  0.4286E+00  0.0000E+00  0.0000E+00  0.1000E+01
       0.0000E+00 -0.2857E+00  0.1486E+02
  -f:  0.0000E+00  0.0000E+00  0.1143E+01  0.0000E+00  0.0000E+00  0.0000E+00
       0.0000E+00 -0.4286E+00  0.3829E+02

minimum value of cs:-0.42857146E+00  variable out of basis:     8
result of pivoting on element:     1     8

 x 8:  0.0000E+00  0.0000E+00  0.2000E+01  0.0000E+00  0.7000E+01  0.0000E+00
       0.0000E+00  0.1000E+01  0.1800E+02
 x 2:  0.0000E+00  0.1000E+01  0.1000E+01  0.0000E+00  0.1000E+01  0.0000E+00
       0.0000E+00  0.0000E+00  0.1800E+02
 x 4:  0.0000E+00  0.0000E+00 -0.1000E+01  0.1000E+01  0.4000E+01  0.0000E+00
       0.0000E+00  0.0000E+00  0.3200E+02
 x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00
       0.0000E+00  0.0000E+00  0.1000E+02
 x 7:  0.0000E+00  0.0000E+00  0.6000E+01  0.0000E+00  0.5000E+01  0.0000E+00
       0.1000E+01  0.0000E+00  0.8600E+02
 x 6:  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.2000E+01  0.1000E+01
       0.0000E+00  0.0000E+00  0.2000E+02
  -f:  0.0000E+00  0.0000E+00  0.2000E+01  0.0000E+00  0.3000E+01  0.0000E+00
       0.0000E+00  0.0000E+00  0.4600E+02
```

3.66 constraints, infeasibility form } same as in Problem 3.65

objective function: $f = x_1 - x_2$

```
PHASE-I OF SIMPLEX METHOD
given system

 x 3: -0.1000E+01  0.1000E+01  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
       0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.8000E+01

 x 4: -0.5000E+01  0.1000E+01  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00
       0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00

 x 9:  0.1000E+01  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.1000E+01
       0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.8000E+01

 x10: -0.1000E+01  0.6000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
      -0.1000E+01  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.1200E+02

 x11:  0.5000E+01  0.2000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
       0.0000E+00 -0.1000E+01  0.0000E+00  0.0000E+00  0.1000E+01  0.6800E+02

 x 5:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00
       0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+02

  -f:  0.1000E+01 -0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
       0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00

  -w: -0.5000E+01 -0.9000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01
       0.1000E+01  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.8800E+02

minimum value of ds:-0.90000000E+01  variable out of basis:     2
result of pivoting on element:     2     2

 x 3:  0.4000E+01  0.0000E+00  0.1000E+01 -0.1000E+01  0.0000E+00  0.0000E+00
       0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.8000E+01

 x 2: -0.5000E+01  0.1000E+01  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00
       0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00

 x 9:  0.6000E+01  0.0000E+00  0.0000E+00 -0.1000E+01  0.0000E+00 -0.1000E+01
       0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.8000E+01

 x10:  0.2900E+02  0.0000E+00  0.0000E+00 -0.6000E+01  0.0000E+00  0.0000E+00
      -0.1000E+01  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.1200E+02

 x11:  0.1500E+02  0.0000E+00  0.0000E+00 -0.2000E+01  0.0000E+00  0.0000E+00
       0.0000E+00 -0.1000E+01  0.0000E+00  0.0000E+00  0.1000E+01  0.6800E+02

 x 5:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00
       0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+02

  -f: -0.4000E+01  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00
       0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00

  -w: -0.5000E+02  0.0000E+00  0.0000E+00  0.9000E+01  0.0000E+00  0.1000E+01
       0.1000E+01  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.8800E+02
```

```
minimum value of ds:-0.50000000E+02  variable out of basis:     1
result of pivoting on element:     4     1

  x 3:  0.0000E+00  0.0000E+00  0.1000E+01 -0.1724E+00  0.0000E+00  0.0000E+00
        0.1379E+00  0.0000E+00  0.0000E+00 -0.1379E+00  0.0000E+00  0.6345E+01

  x 2:  0.0000E+00  0.1000E+01  0.0000E+00 -0.3448E-01  0.0000E+00  0.0000E+00
       -0.1724E+00  0.0000E+00  0.0000E+00  0.1724E+00  0.0000E+00  0.2069E+01

  x 9:  0.0000E+00  0.0000E+00  0.0000E+00  0.2414E+00  0.0000E+00 -0.1000E+01
        0.2069E+00  0.0000E+00  0.1000E+01 -0.2069E+00  0.0000E+00  0.5517E+01

  x 1:  0.1000E+01  0.0000E+00  0.0000E+00 -0.2069E+00  0.0000E+00  0.0000E+00
       -0.3448E-01  0.0000E+00  0.0000E+00  0.3448E-01  0.0000E+00  0.4138E+00

 x11:  0.0000E+00  0.0000E+00  0.0000E+00  0.1103E+01  0.0000E+00  0.0000E+00
        0.5172E+00 -0.1000E+01  0.0000E+00 -0.5172E+00  0.1000E+01  0.6179E+02

  x 5:  0.0000E+00  0.0000E+00  0.0000E+00  0.2069E+00  0.1000E+01  0.0000E+00
        0.3448E-01  0.0000E+00  0.0000E+00 -0.3448E-01  0.0000E+00  0.9586E+01

   -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.1724E+00  0.0000E+00  0.0000E+00
       -0.1379E+00  0.0000E+00  0.0000E+00  0.1379E+00  0.0000E+00  0.1655E+01

   -w:  0.0000E+00  0.0000E+00  0.0000E+00 -0.1345E+01  0.0000E+00  0.1000E+01
       -0.7241E+00  0.1000E+01  0.0000E+00  0.1724E+01  0.0000E+00 -0.6731E+02

minimum value of ds:-0.13448277E+01  variable out of basis:     4
result of pivoting on element:     3     4

  x 3:  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00 -0.7143E+00
        0.2857E+00  0.0000E+00  0.7143E+00 -0.2857E+00  0.0000E+00  0.1029E+02

  x 2:  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.1429E+00
       -0.1429E+00  0.0000E+00  0.1429E+00  0.1429E+00  0.0000E+00  0.2857E+01

  x 4:  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00 -0.4143E+01
        0.8571E+00  0.0000E+00  0.4143E+01 -0.8571E+00  0.0000E+00  0.2286E+02

  x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.8571E+00
        0.1429E+00  0.0000E+00  0.8571E+00 -0.1429E+00  0.0000E+00  0.5143E+01

 x11:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.4571E+01
       -0.4286E+00 -0.1000E+01 -0.4571E+01  0.4286E+00  0.1000E+01  0.3657E+02

  x 5:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.8571E+00
       -0.1429E+00  0.0000E+00 -0.8571E+00  0.1429E+00  0.0000E+00  0.4857E+01

   -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.7143E+00
       -0.2857E+00  0.0000E+00 -0.7143E+00  0.2857E+00  0.0000E+00 -0.2286E+01

   -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.4571E+01
        0.4286E+00  0.1000E+01  0.5571E+01  0.5714E+00  0.0000E+00 -0.3657E+02
```

```
minimum value of ds:-0.45714273E+01  variable out of basis:     6
result of pivoting on element:     6     6

 x 3:  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.8333E+00  0.0000E+00
       0.1667E+00  0.0000E+00  0.0000E+00 -0.1667E+00  0.0000E+00  0.1433E+02

 x 2:  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.1667E+00  0.0000E+00
      -0.1667E+00  0.0000E+00  0.0000E+00  0.1667E+00  0.0000E+00  0.3667E+01

 x 4:  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.4833E+01  0.0000E+00
       0.1667E+00  0.0000E+00  0.0000E+00 -0.1667E+00  0.0000E+00  0.4633E+02

 x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00
       0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+02

 x11:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.5333E+01  0.0000E+00
       0.3333E+00 -0.1000E+01  0.0000E+00 -0.3333E+00  0.1000E+01  0.1067E+02

 x 6:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1167E+01  0.1000E+01
      -0.1667E+00  0.0000E+00 -0.1000E+01  0.1667E+00  0.0000E+00  0.5667E+01

  -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.8333E+00  0.0000E+00
      -0.1667E+00  0.0000E+00  0.0000E+00  0.1667E+00  0.0000E+00 -0.6333E+01

  -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.5333E+01  0.0000E+00
      -0.3333E+00  0.1000E+01  0.1000E+01  0.1333E+01  0.0000E+00 -0.1067E+02

minimum value of ds:-0.33333331E+00  variable out of basis:     7
result of pivoting on element:     5     7

 x 3:  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.3500E+01  0.0000E+00
       0.0000E+00  0.5000E+00  0.0000E+00  0.0000E+00 -0.5000E+00  0.9000E+01

 x 2:  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00 -0.2500E+01  0.0000E+00
       0.0000E+00 -0.5000E+00  0.0000E+00  0.0000E+00  0.5000E+00  0.9000E+01

 x 4:  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.7500E+01  0.0000E+00
       0.0000E+00  0.5000E+00  0.0000E+00  0.0000E+00 -0.5000E+00  0.4100E+02

 x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00
       0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+02

 x 7:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.1600E+02  0.0000E+00
       0.1000E+01 -0.3000E+01  0.0000E+00 -0.1000E+01  0.3000E+01  0.3200E+02

 x 6:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.1500E+01  0.1000E+01
       0.0000E+00 -0.5000E+00 -0.1000E+01  0.0000E+00  0.5000E+00  0.1100E+02

  -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.3500E+01  0.0000E+00
       0.0000E+00 -0.5000E+00  0.0000E+00  0.0000E+00  0.5000E+00 -0.1000E+01

  -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.4768E-06  0.0000E+00
       0.0000E+00  0.0000E+00  0.1000E+01  0.1000E+01  0.1000E+01 -0.1144E-04
```

```
minimum value of ds:-0.47683716E-06  variable out of basis:     5
result of pivoting on element:     1     5

  x 5:  0.0000E+00  0.0000E+00  0.2857E+00  0.0000E+00  0.1000E+01  0.0000E+00
        0.0000E+00  0.1429E+00  0.0000E+00  0.0000E+00 -0.1429E+00  0.2571E+01

  x 2:  0.0000E+00  0.1000E+01  0.7143E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00 -0.1429E+00  0.0000E+00  0.0000E+00  0.1429E+00  0.1543E+02

  x 4:  0.0000E+00  0.0000E+00 -0.2143E+01  0.1000E+01  0.0000E+00  0.0000E+00
        0.0000E+00 -0.5714E+00  0.0000E+00  0.0000E+00  0.5714E+00  0.2171E+02

  x 1:  0.1000E+01  0.0000E+00 -0.2857E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00 -0.1429E+00  0.0000E+00  0.0000E+00  0.1429E+00  0.7429E+01

  x 7:  0.0000E+00  0.0000E+00  0.4571E+01  0.0000E+00  0.0000E+00  0.0000E+00
        0.1000E+01 -0.7143E+00  0.0000E+00 -0.1000E+01  0.7143E+00  0.7314E+02

  x 6:  0.0000E+00  0.0000E+00  0.4286E+00  0.0000E+00  0.0000E+00  0.1000E+01
        0.0000E+00 -0.2857E+00 -0.1000E+01  0.0000E+00  0.2857E+00  0.1486E+02

    -f: 0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.8000E+01

    -w: 0.0000E+00  0.0000E+00  0.1362E-06  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.6812E-07  0.1000E+01  0.1000E+01  0.1000E+01 -0.1022E-04

PHASE-II OF SIMPLEX METHOD
system at start of phase-II

  x 5:  0.0000E+00  0.0000E+00  0.2857E+00  0.0000E+00  0.1000E+01  0.0000E+00
        0.0000E+00  0.1429E+00  0.2571E+01
  x 2:  0.0000E+00  0.1000E+01  0.7143E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00 -0.1429E+00  0.1543E+02
  x 4:  0.0000E+00  0.0000E+00 -0.2143E+01  0.1000E+01  0.0000E+00  0.0000E+00
        0.0000E+00 -0.5714E+00  0.2171E+02
  x 1:  0.1000E+01  0.0000E+00 -0.2857E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00 -0.1429E+00  0.7429E+01
  x 7:  0.0000E+00  0.0000E+00  0.4571E+01  0.0000E+00  0.0000E+00  0.0000E+00
        0.1000E+01 -0.7143E+00  0.7314E+02
  x 6:  0.0000E+00  0.0000E+00  0.4286E+00  0.0000E+00  0.0000E+00  0.1000E+01
        0.0000E+00 -0.2857E+00  0.1486E+02
    -f: 0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00  0.8000E+01
```

3.67

Minimize $f = -x - 3y$

subject to $-4x + 3y + z_1 = 12$

$x + y + z_2 = 7$

$x - 4y + z_3 = 2$

with $x \geq 0,\ y \geq 0,\ z_i \geq 0\ (i = 1, 2, 3)$

Simplex method:

Basic variables	x	y	z_1	z_2	z_3	b_i''	(b_i''/a_{is}'') for $a_{is}'' > 0$
z_1	-4	(3)	1	0	0	12	4 ←
z_2	1	1	0	1	0	7	7
z_3	1	-4	0	0	1	2	
$-f$	-1	-3	0	0	0	0	
		↑					
y	$-4/3$	1	$1/3$	0	0	4	
z_2	$7/3$	0	$-1/3$	1	0	3	$9/7$ ←
z_3	$-13/3$	0	$4/3$	0	1	18	
$-f$	-5	0	1	0	0	12	
	↑						
y	0	1	$1/7$	$4/7$	0	$40/7$	
x	1	0	$-1/7$	$3/7$	0	$9/7$	
z_3	0	0	$5/7$	$13/7$	1	$165/7$	
$-f$	0	0	$2/7$	$15/7$	0	$129/7$	

Optimum solution: $x^* = 9/7,\ y^* = 40/7,\ z_3^* = 165/7$

$f^* = f_{min} = -129/7$

3.68

$x = x_1 - x_2$, $y = y_1 - y_2$

Minimize $f = x_1 - x_2 + 3y_1 - 3y_2$

subject to

$$-4x_1 + 4x_2 + 3y_1 - 3y_2 + z_1 = 12$$
$$x_1 - x_2 + y_1 - y_2 + z_2 = 7$$
$$x_1 - x_2 - 4y_1 + 4y_2 + z_3 = 2$$

with all x_i, y_i and $z_i \geq 0$

Phase II solution:

Basic variables	x_1	x_2	y_1	y_2	z_1	z_2	z_3	b_i''	$\left(\frac{b_i''}{a_{is}''}\right)$ for $a_{is}'' > 0$
z_1	−4	4	3	−3	1	0	0	12	
z_2	1	−1	1	−1	0	1	0	7	
z_3	1	−1	−4	[4] ↑	0	0	1	2	1/2
$-f$	1	−1	3	−3	0	0	0	0	
z_1	−13/4	[13/4]	0	0	1	0	3/4	27/2	54/13
z_2	5/4	−5/4	0	0	0	1	1/4	15/2	
y_2	1/4	−1/4	−1	1	0	0	1/4	1/2	
$-f$	7/4	−7/4	0	0	0	0	3/4	3/2	
x_2	−1	1 ↑	0	0	1	0	3/13	54/13	
z_2	0	0	0	0	5/13	1	7/13	165/13	
y_2	0	0	−1	1	1/13	0	4/13	20/13	
$-f$	0	0	0	0	7/13	0	17/26	114/13	

Optimum solution: $x^* = -54/13$, $y^* = -20/13$,

$z_2^* = 165/13$

$f^* = -114/13 = -8.769$

3.69

$$-4x_1 + 3x_2 + x_3 = 12$$
$$x_1 + x_2 + x_4 = 7$$
$$x_1 - 4x_2 + x_5 = 2$$
$$-x_1 - x_2 = f$$

given system

x3:	-4.000	3.000	1.000	0.000	0.000	12.000
x4:	1.000	1.000	0.000	1.000	0.000	7.000
x5:	1.000	-4.000	0.000	0.000	1.000	2.000
-f:	-1.000	-1.000	0.000	0.000	0.000	0.000

minimum value of c1:-0.10000000E+01 variable out of basis: 1
result of pivoting on element: 3 1

x3:	0.000	-13.000	1.000	0.000	4.000	20.000
x4:	0.000	5.000	0.000	1.000	-1.000	5.000
x1:	1.000	-4.000	0.000	0.000	1.000	2.000
-f:	0.000	-5.000	0.000	0.000	1.000	2.000

minimum value of c1:-0.50000000E+01 variable out of basis: 2
result of pivoting on element: 2 2

x3:	0.000	0.000	1.000	2.600	1.400	33.000
x2:	0.000	1.000	0.000	0.200	-0.200	1.000
x1:	1.000	0.000	0.000	0.800	0.200	6.000
-f:	0.000	0.000	0.000	1.000	0.000	7.000

3.70

$$-4x_1 + 3x_2 + x_3 = 12$$
$$x_1 + x_2 + x_4 = 7$$
$$x_1 + 4x_2 - x_5 + x_6 = 2$$
$$-x_1 - 3x_2 = f$$
$$x_6 = w$$

or $$-x_1 + 4x_2 + x_5 + 2 = w$$

```
PHASE-I OF SIMPLEX METHOD
given system

  x 3: -0.4000E+01  0.3000E+01  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
       0.1200E+02
  x 4:  0.1000E+01  0.1000E+01  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00
       0.7000E+01
  x 6:  0.1000E+01 -0.4000E+01  0.0000E+00  0.0000E+00 -0.1000E+01  0.1000E+01
       0.2000E+01
   -f: -0.1000E+01 -0.3000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
       0.0000E+00
   -w: -0.1000E+01  0.4000E+01  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00
      -0.2000E+01

minimum value of ds:-0.10000000E+01  variable out of basis:     1
result of pivoting on element:     3     1

  x 3:  0.0000E+00 -0.1300E+02  0.1000E+01  0.0000E+00 -0.4000E+01  0.4000E+01
       0.2000E+02
  x 4:  0.0000E+00  0.5000E+01  0.0000E+00  0.1000E+01  0.1000E+01 -0.1000E+01
       0.5000E+01
  x 1:  0.1000E+01 -0.4000E+01  0.0000E+00  0.0000E+00 -0.1000E+01  0.1000E+01
       0.2000E+01
   -f:  0.0000E+00 -0.7000E+01  0.0000E+00  0.0000E+00 -0.1000E+01  0.1000E+01
       0.2000E+01
   -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01
       0.0000E+00

PHASE-II OF SIMPLEX METHOD
system at start of phase-II

  x 3:  0.0000E+00 -0.1300E+02  0.1000E+01  0.0000E+00 -0.4000E+01  0.2000E+02

  x 4:  0.0000E+00  0.5000E+01  0.0000E+00  0.1000E+01  0.1000E+01  0.5000E+01

  x 1:  0.1000E+01 -0.4000E+01  0.0000E+00  0.0000E+00 -0.1000E+01  0.2000E+01

   -f:  0.0000E+00 -0.7000E+01  0.0000E+00  0.0000E+00 -0.1000E+01  0.2000E+01

minimum value of cs:-0.70000000E+01  variable out of basis:     2
result of pivoting on element:     2     2

  x 3:  0.0000E+00  0.0000E+00  0.1000E+01  0.2600E+01 -0.1400E+01  0.3300E+02

  x 2:  0.0000E+00  0.1000E+01  0.0000E+00  0.2000E+00  0.2000E+00  0.1000E+01

  x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.8000E+00 -0.2000E+00  0.6000E+01

   -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.1400E+01  0.4000E+00  0.9000E+01
```

3.71

$$3x_1 + 2x_2 - x_3 + x_8 = 6$$
$$x_1 - x_2 + x_4 = 6$$
$$9x_1 + 7x_2 + x_5 = 108$$
$$3x_1 + 7x_2 + x_6 = 70$$
$$-2x_1 + 5x_2 + x_7 = 35$$
$$+x_1 - 8x_2 = f$$
$$x_8 = w$$

or $$-3x_1 - 2x_2 + x_3 + 6 = w$$

```
PHASE-I OF SIMPLEX METHOD
given system

  x 8:  0.3000E+01  0.2000E+01 -0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.1000E+01  0.6000E+01
  x 4:  0.1000E+01 -0.1000E+01  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00  0.6000E+01
  x 5:  0.9000E+01  0.7000E+01  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00
        0.0000E+00  0.0000E+00  0.1080E+03
  x 6:  0.3000E+01  0.7000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01
        0.0000E+00  0.0000E+00  0.7000E+02
  x 7: -0.2000E+01  0.5000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.1000E+01  0.0000E+00  0.3500E+02
   -f:  0.1000E+01 -0.8000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00  0.0000E+00
   -w: -0.3000E+01 -0.2000E+01  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00 -0.6000E+01

minimum value of ds:-0.30000000E+01  variable out of basis:     1
result of pivoting on element:     1     1


  x 1:  0.1000E+01  0.6667E+00 -0.3333E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.3333E+00  0.2000E+01
  x 4:  0.0000E+00 -0.1667E+01  0.3333E+00  0.1000E+01  0.0000E+00  0.0000E+00
        0.0000E+00 -0.3333E+00  0.4000E+01
  x 5:  0.0000E+00  0.1000E+01  0.3000E+01  0.0000E+00  0.1000E+01  0.0000E+00
        0.0000E+00 -0.3000E+01  0.9000E+02
  x 6:  0.0000E+00  0.5000E+01  0.1000E+01  0.0000E+00  0.0000E+00  0.1000E+01
        0.0000E+00 -0.1000E+01  0.6400E+02
  x 7:  0.0000E+00  0.6333E+01 -0.6667E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.1000E+01  0.6667E+00  0.3900E+02
   -f:  0.0000E+00 -0.8667E+01  0.3333E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00 -0.3333E+00 -0.2000E+01
   -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.1000E+01  0.0000E+00

PHASE-II OF SIMPLEX METHOD
system at start of phase-II

  x 1:  0.1000E+01  0.6667E+00 -0.3333E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.2000E+01
  x 4:  0.0000E+00 -0.1667E+01  0.3333E+00  0.1000E+01  0.0000E+00  0.0000E+00
        0.0000E+00  0.4000E+01
```

```
 x 5:  0.0000E+00  0.1000E+01  0.3000E+01  0.0000E+00  0.1000E+01  0.0000E+00
       0.0000E+00  0.9000E+02
 x 6:  0.0000E+00  0.5000E+01  0.1000E+01  0.0000E+00  0.0000E+00  0.1000E+01
       0.0000E+00  0.6400E+02
 x 7:  0.0000E+00  0.6333E+01 -0.6667E+00  0.0000E+00  0.0000E+00  0.0000E+00
       0.1000E+01  0.3900E+02
  -f:  0.0000E+00 -0.8667E+01  0.3333E+00  0.0000E+00  0.0000E+00  0.0000E+00
       0.0000E+00 -0.2000E+01

minimum value of cs:-0.86666670E+01  variable out of basis:    2
result of pivoting on element:    1    2


 x 2:  0.1500E+01  0.1000E+01 -0.5000E+00  0.0000E+00  0.0000E+00  0.0000E+00
       0.0000E+00  0.3000E+01
 x 4:  0.2500E+01  0.0000E+00 -0.5000E+00  0.1000E+01  0.0000E+00  0.0000E+00
       0.0000E+00  0.9000E+01
 x 5: -0.1500E+01  0.0000E+00  0.3500E+01  0.0000E+00  0.1000E+01  0.0000E+00
       0.0000E+00  0.8700E+02
 x 6: -0.7500E+01  0.0000E+00  0.3500E+01  0.0000E+00  0.0000E+00  0.1000E+01
       0.0000E+00  0.4900E+02
 x 7: -0.9500E+01  0.0000E+00  0.2500E+01  0.0000E+00  0.0000E+00  0.0000E+00
       0.1000E+01  0.2000E+02
  -f:  0.1300E+02  0.0000E+00 -0.4000E+01  0.0000E+00  0.0000E+00  0.0000E+00
       0.0000E+00  0.2400E+02

minimum value of cs:-0.40000000E+01  variable out of basis:    3
result of pivoting on element:    5    3


 x 2: -0.4000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
       0.2000E+00  0.7000E+01
 x 4:  0.6000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00
       0.2000E+00  0.1300E+02
 x 5:  0.1180E+02  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00
      -0.1400E+01  0.5900E+02
 x 6:  0.5800E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01
      -0.1400E+01  0.2100E+02
 x 3: -0.3800E+01  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
       0.4000E+00  0.8000E+01
  -f: -0.2200E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
       0.1600E+01  0.5600E+02

minimum value of cs:-0.21999998E+01  variable out of basis:    1
result of pivoting on element:    4    1


 x 2:  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.6897E-01
       0.1034E+00  0.8448E+01
 x 4:  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00 -0.1034E+00
       0.3448E+00  0.1083E+02
 x 5:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01 -0.2034E+01
       0.1448E+01  0.1628E+02
 x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1724E+00
      -0.2414E+00  0.3621E+01
 x 3:  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.6552E+00
      -0.5172E+00  0.2176E+02
  -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.3793E+00
       0.1069E+01  0.6397E+02
```

3.72 Minimize $f = -x + 8y$

subject to
$$3x + 2y - z_1 = 6 \qquad (3x + 2y - z_1 + u_1 = 6)$$
$$x - y + z_2 = 6$$
$$9x + 7y + z_3 = 108$$
$$3x + 7y + z_4 = 70$$
$$-2x + 5y + z_5 = 35$$
$$x \geq 0,\ y \geq 0,\ z_i \geq 0,\ i = 1, 2, \ldots, 5$$

Since the equations are not in canonical form, we add an artificial variable $u_1 \geq 0$ to the first constraint equation.

Infeasibility form: $w = u_1$ or $-3x - 2y + z_1 - w = -6$

Phase I solution:

Basic variables	Admissible variables x	y	z_1	z_2	z_3	z_4	z_5	Artificial variable u_1	b_i''	$\left(\frac{b_i''}{a_{is}''}\right)$ for $a_{is}'' > 0$
u_1	3	2	−1	0	0	0	0	1	6	2 ←
z_2	+1	−1	0	1	0	0	0	0	6	6
z_3	9	7	0	0	1	0	0	0	108	12
z_4	3	7	0	0	0	1	0	0	70	23.33
z_5	−2	5	0	0	0	0	1	0	35	
$-f$	−1	8	0	0	0	0	0	0	0	
$-w$	−3	−2	1	0	0	0	0	0	−6	
	↑									
x	1	2/3	−1/3	0	0	0	0		2	
z_2	0	−5/3	1/3	1	0	0	0		4	
z_3	0	1	3	0	1	0	0		90	
z_4	0	5	1	0	0	1	0		64	
z_5	0	19/3	−2/3	0	0	0	1		39	
$-f$	0	26/3	−1/3	0	0	0	0		2	
$-w$	0	0	0	0	0	0	0		0	

Since $w = 0$, drop w-row and go to Phase II.

Phase II solution:

x	1	$2/3$	$-1/3$	0	0	0	0	2	
z_2	0	$-5/3$	$1/3$	1	0	0	0	4	12 ←
z_3	0	1	3	0	1	0	0	90	30
z_4	0	5	1	0	0	1	0	64	64
z_5	0	$19/3$	$-2/3$	0	0	0	1	39	
$-f$	0	$26/3$	$-1/3$	0	0	0	0	2	
			↑						
x	1	-1	0	1	0	0	0	6	
z_1	0	-5	1	3	0	0	0	12	
z_3	0	16	0	-9	1	0	0	78	
z_4	0	10	0	-3	0	1	0	52	
z_5	0	3	0	2	0	0	1	47	
$-f$	0	7	0	1	0	0	0	6	

Since all $c_i'' \geq 0$, Phase II is complete.

Optimum solution:

$x^* = 6$, $z_1^* = 12$, $z_2^* = 0$, $z_3^* = 78$, $z_4^* = 52$, $z_5^* = 47$, $y^* = 0$,

$f_{min} = -6$

3.73

$$3x_1 + 2x_2 - 2x_3 - x_4 + x_9 = 6$$
$$x_1 - x_2 + x_3 + x_5 = 6$$
$$9x_1 + 7x_2 - 7x_3 + x_6 = 108$$
$$3x_1 + 7x_2 - 7x_3 + x_7 = 70$$
$$-2x_1 + 5x_2 - 5x_3 + x_8 = 35$$
$$-x_1 + 8x_2 - 8x_3 = f$$
$$x_9 = w$$

or $-14x_1 - 20x_2 + 20x_3 + x_4 - x_5 - x_6 - x_7 - x_8 + 225 = w$

```
PHASE-I OF SIMPLEX METHOD
given system

  x 9:  0.3000E+01  0.2000E+01 -0.2000E+01 -0.1000E+01  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00  0.1000E+01  0.6000E+01
  x 5:  0.1000E+01 -0.1000E+01  0.1000E+01  0.0000E+00  0.1000E+01  0.0000E+00
        0.0000E+00  0.0000E+00  0.0000E+00  0.6000E+01
  x 6:  0.9000E+01  0.7000E+01 -0.7000E+01  0.0000E+00  0.0000E+00  0.1000E+01
        0.0000E+00  0.0000E+00  0.0000E+00  0.1080E+03
  x 7:  0.3000E+01  0.7000E+01 -0.7000E+01  0.0000E+00  0.0000E+00  0.0000E+00
        0.1000E+01  0.0000E+00  0.0000E+00  0.7000E+02
  x 8: -0.2000E+01  0.5000E+01 -0.5000E+01  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.1000E+01  0.0000E+00  0.3500E+02
   -f: -0.1000E+01  0.8000E+01 -0.8000E+01  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
   -w: -0.1400E+02 -0.2000E+02  0.2000E+02  0.1000E+01 -0.1000E+01 -0.1000E+01
       -0.1000E+01 -0.1000E+01  0.0000E+00 -0.2250E+03

minimum value of ds:-0.20000000E+02  variable out of basis:     2
result of pivoting on element:     1     2


  x 2:  0.1500E+01  0.1000E+01 -0.1000E+01 -0.5000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00  0.5000E+00  0.3000E+01
  x 5:  0.2500E+01  0.0000E+00  0.0000E+00 -0.5000E+00  0.1000E+01  0.0000E+00
        0.0000E+00  0.0000E+00  0.5000E+00  0.9000E+01
  x 6: -0.1500E+01  0.0000E+00  0.0000E+00  0.3500E+01  0.0000E+00  0.1000E+01
        0.0000E+00  0.0000E+00 -0.3500E+01  0.8700E+02
  x 7: -0.7500E+01  0.0000E+00  0.0000E+00  0.3500E+01  0.0000E+00  0.0000E+00
        0.1000E+01  0.0000E+00 -0.3500E+01  0.4900E+02
  x 8: -0.9500E+01  0.0000E+00  0.0000E+00  0.2500E+01  0.0000E+00  0.0000E+00
        0.0000E+00  0.1000E+01 -0.2500E+01  0.2000E+02
   -f: -0.1300E+02  0.0000E+00  0.0000E+00  0.4000E+01  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00 -0.4000E+01 -0.2400E+02
   -w:  0.1600E+02  0.0000E+00  0.0000E+00 -0.9000E+01 -0.1000E+01 -0.1000E+01
       -0.1000E+01 -0.1000E+01  0.1000E+02 -0.1650E+03

minimum value of ds:-0.90000000E+01  variable out of basis:     4
result of pivoting on element:     5     4


  x 2: -0.4000E+00  0.1000E+01 -0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.2000E+00  0.0000E+00  0.7000E+01
  x 5:  0.6000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00
        0.0000E+00  0.2000E+00  0.0000E+00  0.1300E+02
  x 6:  0.1180E+02  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01
        0.0000E+00 -0.1400E+01  0.0000E+00  0.5900E+02
  x 7:  0.5800E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.1000E+01 -0.1400E+01  0.0000E+00  0.2100E+02
  x 4: -0.3800E+01  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00
        0.0000E+00  0.4000E+00 -0.1000E+01  0.8000E+01
   -f:  0.2200E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00 -0.1600E+01  0.0000E+00 -0.5600E+02
   -w: -0.1820E+02  0.0000E+00  0.0000E+00  0.0000E+00 -0.1000E+01 -0.1000E+01
       -0.1000E+01  0.2600E+01  0.1000E+01 -0.9300E+02

minimum value of ds:-0.18200001E+02  variable out of basis:     1
result of pivoting on element:     4     1


  x 2:  0.0000E+00  0.1000E+01 -0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
        0.6897E-01  0.1034E+00  0.0000E+00  0.8448E+01
  x 5:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00
       -0.1034E+00  0.3448E+00  0.0000E+00  0.1083E+02
  x 6:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01
```

```
      -0.2034E+01  0.1448E+01  0.0000E+00  0.1628E+02
 x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
       0.1724E+00 -0.2414E+00  0.0000E+00  0.3621E+01
 x 4:  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00
       0.6552E+00 -0.5172E+00 -0.1000E+01  0.2176E+02
  -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
      -0.3793E+00 -0.1069E+01  0.0000E+00 -0.6397E+02
  -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.1000E+01 -0.1000E+01
       0.2138E+01 -0.1793E+01  0.1000E+01 -0.2710E+02

minimum value of ds:-0.17931035E+01  variable out of basis:     8
result of pivoting on element:     3     8


 x 2:  0.0000E+00  0.1000E+01 -0.1000E+01  0.0000E+00  0.0000E+00 -0.7143E-01
       0.2143E+00  0.0000E+00  0.0000E+00  0.7286E+01
 x 5:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01 -0.2381E+00
       0.3810E+00  0.0000E+00  0.0000E+00  0.6952E+01
 x 8:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.6905E+00
      -0.1405E+01  0.1000E+01  0.0000E+00  0.1124E+02
 x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1667E+00
      -0.1667E+00  0.0000E+00  0.0000E+00  0.6333E+01
 x 4:  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.3571E+00
      -0.7143E-01  0.0000E+00 -0.1000E+01  0.2757E+02
  -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.7381E+00
      -0.1881E+01  0.0000E+00  0.0000E+00 -0.5195E+02
  -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.1000E+01  0.2381E+00
      -0.3810E+00  0.0000E+00  0.1000E+01 -0.6952E+01

minimum value of ds:-0.10000000E+01  variable out of basis:     5
result of pivoting on element:     2     5


 x 2:  0.0000E+00  0.1000E+01 -0.1000E+01  0.0000E+00  0.0000E+00 -0.7143E-01
       0.2143E+00  0.0000E+00  0.0000E+00  0.7286E+01
 x 5:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01 -0.2381E+00
       0.3810E+00  0.0000E+00  0.0000E+00  0.6952E+01
 x 8:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.6905E+00
      -0.1405E+01  0.1000E+01  0.0000E+00  0.1124E+02
 x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1667E+00
      -0.1667E+00  0.0000E+00  0.0000E+00  0.6333E+01
 x 4:  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.3571E+00
      -0.7143E-01  0.0000E+00 -0.1000E+01  0.2757E+02
  -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.7381E+00
      -0.1881E+01  0.0000E+00  0.0000E+00 -0.5195E+02
  -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.7451E-07
       0.0000E+00  0.0000E+00  0.1000E+01 -0.1431E-05

minimum value of ds:-0.74505806E-07  variable out of basis:     6
result of pivoting on element:     3     6


 x 2:  0.0000E+00  0.1000E+01 -0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
       0.6897E-01  0.1034E+00  0.0000E+00  0.8448E+01
 x 5:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00
      -0.1034E+00  0.3448E+00  0.0000E+00  0.1083E+02
 x 6:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01
      -0.2034E+01  0.1448E+01  0.0000E+00  0.1628E+02
 x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
       0.1724E+00 -0.2414E+00  0.0000E+00  0.3621E+01
 x 4:  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00
       0.6552E+00 -0.5172E+00 -0.1000E+01  0.2176E+02
  -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
      -0.3793E+00 -0.1069E+01  0.0000E+00 -0.6397E+02
  -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
      -0.1516E-06  0.1079E-06  0.1000E+01 -0.2179E-06
```

```
minimum value of ds:-0.15158078E-06  variable out of basis:     7
result of pivoting on element:     4     7

  x 2: -0.4000E+00  0.1000E+01 -0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
       0.0000E+00  0.2000E+00  0.0000E+00  0.7000E+01
  x 5:  0.6000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00
       0.0000E+00  0.2000E+00  0.0000E+00  0.1300E+02
  x 6:  0.1180E+02  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01
       0.0000E+00 -0.1400E+01  0.0000E+00  0.5900E+02
  x 7:  0.5800E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
       0.1000E+01 -0.1400E+01  0.0000E+00  0.2100E+02
  x 4: -0.3800E+01  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00
       0.0000E+00  0.4000E+00 -0.1000E+01  0.8000E+01
   -f:  0.2200E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
       0.0000E+00 -0.1600E+01  0.0000E+00 -0.5600E+02
   -w:  0.8792E-06  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
       0.0000E+00 -0.1043E-06  0.1000E+01  0.2965E-05

minimum value of ds:-0.10430814E-06  variable out of basis:     8
result of pivoting on element:     5     8

  x 2:  0.1500E+01  0.1000E+01 -0.1000E+01 -0.5000E+00  0.0000E+00  0.0000E+00
       0.0000E+00  0.0000E+00  0.5000E+00  0.3000E+01
  x 5:  0.2500E+01  0.0000E+00  0.0000E+00 -0.5000E+00  0.1000E+01  0.0000E+00
       0.0000E+00  0.0000E+00  0.5000E+00  0.9000E+01
  x 6: -0.1500E+01  0.0000E+00  0.0000E+00  0.3500E+01  0.0000E+00  0.1000E+01
       0.0000E+00  0.0000E+00 -0.3500E+01  0.8700E+02
  x 7: -0.7500E+01  0.0000E+00  0.0000E+00  0.3500E+01  0.0000E+00  0.0000E+00
       0.1000E+01  0.0000E+00 -0.3500E+01  0.4900E+02
  x 8: -0.9500E+01  0.0000E+00  0.0000E+00  0.2500E+01  0.0000E+00  0.0000E+00
       0.0000E+00  0.1000E+01 -0.2500E+01  0.2000E+02
   -f: -0.1300E+02  0.0000E+00  0.0000E+00  0.4000E+01  0.0000E+00  0.0000E+00
       0.0000E+00  0.0000E+00 -0.4000E+01 -0.2400E+02
   -w: -0.1118E-06  0.0000E+00  0.0000E+00  0.2608E-06  0.0000E+00  0.0000E+00
       0.0000E+00  0.0000E+00  0.1000E+01  0.5051E-05

minimum value of ds:-0.11175854E-06  variable out of basis:     1
result of pivoting on element:     1     1

  x 1:  0.1000E+01  0.6667E+00 -0.6667E+00 -0.3333E+00  0.0000E+00  0.0000E+00
       0.0000E+00  0.0000E+00  0.3333E+00  0.2000E+01
  x 5:  0.0000E+00 -0.1667E+01  0.1667E+01  0.3333E+00  0.1000E+01  0.0000E+00
       0.0000E+00  0.0000E+00 -0.3333E+00  0.4000E+01
  x 6:  0.0000E+00  0.1000E+01 -0.1000E+01  0.3000E+01  0.0000E+00  0.1000E+01
       0.0000E+00  0.0000E+00 -0.3000E+01  0.9000E+02
  x 7:  0.0000E+00  0.5000E+01 -0.5000E+01  0.1000E+01  0.0000E+00  0.0000E+00
       0.1000E+01  0.0000E+00 -0.1000E+01  0.6400E+02
  x 8:  0.0000E+00  0.6333E+01 -0.6333E+01 -0.6667E+00  0.0000E+00  0.0000E+00
       0.0000E+00  0.1000E+01  0.6667E+00  0.3900E+02
   -f:  0.0000E+00  0.8667E+01 -0.8667E+01 -0.3333E+00  0.0000E+00  0.0000E+00
       0.0000E+00  0.0000E+00  0.3333E+00  0.2000E+01
   -w:  0.0000E+00  0.7451E-07 -0.7451E-07  0.2235E-06  0.0000E+00  0.0000E+00
       0.0000E+00  0.0000E+00  0.1000E+01  0.5275E-05

minimum value of ds:-0.74505706E-07  variable out of basis:     3
result of pivoting on element:     2     3

  x 1:  0.1000E+01  0.0000E+00  0.0000E+00 -0.2000E+00  0.4000E+00  0.0000E+00
       0.0000E+00  0.0000E+00  0.2000E+00  0.3600E+01
  x 3:  0.0000E+00 -0.1000E+01  0.1000E+01  0.2000E+00  0.6000E+00  0.0000E+00
       0.0000E+00  0.0000E+00 -0.2000E+00  0.2400E+01
```

```
 x 6:  0.0000E+00  0.0000E+00  0.0000E+00  0.3200E+01  0.6000E+00  0.1000E+01
       0.0000E+00  0.0000E+00 -0.3200E+01  0.9240E+02
 x 7:  0.0000E+00  0.0000E+00  0.0000E+00  0.2000E+01  0.3000E+01  0.0000E+00
       0.1000E+01  0.0000E+00 -0.2000E+01  0.7600E+02
 x 8:  0.0000E+00  0.0000E+00  0.0000E+00  0.6000E+00  0.3800E+01  0.0000E+00
       0.0000E+00  0.1000E+01 -0.6000E+00  0.5420E+02
  -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.1400E+01  0.5200E+01  0.0000E+00
       0.0000E+00  0.0000E+00 -0.1400E+01  0.2280E+02
  -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.2384E-06  0.4470E-07  0.0000E+00
       0.0000E+00  0.0000E+00  0.1000E+01  0.5454E-05

PHASE-II OF SIMPLEX METHOD
system at start of phase-II

 x 1:  0.1000E+01  0.0000E+00  0.0000E+00 -0.2000E+00  0.4000E+00  0.0000E+00
       0.0000E+00  0.0000E+00  0.3600E+01
 x 3:  0.0000E+00 -0.1000E+01  0.1000E+01  0.2000E+00  0.6000E+00  0.0000E+00
       0.0000E+00  0.0000E+00  0.2400E+01
 x 6:  0.0000E+00  0.0000E+00  0.0000E+00  0.3200E+01  0.6000E+00  0.1000E+01
       0.0000E+00  0.0000E+00  0.9240E+02
 x 7:  0.0000E+00  0.0000E+00  0.0000E+00  0.2000E+01  0.3000E+01  0.0000E+00
       0.1000E+01  0.0000E+00  0.7600E+02
 x 8:  0.0000E+00  0.0000E+00  0.0000E+00  0.6000E+00  0.3800E+01  0.0000E+00
       0.0000E+00  0.1000E+01  0.5420E+02
  -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.1400E+01  0.5200E+01  0.0000E+00
       0.0000E+00  0.0000E+00  0.2280E+02
```

3.74

$$3x_1 + 2x_2 - x_3 + x_8 = 6$$

$$x_1 - x_2 + x_4 = 6$$

$$9x_1 + 7x_2 + x_5 = 108$$

$$3x_1 + 7x_2 + x_6 = 70$$

$$-2x_1 + 5x_2 + x_7 = 35$$

$$-5x_1 + 2x_2 = f$$

$$x_8 = w$$

or $-3x_1 - 2x_2 + x_3 + 6 = w$

```
PHASE-I OF SIMPLEX METHOD
given system

  x 8:  0.3000E+01  0.2000E+01 -0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.1000E+01  0.6000E+01
  x 4:  0.1000E+01 -0.1000E+01  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00  0.6000E+01
  x 5:  0.9000E+01  0.7000E+01  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00
        0.0000E+00  0.0000E+00  0.1080E+03
  x 6:  0.3000E+01  0.7000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01
        0.0000E+00  0.0000E+00  0.7000E+02
  x 7: -0.2000E+01  0.5000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.1000E+01  0.0000E+00  0.3500E+02
   -f: -0.5000E+01  0.2000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00  0.0000E+00
   -w: -0.3000E+01 -0.2000E+01  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00 -0.6000E+01

minimum value of ds:-0.30000000E+01  variable out of basis:     1
result of pivoting on element:     1     1


  x 1:  0.1000E+01  0.6667E+00 -0.3333E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.3333E+00  0.2000E+01
  x 4:  0.0000E+00 -0.1667E+01  0.3333E+00  0.1000E+01  0.0000E+00  0.0000E+00
        0.0000E+00 -0.3333E+00  0.4000E+01
  x 5:  0.0000E+00  0.1000E+01  0.3000E+01  0.0000E+00  0.1000E+01  0.0000E+00
        0.0000E+00 -0.3000E+01  0.9000E+02
  x 6:  0.0000E+00  0.5000E+01  0.1000E+01  0.0000E+00  0.0000E+00  0.1000E+01
        0.0000E+00 -0.1000E+01  0.6400E+02
  x 7:  0.0000E+00  0.6333E+01 -0.6667E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.1000E+01  0.6667E+00  0.3900E+02
   -f:  0.0000E+00  0.5333E+01 -0.1667E+01  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.1667E+01  0.1000E+02
   -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.1000E+01  0.0000E+00

PHASE-II OF SIMPLEX METHOD
system at start of phase-II

  x 1:  0.1000E+01  0.6667E+00 -0.3333E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.2000E+01
  x 4:  0.0000E+00 -0.1667E+01  0.3333E+00  0.1000E+01  0.0000E+00  0.0000E+00
        0.0000E+00  0.4000E+01
  x 5:  0.0000E+00  0.1000E+01  0.3000E+01  0.0000E+00  0.1000E+01  0.0000E+00
        0.0000E+00  0.9000E+02
  x 6:  0.0000E+00  0.5000E+01  0.1000E+01  0.0000E+00  0.0000E+00  0.1000E+01
        0.0000E+00  0.6400E+02
  x 7:  0.0000E+00  0.6333E+01 -0.6667E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.1000E+01  0.3900E+02
   -f:  0.0000E+00  0.5333E+01 -0.1667E+01  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.1000E+02

minimum value of cs:-0.16666667E+01  variable out of basis:     3
result of pivoting on element:     2     3


  x 1:  0.1000E+01 -0.1000E+01  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00
        0.0000E+00  0.6000E+01
  x 3:  0.0000E+00 -0.5000E+01  0.1000E+01  0.3000E+01  0.0000E+00  0.0000E+00
        0.0000E+00  0.1200E+02
  x 5:  0.0000E+00  0.1600E+02  0.0000E+00 -0.9000E+01  0.1000E+01  0.0000E+00
        0.0000E+00  0.5400E+02
  x 6:  0.0000E+00  0.1000E+02  0.0000E+00 -0.3000E+01  0.0000E+00  0.1000E+01
        0.0000E+00  0.5200E+02
```

```
 x 7:  0.0000E+00  0.3000E+01  0.0000E+00  0.2000E+01  0.0000E+00  0.0000E+00
       0.1000E+01  0.4700E+02
  -f:  0.0000E+00 -0.3000E+01  0.0000E+00  0.5000E+01  0.0000E+00  0.0000E+00
       0.0000E+00  0.3000E+02

minimum value of cs:-0.30000005E+01  variable out of basis:    2
result of pivoting on element:    3    2

 x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.4375E+00  0.6250E-01  0.0000E+00
       0.0000E+00  0.9375E+01
 x 3:  0.0000E+00  0.0000E+00  0.1000E+01  0.1875E+00  0.3125E+00  0.0000E+00
       0.0000E+00  0.2888E+02
 x 2:  0.0000E+00  0.1000E+01  0.0000E+00 -0.5625E+00  0.6250E-01  0.0000E+00
       0.0000E+00  0.3375E+01
 x 6:  0.0000E+00  0.0000E+00  0.0000E+00  0.2625E+01 -0.6250E+00  0.1000E+01
       0.0000E+00  0.1825E+02
 x 7:  0.0000E+00  0.0000E+00  0.0000E+00  0.3688E+01 -0.1875E+00  0.0000E+00
       0.1000E+01  0.3688E+02
  -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.3312E+01  0.1875E+00  0.0000E+00
       0.0000E+00  0.4012E+02
```

3.75 Minimize $f = x - 4y$

subject to $-x + y + z_1 = 4$

$4x + 5y + z_2 = 45$

$5x - 2y + z_3 = 20$

$5x + 2y + z_4 = 10$

with $x \geq 0,\ y \geq 0,\ z_i \geq 0 \quad (i = 1, 2, 3, 4)$

Phase II solution:

Basic variables	x	y	z_1	z_2	z_3	z_4	b_i''	(b_i''/a_{is}'') for $a_{is}'' > 0$
z_1	-1	[1]	1	0	0	0	4	4 ←
z_2	4	5	0	1	0	0	45	9
z_3	5	-2	0	0	1	0	20	
z_4	5	2	0	0	0	1	10	5
$-f$	1	-4	0	0	0	0	0	
		↑						
y	-1	1	1	0	0	0	4	
z_2	9	0	-5	1	0	0	25	$25/9$
z_3	3	0	2	0	1	0	28	$28/3$
z_4	[7]	0	-2	0	0	1	2	$2/7$ ←
$-f$	-3	0	4	0	0	0	16	
	↑							
y	0	1	$5/7$	0	0	$1/7$	$30/7$	
z_2	0	0	$-17/7$	1	0	$-9/7$	$157/7$	
z_3	0	0	$20/7$	0	1	$-3/7$	$190/7$	
x	1	0	$-2/7$	0	0	$1/7$	$2/7$	
$-f$	0	0	$22/7$	0	0	$3/7$	$118/7$	

Optimum solution: $x^* = 2/7,\ y^* = 30/7,$

$z_1^* = 0,\ z_2^* = 157/7,\ z_3^* = 190/7,\ z_4^* = 0$

$f_{min} = -118/7$

3.76

$$
\begin{aligned}
-x_1 + x_2 - x_3 + x_4 &= 4 \\
4x_1 + 5x_2 - 5x_3 + x_5 &= 45 \\
5x_1 - 2x_2 + 2x_3 + x_6 &= 20 \\
-5x_1 - 2x_2 + 2x_3 + x_7 &= 10 \\
-x_1 + 4x_2 &= f
\end{aligned}
$$

given system

x5:	-1.000	1.000	-1.000	-1.000	1.000	0.000	0.000	0.000	4.000
x6:	4.000	5.000	-5.000	-5.000	0.000	1.000	0.000	0.000	45.000
x7:	5.000	-2.000	2.000	2.000	0.000	0.000	1.000	0.000	20.000
x8:	-5.000	-2.000	2.000	2.000	0.000	0.000	0.000	1.000	10.000
-f:	-1.000	4.000	-4.000	0.000	0.000	0.000	0.000	0.000	0.000

minimum value of c1:-0.40000000E+01 variable out of basis: 3 result of pivoting on element: 4 3

x5:	-3.500	0.000	0.000	0.000	1.000	0.000	0.000	0.500	9.000
x6:	-8.500	0.000	0.000	0.000	0.000	1.000	0.000	2.500	70.000
x7:	10.000	0.000	0.000	0.000	0.000	0.000	1.000	-1.000	10.000
x3:	-2.500	-1.000	1.000	1.000	0.000	0.000	0.000	0.500	5.000
-f:	-11.000	0.000	0.000	4.000	0.000	0.000	0.000	2.000	20.000

minimum value of c1:-0.11000000E+02 variable out of basis: 1 result of pivoting on element: 3 1

x5:	0.000	0.000	0.000	0.000	1.000	0.000	0.350	0.150	12.500
x6:	0.000	0.000	0.000	0.000	0.000	1.000	0.850	1.650	78.500
x1:	1.000	0.000	0.000	0.000	0.000	0.000	0.100	-0.100	1.000
x3:	0.000	-1.000	1.000	1.000	0.000	0.000	0.250	0.250	7.500
-f:	0.000	0.000	0.000	4.000	0.000	0.000	1.100	0.900	31.000

3.77

$$
\begin{aligned}
-x_1 + x_2 + x_3 &= 4 \\
4x_1 + 5x_2 + x_4 &= 45 \\
5x_1 - 2x_2 + x_5 &= 20 \\
-5x_1 - 2x_2 + x_6 &= 10 \\
x_1 - 4x_2 &= f
\end{aligned}
$$

given system

x3:	-1.000	1.000	1.000	0.000	0.000	0.000	4.000
x4:	4.000	5.000	0.000	1.000	0.000	0.000	45.000
x5:	5.000	-2.000	0.000	0.000	1.000	0.000	20.000
x6:	-5.000	-2.000	0.000	0.000	0.000	1.000	10.000
-f:	1.000	-4.000	0.000	0.000	0.000	0.000	0.000

minimum value of c1:-0.40000000E+01 variable out of basis: 2
result of pivoting on element: 1 2

x2:	-1.000	1.000	1.000	0.000	0.000	0.000	4.000
x4:	9.000	0.000	-5.000	1.000	0.000	0.000	25.000
x5:	3.000	0.000	2.000	0.000	1.000	0.000	28.000
x6:	-7.000	0.000	2.000	0.000	0.000	1.000	18.000
-f:	-3.000	0.000	4.000	0.000	0.000	0.000	16.000

minimum value of c1:-0.30000000E+01 variable out of basis: 1
result of pivoting on element: 2 1

x2:	0.000	1.000	0.444	0.111	0.000	0.000	6.778
x1:	1.000	0.000	-0.556	0.111	0.000	0.000	2.778
x5:	0.000	0.000	3.667	-0.333	1.000	0.000	19.667
x6:	0.000	0.000	-1.889	0.778	0.000	1.000	37.444
-f:	0.000	0.000	2.333	0.333	0.000	0.000	24.333

3.78

$$
\begin{aligned}
-x_1 + x_2 + x_3 &= 4 \\
4x_1 + 5x_2 + x_4 &= 45 \\
5x_1 - 2x_2 + x_5 &= 20 \\
-5x_1 - 2x_2 + x_6 &= 10 \\
-x_1 + 4x_2 &= f
\end{aligned}
$$

given system

x3:	-1.000	1.000	1.000	0.000	0.000	0.000	4.000
x4:	4.000	5.000	0.000	1.000	0.000	0.000	45.000
x5:	5.000	-2.000	0.000	0.000	1.000	0.000	20.000
x6:	-5.000	-2.000	0.000	0.000	0.000	1.000	10.000
-f:	-1.000	4.000	0.000	0.000	0.000	0.000	0.000

minimum value of c1:-0.10000000E+01 variable out of basis: 1
result of pivoting on element: 3 1

x3:	0.000	0.600	1.000	0.000	0.200	0.000	8.000
x4:	0.000	6.600	0.000	1.000	-0.800	0.000	29.000
x1:	1.000	-0.400	0.000	0.000	0.200	0.000	4.000
x6:	0.000	-4.000	0.000	0.000	1.000	1.000	30.000
-f:	0.000	3.600	0.000	0.000	0.200	0.000	4.000

3.79 Minimize $f = 4x + 5y$

subject to $10x + y - z_1 = 10 \Rightarrow 10x + y - z_1 + s_1 = 10$

$5x + 4y - z_2 = 20 \Rightarrow 5x + 4y - z_2 + s_2 = 20$

$3x + 7y - z_3 = 21 \Rightarrow 3x + 7y - z_3 + s_3 = 21$

$x + 12y - z_4 = 12 \Rightarrow x + 12y - z_4 + s_4 = 12$

with $x \geq 0,\ y \geq 0,\ z_i \geq 0\ (i = 1, 2, 3, 4)$

Introducing artificial variables s_i ($i = 1$ to 4) and the infeasibility form

$w = s_1 + s_2 + s_3 + s_4 \Rightarrow -23x - 29y - w = -63$

Basic variables	x	y	z_1	z_2	z_3	z_4	s_1	s_2	s_3	s_4	$-f$	$-w$	b_i''	(b_i''/a_{is}'') for $a_{is}'' > 0$
s_1	10	1	−1	0	0	0	1	0	0	0	0	0	10	1
s_2	5	4	0	−1	0	0	0	1	0	0	0	0	20	4
s_3	3	7	0	0	−1	0	0	0	1	0	0	0	21	7
s_4	[1]	12	0	0	0	−1	0	0	0	1	0	0	0	0 ←
$-f$	4	5	0	0	0	0	0	0	0	0	1	0	0	
$-w$	−23	−9	0	0	0	0	0	0	0	0	0	1	−63	
	↑													
s_1	0	−119	−1	0	0	[10]	1	0	0		0	0	10	1 ←
s_2	0	−56	0	−1	0	5	0	1	0		0	0	20	4
s_3	0	−29	0	0	−1	3	0	0	1		0	0	21	7
x	1	12	0	0	0	−1	0	0	0		0	0	0	
$-f$	0	−43	0	0	0	4	0	0	0		1	0	0	
$-w$	0	267	0	0	0	−23	0	0	0		0	1	−63	
						↑								
z_4	0	−119/10	−1/10	0	0	1		0	0		0	0	1	
s_2	0	7/2	1/2	−1	0	0		1	0		0	0	15	30/7
s_3	0	[67/10]	3/10	0	−1	0		0	1		0	0	18	180/67 ←
x	1	1/10	−1/10	0	0	0		0	0		0	0	1	10
$-f$	0	46/10	4/10	0	0	0		0	0		1	0	−4	
$-w$	0	−67/10	−23/10	0	0	0		0	0		0	1	−40	
		↑												

z_4	0	0	$29/67$	0	$-119/67$	1	0	0	0	$2209/67$	76
s_2	0	0	$\boxed{23/67}$	-1	$35/67$	0	1	0	0	$375/67$	$16.3\leftarrow$
y	0	1	$3/67$	0	$-10/67$	0	0	0	0	$180/67$	60
x	1	0	$-7/67$	0	$1/67$	0	0	0	0	$49/167$	
$-f$	0	0	$13/67$	0	$46/67$	0	0	1	0	$-1096/67$	
$-w$	0	0	-2	0	-1	0	0	0	1	-22	

↑

z_4	0	0	0	$\frac{29}{23}$	$-\frac{56}{23}$	1	0	0	$\frac{596}{23}$	
z_1	0	0	1	$-67/63$	$35/23$	0	0	0	$375/23$	
y	0	1	0	$3/23$	$105/1541$	0	0	0	$45/23$	
x	1	0	0	$-7/23$	$295/1541$	0	0	0	$56/23$	
$-f$	0	0	0	$13/23$	$455/1541$	0	1	0	$-449/23$	
$-w$	0	0	0	$-134/23$	$47/23$	0	0	1	$244/23$	

Optimum solution:

$$x^* = 56/23 = 2.4348$$

$$y^* = 45/23 = 1.9565$$

$$f^* = 449/23 = 19.5217$$

3.80

$$10x_1 + x_2 - x_3 + x_7 = 10$$
$$5x_1 + 4x_2 - x_4 + x_8 = 20$$
$$3x_1 + 7x_2 - x_5 + x_9 = 21$$
$$x_1 + 12x_2 - x_6 + x_{10} = 12$$
$$-4x_1 - 5x_2 = f$$

$$x_7 + x_8 + x_9 + x_{10} = w$$

or $$-19x_1 - 24x_2 + x_3 + x_4 + x_5 + x_6 + 63 = w$$

```
PHASE-I OF SIMPLEX METHOD
given system

  x 7:  0.1000E+02  0.1000E+01 -0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
        0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+02
  x 8:  0.5000E+01  0.4000E+01  0.0000E+00 -0.1000E+01  0.0000E+00  0.0000E+00
        0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.2000E+02
  x 9:  0.3000E+01  0.7000E+01  0.0000E+00  0.0000E+00 -0.1000E+01  0.0000E+00
        0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.2100E+02
  x10:  0.1000E+01  0.1200E+02  0.0000E+00  0.0000E+00  0.0000E+00 -0.1000E+01
        0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.1200E+02
   -f: -0.4000E+01 -0.5000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
   -w: -0.1900E+02 -0.2400E+02  0.1000E+01  0.1000E+01  0.1000E+01  0.1000E+01
        0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.6300E+02

minimum value of ds:-0.24000000E+02  variable out of basis:    2
result of pivoting on element:    4    2


  x 7:  0.9917E+01  0.0000E+00 -0.1000E+01  0.0000E+00  0.0000E+00  0.8333E-01
        0.1000E+01  0.0000E+00  0.0000E+00 -0.8333E-01  0.9000E+01
  x 8:  0.4667E+01  0.0000E+00  0.0000E+00 -0.1000E+01  0.0000E+00  0.3333E+00
        0.0000E+00  0.1000E+01  0.0000E+00 -0.3333E+00  0.1600E+02
  x 9:  0.2417E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.1000E+01  0.5833E+00
        0.0000E+00  0.0000E+00  0.1000E+01 -0.5833E+00  0.1400E+02
  x 2:  0.8333E-01  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.8333E-01
        0.0000E+00  0.0000E+00  0.0000E+00  0.8333E-01  0.1000E+01
   -f: -0.3583E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.4167E+00
        0.0000E+00  0.0000E+00  0.0000E+00  0.4167E+00  0.5000E+01
   -w: -0.1700E+02  0.0000E+00  0.1000E+01  0.1000E+01  0.1000E+01 -0.1000E+01
        0.0000E+00  0.0000E+00  0.0000E+00  0.2000E+01 -0.3900E+02

minimum value of ds:-0.17000000E+02  variable out of basis:    1
result of pivoting on element:    1    1


  x 1:  0.1000E+01  0.0000E+00 -0.1008E+00  0.0000E+00  0.0000E+00  0.8403E-02
        0.1008E+00  0.0000E+00  0.0000E+00 -0.8403E-02  0.9076E+00
  x 8:  0.0000E+00  0.0000E+00  0.4706E+00 -0.1000E+01  0.0000E+00  0.2941E+00
       -0.4706E+00  0.1000E+01  0.0000E+00 -0.2941E+00  0.1176E+02
  x 9:  0.0000E+00  0.0000E+00  0.2437E+00  0.0000E+00 -0.1000E+01  0.5630E+00
       -0.2437E+00  0.0000E+00  0.1000E+01 -0.5630E+00  0.1181E+02
  x 2:  0.0000E+00  0.1000E+01  0.8403E-02  0.0000E+00  0.0000E+00 -0.8403E-01
       -0.8403E-02  0.0000E+00  0.0000E+00  0.8403E-01  0.9244E+00
```

```
   -f:  0.0000E+00  0.0000E+00 -0.3613E+00  0.0000E+00  0.0000E+00 -0.3866E+00
        0.3613E+00  0.0000E+00  0.0000E+00  0.3866E+00  0.8252E+01
   -w:  0.0000E+00  0.0000E+00 -0.7143E+00  0.1000E+01  0.1000E+01 -0.8571E+00
        0.1714E+01  0.0000E+00  0.0000E+00  0.1857E+01 -0.2357E+02

minimum value of ds:-0.85714287E+00  variable out of basis:     6
result of pivoting on element:     3     6


  x 1:  0.1000E+01  0.0000E+00 -0.1045E+00  0.0000E+00  0.1493E-01  0.0000E+00
        0.1045E+00  0.0000E+00 -0.1493E-01  0.0000E+00  0.7313E+00
  x 8:  0.0000E+00  0.0000E+00  0.3433E+00 -0.1000E+01  0.5224E+00  0.0000E+00
       -0.3433E+00  0.1000E+01 -0.5224E+00  0.0000E+00  0.5597E+01
  x 6:  0.0000E+00  0.0000E+00  0.4328E+00  0.0000E+00 -0.1776E+01  0.1000E+01
       -0.4328E+00  0.0000E+00  0.1776E+01 -0.1000E+01  0.2097E+02
  x 2:  0.0000E+00  0.1000E+01  0.4478E-01  0.0000E+00 -0.1493E+00  0.0000E+00
       -0.4478E-01  0.0000E+00  0.1493E+00  0.0000E+00  0.2687E+01
   -f:  0.0000E+00  0.0000E+00 -0.1940E+00  0.0000E+00 -0.6866E+00  0.0000E+00
        0.1940E+00  0.0000E+00  0.6866E+00  0.0000E+00  0.1636E+02
   -w:  0.0000E+00  0.0000E+00 -0.3433E+00  0.1000E+01 -0.5224E+00  0.0000E+00
        0.1343E+01  0.0000E+00  0.1522E+01  0.1000E+01 -0.5597E+01

minimum value of ds:-0.52238798E+00  variable out of basis:     5
result of pivoting on element:     2     5


  x 1:  0.1000E+01  0.0000E+00 -0.1143E+00  0.2857E-01  0.0000E+00  0.0000E+00
        0.1143E+00 -0.2857E-01  0.0000E+00  0.0000E+00  0.5714E+00
  x 5:  0.0000E+00  0.0000E+00  0.6571E+00 -0.1914E+01  0.1000E+01  0.0000E+00
       -0.6571E+00  0.1914E+01 -0.1000E+01  0.0000E+00  0.1071E+02
  x 6:  0.0000E+00  0.0000E+00  0.1600E+01 -0.3400E+01  0.0000E+00  0.1000E+01
       -0.1600E+01  0.3400E+01  0.0000E+00 -0.1000E+01  0.4000E+02
  x 2:  0.0000E+00  0.1000E+01  0.1429E+00 -0.2857E+00  0.0000E+00  0.0000E+00
       -0.1429E+00  0.2857E+00  0.0000E+00  0.0000E+00  0.4286E+01
   -f:  0.0000E+00  0.0000E+00  0.2571E+00 -0.1314E+01  0.0000E+00  0.0000E+00
       -0.2571E+00  0.1314E+01  0.0000E+00  0.0000E+00  0.2371E+02
   -w:  0.0000E+00  0.0000E+00 -0.2980E-07  0.1192E-06  0.0000E+00  0.0000E+00
        0.1000E+01  0.1000E+01  0.1000E+01  0.1000E+01 -0.3338E-05

minimum value of ds:-0.29802322E-07  variable out of basis:     3
result of pivoting on element:     2     3


  x 1:  0.1000E+01  0.0000E+00  0.0000E+00 -0.3043E+00  0.1739E+00  0.0000E+00
        0.0000E+00  0.3043E+00 -0.1739E+00  0.0000E+00  0.2435E+01
  x 3:  0.0000E+00  0.0000E+00  0.1000E+01 -0.2913E+01  0.1522E+01  0.0000E+00
       -0.1000E+01  0.2913E+01 -0.1522E+01  0.0000E+00  0.1630E+02
  x 6:  0.0000E+00  0.0000E+00  0.0000E+00  0.1261E+01 -0.2435E+01  0.1000E+01
        0.0000E+00 -0.1261E+01  0.2435E+01 -0.1000E+01  0.1391E+02
  x 2:  0.0000E+00  0.1000E+01  0.0000E+00  0.1304E+00 -0.2174E+00  0.0000E+00
        0.0000E+00 -0.1304E+00  0.2174E+00  0.0000E+00  0.1957E+01
   -f:  0.0000E+00  0.0000E+00  0.0000E+00 -0.5652E+00 -0.3913E+00  0.0000E+00
        0.0000E+00  0.5652E+00  0.3913E+00  0.0000E+00  0.1952E+02
   -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.3239E-07  0.4535E-07  0.0000E+00
        0.1000E+01  0.1000E+01  0.1000E+01  0.1000E+01 -0.2852E-05
```

```
PHASE-II OF SIMPLEX METHOD
system at start of phase-II

  x 1:  0.1000E+01  0.0000E+00  0.0000E+00 -0.3043E+00  0.1739E+00  0.0000E+00
        0.2435E+01
  x 3:  0.0000E+00  0.0000E+00  0.1000E+01 -0.2913E+01  0.1522E+01  0.0000E+00
        0.1630E+02
  x 6:  0.0000E+00  0.0000E+00  0.0000E+00  0.1261E+01 -0.2435E+01  0.1000E+01
        0.1391E+02
  x 2:  0.0000E+00  0.1000E+01  0.0000E+00  0.1304E+00 -0.2174E+00  0.0000E+00
        0.1957E+01
   -f:  0.0000E+00  0.0000E+00  0.0000E+00 -0.5652E+00 -0.3913E+00  0.0000E+00
        0.1952E+02

minimum value of cs:-0.56521744E+00  variable out of basis:     4
result of pivoting on element:     3     4


  x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.4138E+00  0.2414E+00
        0.5793E+01
  x 3:  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00 -0.4103E+01  0.2310E+01
        0.4845E+02
  x 4:  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01 -0.1931E+01  0.7931E+00
        0.1103E+02
  x 2:  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.3448E-01 -0.1034E+00
        0.5172E+00
   -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.1483E+01  0.4483E+00
        0.2576E+02

minimum value of cs:-0.14827588E+01  variable out of basis:     5
result of pivoting on element:     4     5


  x 1:  0.1000E+01  0.1200E+02  0.0000E+00  0.0000E+00  0.0000E+00 -0.1000E+01
        0.1200E+02
  x 3:  0.0000E+00  0.1190E+03  0.1000E+01  0.0000E+00  0.0000E+00 -0.1000E+02
        0.1100E+03
  x 4:  0.0000E+00  0.5600E+02  0.0000E+00  0.1000E+01  0.0000E+00 -0.5000E+01
        0.4000E+02
  x 5:  0.0000E+00  0.2900E+02  0.0000E+00  0.0000E+00  0.1000E+01 -0.3000E+01
        0.1500E+02
   -f:  0.0000E+00  0.4300E+02  0.0000E+00  0.0000E+00  0.0000E+00 -0.4000E+01
        0.4800E+02

minimum value of cs:-0.39999986E+01  variable out of basis:     6

solution is unbounded
```

3.81

$$
\begin{aligned}
10x_1 + x_2 - x_3 + x_7 &= 10 \\
5x_1 + 4x_2 - x_4 + x_8 &= 20 \\
3x_1 + 7x_2 - x_5 + x_9 &= 21 \\
x_1 + 12x_2 - x_6 + x_{10} &= 12 \\
6x_1 + 2x_2 &= f
\end{aligned}
$$

$$x_7 + x_8 + x_9 + x_{10} = w$$

or

$$-19x_1 - 24x_2 + x_3 + x_4 + x_5 + x_6 + 63 = w$$

```
PHASE-I OF SIMPLEX METHOD
given system

  x 7:  0.1000E+02  0.1000E+01 -0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
        0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+02
  x 8:  0.5000E+01  0.4000E+01  0.0000E+00 -0.1000E+01  0.0000E+00  0.0000E+00
        0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.2000E+02
  x 9:  0.3000E+01  0.7000E+01  0.0000E+00  0.0000E+00 -0.1000E+01  0.0000E+00
        0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.2100E+02
  x10:  0.1000E+01  0.1200E+02  0.0000E+00  0.0000E+00  0.0000E+00 -0.1000E+01
        0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.1200E+02
   -f:  0.6000E+01  0.2000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
   -w: -0.1900E+02 -0.2400E+02  0.1000E+01  0.1000E+01  0.1000E+01  0.1000E+01
        0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.6300E+02

minimum value of ds:-0.24000000E+02  variable out of basis:    2
result of pivoting on element:    4    2


  x 7:  0.9917E+01  0.0000E+00 -0.1000E+01  0.0000E+00  0.0000E+00  0.8333E-01
        0.1000E+01  0.0000E+00  0.0000E+00 -0.8333E-01  0.9000E+01
  x 8:  0.4667E+01  0.0000E+00  0.0000E+00 -0.1000E+01  0.0000E+00  0.3333E+00
        0.0000E+00  0.1000E+01  0.0000E+00 -0.3333E+00  0.1600E+02
  x 9:  0.2417E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.1000E+01  0.5833E+00
        0.0000E+00  0.0000E+00  0.1000E+01 -0.5833E+00  0.1400E+02
  x 2:  0.8333E-01  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.8333E-01
        0.0000E+00  0.0000E+00  0.0000E+00  0.8333E-01  0.1000E+01
   -f:  0.5833E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1667E+00
        0.0000E+00  0.0000E+00  0.0000E+00 -0.1667E+00 -0.2000E+01
   -w: -0.1700E+02  0.0000E+00  0.1000E+01  0.1000E+01  0.1000E+01 -0.1000E+01
        0.0000E+00  0.0000E+00  0.0000E+00  0.2000E+01 -0.3900E+02

minimum value of ds:-0.17000000E+02  variable out of basis:    1
result of pivoting on element:    1    1


  x 1:  0.1000E+01  0.0000E+00 -0.1008E+00  0.0000E+00  0.0000E+00  0.8403E-02
        0.1008E+00  0.0000E+00  0.0000E+00 -0.8403E-02  0.9076E+00
  x 8:  0.0000E+00  0.0000E+00  0.4706E+00 -0.1000E+01  0.0000E+00  0.2941E+00
       -0.4706E+00  0.1000E+01  0.0000E+00 -0.2941E+00  0.1176E+02
  x 9:  0.0000E+00  0.0000E+00  0.2437E+00  0.0000E+00 -0.1000E+01  0.5630E+00
       -0.2437E+00  0.0000E+00  0.1000E+01 -0.5630E+00  0.1181E+02
  x 2:  0.0000E+00  0.1000E+01  0.8403E-02  0.0000E+00  0.0000E+00 -0.8403E-01
       -0.8403E-02  0.0000E+00  0.0000E+00  0.8403E-01  0.9244E+00
   -f:  0.0000E+00  0.0000E+00  0.5882E+00  0.0000E+00  0.0000E+00  0.1176E+00
       -0.5882E+00  0.0000E+00  0.0000E+00 -0.1176E+00 -0.7294E+01
   -w:  0.0000E+00  0.0000E+00 -0.7143E+00  0.1000E+01  0.1000E+01 -0.8571E+00
        0.1714E+01  0.0000E+00  0.0000E+00  0.1857E+01 -0.2357E+02

minimum value of ds:-0.85714287E+00  variable out of basis:    6
result of pivoting on element:    3    6


  x 1:  0.1000E+01  0.0000E+00 -0.1045E+00  0.0000E+00  0.1493E-01  0.0000E+00
        0.1045E+00  0.0000E+00 -0.1493E-01  0.0000E+00  0.7313E+00
  x 8:  0.0000E+00  0.0000E+00  0.3433E+00 -0.1000E+01  0.5224E+00  0.0000E+00
       -0.3433E+00  0.1000E+01 -0.5224E+00  0.0000E+00  0.5597E+01
  x 6:  0.0000E+00  0.0000E+00  0.4328E+00  0.0000E+00 -0.1776E+01  0.1000E+01
       -0.4328E+00  0.0000E+00  0.1776E+01 -0.1000E+01  0.2097E+02
  x 2:  0.0000E+00  0.1000E+01  0.4478E-01  0.0000E+00 -0.1493E+00  0.0000E+00
       -0.4478E-01  0.0000E+00  0.1493E+00  0.0000E+00  0.2687E+01
   -f:  0.0000E+00  0.0000E+00  0.5373E+00  0.0000E+00  0.2090E+00  0.0000E+00
       -0.5373E+00  0.0000E+00 -0.2090E+00  0.0000E+00 -0.9761E+01
   -w:  0.0000E+00  0.0000E+00 -0.3433E+00  0.1000E+01 -0.5224E+00  0.0000E+00
```

```
        0.1343E+01  0.0000E+00  0.1522E+01  0.1000E+01 -0.5597E+01

minimum value of ds:-0.52238798E+00  variable out of basis:     5
result of pivoting on element:     2     5


  x 1:  0.1000E+01  0.0000E+00 -0.1143E+00  0.2857E-01  0.0000E+00  0.0000E+00
        0.1143E+00 -0.2857E-01  0.0000E+00  0.0000E+00  0.5714E+00
  x 5:  0.0000E+00  0.0000E+00  0.6571E+00 -0.1914E+01  0.1000E+01  0.0000E+00
       -0.6571E+00  0.1914E+01 -0.1000E+01  0.0000E+00  0.1071E+02
  x 6:  0.0000E+00  0.0000E+00  0.1600E+01 -0.3400E+01  0.0000E+00  0.1000E+01
       -0.1600E+01  0.3400E+01  0.0000E+00 -0.1000E+01  0.4000E+02
  x 2:  0.0000E+00  0.1000E+01  0.1429E+00 -0.2857E+00  0.0000E+00  0.0000E+00
       -0.1429E+00  0.2857E+00  0.0000E+00  0.0000E+00  0.4286E+01
   -f:  0.0000E+00  0.0000E+00  0.4000E+00  0.4000E+00  0.0000E+00  0.0000E+00
       -0.4000E+00 -0.4000E+00  0.0000E+00  0.0000E+00 -0.1200E+02
   -w:  0.0000E+00  0.0000E+00 -0.2980E-07  0.1192E-06  0.0000E+00  0.0000E+00
        0.1000E+01  0.1000E+01  0.1000E+01  0.1000E+01 -0.3338E-05

minimum value of ds:-0.29802322E-07  variable out of basis:     3
result of pivoting on element:     2     3


  x 1:  0.1000E+01  0.0000E+00  0.0000E+00 -0.3043E+00  0.1739E+00  0.0000E+00
        0.0000E+00  0.3043E+00 -0.1739E+00  0.0000E+00  0.2435E+01
  x 3:  0.0000E+00  0.0000E+00  0.1000E+01 -0.2913E+01  0.1522E+01  0.0000E+00
       -0.1000E+01  0.2913E+01 -0.1522E+01  0.0000E+00  0.1630E+02
  x 6:  0.0000E+00  0.0000E+00  0.0000E+00  0.1261E+01 -0.2435E+01  0.1000E+01
        0.0000E+00 -0.1261E+01  0.2435E+01 -0.1000E+01  0.1391E+02
  x 2:  0.0000E+00  0.1000E+01  0.0000E+00  0.1304E+00 -0.2174E+00  0.0000E+00
        0.0000E+00 -0.1304E+00  0.2174E+00  0.0000E+00  0.1957E+01
   -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.1565E+01 -0.6087E+00  0.0000E+00
        0.0000E+00 -0.1565E+01  0.6087E+00  0.0000E+00 -0.1852E+02
   -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.3239E-07  0.4535E-07  0.0000E+00
        0.1000E+01  0.1000E+01  0.1000E+01  0.1000E+01 -0.2852E-05

PHASE-II OF SIMPLEX METHOD
system at start of phase-II

  x 1:  0.1000E+01  0.0000E+00  0.0000E+00 -0.3043E+00  0.1739E+00  0.0000E+00
        0.2435E+01
  x 3:  0.0000E+00  0.0000E+00  0.1000E+01 -0.2913E+01  0.1522E+01  0.0000E+00
        0.1630E+02
  x 6:  0.0000E+00  0.0000E+00  0.0000E+00  0.1261E+01 -0.2435E+01  0.1000E+01
        0.1391E+02
  x 2:  0.0000E+00  0.1000E+01  0.0000E+00  0.1304E+00 -0.2174E+00  0.0000E+00
        0.1957E+01
   -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.1565E+01 -0.6087E+00  0.0000E+00
       -0.1852E+02

minimum value of cs:-0.60869569E+00  variable out of basis:     5
result of pivoting on element:     2     5


  x 1:  0.1000E+01  0.0000E+00 -0.1143E+00  0.2857E-01  0.0000E+00  0.0000E+00
        0.5714E+00
  x 5:  0.0000E+00  0.0000E+00  0.6571E+00 -0.1914E+01  0.1000E+01  0.0000E+00
        0.1071E+02
  x 6:  0.0000E+00  0.0000E+00  0.1600E+01 -0.3400E+01  0.0000E+00  0.1000E+01
        0.4000E+02
  x 2:  0.0000E+00  0.1000E+01  0.1429E+00 -0.2857E+00  0.0000E+00  0.0000E+00
        0.4286E+01
   -f:  0.0000E+00  0.0000E+00  0.4000E+00  0.4000E+00  0.0000E+00  0.0000E+00
       -0.1200E+02
```

3.82 $x = x_1 - x_2$, $y = y_1 - y_2$; Minimize $f = 6x_1 - 6x_2 + 2y_1 - 2y_2$

subject to :

$$10x_1 - 10x_2 + y_1 - y_2 - z_1 = 10$$
$$5x_1 - 5x_2 + 4y_1 - 4y_2 - z_2 = 20$$
$$3x_1 - 3x_2 + 7y_1 - 7y_2 - z_3 = 21$$
$$x_1 - x_2 + 12y_1 - 12y_2 - z_4 = 12$$

with all $x_i \geq 0$, $y_i \geq 0$, $z_i \geq 0$

Introduce artificial variables s_1, s_2, s_3 and s_4 into the constraints and the infeasibility form:

$$w = s_1 + s_2 + s_3 + s_4$$

or $$-w - 19x_1 + 19x_2 - 24y_1 + 24y_2 + z_1 + z_2 + z_3 + z_4 = -63$$

Phase I solution :

Basic variables	Admissible variables								Artificial variables				b_i''	(b_i''/a_{is}'') for $a_{is}'' > 0$
	x_1	x_2	y_1	y_2	z_1	z_2	z_3	z_4	s_1	s_2	s_3	s_4		
s_1	10	−10	1	−1	−1	0	0	0	1	0	0	0	10	10
s_2	5	−5	4	−4	0	−1	0	0	0	1	0	0	20	5
s_3	3	−3	7	−7	0	0	−1	0	0	0	1	0	21	3
s_4	1	−1	(12)	−12	0	0	0	−1	0	0	0	1	12	1 ←
$-f$	6	−6	2	−2	0	0	0	0	0	0	0	0	0	
$-w$	−19	19	−24	24	1	1	1	1	0	0	0	0	−63	
			↑											
s_1	(119/12)	−119/12	0	0	−1	0	0	1/12	1	0	0	−1/12	9	108/119 ←
s_2	14/3	−14/3	0	0	0	−1	0	1/3	0	1	0	−1/3	16	48/14
s_3	29/12	−29/12	0	0	0	0	−1	7/12	0	0	1	−7/12	14	168/29
y_1	1/12	−1/12	1	−1	0	0	0	−1/12	0	0	0	1/12	1	12
$-f$	35/6	−35/6	0	0	0	0	0	1/6	0	0	0	−1/6	−2	
$-w$	−17	17	0	0	1	1	1	−1	0	0	0	2	−39	
	↑													

x_1	1	−1	0	0	$-\frac{12}{119}$	0	0	$\frac{1}{119}$	$\frac{12}{119}$	0	0	$-\frac{1}{119}$	$\frac{108}{119}$	108
s_2	0	0	0	0	$\frac{56}{119}$	−1	0	$\frac{35}{119}$					$\frac{1400}{119}$	40
s_3	0	0	0	0	$\frac{29}{119}$	0	−1	$\frac{67}{119}$					$\frac{1405}{119}$	$\frac{1405}{67}$ ←
y_1	0	0	1	−1	$\frac{1}{119}$	0	0	$-\frac{10}{119}$					$\frac{110}{119}$	
$-f$	0	0	0	0	$\frac{70}{119}$	0	0	$\frac{14}{119}$					$-\frac{868}{119}$	
$-w$	0	0	0	0	$-\frac{35}{119}$	1	1	$-\frac{102}{119}$					$-\frac{2805}{119}$	

↑

x_1	1	−1	0	0	$-\frac{7}{67}$	0	$\frac{1}{67}$	0	$\frac{49}{67}$	49
s_2	0	0	0	0	$-\frac{23}{67}$	−1	$\frac{35}{67}$	0	$\frac{375}{67}$	$\frac{375}{35}$ ←
s_4	0	0	0	0	$\frac{29}{67}$	0	$-\frac{119}{67}$	1	$\frac{1405}{67}$	
y_1	0	0	1	−1	$\frac{3}{67}$	0	$-\frac{10}{67}$	0	$\frac{180}{67}$	
$-f$	0	0	0	0	$\frac{36}{67}$	0	$\frac{14}{67}$	0	$\frac{654}{67}$	
$-w$	0	0	0	0	$-\frac{23}{67}$	1	$-\frac{35}{67}$	0	$-\frac{375}{67}$	

↑

x_1	1	−1	0	0	$-\frac{4}{35}$	$\frac{1}{35}$	0	0	$\frac{4}{7}$
s_3	0	0	0	0	$\frac{23}{35}$	$-\frac{67}{35}$	1	0	$\frac{75}{7}$
s_4	0	0	0	0	$\frac{56}{35}$	$-\frac{119}{35}$	0	1	40
y_1	0	0	1	−1	$\frac{1}{7}$	$-\frac{2}{7}$	0	0	$\frac{30}{7}$
$-f$	0	0	0	0	$\frac{2}{5}$	$\frac{2}{5}$	0	0	−12
$-w$	0	0	0	0	0	0	0	0	0

This completes Phase I and Phase II.

Optimal solution: $x_1^* = \frac{4}{7}$, $y_1^* = \frac{30}{7}$, $s_3^* = \frac{75}{7}$, $s_4^* = 40$

$\therefore x^* = \frac{4}{7}$, $y^* = \frac{30}{7}$

$f_{min} = 12$

3.83

$$
\begin{aligned}
3x_1 + 4x_2 + x_3 &= 24 \\
x_1 - x_2 + x_4 &= 3 \\
x_1 + 4x_2 - x_5 + x_7 &= 4 \\
3x_1 + x_2 - x_6 + x_8 &= 3 \\
5x_1 + 2x_2 &= f \\
x_7 + x_8 &= w
\end{aligned}
$$

or

$$-4x_1 - 5x_2 + x_5 + x_6 + 7 = w$$

3.84

$$
\begin{aligned}
3x_1 + 4x_2 + x_3 &= 24\\
x_1 - x_2 + x_4 &= 3\\
x_1 + 4x_2 - x_5 + x_7 &= 4\\
3x_1 + x_2 - x_6 + x_8 &= 3\\
-5x_1 - 2x_2 &= f\\
x_7 + x_8 &= w
\end{aligned}
$$

or

$$-4x_1 - 5x_2 + x_5 + x_6 + 7 = w$$

PHASE-I OF SIMPLEX METHOD
given system

x 3:	0.3000E+01	0.4000E+01	0.1000E+01	0.0000E+00	0.0000E+00	0.0000E+00
	0.0000E+00	0.0000E+00	0.2400E+02			
x 4:	0.1000E+01	-0.1000E+01	0.0000E+00	0.1000E+01	0.0000E+00	0.0000E+00
	0.0000E+00	0.0000E+00	0.3000E+01			
x 7:	0.1000E+01	0.4000E+01	0.0000E+00	0.0000E+00	-0.1000E+01	0.0000E+00
	0.1000E+01	0.0000E+00	0.4000E+01			
x 8:	0.3000E+01	0.1000E+01	0.0000E+00	0.0000E+00	0.0000E+00	-0.1000E+01
	0.0000E+00	0.1000E+01	0.3000E+01			
-f:	-0.5000E+01	-0.2000E+01	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
	0.0000E+00	0.0000E+00	0.0000E+00			
-w:	-0.4000E+01	-0.5000E+01	0.0000E+00	0.0000E+00	0.1000E+01	0.1000E+01
	0.0000E+00	0.0000E+00	-0.7000E+01			

minimum value of ds:-0.50000000E+01 variable out of basis: 2
result of pivoting on element: 3 2

x 3:	0.2000E+01	0.0000E+00	0.1000E+01	0.0000E+00	0.1000E+01	0.0000E+00
	-0.1000E+01	0.0000E+00	0.2000E+02			
x 4:	0.1250E+01	0.0000E+00	0.0000E+00	0.1000E+01	-0.2500E+00	0.0000E+00
	0.2500E+00	0.0000E+00	0.4000E+01			
x 2:	0.2500E+00	0.1000E+01	0.0000E+00	0.0000E+00	-0.2500E+00	0.0000E+00
	0.2500E+00	0.0000E+00	0.1000E+01			
x 8:	0.2750E+01	0.0000E+00	0.0000E+00	0.0000E+00	0.2500E+00	-0.1000E+01
	-0.2500E+00	0.1000E+01	0.2000E+01			
-f:	-0.4500E+01	0.0000E+00	0.0000E+00	0.0000E+00	-0.5000E+00	0.0000E+00
	0.5000E+00	0.0000E+00	0.2000E+01			
-w:	-0.2750E+01	0.0000E+00	0.0000E+00	0.0000E+00	-0.2500E+00	0.1000E+01
	0.1250E+01	0.0000E+00	-0.2000E+01			

minimum value of ds:-0.27500000E+01 variable out of basis: 1
result of pivoting on element: 4 1

x 3:	0.0000E+00	0.0000E+00	0.1000E+01	0.0000E+00	0.8182E+00	0.7273E+00
	-0.8182E+00	-0.7273E+00	0.1855E+02			
x 4:	0.0000E+00	0.0000E+00	0.0000E+00	0.1000E+01	-0.3636E+00	0.4545E+00
	0.3636E+00	-0.4545E+00	0.3091E+01			
x 2:	0.0000E+00	0.1000E+01	0.0000E+00	0.0000E+00	-0.2727E+00	0.9091E-01
	0.2727E+00	-0.9091E-01	0.8182E+00			

```
 x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.9091E-01 -0.3636E+00
      -0.9091E-01  0.3636E+00  0.7273E+00
  -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.9091E-01 -0.1636E+01
       0.9091E-01  0.1636E+01  0.5273E+01
  -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
       0.1000E+01  0.1000E+01  0.0000E+00

PHASE-II OF SIMPLEX METHOD
system at start of phase-II

 x 3:  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.8182E+00  0.7273E+00
       0.1855E+02
 x 4:  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01 -0.3636E+00  0.4545E+00
       0.3091E+01
 x 2:  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00 -0.2727E+00  0.9091E-01
       0.8182E+00
 x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.9091E-01 -0.3636E+00
       0.7273E+00
  -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.9091E-01 -0.1636E+01
       0.5273E+01

minimum value of cs:-0.16363637E+01  variable out of basis:    6
result of pivoting on element:    2    6

 x 3:  0.0000E+00  0.0000E+00  0.1000E+01 -0.1600E+01  0.1400E+01  0.0000E+00
       0.1360E+02
 x 6:  0.0000E+00  0.0000E+00  0.0000E+00  0.2200E+01 -0.8000E+00  0.1000E+01
       0.6800E+01
 x 2:  0.0000E+00  0.1000E+01  0.0000E+00 -0.2000E+00 -0.2000E+00  0.0000E+00
       0.2000E+00
 x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.8000E+00 -0.2000E+00  0.0000E+00
       0.3200E+01
  -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.3600E+01 -0.1400E+01  0.0000E+00
       0.1640E+02

minimum value of cs:-0.14000001E+01  variable out of basis:    5
result of pivoting on element:    1    5

 x 5:  0.0000E+00  0.0000E+00  0.7143E+00 -0.1143E+01  0.1000E+01  0.0000E+00
       0.9714E+01
 x 6:  0.0000E+00  0.0000E+00  0.5714E+00  0.1286E+01  0.0000E+00  0.1000E+01
       0.1457E+02
 x 2:  0.0000E+00  0.1000E+01  0.1429E+00 -0.4286E+00  0.0000E+00  0.0000E+00
       0.2143E+01
 x 1:  0.1000E+01  0.0000E+00  0.1429E+00  0.5714E+00  0.0000E+00  0.0000E+00
       0.5143E+01
  -f:  0.0000E+00  0.0000E+00  0.1000E+01  0.2000E+01  0.0000E+00  0.0000E+00
       0.3000E+02
```

3.85 Minimize $f = 5x + 2y$

subject to $6x + 8y \leq 48$

$x - y \leq 3$

$x + 4y \geq 4$

$3x + y \geq 3$

x unrestricted in sign, $y \geq 0$

Let $x = x_1 - x_2$, $y = x_3$

Adding slack, surplus, and artificial variables, the problem can be stated as:

Minimize $f = 5x_1 - 5x_2 + 2x_3$

subject to $6x_1 - 6x_2 + 8x_3 + x_6 = 48$

$x_1 - x_2 - x_3 + x_7 = 3$

$x_1 - x_2 + 4x_3 - x_4 + y_1 = 4$

$3x_1 - 3x_2 + x_3 - x_5 + y_2 = 3$

with $x_i \geq 0$ $(i = 1, \cdots, 7)$; x_4, x_5 = surplus variables, x_6, x_7 = slack variables, and y_1, y_2 = artificial variables.

Infeasibility form:

$w = y_1 + y_2 = 7 - 4x_1 + 4x_2 - 5x_3 + x_4 + x_5$

Simplex method:

Basic Variables	original variables							Artificial variables		b_i''	$\left(\frac{b_i''}{a_{is}''}\right)$ for $a_{is}'' > 0$
	x_1	x_2	x_3	x_4	x_5	x_6	x_7	y_1	y_2		
x_6	6	−6	8	0	0	1	0	0	0	48	6
x_7	1	−1	−1	0	0	0	1	0	0	3	
y_1	1	−1	(4)	0	0	0	0	1	0	4	1 ←
y_2	3	−3	1	0	0	0	0	0	1	3	
$-f$	5	−5	2	0	0	0	0	0	0	0	
$-w$	−4	4	−5	1	1	0	0	0	0	−7	

↑

x_6	4	−4	0	0	0	1	0	−2	0	40	10
x_7	5/4	−5/4	0	0	0	0	1	1/4	0	4	16/5
x_3	1/4	−1/4	1	0	0	0	0	1/4	0	1	4
y_2	(11/4)	−11/4	0	0	0	0	0	−1/4	1	2	8/11 ←
$-f$	9/2	−9/2	0	0	0	0	0	−1/2	0	−2	
$-w$	−11/4	11/4	0	1	1	0	0	5/4	0	−2	
	↑										
x_6	0	0	0	0	0	1	0	−18/11	−16/11	408/11	
x_7	0	0	0	0	0	0	1	4/11	−5/11	34/11	
x_3	0	0	1	0	0	0	0	3/11	−1/11	9/11	
x_1	1	−1	0	0	0	0	0	−1/11	4/11	8/11	
$-f$	0	0	0	0	0	0	0	−1/11	−18/11	−58/11	
$-w$	0	0	0	1	1	0	0	1	1	0	

optimum solution: $x_6^* = \frac{408}{11}$, $x_7^* = \frac{34}{11}$, $x_3^* = \frac{9}{11}$, $x_1^* = \frac{8}{11}$

$x_2^* = 0$, $x_4^* = 0$, $x_5^* = 0$

$x^* = x_1^* - x_2^* = \frac{8}{11}$

$y^* = x_3^* = \frac{9}{11}$

$f^* = f_{min} = \frac{58}{11}$

3.86

$$3x_1 + 4x_2 + x_3 = 24$$
$$x_1 - x_2 + x_4 = 3$$
$$x_1 + 4x_2 + x_5 = 4$$
$$3x_1 + x_2 - x_6 + x_7 = 3$$
$$-5x_1 - 2x_2 = f$$
$$x_7 = w$$

or

$$-3x_1 - x_2 + x_6 + 3 = w$$

```
PHASE-I OF SIMPLEX METHOD
given system

  x 3:  0.3000E+01  0.4000E+01  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.2400E+02
  x 4:  0.1000E+01 -0.1000E+01  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00
        0.0000E+00  0.3000E+01
  x 5:  0.1000E+01  0.4000E+01  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00
        0.0000E+00  0.4000E+01
  x 7:  0.3000E+01  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.1000E+01
        0.1000E+01  0.3000E+01
   -f: -0.5000E+01 -0.2000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00
   -w: -0.3000E+01 -0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01
        0.0000E+00 -0.3000E+01

minimum value of ds:-0.30000000E+01  variable out of basis:    1
result of pivoting on element:    4    1


  x 3:  0.0000E+00  0.3000E+01  0.1000E+01  0.0000E+00  0.0000E+00  0.1000E+01
       -0.1000E+01  0.2100E+02
  x 4:  0.0000E+00 -0.1333E+01  0.0000E+00  0.1000E+01  0.0000E+00  0.3333E+00
       -0.3333E+00  0.2000E+01
  x 5:  0.0000E+00  0.3667E+01  0.0000E+00  0.0000E+00  0.1000E+01  0.3333E+00
       -0.3333E+00  0.3000E+01
  x 1:  0.1000E+01  0.3333E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.3333E+00
        0.3333E+00  0.1000E+01
   -f:  0.0000E+00 -0.3333E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.1667E+01
        0.1667E+01  0.5000E+01
   -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.1000E+01  0.0000E+00

PHASE-II OF SIMPLEX METHOD
system at start of phase-II

  x 3:  0.0000E+00  0.3000E+01  0.1000E+01  0.0000E+00  0.0000E+00  0.1000E+01
        0.2100E+02
  x 4:  0.0000E+00 -0.1333E+01  0.0000E+00  0.1000E+01  0.0000E+00  0.3333E+00
        0.2000E+01
  x 5:  0.0000E+00  0.3667E+01  0.0000E+00  0.0000E+00  0.1000E+01  0.3333E+00
        0.3000E+01
  x 1:  0.1000E+01  0.3333E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.3333E+00
        0.1000E+01
   -f:  0.0000E+00 -0.3333E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.1667E+01
        0.5000E+01

minimum value of cs:-0.16666667E+01  variable out of basis:    6
result of pivoting on element:    2    6


  x 3:  0.0000E+00  0.7000E+01  0.1000E+01 -0.3000E+01  0.0000E+00  0.0000E+00
        0.1500E+02
  x 6:  0.0000E+00 -0.4000E+01  0.0000E+00  0.3000E+01  0.0000E+00  0.1000E+01
        0.6000E+01
  x 5:  0.0000E+00  0.5000E+01  0.0000E+00 -0.1000E+01  0.1000E+01  0.0000E+00
        0.1000E+01
  x 1:  0.1000E+01 -0.1000E+01  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00
        0.3000E+01
   -f:  0.0000E+00 -0.7000E+01  0.0000E+00  0.5000E+01  0.0000E+00  0.0000E+00
        0.1500E+02

minimum value of cs:-0.70000000E+01  variable out of basis:    2
result of pivoting on element:    3    2
```

```
x 3:  0.0000E+00  0.0000E+00  0.1000E+01 -0.1600E+01 -0.1400E+01  0.0000E+00
      0.1360E+02
x 6:  0.0000E+00  0.0000E+00  0.0000E+00  0.2200E+01  0.8000E+00  0.1000E+01
      0.6800E+01
x 2:  0.0000E+00  0.1000E+01  0.0000E+00 -0.2000E+00  0.2000E+00  0.0000E+00
      0.2000E+00
x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.8000E+00  0.2000E+00  0.0000E+00
      0.3200E+01
 -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.3600E+01  0.1400E+01  0.0000E+00
      0.1640E+02
```

3.87

$$9x_1 + 10x_2 + x_3 = 330$$

$$-21x_1 + 4x_2 + x_4 = 36$$

$$x_1 + 2x_2 - x_5 + x_8 = 6$$

$$6x_1 - x_2 + x_6 = 72$$

$$3x_1 + x_2 + x_7 = 54$$

$$-3x_1 - 2x_2 = f$$

$$x_8 = w$$

or

$$-x_1 - 2x_2 + x_5 + 6 = w$$

```
PHASE-I OF SIMPLEX METHOD
given system

  x 3:  0.9000E+01  0.1000E+02  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00  0.3300E+03
  x 4: -0.2100E+02  0.4000E+01  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00  0.3600E+02
  x 8:  0.1000E+01  0.2000E+01  0.0000E+00  0.0000E+00 -0.1000E+01  0.0000E+00
        0.0000E+00  0.1000E+01  0.6000E+01
  x 6:  0.6000E+01 -0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01
        0.0000E+00  0.0000E+00  0.7200E+02
  x 7:  0.3000E+01  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.1000E+01  0.0000E+00  0.5400E+02
   -f: -0.3000E+01 -0.2000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00  0.0000E+00
   -w: -0.1000E+01 -0.2000E+01  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00
        0.0000E+00  0.0000E+00 -0.6000E+01

minimum value of ds:-0.20000000E+01  variable out of basis:    2
result of pivoting on element:    3    2


  x 3:  0.4000E+01  0.0000E+00  0.1000E+01  0.0000E+00  0.5000E+01  0.0000E+00
        0.0000E+00 -0.5000E+01  0.3000E+03
  x 4: -0.2300E+02  0.0000E+00  0.0000E+00  0.1000E+01  0.2000E+01  0.0000E+00
        0.0000E+00 -0.2000E+01  0.2400E+02
  x 2:  0.5000E+00  0.1000E+01  0.0000E+00  0.0000E+00 -0.5000E+00  0.0000E+00
        0.0000E+00  0.5000E+00  0.3000E+01
  x 6:  0.6500E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.5000E+00  0.1000E+01
        0.0000E+00  0.5000E+00  0.7500E+02
  x 7:  0.2500E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.5000E+00  0.0000E+00
        0.1000E+01 -0.5000E+00  0.5100E+02
   -f: -0.2000E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.1000E+01  0.0000E+00
        0.0000E+00  0.1000E+01  0.6000E+01
   -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.1000E+01  0.0000E+00

PHASE-II OF SIMPLEX METHOD
system at start of phase-II

  x 3:  0.4000E+01  0.0000E+00  0.1000E+01  0.0000E+00  0.5000E+01  0.0000E+00
        0.0000E+00  0.3000E+03
  x 4: -0.2300E+02  0.0000E+00  0.0000E+00  0.1000E+01  0.2000E+01  0.0000E+00
        0.0000E+00  0.2400E+02
  x 2:  0.5000E+00  0.1000E+01  0.0000E+00  0.0000E+00 -0.5000E+00  0.0000E+00
        0.0000E+00  0.3000E+01
  x 6:  0.6500E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.5000E+00  0.1000E+01
        0.0000E+00  0.7500E+02
  x 7:  0.2500E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.5000E+00  0.0000E+00
        0.1000E+01  0.5100E+02
   -f: -0.2000E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.1000E+01  0.0000E+00
        0.0000E+00  0.6000E+01

minimum value of cs:-0.20000000E+01  variable out of basis:    1
result of pivoting on element:    3    1


  x 3:  0.0000E+00 -0.8000E+01  0.1000E+01  0.0000E+00  0.9000E+01  0.0000E+00
        0.0000E+00  0.2760E+03
  x 4:  0.0000E+00  0.4600E+02  0.0000E+00  0.1000E+01 -0.2100E+02  0.0000E+00
        0.0000E+00  0.1620E+03
  x 1:  0.1000E+01  0.2000E+01  0.0000E+00  0.0000E+00 -0.1000E+01  0.0000E+00
        0.0000E+00  0.6000E+01
  x 6:  0.0000E+00 -0.1300E+02  0.0000E+00  0.0000E+00  0.6000E+01  0.1000E+01
        0.0000E+00  0.3600E+02
```

```
 x 7:  0.0000E+00 -0.5000E+01  0.0000E+00  0.0000E+00  0.3000E+01  0.0000E+00
       0.1000E+01  0.3600E+02
  -f:  0.0000E+00  0.4000E+01  0.0000E+00  0.0000E+00 -0.3000E+01  0.0000E+00
       0.0000E+00  0.1800E+02

minimum value of cs:-0.30000000E+01  variable out of basis:     5
result of pivoting on element:     4     5


 x 3:  0.0000E+00  0.1150E+02  0.1000E+01  0.0000E+00  0.0000E+00 -0.1500E+01
       0.0000E+00  0.2220E+03
 x 4:  0.0000E+00  0.5000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.3500E+01
       0.0000E+00  0.2880E+03
 x 1:  0.1000E+01 -0.1667E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1667E+00
       0.0000E+00  0.1200E+02
 x 5:  0.0000E+00 -0.2167E+01  0.0000E+00  0.0000E+00  0.1000E+01  0.1667E+00
       0.0000E+00  0.6000E+01
 x 7:  0.0000E+00  0.1500E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.5000E+00
       0.1000E+01  0.1800E+02
  -f:  0.0000E+00 -0.2500E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.5000E+00
       0.0000E+00  0.3600E+02

minimum value of cs:-0.25000000E+01  variable out of basis:     2
result of pivoting on element:     5     2


 x 3:  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.2333E+01
      -0.7667E+01  0.8400E+02
 x 4:  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.3667E+01
      -0.3333E+00  0.2820E+03
 x 1:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1111E+00
       0.1111E+00  0.1400E+02
 x 5:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01 -0.5556E+00
       0.1444E+01  0.3200E+02
 x 2:  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00 -0.3333E+00
       0.6667E+00  0.1200E+02
  -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00 -0.3333E+00
       0.1667E+01  0.6600E+02

minimum value of cs:-0.33333337E+00  variable out of basis:     6
result of pivoting on element:     1     6


 x 6:  0.0000E+00  0.0000E+00  0.4286E+00  0.0000E+00  0.0000E+00  0.1000E+01
      -0.3286E+01  0.3600E+02
 x 4:  0.0000E+00  0.0000E+00 -0.1571E+01  0.1000E+01  0.0000E+00  0.0000E+00
       0.1171E+02  0.1500E+03
 x 1:  0.1000E+01  0.0000E+00 -0.4762E-01  0.0000E+00  0.0000E+00  0.0000E+00
       0.4762E+00  0.1000E+02
 x 5:  0.0000E+00  0.0000E+00  0.2381E+00  0.0000E+00  0.1000E+01  0.0000E+00
      -0.3810E+00  0.5200E+02
 x 2:  0.0000E+00  0.1000E+01  0.1429E+00  0.0000E+00  0.0000E+00  0.0000E+00
      -0.4286E+00  0.2400E+02
  -f:  0.0000E+00  0.0000E+00  0.1429E+00  0.0000E+00  0.0000E+00  0.0000E+00
       0.5714E+00  0.7800E+02
```

3.88 Minimize $f = -3x_1 - 2x_2$

subject to

$$9x_1 + 10x_2 + x_3 = 330$$
$$-21x_1 + 4x_2 + x_4 = 36$$
$$3x_1 + 6x_2 + x_5 = 18$$
$$6x_1 - x_2 + x_6 = 72$$
$$3x_1 + x_2 + x_7 = 54$$

with all $x_i \geq 0$

Simplex method:

Basic Variables	x_1	x_2	x_3	x_4	x_5	x_6	x_7	b_i''	(b_i''/a_{is}'') for $a_{is}'' > 0$
x_3	9	10	1	0	0	0	0	330	$36\frac{2}{3}$
x_4	−21	4	0	1	0	0	0	36	
x_5	(3)	6	0	0	1	0	0	18	6 ←
x_6	6	−1	0	0	0	1	0	72	12
x_7	3	1	0	0	0	0	1	54	18
$-f$	−3	−2	0	0	0	0	0	0	
	↑								
x_3	0	−8	1	0	−3	0	0	276	
x_4	0	46	0	1	7	0	0	162	
x_1	1	2	0	0	1/3	0	0	6	
x_6	0	−13	0	0	−2	1	0	36	
x_7	0	−5	0	0	−1	0	1	36	
$-f$	0	4	0	0	1	0	0	18	

Optimum solution: $x_2^* = x_5^* = 0$

$x_1^* = 6$, $x_3^* = 276$, $x_4^* = 162$, $x_6^* = x_7^* = 36$

$f^* = f_{min} = -18$

$(x^* = 6, y^* = 0)$

3.89

$$9x_1 + 10x_2 + x_3 = 330$$
$$-21x_1 + 4x_2 + x_4 = 36$$
$$x_1 + 2x_2 + x_5 = 6$$
$$6x_1 - x_2 + x_6 = 72$$
$$3x_1 + x_2 - x_7 + x_8 = 54$$
$$-3x_1 - 2x_2 = f$$
$$x_8 = w$$

or $$-3x_1 - x_2 + x_7 + 54 = w$$

```
PHASE-I OF SIMPLEX METHOD
given system

  x 3:  0.9000E+01  0.1000E+02  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00  0.3300E+03
  x 4: -0.2100E+02  0.4000E+01  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00  0.3600E+02
  x 5:  0.1000E+01  0.2000E+01  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00
        0.0000E+00  0.0000E+00  0.6000E+01
  x 6:  0.6000E+01 -0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01
        0.0000E+00  0.0000E+00  0.7200E+02
  x 8:  0.3000E+01  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
       -0.1000E+01  0.1000E+01  0.5400E+02
   -f: -0.3000E+01 -0.2000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00  0.0000E+00
   -w: -0.3000E+01 -0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.1000E+01  0.0000E+00 -0.5400E+02

minimum value of ds:-0.30000000E+01  variable out of basis:    1
result of pivoting on element:    3    1


  x 3:  0.0000E+00 -0.8000E+01  0.1000E+01  0.0000E+00 -0.9000E+01  0.0000E+00
        0.0000E+00  0.0000E+00  0.2760E+03
  x 4:  0.0000E+00  0.4600E+02  0.0000E+00  0.1000E+01  0.2100E+02  0.0000E+00
        0.0000E+00  0.0000E+00  0.1620E+03
  x 1:  0.1000E+01  0.2000E+01  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00
        0.0000E+00  0.0000E+00  0.6000E+01
  x 6:  0.0000E+00 -0.1300E+02  0.0000E+00  0.0000E+00 -0.6000E+01  0.1000E+01
        0.0000E+00  0.0000E+00  0.3600E+02
  x 8:  0.0000E+00 -0.5000E+01  0.0000E+00  0.0000E+00 -0.3000E+01  0.0000E+00
       -0.1000E+01  0.1000E+01  0.3600E+02
   -f:  0.0000E+00 -0.5000E+01  0.0000E+00  0.0000E+00 -0.3000E+01  0.0000E+00
       -0.1000E+01  0.1000E+01  0.3600E+02
   -w:  0.0000E+00  0.5000E+01  0.0000E+00  0.0000E+00  0.3000E+01  0.0000E+00
        0.1000E+01  0.0000E+00 -0.3600E+02

no feasible solution exists
```

3.90 Maximize $f = 3x + 2y$

subject to $21x - 4y \geq -36$

$3x + 6y \geq 18$

$6x - y \leq 72$

$x \geq 0, \quad y \geq 0$

or Minimize $f = -3x_1 - 2x_2$

subject to $-21x_1 + 4x_2 + y_1 = 36$

$3x_1 + 6x_2 - x_3 + y_2 = 18$

$6x_1 - x_2 + y_3 = 72$

where $x = x_1$, $y = x_2$, x_3 = surplus variable, y_2 = artificial variable; y_1, y_3 = slack variables.

Infeasibility form: $w = y_2$ or $-w + 12x_1 - 9x_2 + x_3 + 126 = 0$

Simplex method:

Basic variables	x_1	x_2	x_3	y_1	y_2	y_3	b_i''	(b_i''/a_{is}'') for $a_{is}'' > 0$
y_1	-21	4	0	1	0	0	36	9
y_2	3	(6)	-1	0	1	0	18	3 ←
y_3	6	-1	0	0	0	1	72	
$-f$	-3	-2	0	0	0	0	0	
$-w$	12	-9	1	0	0	0	-126	
		↑						
y_1	-23	0	(2/3)	1	-2/3	0	24	36 ←
x_2	1/2	1	-1/6	0	1/6	0	3	
y_3	13/2	0	-1/6	0	1/6	1	75	
$-f$	-2	0	-1/3	0	1/3	0	6	
$-w$	33/2	0	-1/2	0	3/2	0	-99	
			↑					

x_3	$-69/2$	0	1	$3/2$	-1	0	36	
x_2	$-21/4$	1	0	$1/4$	0	0	9	
y_3	(3/4)	0	0	$1/4$	0	1	81	108 ←
$-f$	$-27/2$	0	0	$+1/2$	0	0	18	
$-w$	$-3/4$	0	0	$3/4$	1	0	-81	
	↑							
x_3	0	0	1	13	-1	46	3762	
x_2	0	1	0	2	0	7	576	
x_1	1	0	0	$1/3$	0	$4/3$	108	
$-f$	0	0	0	5	0	-18	1476	
$-w$	0	0	0	$4/3$	$4/3$	$4/3$	0	

Optimum solution:

$$x^* = x_1^* = 108 , \quad y^* = x_2^* = 576 , \quad x_3^* = 3762$$

$$f^* = f_{min} = -1476$$

$$y_1^* = y_2^* = y_3^* = 0$$

3.91

Find a and b to minimize

$$f = \sum_{i=1}^{6} |a + b x_i - t_i|$$

$$= |a - 400| + |a + 20b - 350| + |a + 40b - 250|$$
$$+ |a + 60b - 175| + |a + 80b - 100| + |a + 100b - 50|$$

Since $a + b x_i - t_i$ can be positive, negative, or zero, we express it as a difference of two nonnegative numbers as

$$a + b x_i - t_i = y_i - z_i \quad ; \quad y_i \geq 0, \quad z_i \geq 0$$

If we enforce $y_i z_i = 0$, then

$$|a + b x_i - t_i| = y_i + z_i$$

Thus the LP problem can be stated as follows:

Find a, b, y_i $(i = 1, 2, \ldots, 6)$ and z_i $(i = 1, 2, \ldots, 6)$ which minimizes

$$f = \sum_{i=1}^{6} (y_i + z_i)$$

subject to

$$a + b x_i - t_i = y_i - z_i \quad \text{or} \quad a + b x_i - y_i + z_i = t_i \; ; \; i = 1, 2, \ldots, 6$$

since a and b are unrestricted, we define

$$a = a^+ - a^- \quad \text{and} \quad b = b^+ - b^-$$

with $a^+ \geq 0$, $a^- \geq 0$, $b^+ \geq 0$, $b^- \geq 0$

(b) Find a and b to minimize

$$f(a, b) = \max_i \{ |a + b x_i - t_i| \}$$

This problem can be restated as follows:

Find a, b and c to minimize $f = c$

subject to

$$|a + b x_i - t_i| \leq c \; ; \; i = 1, 2, \ldots, 6$$

As before, we can define

$$a + b x_i - t_i = y_i - z_i$$

so that

$$|a + b x_i - t_i| = y_i + z_i$$

This permits the statement of the problem as follows:

Find $a^+, a^-, b^+, b^-, c, y_i\ (i = 1, 2, \ldots, 6)$ and $z_i\ (i = 1, 2, \ldots, 6)$

to minimize $f = c$

subject to

$a + b x_i - y_i + z_i = t_i \quad ; \quad i = 1, 2, \ldots, 6$

$y_i + z_i \leq c \quad ; \quad i = 1, 2, \ldots, 6$

$y_i \geq 0, \quad i = 1, 2, \ldots, 6$

$z_i \geq 0, \quad i = 1, 2, \ldots, 6$

$a^+ \geq 0, \quad a^- \geq 0, \quad b^+ \geq 0, \quad b^- \geq 0, \quad c \geq 0$

3.92 Let x_1 = amount of mixed nuts A used (lb)

x_2 = amount of mixed nuts B used (lb)

Minimize $f = 2.5 x_1 + 3.0 x_2$

subject to

$$0.2 x_1 + 0.1 x_2 \geq 4$$
$$0.1 x_1 + 0.2 x_2 \geq 5$$
$$0.15 x_1 + 0.25 x_2 \geq 6$$
$$x_1 \geq 0, \quad x_2 \geq 0$$

Plotting of constraints:

$g_1 = 0.2 x_1 + 0.1 x_2 - 4 \geq 0$

$g_1 = 0$ at $(0, 40), (20, 0)$

$g_2 = 0.1 x_1 + 0.2 x_2 - 5 \geq 0$

$g_2 = 0$ at $(0, 25), (50, 0)$

$g_3 = 0.15 x_1 + 0.25 x_2 - 6 \geq 0$

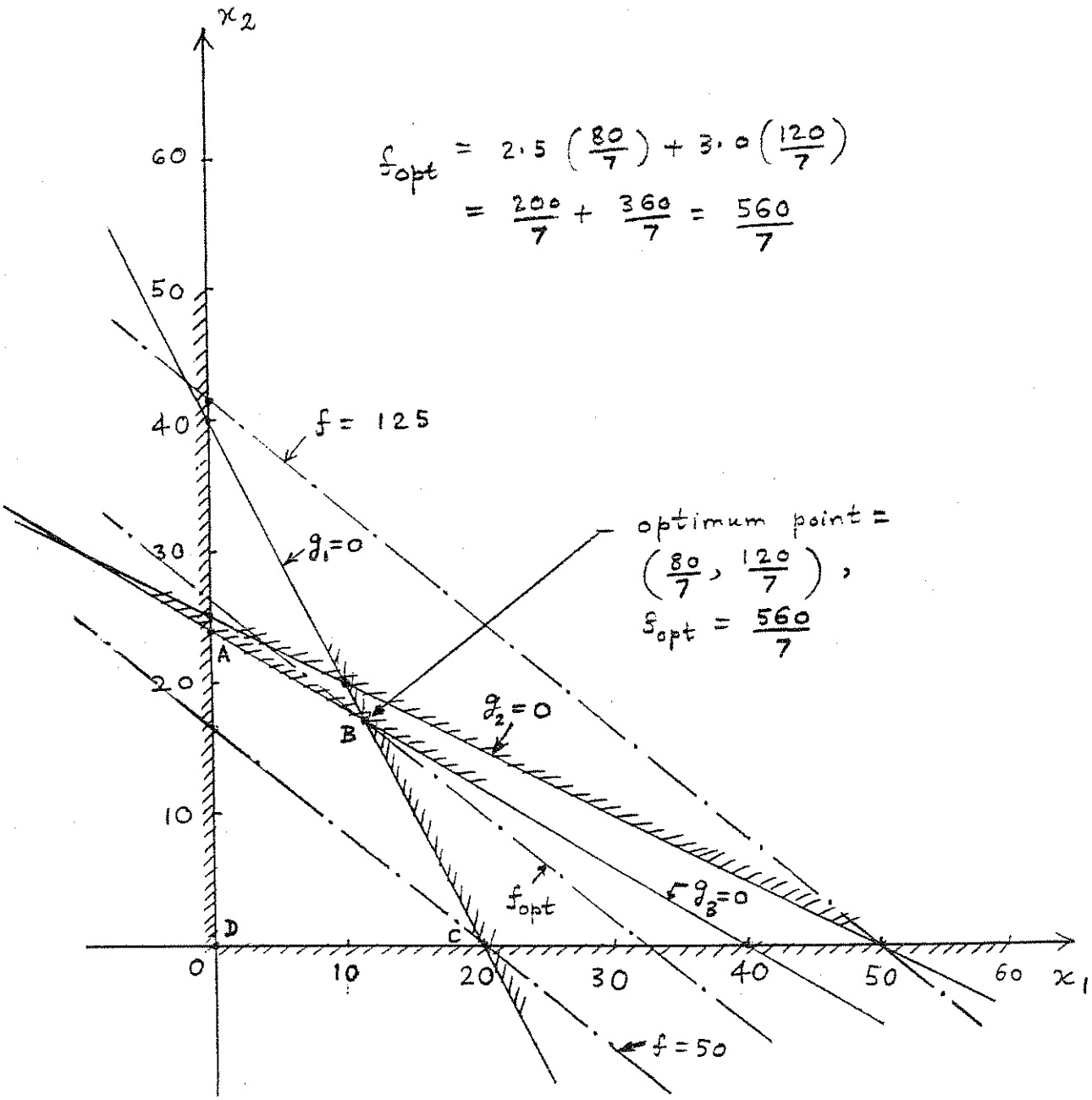
$g_3 = 0$ at $(0, 24), (40, 0)$

Contours of $f = 2.5 x_1 + 3.0 x_2$:

$f = 50$ at $(20, 0), (0, 16\frac{2}{3})$

$f = 125$ at $(50, 0), (0, 41\frac{2}{3})$

Feasible space = ABCDA ; optimum point = B



$$f_{opt} = 2.5\left(\frac{80}{7}\right) + 3.0\left(\frac{120}{7}\right)$$
$$= \frac{200}{7} + \frac{360}{7} = \frac{560}{7}$$

3.93 Let x_i = number of bearings B_i produced per day $(i = 1, 2, 3)$

Maximize $f = 4x_1 + 2x_2 + 3x_3$

subject to

$$10x_1 + 6x_2 + 12x_3 \leq 1200$$

$$8x_1 + 4x_2 + 4x_3 \leq 1000$$

$$0 \leq x_1 \leq 500$$

$$0 \leq x_2 \leq 400$$

$$0 \leq x_3 \leq 600$$

given system

x4:	10.000	6.000	12.000	1.000	0.000	0.000	0.000	0.000	1200.000
x5:	8.000	4.000	4.000	0.000	1.000	0.000	0.000	0.000	1000.000
x6:	1.000	0.000	0.000	0.000	0.000	1.000	0.000	0.000	500.000
x7:	0.000	1.000	0.000	0.000	0.000	0.000	1.000	0.000	400.000
x8:	0.000	0.000	1.000	0.000	0.000	0.000	0.000	1.000	600.000
-f:	-4.000	-2.000	-3.000	0.000	0.000	0.000	0.000	0.000	0.000

minimum value of c1:-0.40000000E+01 variable out of basis: 1 result of pivoting on element: 1 1

x1:	1.000	0.600	1.200	0.100	0.000	0.000	0.000	0.000	120.000
x5:	0.000	-0.800	-5.600	-0.800	1.000	0.000	0.000	0.000	40.000
x6:	0.000	-0.600	-1.200	-0.100	0.000	1.000	0.000	0.000	380.000
x7:	0.000	1.000	0.000	0.000	0.000	0.000	1.000	0.000	400.000
x8:	0.000	0.000	1.000	0.000	0.000	0.000	0.000	1.000	600.000
-f:	0.000	0.400	1.800	0.400	0.000	0.000	0.000	0.000	480.000

Solution: $\vec{X}^* = \begin{Bmatrix} x_1^* \\ x_2^* \\ x_3^* \end{Bmatrix} = \begin{Bmatrix} 120 \\ 0 \\ 0 \end{Bmatrix}$ with $f^* = 480$

3.94

Let x_1 = number of units of A produced per day

x_2 = number of units of B produced per day

Maximize $f = 10x_1 + 15x_2$

subject to

$$g_1 = 16x_1 + 10x_2 - 1500 \leq 0$$

$$g_2 = 10x_1 + 12x_2 - 1000 \leq 0$$

$$g_3 = 4x_1 + 8x_2 - 500 \leq 0$$

$$g_4 = x_1 \geq 0 \quad , \quad g_5 = x_2 \geq 0$$

Plotting of constraints:

$g_1 = 0$ at $(0, 150)$, $(93\frac{3}{4}, 0)$

$g_2 = 0$ at $(0, 83\frac{1}{3})$, $(0, 100)$

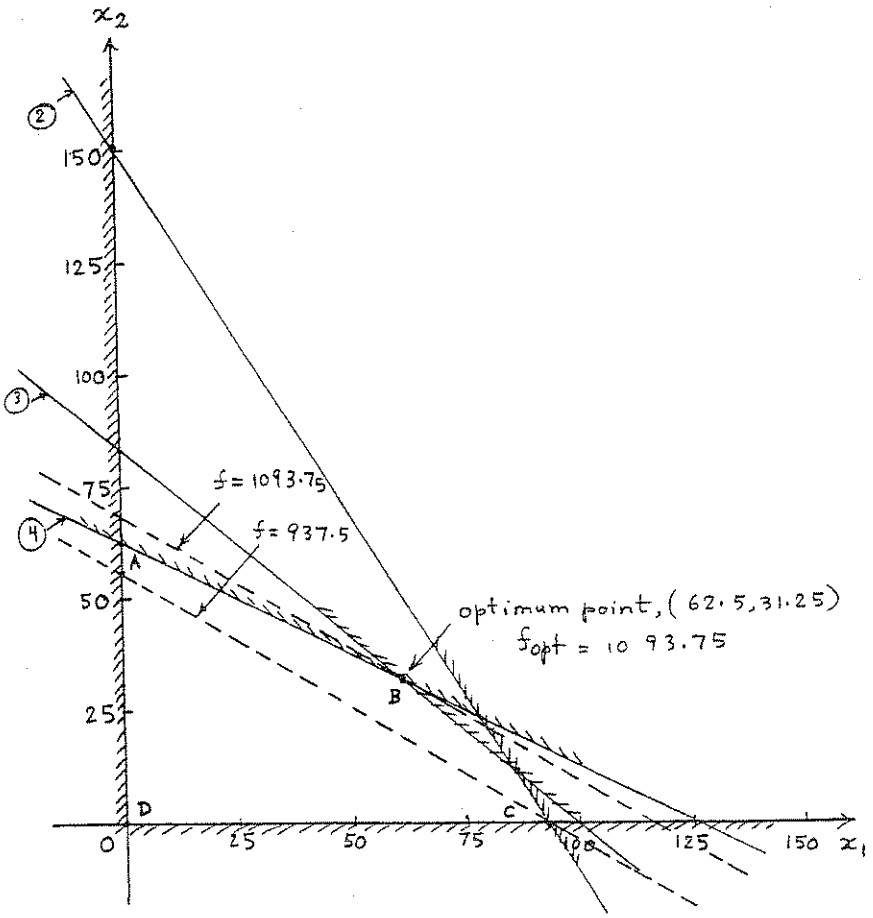
$g_3 = 0$ at $(0, 62\frac{1}{2})$, $(125, 0)$

Feasible space = ABCDA

optimum point = B

Contours of f:

$f = 937.5$ at $(93\frac{3}{4}, 0)$, $(0, 62\frac{1}{2})$



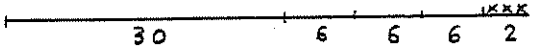
3.95

Roll of 50" – width:

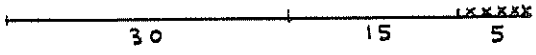
Scheme 1, number of rolls cut according to scheme: x_1



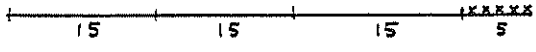
Scheme 2, " x_2



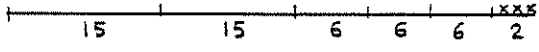
Scheme 3, " x_3



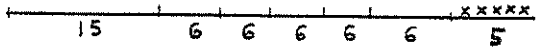
Scheme 4, " x_4



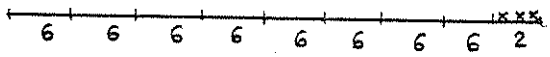
Scheme 5, " x_5



Scheme 6, " x_6

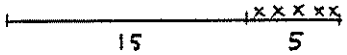


Scheme 7, " x_7

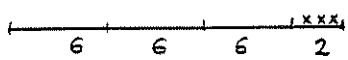


Roll of 20" – width:

Scheme 1, number of rolls cut according to this scheme: y_1



Scheme 2, " y_2



Number of 6" – wide rolls produced:

$$w_1 = x_1 + 3x_2 + 3x_5 + 5x_6 + 8x_7 + 3y_2$$

Number of 15" – wide rolls produced:

$$w_2 = x_3 + 2x_4 + 2x_5 + x_6 + y_1$$

Number of 30" – wide rolls produced:

$$w_3 = x_2 + x_3$$

Number of 40" – wide rolls produced:

$$w_4 = x_1$$

Number of surplus rolls of width 6" produced:

$$g_1 = w_1 - 100$$

Number of surplus rolls of width 15" produced:

$$g_2 = w_2 - 50$$

Number of surplus rolls of width 30" produced:

$$g_3 = w_3 - 200$$

Number of surplus rolls of width 40" produced:

$$g_4 = w_4 - 150$$

$$\vec{x} = \{x_1, \ldots, x_7 ; y_1, y_2 ; g_1, g_2, g_3, g_4\}^T$$

Minimize $f = 4x_1 + 2x_2 + 5x_3 + 5x_4 + 2x_5 + 5x_6 + 2x_7 + 5y_1 + 2y_2 + 6g_1 + 15g_2 + 30g_3 + 40g_4$

with $x_i \geq 0$ $(i = 1, \ldots, 7)$, $y_j \geq 0$ $(j = 1, 2)$, $g_k \geq 0$ $(k = 1, \ldots, 4)$.

3.96 Let (x_1, x_2) = number of units of (P_1, P_2) produced per week.

Maximize $f = 200\,x_1 + 300\,x_2$

subject to

$g_1 = 5x_1 + 4x_2 - 500 \le 0$ (time on lathes)

$g_2 = 2x_1 + 4x_2 - 400 \le 0$ (time on milling machines)

$g_3 = -x_1 \le 0, \quad g_4 = -x_2 \le 0$

$g_1 = 0$ at $(0, 125)$ and $(100, 0)$

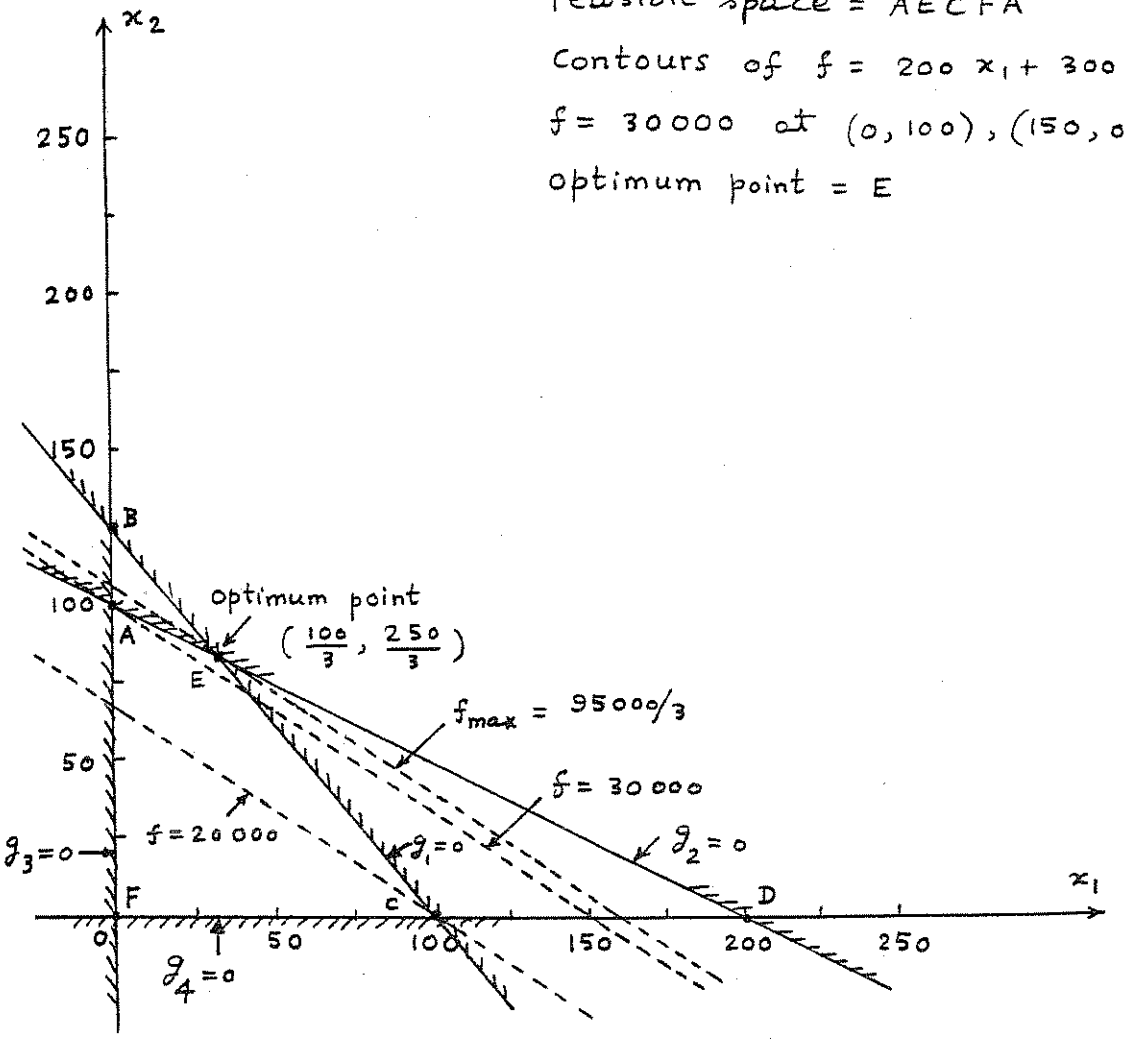
$g_2 = 0$ at $(0, 100)$ and $(200, 0)$

Feasible space = AECFA

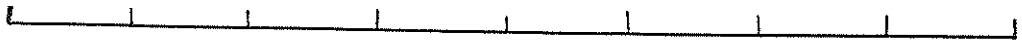
Contours of $f = 200\,x_1 + 300\,x_2$

$f = 30000$ at $(0, 100), (150, 0)$

optimum point = E



3.97



1	2	3	4	5	6	7	8	6-month Period #
x_{11}	x_{21}	x_{31}	x_{41}	x_{51}	x_{61}	x_{71}	x_{81}	Amount invested in: 6 month CD
x_{12}	x_{22}	x_{32}	x_{42}	x_{52}	x_{62}	x_{72}		1 year CD
x_{13}	x_{23}	x_{33}	x_{43}	x_{53}				2 year CD
x_{14}								4 year CD

Assumption:

All CD's mature at end of the 4th year.

Let

x_{ij} = amount invested at beginning of period i in CD type j

($i = 1, 2, \ldots, 8$ for $j = 1$; $i = 1, 2, \ldots, 7$ for $j = 2$;

$i = 1, 2, \ldots, 5$ for $j = 3$; $i = 1$ for $j = 4$)

Money available at beginning of period i:

$i = 1 \rightarrow$ 50,000

$i = 2 \rightarrow$ $x_{11} + 0.05\, x_{11} = 1.05\, x_{11}$

$i = 3 \rightarrow$ $1.05\, x_{21} + 1.07\, x_{12}$

$i = 4 \rightarrow$ $1.05\, x_{31} + 1.07\, x_{22}$

$i = 5 \rightarrow$ $1.05\, x_{41} + 1.07\, x_{32} + 1.10\, x_{13}$

$i = 6 \rightarrow$ $1.05\, x_{51} + 1.07\, x_{42} + 1.10\, x_{23}$

$i = 7 \rightarrow$ $1.05\, x_{61} + 1.07\, x_{52} + 1.10\, x_{33}$

$i = 8 \rightarrow$ $1.05\, x_{71} + 1.07\, x_{62} + 1.10\, x_{43}$

Money available at end of period 8:

$$1.05\, x_{81} + 1.07\, x_{72} + 1.10\, x_{53} + 1.15\, x_{14}$$

Problem:

$$\vec{x} = \{x_{11}, x_{21}, \ldots, x_{81};\ x_{12}, x_{22}, \ldots, x_{72};\ x_{13}, x_{23}, \ldots, x_{53};\ x_{14}\}^T$$

Maximize

$$f = 1.05\, x_{81} + 1.07\, x_{72} + 1.10\, x_{53} + 1.15\, x_{14} - 50000$$

subject to

$$x_{11} + x_{12} + x_{13} + x_{14} = 50\,000$$

$$x_{21} + x_{22} + x_{23} = 1.05\ x_{11}$$

$$x_{31} + x_{32} + x_{33} = 1.05\ x_{21} + 1.07\ x_{12}$$

$$x_{41} + x_{42} + x_{43} = 1.05\ x_{31} + 1.07\ x_{22}$$

$$x_{51} + x_{52} + x_{53} = 1.05\ x_{41} + 1.07\ x_{32} + 1.10\ x_{13}$$

$$x_{61} + x_{62} = 1.05\ x_{51} + 1.07\ x_{42} + 1.10\ x_{23}$$

$$x_{71} + x_{72} = 1.05\ x_{61} + 1.07\ x_{52} + 1.10\ x_{33}$$

$$x_{81} = 1.05\ x_{71} + 1.07\ x_{62} + 1.10\ x_{43}$$

all $x_{ij} \geq 0$

3.98 Let (x_1, x_2) = number of parts (A, B) produced in a month

Maximize $f = x_1 + x_2$

subject to

$$0.6x_1 + 0.9x_2 \leq 10*8*30 = 2400 \qquad (g_1 \leq 0)$$

$$0.4x_1 + 0.1x_2 \leq 3*8*30 = 720 \qquad (g_2 \leq 0)$$

$$0.1x_1 + 0.2x_2 \leq 4*8*30 = 960 \qquad (g_3 \leq 0)$$

$$0.5x_1 + 0.3x_2 \leq 6*8*30 = 1440 \qquad (g_4 \leq 0)$$

$$0.2x_1 + 0.3x_2 \leq 5*8*30 = 1200 \qquad (g_5 \leq 0)$$

$$x_1 \geq 0, \quad x_2 \geq 0 \qquad (g_6 \leq 0, \ g_7 \leq 0)$$

$g_1 = 0$ at $(0, 2666.7)$, $(4000.0, 0)$

$g_2 = 0$ at $(0, 7200)$, $(1800, 0)$

$g_3 = 0$ at $(0, 4800)$, $(9600, 0)$

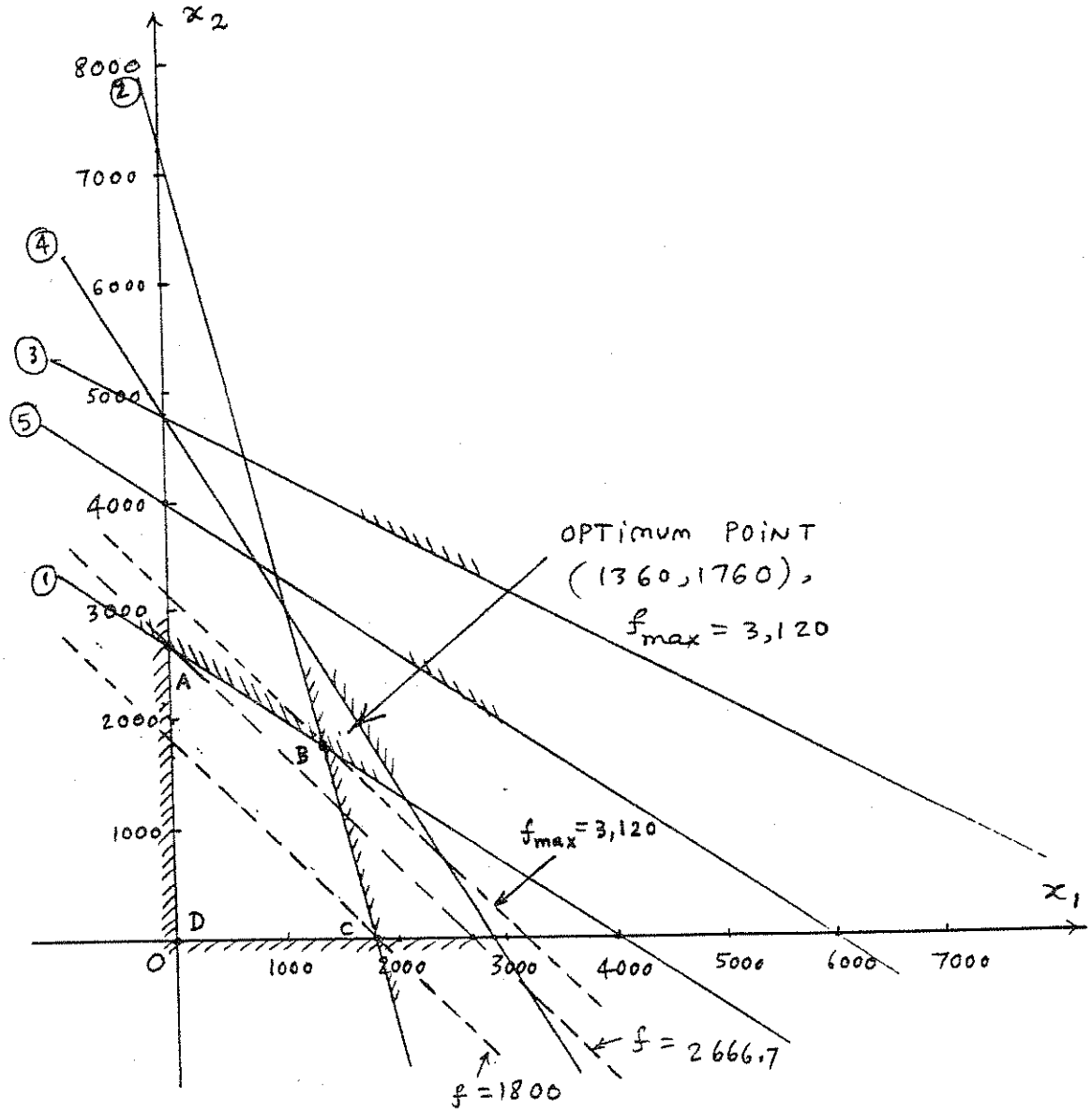
$g_4 = 0$ at $(0, 4800)$, $(2880, 0)$

$g_5 = 0$ at $(0, 4000)$, $(6000, 0)$

Feasible space = ABCDA

Contours of f:

$f = 1800$ at $(1800, 0)$, $(0, 1800)$


x_2
8000
7000
6000
5000
4000
3000
2000
1000
O
1000
2000
3000
4000
5000
6000
7000
x_1
A
B
C
D
OPTimum POiNT
(1360, 1760),
$f_{max} = 3,120$
$f_{max} = 3,120$
$f = 2666.7$
$f = 1800$

(b) If $x_1 = x_2 = x$, problem becomes:

Maximize $f = 2x$

subject to

$1.5x \le 2400$ or $x \le 1600$

$0.5x \le 720$ or $x \le 1440$

$0.3x \le 960$ or $x \le 3200$

$0.8x \le 1440$ or $x \le 1800$

$0.5x \le 1200$ or $x \le 2400$

$x \ge 0$

$\therefore x^* = 1440, \quad f_{max} = 2880.$

3.99 Let (x_1, x_2) = number of units of (A, B) sold per month and number of potential customers for $(x_1, x_2) = (y_1, y_2)$

$x_1 = 0.4\, y_1$, $x_2 = 0.8\, y_2 \Rightarrow y_1 = 2.5\, x_1$, $y_2 = 1.25\, x_2$

Money to be spent per month ($) $= 3(2.5\, x_1) + 1(1.25\, x_2)$

$= 7.5\, x_1 + 1.25\, x_2$

Time to be spent per month (hrs) $= 3(2.5\, x_1) + 1(1.25\, x_2)$

$= 7.5\, x_1 + 1.25\, x_2$

Income = income from sales − expenses

$= 0.2(250\, x_1) + 0.2(100\, x_2) - 7.5\, x_1 - 1.25\, x_2$

$= 42.5\, x_1 + 18.75\, x_2$

Problem: Maximize $f = 42.5\, x_1 + 18.75\, x_2$

subject to $g_1 = 7.5 x_1 + 1.25\, x_2 - 200 \le 0$

g_2, g_3: $10 \le x_1 \le 25$

g_4, g_5: $10 \le x_2 \le 25$

$g_1 = 0$ at $(0, 160)$, $(26.67, 0)$, $(10, 100)$, $(20, 40)$

Feasible space = APQBA in figure

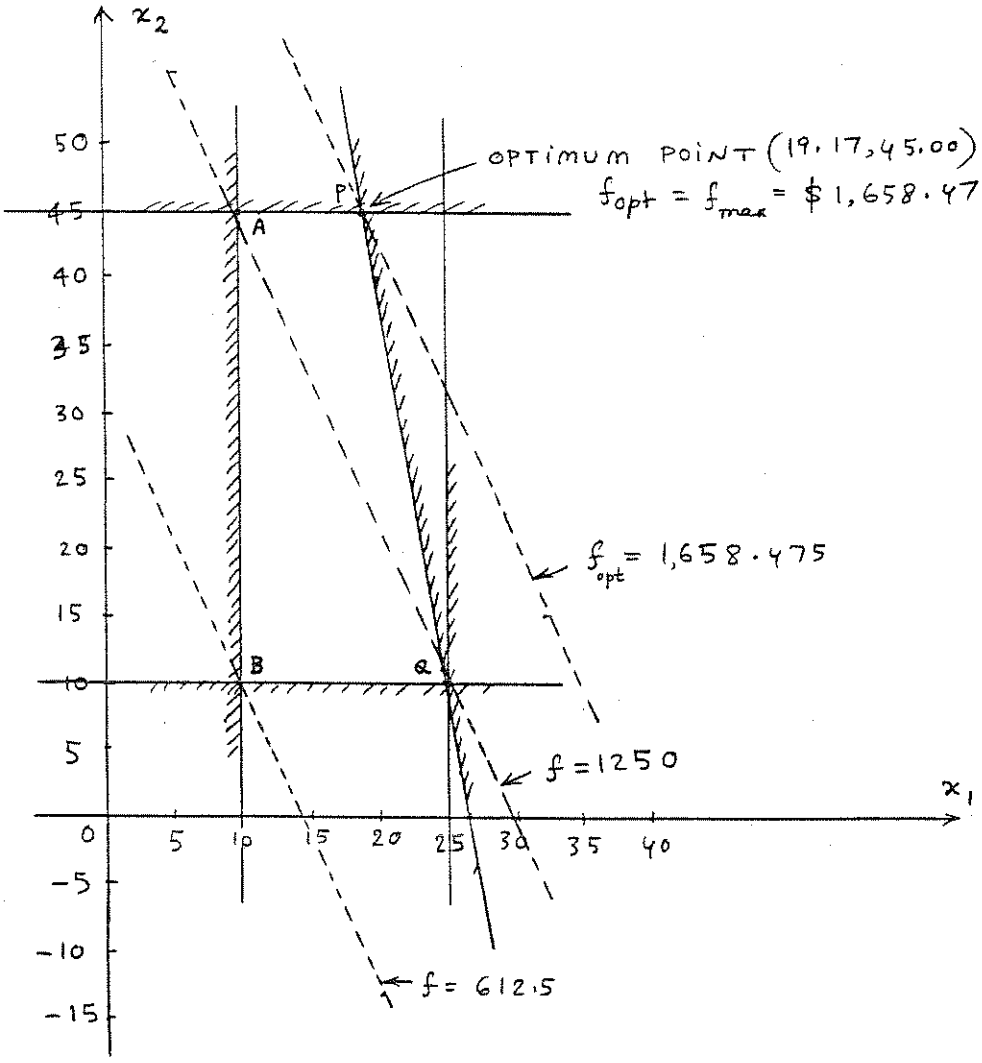
Contours of f:

$f = 612.5$ at $(10, 10)$, $(20, -12.67)$

$f = 1250$ at $(25, 10)$, $(10, 44)$

$f = 1658.475$ at $P(19.17, 45.00)$

optimum point = P.



3.100 Let (x_1, x_2) = power generated at plant (A, B) in Mwh

Pollution level at A = $(2.5\,x_1)\,1.0 + (1.0\,x_1)\,1.5 + (3.0\,x_1)\,2.0 = 10\,x_1$

Pollution level at B = $(1.5\,x_2)\,1.5 + (2.0\,x_2)\,2.0 + (2.5\,x_2)\,2.5 = 12.5\,x_2$

Cost of operating plant A = $(2.5\,x_1)\,20 + (1.0\,x_1)\,25 + (3.0\,x_1)\,18$

$= 129\,x_1$

Cost of operating plant B = $(1.5\,x_2)\,18 + (2.0\,x_2)\,28 + (2.5\,x_2)\,12$

$= 113\,x_2$

(a) Find (x_1, x_2) which minimizes $f = 10\,x_1 + 12.5\,x_2$

subject to $x_1 \geq 30, \quad x_2 \geq 80$

(b) Find (x_1, x_2) which minimizes $f = 129\,x_1 + 113\,x_2$

subject to $x_1 \geq 30, \quad x_2 \geq 80$

3.101

Design variables:

x_{ij} = quantity of vegetable type i (i = 1 to 5) bought from farm j (j = 1 to 4) in tons

objective function: (Minimize the cost):

$$f = 200\,x_{11} + 600\,x_{21} + 1600\,x_{31} + 800\,x_{41} + 1200\,x_{51} + 300\,x_{12} + 550\,x_{22} + 1400\,x_{32} + 850\,x_{42} + 1100\,x_{52} + 250\,x_{13} + 650\,x_{23} + 1500\,x_{33} + 700\,x_{43} + 1000\,x_{53} + 150\,x_{14} + 500\,x_{24} + 1700\,x_{34} + 900\,x_{44} + 1300\,x_{54}$$

Constraints:

$$x_{11} + x_{21} + x_{31} + x_{41} + x_{51} \leq 180$$

$$x_{12} + x_{22} + x_{32} + x_{42} + x_{52} \leq 200$$

$$x_{13} + x_{23} + x_{33} + x_{43} + x_{53} \leq 100$$

$$x_{14} + x_{24} + x_{34} + x_{44} + x_{54} \leq 120$$

$$x_{11} + x_{12} + x_{13} + x_{14} \geq 100$$

$$x_{21} + x_{22} + x_{23} + x_{24} \geq 60$$

$$x_{31} + x_{32} + x_{33} + x_{34} \geq 20$$

$$x_{41} + x_{42} + x_{43} + x_{44} \geq 80$$

$$x_{51} + x_{52} + x_{53} + x_{54} \geq 40$$

all $x_{ij} \geq 0$

3.102

Let x_i = number of days used for process type i in a month (i = 1, 2, 3, 4)

Cost per month for process 1 = $(50*5 + 10*3 + 150*6)\,x_1 = 1180\,x_1$

Cost per month for process 2 = $(50*8 + 10*5 + 150*12)\,x_2 = 2250\,x_2$

Cost per month for process 3 = $(50*3 + 10*2 + 150*5)\,x_3 = 920\,x_3$

Cost per month for process 4 = $(50*10 + 10*7 + 150*12)\,x_4 = 2370\,x_4$

Income from steel = $350*4\,x_1 + 350*6x_2 + 350*2x_3 + 350*6\,x_4$

Income from side products = $100*1\,x_1 + 100*2\,x_2 + 100*1\,x_3 + 100*4\,x_4$

Profit = Total income − total cost

$$f = 320x_1 + 50x_2 - 120x_3 + 130x_4$$

Constraints:

Availability of iron ore: $5x_1 + 8x_2 + 3x_3 + 10x_4 \leq 600$

Availability of coal: $3x_1 + 5x_2 + 2x_3 + 7x_4 \leq 250$

Limitation of side products: $x_1 + 2x_2 + x_3 + 4x_4 \leq 200$

Limitation of number of days: $x_1 + x_2 + x_3 + x_4 \leq 30$

$x_i \geq 0$, $i = 1, 2, 3, 4$

given system

x5:	5.000	8.000	3.000	10.000	1.000	0.000	0.000	0.000	600.000
x6:	3.000	5.000	2.000	7.000	0.000	1.000	0.000	0.000	250.000
x7:	1.000	2.000	1.000	4.000	0.000	0.000	1.000	0.000	200.000
x8:	1.000	1.000	1.000	1.000	0.000	0.000	0.000	1.000	30.000
-f:	-320.000	-50.000	120.000	-130.000	0.000	0.000	0.000	0.000	0.000

minimum value of c1:-0.32000000E+03 variable out of basis: 1 result of pivoting on element: 4 1

x5:	0.000	3.000	-2.000	5.000	1.000	0.000	0.000	-5.000	450.000
x6:	0.000	2.000	-1.000	4.000	0.000	1.000	0.000	-3.000	160.000
x7:	0.000	1.000	0.000	3.000	0.000	0.000	1.000	-1.000	170.000
x1:	1.000	1.000	1.000	1.000	0.000	0.000	0.000	1.000	30.000
-f:	0.000	270.000	440.000	190.000	0.000	0.000	0.000	320.000	9600.000

Solution:

$$\vec{x}^* = \begin{Bmatrix} 30 \\ 0 \\ 0 \\ 0 \end{Bmatrix} , \quad f^* = 9600$$

3.103

M-file: LP02.m

```
clear all; clc; close all;
f = [2 3 2 -1 1]';
A = [];
b = []';
lb = zeros(5,1);
Aeq = [3 -3 4 2 -1;1 1 1 3 1];
beq = [0 2]';
options = optimset('largeScale','off','simplex','on');
[x,fval,exitflag,output] = linprog(f,A,b,Aeq,beq,lb,[],[],optimset('Display','iter'))
```

Output (Results):

```
Residuals:    Primal      Dual      Duality     Total
              Infeas     Infeas       Gap        Rel
              A*x-b    A'*y+z-f     x'*z       Error
-------------------------------------------------------
  Iter    0:  8.59e+002 6.10e+000 2.05e+003 7.00e+002
  Iter    1:  7.20e+001 1.35e-015 2.18e+002 3.60e+001
  Iter    2:  5.07e-015 8.01e-016 6.64e+000 1.89e+000
  Iter    3:  4.97e-016 4.25e-015 3.10e+000 1.24e+000
  Iter    4:  1.64e-013 8.88e-016 9.25e-001 9.25e-001
  Iter    5:  2.22e-016 9.85e-015 6.09e-002 6.09e-002
  Iter    6:  5.53e-014 1.68e-015 1.09e-003 1.09e-003
  Iter    7:  6.75e-016 9.93e-016 5.52e-008 5.52e-008
  Iter    8:  4.44e-016 6.66e-016 5.66e-015 5.72e-015
Optimization terminated.

x =

    0.0000
    0.0000
    0.0000
    0.4000
    0.8000


fval =

    0.4000


exitflag =

     1


output =                          algorithm: 'large-scale: interior point'
                                cgiterations: 0
      iterations: 8                  message: 'Optimization terminated.'
```

3.104

M-file: LP03.m

```
clear all; clc; close all;
f = [-6 -7]';
A = [7 6;5 9;1 -1];
b = [42 45 4]';
lb = zeros(2,1);
Aeq = [];
beq = [];
options = optimset('largeScale','off','simplex','on');
[x,fval,exitflag,output] = linprog(f,A,b,Aeq,beq,lb,[],[],optimset('Display','iter'))
```

Output (Results):

```
 Residuals:   Primal      Dual      Duality    Total
              Infeas     Infeas      Gap       Rel
              A*x-b    A'*y+z-f     x'*z      Error
 ---------------------------------------------------------
 Iter    0:  1.99e+003 1.96e+001 2.48e+003 1.30e+003
 Iter    1:  1.98e+002 1.40e-015 2.89e+002 3.21e+000
 Iter    2:  2.03e-014 2.52e-015 5.04e+001 6.29e-001
 Iter    3:  8.88e-016 4.26e-014 1.31e+001 2.77e-001
 Iter    4:  9.29e-013 8.95e-016 1.88e+001 3.12e-001
 Iter    5:  1.02e-014 9.49e-016 6.39e-001 1.50e-002
 Iter    6:  2.29e-013 8.95e-016 3.00e-004 7.16e-006
 Iter    7:  7.12e-015 1.78e-015 3.00e-009 7.16e-011
Optimization terminated.

x =

    3.2727
    3.1818


fval =

  -41.9091


exitflag =

     1


output =

      iterations: 7
       algorithm: 'large-scale: interior point'
    cgiterations: 0
         message: 'Optimization terminated.'
```

3.105

M-file: LP04.m

```
clear all; clc; close all;
f = [1 -8]';
A = [-3 -2;1 -1;9 7;3 7;-2 5];
b = [-6 6 108 70 35]';
lb = zeros(2,1);
Aeq = [];
beq = [];
options = optimset('largeScale','off','simplex','on');
[x,fval,exitflag,output] = linprog(f,A,b,Aeq,beq,lb,[],[],optimset('Display','iter'))
```

Output (Results):

```
 Residuals:    Primal      Dual      Duality     Total
               Infeas     Infeas       Gap        Rel
               A*x-b     A'*y+z-f     x'*z       Error
 ---------------------------------------------------
 Iter     0:  1.97e+003 1.81e+001 2.96e+003 7.00e+002
 Iter     1:  3.94e-013 2.27e-015 2.33e+002 9.07e-001
 Iter     2:  3.18e-014 3.66e-015 4.02e+001 4.85e-001
 Iter     3:  5.66e-011 9.18e-016 6.14e+000 9.17e-002
 Iter     4:  1.65e-011 1.55e-014 6.90e-001 1.07e-002
 Iter     5:  4.53e-013 9.24e-016 1.02e-004 1.60e-006
 Iter     6:  4.17e-014 4.58e-016 1.02e-009 1.60e-011
Optimization terminated.

x =

    3.6207
    8.4483


fval =

  -63.9655


exitflag =

     1


output =

      iterations: 6
       algorithm: 'large-scale: interior point'
    cgiterations: 0
         message: 'Optimization terminated.'
```

3.106

M-file: LP05.m

```
clear all; clc; close all;
f = [-240 -104 -60 -19]';
A = [20 9 6 1;10 4 2 1];
b = [20 10]';
lb = zeros(4,1);
Aeq = [];
beq = [];
options = optimset('largeScale','off','simplex','on');
[x,fval,exitflag,output] = linprog(f,A,b,Aeq,beq,lb,[],[],optimset('Display','iter'))
```

Output (Results):

```
 Residuals:    Primal      Dual      Duality     Total
               Infeas     Infeas       Gap        Rel
               A*x-b     A'*y+z-f     x'*z       Error
 ---------------------------------------------------------
 Iter     0:  4.09e+003 5.94e+002 8.12e+004 4.23e+004
 Iter     1:  3.73e+002 1.46e-014 6.71e+003 1.67e+001
 Iter     2:  4.14e-014 1.16e-013 3.41e+002 6.21e-001
 Iter     3:  3.55e-015 1.42e-013 2.31e+001 9.00e-002
 Iter     4:  7.58e-012 1.87e-013 2.50e+000 1.01e-002
 Iter     5:  4.37e-012 1.48e-014 1.90e+000 7.70e-003
 Iter     6:  8.88e-015 1.60e-014 8.93e-003 3.63e-005
 Iter     7:  1.68e-014 3.66e-015 4.47e-007 1.82e-009
Optimization terminated.

x =

    0.0000
    2.0000
    0.0000
    2.0000


fval =

 -246.0000


exitflag =

     1


output =

      iterations: 7
       algorithm: 'large-scale: interior point'
    cgiterations: 0
         message: 'Optimization terminated.'
```

3.107

M-file: LP06.m

```
clear all; clc; close all;
f = [200 600 1600 800 1200 300 550 1400 850 1100 250 650 1500 700 1000 150 500 1700 900 1300]';
A = [1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0;...
   0 0 0 0 0 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0;...
   0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 0 0 0 0 0;...
   0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1;...
   -1 0 0 0 0 -1 0 0 0 0 -1 0 0 0 0 -1 0 0 0 0;...
   0 -1 0 0 0 0 -1 0 0 0 0 -1 0 0 0 0 -1 0 0 0;...
   0 0 -1 0 0 0 0 -1 0 0 0 0 -1 0 0 0 0 -1 0 0;...
   0 0 0 -1 0 0 0 0 -1 0 0 0 0 -1 0 0 0 0 -1 0;...
   0 0 0 0 -1 0 0 0 0 -1 0 0 0 0 -1 0 0 0 0 -1];
b = [180 200 100 120 -100 -60 -20 -80 -40]';
lb = zeros(20,1);
Aeq = [];
beq = [];
options = optimset('largeScale','off','simplex','on');
[x,fval,exitflag,output] = linprog(f,A,b,Aeq,beq,lb,[],[],optimset('Display','iter'))
```

Output (Results):

```
Residuals:    Primal      Dual      Duality     Total
              Infeas     Infeas       Gap        Rel
              A*x-b     A'*y+z-f     x'*z       Error
-------------------------------------------------------
Iter    0:  1.05e+003 5.39e+003 4.63e+006 1.73e+006
Iter    1:  1.68e-013 7.71e-013 6.46e+005 1.81e+000
Iter    2:  7.72e-014 7.15e-013 1.52e+005 6.77e-001
Iter    3:  9.07e-012 7.24e-011 5.88e+004 2.87e-001
Iter    4:  2.19e-012 8.47e-012 8.63e+003 4.86e-002
Iter    5:  3.29e-011 6.96e-013 1.14e+003 6.54e-003
Iter    6:  1.94e-013 5.47e-013 1.89e+000 1.09e-005
Iter    7:  1.23e-013 5.92e-013 9.45e-005 5.46e-010
Optimization terminated.

x =

   20.7229
    0.0000
    0.0000
    9.5927
    0.0000
    0.0000
   19.2771
   20.0000
    0.0000
   10.4073
    0.0000
    0.0000
    0.0000
```

```
   70.4073
   29.5927
   79.2771
   40.7229
    0.0000
    0.0000
    0.0000


fval =

  1.7300e+005


exitflag =

     1


output =

      iterations: 7
       algorithm: 'large-scale: interior point'
    cgiterations: 0
         message: 'Optimization terminated.'
```

Chapter 4
Linear Programming II:
Additional Topics and Extensions

4.1

Minimize $f = -5x_1 + 2x_2 + 5x_3 - 3x_4$

subject to

$$2x_1 + x_2 - x_3 = 6$$

$$3x_1 + 8x_3 + x_4 = 7$$

$$x_i \geq 0, \quad i = 1, 2, 3, 4$$

$$\vec{X}^* = \begin{Bmatrix} x_1^* \\ x_2^* \\ x_3^* \\ x_4^* \end{Bmatrix} = \begin{Bmatrix} 2.333 \\ 1.333 \\ 0 \\ 0 \end{Bmatrix}, \quad f^* = -9.0$$

4.2

Maximize $f = 15x_1 + 6x_2 + 9x_3 + 2x_4$

subject to

$$10x_1 + 5x_2 + 25x_3 + 3x_4 \leq 50$$

$$12x_1 + 4x_2 + 12x_3 + x_4 \leq 48$$

$$7x_1 + x_4 \leq 35$$

$$x_i \geq 0, \quad i = 1, 2, 3, 4$$

$$\vec{X}^* = \begin{Bmatrix} x_1^* \\ x_2^* \\ x_3^* \\ x_4^* \end{Bmatrix} = \begin{Bmatrix} 2 \\ 6 \\ 0 \\ 0 \end{Bmatrix}, \quad f^* = 66.0$$

4.3

Minimize $f=2x_1+3x_2+2x_3-x_4+x_5$ subject to

$$3x_1-3x_2+4x_3+2x_4-x_5=0$$
$$x_1+x_2+x_3+3x_4+x_5=2$$
$$x_i\geq 0,\ i=1 \text{ to } 5$$

Solution

STEP 1: Since the given constraint equations are not in canonical form, we start with Phase I. Thus we add the artificial variables x_6 and x_7, introduce the infeasibility form

$$w=x_6+x_7,$$

and write the system of equations (including f and w equations) in canonical form as

$$\left.\begin{aligned} 3x_1-3x_2+4x_3+2x_4-x_5+x_6 &= b_1=0\\ x_1+x_2+x_3+3x_4+x_5+x_7 &= b_2=2\\ c_1x_1+c_2x_2+c_3x_3+c_4x_4+c_5x_5-f &= 0\\ d_1x_1+d_2x_2+d_3x_3+d_4x_4+d_5x_5-w &= -w_0 \end{aligned}\right\} \quad (E_1)$$

where

$$c_1=2,\ c_2=3,\ c_3=2,\ c_4=-1,\ c_5=1;$$
$$d_1=-a_{11}-a_{21}=-4$$
$$d_2=-a_{12}-a_{22}=2$$
$$d_3=-a_{13}-a_{23}=-5$$
$$d_4=-a_{14}-a_{24}=-5$$
$$d_5=-a_{15}-a_{25}=0$$

and

$$w_0=b_1+b_2=0+2=2.$$

Equations (E_1) are written in tableau form as shown in Table 4.1.

TABLE 4.1

ORIGINAL SYSTEM OF EQUATIONS

Admissible variables (original)					Artificial variables		Objective variables		Constants
x_1	x_2	x_3	x_4	x_5	x_6	x_7	$-f$	$-w$	
3	−3	4	2	−1	1	0	0	0	0
1	1	1	3	1	0	1	0	0	2
2	3	2	−1	1	0	0	1	0	0
−4	2	−5	−5	0	0	0	0	1	−2

STEP 2: The iterative procedure (cycle 0) starts with x_6, x_7, $-f$ and $-w$ as basic variables. A tableau is opened by entering the coefficients of the basic variables and the constant terms as shown in Table 4.2. The inverse of the basis, $[\beta_{ij}]$ will be an identity matrix, and the simplex multipliers corresponding to $f(\pi_i)$ and $w(\sigma_i)$, $i=1, 2$, will be zero.

TABLE 4.2

TABLEAU AT THE BEGINNING OF CYCLE 0

Basic variables	Columns of the canonical form				Values of the basic variables (constants)	x_3†
	x_6	x_7	$-f$	$-w$		
x_6	1	0	0	0	0	$\boxed{\bar{a}_{63}=4}$ Pivot
x_7	0	1	0	0	2	$\bar{a}_{73}=1$
	Inverse of the basis $=[\beta_{ij}]$					
$-f$	$-\pi_1=0$	$-\pi_2=0$	1	0	0	$\bar{c}_3=2$
$-w$	$-\sigma_1=0$	$-\sigma_2=0$	0	1	-2	$\bar{d}_3=-5$

† This column is filled at the end of step 5.

STEP 3: The relative cost factors $\bar{d}_j$ are computed as

$$\bar{d}_j=d_j-\sigma^T A_j\equiv d_j,\ j=1 \text{ to } 7$$

since all σ_i are zero.

$$\therefore\ \bar{d}_1=-4,\ \bar{d}_2=2,\ \bar{d}_3=-5,\ \bar{d}_4=-5,\ \bar{d}_5=0,\ \bar{d}_6=0,\ \bar{d}_7=0.$$

These coefficients are entered in the first row of Table 4.3 as shown.

TABLE 4.3

RELATIVE COST FACTORS $\bar{d}_j$ AND $\bar{c}_j$

Cycle number		Variable x_j						
		x_1	x_2	x_3	x_4	x_5	x_6	x_7
$(\bar{d}_j)$	Phase I							
	Cycle 0	-4	2	-5	-5	0	0	0
	Cycle 1	$-1/4$	$-7/4$	0	$-5/2$	$-5/4$	5/4	0
	Cycle 2	7/2	$-11/2$	5	0	$-5/2$	5/2	0
	Cycle 3	0	0	0	0	0	1	1
$(\bar{c}_j)$	Phase II							
	Cycle 3	49/11	0	59/11	0	$-2/11$	—	—
	Cycle 4	21/5	2/5	5	0	0	—	—

STEP 4: Since all $\bar{d}_j$ are not ≥ 0, the present solution is not optimal. Hence select a variable x_s to be brought into the basis in the next cycle such that

$$\bar{d}_s=\min(\bar{d}_j<0)$$

Since there is a tie between $\bar{d}_3$ and $\bar{d}_4$, we take arbitrarily that $\bar{d}_s=\bar{d}_3$ and hence $x_s=x_3$.

STEP 5: Compute the elements of x_s column as

$$\bar{A}_s=[\beta_{ij}]\,A_s$$

where $[\beta_{ij}]$ is available in Table 4.2 and A_s in Table 4.1.

$$\therefore \quad \bar{A}_3=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\begin{Bmatrix}4\\1\end{Bmatrix}=\begin{Bmatrix}4\\1\end{Bmatrix}$$

Enter this vector along with the values of $\bar{d}_3$ and $\bar{c}_3$, in the last column of Table 4.2. The value of $\bar{c}_3$ is given by

$$\bar{c}_3=c_3-\pi^T A_3=c_3=2.$$

STEP 6: Select the variable (x_r) to be dropped from the current basic set as:

$$\frac{\bar{b}_r}{\bar{a}_{rs}}=\min_{\bar{a}_{is}>0}\left(\frac{\bar{b}_i}{\bar{a}_{is}}\right)$$

Here

$$(\bar{b}_6/\bar{a}_{63})=0/4=0 \text{ and } (b_7/\bar{a}_{73})=2/1=2$$

and hence

$$x_r=x_6$$

STEP 7: To bring x_3 into the basis in place of x_6, pivot on $\bar{a}_{rs}=\bar{a}_{63}$ in Table 4.2. Enter the result in Table 4.4 keeping its last column as blank. We now go to step 3 to start cycle 1.

TABLE 4.4

TABLEAU AT THE BEGINNING OF CYCLE 1

Basic variables	Columns of the original canonical form				Value of the basic variable	$x_s=x_4$†
	x_6	x_7	$-f$	$-w$		
x_3	1/4	0	0	0	0	$\boxed{\frac{1}{4}=\bar{c}_{34}}$ Pivot
x_7	−1/4	1	0	0	2	$5/2=\bar{a}_{74}$
	← β_{ij} →					
$-f$	$-1/2=-\pi_1$	$0=-\pi_2$	1	0	0	$-2=\bar{c}_4$
$-w$	$5/4=-\sigma_1$	$0=-\sigma_2$	0	1	−2	$-5/2=\bar{d}_4$

† This column is entered at the end of step 5.

STEP 3: The relative cost factors are calculated as

$$\bar{d}_j=d_j-\sigma^T A_j$$

where $-\sigma_i$ are given by the row of $-w$ in Table 4.4 and A_j are given in Table 4.1.

$$\therefore \quad \bar{d}_1=d_1-\sigma_1 a_{11}-\sigma_2 a_{21}=-4-(-5/4)(3)-(0)(1)=-1/4$$

$$\bar{d}_2=d_2-\sigma_1 a_{12}-\sigma_2 a_{22}=2-(-5/4)(-3)-(0)(1)=-7/4$$

$$\bar{d}_3=d_3-\sigma_1 a_{13}-\sigma_2 a_{23}=-5-(-5/4)(4)-(0)(1)=0$$

$$\bar{d}_4=d_4-\sigma_1 a_{14}-\sigma_2 a_{24}=-5-(-5/4)(2)-(0)(3)=-5/2$$

$$\bar{d}_5=d_5-\sigma_1 a_{15}-\sigma_2 a_{25}=0-(-5/4)(-1)-(0)(1)=-5/4$$

$$\bar{d}_6=d_6-\sigma_1 a_{16}-\sigma_2 a_{26}=0-(-5/4)(1)-(0)(0)=5/4$$

$$\bar{d}_7=d_7-\sigma_1 a_{17}-\sigma_2 a_{27}=0-(-5/4)(0)-(0)(1)=0$$

Enter these values in the second row of Table 4.3.

STEP 4: Since all $\bar{d}_j$ are not ≥ 0, the current solution is not optimum. Hence select the variable x_s that enters the basis in the next cycle as

$$\bar{d}_s = \min\,(\bar{d}_j < 0) = \bar{d}_4 \text{ in this case.}$$

STEP 5: Compute the elements of x_s column as

$$\bar{\mathbf{A}}_s = [\beta_{ij}]\,\mathbf{A}_s$$

where $[\beta_{ij}]$ is available in Table 4.4 and $\mathbf{A}_s$ in Table 4.1.

$$\therefore\quad \bar{\mathbf{A}}_4 = \begin{bmatrix} 1/4 & 0 \\ -1/4 & 1 \end{bmatrix} \begin{Bmatrix} 2 \\ 3 \end{Bmatrix} = \begin{Bmatrix} 1/2 \\ 5/2 \end{Bmatrix}$$

Enter these values along with the values of $\bar{c}_4 = c_4 - \pi^T \mathbf{A}_4 = -1 - (\tfrac{1}{2})(2) - (0)(3) = -2$ and $\bar{d}_4 = -5/2$ in the last column of Table 4.4.

STEP 6: Find the variable x_r to be dropped from the basis as

$$\frac{\bar{b}_r}{\bar{a}_{rs}} = \min_{\bar{a}_{is}>0} \left(\frac{\bar{b}_i}{\bar{a}_{is}}\right)$$

Since $\bar{b}_3/\bar{a}_{34} = 0/(1/2) = 0$, and $\bar{b}_7/\bar{a}_{74} = 2/(5/2) = 4/5$, we find that

$$x_r = x_3.$$

STEP 7: To bring x_4 into the basis in place of x_3, pivot on $\bar{a}_{34}$ in Table 4.4. Enter the result in Table 4.5 by keeping its last column blank. We now go to step 3 to start the next cycle.

TABLE 4.5

TABLEAU AT THE BEGINNING OF CYCLE 2

Basic variables	Columns of the original canonical form				Value of the basic variable	$x_s = x_2^{\dagger}$
	x_6	x_7	$-f$	$-w$		
x_4	1/2	0	0	0	0	$\bar{a}_{42} = -3/2$
x_7	−3/2	1	0	0	2	$\boxed{\bar{a}_{72} = 11/2}$ Pivot
	← $[\beta_{ij}]$ →					
$-f$	$-\pi_1 = 1/2$	$-\pi_2 = 0$	1	0	0	$\bar{c}_2 = 3/2$
$-w$	$-\sigma_1 = 5/2$	$-\sigma_2 = 0$	0	1	−2	$\bar{d}_2 = -11/2$

†This column is filled at the end of step 5.

STEP 3: The relative cost factors are computed as

$$\bar{d}_j = d_j - \sigma^T \mathbf{A}_j$$

$$\therefore\quad \bar{d}_1 = d_1 - \sigma_1 a_{11} - \sigma_2 a_{21} = -4 - \left(-\frac{5}{2}\right)(3) - (0)(1) = \frac{7}{2}$$

$$\bar{d}_2 = d_2 - \sigma_1 a_{12} - \sigma_2 a_{22} = 2 - \left(-\frac{5}{2}\right)(-3) - (0)(1) = -\frac{11}{2}$$

$$\bar{d}_3=d_3-\sigma_1 a_{13}-\sigma_2 a_{23}=-5-\left(-\frac{5}{2}\right)(4)-(0)(1)=5$$

$$\bar{d}_4=d_4-\sigma_1 a_{14}-\sigma_2 a_{24}=-5-\left(-\frac{5}{2}\right)(2)-(0)(3)=0$$

$$\bar{d}_5=d_5-\sigma_1 a_{15}-\sigma_2 a_{25}=0-\left(-\frac{5}{2}\right)(-1)-(0)(1)=-\frac{5}{2}$$

$$\bar{d}_6=d_6-\sigma_1 a_{16}-\sigma_2 a_{26}=0-\left(-\frac{5}{2}\right)(1)-(0)(0)=\frac{5}{2}$$

$$\bar{d}_7=d_7-\sigma_1 a_{17}-\sigma_2 a_{27}=0-\left(-\frac{5}{2}\right)(0)-(0)(1)=0$$

Enter these values as third row in Table 4.3.

STEP 4: Since all $\bar{d}_j$ are not ≥ 0, the present solution is not optimal. Hence we select a variable x_s to enter the next basis as

$$x_s=\min\,(\bar{d}_j<0)=x_2$$

STEP 5: Compute the elements of x_s column as

$$\bar{A}_2=[\beta_{ij}]\,A_2=\begin{bmatrix}1/2 & 0\\ -3/2 & 1\end{bmatrix}\begin{Bmatrix}-3\\ 1\end{Bmatrix}=\begin{Bmatrix}-3/2\\ 11/2\end{Bmatrix}$$

Enter this column, along with the values of

$$\bar{c}_2=c_2-\pi^T A_2=3-(-\tfrac{1}{2})(-3)-(0)(1)=3/2,$$

and $\bar{d}_2=-11/2$ in the last column of Table 4.5.

STEP 6: Select a variable (x_r) to be dropped from the basis using the criteria

$$\frac{\bar{b}_r}{\bar{a}_{rs}}=\min_{\bar{a}_{is}>0}\left(\frac{\bar{b}_i}{\bar{a}_{is}}\right)$$

Since $\bar{a}_{72}$ is the only positive element, $x_r=x_7$.

STEP 7: Pivot on the element $\bar{a}_{rs}=\bar{a}_{72}$ to bring x_2 into the basis in place of x_7 in Table 4.5. Enter the result in Table 4.6 by keeping its last column as blank. Now go to step 3 to start the next cycle.

TABLE 4.6

TABLEAU AT THE BEGINNING OF CYCLE 3

Basic variables	Columns of the original canonical form				Value of the basic variable	$x_s=x_5$†
	x_6	x_7	$-f$	$-w$		
x_4	1/11	3/11	0	0	6/11	$\bar{a}_{45}=2/11$
x_2	−3/11	2/11	0	0	4/11	$\bar{a}_{25}=5/11$ Pivot
	← $[\beta_{ij}]$ →					
$-f$	$10/11=-\pi_1$	$-3/11=-\pi_2$	1	0	−6/11	$\bar{c}_5=-2/11$
$-w$	$1=-\sigma_1$	$1=-\sigma_2$	0	1	0	

†This column is filled at the end of step 5.

STEP 3: Since the basic set consists of the original or admissible variables x_4 and x_2 (without any artificial variable), we now start Phase II computations. As a verification, we compute the relative cost factors $\bar{d}_j$ as

$$\bar{d}_1=d_1-\sigma_1 a_{11}-\sigma_2 a_{21}=-4-(-1)(3)-(-1)(1)=0$$
$$\bar{d}_2=d_2-\sigma_1 a_{12}-\sigma_2 a_{22}=2-(-1)(-3)-(-1)(1)=0$$
$$\bar{d}_3=d_3-\sigma_1 a_{13}-\sigma_2 a_{23}=-5-(-1)(4)-(-1)(1)=0$$
$$\bar{d}_4=d_4-\sigma_1 a_{14}-\sigma_2 a_{24}=-5-(-1)(2)-(-1)(3)=0$$
$$\bar{d}_5=d_5-\sigma_1 a_{15}-\sigma_2 a_{25}=0-(-1)(-1)-(-1)(1)=0$$
$$\bar{d}_6=d_6-\sigma_1 a_{16}-\sigma_2 a_{26}=0-(-1)(1)-(-1)(0)=1$$
$$\bar{d}_7=d_7-\sigma_1 a_{17}-\sigma_2 a_{27}=0-(-1)(0)-(-1)(1)=1$$

Since all $\bar{d}_j$ corresponding to the original variables are zero, this checks the conclusion that Phase I has been successfully completed. Also the value of w can be seen to be zero.

Since we are starting with Phase II, we compute the relative cost factors corresponding to $f(\bar{c}_j)$ as

$$\bar{c}_1=c_1-\pi_1 a_{11}-\pi_2 a_{21}=2-\left(-\frac{10}{11}\right)(3)-\left(\frac{3}{11}\right)(1)=\frac{49}{11}$$

$$\bar{c}_2=c_2-\pi_1 a_{12}-\pi_2 a_{22}=3-\left(-\frac{10}{11}\right)(-3)-\left(\frac{3}{11}\right)(1)=0$$

$$\bar{c}_3=c_3-\pi_1 a_{13}-\pi_2 a_{23}=2-\left(-\frac{10}{11}\right)(4)-\left(\frac{3}{11}\right)(1)=\frac{59}{11}$$

$$\bar{c}_4=c_4-\pi_1 a_{14}-\pi_2 a_{24}=-1-\left(-\frac{10}{11}\right)(2)-\left(\frac{3}{11}\right)(3)=0$$

$$\bar{c}_5=c_5-\pi_1 a_{15}-\pi_2 a_{25}=1-\left(-\frac{10}{11}\right)(-1)-\left(\frac{3}{11}\right)(1)=-\frac{2}{11}$$

Since the row and the column corresponding to w will be dropped, we don't have to compute $\bar{c}_6$ and $\bar{c}_7$ corresponding to artificial variables.

Enter the values of $\bar{d}_j$ and $\bar{c}_j$ as shown in Table 4.3.

STEP 4: Since all $\bar{c}_j$ are not ≥ 0, the current solution is not optimal. Hence select a variable x_s to enter the next basis as

$$\bar{c}_s=\min\ (\bar{c}_j<0)=\bar{c}_5 \text{ in this case.}$$

$$\therefore \qquad x_s=x_5$$

STEP 5: Compute the elements of $\bar{A}_s$ as

$$\bar{A}_5=[\beta_{ij}]\,A_5=\begin{bmatrix} 1/11 & 3/11 \\ -3/11 & 2/11 \end{bmatrix}\begin{Bmatrix} -1 \\ 1 \end{Bmatrix}=\begin{Bmatrix} 2/11 \\ 5/11 \end{Bmatrix}$$

Enter this vector, along with the value of $\bar{c}_5$, in the last column of Table 4.6.

STEP 6: Select a variable (x_r) to be dropped from the basis in the next cycle such that

$$\frac{\bar{b}_r}{\bar{a}_{rs}}=\min_{\bar{a}_{is}<0}\left(\frac{\bar{b}_i}{\bar{a}_{is}}\right)$$

Since $\bar{b}_4/\bar{a}_{45}=3$, and $\bar{b}_2/\bar{a}_{25}=4/5$, $x_r=x_2$.

STEP 7: To bring x_5 into the basis in place of x_2, pivot on the element $\bar{a}_{25}$ in Table 4.6 and enter the result in Table 4.7.

TABLE 4.7

TABLEAU AT THE BEGINNING OF CYCLE 4

Basic variables	Columns of the original canonical form			Value of the basic variable	x_s†
	x_6	x_7	$-f$		
x_4	1/5	1/5	0	2/5	
x_5	−3/5	2/5	0	4/5	
	← $[\beta_{ij}]$ →				
$-f$	$-\pi_1=4/5$	$-\pi_2=-1/5$	1	−2/5	

†This column is blank at the beinning of cycle 4.

STEP 3: Compute the relative cost coefficients $\bar{c}_j$ as

$$\bar{c}_j=c_j-\pi^T A_j$$

$$\therefore\ \bar{c}_1=c_1-\pi_1 a_{11}-\pi_2 a_{21}=2-\left(-\frac{4}{5}\right)(3)-\left(\frac{1}{5}\right)(1)=21/5$$

$$\bar{c}_2=c_2-\pi_1 a_{12}-\pi_2 a_{22}=3-\left(-\frac{4}{5}\right)(-3)-\left(\frac{1}{5}\right)(1)=2/5$$

$$\bar{c}_3=c_3-\pi_1 a_{13}-\pi_2 a_{23}=2-\left(-\frac{4}{5}\right)(4)-\left(\frac{1}{5}\right)(1)=5$$

$$\bar{c}_4=c_4-\pi_1 a_{14}-\pi_2 a_{24}=-1-\left(-\frac{4}{5}\right)(2)-\left(\frac{1}{5}\right)(3)=0$$

$$\bar{c}_5=c_5-\pi_1 a_{15}-\pi_2 a_{25}=1-\left(-\frac{4}{5}\right)(-1)-\left(\frac{1}{5}\right)(1)=0$$

Enter these value in Table 4.3 .

STEP 4: Since all $\bar{c}_j>0$, the current solution is optimum with

$$x_4=2/5,\ x_5=4/5 \quad \text{(basic variables)},$$

$$x_1=x_2=x_3=0 \quad \text{(nonbasic variables)},$$

and

$$f_{\min}=2/5.$$

4.4 The regular simplex method is not computationally efficient for implementation on digital computers. It computes and stores much information that is not required in the current and subsequent iterations. The revised simplex method computes and stores only the information that is needed in the current iteration. The inverse of the next basis is obtained directly from the inverse of the current basis, without inverting the basis matrix directly.

4.5 (a) Maximize $f = x_2$

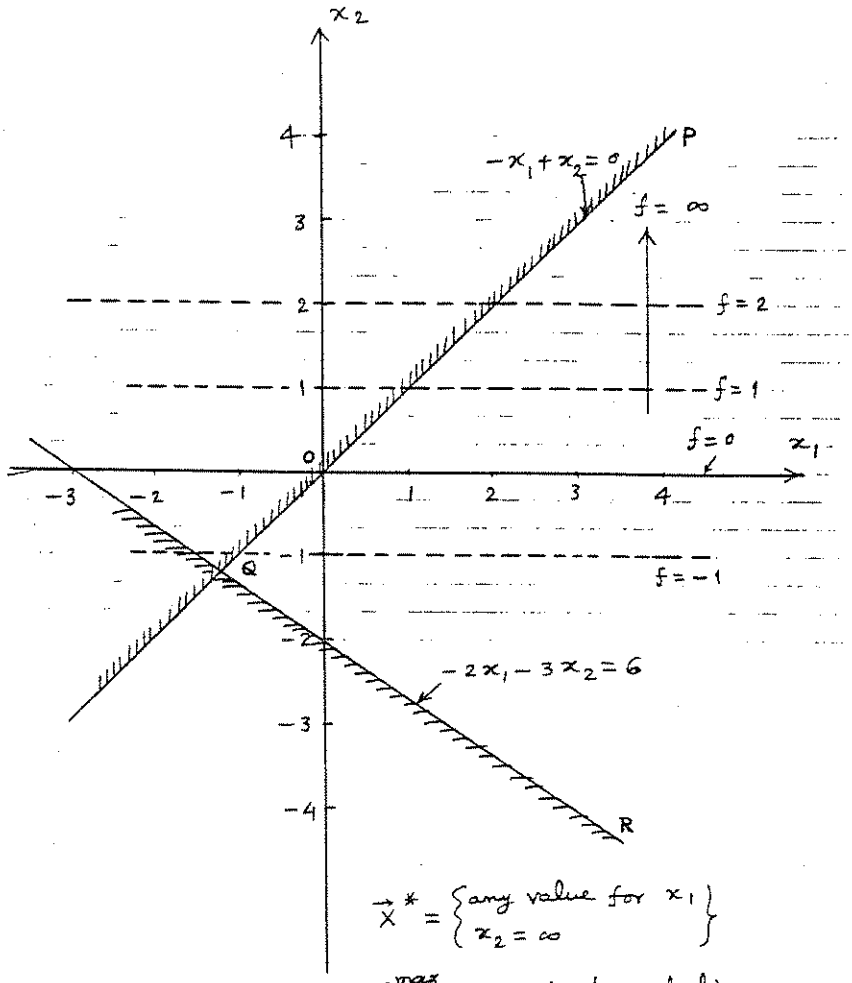
subject to $g_1 = -x_1 + x_2 \le 0$, $g_2 = -2x_1 - 3x_2 - 6 \le 0$

x_1, x_2 unrestricted in sign

$g_1 = 0$ at $x_1 = x_2$

$g_2 = 0$ at $(0, -2), (-3, 0)$

Solution is unbounded.



$\vec{X}^* = \begin{cases} \text{any value for } x_1 \\ x_2 = \infty \end{cases}$

$f^{max} = \infty$ (unbounded)

(b) Let $x_1 = y_1 - y_2$, $x_2 = y_3 - y_4$; $y_i \ge 0$ $(i = 1, 2, 3, 4)$

Maximize $f = y_3 - y_4$

subject to $-y_1 + y_2 + y_3 - y_4 \le 0$

$-2y_1 + 2y_2 - 3y_3 + 3y_3 - 6 \le 0$

Solution : Unbounded.

4.6

Minimize $f = 3x_1 + x_3 + 2x_5$

subject to $-x_1 - x_3 + x_4 - x_5 = 1$

$-x_2 + 2x_3 - 3x_4 - 2x_5 = 2$

$x_i \geq 0, \quad i = 1 \text{ to } 5$

No feasible solution exists.

4.7

(a)

Primal problem

Minimize $f = -4x_1 - 2x_2$

subject to

$x_1 + 2x_2 = 8$

$x_1 - 2x_2 \geq 2$

$-x_1 + x_2 \geq -11$

$x_1 \geq 0$

x_2 = unrestricted in sign

Dual problem

Maximize $v = 8y_1 + 2y_2 - 11y_3$

subject to

$2y_1 - 2y_2 + y_3 = -2$

$y_1 + y_2 - y_3 \leq -4$

y_1 = unrestricted in sign

$y_2 \geq 0, \quad y_3 \geq 0$

(b) solution of dual problem:

```
PHASE-I OF SIMPLEX METHOD
given system

  x 6: -0.2000E+01  0.2000E+01 -0.1000E+01  0.2000E+01  0.0000E+00  0.1000E+01
       0.0000E+00  0.2000E+01
  x 7: -0.1000E+01 -0.1000E+01  0.1000E+01  0.1000E+01 -0.1000E+01  0.0000E+00
       0.1000E+01  0.4000E+01
   -f: -0.8000E+01 -0.2000E+01  0.1100E+02  0.8000E+01  0.0000E+00  0.0000E+00
       0.0000E+00  0.0000E+00
   -w:  0.3000E+01 -0.1000E+01  0.0000E+00 -0.3000E+01  0.1000E+01  0.0000E+00
       0.0000E+00 -0.6000E+01

minimum value of ds:-0.30000000E+01  variable out of basis:     4
result of pivoting on element:     1     4


  x 4: -0.1000E+01  0.1000E+01 -0.5000E+00  0.1000E+01  0.0000E+00  0.5000E+00
       0.0000E+00  0.1000E+01
  x 7:  0.0000E+00 -0.2000E+01  0.1500E+01  0.0000E+00 -0.1000E+01 -0.5000E+00
       0.1000E+01  0.3000E+01
   -f:  0.0000E+00 -0.1000E+02  0.1500E+02  0.0000E+00  0.0000E+00 -0.4000E+01
       0.0000E+00 -0.8000E+01
   -w:  0.0000E+00  0.2000E+01 -0.1500E+01  0.0000E+00  0.1000E+01  0.1500E+01
       0.0000E+00 -0.3000E+01

minimum value of ds:-0.15000000E+01  variable out of basis:     3
result of pivoting on element:     2     3
```

```
 x 4: -0.1000E+01  0.3333E+00  0.0000E+00  0.1000E+01 -0.3333E+00  0.3333E+00
       0.3333E+00  0.2000E+01
 x 3:  0.0000E+00 -0.1333E+01  0.1000E+01  0.0000E+00 -0.6667E+00 -0.3333E+00
       0.6667E+00  0.2000E+01
  -f:  0.0000E+00  0.1000E+02  0.0000E+00  0.0000E+00  0.1000E+02  0.1000E+01
      -0.1000E+02 -0.3800E+02
  -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01
       0.1000E+01  0.0000E+00

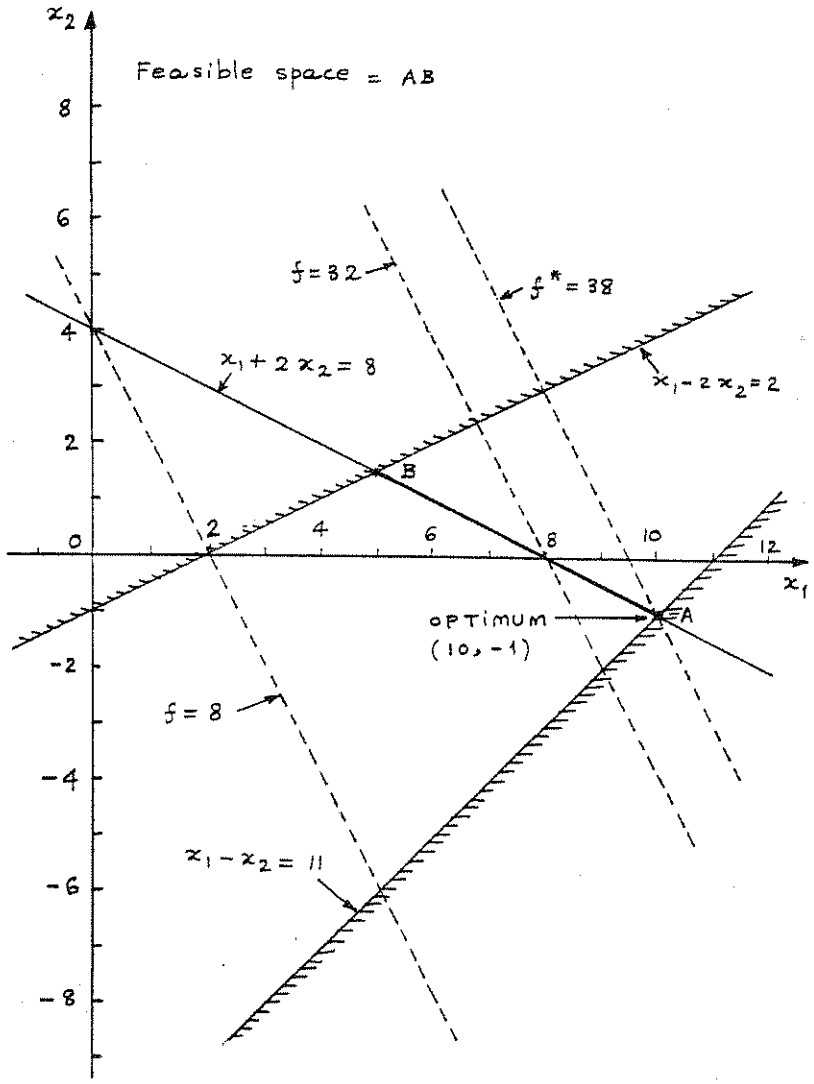
PHASE-II OF SIMPLEX METHOD
system at start of phase-II

 x 4: -0.1000E+01  0.3333E+00  0.0000E+00  0.1000E+01 -0.3333E+00  0.2000E+01

 x 3:  0.0000E+00 -0.1333E+01  0.1000E+01  0.0000E+00 -0.6667E+00  0.2000E+01

  -f:  0.0000E+00  0.1000E+02  0.0000E+00  0.0000E+00  0.1000E+02 -0.3800E+02
```

(c) Graphical solution:



4.8

Problem: Maximize $f = 2 \times 10^6 x_1 + 3 \times 10^6 x_2$

subject to

$$x_1 + x_2 \leq 5$$
$$x_1 \leq 4$$
$$x_2 \leq 4$$
$$x_i \geq 0, \quad i = 1, 2$$

Solution: $x_1^* = 1, \quad x_2^* = 4, \quad f^* = 14 \times 10^6$

4.9

$$\vec{X}^* = \{ x_1^*, x_2^*, x_3^*, x_4^*, x_5^*, x_6^*, x_7^* \}^T$$
$$= \{ 0, 0, 0.5, 1.5, 0, 0, 0 \}^T$$
$$f^* = f_{min} = 24.0$$

4.10

Primal problem:

Minimize
$$f = 2x_1 + x_2 - 5x_3$$
subject to
$$-x_1 + 2x_2 - x_3 \geq -8$$
$$3x_1 - 2x_2 \geq -18$$
$$-2x_1 - x_2 + 2x_3 \geq 4$$
x_1, x_2, x_3 unrestricted in sign

Dual problem:

Maximize
$$v = -8y_1 - 18y_2 + 4y_3$$
subject to
$$-y_1 + 3y_2 - 2y_3 = 2$$
$$2y_1 - 2y_2 - y_3 = 1$$
$$-y_1 + 2y_3 = -5$$
$$y_i \geq 0, \quad i = 1, 2, 3$$

No feasible solution exists.

4.11

Primal problem:

Minimize $f = x_1 - 2x_2$

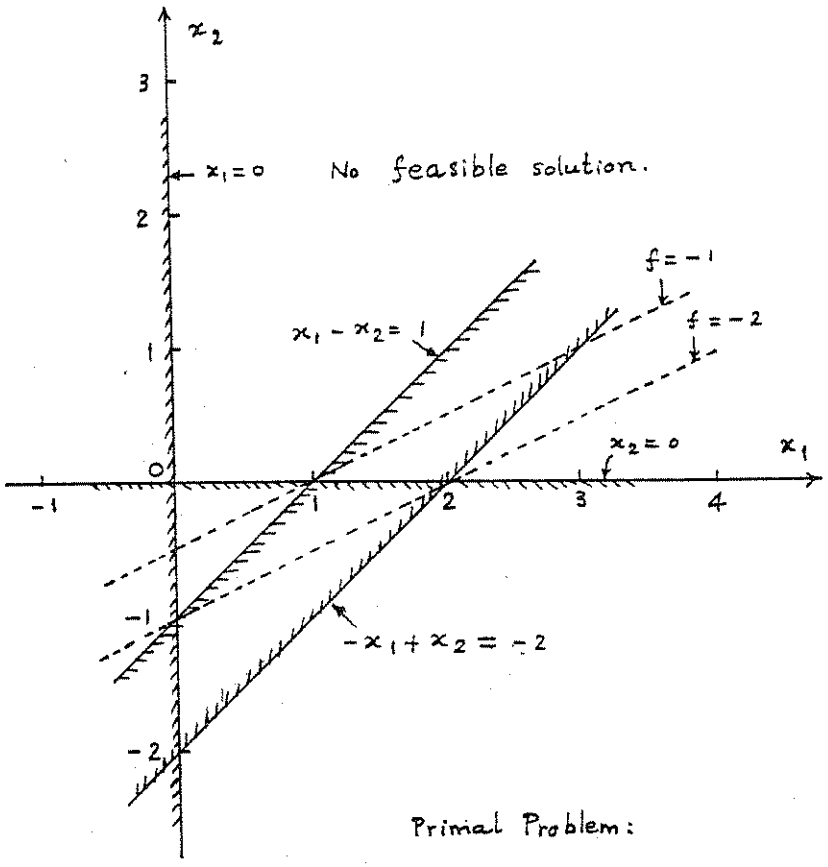
subject to
$$x_1 - x_2 \geq 2$$
$$-x_1 + x_2 \geq -1 \quad \text{or} \quad x_1 - x_2 \leq 1$$
$$x_1 \geq 0, \quad x_2 \geq 0$$

Dual problem:

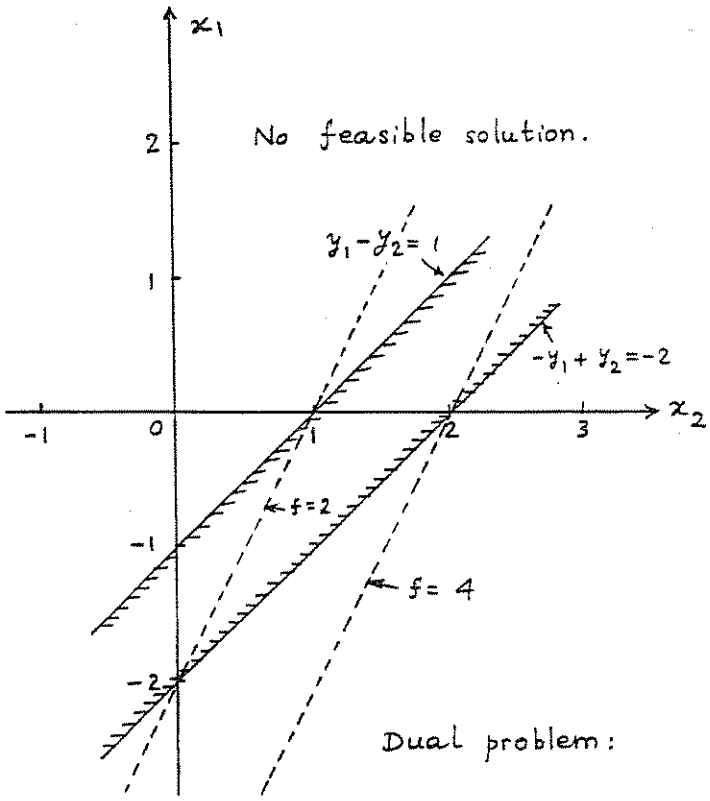
Maximize $v = 2y_1 - y_2$

subject to
$$y_1 - y_2 \leq 1$$
$$-y_1 + y_2 \leq -2 \quad \text{or} \quad y_1 - y_2 \geq 2$$
$$y_1 \geq 0, \quad y_2 \geq 0$$

No feasible solution due to conflicting constraints.



Primal Problem:



Dual problem:

4.12 Problem: Maximize $f = 8x_1 + 3x_2 + 8x_3 + 6x_4$

subject to

$$\begin{aligned} 4x_1 + 3x_2 + x_3 + 3x_4 &\le 16 \\ 4x_1 - x_2 + x_3 &\le 12 \end{aligned}$$

$$\begin{aligned} x_1 + 2x_2 &\le 8 \\ 3x_1 + x_2 &\le 10 \end{aligned}$$

$$\begin{aligned} 2x_3 + 3x_4 &\le 9 \\ 4x_3 + x_4 &\le 12 \end{aligned}$$

$$x_i \ge 0, \quad i = 1, 2, 3, 4$$

Solution: $f^* = 47.9$, $x_1^* = 2.35$, $x_2^* = 0.10$, $x_3^* = 2.70$, $x_4^* = 1.20$

4.13 Problem: Minimize $f = 10x_1 + 2x_2 + 4x_3 + 8x_4 + x_5$

subject to

$$\begin{aligned} 3x_1 + x_4 + x_5 &\ge 8 \\ x_1 + 2x_4 - x_5 &\ge 20 \end{aligned}$$

$$\begin{aligned} x_1 + 4x_2 - x_3 &\ge 16 \\ 2x_1 + x_2 + x_3 &\ge 4 \end{aligned}$$

$$x_i \ge 0, \quad i = 1 \text{ to } 5$$

Solution: $x_1^* = 0.0$

$x_2^* = 4.0$

$x_3^* = 0.0$

$x_4^* = 10.0$

$x_5^* = 0.0$

$f^* = 88.0$

4.14

Let the primal problem be given by

Maximize $f=2x_1+x_2$ subject to

$$x_1-2x_2\geqslant 2$$
$$x_1+2x_2=8$$
$$x_1-x_2\leqslant 11$$

where

$$x_1\geqslant 0$$

and x_2 is unrestricted in sign.

In order to write the corresponding dual problem using the relations of Table 4.13, rewrite the primal problem as:

Minimize $f=-2x_1-x_2$ subject to

$$x_1-2x_2\geqslant 2$$
$$x_1+2x_2=8$$
$$-x_1+x_2\geqslant -11$$

where $x_1\geqslant 0$ and x_2 is unrestricted in sign.

The dual problem then becomes

Maximize $v=2y_1+8y_2-11y_3$ subject to

$$y_1+y_2-y_3\leqslant -2$$
$$-2y_1+2y_2+y_3=-1$$

where $y_1\geqslant 0$, y_2 is unrestricted in sign, and $y_3\geqslant 0$.

4.15

In this case $\Delta\mathbf{b}=\begin{Bmatrix} -20 \\ -680 \end{Bmatrix}$ and the new values of the basic variables are given by

$$\mathbf{X}'_B=\begin{Bmatrix} x'_3 \\ x'_2 \end{Bmatrix}=\mathbf{X}_B+\mathbf{B}^{-1}\,\Delta\mathbf{b}=\begin{Bmatrix} 800/3 \\ 40/3 \end{Bmatrix}+\begin{bmatrix} 4/15 & -1/15 \\ -1/150 & 2/75 \end{bmatrix}\begin{Bmatrix} -20 \\ -680 \end{Bmatrix}$$
$$=\begin{Bmatrix} 920/3 \\ -14/3 \end{Bmatrix}.$$

Since x'_2 is negative, the original optimal basis will no longer be optimal for the new problem. Hence the new optimum solution is found as follows:

STEP (i): Replace the column of $\bar{b}_i$ in the original optimal tableau by

$$\bar{\mathbf{b}}'=\begin{Bmatrix} \bar{b}'_3 \\ \bar{b}'_2 \end{Bmatrix}=\mathbf{B}^{-1}\,(\mathbf{b}+\Delta\mathbf{b})=\mathbf{X}'_B=\begin{Bmatrix} 920/3 \\ -14/3 \end{Bmatrix}$$

the value of $f_{\min}$ by

$$f'_{\min}=\mathbf{c}_B^T\,\mathbf{X}'_B=(-30 \quad -100)\begin{Bmatrix} 920/3 \\ -14/3 \end{Bmatrix}=-\frac{26200}{3}$$

and change the signs of all the numbers lying in the rows in which the value of $\bar{b}'_i$ are negative ($\bar{b}'_2$ in the present case). For the present problem, this gives the following tableau:

Basic variable	Variables						$-f$	$\bar{b}_i$	ratio $(\bar{b}_i/\bar{a}_{is})$ for $\bar{a}_{is}>0$
	x_1	x_2	x_3	x_4	x_5	x_6			
x_3	5/3	0	1	7/3	4/15	−1/15	0	920/3	
x_2	−1/30	−1	0	1/30	1/150	−2/75	0	14/3	
$-f$	175/21	0	0	50/3	22/3	2/3	1	26200/3	

STEPS (ii) AND (iii): Introduce an artificial variable $x_7 \geq 0$ and go through Phase I computations as shown below:

Basic variable	Variables							$-f$	$-w$	$\bar{b}_i$	ratio $(\bar{b}_i/\bar{a}_{is})$ for $\bar{a}_{is}>0$
	x_1	x_2	x_3	x_4	x_5	x_6	x_7				
x_3	5/3	0	1	7/3	4/15	−1/15	0	0	0	920/3	1150
x_7	−1/30	−1	0	1/30	$\boxed{1/150}$ Pivot element	−2/75	1	0	0	14/3	700←smaller one, x_7 drops from basis
$-f$	175/21	0	0	50/3	22/3	2/3	0	1	0	26200/3	
$-w$	1/30	1	0	−1/30	−1/150 ↑	2/75	0	0	1	−14/3	

smallest $\bar{d}_j < 0$, x_5 enters the basis

Result of pivot operation:

x_3	3	40	1	$\boxed{1}$ Pivot element	0	1	—	0	0	120	120
x_5	−5	−150	0	5	1	−4	—	0	0	700	140
$-f$	45	1100	0	−20 ↑	0	30	—	1	0	3600	
$-w$	0	0	0	0	0	0	—	0	1	0	

STEP (iv): Since x_7 is dropped from basis, the column corresponding to it is not filled. Now Phase II computations are performed as follows.

Result of pivoting on 1 in the above tableau:

x_4	3	40	1	1	0	1	—	0	—	120
x_5	−20	−350	−5	0	1	−9	—	0	—	100
$-f$	105	1900	20	0	0	50	—	1	—	6000

Thus the optimal solution is given by

$x_4=120$, $x_5=100$ (slack variable),

$x_1=x_2=x_3=x_6=0$ (nonbasic variables),

$f_{\min}=-6000$ and maximum profit=Rs 6,000.

4.16

Since only the cost coefficients corresponding to the nonbasic variables x_1 and x_4 are changed, the original optimal solution remains optimal if

$$\bar{c}_j+\Delta c_j\geq 0, \quad j=\text{nonbasic}=1 \text{ and } 4$$

In the present case,

$$\bar{c}_1=25/3,\ \bar{c}_4=50/3,\ \Delta c_1=5 \text{ and } \Delta c_4=-10$$

Thus

$$\bar{c}_1+\Delta c_1=40/3\geq 0 \text{ and } \bar{c}_4+\Delta c_4=20/3\geq 0$$

and hence the optimum solution given in Example 4.5 remains optimal for the present case also. Since the cost coefficients corresponding to the basic variables are not changed, $\Delta f=0$ and hence the optimum value of the objective function also remains unchanged.

4.17

Here $\Delta c_1=5$ (x_1 is nonbasic), $\Delta c_2=10$ (x_2 is basic), $\Delta c_3=0$ (x_3 is basic), and $\Delta c_4=0$ (x_4 is nonbasic). Equation (4.43) gives the new cost coefficients of nonbasic variables as

$$\bar{c}_j'=\bar{c}_j+\Delta c_j-\sum_k \Delta c_k \left(\sum_i a_{ij}\,\beta_{ki}\right)$$

where j represents nonbasic variables ($j=1, 4$) and k and i represent basic variables (2 and 3). Thus

$$\bar{c}_1'=\bar{c}_1+\Delta c_1-[\Delta c_2(a_{21}\beta_{22}+a_{31}\beta_{23})+\Delta c_3\,(a_{21}\beta_{32}+a_{31}\beta_{33})]$$

and

$$\bar{c}_4'=\bar{c}_4+\Delta c_4-[\Delta c_2\,(a_{24}\beta_{22}+a_{34}\beta_{23})+\Delta c_3\,(a_{24}\beta_{32}+a_{34}\beta_{33})]$$

Instead of renumbering the basic variables from 1 to m, we can note that x_3 is the first and x_2 is the second basic variable and hence

$$a_{21}=\text{original } a_{21}=3,\ a_{31}=\text{original } a_{11}=7,\ a_{24}=\text{original } a_{24}=1,$$

and $\quad a_{34}=\text{original } a_{14}=9.$

From Eq. (E_4) of Example 4.5, we have

$$\beta_{22}=2/75,\ \beta_{23}=-1/150,\ \beta_{32}=-1/15 \text{ and } \beta_{33}=4/15$$

Further $\bar{c}_1=25/3$ and $\bar{c}_4=50/3$ and hence

$$\bar{c}_1'=\frac{25}{3}+5-\left[+10\left(3.\frac{2}{75}-7.\frac{1}{150}\right)+0\right]=13$$

and

$$\bar{c}_4'=\frac{50}{3}+0-\left[+10\left(1.\frac{2}{75}-9.\frac{1}{150}\right)+0\right]=\frac{49}{3}$$

Since both $\bar{c}_1'$ and $\bar{c}_4'$ are nonnegative ($\bar{c}_5'$ and $\bar{c}_6'$ can also be verified to be nonnegative), the previous optimal solution remains optimal with the new optimum value of the objective function given by

$$f_{\min}'=f_{\min}+\Delta f=f_{\min}+\sum_{i=\text{basic}} x_i\,\Delta c_i$$

$$=-\frac{28000}{3}+\frac{40}{3}(+10)+\frac{800}{3}(0)=-\frac{27600}{3}$$

4.18

Let x_k be the number of units of E manufactured per day. Then

$$c_k=-120,\ a_{1k}=10 \text{ and } a_{2k}=10.$$

$$\therefore\quad \bar{c}_k=c_k-\pi_1 a_{1k}-\pi_2 a_{2k}=-120+\left(\frac{22}{3}\right)(10)+\left(\frac{2}{3}\right)(10)=-40.$$

Since $\bar{c}_k$ is negative, the objective function can be reduced further by bringing x_k into the basis.

To bring x_k into the basis, we first transform A_k as

$$\bar{A}_k=B^{-1}A_k=\begin{bmatrix} 4/15 & -1/15 \\ -1/150 & 2/75 \end{bmatrix}\begin{Bmatrix} 10 \\ 10 \end{Bmatrix}=\begin{Bmatrix} 2 \\ 1/5 \end{Bmatrix}$$

Now we start with the optimal tableau of the original problem (in Example 4.5) and add entries corresponding to x_k as follows.

Basic variables	Variables							$-f$	$\bar{b}_i$	ratio $(\bar{b}_i/\bar{a}_{is})$ for $\bar{a}_{is}>0$
	x_1	x_2	x_3	x_4	x_5	x_6	x_k			
x_3	5/3	0	1	7/3	4/15	−1/15	2	0	800/3	400/3
x_2	1/30	1	0	−1/30	−1/150	2/75	$\boxed{1/5}$	0	40/3	200/3←
$-f$	25/3	0	0	50/3	22/3	2/3	−40	1	28000/3	

↑

We carry out the pivot operation on the element 1/5 according to the regular simplex method and obtain the following tableau:

x_3	4/3	−10	1	8/3	1/3	−1/3	0	0	400/3
x_k	1/6	5	0	−1/6	−1/30	2/15	1	0	200/3
$-f$	15	200	0	10	6	6	0	1	12000

Since all $\bar{c}_j>0$, the present solution is optimum with $x_3=400/3$, $x_k=200/3$ (basic variables),

$$x_1=x_2=x_4=x_5=x_6=0 \quad \text{(nonbasic variables)},$$

$f_{\min}=-12000$, and maximum profit = \$ 12,000.

4.19 Let (x_1, x_2, x_3, x_4) = number of units of products (A, B, C, D) produced

Maximize $f = 15x_1 + 25x_2 + 20x_3 + 60x_4$

subject to

g_1: $4x_1 + 9x_2 + 7x_3 + 10x_4 \le 6000$ (copper)

g_2: $2x_1 + x_2 + 3x_3 + 20x_4 \le 4000$ (zinc)

with $x_i \ge 0$ $(i = 1, 2, 3, 4)$

Basic variables	x_1	x_2	x_3	x_4	x_5	x_6	$-f$	$\bar{b}_i$	$(\bar{b}_i/\bar{a}_{is})$ for $\bar{a}_{is} > 0$
x_5	4	9	7	10	1	0	0	6000	600
x_6	2	1	3	(20)	0	1	0	4000	200 ←
$-f$	−15	−25	−20	−60 ↑	0	0	1	0	
x_5	3	(17/2)	11/2	0	1	−1/2	0	4000	8000/17 ←
x_4	1/10	1/20	3/20	1	0	1/20	0	200	4000
$-f$	−9	−22 ↑	−11	0	0	3	1	12000	
x_2	(6/17)	1	11/17	0	2/7	−1/7	0	8000/17	8000/6 ←
x_4	7/85	0	2/17	1	−1/170	9/170	0	3000/17	15000/7
$-f$	−21/17	0	55/17	0	44/17	29/17	1	380000/17	
x_1	1 ↑	17/6	11/6	0	1/3	−1/6	0	8000/6	
x_4	0	−7/30	−1/30	1	−1/30	1/15	0	3400/51	
$-f$	0	7/2	11/2	0	3	3/2	1	24000	

optimal solution: $\vec{X}_B = \begin{Bmatrix} x_1 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 4000/3 \\ 200/3 \end{Bmatrix}$

$f_{min} = -24{,}000$

$$\vec{c}_B = \begin{Bmatrix} c_1 \\ c_4 \end{Bmatrix} = \begin{Bmatrix} -15 \\ -60 \end{Bmatrix}$$

$$[B] = \begin{bmatrix} 4 & 10 \\ 2 & 20 \end{bmatrix}, \quad [B]^{-1} = \begin{bmatrix} \beta_{11} & \beta_{14} \\ \beta_{41} & \beta_{44} \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & -\frac{1}{6} \\ -\frac{1}{30} & \frac{1}{15} \end{bmatrix}$$

$$\vec{\pi} = \vec{c}_B^{\,T} [B]^{-1} = \{-15 \quad -60\} \begin{bmatrix} 1/3 & -1/6 \\ -1/30 & 1/15 \end{bmatrix} = \begin{Bmatrix} -3 \\ -3/2 \end{Bmatrix}$$

4.20

$c_4 = -60$, $(c_4)_{new} = -30$; $\Delta c_4 = +30$

$f = 15x_1 + 25x_2 + 20x_3 + 30x_4$ (maximize)

Solution: $x_1^* = 1500.0$, $x_2^* = x_3^* = x_4^* = 0$

$f_{max} = 22,500.0$

4.21

$(c_1)_{new} = 10$, $(c_2)_{new} = 20$

Maximize $f = 10x_1 + 20x_2 + 20x_3 + 60x_4$

Solution: $x_2^* = 470.588$, $x_4^* = 176.471$

$x_1^* = x_3^* = 0$

$f_{max} = 20,000.0$

4.22

$(c_2)_{new} = 30$, $(c_3)_{new} = 25$

Maximize $f = 15x_1 + 30x_2 + 25x_3 + 60x_4$

Solution: $x_2^* = 470.588$, $x_4^* = 176.471$

$x_1^* = x_3^* = 0$

$f_{max} = 24,705.883$

4.23

$(b_1)_{new} = 4000$, $(b_2)_{new} = 6000$

Solution: $x_1^* = 333.333$, $x_4^* = 266.667$

$x_2^* = x_3^* = 0$

$f_{max} = 21,000.0$

4.24

x_5 = number of units of product E produced

New problem:

Maximize $f = 15x_1 + 25x_2 + 20x_3 + 60x_4 + 30x_5$

subject to

(copper): $4x_1 + 9x_2 + 7x_3 + 10x_4 + 6x_5 \leq 6000$

(zinc): $2x_1 + x_2 + 3x_3 + 20x_4 + 3x_5 \leq 4000$

$x_i \geq 0$, $i = 1$ to 5

Solution: $x_4^* = 66.667$, $x_5^* = 888.888$

$x_1^* = x_2^* = x_3^* = 0$

$f_{max} = 30,666.667$

4.25

New problem: Maximize $f = 15x_1 + 25x_2 + 20x_3 + 60x_4$

subject to

(copper): $4x_1 + 9x_2 + 7x_3 + 10x_4 \leq 6000$

(zinc): $2x_1 + x_2 + 3x_3 + 20x_4 \leq 4000$

(Nickel): $4x_1 + 3x_2 + 2x_3 + 5x_4 \leq 2000$

$x_i \geq 0$, $i = 1$ to 4

Solution: $x_2^* = 111.111$, $x_3^* = 555.55$, $x_4^* = 111.111$

$x_1^* = x_5^* = 0$

$f_{max} = 20,555.555$

4.26

New problem: Maximize $f = 15x_1 + 25x_2 + 20x_3 + 60x_4$

subject to

$5x_1 + 9x_2 + 7x_3 + 10x_4 \leq 6000$

$3x_1 + x_2 + 3x_3 + 20x_4 \leq 4000$

$x_i \geq 0$, $i = 1$ to 4

Solution: $x_2^* = 470.588$, $x_4^* = 176.471$

$x_1^* = x_3^* = 0$

$f_{max} = 22,352.941$

4.27 New problem: Maximize $f = 15x_1 + 25x_2 + 20x_3 + 60x_4$

subject to

$$4x_1 + 9x_2 + 5x_3 + 10x_4 \leq 6000$$

$$2x_1 + x_2 + 4x_3 + 20x_4 \leq 4000$$

$$x_i \geq 0, \quad i = 1, 2, 3, 4$$

Solution: $x_1^* = 1333.333$, $x_4^* = 66.667$

$x_2^* = x_3^* = 0$

$f_{max} = 24{,}000.0$

4.28 $(b_1)_{new} = 8000$, $(b_2)_{new} = 5000$

New problem: Maximize $f = 15x_1 + 25x_2 + 20x_3 + 60x_4$

subject to

$$4x_1 + 9x_2 + 5x_3 + 10x_4 \leq 8000$$

$$2x_1 + x_2 + 4x_3 + 20x_4 \leq 5000$$

$$x_i \geq 0, \quad i = 1, 2, 3, 4$$

Solution: $x_1^* = 1833.333$, $x_4^* = 66.667$, $x_2^* = x_3^* = 0$

$f_{max} = 31{,}500.0$

4.29 Minimize $f = 8x_1 - 2x_2$

subject to $g_1 = -4x_1 + 2x_2 - 1 \leq 0$

$$g_2 = 5x_1 - 4x_2 - 3 \leq 0$$

$$g_3 = -x_1 \leq 0, \quad g_4 = -x_2 \leq 0$$

Graphical solution:

$g_1 = 0$ at $(0, \frac{1}{2})$, $(-\frac{1}{4}, 0)$

$g_2 = 0$ at $(0, -\frac{3}{4})$, $(\frac{3}{5}, 0)$

Contours of f:

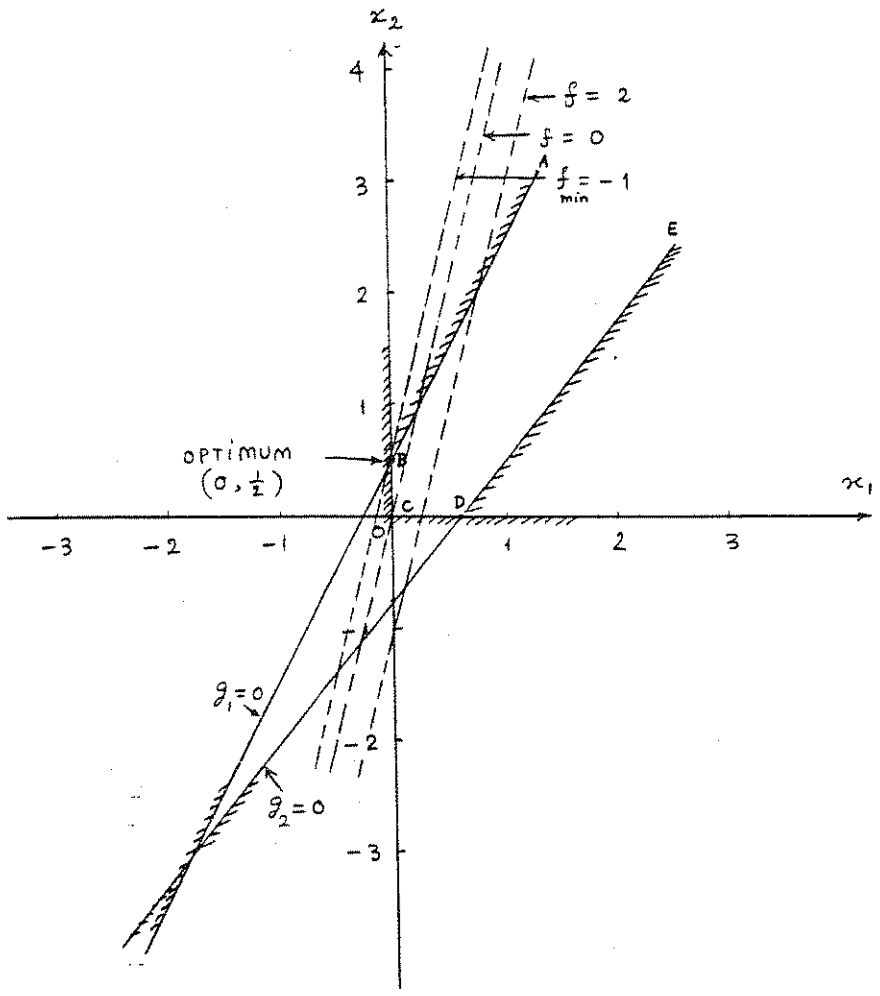
$f = 0$ at $(0,0)$, $(1,4)$

$f = 2$ at $(0,-1)$, $(\frac{1}{4}, 0)$

Feasible space = ABCDE in figure

Optimum point = B

with $\vec{x}^* = \begin{Bmatrix} 0 \\ \frac{1}{2} \end{Bmatrix}$ and $f^* = -1$



Simplex method:

Let $x_3 \geq 0$ and $x_4 \geq 0$ be the slack variables for g_1 and g_2.

Basic variables	Variables x_1	x_2	x_3	x_4	$-f$	$\bar{b}_i$	$(\bar{b}_i/\bar{a}_{is})$ for $\bar{a}_{is} > 0$
x_3	-4	(2)	1	0	0	1	$\frac{1}{2}$ ←
x_4	5	-4	0	1	0	3	
$-f$	8	-2	0	0	1		
		↑					

x_2	-2	1	$\frac{1}{2}$	0	0	$\frac{1}{2}$
x_4	-3	0	2	1	0	5
$-f$	4	0	1	0	1	1

optimum solution: $x_1^* = x_3^* = 0$, $x_2^* = \frac{1}{2}$, $x_4^* = 5$

$$f_{min} = -1$$

$$\vec{x}_B = \begin{Bmatrix} x_2 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} \frac{1}{2} \\ 5 \end{Bmatrix}, \quad \vec{c}_B = \begin{Bmatrix} -2 \\ 0 \end{Bmatrix}$$

$$[B] = \begin{bmatrix} 2 & 0 \\ -4 & 1 \end{bmatrix}, \quad [B]^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} \beta_{22} & \beta_{24} \\ \beta_{42} & \beta_{44} \end{bmatrix}$$

$$\vec{\pi} = \vec{c}_B^{\,T} [B]^{-1} = (-2 \quad 0) \begin{bmatrix} \frac{1}{2} & 0 \\ 2 & 1 \end{bmatrix} = \begin{Bmatrix} -1 \\ 0 \end{Bmatrix} = \text{simplex multipliers}$$

$\bar{c}_1 = 4$, $\bar{c}_3 = 1$ $\Rightarrow$ cost coefficients of nonbasic variables

---

4.30

$$\vec{b}_{new} = \begin{Bmatrix} 2 \\ 3 \end{Bmatrix}, \quad \Delta\vec{b} = \begin{Bmatrix} 1 \\ 0 \end{Bmatrix}$$

$$\vec{x}_B' = \vec{x}_B + [B]^{-1} \Delta\vec{b} = \begin{Bmatrix} \frac{1}{2} \\ 5 \end{Bmatrix} + \begin{bmatrix} \frac{1}{2} & 0 \\ 2 & 1 \end{bmatrix} \begin{Bmatrix} 1 \\ 0 \end{Bmatrix} = \begin{Bmatrix} 1 \\ 7 \end{Bmatrix}$$

Since $x_2' = 1$ and $x_4' = 7$ (both ≥ 0), the original basis [B] remains optimal.

$$f_{min}' = f_{min} + \Delta f = f_{min} + \vec{c}_B^{\,T} \Delta\vec{x}_B = -1 + (-2 \quad 0) \begin{Bmatrix} \frac{1}{2} \\ 2 \end{Bmatrix} = -2$$

Graphical solution:

$$g_1 = -4x_1 + 2x_2 - 2 \leq 0$$

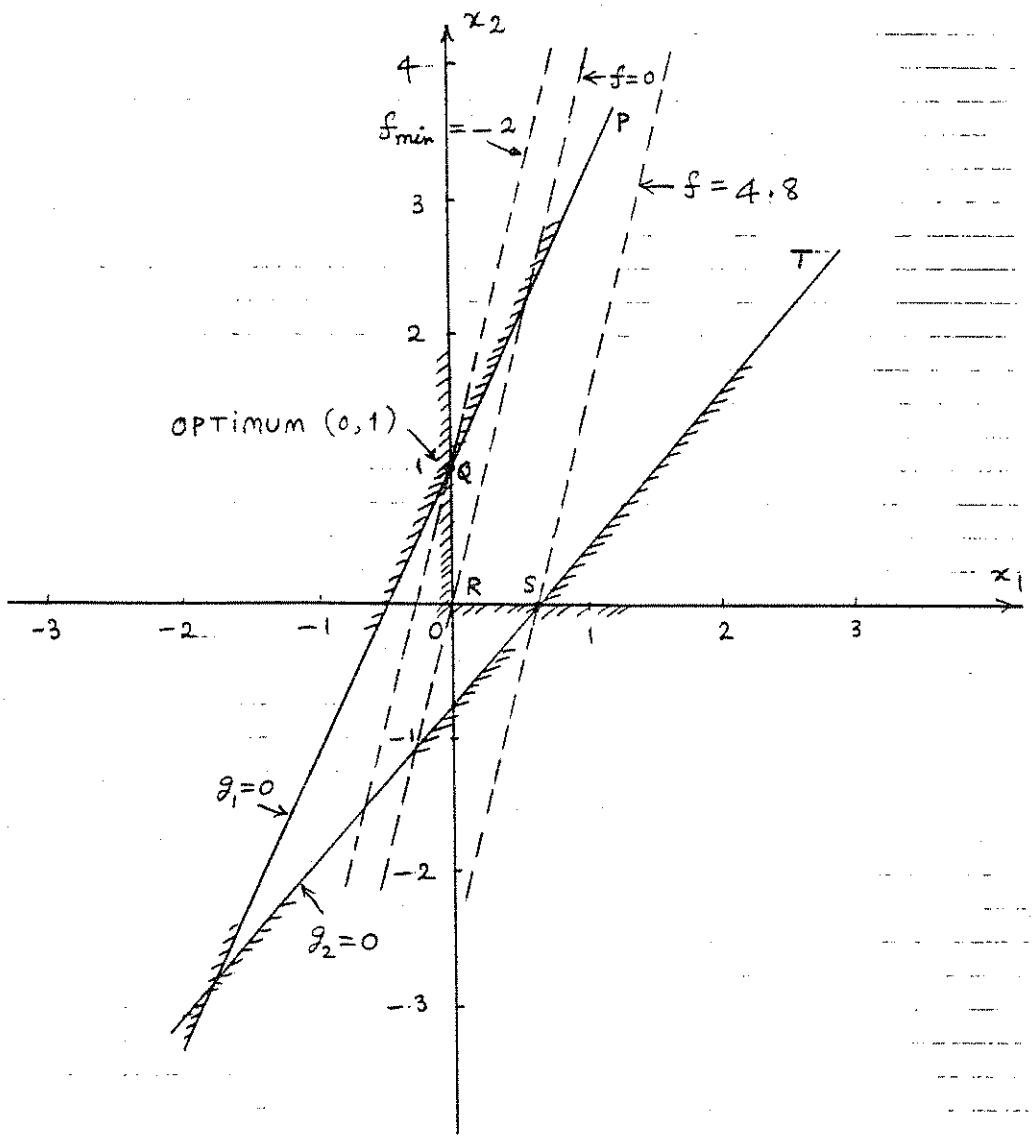
$g_1 = 0$ at $(0, 1)$, $(-\frac{1}{2}, 0)$

$$g_2 = 5x_1 - 4x_2 - 3 \leq 0$$

$g_2 = 0$ at $(0, -\frac{3}{4})$, $(\frac{3}{5}, 0)$

Feasible space = PQRST

optimum point = Q $= \begin{Bmatrix} 0 \\ 1 \end{Bmatrix}$, $f^* = -2$.


x_2
4
$f_{min} = -2$
3
2
1
OPTimum (0,1)
Q
R
S
P
T
$f=0$
$f=4.8$
x_1
-3
-2
-1
0
1
2
3
-1
-2
-3
$g_1=0$
$g_2=0$

4.31

$\vec{b}_{new} = \begin{Bmatrix} 1 \\ 4 \end{Bmatrix}$, $\Delta\vec{b} = \begin{Bmatrix} 0 \\ 1 \end{Bmatrix}$

$\vec{x}_B' = \vec{x}_B + [B]^{-1}\Delta\vec{b} = \begin{Bmatrix} 1/2 \\ 5 \end{Bmatrix} + \begin{bmatrix} 1/2 & 0 \\ 2 & 1 \end{bmatrix}\begin{Bmatrix} 0 \\ 1 \end{Bmatrix} = \begin{Bmatrix} 1/2 \\ 6 \end{Bmatrix} > \vec{0}$

∴ original basis [B] remains optimal with $x_2' = \frac{1}{2}$ and $x_4' = 6$.

$f'_{min} = f_{min} + \Delta f = f_{min} + \vec{c}_B^T \Delta\vec{x}_B = -1 + (-2 \quad 0)\begin{Bmatrix} 0 \\ 1 \end{Bmatrix} = -1$

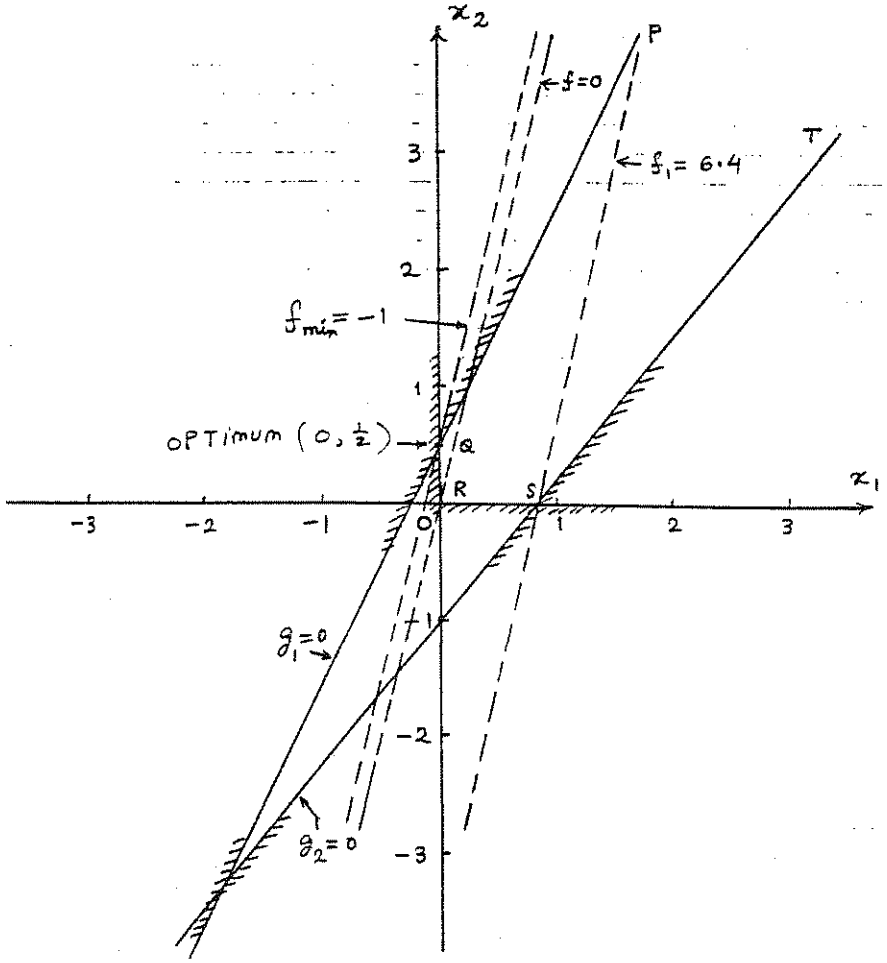
Graphical solution:

$g_1 = -4x_1 + 2x_2 - 1 \le 0$; $g_1 = 0$ at $(0, \frac{1}{2})$, $(-\frac{1}{4}, 0)$

$g_2 = 5x_1 - 4x_2 - 4 \le 0$; $g_2 = 0$ at $(0, -1)$, $(\frac{4}{5}, 0)$

Feasible space = PQRST in figure

Optimum point = $Q = \begin{Bmatrix} 0 \\ 1/2 \end{Bmatrix}$ with $f^* = -1$.



4.32 $\Delta c_1 = 2, \quad \Delta c_2 = 0$

New cost coefficients of nonbasic variables:

$\bar{c}_1 + \Delta c_1 = 4 + 2 = 6 > 0$

$\bar{c}_3 + \Delta c_3 = 1 + 0 = 1 > 0$

Since $\bar{c}_1' > 0$ and $\bar{c}_3' > 0$, original optimum remains optimum.

$\Delta f = 0$ and $f_{new} = f_{old} = -1$

$$\vec{x}^*_{new} = \vec{x}^*_{old} = \begin{Bmatrix} x_2^* \\ x_4^* \end{Bmatrix} = \begin{Bmatrix} 1/2 \\ 5 \end{Bmatrix}$$

Graphical solution:

Contours of $f = 10x_1 - 2x_2$:

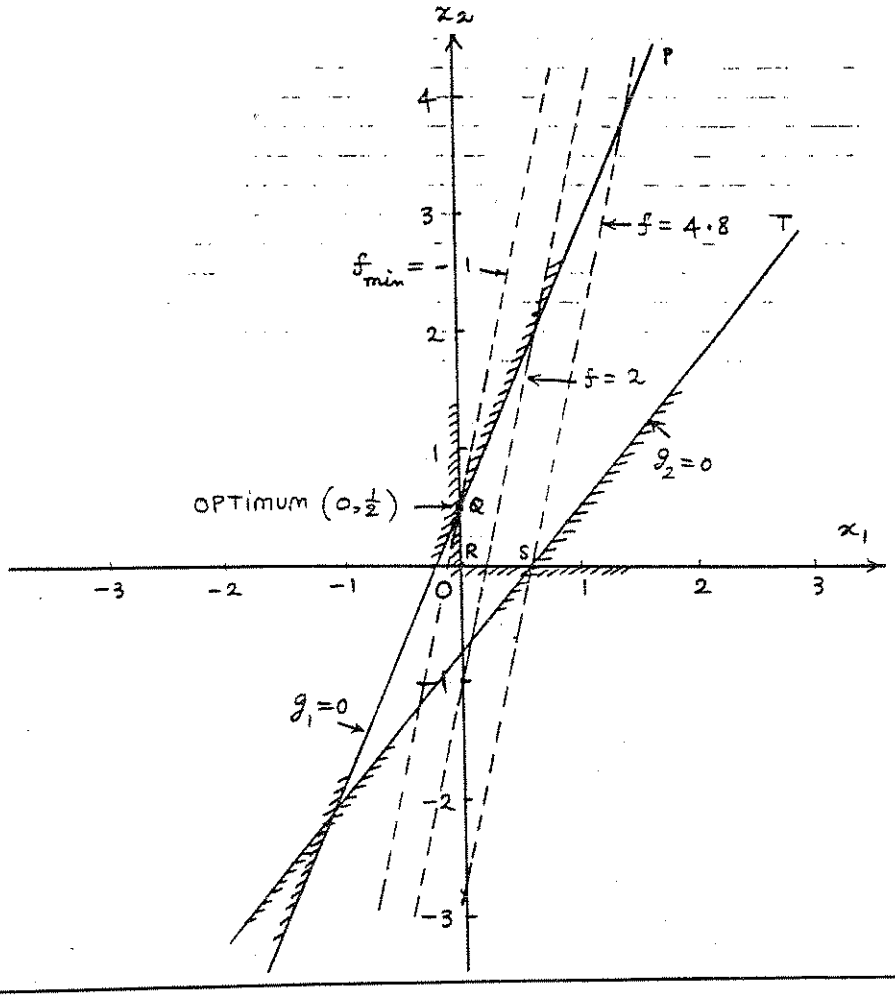
$f = 0$ at $(0,0)$, $(1,5)$

$f = 2$ at $(0,-1)$, $(\frac{1}{5},0)$

Feasible space = PQRST

Optimum point = Q

$= \begin{Bmatrix} 0 \\ 1/2 \end{Bmatrix}$ with $f^* = -1$



4.33 $\Delta c_1 = 0, \quad \Delta c_2 = -2, \quad \Delta c_3 = 0, \quad \Delta c_4 = 0$

Basic variables = 2, 4 ; Nonbasic variables = 1, 3

$$\bar{c}_j' = \bar{c}_j + \Delta c_j - \sum_{\substack{k \\ \text{basic} \\ 2,4}} \Delta c_k \left(\sum_{\substack{i \\ \text{basic} \\ 2,4}} a_{ij} \beta_{ki} \right)$$

$\bar{c}_j'$: nonbasic 1, 3

$$\bar{c}_j' = \bar{c}_j + \Delta c_j - \Delta c_2 \left(a_{2j} \beta_{22} + a_{4j} \beta_{24} \right) - \overset{0}{\Delta c_4} \left(a_{2j} \beta_{42} + a_{4j} \beta_{44} \right)$$

$$\bar{c}_1' = \bar{c}_1 + \Delta c_1 - \Delta c_2 \left(a_{21} \beta_{22} + a_{41} \beta_{24} \right)$$

$$= 4 + 0 - (-2)(-2 + 0) = 0$$

$$\bar{c}_3' = \bar{c}_3 + \Delta c_3 - \Delta c_2 \left(a_{23} \beta_{22} + a_{43} \beta_{24} \right)$$

$$= 1 + 0 - (-2)\left(1 * \tfrac{1}{2} + 0 * 0 \right) = 2 > 0$$

Since $\bar{c}_1' = 0$ (cost coefficient of nonbasic variable = 0), alternate solutions exist.

One solution is:

Previous optimum solution $= \vec{x}^* = \begin{Bmatrix} x_2^* \\ x_4^* \end{Bmatrix} = \begin{Bmatrix} \frac{1}{2} \\ 5 \end{Bmatrix}$

$$f^* = f_{min} = -1$$

Graphical solution:

Contours of $f = 8x_1 - 4x_2$:

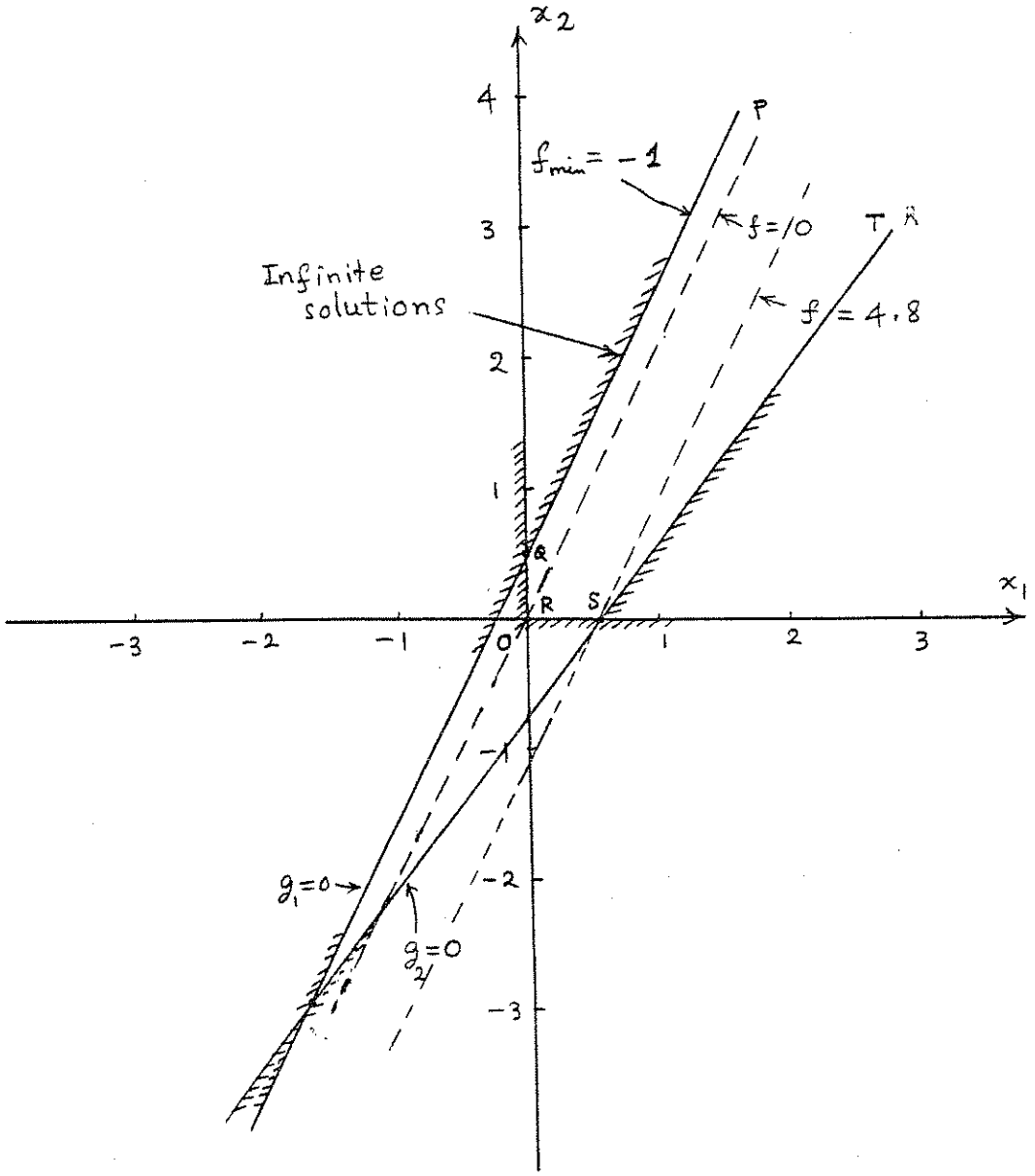
$f = 0$ at $(0, 0)$, $(1, 2)$

$f = 2$ at $(0, -\frac{1}{2})$, $(\frac{1}{4}, 0)$

Feasible space = PQRST in figure

Optimum point = All points on PQ

$$f^* = -1$$


x_2
4
3
2
1
-1
-2
-3
P
$f_{min} = -1$
$f = 10$
T
$f = 4.8$
Infinite solutions
Q
R
S
x_1
-3
-2
-1
0
1
2
3
$g_1 = 0$
$g_2 = 0$

4.34

$\Delta a_{11} = -1$ (for nonbasic variable)

$\bar{c}_j = c_j - \vec{\pi}^T \vec{A}_j$ (j = nonbasic; 1, 2)

New $\vec{A}_1 = \begin{Bmatrix} -5 \\ 5 \end{Bmatrix}$ (only $\vec{A}_1$ is changed)

$\bar{c}_1 = 8 - (-1 \quad 0) \begin{Bmatrix} -5 \\ 5 \end{Bmatrix} = 3 > 0$

$\therefore$ original optimum remains optimum.

Graphical solution:

Feasible space: PQRST

optimum point = Q

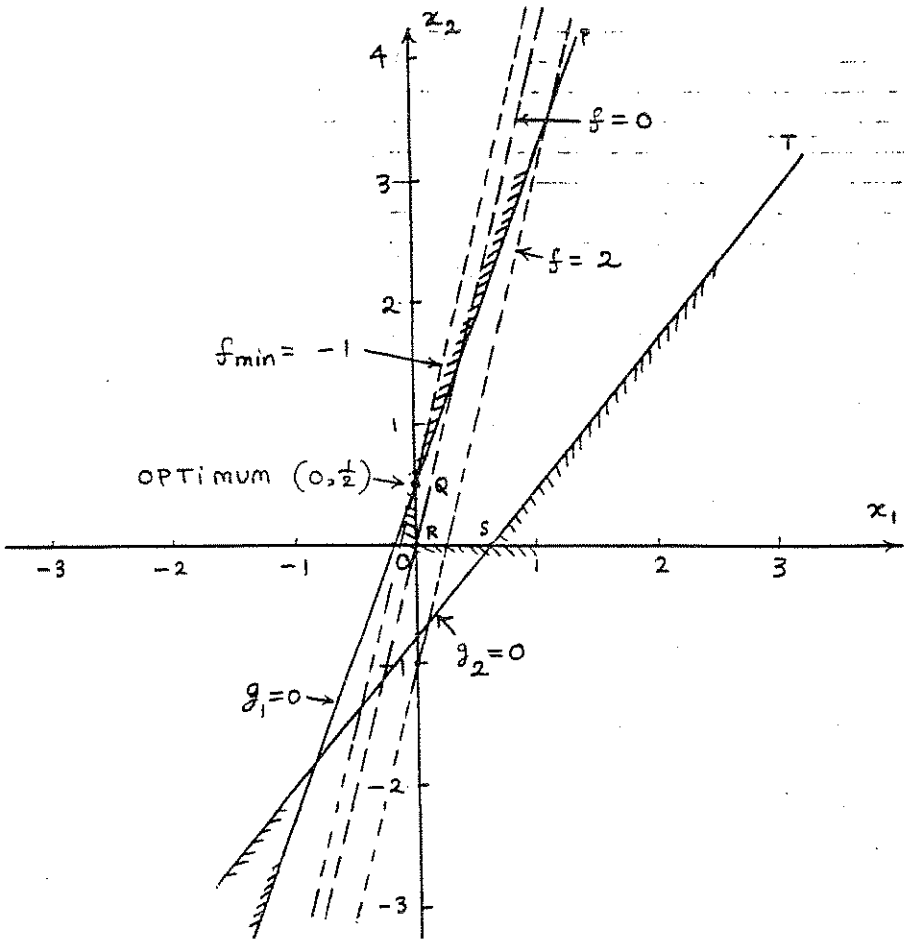
$\vec{x}^* = \begin{Bmatrix} 0 \\ 1/2 \end{Bmatrix}$, $f^* = -1$

$g_1 = -5x_1 + 2x_2 - 1 \leq 0$

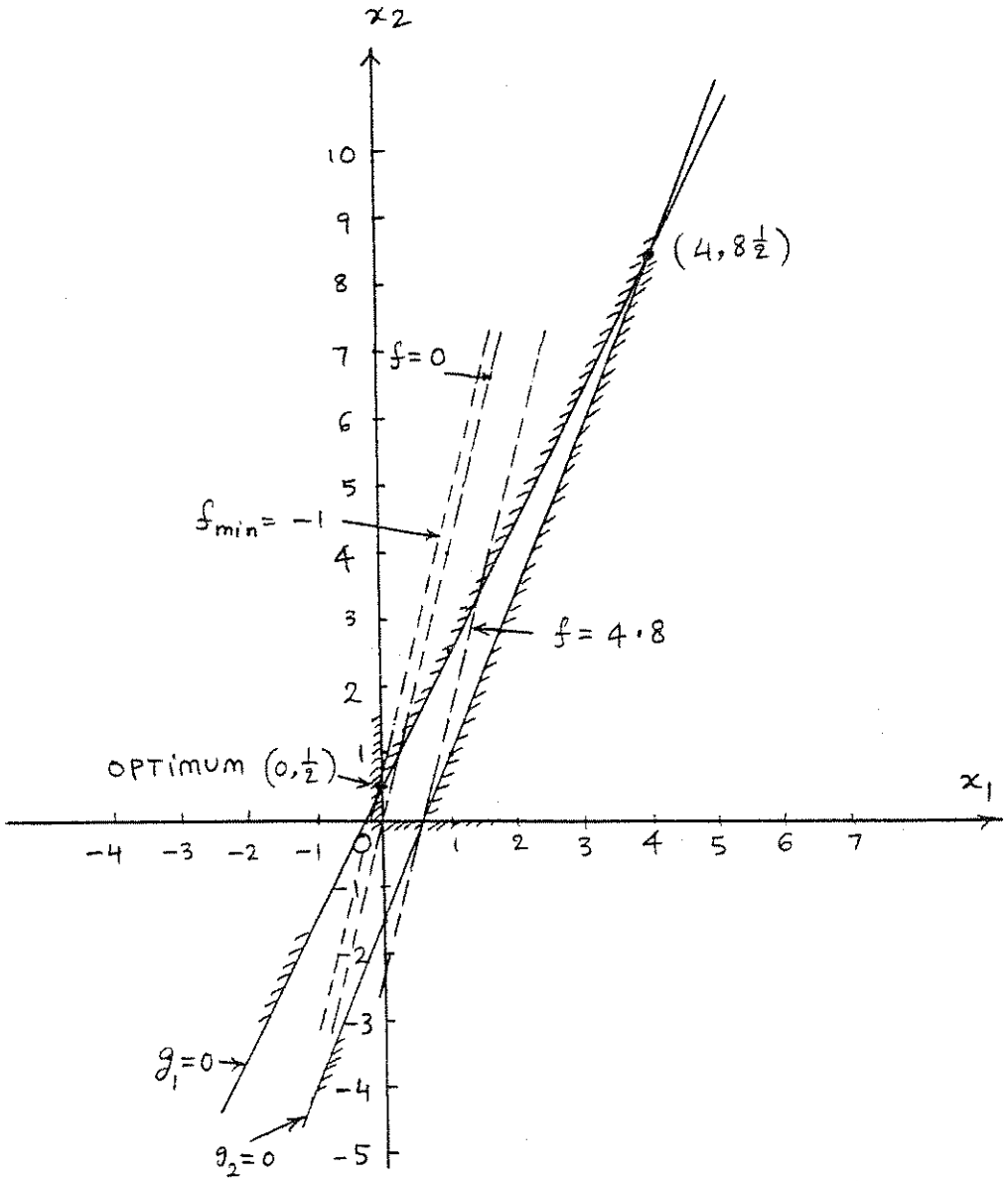
$= 0$ at $(0, \frac{1}{2})$, $(-\frac{1}{5}, 0)$

$g_2 = 5x_1 - 4x_2 - 3 \leq 0$

$= 0$ at $(0, -\frac{3}{4})$, $(\frac{3}{5}, 0)$



4.35 Graphical solution:



4.36 Minimize $f = 2x_1 - 2x_2 + 5x_3$

subject to $x_1 - x_2 = 0$

$x_1 + x_2 + x_3 = 1$

$x_i \geq 0 \;,\; i = 1, 2, 3$

Let $\epsilon = 0.05$ for testing convergence.

Problem is already in the required form of Eq. (4.59).

1. Let $\vec{X}^{(1)} = \begin{Bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{Bmatrix}$ = initial feasible point.

   set $k = 1$.

2. $\left| f(\vec{X}^{(1)}) \right| = \left| \frac{5}{3} \right| > \epsilon$. Hence go to step 3.

3. $[a] = [1 \quad -1 \quad 0]$, $\vec{c} = \{2 \quad -2 \quad 5\}^T$

   $\| \vec{c} \| = \{(2)^2 + (-2)^2 + (5)^2\}^{\frac{1}{2}} = \sqrt{33}$

   $[D(\vec{X}^{(1)})] = \begin{bmatrix} 1/3 & 0 & 0 \\ 0 & 1/3 & 0 \\ 0 & 0 & 1/3 \end{bmatrix}$, $\underset{1\times 3}{[a]} \underset{3\times 3}{[D(\vec{X}^{(1)})]} = \{\frac{1}{3} \quad -\frac{1}{3} \quad 0\}$

   $[P] = \begin{bmatrix} [a][D(\vec{X}^{(1)})] \\ 1 \quad 1 \quad 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & -\frac{1}{3} & 0 \\ 1 & 1 & 1 \end{bmatrix}$

   $\left([P][P]^T\right)^{-1} = \begin{bmatrix} 2/9 & 0 \\ 0 & 3 \end{bmatrix}^{-1} = \begin{bmatrix} 9/2 & 0 \\ 0 & 1/3 \end{bmatrix}$

   $[D(\vec{X}^{(1)})]\,\vec{c} = \{2/3 \quad -2/3 \quad 5/3\}^T$

   $\left([I] - [P]^T\left([P][P]^T\right)^{-1}[P]\right)[D(\vec{X}^{(1)})]\,\vec{c}$

   $= \left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} \frac{1}{3} & 1 \\ -\frac{1}{3} & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 9/2 & 0 \\ 0 & 1/3 \end{bmatrix}\begin{bmatrix} \frac{1}{3} & -\frac{1}{3} & 0 \\ 1 & 1 & 1 \end{bmatrix}\right)\begin{Bmatrix} 2/3 \\ -2/3 \\ 5/3 \end{Bmatrix}$

   $= \begin{Bmatrix} -5/9 \\ -11/9 \\ 10/9 \end{Bmatrix}$

Using $\alpha = \frac{1}{4}$, Eq. (4.68) gives

$$\vec{Y}^{(2)} = \begin{Bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{Bmatrix} - \frac{1}{4} \begin{Bmatrix} -5/9 \\ -11/9 \\ 10/9 \end{Bmatrix} \frac{1}{\sqrt{3(2)}\ \sqrt{33}} = \begin{Bmatrix} 0.3432 \\ 0.3550 \\ 0.3136 \end{Bmatrix}$$

$$\sum_{r=1}^{n} x_r^{(1)}\, y_r^{(2)} = \frac{1}{3}\{0.3432 + 0.3550 + 0.3136\} = 0.3373$$

Eq. (4.71) gives

$$\{x_i^{(2)}\} = \left\{ \frac{x_i^{(1)}\, y_i^{(2)}}{\sum_{r=1}^{n} x_r^{(1)}\, y_r^{(2)}} \right\} = \frac{1}{0.3373} \begin{Bmatrix} \frac{1}{3}(0.3432) \\ \frac{1}{3}(0.3550) \\ \frac{1}{3}(0.3136) \end{Bmatrix}$$

$$\therefore \quad \vec{X}^{(2)} = \begin{Bmatrix} 0.3392 \\ 0.3508 \\ 0.3099 \end{Bmatrix}$$

$$f(\vec{X}^{(2)}) = 2(0.3392) - 2(0.3508) + 5(0.3099) = 1.5263$$

$$f(\vec{X}^{(1)}) = 1.6667$$

4.37 Minimize $f = 3x_1 + 5x_2 - 3x_3$

subject to $x_1 - x_3 = 0$

$x_1 + x_2 + x_3 = 1$

$x_i \geq 0, \quad i = 1, 2, 3$

Problem is already in the required form of Eq. (4.59). Choose $\epsilon = 0.05$ for testing the convergence.

1. Let $\vec{X}^{(1)} = \begin{Bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{Bmatrix}$ = initial feasible point.

   set $k = 1$.

2. $\left| f(\vec{X}^{(1)}) \right| = \left| \frac{5}{3} \right| > \epsilon$. Hence go to step 3.

3. $[a] = [1 \quad 0 \quad -1]$, $\vec{c} = \{3 \quad 5 \quad -3\}^T$

   $\| \vec{c} \| = \{(3)^2 + (5)^2 + (-3)^2\}^{\frac{1}{2}} = \sqrt{43}$

$$\left[D\left(\vec{x}^{(1)}\right)\right] = \begin{bmatrix} 1/3 & 0 & 0 \\ 0 & 1/3 & 0 \\ 0 & 0 & 1/3 \end{bmatrix}, \quad [a]\left[D\left(\vec{x}^{(1)}\right)\right] = \left\{\frac{1}{3} \quad 0 \quad -\frac{1}{3}\right\}$$

$$[P] = \begin{bmatrix} [a]\left[D\left(\vec{x}^{(1)}\right)\right] \\ 1 \quad 1 \quad 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & 0 & -\frac{1}{3} \\ 1 & 1 & 1 \end{bmatrix}$$

$$\left([P][P]^T\right)^{-1} = \begin{bmatrix} 2/9 & 0 \\ 0 & 3 \end{bmatrix}^{-1} = \begin{bmatrix} 9/2 & 0 \\ 0 & 1/3 \end{bmatrix}$$

$$\left[D\left(\vec{x}^{(1)}\right)\right]\vec{c} = \left\{1 \quad 5/3 \quad -1\right\}^T$$

$$[P]^T\left([P][P]^T\right)^{-1}[P] = \begin{bmatrix} 5/6 & 1/3 & -1/6 \\ 1/3 & 1/3 & 1/3 \\ -1/6 & 1/3 & -1/6 \end{bmatrix}$$

$$\left[[I] - [P]^T\left([P][P]^T\right)^{-1}[P]\right]\left[D\left(\vec{x}^{(1)}\right)\right]\vec{c}$$

$$= \left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 5/6 & 1/3 & -1/6 \\ 1/3 & 1/3 & 1/3 \\ -1/6 & 1/3 & -1/6 \end{bmatrix}\right)\begin{Bmatrix} 1 \\ 5/3 \\ -1 \end{Bmatrix} = \begin{Bmatrix} -5/9 \\ 10/9 \\ -14/9 \end{Bmatrix}$$

Using $\alpha = \frac{1}{4}$, Eq. (4.68) gives

$$\vec{Y}^{(2)} = \begin{Bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{Bmatrix} - \frac{1}{4}\begin{Bmatrix} -5/9 \\ 10/9 \\ -14/9 \end{Bmatrix}\frac{1}{\sqrt{3(2)}\ \sqrt{43}} = \begin{Bmatrix} 0.3419 \\ 0.3160 \\ 0.3575 \end{Bmatrix}$$

$$\sum_{r=1}^{n} x_r^{(1)} y_r^{(2)} = \frac{1}{3}\left(0.3419 + 0.3160 + 0.3575\right) = 0.3385$$

Eq. (4.71) $\Rightarrow$

$$\left\{x_i^{(2)}\right\} = \left\{\frac{x_i^{(1)}\ y_i^{(2)}}{\sum_{r=1}^{n} x_r^{(1)}\ y_r^{(2)}}\right\} = \frac{1}{0.3385}\begin{Bmatrix} 0.3419/3 \\ 0.3160/3 \\ 0.3575/3 \end{Bmatrix} = \begin{Bmatrix} 0.3367 \\ 0.3112 \\ 0.3520 \end{Bmatrix}$$

$$f\left(\vec{x}^{(2)}\right) = 1.5101 \quad ; \quad f\left(\vec{x}^{(1)}\right) = 1.6667$$

4.38 Minimize $f = x_1 + x_2 + x_3$

subject to $x_1 + x_2 - x_3 = 4$

$3x_1 - x_2 = 0$

$x_i \geq 0, \quad i = 1, 2, 3$

Here $\vec{d} = \{1 \quad 1 \quad 1\}^T$, $[\alpha] = \begin{bmatrix} 1 & 1 & -1 \\ 3 & -1 & 0 \end{bmatrix}$, $\vec{b} = \begin{Bmatrix} 4 \\ 0 \end{Bmatrix}$

$m = 3, \quad n = 6, \quad \beta = 10$

$$\vec{z} = \frac{1}{10} \begin{Bmatrix} z_1 \\ z_2 \\ z_3 \end{Bmatrix}, \qquad \vec{e} = \begin{Bmatrix} 1 \\ 1 \\ 1 \end{Bmatrix}$$

Minimize $F = \{10 \quad 10 \quad 10 \quad 0 \quad 0 \quad M\} \vec{z}$

subject to

$$\left[\begin{bmatrix} 1 & 1 & -1 \\ 3 & -1 & 0 \end{bmatrix} \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} - \frac{6}{10} \begin{Bmatrix} 4 \\ 0 \end{Bmatrix} \left(\frac{6}{10} \begin{Bmatrix} 4 \\ 0 \end{Bmatrix} - \begin{bmatrix} 1 & 1 & -1 \\ 3 & -1 & 0 \end{bmatrix} \begin{Bmatrix} 1 \\ 1 \\ 1 \end{Bmatrix} \right) \right] \vec{z} = \vec{0}$$

i.e.,

$$\begin{bmatrix} 1 & 1 & -1 & 0 & -\frac{12}{5} & \frac{7}{5} \\ 3 & -1 & 0 & 0 & 0 & -2 \end{bmatrix} \vec{z} = \vec{0}$$

$$\{-(1 \quad 1 \quad 1) \quad -1 \quad 5 \quad -1\} \vec{z} = 0$$

$$\sum_{i=1}^{n} z_i = 1$$

This leads to the desired form of the problem:

Minimize $F = \{10 \quad 10 \quad 10 \quad 0 \quad 0 \quad M\} \vec{z}$

subject to

$$z_1 + z_2 - z_3 - \frac{12}{5} z_5 + \frac{7}{5} z_6 = 0$$

$$3z_1 - z_2 - 2z_6 = 0$$

$$-z_1 - z_2 - z_3 - z_4 + 5z_5 - z_6 = 0$$

$$z_1 + z_2 + z_3 + z_4 + z_5 + z_6 = 1$$

$$\vec{z} \geq \vec{0}$$

4.39

Let x_{ij} = number of sets of equipment sent from city i to city j $(i = 1, 2 \; ; \; j = 1, 2, 3)$

$$\vec{X} = \{ x_{11} \quad x_{12} \quad x_{13} \quad x_{21} \quad x_{22} \quad x_{23} \}^T$$

Minimize $f = 4 x_{11} + 3 x_{12} + x_{13} + 2 x_{21} + 7 x_{22} + 5 x_{23}$

subject to

$$x_{11} + x_{12} + x_{13} = 3$$

$$x_{21} + x_{22} + x_{23} = 3$$

$$x_{11} + x_{21} = 2$$

$$x_{12} + x_{22} = 3$$

$$x_{13} + x_{23} = 1$$

all $x_{ij} \geq 0$

4.40

Minimize $f = 3 x_1^2 + 2 x_2^2 + 5 x_3^2 - 4 x_1 x_2 - 2 x_1 x_3 - 2 x_2 x_3$

subject to

$$3 x_1 + 5 x_2 + 2 x_3 \geq 10$$

$$3 x_1 + 5 x_3 \leq 15$$

$$x_i \geq 0 \; , \; i = 1, 2, 3$$

$$[H] = \left[\frac{\partial^2 f}{\partial x_i \, \partial x_j} \right] = \begin{bmatrix} 6 & -4 & -2 \\ -4 & 4 & -2 \\ -2 & -2 & 10 \end{bmatrix}$$

constraints:

$$[A] \vec{X} \leq \vec{B}$$

with

$$[A] = \begin{bmatrix} -3 & -5 & -2 \\ 3 & 0 & 5 \end{bmatrix} \; , \quad \vec{B} = \begin{Bmatrix} -10 \\ 15 \end{Bmatrix}$$

Solution:

$$\vec{X}^* = \begin{Bmatrix} x_1^* \\ x_2^* \\ x_3^* \end{Bmatrix} = \begin{Bmatrix} 0.981461 \\ 1.232280 \\ 0.447110 \end{Bmatrix} \; , \quad f^* = 0.109052$$

4.41

Minimize $f = (x_1 + 0.5\,x_2)(0.375 - 0.00005\{x_1 + 0.5\,x_2\})$
$+ (0.2\,x_1 + 0.5\,x_2)(0.750 - 0.0001\{0.2\,x_1 + 0.5\,x_2\})$
$- x_1(2.00 - 0.0005\,x_1 - 0.00015\,x_2)$
$- x_2(3.50 - 0.0002\,x_1 - 0.0015\,x_2)$

$= -1.475\,x_1 - 2.9375\,x_2 + 0.000446\,x_1^2$
$+ 0.0014625\,x_2^2 + 0.000280\,x_1 x_2$

$$\frac{\partial f}{\partial x_1} = -1.475 + 2(0.000446)\,x_1 + 0.000280\,x_2$$

$$\frac{\partial^2 f}{\partial x_1^2} = 0.000892, \quad \frac{\partial^2 f}{\partial x_1 \partial x_2} = 0.000280$$

$$\frac{\partial f}{\partial x_2} = -2.9375 + 2(0.0014625)\,x_2 + 0.000280\,x_1$$

$$\frac{\partial^2 f}{\partial x_2^2} = 0.0029250$$

$$[H] = \begin{bmatrix} 0.000892 & 0.000280 \\ 0.000280 & 0.0029250 \end{bmatrix}$$

Constraints: $x_1 + 0.5\,x_2 \le 1000$

$0.2\,x_1 + 0.5\,x_2 \le 250$

$x_1 \ge 0, \quad x_2 \ge 0$

Solution: $\vec{X}^* = \begin{Bmatrix} 655.810 \\ 237.676 \end{Bmatrix}$, $f^* = -1347.4141$

4.42 $\vec{X} = \{ M_1 \quad M_2 \quad M_3 \quad M_4 \quad M_5 \quad M_6 \quad M_7 \}^T$

Assume: Reserve strengths are positive for safety.

Minimize $f(\vec{X}) = 200(M_1 + M_2 + M_6 + M_7) + 100(M_3 + M_4 + M_5)$

subject to

$Z_1 = M_3 + 2M_4 + M_5 - 10^5 \geq 0$

$Z_2 = M_2 + 2M_4 + M_6 - 10^5 \geq 0$

$Z_3 = M_2 + 2M_4 + M_5 - 10^5 \geq 0$

$Z_4 = M_3 + 2M_4 + M_6 - 10^5 \geq 0$

$Z_5 = M_1 + M_2 + M_6 + M_7 - 75000 \geq 0$

$Z_6 = M_1 + M_3 + M_5 + M_7 - 75000 \geq 0$

$Z_7 = M_1 + M_3 + M_6 + M_7 - 75000 \geq 0$

$Z_8 = M_1 + M_2 + M_5 + M_7 - 75000 \geq 0$

$Z_9 = M_1 + 2M_4 + 2M_6 + M_7 - 175000 \geq 0$

$Z_{10} = M_1 + 2M_4 + 2M_5 + M_7 - 175000 \geq 0$

with

$0 \leq M_i \leq 2 * 10^5 \quad ; \quad i = 1, 2, \ldots, 7$

This problem has 7 variables and 10 inequality constraints.

Dual problem:

will have 7 constraints and 10 variables

∴ Solve the problem using dual simplex method.

4.43

M-file: LP08.m

```
clear all; clc; close all;
f = [2 9 24 8 5 0 0]';
A = [];
b = []';
lb = zeros(7,1);
Aeq = [1 1 2 0 -1 -1 0;-2 0 1 1 1 0 -1];
beq = [1 2]';
options = optimset('Display','iter');
[x,fval,exitflag,output] = linprog(f,A,b,Aeq,beq,lb,[],[],options)
```

Output (Results):

```
Residuals:    Primal      Dual      Duality     Total
              Infeas      Infeas      Gap        Rel
              A*x-b     A'*y+z-f    x'*z       Error
-------------------------------------------------------
Iter    0:  1.99e+002 3.08e+001 1.29e+004 4.80e+003
Iter    1:  4.49e-014 4.80e-015 2.03e+003 9.96e-001
Iter    2:  0.00e+000 3.55e-015 2.29e+001 7.38e-001
Iter    3:  5.55e-016 6.22e-015 8.17e+000 2.72e-001
Iter    4:  6.08e-014 1.99e-015 7.71e-001 3.14e-002
Iter    5:  1.78e-015 5.33e-015 1.47e-002 6.12e-004
Iter    6:  4.58e-016 5.18e-015 7.38e-007 3.08e-008
Iter    7:  9.49e-016 4.17e-015 1.00e-013 4.00e-015
Optimization terminated.

x =

    0.0000
    0.0000
    0.5000
    1.5000
    0.0000
    0.0000
    0.0000


fval =

   24.0000


exitflag =

     1


output =

      iterations: 7
       algorithm: 'large-scale: interior point'
    cgiterations: 0
         message: 'Optimization terminated.'
```

4.44

M-file: LP09.m

```
clear all; clc; close all;
f = [-8 -3 -8 -6]';
A = [4 3 1 3;4 -1 1 0;1 2 0 0; 3 1 0 0;0 0 2 3; 0 0 4 1];
b = [16 12 8 10 9 12]';
lb = zeros(4,1);
Aeq = [];
beq = [];
options = optimset('Display','iter');
[x,fval,exitflag,output] = linprog(f,A,b,Aeq,beq,lb,[],[],options)
```

Output (Results):

```
Residuals:    Primal      Dual      Duality     Total
              Infeas      Infeas      Gap        Rel
              A*x-b     A'*y+z-f     x'*z       Error
 ---------------------------------------------------------
 Iter    0:  1.65e+003 3.03e+001 6.11e+003 2.50e+003
 Iter    1:  1.58e+002 2.80e-015 6.88e+002 5.63e+000
 Iter    2:  2.05e-014 2.78e-015 8.93e+001 8.12e-001
 Iter    3:  9.23e-015 2.49e-014 3.41e+001 5.08e-001
 Iter    4:  2.36e-012 7.34e-015 1.21e+001 2.06e-001
 Iter    5:  4.27e-014 1.96e-015 8.65e-001 1.78e-002
 Iter    6:  1.53e-013 2.05e-015 4.48e-002 9.35e-004
 Iter    7:  6.39e-013 1.28e-015 1.09e-004 2.27e-006
 Iter    8:  1.07e-014 1.28e-015 1.09e-009 2.27e-011
Optimization terminated.

x =

    2.3500
    0.1000
    2.7000
    1.2000


fval =

  -47.9000


exitflag =

     1


output =

      iterations: 8
       algorithm: 'large-scale: interior point'
    cgiterations: 0
         message: 'Optimization terminated.'
```

4.45

M-file: LP10.m

```
clear all; clc; close all;
f = [10 2 4 8 1]';
A = [-1 -4 3 0 0; -2 -1 -1 0 0;-3 0 0 -1 -1;-1 0 0 -2 1];
b = [-16 -4 -8 -20]';
lb = zeros(5,1);
Aeq = [];
beq = [];
options = optimset('Display','iter');
[x,fval,exitflag,output] = linprog(f,A,b,Aeq,beq,lb,[],[],options)
```

Output (Results):

```
  Residuals:    Primal      Dual      Duality     Total
                Infeas     Infeas       Gap        Rel
                A*x-b    A'*y+z-f     x'*z       Error
  ------------------------------------------------------
  Iter    0:  5.05e+002 1.73e+001 7.70e+003 2.50e+003
  Iter    1:  6.70e-014 1.70e+000 1.19e+003 9.71e-001
  Iter    2:  2.65e-011 5.19e-016 2.22e+002 8.47e-001
  Iter    3:  1.32e-012 4.44e-016 2.52e+001 2.36e-001
  Iter    4:  7.16e-015 4.89e-014 5.15e+000 5.58e-002
  Iter    5:  3.35e-013 9.93e-016 4.44e-002 5.04e-004
  Iter    6:  1.62e-012 8.90e-016 2.40e-006 2.72e-008
  Iter    7:  1.18e-014 3.66e-015 3.54e-013 4.84e-015
Optimization terminated.

x =

    0.0000
    4.0000
    0.0000
   10.0000
    0.0000


fval =

   88.0000


exitflag =

     1


output =

      iterations: 7
       algorithm: 'large-scale: interior point'
    cgiterations: 0
         message: 'Optimization terminated.'
```

4.46

M-file: LP11.m

```
clear all; clc; close all;
f = [2 -2 5]';
A = [];
b = [];
lb = zeros(3,1);
Aeq = [1 -1 0;1 1 1];
beq = [0 1]';
options = optimset('Display','iter');
[x,fval,exitflag,output] = linprog(f,A,b,Aeq,beq,lb,[],[],options)
```

Output (Results):

```
 Residuals:    Primal      Dual      Duality    Total
               Infeas     Infeas      Gap       Rel
               A*x-b     A'*y+z-f    x'*z      Error
 ---------------------------------------------------------
 Iter    0:  2.99e+002 5.72e+000 1.45e+003 5.00e+002
 Iter    1:  1.00e-013 8.88e-016 8.21e+000 1.68e+000
 Iter    2:  2.22e-016 1.99e-015 5.09e+000 1.86e+000
 Iter    3:  8.47e-013 1.99e-015 4.84e-001 4.84e-001
 Iter    4:  1.00e-013 8.88e-016 3.19e-004 3.19e-004
 Iter    5:  1.57e-016 4.44e-016 1.59e-008 1.59e-008
 Iter    6:  1.67e-016 8.88e-016 1.59e-015 1.07e-015
Optimization terminated.

x =

    0.5000
    0.5000
    0.0000


fval =

  7.1609e-016


exitflag =

     1


output =

      iterations: 6
       algorithm: 'large-scale: interior point'
    cgiterations: 0
         message: 'Optimization terminated.'
```

4.47

M-file: LP12.m

```
clear all; clc; close all;
f = [3 5 -3]';
A = [];
b = []';
lb = zeros(3,1);
Aeq = [1 0 -1;1 1 1];
beq = [0 1]';
options = optimset('Display','iter');
[x,fval,exitflag,output] = linprog(f,A,b,Aeq,beq,lb,[],[],options)
```

Output (Results):

```
  Residuals:    Primal      Dual      Duality     Total
                Infeas     Infeas       Gap        Rel
                A*x-b     A'*y+z-f     x'*z       Error
  ---------------------------------------------------------
  Iter     0:  2.99e+002 7.18e+000 1.66e+003 5.00e+002
  Iter     1:  1.31e+000 2.04e-015 2.11e+001 1.36e+000
  Iter     2:  2.22e-016 6.28e-016 5.16e+000 1.04e+000
  Iter     3:  4.44e-016 1.91e-014 2.70e+001 1.00e+000
  Iter     4:  4.44e-016 2.51e-015 1.53e-003 1.53e-003
  Iter     5:  4.48e-016 4.44e-016 7.63e-008 7.63e-008
  Iter     6:  2.22e-016 4.44e-016 7.63e-015 8.08e-015
Optimization terminated.

x =

    0.5000
    0.0000
    0.5000


fval =

  1.5543e-015


exitflag =

     1


output =

      iterations: 6
       algorithm: 'large-scale: interior point'
    cgiterations: 0
         message: 'Optimization terminated.'
```

4.48

M-file: QP02.m

```
clear all; clc; close all;
H = [2 0;0 0];
f = [-2 -1];
A = [2 3;2 1];
b = [6 4]';
lb = zeros(2,1);
[x,fval] = quadprog(H,f,A,b,[],[],lb,[])
```

Output (Results):

```
Warning: Large-scale method does not currently solve this problem formulation,
using medium-scale method instead.
> In quadprog at 263
  In QP02 at 7
Optimization terminated.

x =

    0.6667
    1.5556


fval =

   -2.4444
```

4.49

M-file: QP03.m

```
clear all; clc; close all;
H = [2 0;0 2];
f = [-4 -6];
A = [1 1];
b = 2;
lb = zeros(2,1);
[x,fval] = quadprog(H,f,A,b,[],[],lb,[])
```

Output (Results):

```
Warning: Large-scale method does not currently solve this problem formulation,
using medium-scale method instead.
> In quadprog at 263
  In QP03 at 7
Optimization terminated.

x =

    0.5000
    1.5000


fval =

   -8.5000
```

4.50

M-file: QP04.m

```
clear all; clc; close all;
H = [2 0;0 0];
f = [-2 1];
A = [1 1];
b = 2;
Aeq = [-1 1];
beq = 1;
lb = zeros(2,1);
[x,fval] = quadprog(H,f,A,b,Aeq,beq,lb,[])
```

Output (Results):

```
Warning: Large-scale method does not currently solve this problem formulation,
using medium-scale method instead.
> In quadprog at 263
  In QP04 at 9
Optimization terminated.

x =

    0.5000
    1.5000


fval =

    0.7500
```

4.51

M-file: QP05.m

```
clear all; clc; close all;
H = [2 -3;-3 2];
f = [-6 5]';
A = [1 1;3 6];
b = [4 20]';
lb = zeros(2,1);
[x,fval] = quadprog(H,f,A,b,[],[],lb,[])
```

Output (Results):

```
Warning: Large-scale method does not currently solve this problem formulation,
using medium-scale method instead.
> In quadprog at 263
  In QP05 at 7
Optimization terminated.

x =

    3.1000
    0.9000


fval =

  -12.0500
```

Chapter 5
Nonlinear Programming I:
One-Dimensional Minimization Methods

5.1 Minimize the function:

$$f(x) = 0.65 - \frac{0.75}{1+x^2} - 0.65\, x \tan^{-1}\frac{1}{x}$$

(a) Unrestricted search with step size 0.1 from 0.0:

x	$f(x)$	
0.0000	-0.1000000E+00	
0.1000	-0.1881976E+00	
0.2000	-0.2496960E+00	
0.3000	-0.2875446E+00	
0.4000	-0.3060271E+00	} x^* lies in the interval 0.4 to 0.6.
0.5000	-0.3098234E+00	
0.6000	-0.3033176E+00	
0.7000	-0.2901877E+00	
0.8000	-0.2732659E+00	
0.9000	-0.2545837E+00	$x^* \simeq 0.5$
1.0000	-0.2355088E+00	
1.1000	-0.2169043E+00	
1.2000	-0.1992729E+00	
1.3000	-0.1828732E+00	
1.4000	-0.1678054E+00	
1.5000	-0.1540718E+00	
1.6000	-0.1416174E+00	
1.7000	-0.1303571E+00	
1.8000	-0.1201921E+00	
1.9000	-0.1110200E+00	
2.0000	-0.1027418E+00	
2.1000	-0.9526438E-01	
2.2000	-0.8850193E-01	
2.3000	-0.8237725E-01	
2.4000	-0.7682091E-01	
2.5000	-0.7177109E-01	
2.6000	-0.6717324E-01	
2.7000	-0.6297886E-01	
2.8000	-0.5914527E-01	
2.9000	-0.5563456E-01	
3.0000	-0.5241358E-01	

(b) Unrestricted search with accelerated step size:

Step size	x	$f(x)$	
0.1000E+00	0.0000E+00	-0.1000000E+00	
0.2000E+00	0.1000E+00	-0.1881976E+00	
0.4000E+00	0.3000E+00	-0.2875446E+00	} Final interval = (0.3, 1.5)
0.8000E+00	0.7000E+00	-0.2901877E+00	
0.1600E+01	0.1500E+01	-0.1540717E+00	
0.3200E+01	0.3100E+01	-0.4945296E-01	
0.6400E+01	0.6300E+01	-0.1305407E-01	

$$x^* \simeq \frac{0.3+1.5}{2} = 0.9$$

(c) Exhaustive search:

If the midpoint of the final interval is taken as x^*,

$$\frac{1}{n+1} \leq \frac{1}{20} \quad \text{or} \quad n > 19$$

Using $n = 20$, we obtain the following result:

x	$f(x)$	
0.000	-0.1000000E+00	
0.150	-0.2221322E+00	
0.300	-0.2875446E+00	} Interval for x^* = (0.3, 0.6)
0.450	-0.3094738E+00	
0.600	-0.3033176E+00	
0.750	-0.2820564E+00	
0.900	-0.2545837E+00	
1.050	-0.2261094E+00	
1.200	-0.1992729E+00	
1.350	-0.1751709E+00	
1.500	-0.1540718E+00	
1.650	-0.1358441E+00	
1.800	-0.1201920E+00	
1.950	-0.1067752E+00	
2.100	-0.9526443E-01	
2.250	-0.8536446E-01	
2.400	-0.7682091E-01	
2.550	-0.6941885E-01	
2.700	-0.6297880E-01	
2.850	-0.5735165E-01	
3.000	-0.5241364E-01	

$$x^* \simeq \frac{0.3+0.6}{2} = 0.45$$

(d) Dichotomous search:

$\delta = 0.0001$

$$\frac{1}{2}\frac{L_n}{L_0} \leq \frac{1}{20} \quad \text{or} \quad \frac{L_n}{L_0} \leq \frac{1}{10}$$

i.e., $$\frac{1}{2^{n/2}} + \frac{0.0001}{3}\left(1 - \frac{1}{2^{n/2}}\right) \leq \frac{1}{10}$$

i.e., $$\frac{1}{2^{n/2}} \leq 0.09997$$

i.e., $$2^{n/2} \geq 10.0030$$

We select $n = 8$.

$x_1 = 1.5 - \frac{0.0001}{2} = 1.49995$, $f(x_1) = -0.15407827$

$x_2 = 1.5 + \frac{0.0001}{2} = 1.50005$, $f(x_2) = -0.15406522$

New interval = $(0.0, 1.50005)$.

$x_3 = \frac{0.0 + 1.50005}{2} - 0.00005 = 0.749975$

$x_4 = 0.750025 + 0.00005 = 0.750075$

$f(x_3) = -0.28206065$, $f(x_4) = -0.28204364$

New interval = $(0.0, 0.750075)$

$x_5 = (0.0 + 0.750075)/2 - 0.00005 = 0.3749875$

$x_6 = (0.0 + 0.750075)/2 + 0.00005 = 0.3750875$

$f(x_5) = -0.30296370$, $f(x_6) = -0.30297792$

New interval = $(0.3749875, 0.750075)$

$x_7 = (0.3749875 + 0.750075)/2 - 0.00005 = 0.56248125$

$x_8 = (0.3749875 + 0.750075)/2 + 0.00005 = 0.56258125$

$f(x_7) = -0.30671439$, $f(x_8) = -0.30670676$

$\therefore$ Final interval = $(0.3749875, 0.56258125)$

$\therefore x^* \simeq (0.3749875 + 0.56258125)/2 = 0.468784375$

(e) Interval halving method:

$$\frac{1}{2}\ \frac{L_n}{L_0} \leq \frac{1}{20}$$

$$\left(\frac{1}{2}\right)^{\frac{n-1}{2}} \leq \frac{1}{10} \quad ; \quad n = \text{odd}$$

For $n = 5$, $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$

For $n = 7$, $\left(\frac{1}{2}\right)^3 = \frac{1}{8}$

For $n = 9$, $\left(\frac{1}{2}\right)^4 = \frac{1}{16}$

$\therefore n = 9$

$x_1 = 0.75, \quad x_0 = 1.50, \quad x_2 = 2.25$

$f(x_1) = -0.28205642, \quad f(x_0) = -0.15407175, \quad f(x_2) = -0.08536446$

New interval = $(0.0, 1.5)$

$x_1 = 0.375, \quad x_0 = 0.75, \quad x_2 = 1.125$

$f(x_1) = -0.30296549, \quad f(x_0) = -0.28205642, \quad f(x_2) = -0.21239173$

New interval = $(0.0, 0.75)$

$x_1 = 0.1875, \quad x_0 = 0.375, \quad x_2 = 0.5625$

$f(x_1) = -0.24337985, \quad f(x_0) = -0.30296549, \quad f(x_2) = -0.30671299$

New interval = $(0.375, 0.750)$

$x_1 = 0.46875, \quad x_0 = 0.5625, \quad x_2 = 0.65625$

$f(x_1) = -0.30993825, \quad f(x_0) = -0.30671299, \quad f(x_2) = -0.29654604$

Final interval = $(0.375, 0.5625)$

$$x^* \simeq \frac{0.3750 + 0.5625}{2} = 0.46875$$

5.2 Minimize $f(x) = 0.65 - \dfrac{0.75}{1+x^2} - 0.65\, x \tan^{-1}\dfrac{1}{x}$ using Quadratic Interpolation Method.

Let $t_0 = 0.1$ and $A = 0$

$f_A = f(0) = -0.10$, $f_1 = f(t_0 = 0.1) = -0.1882$

Since $f_1 < f_A$, we set $f_B = f_1 = -0.1882$, and find

$f_2 = f(2t_0 = 0.2) = -0.249696$

As $f_2 < f_1$, we set new $t_0 = 0.2$ and $f_1 = -0.249696$.

But $f_1 < f_A$. Set $f_B = f_1 = -0.249696$ and find

$f_2 = f(2t_0 = 0.4) = -0.306027$

As $f_2 < f_1$, we take $t_0 = 0.4$ and $f_1 = -0.306027$.

Noting that $f_1 < f_A$, we set $f_B = f_1 = -0.306027$, $f_2 = f(2t_0 = 0.8) = -0.273266$, and note that $f_2 > f_1$.

Hence we take $f_C = f_2 = -0.273266$ and compute $\tilde{x}^*$ using Eq. (5.40) as

$$\tilde{x}^* = \frac{4(-0.306027) - 3(-0.10) - (-0.273266)}{4(-0.306027) - 2(-0.273266) - 2(-0.10)}(0.4)$$

$$= 0.545121$$

Convergence test:

$A = 0$, $B = 0.4$, $C = 0.8$

$f_A = -0.10$, $f_B = -0.306027$, $f_C = -0.273266$

$a = -0.10$, $b = -0.813553$, $c = 0.746213$

$h(\tilde{x}^*) = h(0.545121) = -0.321742$

$f(\tilde{x}^*) = -0.307924$

$$\left|\frac{h(\tilde{x}^*) - f(\tilde{x}^*)}{f(\tilde{x}^*)}\right| = 0.0449 > \epsilon = 0.001 \text{ (assumed)}$$

Refit 1:

Since $B < \tilde{x}^*$ and $f_B > f(\tilde{x}^*)$, we take

$A = 0.4$, $B = 0.545121$, $C = 0.8$

$f_A = -0.306027$, $f_B = -0.307924$, $f_C = -0.273266$

$a = -0.219548$, $b = -0.365248$, $c = 0.372626$

$\tilde{x}^* = 0.490099$ [Eq. (5.36)]

$h(\tilde{x}^*) = -0.309052$, $f(\tilde{x}^*) = -0.309974$

$$\left| \frac{h(\tilde{x}^*) - f(\tilde{x}^*)}{f(\tilde{x}^*)} \right| = 0.003 > \epsilon = 0.001$$

Refit 2:

Since $\tilde{x}^* < B$ and $f_B > f(\tilde{x}^*)$, we choose

$A = 0.4$, $B = 0.490099$, $C = 0.545121$,

$f_A = -0.306027$, $f_B = -0.309974$, $f_C = -0.307924$

and compute

$a = -0.178998$, $b = -0.541027$, $c = 0.558612$, and

$\tilde{x}^* = 0.484260$ [Eq. (5.36)]

This gives

$h(\tilde{x}^*) = -0.309997$

$f(\tilde{x}^*) = -0.310014$

and

$$\left| \frac{h(\tilde{x}^*) - f(\tilde{x}^*)}{f(\tilde{x}^*)} \right| = 0.000056 < \epsilon = 0.001$$

$\therefore$ we take $x^* = \tilde{x}^* = 0.484260$

5.3

$$f(x) = 0.65 - \frac{0.75}{1+x^2} - 0.65\,x\,\tan^{-1}\left(\frac{1}{x}\right)$$

$$f'(x) = \frac{df}{dx} = \frac{1.5\,x}{(1+x^2)^2} - 0.65\,\tan^{-1}\left(\frac{1}{x}\right) + \frac{0.65\,x}{1+x^2}$$

Let $A = 0$, $t_0 = 0.1$

$f_A = f(0) = -0.10$, $f_A' = f'(0) = -1.02102$

$f_B = f(0.1) = -0.18820$, $f_B' = f'(0.1) = -0.74483$

Since $f_B' < 0$, set $t_0 = 0.2$ and find

$f_B = f(0.2) = -0.24970$ and $f_B' = f'(0.2) = -0.49034$

As $f_B' < 0$, set $t_0 = 0.4$ and find

$f_B = f(0.4) = -0.30603$ and $f_B' = f'(0.4) = -0.10365$

As $f_B' < 0$, set $t_0 = 0.8$ and find

$f_B = f(0.8) = -0.27327$ and $f_B' = f'(0.8) = 0.18080$

Since $f_B' > 0$, we take $A = 0$ and $B = 0.8 = t_0$, and evaluate

$$Z = \frac{3(f_A - f_B)}{(B-A)} + f_A' + f_B'$$

$$= \frac{3(-0.1 + 0.27327)}{(0.8-0)} - 1.02102 + 0.18080 = -0.190$$

$$Q = \left(Z^2 - f_A' f_B'\right)^{\frac{1}{2}} = \left\{(-0.190)^2 - (-1.02102)(0.18080)\right\}^{\frac{1}{2}}$$

$$= 0.470$$

$$\tilde{x}^* = A + (f_A' + Z \pm Q)(B-A) \Big/ (f_A' + f_B' + 2Z)$$

$$= 0 + (-1.02102 - 0.190 \pm 0.470)(0.8-0) \Big/ (-1.02102 + 0.18080 - 2*0.190)$$

$$= 0.486 \quad \text{or} \quad 1.102$$

We select $\tilde{x}^* = 0.486$ since $A < \tilde{x}^* < B$.

Convergence check:

Test 1: $\left|f'(\tilde{x}^*)\right| = |0.005| > \epsilon = 0.001$ (assumed)

Test 2: Using $A=0$, $B=0.8$, $f_A=-0.1$, $f_A'=-1.02102$,

$f_B=-0.27327$ and $f_B'=0.18080$,

Eqs. (5.47) to (5.51) give

$$h = a + bx + cx^2 + dx^3$$

$$= -0.1 - 1.021x + 1.514x^2 - 0.636x^3$$

$h(\tilde{x}^*) = -0.312$; $f(\tilde{x}^*) = -0.310$

$$\left|\frac{h(\tilde{x}^*) - f(\tilde{x}^*)}{f(\tilde{x}^*)}\right| = \left|\frac{-0.312 + 0.310}{-0.310}\right| = 0.006 > \epsilon$$

Refit 1:

Since $f'(\tilde{x}^*) = 0.005 > 0$, we set

$B = \tilde{x}^* = 0.486$, $f_B = -0.310$, $f_B' = 0.005$

$A = 0$, $f_A = -0.10$, $f_A' = -1.02102$

Using Eqs. (5.51), (5.55) and (5.54), we find

$Z = 0.281$, $Q = 0.291$, $\tilde{x}^* = 0.481$ or 1.105

We select $\tilde{x}^* = 0.481$ since $A < \tilde{x}^* < B$.

Convergence check:

Test 1: $f'(\tilde{x}^*) = f'(0.481) = 0.000 < \epsilon$

Test 2: Eqs. (5.47) to (5.51) yield

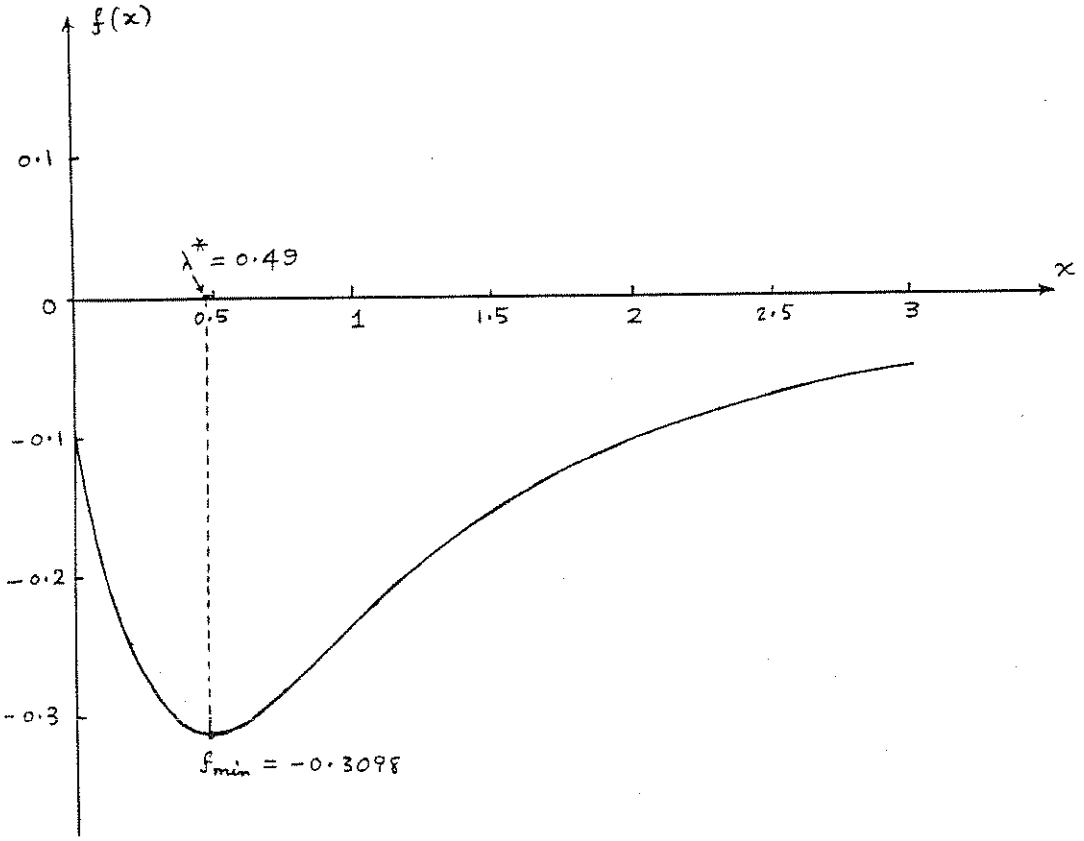
$a = -0.1$, $b = -1.021$, $c = 1.523$, $d = -0.640$

$h(\tilde{x}^*) = -0.310$

$$\left|\frac{h(\tilde{x}^*) - f(\tilde{x}^*)}{f(\tilde{x}^*)}\right| = \left|\frac{-0.310 + 0.310}{-0.310}\right| = 0.0 < \epsilon$$

$\therefore$ $x^* = \tilde{x}^* = 0.481$ with $f(x^*) = -0.31002$.

5.4



5.5

Eq. (1) can be expressed as

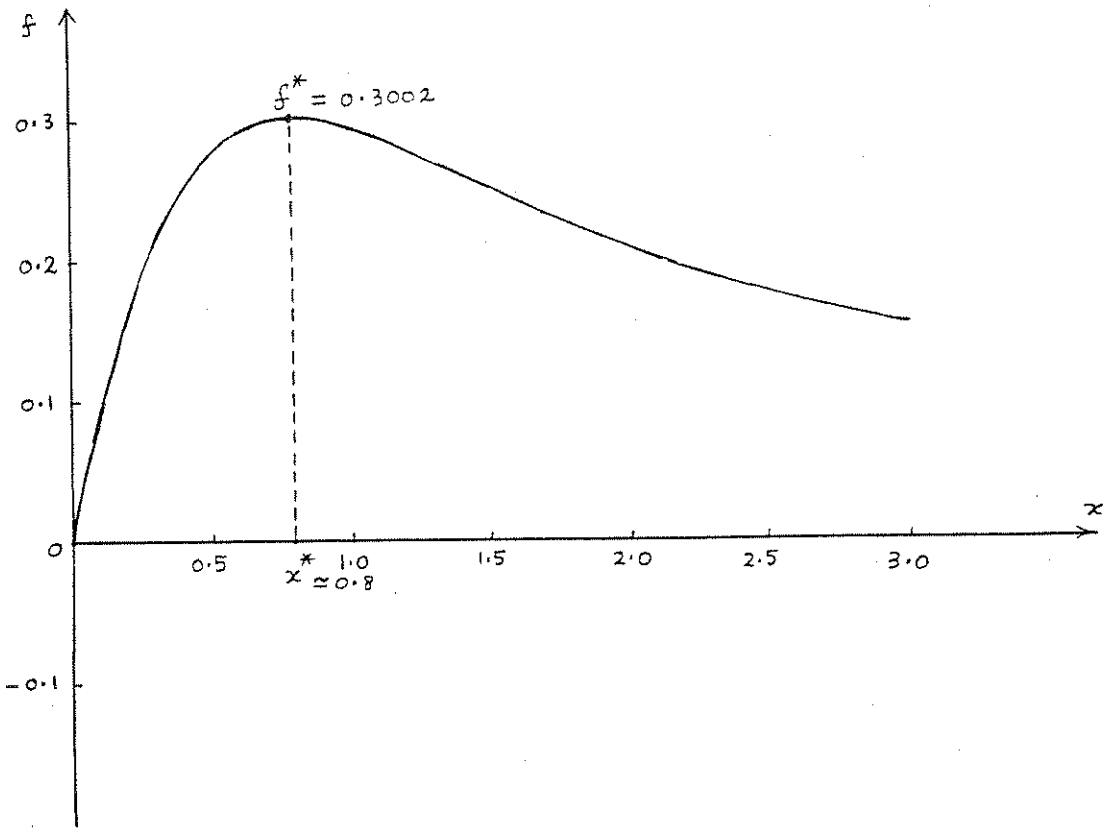
$$f = -\frac{1}{2}\left[-\frac{1}{\sqrt{1+\lambda^2}} + \left(2 - \frac{1}{1+\lambda^2}\right)\sqrt{1+\lambda^2} - 2\lambda\right]$$

with $f = \tau_{zy}/p_{max}$ and $\lambda = z/b$

The above equation can be simplified as

$$f = \frac{0.5}{\sqrt{1+\lambda^2}} - \sqrt{1+\lambda^2}\left(1 - \frac{0.5}{1+\lambda^2}\right) + \lambda$$

5.6



5.7 (a) Maximize the function:

$$f(\lambda) = \frac{0.5}{\sqrt{1+\lambda^2}} - \sqrt{1+\lambda^2}\left(1 - \frac{0.5}{1+\lambda^2}\right) + \lambda = \lambda - \frac{\lambda^2}{\sqrt{1+\lambda^2}}$$

Unrestricted search with fixed step size:

λ	$f(\lambda)$
0.0000	0.0000000E+00
0.1000	0.9004957E-01
0.2000	0.1607769E+00
0.3000	0.2137956E+00
0.4000	0.2514438E+00
0.5000	0.2763932E+00
0.6000	0.2913026E+00
0.7000	0.2985764E+00
0.8000	0.3002439E+00
0.9000	0.2979318E+00
1.0000	0.2928932E+00
1.1000	0.2860659E+00
1.2000	0.2781343E+00
1.3000	0.2695888E+00
1.4000	0.2607733E+00
1.5000	0.2519245E+00
1.6000	0.2432026E+00
1.7000	0.2347118E+00

Interval = (0.7, 0.9)

$$\lambda^* \simeq \frac{0.7 + 0.9}{2} = 0.8$$

1.8000	0.2265168E+00
1.9000	0.2186553E+00
2.0000	0.2111456E+00
2.1000	0.2039928E+00
2.2000	0.1971936E+00
2.3000	0.1907387E+00
2.4000	0.1846154E+00
2.5000	0.1788085E+00
2.6000	0.1733015E+00
2.7000	0.1680784E+00
2.8000	0.1631229E+00
2.9000	0.1584184E+00
3.0000	0.1539502E+00

(b) Unrestricted search with accelerated step:

step size	λ	$f(\lambda)$
0.1000E+00	0.0000E+00	0.0000000E+00
0.2000E+00	0.1000E+00	0.9004957E-01
0.4000E+00	0.3000E+00	0.2137956E+00
0.8000E+00	0.7000E+00	0.2985763E+00
0.1600E+01	0.1500E+01	0.2519245E+00
0.3200E+01	0.3100E+01	0.1497030E+00
0.6400E+01	0.6300E+01	0.7789660E-01

Interval = (0.3, 1.5)

$\lambda^* \simeq \frac{0.3+1.5}{2} = 0.9$

(c) Exhaustive search:

$$\frac{1}{n+1} \leq \frac{1}{20} \quad \text{or} \quad n > 19$$

Using $n = 20$, we obtain the following result:

λ	$f(\lambda)$
0.000	0.0000000E+00
0.150	0.1277490E+00
0.300	0.2137956E+00
0.450	0.2653359E+00
0.600	0.2913026E+00
0.750	0.3000000E+00
0.900	0.2979317E+00
1.050	0.2896553E+00
1.200	0.2781345E+00
1.350	0.2651980E+00
1.500	0.2519244E+00
1.650	0.2389235E+00
1.800	0.2265170E+00
1.950	0.2148559E+00
2.100	0.2039928E+00
2.250	0.1939240E+00
2.400	0.1846154E+00
2.550	0.1760182E+00
2.700	0.1680784E+00
2.850	0.1607399E+00
3.000	0.1539502E+00

(d) Dichotomous search:

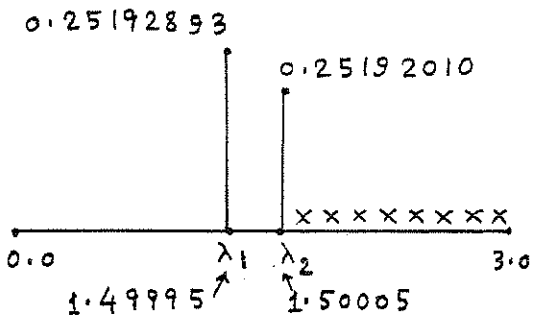
$\delta = 0.0001$

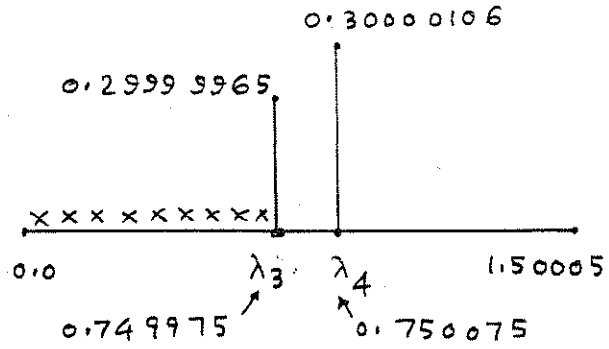
$$\frac{1}{2}\frac{L_n}{L_0} \le \frac{1}{20} \quad \text{or} \quad \frac{L_n}{L_0} \le \frac{1}{10}$$

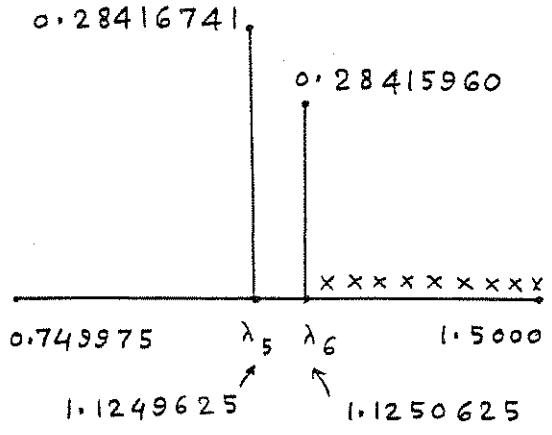
$$\frac{1}{2^{n/2}} + \frac{0.0001}{3}\left(1 - \frac{1}{2^{n/2}}\right) \le \frac{1}{10}$$

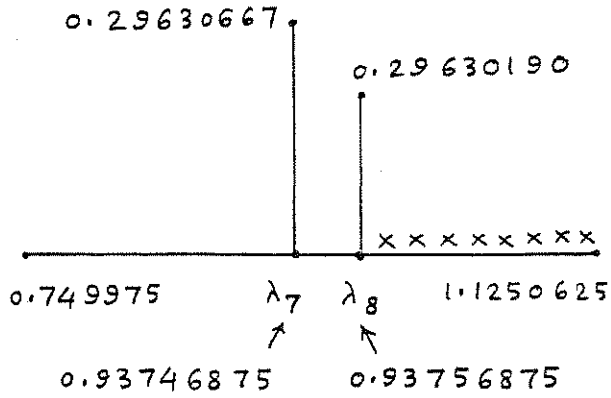
$n = 8$ satisfies this inequality.

Results are shown below:







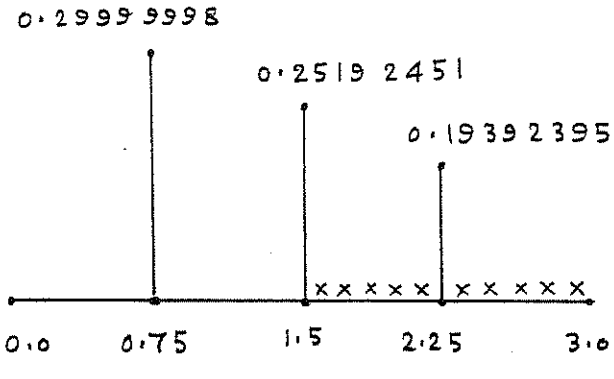


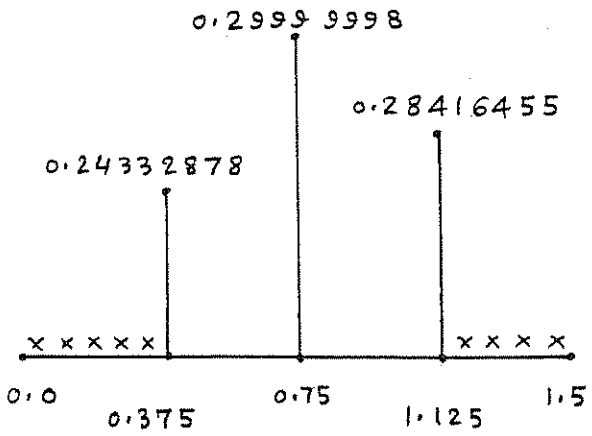
$$\lambda^* \simeq \frac{0.749975 + 0.93756875}{2} = 0.843771875$$

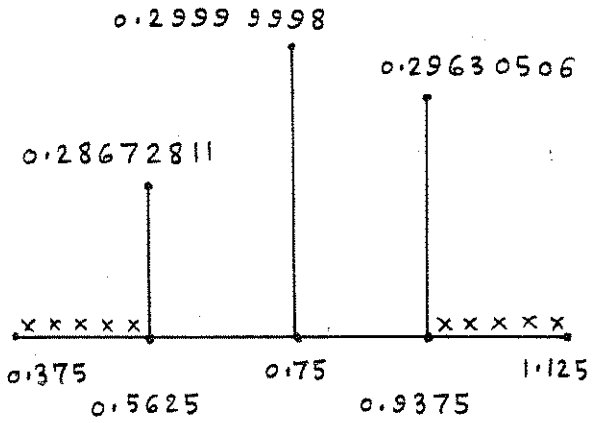
(e) Interval halving method:

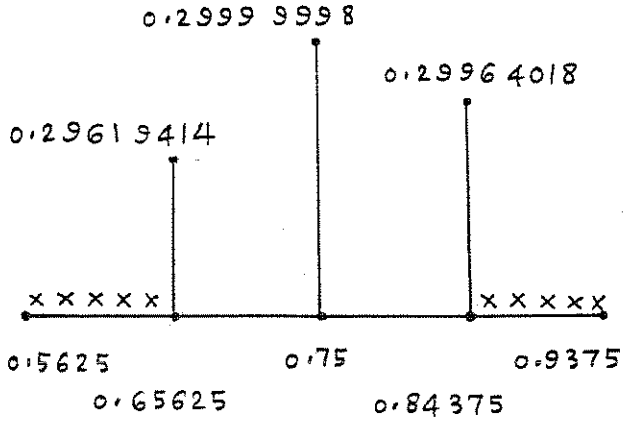
$$\frac{1}{2}\frac{L_n}{L_0} \le \frac{1}{20}$$

$$\left(\frac{1}{2}\right)^{\frac{n-1}{2}} \le \frac{1}{10} \qquad \therefore\ n = 9$$









Final interval = $(0.65625,\ 0.84375)$

$$\lambda^* \simeq \frac{0.65625 + 0.84375}{2} = 0.75$$

5.8

(a) Fibonacci method with $n = 8$: Let $L_0 = (0, 1)$.

$$L_2^* = \frac{F_{n-2}}{F_n} L_0 = \frac{F_6}{F_8} L_0 = \frac{13}{34}(1) = 0.3823529$$

Results are shown below:

i	a	b	x1	x2	f(x1)	f(x2)
2	0.0000000	1.0000000	0.3823529	0.6176471	0.2458005	0.2930779
3	0.3823529	1.0000000	0.6176471	0.7647059	0.2930779	0.3001852
4	0.6176471	1.0000000	0.7647059	0.8529412	0.3001852	0.2994280
5	0.6176471	0.8529412	0.7058824	0.7647058	0.2988119	0.3001852
6	0.7058824	0.8529412	0.7647059	0.7941176	0.3001852	0.3002701
7	0.7647059	0.8529412	0.7941177	0.8235294	0.3002701	0.3000058
8	0.7647059	0.8235294	0.7940677	0.7941676	0.3002703	0.3002700

The final interval of uncertainty is from 0.764706 to 0.794168
The final interval of uncertainty is 2.94617E-02
The estimate of the optimum point is 0.779437
The estimate of the optimum value is 0.300270

(b) Golden section method with $n = 8$: Let $L_0 = (0, 1)$.

i	a	b	x_1	x_2	$f(x_1)$	$f(x_2)$
2	0.0	1.0	0.382	0.618	0.2456833	0.2931111
3	0.382	1.0	0.618	0.764	0.2931111	0.3001786
4	0.618	1.0	0.764	0.854	0.3001786	0.2994017
5	0.618	0.854	0.708	0.764	0.2988920	0.3001786
6	0.708	0.854	0.764	0.798	0.3001786	0.3002544
7	0.764	0.854	0.798	0.820	0.3002544	0.3000547
8	0.764	0.820	0.786	0.798	0.3002831	0.3000547

Final interval of uncertainty = 0.764 to 0.798

$$x^* \simeq \frac{0.764 + 0.798}{2} = 0.781$$

$$f(x^*) \simeq f(0.781) = 0.3002775$$

5.9 Maximize

$$f(x) = \frac{0.5}{\sqrt{1+x^2}} - \sqrt{1+x^2}\left(1 - \frac{0.5}{1+x^2}\right) + x = x - \frac{x^2}{\sqrt{1+x^2}}$$

using quadratic interpolation method.

$A = 0$, $f_A = f(A=0) = 0$; $t_0 = 0.1$, $f_1 = f(t_0) = 0.09$

$f_1 > f_A$

$f_2 = f(0.2) = 0.1608$, $f_2 > f_1$

$f_3 = f(0.4) = 0.2514$, $f_3 > f_2$

$f_4 = f(0.8) = 0.3002$, $f_4 > f_3$

$f_5 = f(1.6) = 0.2432$, $f_5 < f_4$

Choose $A = 0$, $B = 0.8$, $C = 1.6$

$f_A = 0$, $f_B = 0.3002$, $f_C = 0.2432$; $t_0 = 0.8$

$$\tilde{x}^* = \left(\frac{4(0.3002) - 3(0) - 0.2432}{4(0.3022) - 2(0.2432) - 0}\right)(0.8) = 1.0723$$

$f(\tilde{x}^*) = 0.2881 > f_C$

Ignore C for next fit.

Convergence check:

$a = 0$, $b = 0.5985$, $c = -0.2791$

$h(\tilde{x}^*) = 0.3209$, $f(\tilde{x}^*) = 0.2881$

Since $h(\tilde{x}^*)$ is not close to $f(\tilde{x}^*)$, we use refitting.

Refit 1:

$A = 0$, $B = 0.8$, $C = 1.0723$

$f_A = 0$, $f_B = 0.3002$, $f_C = 0.2881$

$a = 0$, $b = 0.5704$, $c = -0.2440$

$\tilde{x}^* = 0.8794$

$h(\tilde{x}^*) = 0.3129$, $f(\tilde{x}^*) = 0.2987$

convergence check:

$$\left| \frac{f(\tilde{x}^*) - h(\tilde{x}^*)}{f(\tilde{x}^*)} \right| = 0.0477 = \text{not small}$$

Refit 2:

Since $f(\tilde{x}^*) > f_C$, we ignore C.

$A = 0$, $B = 0.8$, $C = 0.8794$

$f_A = 0$, $f_B = 0.3002$, $f_C = 0.2987$

$\tilde{x}^* = 0.8186$

with $f(\tilde{x}^*) = 0.3002$

$\therefore$ We take $x^* \simeq \tilde{x}^* = 0.8186$

5.10 Cubic interpolation method:

$$f(x) = x - \frac{x^2}{\sqrt{1+x^2}}$$

$$f'(x) = \frac{df}{dx} = 1 - \frac{2x}{\sqrt{1+x^2}} + \frac{x^3}{(1+x^2)^{3/2}}$$

x	$f(x)$	$f'(x)$
0.0	0.09005	0.801978
0.2	0.160777	0.615311
0.3	0.213796	0.44903
0.4	0.251444	0.308446
0.5	0.276393	0.195016
0.6	0.291303	0.107199
0.7	0.298576	0.041664
0.8	0.300244	−0.00561

We take $A = 0.7$, $f_A' = 0.041664$, $f_A = 0.298576$

$B = 0.8$, $f_B' = -0.00561$, $f_B = 0.300244$

$Z = -0.0139817$

$Q = 0.0207207$

$a = 0.0, \quad b = 0.8251546, \quad c = -0.8424936$

$\tilde{x}^* = 0.7860852$ or 1.298870

We select $\tilde{x}^* = 0.7860852$ and find

$f(\tilde{x}^*) = 0.3002831, \quad f'(\tilde{x}^*) = 0.00002741814$

Since $|f'(\tilde{x}^*)| < \epsilon = 0.0001$ (assumed),

we take

$$x^* \simeq \tilde{x}^* = 0.7860852$$

5.11

$$f(\lambda) = \frac{0.5}{\sqrt{1+\lambda^2}} - \sqrt{1+\lambda^2} + \frac{0.5}{\sqrt{1+\lambda^2}} + \lambda = \lambda - \frac{\lambda^2}{\sqrt{1+\lambda^2}}$$

$$f'(\lambda) = \frac{df}{d\lambda} = 1 - \frac{(2\lambda + \lambda^3)}{(1+\lambda^2)^{3/2}}$$

$$f''(\lambda) = \frac{d^2f}{d\lambda^2} = -\frac{(2+3\lambda^2)}{(1+\lambda^2)^{3/2}} + \frac{3\lambda(2\lambda+\lambda^3)}{(1+\lambda^2)^{5/2}}$$

(a) Newton's method:

$$\lambda_{i+1} = \lambda_i - \frac{f'(\lambda_i)}{f''(\lambda_i)} \quad ; \quad \lambda_1 = 0.6$$

λ_i	$f'(\lambda_i)$	$f''(\lambda_i)$	λ_{i+1}
0.600000E+00	0.107198E+00	-.760320E+00	0.7409912E+00
0.740991E+00	0.203097E-01	-.485815E+00	0.7827967E+00
0.782797E+00	0.140047E-02	-.419968E+00	0.7861314E+00
0.786131E+00	0.822544E-05	-.415011E+00	0.7861513E+00
0.786151E+00	0.000000E+00	-.414981E+00	0.7861513E+00

(b) Quasi-Newton method:

$$\lambda_{i+1} = \lambda_i - \frac{\Delta\lambda\{f(\lambda_i+\Delta\lambda) - f(\lambda_i-\Delta\lambda)\}}{2\{f(\lambda_i+\Delta\lambda) - 2f(\lambda_i) + f(\lambda_i-\Delta\lambda)\}} \quad ;$$

$\lambda_1 = 0.6, \quad \Delta\lambda = 0.001$

λ_i	$f(\lambda_i-\Delta\lambda)$	$f(\lambda_i)$	$f(\lambda_i+\Delta\lambda)$	λ_{i+1}
0.600000E+00	0.291195E+000	.291303E+000	.291409E+00	0.7383462E+00
0.738346E+00	0.299759E+000	.299781E+000	.299802E+00	0.7810227E+00
0.781023E+00	0.300275E+000	.300278E+000	.300280E+00	0.7858560E+00
0.785856E+00	0.300283E+000	.300283E+000	.300283E+00	0.7861417E+00
0.786142E+00	0.300283E+000	.300283E+000	.300283E+00	0.7861417E+00
0.786142E+00	0.300283E+000	.300283E+000	.300283E+00	0.7861417E+00

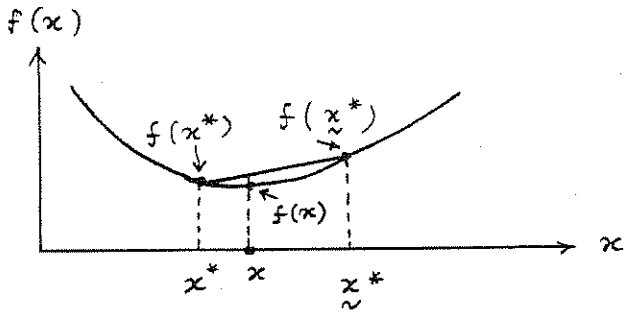
(c) Secant method:

$$\lambda_{i+1} = A - \frac{f'(A)(B-A)}{f'(B)-f'(A)} \quad ; \quad \lambda_1 = 0.0 \,, \quad t_0 = 0.1$$

λ	$f(\lambda)$	$f'(\lambda)$
0.0000E+00	0.000000E+00	0.100000E+01
0.1000E+00	0.900496E-01	0.801978E+00
0.2000E+00	0.160777E+00	0.615311E+00
0.3000E+00	0.213796E+00	0.449030E+00
0.4000E+00	0.251444E+00	0.308445E+00
0.5000E+00	0.276393E+00	0.195016E+00
0.6000E+00	0.291303E+00	0.107198E+00
0.7000E+00	0.298576E+00	0.416635E-01
0.8000E+00	0.300244E+00	-.560665E-02

A	B	$f'(A)$	$f'(B)$	λ_{i+1}	$f'(\lambda_{i+1})$
0.7	0.8	0.0416635	-0.00560665	0.7881392	-0.00008219481
0.7	0.7881392	0.0416635	-0.00008219481	0.7864341	-0.00001171827
0.7	0.7864341	0.0416635	-0.00001171827	0.7861917	-0.00000168085 1

5.12 Let x^* be a local minimum of $f(x)$, and $\underset{\sim}{x}^*$ be another local minimum.



Let $x = \alpha\, x^* + (1-\alpha)\, \underset{\sim}{x}^* \quad ; \quad 0 < \alpha < 1$

Since f is convex, we have

$$f(x) = f\left(\alpha\, x^* + (1-\alpha)\, \underset{\sim}{x}^*\right) < \underbrace{\alpha\, f(x^*) + (1-\alpha)\, f(\underset{\sim}{x}^*)}_{\text{linearly interpolated value}}$$

But $\alpha\, f(x^*) + (1-\alpha)\, f(\underset{\sim}{x}^*) = f(\underset{\sim}{x}^*) + \alpha\left(f(x^*) - f(\underset{\sim}{x}^*)\right)$

$$< f(\underset{\sim}{x}^*)$$

since $\alpha > 0$ and $f(\underset{\sim}{x}^*) > f(x^*)$

This shows that $f(x) < f(\underset{\sim}{x}^*)$ for all $x^* < x < \underset{\sim}{x}^*$.

Hence $\underset{\sim}{x}^*$ cannot be a different local minimum.

$\therefore$ A convex function is unimodal.

5.13

(a) Exhaustive search:

$R = \frac{L_n}{L_0} = \frac{2}{n+1}$; any n ; n = number of experiments

$n =$	2	3	4	5	6	7	8	9	10
$R =$	0.6667	0.5	0.4	0.3333	0.2857	0.25	0.2222	0.2	0.1818

(b) Dichotomous search:

$R = \frac{L_n}{L_0} = \frac{1}{2^{n/2}} + \frac{\delta}{L_0}\left(1 - \frac{1}{2^{n/2}}\right)$; $\delta = 10^{-4}$; $L_0 = 1$ (assume); n = even

$= \frac{1}{2^{n/2}} + 0.0001\left(1 - \frac{1}{2^{n/2}}\right)$

$n =$	2	4	6	8	10
$R =$	0.500 05	0.250075	0.1250875	0.0625 9375	0.031346875

(c) Interval halving method:

$R = \frac{L_n}{L_0} = \left(\frac{1}{2}\right)^{\frac{n-1}{2}}$; n = odd ≥ 3

$n =$	3	5	7	9	11
$R =$	0.5	0.25	0.125	0.0625	0.03125

(d) Fibonacci method:

$R = \frac{L_n}{L_0} = \frac{1}{F_n}$; any $n \geq 2$

$n =$	2	3	4	5	6	7	8	9
$F_n =$	2	3	5	8	13	21	34	55
$R =$	0.5	0.3333	0.2	0.125	0.07692	0.04762	0.02941	0.01818

(e) Golden section method:

$$R = \frac{L_n}{L_0} = 0.618^{n-1} \quad ; \quad \text{any} \quad n \geq 2$$

$n =$	2	3	4	5	6	7	8	9
$R =$	0.618	0.3819	0.23603	0.14587	0.09014	0.05571	0.03443	0.02128

5.14 $R = \frac{L_n}{L_0} = 0.001 = 10^{-3}$

(a) Exhaustive search:

$$R = \frac{2}{n+1} = \frac{1}{1000} \quad ; \quad n = 1999$$

(b) Dichotomous search: ($L_0 = 1$, $\delta = 10^{-4}$, n = even)

$$R = \frac{1}{2^{n/2}} + 0.0001\left(1 - \frac{1}{2^{n/2}}\right) = \frac{1}{1000}$$

$$\frac{0.9999}{2^{n/2}} = 0.001 - 0.0001 = 0.0009 \quad ; \quad 2^{n/2} = 1111$$

$$n = 22$$

(c) Interval halving method: (n = odd)

$$R = \left(\frac{1}{2}\right)^{\frac{n-1}{2}} = \frac{1}{1000} \quad ; \quad 2^{\frac{n-1}{2}} = 1000$$

$$n = 21$$

(d) Fibonacci method:

$$R = \frac{1}{F_n} = \frac{1}{1000}$$

$n = 16$ ($F_{16} = 1597$)

(e) Golden section method:

$$R = (0.618)^{n-1} = \frac{1}{1000} = 0.001$$

$n = 14 \rightarrow R = 0.001918$

$n = 15 \rightarrow R = 0.001185$

$n = 16 \rightarrow R = 0.0007325$; $\therefore n = 16$

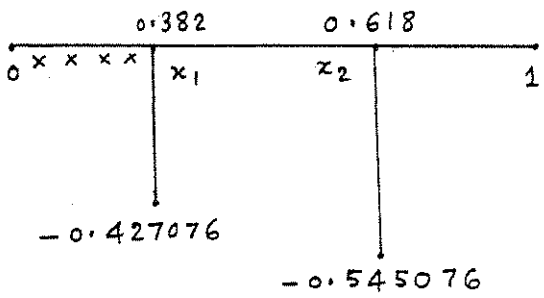
5.15 $f = x(x - 1.5)$

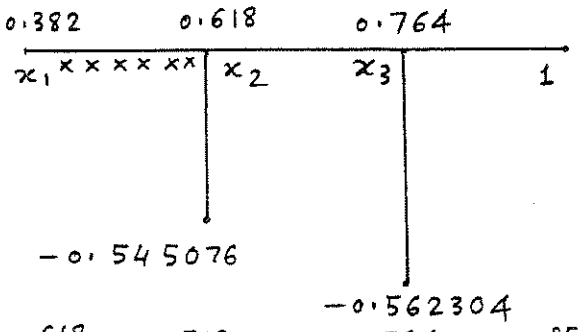
If midpoint of final interval of uncertainty is taken as the optimum, $\frac{1}{2}\frac{L_n}{L_0} \le \frac{1}{20}$ or $\frac{L_n}{L_0} \le \frac{1}{10}$

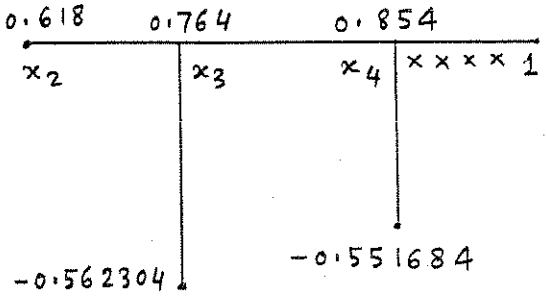
(a) Golden section method:

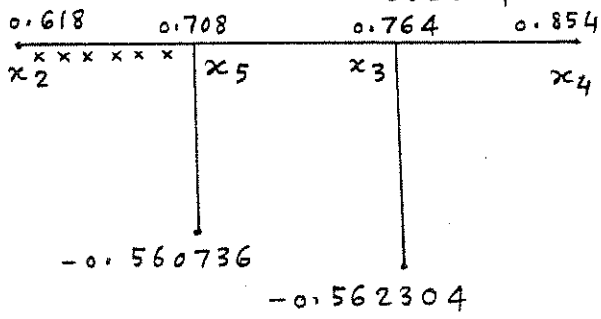
$$\frac{L_n}{L_0} = (0.618)^{n-1} \le 0.1$$

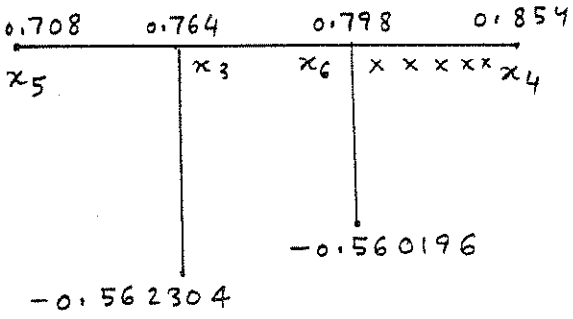
$\therefore n = 6$











Final interval:

$L_n = (0.708, 0.798)$

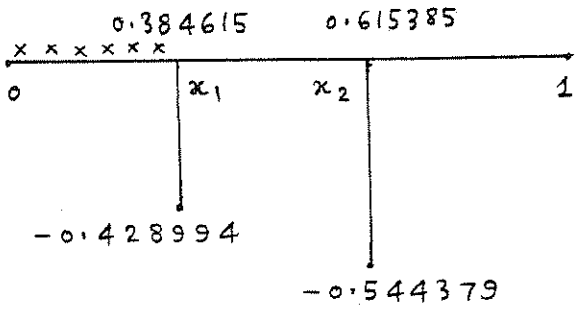
$x^* \simeq \frac{x_5 + x_6}{2} = 0.753$

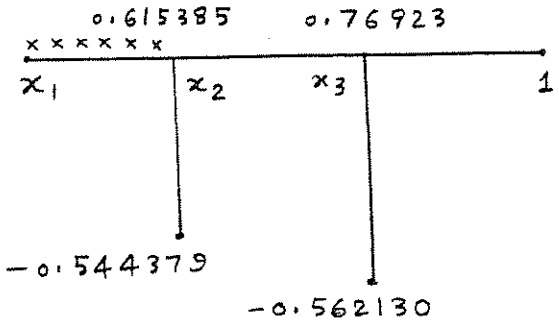
$f^* \simeq f(x^*) = -0.562491$

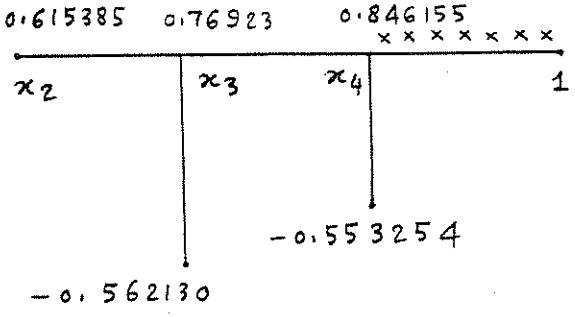
(b) Fibonacci method:

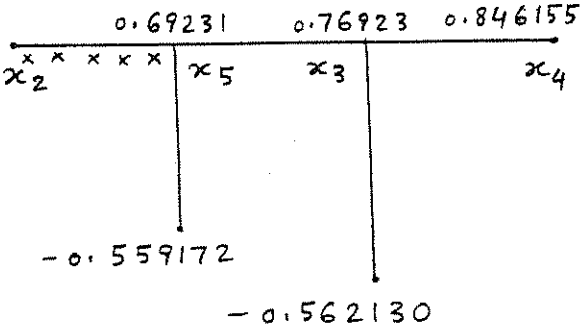
$$\frac{L_n}{L_0} = \frac{1}{F_n} \le 0.1 \quad ; \quad n = 6 \quad (F_6 = 13)$$

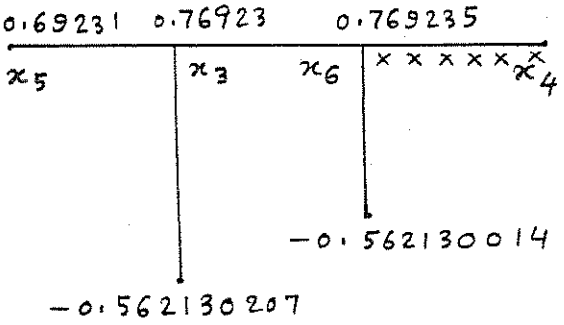
$$L_2^* = \frac{F_4}{F_6} L_0 = \frac{5}{13}(1) = 0.384615$$











Final interval = (0.69231, 0.769235)

$$x^* \simeq \frac{x_5 + x_6}{2} = 0.7307725$$

$$f^* \simeq f(x^*) = -0.562130303$$

5.16

(a)

$f = \lambda^5 - 5\lambda^3 - 20\lambda + 5$ (Minimize)

Unrestricted search with fixed step size: ($\lambda_1 = 0$, $\delta = 0.1$)

i	$\lambda_i = \lambda_1 + (i-1)\delta$	$f_i = f(\lambda_i)$	Is $f_i \geq f_{i-1}$?
1	0.0	5.00	—
2	0.1	2.99	NO
3	0.2	0.96	NO
4	0.3	−1.13	NO
5	0.4	−3.31	NO
6	0.5	−5.59	NO
7	0.6	−8.00	NO
8	0.7	−10.55	NO
9	0.8	−13.23	NO
10	0.9	−16.91	NO
11	1.0	−19.00	NO
12	1.1	−22.04	NO
13	1.2	−25.15	NO
14	1.3	−28.27	NO
15	1.4	−31.34	NO
16	1.5	−34.28	NO
17	1.6	−36.99	NO
18	1.7	−39.37	NO
19	1.8	−41.26	NO
20	1.9	−42.53	NO
21	2.0	−43.00	NO
22	2.1	−42.46	YES

λ^* lies in the interval (1.9, 2.1)

$$\lambda^* \simeq \frac{1.9 + 2.1}{2} = 2.0$$

$$f^* \simeq f(\lambda^*) = -43.0$$

(b) Unrestricted search with accelerated step size:

i	δ	$\lambda_i = \lambda_1 + \delta$	$f_i = f(\lambda_i)$	Is $f_i \geq f_{i-1}$?
1	—	0.0	5.00	—
2	0.1	0.1	2.99	No
3	0.2	0.2	0.96	NO
4	0.4	0.4	−3.31	NO
5	0.8	0.8	−13.23	NO
6	1.6	1.6	−36.99	NO
7	3.2	3.2	112.70	YES

λ^* lies in the interval (0.8, 3.2)

$\lambda^* \simeq x_6 = 1.6$, $f^* \simeq f(x_6) = -36.99$

(c) Exhaustive search: (Let $n = 9$)

i	λ_i	f_i
1	0.5	−5.59
2	1.0	−19.00
3	1.5	−34.28
4	2.0	−43.00
5	2.5	−25.47
6	3.0	53.00
7	3.5	245.84
8	4.0	669.00
9	4.5	1304.65

λ^* lies in the interval (1.5, 2.5)

$\lambda^* \simeq 2.0$ with $f^* \simeq -43.00$

(d) Dichotomous search:

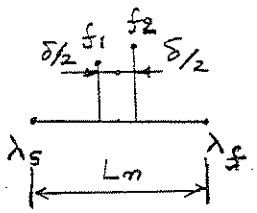
$L_0 = (0, 5)$, $\delta = 0.0001$. Let $\frac{L_n}{L_0} = 0.05$

$$\frac{L_n}{L_0} = \frac{1}{2^{n/2}} + \frac{\delta}{L_0}\left(1 - \frac{1}{2^{n/2}}\right) = \frac{1}{20}$$

i.e., $\frac{1}{2^{n/2}} = \frac{49999}{2499} \simeq 20$; $n = 10$

Number of experiments	L_n	λ_s	λ_f	λ_1	λ_2	f_1	f_2	Result
2	–	0.	5.	2.49995	2.50005	-25.472	-25.464	$f_2 > f_1$
4	2.50005	0.	2.50005	1.249975	1.250075	-26.71309	-26.7162	$f_1 > f_2$
6	1.250075	1.249975	2.50005	1.8749625	1.8750625	-42.2843	-42.2854	$f_1 > f_2$
8	0.62065	1.8749625	2.50005	2.18745625	2.18755625	-40.99988	-40.9976	$f_2 > f_1$
10	0.312594	1.8749625	2.18755625	2.0312094	2.0313094	-42.9502253	-42.9499 0225	$f_2 > f_1$
	0.156347	1.8749625	2.0313094					

$L_n = L_{10} = 0.156347 = 0.03127\, L_0 < 0.05\, L_0 \rightarrow$ O.K.



∴ optimum value lies

between $1.8749625 < \lambda^* < 2.0313094$

$$f^* \simeq f(\lambda^*) \approx \frac{1}{2}(-42.9502253 - 42.94990225)$$

$$= -42.950064$$

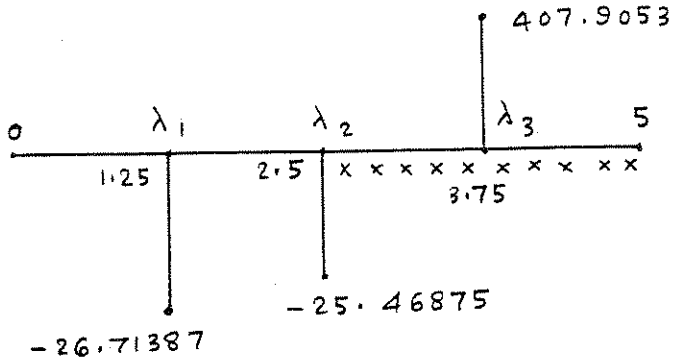
(e) Interval halving method:

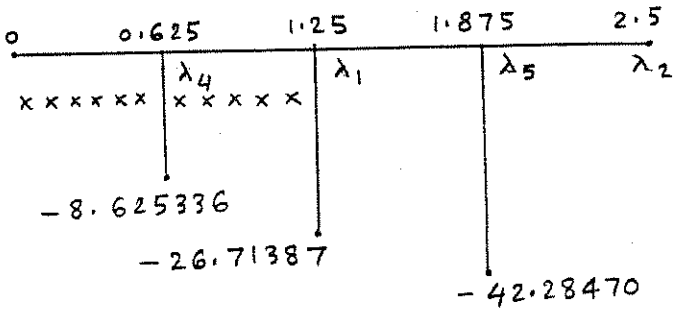
Assume: optimum point is required within 10% of exact value

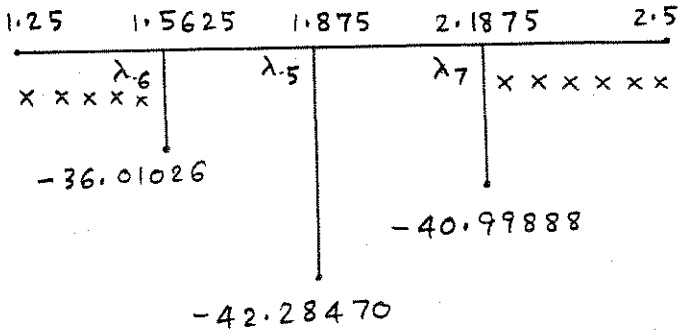
$$\frac{1}{2}\,\frac{L_n}{L_0} = \frac{1}{2}\left(\frac{1}{2}\right)^{\frac{n-1}{2}} \leq \frac{1}{10}$$

i.e., $$\frac{1}{2^{\frac{n-1}{2}}} \leq \frac{1}{5}$$

$\therefore$ $n = 7$ (odd)







$L_n = L_7 = (1.5625, 2.1875)$

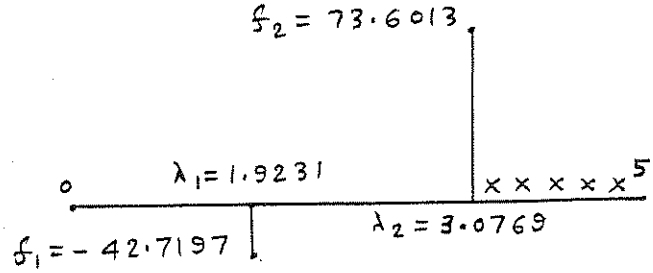
$\lambda^* \simeq 1.875$, $f^* \simeq -42.28470$

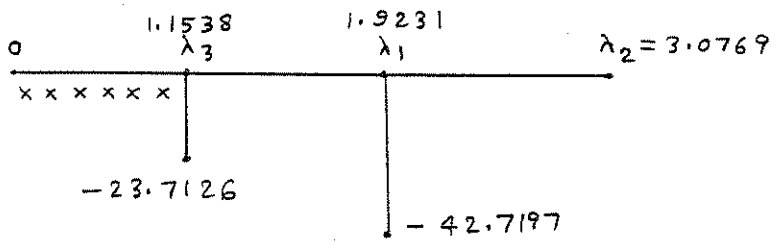
(f) Fibonacci search:

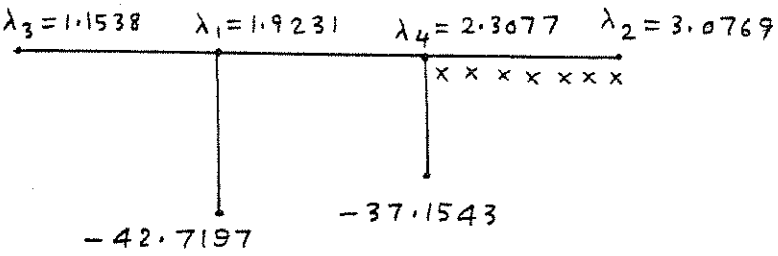
Let $n = 6$

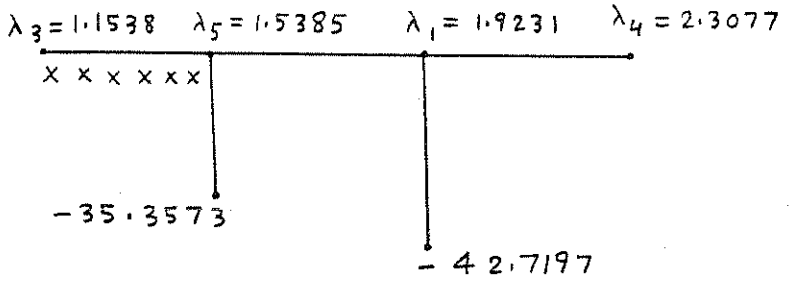
$L_2^* = \frac{F_4}{F_6} L_0 = \frac{5}{13}(5) = 1.9231$

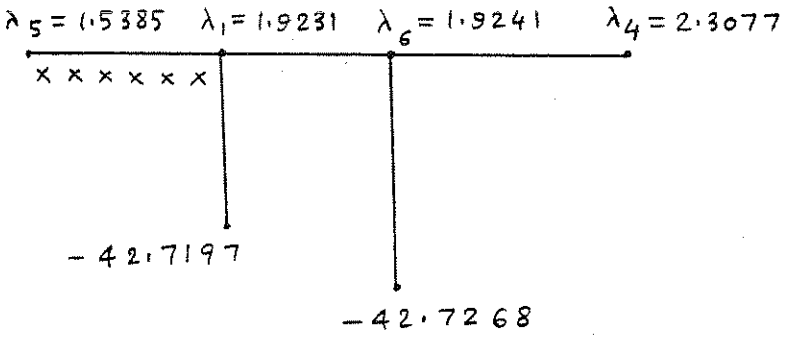
$\lambda_1 = 1.9231$, $\lambda_2 = 3.0769$











Final interval = $L_6 = (1.9231, 2.3077)$

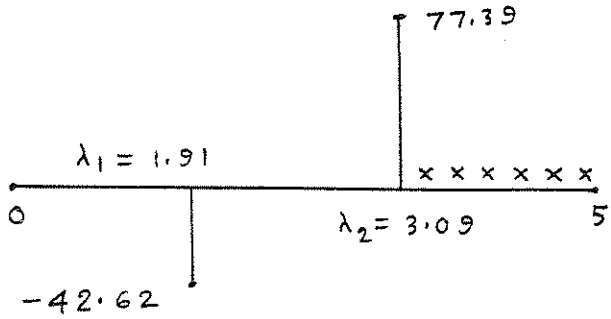
$\lambda^* \simeq 2.115$

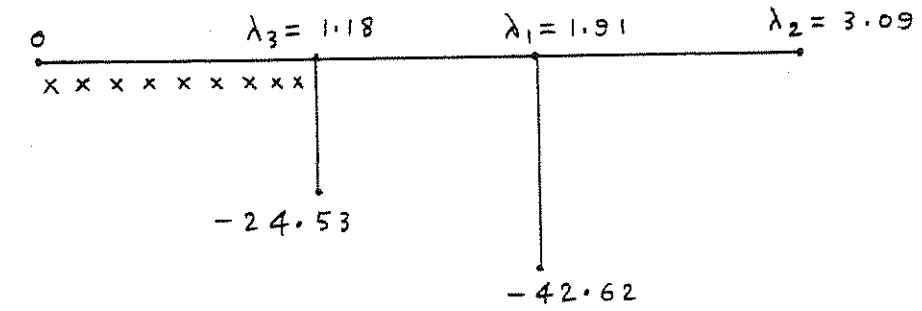
(g) Golden section method:

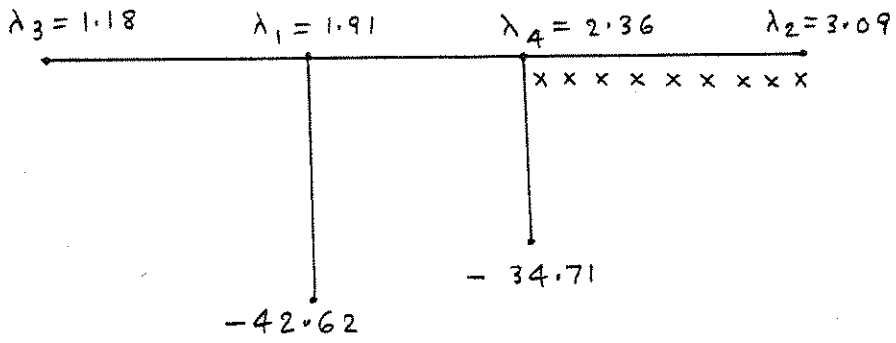
Let $n = 6$

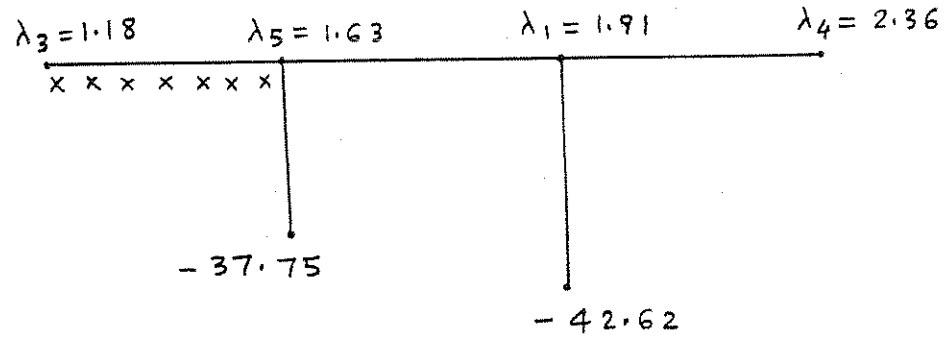
$L_2^* = 0.382\ L_0 = 1.91$

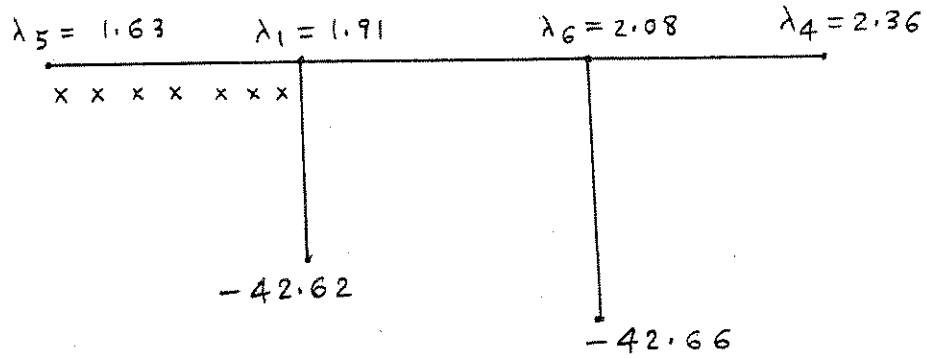
$\lambda_1 = 1.91, \quad \lambda_2 = 3.09$











Final interval = (1.91, 2.36)

$\lambda^* \simeq 2.135$

5.17 $f = \lambda / \log \lambda$ (Minimize)

Classical solution: $\frac{df}{d\lambda} = \frac{\log \lambda - 1}{(\log \lambda)^2} = 0 \Rightarrow \log \lambda = 1$

$$\lambda^* = e^1 = 2.71828$$

(a) Quadratic interpolation method:

Taking $A = 1.1$ and $t_0 = 0.1$, we find

$f(1.1) = 26.5747$

$f(1.2) = 15.155$ $\qquad \log \lambda = \log_{10} \lambda$ assumed.

$f(1.3) = 11.409$

$f(1.5) = 8.518$

$f(1.9) = 6.816$

$f(2.7) = 6.259$

$f(4.3) = 6.788 \quad > f(2.7)$

$A = 1.1, \quad B = 2.7, \quad C = 4.3$

$f_A = 26.5747, \quad f_B = 6.259, \quad f_C = 6.788$

$a = 52.6330, \quad b = -28.1676, \quad c = 4.0712$

$\tilde{\lambda}^* = 3.4594$

$h(\tilde{\lambda}^*) = 3.9113$, $f(\tilde{\lambda}^*) = 6.4182$, convergence not satisfied.

Refit 1:

$A = 1.1, \quad B = 2.7, \quad C = 3.4594$

$f_A = 26.5747, \quad f_B = 6.259, \quad f_C = 6.4182$

$a = 56.7883, \quad b = -33.4841, \quad c = 5.4702$

$\tilde{\lambda}^* = 3.0606$

$h(\tilde{\lambda}^*) = 5.5480$, $f(\tilde{\lambda}^*) = 6.3000$, convergence not satisfied.

Refit 2:

$A = 1.1, \quad B = 2.7, \quad C = 3.0606$

$f_A = 26.5747, \quad f_B = 6.259, \quad f_C = 6.3000$

$a = 59.9477, \quad b = -37.5264, \quad c = 6.5340$

$\tilde{\lambda}^* = 2.8716$

$h(\tilde{\lambda}^*) = 6.0668, \quad f(\tilde{\lambda}^*) = 6.2682$

h and f are assumed to be sufficiently close.

$\therefore \lambda^* \simeq 2.8716$

(b) Cubic interpolation method:

Let $A = 2.7, \quad f_A' = -0.01575$ $\quad \log \lambda = \log_{10} \lambda$ assumed.

$B = 4.3, \quad f_B' = 0.49634$

$f_A = 6.25922, \quad f_B = 6.78802$

$z = -0.51092, \quad Q = 0.51851$

$\tilde{\lambda}^* = 2.72411, \quad f(\tilde{\lambda}^*) = 0.0049113$

$f'(\tilde{\lambda}^*) = 0.0049113$

Refit 1:

$A = 2.7, \quad B = 2.72411$

$f_A = 6.25922, \quad f_B = 6.25909$

$f_A' = -0.01575, \quad f_B' = 0.0049113$

$z = 0.005189, \quad Q = 0.010212$

$\tilde{\lambda}^* = 2.71828$

$f'(\tilde{\lambda}^*) = -4.12758 * 10^{-7}$ (converged)

$\therefore \lambda^* \simeq 2.71828$

5.18

$$f = \lambda / \log_e \lambda$$

$$f'(\lambda) = \frac{df}{d\lambda} = \frac{\log \lambda - 1}{(\log \lambda)^2}$$

$$f''(\lambda) = \frac{d^2 f}{d\lambda^2} = \frac{2 - \log \lambda}{\lambda (\log \lambda)^3}$$

(a) Newton method: $(\lambda_1 = 1.5)$

λ_i	f'	f''	λ_{i+1}
0.150000E+01	-.361635E+010	.159471E+02	0.1726771E+01
0.172677E+01	-.152064E+010	.516502E+01	0.2021182E+01
0.202118E+01	-.598417E+000	.184067E+01	0.2346290E+01
0.234629E+01	-.202336E+000	.788221E+00	0.2602989E+01
0.260299E+01	-.473552E-010	.457804E+00	0.2706429E+01
0.270643E+01	-.440830E-020	.376013E+00	0.2718153E+01
0.271815E+01	-.473902E-040	.367967E+00	0.2718282E+01
0.271828E+01	0.000000E+000	.367879E+00	0.2718282E+01

(b) Quasi-Newton method: $(\lambda_1 = 1.5)$

λ_i	$f(\lambda_i - \Delta\lambda)$	$f(\lambda_i)$	$f(\lambda_i + \Delta\lambda)$	λ_{i+1}	$f'(\lambda_i)$
0.150000E+01	0.370308E+01	0.369946E+01	0.369585E+01	0.1729833E+01	con= 0.36166E+01
0.172983E+01	0.315799E+01	0.315649E+01	0.315498E+01	0.2030429E+01	con= 0.15050E+01
0.203043E+01	0.286742E+01	0.286684E+01	0.286626E+01	0.2301484E+01	con= 0.58162E+00
0.230148E+01	0.276129E+01	0.276105E+01	0.276081E+01	0.2502384E+01	con= 0.23949E+00
0.250238E+01	0.272825E+01	0.272816E+01	0.272806E+01	0.2914884E+01	con= 0.98348E-01
0.291488E+01	0.272456E+01	0.272462E+01	0.272468E+01	0.2786884E+01	con= 0.61035E-01
0.278688E+01	0.271909E+01	0.271911E+01	0.271914E+01	0.2736884E+01	con= 0.23842E-01
0.273688E+01	0.271834E+01	0.271834E+01	0.271835E+01	0.2722384E+01	con= 0.69141E-02
0.272238E+01	0.271828E+01	0.271829E+01	0.271829E+01	0.2728884E+01	con= 0.15497E-02

(c) Secant method:

λ_i	$f(\lambda_i)$	$f'(\lambda_i)$
0.1500E+01	0.369946E+01	-.361635E+01
0.1700E+01	0.320375E+01	-.166700E+01
0.1900E+01	0.296018E+01	-.869336E+00
0.2100E+01	0.283043E+01	-.468803E+00
0.2300E+01	0.276141E+01	-.240856E+00
0.2500E+01	0.272839E+01	-.997026E-01
0.2700E+01	0.271834E+01	-.684014E-02
0.2900E+01	0.272374E+01	0.570839E-01
0.3100E+01	0.273996E+01	0.102652E+00
0.3300E+01	0.276400E+01	0.136043E+00

Using $\lambda_1 = A = 2.7$, $f'_A = -0.00684014$,

$\lambda_2 = B = 2.9$, $f'_B = 0.0570839$,

we obtain $\tilde{\lambda}^{(1)} = 2.721401$, $f'(\tilde{\lambda}^{(1)}) = 0.00114468$

Next, $\lambda_1 = A = 2.7$, $f_A' = -0.00684014$,

$\lambda_2 = B = 2.721401$, $f_B' = 0.00114416 8$

$\tilde{\lambda}^{(2)} = 2.718334$, $f'(\tilde{\lambda}^{(2)}) = 0.00001931116$

$\therefore \lambda^* \simeq 2.718334$

5.19

$$\vec{X} = \vec{X}_1 + \lambda \vec{S} = \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} x_{11} + \lambda s_1 \\ x_{12} + \lambda s_2 \\ x_{13} + \lambda s_3 \end{Bmatrix}$$

with $\vec{X}_1 = \begin{Bmatrix} x_{11} \\ x_{12} \\ x_{13} \end{Bmatrix}$ and $\vec{S} = \begin{Bmatrix} s_1 \\ s_2 \\ s_3 \end{Bmatrix}$

$$\text{Numerator} = 2(x_{11} + \lambda s_1)^2 + 2(x_{12} + \lambda s_2)^2 + 3(x_{13} + \lambda s_3)^2 - 2(x_{11} + \lambda s_1)(x_{12} + \lambda s_2) - 2(x_{12} + \lambda s_2)(x_{13} + \lambda s_3)$$

$$= \lambda^2 \underbrace{(2 s_1^2 + 2 s_2^2 + 3 s_3^2 - 2 s_1 s_2 - 2 s_2 s_3)}_{a} + \lambda \underbrace{(4 x_{11} s_1 + 4 x_{12} s_2 + 6 x_{13} s_3 - 2 x_{12} s_1 - 2 x_{11} s_2 - 2 x_{13} s_2 - 2 x_{12} s_3)}_{b} + \underbrace{(2 x_{11}^2 + 2 x_{12}^2 + 3 x_{13}^2 - 2 x_{11} x_{12} - 2 x_{12} x_{13})}_{c}$$

$$\equiv a \lambda^2 + b \lambda + c$$

$$\text{Denominator} = (x_{11} + \lambda s_1)^2 + (x_{12} + \lambda s_2)^2 + 2(x_{13} + \lambda s_3)^2$$

$$= \lambda^2 \underbrace{(s_1^2 + s_2^2 + 2 s_3^2)}_{d} + \lambda \underbrace{(2 x_{11} s_1 + 2 x_{12} s_2 + 4 x_{13} s_3)}_{e} + \underbrace{(x_{11}^2 + x_{12}^2 + 2 x_{13}^2)}_{g}$$

$$\equiv d \lambda^2 + e \lambda + g$$

$$\frac{df}{d\lambda} = \frac{(Nr)\frac{d}{d\lambda}(Dr) - (Dr)\frac{d}{d\lambda}(Nr)}{(Dr)^2} = 0$$

with $\frac{d}{d\lambda}(Dr) = 2\lambda d + e$, $\frac{d}{d\lambda}(Nr) = 2\lambda a + b$

$$\frac{df}{d\lambda} = 0 \Rightarrow$$

$$(a\lambda^2 + b\lambda + c)(2\lambda d + e) - (d\lambda^2 + e\lambda + g)(2\lambda a + b) = 0$$

i.e.,

$$\lambda^2 (bd - ae) + 2\lambda (cd - ag) + (ce - bg) = 0$$

$$\therefore \lambda^* = \frac{-2(cd - ag) \pm \sqrt{4(cd - ag)^2 - 4(bd - ae)(ce - bg)}}{2(bd - ae)}$$

We need to use the smaller positive value for λ^*.

5.20

$$f = x_1 - x_2 + 2x_1^2 + 2x_1 x_2 + x_2^2$$

$$\vec{X}_1 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}, \quad \vec{S} = \begin{Bmatrix} -1 \\ 0 \end{Bmatrix}, \quad \vec{X} = \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \vec{X}_1 + \lambda \vec{S} = \begin{Bmatrix} -\lambda \\ 0 \end{Bmatrix}$$

$$f(\lambda) = -\lambda - 0 + 2\lambda^2 + 0 + 0 = -\lambda + 2\lambda^2$$

Let $A = 0$, $t_0 = 0.1$

$f(0) = 0$, $f(0.5) = 0$, $f(1.0) = 1.0$

We take $A = 0$, $B = 0.5$, $C = 1.0$, $t = 0.5$

$$f_A = 0, \quad f_B = 0, \quad f_C = 1.0$$

Eqs. (5.37) – (5.40) give:

$$a = 0, \quad b = -1.0, \quad c = 2.0$$

$$\tilde{\lambda}^* = \frac{4(0) - 3(0) - 1.0}{4(0) - 2(1) - 2(0)} (0.5) = 0.25$$

$$\therefore \vec{X}_2 = \vec{X}_1 + \tilde{\lambda}^* \vec{S} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} + \frac{1}{4} \begin{Bmatrix} -1 \\ 0 \end{Bmatrix} = \begin{Bmatrix} -0.25 \\ 0 \end{Bmatrix}$$

$$h(\lambda) = a + b\lambda + c\lambda^2 = 0 + (-1.0)(\lambda) + 2.0\lambda^2 = -\lambda + 2\lambda^2$$

Since $h(\lambda) \equiv f(\lambda)$, $\tilde{\lambda}^*$ is the exact solution.

$$\therefore \lambda^* = 0.25$$

5.21

$$f = 100\,(x_2 - x_1^2)^2 + (1 - x_1)^2$$

$$\vec{x}_1 = \begin{Bmatrix} -1 \\ 1 \end{Bmatrix}, \quad \vec{s} = \begin{Bmatrix} 4 \\ 0 \end{Bmatrix}, \quad \vec{x} = \vec{x}_1 + \lambda \vec{s} = \begin{Bmatrix} -1 + 4\lambda \\ 1 \end{Bmatrix}$$

$$f(x_1, x_2) = f(-1 + 4\lambda, 1)$$

$$= 100\left(1 - [-1 + 4\lambda]^2\right)^2 + (1 + 1 - 4\lambda)^2$$

$$= 100\left(8\{\lambda - 2\lambda^2\}\right)^2 + 2^2(1 - 2\lambda)^2$$

$$f(\lambda) = 4 - 16\lambda + 6416\lambda^2 - 25600\lambda^3 + 25600\lambda^4$$

Let $t_0 = 0.001$

$A = 0$, $f_A = f(\lambda = 0) = 4$

$\lambda = 0.001$, $f = 3.9903905$

$\lambda = 0.002$, $f = 3.9934597$

$\therefore$ $A = 0$, $B = 0.001$, $C = 0.002$, $t = 0.001$

$f_A = 4.0$, $f_B = 3.9903905$, $f_C = 3.9934597$

$b = -15.9488$, $c = 6339.31$, $\tilde{\lambda}^* = 0.0012579$,

$f(\tilde{\lambda}^*) = 3.9899747$

convergence check:

$$f'\Big|_{\tilde{\lambda}^*} = \frac{f(\tilde{\lambda}^* + \Delta\lambda) - f(\tilde{\lambda}^* - \Delta\lambda)}{2\,\Delta\lambda} = 0.32544136$$

Next, we take

$A = 0.001$, $B = \tilde{\lambda}^* = 0.0012579$, $C = 0.002$

$f_A = 3.9903905$, $f_B = 3.9899747$, $f_C = 3.9934597$

Using Eq. (5.36), we obtain

$\tilde{\lambda}^* = 0.001257$, $f(\tilde{\lambda}^*) = 3.9899750$

$b = -15.8554$, $c = 6308.37$

$f'(\tilde{\lambda}^*) = 0.32067299$

5.22 Minimize $f = 100(x_1 - x_2^2)^2 + (1 - x_1)^2$

$\vec{x}_1 = \begin{Bmatrix} -1 \\ 1 \end{Bmatrix}$, $\vec{s} = \begin{Bmatrix} 4 \\ 0 \end{Bmatrix}$, $\vec{x} = \vec{x}_1 + \lambda \vec{s} = \begin{Bmatrix} -1 + 4\lambda \\ 1 \end{Bmatrix}$

$f(\lambda) = 4 - 16\lambda + 6416\lambda^2 - 25600\lambda^3 + 25600\lambda^4$

$f'(\lambda) = -16 + 12832\lambda - 76800\lambda^2 + 102400\lambda^3$

Let $t_0 = 0.1$

$A = 0$, $f(A) = 4$, $f'(A) = -16$

$B = 0.1$, $f(B) = 43.52$, $f'(B) = 601.6$

$Z = -600.0$, $a = 0.0$, $b = -16.0$, $c = 6160.0$

$Q = 607.9684$, $\tilde{\lambda}^* = 0.1992136, 0.001307219$

We select $\tilde{\lambda}^* = 0.001307219$ since $A < \tilde{\lambda}^* < B$

$f(\tilde{\lambda}^*) = 3.989991$, $f'(\tilde{\lambda}^*) = 0.6432279$

Next, we choose:

$A = 0$, $f_A = 4$, $f_A' = -16$

$B = 0.001307219$, $f_B = 3.989991$, $f_B' = 0.6432279$

$Z = 7.613481$, $a = 0.0$, $b = -16.0$, $c = 6415.542$

$Q = 8.261764$, $\tilde{\lambda}^* = 0.1676519, 0.00125631$

We use $\tilde{\lambda}^* = 0.00125631$, $f'(\tilde{\lambda}^*) = -0.00004363726$

Next, we select:

$A = 0.00125631$, $f_A = 3.989974$, $f_A' = -0.00004363726$

$B = 0.001307219$, $f_B = 3.989991$, $f_B' = 0.6432279$

$Z = -0.3543438$, $a = 0.0$, $b = -57.3812$, $c = 38713.19$

$Q = 0.3543834$, $\tilde{\lambda}^* = 0.001807165, 0.001256313$

Using $\tilde{\lambda}^* = 0.001256313$, we find

$f'(\tilde{\lambda}^*) = -0.23147 * 10^{-5}$; convergence achieved.

$\therefore \lambda^* = 0.001256313$

5.23 Minimize $f = 100\,(x_1 - x_2^2)^2 + (1 - x_1)^2$

$\vec{X}_1 = \begin{Bmatrix} -1 \\ 1 \end{Bmatrix}$, $\vec{S} = \begin{Bmatrix} 4 \\ 0 \end{Bmatrix}$, $\vec{X} = \vec{X}_1 + \lambda \vec{S} = \begin{Bmatrix} -1 + 4\lambda \\ 1 \end{Bmatrix}$

$f(\lambda) = 4 - 16\lambda + 6416\,\lambda^2 - 25600\,\lambda^3 + 25600\,\lambda^4$

$f'(\lambda) = \dfrac{df}{d\lambda} = -16 + 12832\,\lambda - 76800\,\lambda^2 + 102400\,\lambda^3$

Let $t_0 = 0.1$

$A = 0$, $f(A) = 4$, $f'(A) = -16$

$f'(t_0) = f'(0.1) = 601.6 > 0$

$B = t_0 = 0.1$, $f(B) = 43.52$, $f'(B) = 601.6$

$a = -16$, $b = 6176$, $\tilde{\lambda}^* = 0.00259$,

$f(\tilde{\lambda}^*) = 4.00117$, $f'(\tilde{\lambda}^*) = 16.72985$

Next, we choose $A = 0$, $f(A) = 4$, $f'(A) = -16$

$B = \tilde{\lambda}^* = 0.00259$, $f(B) = 4.00117$, $f'(B) = 16.7298$

$\tilde{\lambda}^* = 0.00127$, $f(\tilde{\lambda}^*) = 3.9899$, $f'(\tilde{\lambda}^*) = 0.12814$

Next, we select

$A = 0$, $f(A) = 4$, $f'(A) = -16$

$B = \tilde{\lambda}^* = 0.00127$, $f(B) = 3.9899$, $f'(B) = 0.12814$

$\tilde{\lambda}^* = 0.00126$, $f(\tilde{\lambda}^*) = 3.9899$, $f'(\tilde{\lambda}^*) = 9.67605 * 10^{-4}$

$\therefore \quad \vec{X}_2 = \vec{X}_1 + \lambda^* \vec{S} = \begin{Bmatrix} -1 \\ 1 \end{Bmatrix} + 0.00126 \begin{Bmatrix} 4 \\ 0 \end{Bmatrix}$

$= \begin{Bmatrix} -0.99497 \\ 1.0 \end{Bmatrix}$

5.24 $\lambda_{i+1} = \lambda_i - \dfrac{f'(\lambda_i)}{f''(\lambda_i)}$

where $f'(\lambda) = -16 + 12832\lambda - 76800\lambda^2 + 102400\lambda^3$

and $f''(\lambda) = 12832 - 153600\lambda + 307200\lambda^2$

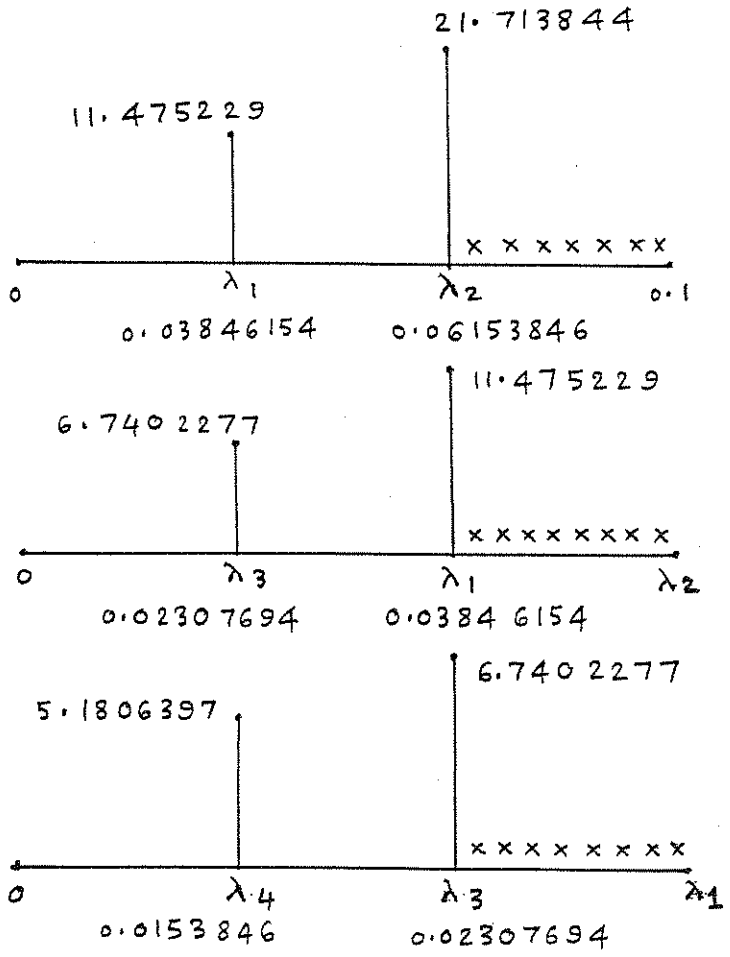
Using $\lambda_1 = 0.0025$, we obtain the following results:

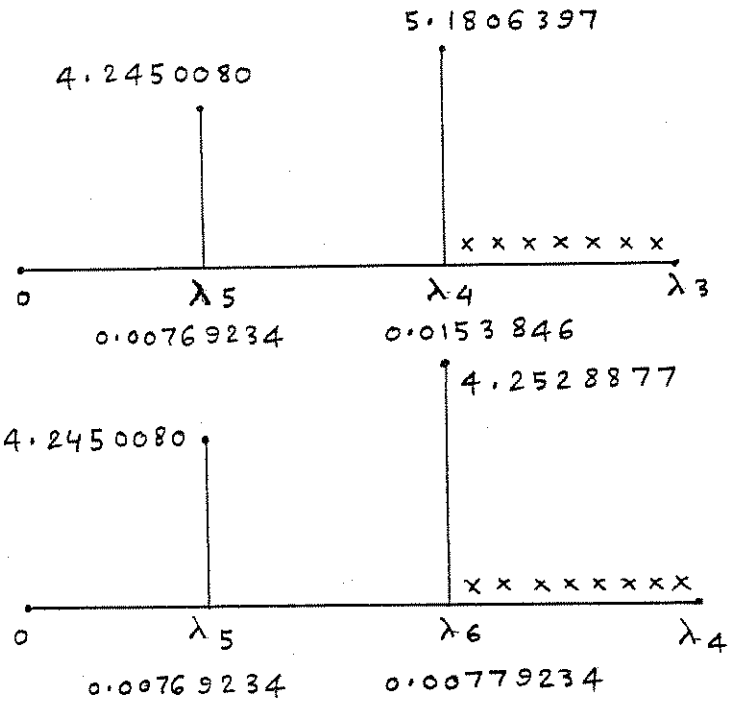
λ_i	$f'(\lambda_i)$	$f''(\lambda_i)$	λ_{i+1}
0.250000E-02	0.156016E+02	0.124499E+05	0.1246851E-02
0.124685E-02	-.119600E+00	0.126410E+05	0.1256313E-02
0.125631E-02	-.795499E-05	0.126395E+05	0.1256313E-02
0.125631E-02	-.437096E-06	0.126395E+05	0.1256313E-02

5.25 $f(\lambda) = 4 - 16\lambda + 6416\lambda^2 - 25600\lambda^3 + 25600\lambda^4$

Let $n = 6$, $L_0 = (0, 0.1)$

$L_2^* = \dfrac{F_4}{F_6} L_0 = \dfrac{5}{13}(0.1) = 0.03846154$





Final interval = $(0, 0.00779234)$

$\lambda^* \simeq \frac{0+\lambda_6}{2} = 0.0038 9617$

$f(\lambda^*) \simeq 4.0335488$

5.31

```
function p1
clear;clc;warning off;

a=0;b=3;

% option=optimset('Display',iter);
[vx,fval,exitflag,output]=fminbnd(@objfun,0,3)
fprintf('optimization area\n');
fprintf('from %6.4f  to %6.4f\n',a, b);
fprintf('optimal x:  %6.4d\n',vx);
fprintf('function value at x* = %6.4d\n', fval);
end

function f=objfun(x)
r=sqrt(1+x^2);
ff=0.5/r-r*(1-0.5/(r^2))+x;
f=-ff;
end
```

Output

```
vx =   0.7861

fval =   -0.3003

exitflag =    1

output =

    iterations: 10

     funcCount: 11

     algorithm: 'golden section search, parabolic interpolation'

       message: [1x112 char]

optimization area

from 0.0000  to 3.0000

optimal x:  7.8614e-001

function value at x* = -3.0028e-001
```

5.32

```
function p2
clear;clc;warning off;

a=0;%%%%% search area between a and b
b=5;

% option=optimset('Display',iter);
[vx,fval,exitflag,output]=fminbnd(@objfun,a,b);
fprintf('optimization area\n');
fprintf('from %6.4f  to %6.4f\n',a, b);
fprintf('optimal x:  %6.4d\n',vx);
fprintf('function value at x* = %6.4d\n', fval);
end

function f=objfun(x)
% r=sqrt(1+x^2);
f=(x^5)-5*(x^3)-20*x+5;
end
```

Output

```
optimization area

from 0.0000  to 5.0000

optimal x:  2.0000e+000

function value at x* = -4.3000e+001
```

5.33

```
function p3
clear;clc;warning off;

a=0;%%%%% search area between a and b
b=1;

% option=optimset('Display',iter);
[vx,fval,exitflag,output]=fminbnd(@objfun,a,b);
fprintf('optimization area\n');
fprintf('from %6.4f  to %6.4f\n',a, b);
fprintf('optimal x:  %6.4d\n',vx);
fprintf('function value at x* = %6.4d\n', fval);
end

function f=objfun(x)
f=x*(x-1.5);
end
```

Output

```
optimization area

from 0.0000  to 1.0000

optimal x:  7.5000e-001

function value at x* = -5.6250e-001
```

5.34

```
function p4
clear;clc;warning off;

a=0;%%%%% search area between a and b
b=10;

% option=optimset('Display',iter);
[vx,fval,exitflag,output]=fminbnd(@objfun,a,b);
fprintf('optimization area\n');
fprintf('from %6.4f  to %6.4f\n',a, b);
fprintf('optimal x:  %6.4d\n',vx);
fprintf('function value at x* = %6.4d\n', fval);
end

function f=objfun(x)
f=(x^3)/16-27*x/4;
end
```

Output

```
optimization area

from 0.0000  to 10.0000

optimal x:  6.0000e+000

function value at x* = -2.7000e+001
```

5.35

```
function p5
clear;clc;warning off;

a=-4;%%%%% search area between a and b
b=4;

% option=optimset('Display',iter);
[vx,fval,exitflag,output]=fminbnd(@objfun,a,b)
fprintf('optimization area\n');
fprintf('from %6.4f  to %6.4f\n',a, b);
fprintf('optimal x:  %6.4d\n',vx);
fprintf('function value at x* = %6.4d\n', fval);
end

function f=objfun(x)
f=(x^3)+(x^2)-x-2;
end
```

Output

```
optimization area

from -4.0000  to 4.0000

optimal x:  3.3334e-001

function value at x* = -2.1852e+000
```

5.36

```
function p6
clear;clc;warning off;

a=-4;%%%%% search area between a and b
b=4;

% option=optimset('Display',iter);
[vx,fval,exitflag,output]=fminbnd(@objfun,a,b)
fprintf('optimization area\n');
fprintf('from %6.4f  to %6.4f\n',a, b);
fprintf('optimal x:  %6.4d\n',vx);
fprintf('function value at x* = %6.4d\n', fval);
end

function f=objfun(x)
f=(-1.5/x)+6*(10e-6)*(x^-9);
end
```

Output

```
optimization area

from -4.0000  to 4.0000

optimal x:  3.7114e-001

function value at x* = -3.5925e+000
```

Chapter 6
Nonlinear Programming II:
Unconstrained Optimization Techniques

6.1

$$u(x) = \left(1 - \frac{x}{\ell}\right) u_1 + \frac{x}{\ell} u_2 \qquad (1)$$

$$\frac{\partial u}{\partial x} = -\frac{1}{\ell} u_1 + \frac{1}{\ell} u_2$$

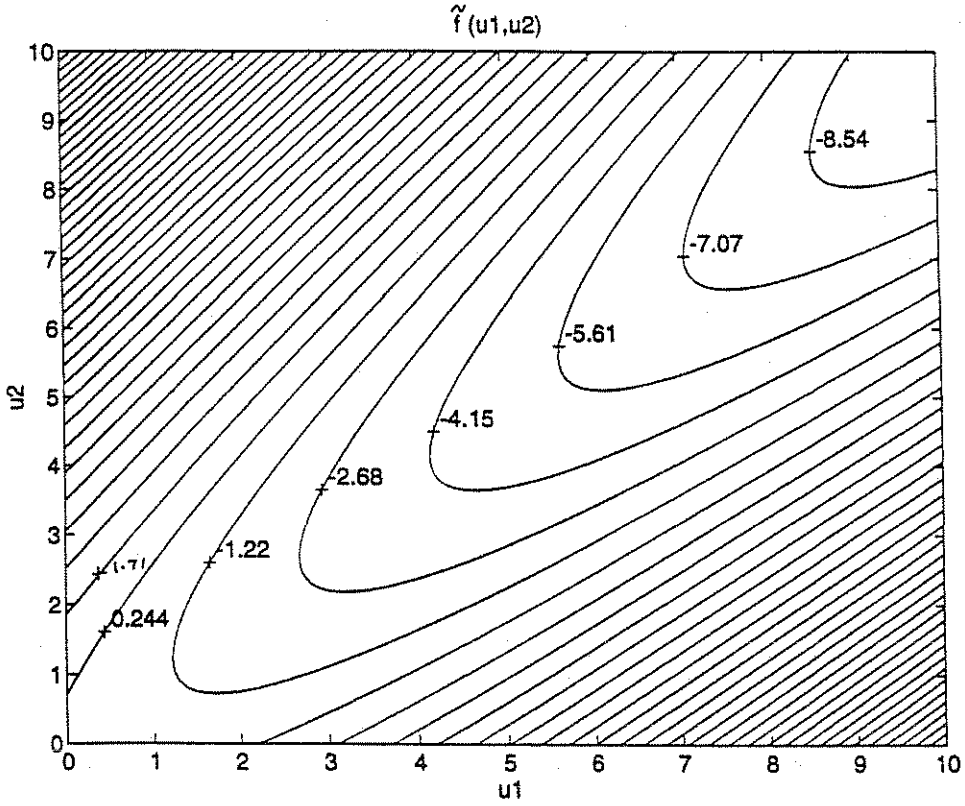
$$f = \frac{1}{2} \int_0^{\ell} EA \left(\frac{\partial u}{\partial x}\right)^2 dx - P_0 u_2$$

$$= \frac{EA}{2} \int_0^{\ell} \left(-\frac{u_1}{\ell} + \frac{u_2}{\ell}\right)^2 dx - P_0 u_2 \qquad (2)$$

$$\tilde{f} = \frac{f}{P_0} = \left(\frac{EA}{P_0 \ell}\right) \frac{1}{2} (u_2 - u_1)^2 - u_2 = \frac{1}{2} (u_2 - u_1)^2 - u_2$$

$$= (0.5\, u_1^2 + 0.5\, u_2^2 - u_1 u_2 - u_2) \qquad (3)$$

Contours of $\tilde{f}$ are shown below (in the range $0 \le u_i \le 10$).



6.2 f can be expressed as

$$f = \frac{X}{Y} \tag{1}$$

where $$X = \frac{E h^3}{3 l^2}\left(\frac{c_1^2}{4} + \frac{c_2^2}{10} + \frac{c_1 c_2}{5}\right) \tag{2}$$

and $$Y = \rho h l \left(\frac{c_1^2}{30} + \frac{c_2^2}{280} + \frac{2 c_1 c_2}{105}\right) \tag{3}$$

Analytical solution:

$\frac{\partial f}{\partial c_1} = 0$, $\frac{\partial f}{\partial c_2} = 0$ lead to

$$\begin{bmatrix} \left(\frac{1}{2} - \frac{\tilde{f}}{15}\right) & \left(\frac{1}{5} - \frac{2\tilde{f}}{105}\right) \\ \left(\frac{1}{5} - \frac{2\tilde{f}}{105}\right) & \left(\frac{1}{5} - \frac{\tilde{f}}{140}\right) \end{bmatrix} \begin{Bmatrix} c_1 \\ c_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \tag{4}$$

where $\tilde{f} = \frac{3 f \rho l^4}{E h^2}$. Setting the determinant of the matrix on the left side of Eq. (4) equal to zero, we obtain:

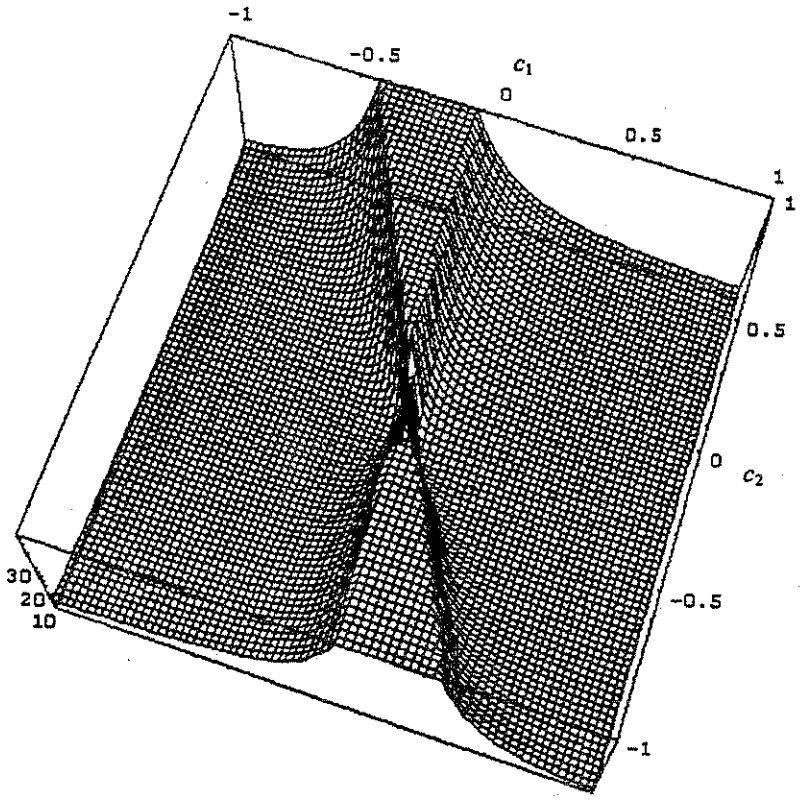
$$\tilde{f}_1 = 7.0751, \quad \tilde{f}_2 = 74.8087 \tag{5}$$

Note: This is an eigenvalue problem. Unless we fix either c_1 or c_2 at a particular value, we can not solve the problem.

Graphical solution:

Plot of $\tilde{f} = \left(3 f \rho l^3 / E h^2\right)$ is shown in the following figure. The minimum value appears to be along a line.

$g_{min} = 7.0722418$.



c1	c2	function
100000	-43665.225	7.0722419
10000	-4367.7335	7.0722418
1000	-436.77335	7.0722418
100	-43.675241	7.0722418
10	-4.3677338	7.0722418
1	-0.4367734	7.0722418
0.9	-0.3930939	7.0722418
0.8	-0.3494182	7.0722418
0.7	-0.3057414	7.0722418
0.6	-0.2620641	7.0722418
0.5	-0.2183867	7.0722418
0.4	-0.1747094	7.0722418
0.3	-0.131032	7.0722418
0.2	-0.0873547	7.0722418
0.1	-0.0436773	7.0722418
0	0.00001	28
0	0.001	28
0	0.1	28
0	10	28
0	1000	28
0	100000	28

6.3

$$R(\vec{x}) = \frac{(x_1 \quad x_2 \quad x_3)\; k \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix}}{(x_1 \quad x_2 \quad x_3)\; m \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix}}$$

$$= \frac{k\,(2x_1^2 - 2x_1x_2 + 2x_2^2 - 2x_2x_3 + x_3^2)}{m\,(x_1^2 + x_2^2 + x_3^2)}$$

Solution method: Since $R(\vec{x})$ is differentiable, it can be minimized by DFP / BFGS / Newton method.

Solution: $\vec{x}^* = \begin{Bmatrix} 1.0 \\ 1.8019 \\ 2.2470 \end{Bmatrix}$ (with $\frac{k}{m}$ fixed at a constant value).

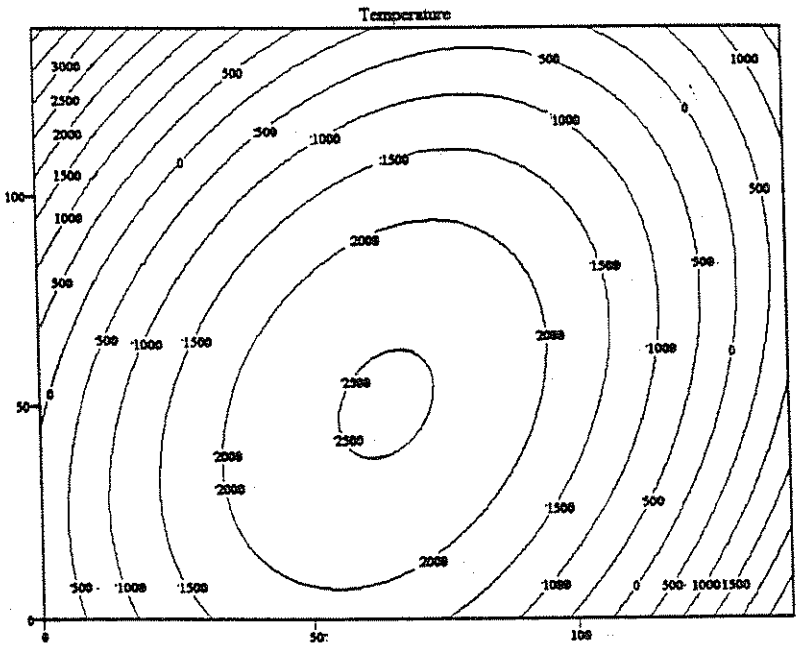
6.4

$f = 0.6382\,x_1^2 + 0.3191\,x_2^2 - 0.2809\,x_1x_2 - 67.906\,x_1 - 14.290\,x_2$

Analytical solution: $x_1 = 65.567$, $x_2 = 52.974$

Graphical solution:

$\simeq (x_1 = 65,\ x_2 = 52)$.



6.5

f = potential energy = strain energy − potential of loads

$$= \tfrac{1}{2}(300)\,x_1^2 + \tfrac{1}{2}(200)\,x_2^2 + \tfrac{1}{2}(400)\,(x_2 - x_1)^2 + \tfrac{1}{2}(100)\,x_2^2 - 1000\,x_1 - 2000\,x_2$$

$$= 350\,x_1^2 + 350\,x_2^2 - 400\,x_1 x_2 - 1000\,x_1 - 2000\,x_2$$

Solution by Newton's method:

Let $\vec{X}_1 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$

$$\nabla f_1 = \begin{Bmatrix} \partial f/\partial x_1 \\ \partial f/\partial x_2 \end{Bmatrix}_{\vec{X}_1} = \begin{Bmatrix} 700\,x_1 - 400\,x_2 - 1000 \\ 700\,x_2 - 400\,x_1 - 2000 \end{Bmatrix}_{\vec{X}_1} = \begin{Bmatrix} -1000 \\ -2000 \end{Bmatrix}$$

$$[J_1] = \begin{bmatrix} \dfrac{\partial^2 f}{\partial x_1^2} & \dfrac{\partial^2 f}{\partial x_1 \partial x_2} \\ \dfrac{\partial^2 f}{\partial x_1 \partial x_2} & \dfrac{\partial^2 f}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} 700 & -400 \\ -400 & 700 \end{bmatrix}$$

$$[J_1]^{-1} = \begin{bmatrix} 7/3300 & 4/3300 \\ 4/3300 & 7/3300 \end{bmatrix}$$

$$\vec{X}_2 = \vec{X}_1 - [J_1]^{-1}\nabla f_1 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} - \frac{1}{3300}\begin{bmatrix} 7 & 4 \\ 4 & 7 \end{bmatrix}\begin{Bmatrix} -1000 \\ -2000 \end{Bmatrix}$$

$$= \begin{Bmatrix} 4.5455 \\ 5.4545 \end{Bmatrix}$$

$\nabla f_2 = \nabla f(\vec{X}_2) = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \Rightarrow \vec{X}_2$ is optimum

$$\therefore \vec{X}^* = \begin{Bmatrix} 4.5455 \\ 5.4545 \end{Bmatrix}$$

6.6

$$\frac{EA}{s} = \frac{207 * 10^9\,(10^{-5})}{\sqrt{h^2 + (l/2)^2}} = \frac{207 * 10^4}{4.0697} = 50.8637 * 10^4$$

(a)

$$\left(\frac{l}{2s}\right)^2 = \left(\frac{1.5}{2 * 4.0697}\right)^2 = 0.033962$$

$$(h/s)^2 = (4/4.0697)^2 = 0.966040$$

$$f = 1.7274 * 10^4 \; x_1^2 + 49.1364 * 10^4 \; x_2^2 - 8.66 * 10^3 \; x_1 - 5.00 * 10^3 \; x_2$$

(b) $$\vec{S}_1 = -\nabla f_1 = -\nabla f(\vec{X}_1) = -\begin{Bmatrix} \partial f/\partial x_1 \\ \partial f/\partial x_2 \end{Bmatrix}_{\vec{X}_1}$$

$$= -\begin{Bmatrix} 3.4548 * 10^4 x_1 - 8.66 * 10^3 \\ 98.2728 * 10^4 \; x_2 - 5.00 * 10^3 \end{Bmatrix}_{\vec{X}_1} = + \begin{Bmatrix} 8660 \\ 5000 \end{Bmatrix} = \begin{Bmatrix} 0.866 \\ 0.500 \end{Bmatrix}$$

normalized.

(c) $$\vec{X} = \vec{X}_1 + \lambda \vec{S}_1 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} + \lambda \begin{Bmatrix} 0.8660 \\ 0.5000 \end{Bmatrix} = \begin{Bmatrix} 0.866\,\lambda \\ 0.5\,\lambda \end{Bmatrix}$$

$$f(\lambda) = 1.7274 * 10^4 (0.866\,\lambda)^2 + 49.1364 * 10^4 (0.5\,\lambda)^2$$
$$- 8.66 * 10^3 (0.866\,\lambda) - 5.00 * 10^3 (0.5\,\lambda)$$
$$= 1.2955 * 10^4 \lambda^2 + 12.2841 * 10^4 \lambda^2 - 7499.56\,\lambda$$
$$- 2500.0\,\lambda = 13.5796 * 10^4 \lambda^2 - 9999.56\,\lambda$$

(d) Calculus method of solution :

$$\frac{df}{d\lambda} = 27.1592 * 10^4 \lambda - 9999.56 = 0$$

$$\lambda^* = 0.03682$$

$$\vec{X}_2 = \vec{X}_1 + \lambda^* \vec{S}_1 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} + 0.03682 \begin{Bmatrix} 0.866 \\ 0.500 \end{Bmatrix} = \begin{Bmatrix} 0.03189 \\ 0.01841 \end{Bmatrix}$$

$$f(\vec{X}_1) = f(x_1 = 0, \; x_2 = 0) = 0$$

$$f(\vec{X}_2) = f(x_1 = 0.03189, \; x_2 = 0.01841)$$
$$= 1.7274 * 10^4 (0.03189)^2 + 49.1364 * 10^4 (0.01841)^2$$
$$- 8.66 * 10^3 (0.03189) - 5.00 * 10^3 (0.01841)$$
$$= 17.5672 + 166.5371 - 276.1674 - 92.0500$$
$$= -184.1131$$

6.7 $[K] = \begin{bmatrix} 8500 & -1000 & -2500 \\ -1000 & 3000 & -500 \\ -2500 & -500 & 11{,}500 \end{bmatrix}$, $\vec{P} = \begin{Bmatrix} 1000 \\ 2000 \\ 3000 \end{Bmatrix}$

$$f = \tfrac{1}{2}(k_1 + k_4 + k_5)x_1^2 - k_4 x_1 x_2 - k_5 x_1 x_3 - k_6 x_2 x_3 + \tfrac{1}{2}(k_2 + k_4 + k_6)x_2^2 + \tfrac{1}{2}(k_3 + k_5 + k_6 + k_7 + k_8)x_3^2 - P_1 x_1 - P_2 x_2 - P_3 x_3$$

$$= 4250\,x_1^2 - 1000\,x_1 x_2 - 2500\,x_1 x_3 - 500\,x_2 x_3 + 1500\,x_2^2 + 5750\,x_3^2 - 1000\,x_1 - 2000\,x_2 - 3000\,x_3$$

$$\frac{\partial f}{\partial x_1} = 8500\,x_1 - 1000\,x_2 - 2500\,x_3 - 1000$$

$$\frac{\partial f}{\partial x_2} = 3000\,x_2 - 1000\,x_1 - 500\,x_3 - 2000$$

$$\frac{\partial f}{\partial x_3} = 11500\,x_3 - 2500\,x_1 - 500\,x_2 - 3000$$

$$\frac{\partial^2 f}{\partial x_1^2} = 8500, \quad \frac{\partial^2 f}{\partial x_2^2} = 3000, \quad \frac{\partial^2 f}{\partial x_3^2} = 11500,$$

$$\frac{\partial^2 f}{\partial x_1 \partial x_2} = -1000, \quad \frac{\partial^2 f}{\partial x_1 \partial x_3} = -2500, \quad \frac{\partial^2 f}{\partial x_2 \partial x_3} = -500$$

Let $\vec{x}_1 = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}$

$$\nabla f_1 = \nabla f \Big|_{\vec{x}_1} = \begin{Bmatrix} -1000 \\ -2000 \\ -3000 \end{Bmatrix}$$

$$[J_1] = \begin{bmatrix} 8500 & -1000 & -2500 \\ -1000 & 3000 & -500 \\ -2500 & -500 & 11500 \end{bmatrix}$$

$$[J_1]^{-1} = 10^{-5} \begin{bmatrix} 13.26 & 4.935 & 3.096 \\ 4.935 & 35.41 & 2.612 \\ 3.096 & 2.612 & 9.482 \end{bmatrix}$$

$$\vec{x}_2 = \vec{x}_1 + \vec{s}_1 = \vec{x}_1 - [J_1]^{-1} \nabla f_1$$

$$= \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix} + 10^{-5} \begin{bmatrix} 13.26 & 4.935 & 3.096 \\ 4.935 & 35.41 & 2.612 \\ 3.096 & 2.612 & 9.482 \end{bmatrix} \begin{Bmatrix} 1000 \\ 2000 \\ 3000 \end{Bmatrix}$$

$$= \begin{Bmatrix} 0.3241 \\ 0.8360 \\ 0.3677 \end{Bmatrix} = \vec{x}^* = \text{Equilibrium configuration of carts}$$

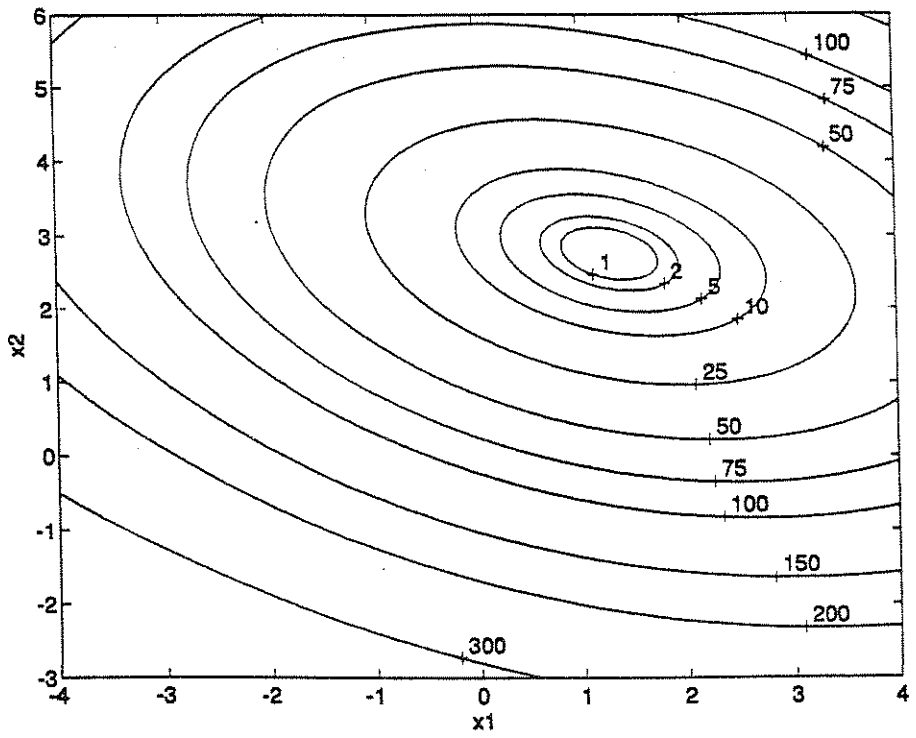
Note: Since f is a quadratic, $\vec{x}^* = \vec{x}_2$.

$$f(\vec{x}_1) = 0, \quad f(\vec{x}_2) = -1549.6$$

6.8 Contours of $f = (x_1 + 2x_2 - 7)^2 + (2x_1 + x_2 - 5)^2$

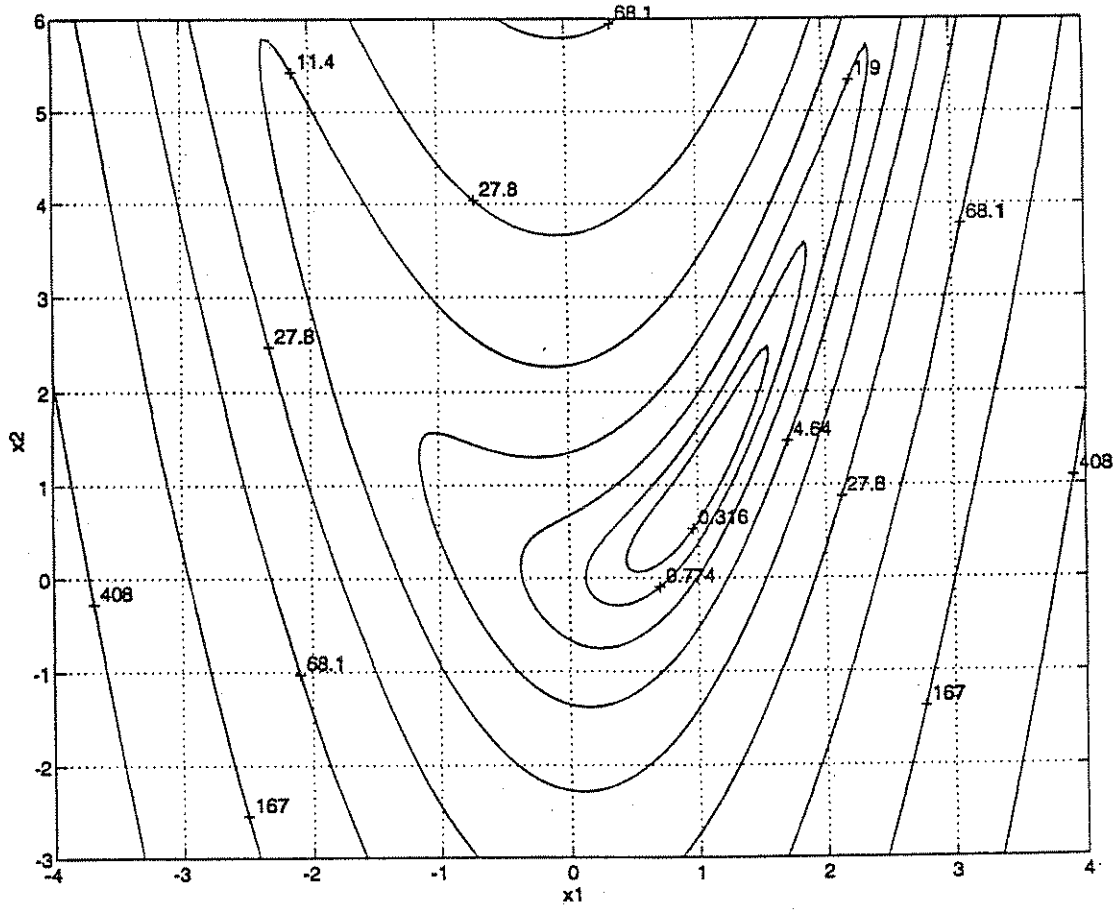
series of values given to x_1 in the range $-4 \le x_1 \le 4$.

For fixed x_1, the resulting quadratic solved to find values of x_2.



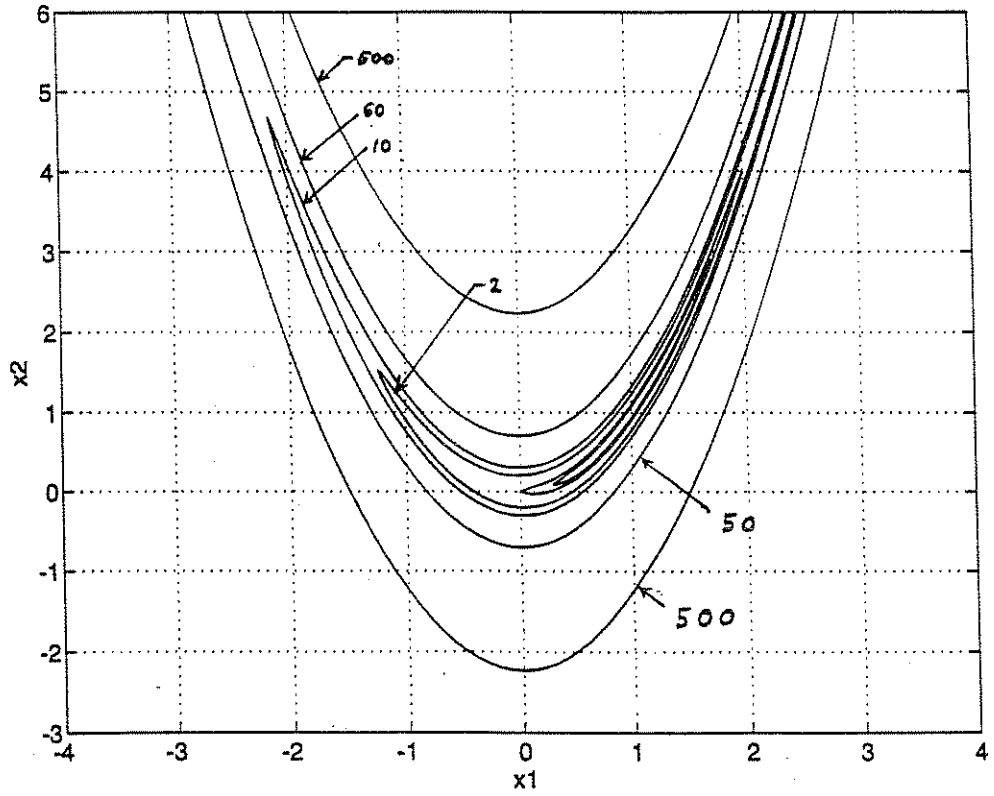
6.9 Contours of $f = 2(x_2 - x_1^2)^2 + (1 - x_1)^2$

Series of values are given to x_1 in the range $-4 \leq x_1 \leq 4$. The resulting quadratic equation is solved to find the values of x_2.



6.10 Contours of $f = 100\,(x_2 - x_1^2)^2 + (1 - x_1)^2$

A series of values are given to x_1 in the range $-4 \le x_1 \le 4$. For each value of x_1, the resulting quadratic is solved to find the corresponding values of x_2.



6.11 Given equations: $[A]\,\vec{x} = \vec{b}$

or
$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$
$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$
$$\vdots$$
$$a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n = b_n$$

For any arbitrary design vector $\vec{x}$, the above equations may not be satisfied. In such a case, we define errors or residuals for the equations as

$$e_i = a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n - b_i \quad ; \quad i = 1, 2, \ldots, n$$

Since the errors, e_i, will be zero at the correct solution, we pose an unconstrained optimization problem as follows:

Find $\vec{x} = \{ x_1, x_2, \ldots, x_n \}^T$

to minimize $f = e_1^2 + e_2^2 + \cdots + e_n^2$

$$= \sum_{i=1}^{n} \left(a_{i1} x_1 + a_{i2} x_2 + \cdots + a_{in} x_n - b_i \right)^2$$

$$= \frac{1}{2} \vec{x}^T [A] \vec{x} - \vec{x}^T \vec{b}$$

6.12

$$f = (2x_1 + x_2 - 4)^2 + (x_1 + 2x_2 + x_3 - 8)^2 + (x_2 + 3x_3 - 11)^2$$

$$\frac{\partial f}{\partial x_1} = 10 x_1 + 8 x_2 + 2 x_3 - 32$$

$$\frac{\partial f}{\partial x_2} = 8 x_1 + 12 x_2 + 10 x_3 - 62$$

$$\frac{\partial f}{\partial x_3} = 2 x_1 + 10 x_2 + 20 x_3 - 82$$

$$\vec{x}_1 = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}$$

$$f(\vec{x}_1) = 201.0$$

$$\nabla f_1 = \nabla f(\vec{x}_1) = \begin{Bmatrix} -32 \\ -62 \\ -82 \end{Bmatrix}, \quad \vec{s}_1 = -\nabla f_1 = \begin{Bmatrix} 32 \\ 62 \\ 82 \end{Bmatrix} = \begin{Bmatrix} 0.2972 \\ 0.5758 \\ 0.7616 \end{Bmatrix}$$

normalized.

$$\vec{x} = \vec{x}_1 + \lambda \vec{s}_1 = \begin{Bmatrix} 0.2972\ \lambda \\ 0.5758\ \lambda \\ 0.7616\ \lambda \end{Bmatrix}$$

$$f(\lambda) = (0.5944\ \lambda + 0.5758\ \lambda - 4)^2 + (0.2972\lambda + 1.1516\ \lambda + 0.7616\ \lambda - 8)^2$$

$$+ (0.5758\ \lambda + 2.2848\ \lambda - 11)^2$$

$$= (1.1702\ \lambda - 4)^2 + (2.2104\ \lambda - 8)^2 + (2.8606\ \lambda - 11)^2$$

For optimal λ^*, we use the calculus approach.

$$\frac{df}{d\lambda} = 2(1.1702)(1.1702\lambda - 4) + 2(2.2104)(2.2104\lambda - 8)$$
$$+ 2(2.8606)(2.8606\lambda - 11) = 0$$

i.e., $\lambda^* = 3.7283$

$$\vec{X}_2 = \vec{X}_1 + \lambda^* \vec{S}_1 = \begin{Bmatrix} 1.1081 \\ 2.1467 \\ 2.8395 \end{Bmatrix}, \qquad f(\vec{X}_2) = 0.3018684$$

$$\vec{S}_2 = \begin{Bmatrix} -0.7251 \\ -0.3826 \\ 0.5726 \end{Bmatrix}$$

$$\nabla f_2 = \begin{Bmatrix} 1.933601 \\ 1.020199 \\ -1.526802 \end{Bmatrix}, \qquad |\nabla f_2| = 2.666598 \neq 0$$

$\Rightarrow$ solution not converged.

$$\vec{X} = \vec{X}_2 + \lambda \vec{S}_2 = \begin{Bmatrix} 1.1081 \\ 2.1467 \\ 2.8395 \end{Bmatrix} + \lambda \begin{Bmatrix} -0.7251 \\ -0.3826 \\ 0.5726 \end{Bmatrix}$$

$$= \begin{Bmatrix} 1.1081 - 0.7251\lambda \\ 2.1467 - 0.3826\lambda \\ 2.8395 + 0.5726\lambda \end{Bmatrix}$$

$$f(\lambda) = (2.2162 - 1.4502\lambda + 2.1467 - 0.3826\lambda - 4)^2$$
$$+ (1.1081 - 0.7251\lambda + 4.2934 - 0.7652\lambda + 2.8395$$
$$+ 0.5726\lambda - 8)^2 + (2.1467 - 0.3826\lambda + 8.5185 + 1.7178\lambda - 11)^2$$

$$= (0.3629 - 1.8328\lambda)^2 + (0.2410 - 0.9177\lambda)^2$$
$$+ (-0.3348 + 1.3352\lambda)^2$$

$$\frac{df}{d\lambda} = 2(-1.8326)(-1.8326\lambda + 0.3629)$$
$$+ 2(-0.9177)(-0.9177\lambda + 0.2410)$$
$$+ 2(1.3352)(1.3352\lambda - 0.3348) = 0$$

i.e., $\lambda^* = 0.2228$

$$\vec{X}_3 = \vec{X}_2 + \lambda^* \vec{S}_2 = \begin{Bmatrix} 0.9465 \\ 2.0615 \\ 2.9671 \end{Bmatrix}$$

$$f(\vec{X}_3) = 0.00479371$$

$$\nabla f(\vec{X}_3) = \begin{Bmatrix} -0.108799 \\ -0.018997 \\ -0.150001 \end{Bmatrix}$$

Since $|\nabla f(\vec{X}_3)| \neq 0$, we need to continue the iterative process.

Exact solution: $\vec{X}^* = \begin{Bmatrix} 1 \\ 2 \\ 3 \end{Bmatrix}$

6.13 Total cost = (investment in equipment) + (cost of conductors) + (cost due to resistance loss)

$$f = 500\, e + 2*10^6\, c + \frac{0.15\, i^2}{c}$$

where $100*10^6$ watts $= \sqrt{3}\, e\, i$ or $i = \dfrac{100*10^6}{\sqrt{3}\, e}$

$$f = k_1 e + k_2 c + \frac{k_3}{c\, e^2} \qquad (1)$$

with $k_1 = 500$, $k_2 = 2*10^6$, $k_3 = 0.05*10^{16}$

Analytical solution:

$$\frac{\partial f}{\partial c} = k_2 - \frac{k_3}{c^2 e^2} = 0 \quad \text{or} \quad c^2 e^2 = k_3/k_2 \qquad (2)$$

$$\frac{\partial f}{\partial e} = k_1 - \frac{2 k_3}{c e^3} = 0 \quad \text{or} \quad c e^3 = 2 k_3/k_1 \qquad (3)$$

Eqs. (2) and (3) give:

$$e = \left(\frac{4\, k_2\, k_3}{k_1^2}\right)^{\frac{1}{4}}, \qquad c = \left(\frac{k_3\, k_1^2}{4\, k_3^3}\right)^{\frac{1}{4}} \qquad (4)$$

Using the values of k_1, k_2 and k_3, we obtain

$e = 11,246.83$ volts, $\quad c = 1.405854$ amperes

Solution by Newton's method:

$$\frac{\partial f}{\partial e} = k_1 - \frac{2k_3}{c\,e^3}, \quad \frac{\partial f}{\partial c} = k_2 - \frac{k_3}{c^2 e^2}$$

$$\frac{\partial^2 f}{\partial e^2} = \frac{6k_3}{c\,e^4}, \quad \frac{\partial^2 f}{\partial e\,\partial c} = \frac{2k_3}{c^2 e^3}, \quad \frac{\partial^2 f}{\partial c^2} = \frac{2k_3}{c^3 e^2}$$

Using e and c as 1st and 2nd variable, we obtain

$$[J] = \frac{2k_3}{c^2 e^2}\begin{bmatrix} \frac{1}{c} & \frac{1}{e} \\ \frac{1}{e} & \frac{3c}{e^2} \end{bmatrix}; \quad \text{determinant } [J] = \frac{8k_3^2}{c^4 e^6}$$

$$[J]^{-1} = \frac{c^2 e^4}{4k_3}\begin{bmatrix} \frac{3c}{e^2} & -\frac{1}{e} \\ -\frac{1}{e} & \frac{1}{c} \end{bmatrix}$$

$$\nabla f = \begin{Bmatrix} \partial f/\partial c \\ \partial f/\partial e \end{Bmatrix} = \begin{Bmatrix} \left(k_2 - \frac{k_3}{c^2 e^2}\right) \\ \left(k_1 - \frac{2k_3}{c e^3}\right) \end{Bmatrix}$$

Let $\quad \vec{x}_1 = \begin{Bmatrix} c \\ e \end{Bmatrix}_1 = \begin{Bmatrix} 1 \\ 10^4 \end{Bmatrix}$

$$\nabla f_1 = \nabla f(\vec{x}_1) = \begin{Bmatrix} -3 \times 10^6 \\ -500 \end{Bmatrix}$$

$$[J]_1^{-1} = [J]^{-1}(\vec{x}_1) = 5.0\begin{bmatrix} \frac{3}{10^8} & -\frac{1}{10^4} \\ -\frac{1}{10^4} & 1 \end{bmatrix}$$

$$\vec{x}_2 = \vec{x}_1 - [J_1]^{-1}\nabla f_1 = \begin{Bmatrix} 0.8 \\ 9000.0 \end{Bmatrix} = \begin{Bmatrix} c \\ e \end{Bmatrix}_2$$

6.14

$$f = 2x_1^2 + 16x_2^2 - 2x_1x_2 - x_1 - 6x_2 - 5$$

$$= c + \vec{B}^T\vec{X} + \frac{1}{2}\vec{X}^T[A]\vec{X}$$

with $c = -5$, $\vec{B} = \begin{Bmatrix} -1 \\ -6 \end{Bmatrix}$, $[A] = \begin{bmatrix} 4 & -2 \\ -2 & 32 \end{bmatrix}$, $\vec{x} = \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix}$

stage 1: Reduce $[A]$ to diagonal form:

$$\left|[A] - \lambda[I]\right| = 0$$

$$\begin{vmatrix} 4-\lambda & -2 \\ -2 & 32-\lambda \end{vmatrix} = \lambda^2 - 36\lambda + 124 = 0 \Rightarrow \lambda_2 = 3.8579, \quad \lambda_1 = 32.1421$$

$$\begin{bmatrix} (4-\lambda_i) & -2 \\ -2 & (32-\lambda_i) \end{bmatrix} \begin{Bmatrix} u_{1i} \\ u_{2i} \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \Rightarrow$$

$$\vec{u}_1 = \begin{Bmatrix} u_{11} \\ u_{21} \end{Bmatrix} = \begin{Bmatrix} 1.0 \\ -14.07105 \end{Bmatrix}, \quad \vec{u}_2 = \begin{Bmatrix} u_{12} \\ u_{22} \end{Bmatrix} = \begin{Bmatrix} 1.0 \\ 0.07105 \end{Bmatrix}$$

$$\vec{X} = [R]\vec{Y} \text{ and } [\tilde{A}] = [R]^T[A][R]$$

$$[\tilde{A}] = \begin{bmatrix} 1 & -14.07105 \\ 1 & 0.07105 \end{bmatrix} \begin{bmatrix} 4 & -2 \\ -2 & 32 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -14.07105 & 0.07105 \end{bmatrix}$$

$$= \begin{bmatrix} 6396.1518 & 0 \\ 0 & 3.87737 \end{bmatrix}$$

$$f(y_1, y_2) = c + \vec{B}^T[R]\vec{Y} + \frac{1}{2}\vec{Y}^T[\tilde{A}]\vec{Y}$$

$$= -5 + 83.4263\,y_1 - 1.4263\,y_2 + \frac{1}{2}(6396.1518)\,y_1^2 + \frac{1}{2}(3.87737)\,y_2^2$$

stage 2: Reduce $[\tilde{A}]$ to a unit matrix:

$$\vec{Y} = [S]\vec{Z} \quad \text{where} \quad [S] = \begin{bmatrix} \frac{1}{\sqrt{6396.1518}} & 0 \\ 0 & \frac{1}{\sqrt{3.87737}} \end{bmatrix}$$

i.e., $$[S] = \begin{bmatrix} 0.0125 & 0 \\ 0 & 0.5078 \end{bmatrix}$$

stage 3: complete transformation:

$$\vec{x} = [R]\vec{Y} = [R][S]\vec{z} = [T]\vec{z}$$

with $$[T] = \begin{bmatrix} 1 & 1 \\ -14.07105 & 0.07105 \end{bmatrix}\begin{bmatrix} 0.0125 & 0 \\ 0 & 0.5078 \end{bmatrix}$$

$$= \begin{bmatrix} 0.0125 & 0.5078 \\ -0.1759 & 0.0361 \end{bmatrix}$$

or $$x_1 = 0.0125\, z_1 + 0.5078\, z_2$$

$$x_2 = -0.1759\, z_1 + 0.0361\, z_2$$

Final quadratic function:

$$f(z_1, z_2) = c + \vec{B}^T[T]\vec{z} + \frac{1}{2}\vec{z}^T[T][A][T]\vec{z}$$

$$= -5 + 1.0429\, z_1 - 0.7244\, z_2 + \frac{1}{2} z_1^2 + \frac{1}{2} z_2^2$$

6.15

$$f(x_1, x_2) = 4x_1^2 + 3x_2^2 - 5x_1x_2 - 8x_1 + 10$$

$$= \frac{1}{2}\vec{x}^T[A]\vec{x} + \vec{B}^T\vec{x} + c$$

With $$[A] = \begin{bmatrix} 8 & -5 \\ -5 & 6 \end{bmatrix}, \quad \vec{B} = \begin{Bmatrix} -8 \\ 0 \end{Bmatrix}, \quad c = 10$$

stage 1: Reduce [A] to diagonal form:

$$\left|[A] - \lambda[I]\right| = 0 \Rightarrow \lambda^2 - 14\lambda + 23 = 0$$

$$\lambda_1 = 12.0990, \quad \lambda_2 = 1.9000$$

$$\begin{bmatrix} 8-\lambda_1 & -5 \\ -5 & 6-\lambda_1 \end{bmatrix}\begin{Bmatrix} u_{11} \\ u_{21} \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \Rightarrow \vec{u}_1 = \begin{Bmatrix} 1.0 \\ -0.8198 \end{Bmatrix}$$

$$\begin{bmatrix} 8-\lambda_2 & -5 \\ -5 & 6-\lambda_2 \end{bmatrix}\begin{Bmatrix} u_{12} \\ u_{22} \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \Rightarrow \vec{u}_2 = \begin{Bmatrix} 1.0 \\ 1.2198 \end{Bmatrix}$$

$$[R] = [\vec{u}_1 \quad \vec{u}_2] = \begin{bmatrix} 1 & 1 \\ -0.8198 & 1.2198 \end{bmatrix}$$

$$\vec{x} = [R]\,\vec{Y} = \begin{Bmatrix} y_1 + y_2 \\ -0.8198\, y_1 + 1.2198\, y_2 \end{Bmatrix}$$

$$[\tilde{A}] = [R]^T [A][R] = \begin{bmatrix} 1 & -0.8198 \\ 1 & 1.2198 \end{bmatrix}\begin{bmatrix} 8 & -5 \\ -5 & 6 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ -0.8198 & 1.2198 \end{bmatrix}$$

$$= \begin{bmatrix} 20.2304 & 0 \\ 0 & 4.7295 \end{bmatrix}$$

$$f(y_1, y_2) = c + \vec{B}^T [R]\vec{Y} + \tfrac{1}{2}\vec{Y}^T [\tilde{A}]\vec{Y}$$

$$= 10 - 8\, y_1 - 8\, y_2 + 10.1152\, y_1^2 + 2.3648\, y_2^2$$

Stage 2: Reduce $[\tilde{A}]$ to a unit matrix:

$$\vec{Y} = [S]\,\vec{Z} \quad \text{where} \quad [S] = \begin{bmatrix} \frac{1}{\sqrt{20.2304}} & 0 \\ 0 & \frac{1}{\sqrt{4.7295}} \end{bmatrix}$$

$$= \begin{bmatrix} 0.2223 & 0 \\ 0 & 0.4598 \end{bmatrix}$$

Stage 3: Complete transformation:

$$[T] = [R][S] = \begin{bmatrix} 1 & 1 \\ -0.8198 & 1.2198 \end{bmatrix}\begin{bmatrix} 0.2223 & 0 \\ 0 & 0.4598 \end{bmatrix}$$

$$= \begin{bmatrix} 0.2223 & 0.4598 \\ -0.1822 & 0.5609 \end{bmatrix}$$

$$f(z_1, z_2) = c + \vec{B}^T [T]\vec{Z} + \tfrac{1}{2}\vec{Z}^T [T]^T [A][T]\vec{Z}$$

$$= 10 + 1.7784\, z_1 + 3.6784\, z_2 + \tfrac{1}{2} z_1^2 + \tfrac{1}{2} z_2^2$$

and $$\vec{x} = [T]\,\vec{Z} = \begin{Bmatrix} 0.2223\, z_1 + 0.4598\, z_2 \\ -0.1822\, z_1 + 0.5609\, z_2 \end{Bmatrix}$$

6.16

$$f = 2x_1^2 + 16x_2^2 - 2x_1x_2 - x_1 - 6x_2 - 5$$

$$[A] = \begin{bmatrix} \partial^2 f/\partial x_1^2 & \partial^2 f/\partial x_1 \partial x_2 \\ \partial^2 f/\partial x_1 \partial x_2 & \partial^2 f/\partial x_2^2 \end{bmatrix} = \begin{bmatrix} 4 & -2 \\ -2 & 32 \end{bmatrix}$$

(a) $\vec{S}_1^T [A] \vec{S}_2 = (15 \quad -1) \begin{bmatrix} 4 & -2 \\ -2 & 32 \end{bmatrix} \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} = 0$

$\vec{S}_1$ and $\vec{S}_2$ are conjugate.

(b) $\vec{S}_1^T [A] \vec{S}_2 = (-1 \quad 15) \begin{bmatrix} 4 & -2 \\ -2 & 32 \end{bmatrix} \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} = 448 \neq 0$

$\vec{S}_1$ and $\vec{S}_2$ are not conjugate.

6.17

The random numbers generated and the corresponding values of design variables and the objective function are shown in Table 1. To save space, only the trials which produced improvement are shown in this table.

TABLE 1

MINIMIZATION OF f BY RANDOM JUMPING

Trial number	Random numbers		Corresponding design variables		Value of $f(x_1, x_2)$
	(1)	(2)	x_1	x_2	
1	0.12606	0.63946	−7.47887	2.78913	67.65904
3	0.42469	0.84082	−1.50612	6.81649	22.14585
13	0.33082	0.86722	−3.38357	7.34448	16.40928
21	0.29605	0.69995	−4.07908	3.99909	8.56712
23	0.47964	0.51831	−0.40723	0.36612	−0.60582
118	0.42872	0.60919	−1.42552	2.18379	−1.00223
1238	0.43523	0.57343	−1.29534	1.46852	−1.05596
1836	0.43380	0.59430	−1.32398	1.88601	−1.14119
2451	0.44137	0.58901	−1.17266	1.78015	−1.20863
8573	0.44690	0.57240	−1.06193	1.44792	−1.23317
10000[1]	No improvement				
Correct solution	—	—	−1.00000	1.50000	−1.25000

6.18 For each trial design vector, two random numbers, uniformly distributed in the interval $(-1, 1)$, are generated and the design vector is computed as

$$\vec{x} = \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} l_1 + r_1 (u_1 - l_1) \\ l_2 + r_2 (u_2 - l_2) \end{Bmatrix}$$

where r_1, r_2 denote the random numbers; l_1, l_2 represent the lower bounds and u_1, u_2 indicate the upper bounds on x_1, x_2. The random walk procedure with direction exploitation gave the following results (only successful trials that reduced the value of the objective function are given):

Random numbers r_1	r_2	Design variables x_1	x_2	step length	Value of $f(\vec{x})$	Number of trials required
0.684518	-0.728996	0.684518	4.27100	1.00000	21.4393	1
-0.581298	-0.813690	-0.581298	3.83295	1.00000	6.49691	3
-0.581298	-0.813690	-1.16260	3.01926	1.00000	0.616981	3
-0.581298	-0.813690	-1.74390	2.20557	1.00000	-0.695151	3
-0.378351	-0.925662	-0.378351	1.11990	1.00000	-0.805204	19
-0.864545	0.502556	-0.864545	1.98877	1.00000	-0.841999	995
-0.432387	-0.901688	-0.432387	0.885671	1.00000	-0.925633	140
-0.976259	0.216606	-0.976259	1.53579	1.00000	-1.24589	84
-0.968081	0.250637	-0.968081	1.42971	1.00000	-1.24751	2

6.19 (a) $[A] = \begin{bmatrix} 1 & 2 \\ 1.0001 & 2 \end{bmatrix}$, $[A]^{-1} = \begin{bmatrix} -10\,000 & 10\,000 \\ 5000.5 & -5000 \end{bmatrix}$

$$\|[A]\|_\infty = \max \begin{Bmatrix} |1|+|2| \\ |1.0001|+|2| \end{Bmatrix} = 3.0001$$

$$\|[A]^{-1}\|_\infty = \max \begin{Bmatrix} |-10\,000| + |10\,000| \\ |5000.5| + |-5000| \end{Bmatrix} = 20\,000.0$$

$$\text{cond}\,[A] = 3.0001\,(20\,000.0) = 60\,002.0$$

(b) $[B] = \begin{bmatrix} 3.9 & 1.6 \\ 6.8 & 2.9 \end{bmatrix}$, $[B]^{-1} = \begin{bmatrix} 6.7442 & -3.7209 \\ -15.8140 & 9.0698 \end{bmatrix}$

$$\|[B]\|_\infty = \max \begin{Bmatrix} |3.9|+|1.6| \\ |6.8| + |2.9| \end{Bmatrix} = 9.7$$

$$\|[B]^{-1}\|_\infty = \max \begin{Bmatrix} |6.7442| + |-3.7209| \\ |-15.8140| + |9.0698| \end{Bmatrix} = 24.8838$$

$$\text{Cond}\,[B] = 9.7\,(24.8838) = 241.37286$$

6.20

$$f = 100\,(x_2 - x_1^2)^2 + (1 - x_1)^2$$

$$\frac{\partial f}{\partial x_1} = -400\,x_1\,(x_2 - x_1^2) - 2\,(1 - x_1)$$

$$\frac{\partial f}{\partial x_2} = 200\,(x_2 - x_1^2)$$

$$\frac{\partial^2 f}{\partial x_1^2} = -400\,(x_2 - x_1^2) + 800\,x_1^2 + 2$$

$$\frac{\partial^2 f}{\partial x_2^2} = 200 \quad , \quad \frac{\partial^2 f}{\partial x_1 \partial x_2} = -400\,x_1$$

$$\nabla f = \begin{Bmatrix} -400\,x_1\,(x_2 - x_1^2) - 2\,(1 - x_1) \\ 200\,(x_2 - x_1^2) \end{Bmatrix}$$

$$[J] = \begin{bmatrix} -400\,(x_2 - x_1^2) + 800\,x_1^2 + 2 & -400\,x_1 \\ -400\,x_1 & 200 \end{bmatrix}$$

$$\vec{x}_1 = \begin{Bmatrix} -1.2 \\ 1.0 \end{Bmatrix} \quad , \quad f_1 = 24.2$$

Iteration 1:

$$\nabla f_1 = \begin{Bmatrix} -215.6 \\ -88.0 \end{Bmatrix} \quad , \quad [J_1] = \begin{bmatrix} 1330.0 & 480.0 \\ 480.0 & 200.0 \end{bmatrix}$$

$$[J_1]^{-1} = \begin{bmatrix} 0.00561798 & -0.013483 \\ -0.013483 & 0.037359 \end{bmatrix}$$

$$\vec{x}_2 = \vec{x}_1 - [J_1]^{-1} \nabla f_1 = \begin{Bmatrix} -1.17528 \\ 1.38067 \end{Bmatrix}$$

$$f_2 = f(\vec{x}_2) = 4.731885$$

Iteration 2:

$$\nabla f_2 = \begin{Bmatrix} -4.637888 \\ -0.122237 \end{Bmatrix} \quad , \quad [J_2] = \begin{bmatrix} 1107.2729 & 470.11243 \\ 470.11243 & 200.0 \end{bmatrix}$$

$$[J_2]^{-1} = \begin{bmatrix} 0.445527 & -1.047240 \\ -1.047240 & 2.466602 \end{bmatrix}$$

$$\vec{x}_3 = \vec{x}_2 - [J_2]^{-1} \nabla f_2 = \begin{Bmatrix} 0.763013 \\ -3.174796 \end{Bmatrix}$$

$$f_3 = f(\vec{x}_3) = 1411.5499$$

Iteration 3:

$$\nabla f_3 = \begin{Bmatrix} 1146.1776 \\ -751.3970 \end{Bmatrix}, \quad [J_3] = \begin{bmatrix} 1970.5453 & -305.2053 \\ -305.2053 & 200.0 \end{bmatrix}$$

$$[J_3]^{-1} = \begin{bmatrix} 0.000664543 & 0.001014110 \\ 0.001014110 & 0.006547558 \end{bmatrix}$$

$$\vec{x}_4 = \vec{x}_3 - [J_3]^{-1} \nabla f_3 = \begin{Bmatrix} 0.763328 \\ 0.582670 \end{Bmatrix}$$

$$f_4 = f(\vec{x}_4) = 0.0560136$$

6.21 Minimize $f = 100\,(x_2 - x_1^2)^2 + (1 - x_1)^2$

$\vec{X}_1 = \begin{Bmatrix} -1.2 \\ 1.0 \end{Bmatrix}$, $f_1 = 100\,(1.0 - (-1.2)^2)^2 + (1 - (-1.2))^2 = 24.2$

Iteration 1:

Use probe length $\epsilon = 0.0001$ along $\vec{S}_1 = \begin{Bmatrix} 1 \\ 0 \end{Bmatrix}$:

$f^+ = f(\vec{X}_1 + \epsilon \vec{S}_1) = f(-1.1999, 1.0) = 24.178447 < f_1$

$\therefore + \vec{S}_1$ is the correct direction.

$$f(\lambda) = f(\vec{X}_1 + \lambda \vec{S}_1) = f\left(\begin{Bmatrix} -1.2 \\ 1.0 \end{Bmatrix} + \lambda \begin{Bmatrix} 1.0 \\ 0.0 \end{Bmatrix}\right) = f(-1.2 + \lambda, 1)$$

$$= 100\left(1 - (\lambda - 1.2)^2\right)^2 + (1 + 1.2 - \lambda)^2$$

$$= 24.2 - 215.6\,\lambda + 665.0\,\lambda^2 - 480.0\,\lambda^3 + 100.0\,\lambda^4$$

$$\frac{df}{d\lambda} = -215.6 + 1330.0\,\lambda - 1440.0\,\lambda^2 + 400.0\,\lambda^3 = 0$$

Solution of this equation (smallest root) gives

$$\lambda^* = 0.205022$$

$$\vec{X}_2 = \vec{X}_1 + \lambda_1^* \vec{S}_1 = \begin{Bmatrix} -1.2 \\ 1.0 \end{Bmatrix} + 0.205022 \begin{Bmatrix} 1 \\ 0 \end{Bmatrix} = \begin{Bmatrix} -0.994978 \\ 1.0 \end{Bmatrix}$$

$f_2 = f(\vec{X}_2) = 3.989973$

Iteration 2:

We use the probe length $\epsilon = 0.0001$ along $\vec{S}_2 = \begin{Bmatrix} 0 \\ 1 \end{Bmatrix}$:

$$f^+ = f(\vec{X}_2 + \epsilon \vec{S}_2) = f\left(\begin{Bmatrix} -0.994978 \\ 1.0 \end{Bmatrix} + \epsilon \begin{Bmatrix} 0 \\ 1 \end{Bmatrix}\right)$$

$= f(-0.994978, 1.0001) = 3.990176 > f_2$

$$f^- = f(\vec{X}_2 - \epsilon \vec{S}_2) = f\left(\begin{Bmatrix} -0.994978 \\ 1.0 \end{Bmatrix} - \epsilon \begin{Bmatrix} 0 \\ 1 \end{Bmatrix}\right)$$

$= f(-0.994978, 0.9999) = 3.989775 < f_2$

$\therefore -\vec{S}_2$ is the correct direction.

$$f(\lambda) = f(\vec{X}_2 - \lambda \vec{S}_2) = f(-0.994978, 1 - \lambda)$$

$$= 3.989975 - 2.0038\,\lambda + 100\,\lambda^2$$

$$\frac{df}{d\lambda} = -2.0038 + 200\,\lambda = 0$$

$$\lambda^* = 0.010019$$

$$\vec{x}_3 = \vec{x}_2 - \lambda^* \vec{s}_2 = \begin{Bmatrix} -0.994978 \\ 1.0 \end{Bmatrix} - 0.010019 \begin{Bmatrix} 0 \\ 1 \end{Bmatrix}$$

$$= \begin{Bmatrix} -0.994978 \\ 0.989981 \end{Bmatrix}$$

$$f_3 = f(\vec{x}_3) = 3.979937$$

Iteration 3:

We use a probe length $\epsilon = 0.0001$ along $\vec{s}_1 = \begin{Bmatrix} 1 \\ 0 \end{Bmatrix}$:

$$f^+ = f(\vec{x}_3 + \epsilon \vec{s}_1) = f\left(\begin{Bmatrix} -0.994978 \\ 0.989981 \end{Bmatrix} + 0.0001 \begin{Bmatrix} 1 \\ 0 \end{Bmatrix} \right)$$

$$= f(-0.994878, 0.989981) = 3.9795425 < f_3$$

$\therefore +\vec{s}_1$ is the correct direction.

$$f(\lambda) = f(\vec{x}_3 + \lambda \vec{s}_1) = f(-0.994978 + \lambda, 0.989981)$$

$$= 100\lambda^4 - 395.9924\lambda^3 + 393.0249\lambda^2 - 3.989956\lambda + 3.979937$$

$$\frac{df}{d\lambda} = 400\lambda^3 - 1187.9772\lambda^2 + 786.0498\lambda - 3.989956 = 0$$

Smallest root of this equation gives $\lambda^* = 0.00511271$

$$\vec{x}_4 = \vec{x}_3 + \lambda^* \vec{s}_1 = \begin{Bmatrix} -0.98986529 \\ 0.989981 \end{Bmatrix}$$

$$f_4 = f(\vec{x}_4) = 3.9697583$$

Iteration 4:

We use a probe length $\epsilon = 0.0001$ along $\vec{s}_2 = \begin{Bmatrix} 0 \\ 1 \end{Bmatrix}$:

$$f^+ = f(\vec{x}_4 + \epsilon \vec{s}_2) = f\left(\begin{Bmatrix} -0.98986529 \\ 0.989981 \end{Bmatrix} + \epsilon \begin{Bmatrix} 0 \\ 1 \end{Bmatrix} \right)$$

$$= f(-0.98986529, 0.990081) = 3.9700656 > f_4$$

$$f^- = f(\vec{x}_4 - \epsilon \vec{s}_2) = f(-0.98986529, 0.989881)$$

$$= 3.9696596 < f_4$$

$\therefore -\vec{s}_2$ is the correct direction.

$$f(\lambda) = f(\vec{x}_4 - \lambda \vec{s}_2) = f(-0.98986529, 0.989981 - \lambda)$$

$$= 100\,\lambda^2 - 2.029542\,\lambda + 3.96986187$$

$$\frac{df}{d\lambda} = 200\,\lambda - 2.029542 = 0$$

$$\lambda^* = 0.01014771$$

$$\vec{X}_5 = \vec{X}_4 - \lambda^* \vec{S}_2 = \begin{Bmatrix} -0.98986529 \\ 0.97983329 \end{Bmatrix}$$

$$f_5 = f(\vec{X}_5) = 3.959564$$

6.22 Minimize $f = 100\,(x_2 - x_1^2)^2 + (1 - x_1)^2$ using Powell's method.

From the univariate method (Problem 6.21), we find:

$\vec{X}_1 = \begin{Bmatrix} -1.2 \\ 1.0 \end{Bmatrix}$, $f_1 = 24.2$ (starting point)

$\vec{X}_2 = \begin{Bmatrix} -0.994978 \\ 1.0 \end{Bmatrix}$, $f_2 = 3.989973$ (minimum along $\vec{S}_1$)

$\vec{X}_3 = \begin{Bmatrix} -0.994978 \\ 0.989981 \end{Bmatrix}$, $f_3 = 3.979937$ (minimum along $\vec{S}_2$)

$\vec{X}_4 = \begin{Bmatrix} -0.98986529 \\ 0.989981 \end{Bmatrix}$, $f_4 = 3.9697583$ (minimum along $\vec{S}_1$)

$\vec{X}_5 = \begin{Bmatrix} -0.98986529 \\ 0.97983329 \end{Bmatrix}$, $f_5 = 3.959564$ (minimum along $\vec{S}_2$)

Pattern direction:

$$\vec{S}_p^{(1)} = \vec{X}_5 - \vec{X}_3 = \begin{Bmatrix} -0.98986529 \\ 0.97983329 \end{Bmatrix} - \begin{Bmatrix} -0.994978 \\ 0.989981 \end{Bmatrix}$$

$$= \begin{Bmatrix} 0.00511271 \\ -0.01014771 \end{Bmatrix}$$

$$f(\lambda) = f(\vec{X}_5 + \lambda \vec{S}_p^{(1)}) = f(-0.98986529 + 0.00511271\,\lambda,\ 0.97983329 - 0.01014771\,\lambda)$$

$$\lambda^* = 40.0$$

$$\vec{X}_6 = \vec{X}_5 + \lambda^* \vec{S}_p^{(1)} = \begin{Bmatrix} -0.78535688 \\ 0.57392490 \end{Bmatrix}$$

$$f_6 = f(\vec{X}_6) = 3.3712015$$

Next, we discard $\vec{s}_1$ and use probe length $\epsilon = 0.0001$ along $+\vec{s}_2$.

$$\vec{x}^{+} = \vec{x}_6 + \epsilon \vec{s}_2 = \begin{Bmatrix} -0.78535688 \\ 0.57402490 \end{Bmatrix}$$

$$f^{+} = 3.3703454 < f_6$$

$\therefore +\vec{s}_2$ is the correct direction.

$$\vec{x} = \vec{x}_6 + \lambda \vec{s}_2$$

$$f(\vec{x}) = f(\lambda) = f(-0.78535688,\ 0.57392490 + \lambda)$$

$$\lambda^* = 0.045$$

$$\vec{x}_7 = \vec{x}_6 + \lambda^* \vec{s}_2 = \begin{Bmatrix} -0.78535688 \\ 0.61892492 \end{Bmatrix}$$

$$f_7 = 3.1879570$$

Use probe length $\epsilon = 1.0$ along $\vec{s}_p^{(1)}$:

$$\vec{x}^{+} = \vec{x}_7 + \epsilon \vec{s}_p^{(1)} = \begin{Bmatrix} -0.78535688 \\ 0.61892492 \end{Bmatrix} + \epsilon \begin{Bmatrix} 0.00511271 \\ -0.01014771 \end{Bmatrix}$$

$$= \begin{Bmatrix} -0.78024417 \\ 0.60877721 \end{Bmatrix}$$

$$f^{+} = 3.1692691 < f_7$$

$\therefore +\vec{s}_p^{(1)}$ is the correct direction.

$$\vec{x} = \vec{x}_7 + \lambda \vec{s}_p^{(1)} = \begin{Bmatrix} -0.78535688 + 0.00511271\ \lambda \\ 0.61892492 - 0.01014771\ \lambda \end{Bmatrix}$$

$$\lambda^* = 13.5$$

$$\vec{x}_8 = \vec{x}_7 + \lambda^* \vec{s}_p^{(1)} = \begin{Bmatrix} -0.71633530 \\ 0.48193082 \end{Bmatrix}$$

$$f_8 = 3.0431848$$

6.23 Minimize $f = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$ using steepest descent method.

$$\nabla f = \begin{Bmatrix} \partial f/\partial x_1 \\ \partial f/\partial x_2 \end{Bmatrix} = \begin{Bmatrix} -400\, x_1 (x_2 - x_1^2) - 2(1 - x_1) \\ 200\ (x_2 - x_1^2) \end{Bmatrix}$$

$$\vec{X}_1 = \begin{Bmatrix} -1.2 \\ 1.0 \end{Bmatrix}, \quad f_1 = 24.2, \quad \nabla f_1 = \begin{Bmatrix} -215.6 \\ -88.0 \end{Bmatrix}$$

$$\vec{X} = \vec{X}_1 - \lambda \nabla f_1 = \begin{Bmatrix} -1.2 + 215.6\,\lambda \\ 1.0 + 88.0\,\lambda \end{Bmatrix}$$

$f(\vec{X})$ = minimum at $\lambda^* \simeq 0.0008$

$$\vec{X}_2 = \vec{X}_1 - \lambda^* \nabla f_1 = \begin{Bmatrix} -1.0275201 \\ 1.0704000 \end{Bmatrix}, \quad f_2 = f(\vec{X}_2) = 4.1321611$$

$$\nabla f_2 = \begin{Bmatrix} 1.946681729 \\ 2.9204888 \end{Bmatrix}$$

$$\vec{X} = \vec{X}_2 - \lambda \nabla f_2 = \begin{Bmatrix} -1.0275201 - 1.946681729\,\lambda \\ 1.0704 - 2.9204888\,\lambda \end{Bmatrix}$$

$\lambda^* \simeq 0.0013$

$$\vec{X}_3 = \begin{Bmatrix} -1.0300508 \\ 1.0666034 \end{Bmatrix}, \quad f_3 = 4.1242409$$

6.24 Minimize $f = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$ using Fletcher-Reeves method.

$$\vec{X}_1 = \begin{Bmatrix} -1.2 \\ 1.0 \end{Bmatrix}, \quad f_1 = 24.2$$

$$\nabla f = \begin{Bmatrix} \partial f/\partial x_1 \\ \partial f/\partial x_2 \end{Bmatrix} = \begin{Bmatrix} -400\, x_1 (x_2 - x_1^2) - 2(1 - x_1) \\ 200\ (x_2 - x_1^2) \end{Bmatrix}$$

$$\nabla f_1 = \begin{Bmatrix} -215.6 \\ -88.0 \end{Bmatrix}, \quad |\nabla f_1| = \{(-215.6)^2 + (-88)^2\}^{\frac{1}{2}} = 232.867688$$

$$\vec{S}_1 = -\nabla f_1 = \begin{Bmatrix} 215.6 \\ 88.0 \end{Bmatrix}$$

$$\vec{X} = \vec{X}_1 + \lambda_1 \vec{S}_1 = \begin{Bmatrix} -1.2 + 215.6\,\lambda \\ 1.0 + 88.6\,\lambda \end{Bmatrix}$$

$\lambda_1^* \simeq 0.0008$

$\vec{x}_2 = \vec{x}_1 + \lambda_1^* \vec{s}_1 = \begin{Bmatrix} -1.0275201 \\ 1.0704 \end{Bmatrix}$

$f_2 = 4.1321611$

$\nabla f_2 = \begin{Bmatrix} 1.946681729 \\ 2.9204888 \end{Bmatrix}$, $|\nabla f_2| = \{1.946681729^2 + 2.9204888^2\}^{\frac{1}{2}}$

$= 3.509818312$

$\vec{s}_2 = -\nabla f_2 + \dfrac{|\nabla f_2|^2}{|\nabla f_1|^2} \vec{s}_1 = \begin{Bmatrix} -1.946681729 \\ -2.9204888 \end{Bmatrix} + \left(\dfrac{12.31882458}{54,227.36}\right) \begin{Bmatrix} 215.6 \\ 88.0 \end{Bmatrix}$

$= \begin{Bmatrix} -1.897703896 \\ -2.900497784 \end{Bmatrix}$

$\vec{x} = \vec{x}_2 + \lambda_2 \vec{s}_2 = \begin{Bmatrix} -1.0275201 - 1.897703896\,\lambda \\ 1.0704 - 2.900497784\,\lambda \end{Bmatrix}$

$\lambda_2^* \simeq 0.0013$

$\vec{x}_3 = \vec{x}_2 + \lambda_2^* \vec{s}_2 = \begin{Bmatrix} -1.0299871 \\ 1.0666294 \end{Bmatrix}$

$f_3 = 4.1241608$

6.25 Minimize $f = 100(x_2 - x_1^2)^2 + (1-x_1)^2$ using DFP method.

Let $\vec{x}_1 = \begin{Bmatrix} -1.2 \\ 1.0 \end{Bmatrix}$ and $[B_1] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$f_1 = 24.2$, $\nabla f = \begin{Bmatrix} -400 x_1 (x_2 - x_1^2) - 2(1-x_1) \\ 200 (x_2 - x_1^2) \end{Bmatrix}$

$\nabla f_1 = \begin{Bmatrix} -215.6 \\ -88.0 \end{Bmatrix}$

$\vec{s}_1 = -[B_1] \nabla f_1 = \begin{Bmatrix} 215.6 \\ 88.0 \end{Bmatrix}$

$\vec{x} = \vec{x}_1 + \lambda_1 \vec{s}_1 = \begin{Bmatrix} -1.2 + 215.6\,\lambda \\ 1.0 + 88.0\,\lambda \end{Bmatrix}$

$\lambda_1^* \simeq 0.0008$

$\vec{x}_2 = \vec{x}_1 + \lambda_1^* \vec{s}_1 = \begin{Bmatrix} -1.0275201 \\ 1.0704 \end{Bmatrix}$

$$f_2 = 4.1321611, \qquad \nabla f_2 = \begin{Bmatrix} 1.946681729 \\ 2.9204888 \end{Bmatrix}$$

Since $\nabla f_2 \neq \vec{0}$, we proceed to next iteration.

$$\vec{g}_1 = \nabla f_2 - \nabla f_1 = \begin{Bmatrix} 1.946681729 \\ 2.9204888 \end{Bmatrix} - \begin{Bmatrix} -215.6 \\ -88.0 \end{Bmatrix} = \begin{Bmatrix} 217.5466817 \\ 90.9204888 \end{Bmatrix}$$

$$\vec{s}_1^T \vec{g}_1 = \begin{pmatrix} 215.6 & 88.0 \end{pmatrix} \begin{Bmatrix} 217.5466817 \\ 90.9204888 \end{Bmatrix} = 54904.06758$$

$$[B_1]\vec{g}_1 = \begin{Bmatrix} 217.5466817 \\ 90.9204888 \end{Bmatrix}$$

$$\vec{g}_1^T [B_1] \vec{g}_1 = 55593.094$$

$$\vec{s}_1 \vec{s}_1^T = \begin{bmatrix} 46483.36 & 18972.8 \\ 18972.8 & 7744.0 \end{bmatrix}$$

$$[M_1] = \lambda_1^* \frac{\vec{s}_1 \vec{s}_1^T}{\vec{s}_1^T \vec{g}_1} = \frac{0.0008}{54904.06758} \vec{s}_1 \vec{s}_1^T$$

$$= 10^{-7} \begin{bmatrix} 6773.029492 & 2764.501454 \\ 2764.501454 & 1128.368095 \end{bmatrix}$$

$$[N_1] = -\frac{\left([B_1]\vec{g}_1\right)\left([B_1]\vec{g}_1\right)^T}{\vec{g}_1^T [B_1] \vec{g}_1} = -\frac{\vec{g}_1 \vec{g}_1^T}{\vec{g}_1^T [B_1] \vec{g}_1}$$

$$= -10^{-4} \begin{bmatrix} 8513.0285 & 3557.897121 \\ 3557.897121 & 1486.971636 \end{bmatrix}$$

$$[B_2] = [B_1] + [M_1] + [N_1]$$

$$= \begin{bmatrix} 0.14937445 & -0.35551326 \\ -0.35551326 & 0.85141568 \end{bmatrix}$$

$$\vec{s}_2 = -[B_2]\nabla f_2 = \begin{Bmatrix} 0.74748798 \\ -1.79447879 \end{Bmatrix}$$

$$\vec{x} = \vec{x}_2 + \lambda_2 \vec{s}_2$$

$$= \begin{Bmatrix} -1.0275201 + 0.74748798\ \lambda \\ 1.0704 - 1.79447879\ \lambda \end{Bmatrix}$$

$\lambda_2^* \simeq 0.145$

$\vec{x}_3 = \vec{x}_2 + \lambda_2^* \vec{s}_2 = \begin{Bmatrix} -0.91913432 \\ 0.81020057 \end{Bmatrix}$

$f_3 = f(\vec{x}_3) = 3.8028438$

6.26 Minimize $f = 100(x_2 - x_1^2)^2 + (1-x_1)^2$ using BFGS method.

$\vec{x}_1 = \begin{Bmatrix} -1.2 \\ 1.0 \end{Bmatrix}$, $f_1 = 24.2$

$[B_1] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $\nabla f = \begin{Bmatrix} -400\, x_1 (x_2 - x_1^2) - 2(1-x_1) \\ 200\,(x_2 - x_1^2) \end{Bmatrix}$

$\nabla f_1 = \begin{Bmatrix} -215.6 \\ -88.0 \end{Bmatrix}$, $\vec{s}_1 = -[B_1]\nabla f_1 = \begin{Bmatrix} 215.6 \\ 88.0 \end{Bmatrix}$

$\vec{x} = \vec{x}_1 + \lambda_1 \vec{s}_1 = \begin{Bmatrix} -1.2 + 215.6\,\lambda \\ 1.0 + 88.0\,\lambda \end{Bmatrix}$

$\lambda_1^* \simeq 0.0008$, $\vec{x}_2 = \vec{x}_1 + \lambda_1^* \vec{s}_1 = \begin{Bmatrix} -1.0275201 \\ 1.0704 \end{Bmatrix}$

$f_2 = 4.1321611$

Since $\nabla f_2 = \begin{Bmatrix} 1.94668729 \\ 2.9204888 \end{Bmatrix} \neq \vec{0}$, we proceed to next iteration.

$\vec{d}_1 = \vec{x}_2 - \vec{x}_1 = \begin{Bmatrix} -1.0275201 \\ 1.0704 \end{Bmatrix} - \begin{Bmatrix} -1.2 \\ 1.0 \end{Bmatrix} = \begin{Bmatrix} 0.1724799 \\ 0.0704 \end{Bmatrix}$

$\vec{g}_1 = \nabla f_2 - \nabla f_1 = \begin{Bmatrix} 1.94668729 \\ 2.9204888 \end{Bmatrix} - \begin{Bmatrix} -215.6 \\ -88.0 \end{Bmatrix}$

$= \begin{Bmatrix} 217.54668729 \\ 90.9204888 \end{Bmatrix}$

$\vec{d}_1 \vec{d}_1^{\,T} = \begin{bmatrix} 0.02974932 & 0.01214258 \\ 0.01214258 & 0.00495616 \end{bmatrix}$

$$\vec{d}_1^{\,T}\,\vec{g}_1 = 43.92323231$$

$$\vec{d}_1\,\vec{g}_1^{\,T} = \begin{bmatrix} 37.52242 99 & 15.68195682 \\ 15.31528639 & 6.40080241 \end{bmatrix}$$

$$\vec{g}_1\,\vec{d}_1^{\,T} = \begin{bmatrix} 37.52242 99 & 15.31528639 \\ 15.68195682 & 6.40080241 \end{bmatrix}$$

$$\vec{g}_1^{\,T}\,[B_1]\,\vec{g}_1 = 55593.094$$

$$\vec{d}_1\,\vec{g}_1^{\,T}\,[B_1] = \vec{d}_1\,\vec{g}_1^{\,T}$$

$$[B_1]\,\vec{g}_1\,\vec{d}_1^{\,T} = \vec{g}_1\,\vec{d}_1^{\,T}$$

$$[B_2] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \left(1 + \frac{55593.094}{43.92323231}\right) * \frac{1}{43.92323231} *$$

$$\begin{bmatrix} 0.02974932 & 0.01214258 \\ 0.01214258 & 0.00495616 \end{bmatrix}$$

$$- \frac{1}{43.92323231} \begin{bmatrix} 37.52242 99 & 15.68195682 \\ 15.31528639 & 6.40080241 \end{bmatrix}$$

$$- \frac{1}{43.92323231} \begin{bmatrix} 37.52242 99 & 15.31528639 \\ 15.68195682 & 6.40080241 \end{bmatrix}$$

$$= \begin{bmatrix} 0.148708 & -0.3558145 \\ -0.3558145 & 0.8513 6222 \end{bmatrix}$$

$$\vec{s}_2 = -[B_2]\,\nabla f_2 = \begin{Bmatrix} 0.7496 6511 \\ -1.79 3736 24 \end{Bmatrix}$$

$$\vec{x} = \vec{x}_2 + \lambda_2\,\vec{s}_2 = \begin{Bmatrix} -1.0275201 + 0.74966511\,\lambda \\ 1.0704 - 1.79373624\,\lambda \end{Bmatrix}$$

$$\lambda_2^* \simeq 0.145$$

$$\vec{x}_3 = \begin{Bmatrix} -0.91881865 \\ 0.81030822 \end{Bmatrix}, \qquad f_3 = 3.7969186$$

6.27 Minimize $f = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$ using Marquardt's method.

Let $\alpha_1 = 10^4$, $c_1 = \frac{1}{4}$, $c_2 = 2$, $\epsilon = 10^{-2}$.

$$\nabla f = \begin{Bmatrix} -400\, x_1 (x_2 - x_1^2) - 2(1 - x_1) \\ 200\,(x_2 - x_1^2) \end{Bmatrix}$$

$$[J] = \begin{bmatrix} -400\, x_2 + 1200\, x_1^2 + 2 & -400\, x_1 \\ -400\, x_1 & 200 \end{bmatrix}$$

$$\vec{x}_1 = \begin{Bmatrix} -1.2 \\ 1.0 \end{Bmatrix}, \quad f_1 = 24.2, \quad \nabla f_1 = \begin{Bmatrix} -215.6 \\ -88.0 \end{Bmatrix}$$

$$[J_1] = \begin{bmatrix} 1330 & 480 \\ 480 & 200 \end{bmatrix}$$

$$|\nabla f_1| = \{(-215.6)^2 + (-88.0)^2\}^{\frac{1}{2}} = 232.8677 > \epsilon$$

$$[A] = [J_1] + \alpha_1 [I] = \begin{bmatrix} 11330 & 480 \\ 480 & 10200 \end{bmatrix}$$

$$[A]^{-1} = \begin{bmatrix} 8.843757 & -0.41617677 \\ -0.41617677 & 9.8235061 \end{bmatrix} \times 10^{-5}$$

$$\vec{x}_2 = \vec{x}_1 - [A]^{-1} \nabla f_1 = \begin{Bmatrix} -1.181299096 \\ 1.007747 4082 \end{Bmatrix}$$

$f_2 = f(\vec{x}_2) = 19.790754$

As $f_2 < f_1$, we set $\alpha_2 = c_1 \alpha_1 = 2500$ and proceed to next iteration.

$$\nabla f_2 = \begin{Bmatrix} -187.56795 \\ -77.544022 \end{Bmatrix}$$

Since $|\nabla f_2| = \{(-187.56795)^2 + (-77.544022)^2\}^{\frac{1}{2}}$

$= 202.9650492 > \epsilon$, we compute

$$[J_2] = \begin{bmatrix} 1273.4622 & 472.51962 \\ 472.51962 & 200.0 \end{bmatrix}$$

$$[A] = [J_2] + \alpha_2 [I] = \begin{bmatrix} 3773.4622 & 472.51962 \\ 472.51962 & 2700.0 \end{bmatrix}$$

$$\vec{x}_3 = \vec{x}_2 - [A]^{-1} \nabla f_2 = \begin{Bmatrix} -1.134155196 \\ 1.032442248 \end{Bmatrix}$$

since

$$[A]^{-1} = 10^{-4} \begin{bmatrix} 2.7094633 & -0.47417576 \\ -0.47417576 & 3.786688 \end{bmatrix}.$$

$f_3 = f(\vec{x}_3) = 10.999399$

Since $f_3 < f_2$, we set $\alpha_3 = c_1 \alpha_2 = 625$ and proceed to the next iteration.

6.28

$f = x_1^2 + 4x_2^2 = \frac{1}{2} \vec{x}^T [A] \vec{x}$ where $[A] = \begin{bmatrix} 2 & 0 \\ 0 & 8 \end{bmatrix}$

$\nabla f = \begin{Bmatrix} 2x_1 \\ 8x_2 \end{Bmatrix}$

Let $\vec{x}_1 = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$, $f_1 = 5$, $\nabla f_1 = \begin{Bmatrix} 2 \\ 8 \end{Bmatrix}$

$\vec{s}_1 = -\nabla f_1 = \begin{Bmatrix} -2 \\ -8 \end{Bmatrix}$, $\nabla f_1 = \{(2)^2 + (8)^2\}^{\frac{1}{2}} = 8.246211$

$\vec{x} = \vec{x}_1 + \lambda_1 \vec{s}_1 = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} + \lambda_1 \begin{Bmatrix} -2 \\ -8 \end{Bmatrix} = \begin{Bmatrix} 1-2\lambda_1 \\ 1-8\lambda_1 \end{Bmatrix}$

$f(\lambda_1) = (1-2\lambda_1)^2 + 4(1-8\lambda_1)^2 = 260\lambda_1^2 - 68\lambda_1 + 5$

$\frac{df}{d\lambda_1} = 520\lambda_1 - 68 = 0 \Rightarrow \lambda_1^* = 0.130769$

$\vec{x}_2 = \vec{x}_1 + \lambda_1^* \vec{s}_1 = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} + 0.130769 \begin{Bmatrix} -2 \\ -8 \end{Bmatrix} = \begin{Bmatrix} 0.738462 \\ -0.046152 \end{Bmatrix}$

$f_2 = f(\vec{x}_2) = 0.553846$

$\nabla f_2 = \begin{Bmatrix} 1.476924 \\ -0.369216 \end{Bmatrix}$, $|\nabla f_2|^2 = 2.317624$

$\vec{s}_2 = -\nabla f_2 + \frac{|\nabla f_2|^2}{|\nabla f_1^2|} \vec{s}_1$

$$= \begin{Bmatrix} -1.476924 \\ 0.369216 \end{Bmatrix} + 2.317624 \begin{Bmatrix} -2 \\ -8 \end{Bmatrix} = \begin{Bmatrix} -1.545089412 \\ 0.096554352 \end{Bmatrix}$$

$$[A]\,\vec{s}_1 = \begin{bmatrix} 2 & 0 \\ 0 & 8 \end{bmatrix} \begin{Bmatrix} -2 \\ -8 \end{Bmatrix} = \begin{Bmatrix} -4 \\ -64 \end{Bmatrix}$$

$$\vec{s}_2^{\,T}\,[A]\,\vec{s}_1 = (-1.545089412 \quad 0.096554352) \begin{Bmatrix} -4 \\ -64 \end{Bmatrix} \simeq 0$$

$\therefore$ $\vec{s}_1$ and $\vec{s}_2$ are $[A]$- conjugate.

This property can be verified even after carrying out more iterations.

6.29 (a) $\vec{x}_0 = \begin{Bmatrix} 4 \\ -3 \end{Bmatrix}$, size $= a = 4$, $n = 2$

$$p = \frac{4}{2\sqrt{2}}(\sqrt{3} + 2 - 1) = 3.8637 \; ; \quad q = \frac{4}{2\sqrt{2}}(\sqrt{3} - 1) = 1.0352$$

$$\vec{x}_1 = \begin{pmatrix} 4 \\ -3 \end{pmatrix} + 3.8637 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 1.0352 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{Bmatrix} 7.8637 \\ -1.9648 \end{Bmatrix}$$

$$\vec{x}_2 = \begin{pmatrix} 4 \\ -3 \end{pmatrix} + 3.8637 \begin{pmatrix} 0 \\ 1 \end{pmatrix} + 1.0352 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 5.0352 \\ 0.8637 \end{pmatrix}$$

(b) $\vec{x}_0 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$$\vec{x}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 3.8637 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 1.0352 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 4.8637 \\ 2.0352 \end{pmatrix}$$

$$\vec{x}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 3.8637 \begin{pmatrix} 0 \\ 1 \end{pmatrix} + 1.0352 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2.0352 \\ 4.8637 \end{pmatrix}$$

(c) $\vec{x}_0 = \begin{pmatrix} -1 \\ -2 \end{pmatrix}$

$$\vec{x}_1 = \begin{pmatrix} -1 \\ -2 \end{pmatrix} + 3.8637 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 1.0352 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2.8637 \\ -0.9648 \end{pmatrix}$$

$$\vec{x}_2 = \begin{pmatrix} -1 \\ -2 \end{pmatrix} + 3.8637 \begin{pmatrix} 0 \\ 1 \end{pmatrix} + 1.0352 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.0352 \\ 1.8637 \end{pmatrix}$$

6.30 $\vec{x}_1 = \begin{Bmatrix} 2 \\ -1 \\ -8 \end{Bmatrix}$

$\vec{x}_2 = (2.2598 \; , -0.85, -8)^T$

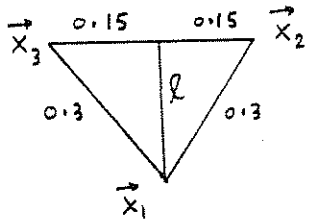
$\vec{x}_3 = (2.2598, \; -1.15 \; , -8)^T$

$\vec{x}_4 = (2.1732 \; , -1 \; , -8.2450)^T$

$\because \vec{x}_0 = \sum_{i=1}^{3} \vec{x}_i \Big/ 3 = \begin{Bmatrix} 2.1732 \\ -1 \\ -8 \end{Bmatrix}$

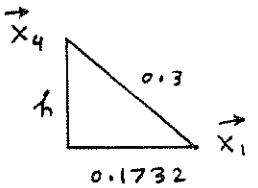
and

$\vec{x}_4 = \vec{x}_0 + \begin{Bmatrix} 0 \\ 0 \\ h \end{Bmatrix}$



$l = \sqrt{0.3^2 - 0.15^2}$

$= 0.2598$



$h = 0.2450$

Distances between $(\vec{x}_1, \vec{x}_2)$, $(\vec{x}_1, \vec{x}_3)$, $(\vec{x}_1, \vec{x}_4)$, $(\vec{x}_2, \vec{x}_3)$, $(\vec{x}_2, \vec{x}_4)$, $(\vec{x}_3, \vec{x}_4)$ can be verified to be equal to 0.3

$n = 3, \quad \alpha = 3$

6.31 $p = \frac{3}{3\sqrt{2}}\left(\sqrt{4} + 3 - 1\right) = 2.8284 \; ; \quad q = \frac{3}{3\sqrt{2}}\left(\sqrt{4} - 1\right) = 0.7071$

(a) $\vec{x}_0 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$, $\vec{x}_1 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + 2.8284 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + 0.7071 \left(\begin{Bmatrix} 0 \\ 1 \\ 0 \end{Bmatrix} + \begin{Bmatrix} 0 \\ 0 \\ 1 \end{Bmatrix} \right)$

$$= \begin{Bmatrix} 2.8284 \\ 0.7071 \\ 0.7071 \end{Bmatrix}$$

$$\vec{x}_2 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + 2.8284 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + 0.7071 \left(\begin{Bmatrix} 1 \\ 0 \\ 0 \end{Bmatrix} + \begin{Bmatrix} 0 \\ 0 \\ 1 \end{Bmatrix} \right) = \begin{pmatrix} 0.7071 \\ 2.8284 \\ 0.7071 \end{pmatrix}$$

$$\vec{x}_3 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + 2.8284 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + 0.7071 \left(\begin{Bmatrix} 1 \\ 0 \\ 0 \end{Bmatrix} + \begin{Bmatrix} 0 \\ 1 \\ 0 \end{Bmatrix} \right) = \begin{pmatrix} 0.7071 \\ 0.7071 \\ 2.8284 \end{pmatrix}$$

(b) $\vec{x}_0 = \begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix}$

$$\vec{x}_1 = \begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix} + 2.8284 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + 0.7071 \left(\begin{Bmatrix} 0 \\ 1 \\ 0 \end{Bmatrix} + \begin{Bmatrix} 0 \\ 0 \\ 1 \end{Bmatrix} \right) = \begin{pmatrix} 6.8284 \\ 3.7071 \\ 2.7071 \end{pmatrix}$$

$$\vec{x}_2 = \begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix} + 2.8284 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + 0.7071 \left(\begin{Bmatrix} 1 \\ 0 \\ 0 \end{Bmatrix} + \begin{Bmatrix} 0 \\ 0 \\ 1 \end{Bmatrix} \right) = \begin{pmatrix} 4.7071 \\ 5.8284 \\ 2.7071 \end{pmatrix}$$

$$\vec{x}_3 = \begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix} + 2.8284 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + 0.7071 \left(\begin{Bmatrix} 1 \\ 0 \\ 0 \end{Bmatrix} + \begin{Bmatrix} 0 \\ 1 \\ 0 \end{Bmatrix} \right) = \begin{pmatrix} 4.7071 \\ 3.7071 \\ 4.8284 \end{pmatrix}$$

(c) $\vec{x}_0 = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$

$$\vec{x}_1 = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} + 2.8284 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + 0.7071 \left(\begin{Bmatrix} 0 \\ 1 \\ 0 \end{Bmatrix} + \begin{Bmatrix} 0 \\ 0 \\ 1 \end{Bmatrix} \right) = \begin{pmatrix} 3.8284 \\ -1.2929 \\ 3.7071 \end{pmatrix}$$

$$\vec{x}_2 = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} + 2.8284 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + 0.7071 \left(\begin{Bmatrix} 1 \\ 0 \\ 0 \end{Bmatrix} + \begin{Bmatrix} 0 \\ 0 \\ 1 \end{Bmatrix} \right) = \begin{pmatrix} 1.7071 \\ 0.8284 \\ 3.7071 \end{pmatrix}$$

$$\vec{x}_3 = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} + 2.8284 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + 0.7071 \left(\begin{Bmatrix} 1 \\ 0 \\ 0 \end{Bmatrix} + \begin{Bmatrix} 0 \\ 1 \\ 0 \end{Bmatrix} \right) = \begin{pmatrix} 1.7071 \\ -1.2929 \\ 5.8284 \end{pmatrix}$$

6.32

$$\vec{S}_1^T [A] \vec{S}_2 = \begin{pmatrix} 2 & -3 & 6 \end{pmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 6 \\ 3 & 6 & 9 \end{bmatrix} \begin{Bmatrix} s_1 \\ s_2 \\ s_3 \end{Bmatrix} = 0$$

$$= \{ 14 \quad 25 \quad 42 \} \begin{Bmatrix} s_1 \\ s_2 \\ s_3 \end{Bmatrix} = 14 s_1 + 25 s_2 + 42 s_3 = 0$$

Let s_1 and s_2 be arbitrary. Then

$$\vec{S}_2 = \begin{Bmatrix} s_1 \\ s_2 \\ s_3 \end{Bmatrix} = \begin{Bmatrix} s_1 \\ s_2 \\ \frac{1}{42}(-14 s_1 - 25 s_2) \end{Bmatrix}$$

6.33

$$f(x_1, x_2) = 100 (x_2 - x_1^2)^2 + (1 - x_1)^2$$

(a)
$$\nabla f = \begin{Bmatrix} \partial f / \partial x_1 \\ \partial f / \partial x_2 \end{Bmatrix} = \begin{Bmatrix} -400 x_1 (x_2 - x_1^2) - 2(1 - x_1) \\ 200 (x_2 - x_1^2) \end{Bmatrix}$$

At $\vec{x} = \begin{Bmatrix} 0.5 \\ 0.5 \end{Bmatrix}$, $\nabla f = \begin{Bmatrix} -51 \\ 50 \end{Bmatrix}$

(b) central differences:

$$\nabla f = \begin{Bmatrix} \partial f / \partial x_1 \\ \partial f / \partial x_2 \end{Bmatrix} = \begin{Bmatrix} \dfrac{f(\vec{x} + \begin{Bmatrix} \Delta x_1 \\ 0 \end{Bmatrix}) - f(\vec{x} - \begin{Bmatrix} \Delta x_1 \\ 0 \end{Bmatrix})}{2 \Delta x_1} \\ \dfrac{f(\vec{x} + \begin{Bmatrix} 0 \\ \Delta x_2 \end{Bmatrix}) - f(\vec{x} - \begin{Bmatrix} 0 \\ \Delta x_2 \end{Bmatrix})}{2 \Delta x_2} \end{Bmatrix}$$

$$= \begin{Bmatrix} \dfrac{6.24630 - 6.50377}{2(0.005)} \\ \dfrac{6.75250 - 6.25250}{2(0.005)} \end{Bmatrix} = \begin{Bmatrix} -25.747442 \\ 49.999954 \end{Bmatrix}$$

(c) Forward differences:

$$\nabla f = \begin{Bmatrix} \dfrac{\partial f}{\partial x_1} \\ \dfrac{\partial f}{\partial x_2} \end{Bmatrix} = \begin{Bmatrix} \dfrac{f(\vec{x} + \begin{Bmatrix} \Delta x_1 \\ 0 \end{Bmatrix}) - f(\vec{x})}{\Delta x_1} \\ \dfrac{f(\vec{x} + \begin{Bmatrix} 0 \\ \Delta x_2 \end{Bmatrix}) - f(\vec{x})}{\Delta x_2} \end{Bmatrix}$$

$$= \begin{Bmatrix} \dfrac{6.24630 - 6.50000}{0.005} \\ \dfrac{6.75250 - 6.50000}{0.005} \end{Bmatrix} = \begin{Bmatrix} -50.739956 \\ 50.499916 \end{Bmatrix}$$

(d) Backward differences:

$$\nabla f = \begin{Bmatrix} \dfrac{\partial f}{\partial x_1} \\ \dfrac{\partial f}{\partial x_2} \end{Bmatrix} = \begin{bmatrix} \dfrac{f(\vec{x}) - f\left(\vec{x} - \begin{Bmatrix} \Delta x_1 \\ 0 \end{Bmatrix}\right)}{\Delta x_1} \\ \dfrac{f(\vec{x}) - f\left(\vec{x} - \begin{Bmatrix} 0 \\ \Delta x_2 \end{Bmatrix}\right)}{\Delta x_2} \end{bmatrix}$$

$$= \begin{Bmatrix} \dfrac{6.50000 - 6.50377}{0.005} \\ \dfrac{6.50000 - 6.25250}{0.005} \end{Bmatrix} = \begin{Bmatrix} -0.75492859 \\ 49.499989 \end{Bmatrix}$$

6.34

step size $(\Delta x_1, \Delta x_2)$	Central difference formula	Forward difference formula	Backward difference formula
0.1	$\frac{\partial f}{\partial x_1}$: −20.0	−43.78	3.80
	$\frac{\partial f}{\partial x_2}$: 50.0	60.00	40.00
0.01	$\frac{\partial f}{\partial x_1}$: −25.49	−50.47	−0.51
	$\frac{\partial f}{\partial x_2}$: 49.99	50.99	48.99
0.005	$\frac{\partial f}{\partial x_1}$: −25.75	−50.74	−0.75
	$\frac{\partial f}{\partial x_2}$: 50.00	50.50	49.50
0.0005	$\frac{\partial f}{\partial x_1}$: −25.97	−50.98	−0.97
	$\frac{\partial f}{\partial x_2}$: 50.00	50.05	49.95
0.00001	$\frac{\partial f}{\partial x_1}$: −26.03	−51.07	−1.00
	$\frac{\partial f}{\partial x_2}$: 50.07	50.07	50.07
0.000 0001	$\frac{\partial f}{\partial x_1}$: −28.61	−57.22	0.00
	$\frac{\partial f}{\partial x_2}$: 50.06	57.22	42.91

Exact (analytical) values: $\frac{\partial f}{\partial x_1} = -51.00$, $\frac{\partial f}{\partial x_2} = 50.00$

Above results indicate that only the forward difference formula gives accurate values for this function at point (0.5, 0.5).

6.35

The gradient g and the Hessian matrix J of f are given by

$$g=\begin{Bmatrix} \partial f/\partial x_1 \\ \partial f/\partial x_2 \end{Bmatrix}=\frac{2}{(x_1^2+x_2^2+2)^2}\begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix}$$

and

$$J=\begin{bmatrix} \partial^2 f/\partial x_1^2 & \partial^2 f/\partial x_1\,\partial x_2 \\ \partial^2 f/\partial x_1\,\partial x_2 & \partial^2 f/\partial x_2^2 \end{bmatrix}$$

$$=\frac{2}{(x_1^2+x_2^2+2)^3}\begin{bmatrix} (-3x_1^2+x_2^2+2) & (-4x_1x_2) \\ (-4x_1x_2) & (-3x_2^2+x_1^2+2) \end{bmatrix}$$

At $X_1=\begin{Bmatrix} 4 \\ 0 \end{Bmatrix}$,

$$g=\begin{Bmatrix} 0.0247 \\ 0 \end{Bmatrix},\quad [J]=\begin{bmatrix} -0.01580 & 0 \\ 0 & 0.00617 \end{bmatrix},$$

and

$$J^{-1}=\frac{1}{(-0.0000975)}\begin{bmatrix} 0.00617 & 0 \\ 0 & -0.01580 \end{bmatrix}=\begin{bmatrix} -63.4 & 0 \\ 0 & 162.0 \end{bmatrix}$$

Hence Eq. (6·98) gives

$$X_2=X_1-J_1^{-1}g_1=\begin{Bmatrix} 5.57 \\ 0 \end{Bmatrix}.$$

If we compare the values of f at X_1 and X_2, we find that $f_1=-0.0556$ and $f_2=-0.0303$. Thus f_2 is greater than f_1 and hence the method is diverging (true minimum point is $X^*=\begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$ with $f^*=-0.5$). However, if we use Eq. (6·99) instead of Eq. (6·98), the method can be made to converge to the minimum point.

6.36 $f = 2(x_2 - x_1^2)^2 + (1 - x_1)^2$

First simplex: $\vec{x}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $\vec{x}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\vec{x}_3 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$f_1 = 1$, $f_2 = 2$, $f_3 = 3$

$f_h = f_3$ and $\vec{x}_h = \vec{x}_3 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$\vec{x}_o = \frac{1}{2}(\vec{x}_1 + \vec{x}_2) = \begin{pmatrix} 0.5 \\ 0 \end{pmatrix}$, $f(\vec{x}_o) = 0.375$

$\vec{x}_r = 2\vec{x}_o - \vec{x}_h = \begin{pmatrix} 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, $f(\vec{x}_r) = 8$

Since $f(\vec{x}_r) > f(\vec{x}_i)$ for all i, use contraction with $\beta = 0.1$:

$\vec{x}_c = 0.1\,\vec{x}_h + (1 - 0.1)\,\vec{x}_o = \begin{pmatrix} 0.45 \\ 0.1 \end{pmatrix}$, $f(\vec{x}_c) = 0.3235$

Second simplex: $\vec{x}_1 = \vec{x}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $\vec{x}_2 = \vec{x}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\vec{x}_3 = \vec{x}_c = \begin{pmatrix} 0.45 \\ 0.1 \end{pmatrix}$

$f_1 = 1$, $f_2 = 2$, $f_3 = 0.3235$

$f_h = f_2 = 2$, $\vec{x}_h = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$\vec{x}_o = \frac{1}{2}(\vec{x}_1 + \vec{x}_3) = \frac{1}{2}\left\{\begin{pmatrix} 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right\} = \begin{pmatrix} 0 \\ 0.5 \end{pmatrix}$

$\vec{x}_r = 2\vec{x}_o - \vec{x}_h = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$, $f(\vec{x}_r) = 4$

Since $f(\vec{x}_r) > f(\vec{x}_i)$ for all i, use contraction with $\beta = 0.1$:

$\vec{x}_c = 0.1\,\vec{x}_h + (1 - 0.1)\,\vec{x}_o = \begin{pmatrix} 0.1 \\ 0.45 \end{pmatrix}$

$f(\vec{x}_c) = 1.1972 > f_1$, but $< f_h$.

Third simplex: $\vec{x}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $\vec{x}_2 = \vec{x}_c = \begin{pmatrix} 0.1 \\ 0.45 \end{pmatrix}$, $\vec{x}_3 = \begin{pmatrix} 0.45 \\ 0.1 \end{pmatrix}$

$f_1 = 0$, $f_2 = 1.1972$, $f_3 = 0.3235$

$f_h = f_2 = 1.1972$, $\vec{x}_h = \begin{pmatrix} 0.1 \\ 0.45 \end{pmatrix}$

$\vec{x}_o = \frac{1}{2}(\vec{x}_1 + \vec{x}_3) = \begin{pmatrix} 0.225 \\ 0.05 \end{pmatrix}$, $f(\vec{x}_o) = 0.600625$

$\vec{x}_r = 2\vec{x}_o - \vec{x}_h = \begin{pmatrix} 0.35 \\ -0.35 \end{pmatrix}$, $f(\vec{x}_r) = 0.869012 > f_1$ and f_3

Use contraction with $\beta = 0.1$:

$\vec{x}_c = 0.1\,\vec{x}_h + (1-0.1)\,\vec{x}_o = \begin{pmatrix} 0.2125 \\ 0.09 \end{pmatrix}$

$f(\vec{x}_c) = 0.624178$

Fourth simplex: $\vec{x}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $\vec{x}_2 = \begin{pmatrix} 0.45 \\ 0.1 \end{pmatrix}$, $\vec{x}_3 = \vec{x}_c = \begin{pmatrix} 0.2125 \\ 0.09 \end{pmatrix}$

$f_1 = 0$, $f_2 = 0.3235$, $f_3 = 0.62478$

$f_h = f_3 = 0.62478$, $\vec{x}_h = \begin{pmatrix} 0.2125 \\ 0.09 \end{pmatrix}$

$\vec{x}_o = \frac{1}{2}\left\{\begin{pmatrix} 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0.45 \\ 0.1 \end{pmatrix}\right\} = \begin{pmatrix} 0.225 \\ 0.05 \end{pmatrix}$

$\vec{x}_r = 2\vec{x}_o - \vec{x}_h = \begin{pmatrix} 0.2375 \\ 0.01 \end{pmatrix}$, $f(\vec{x}_r) = 0.766807 > f_1$ and f_2

Use contraction with $\beta = 0.1$:

$\vec{x}_c = 0.1\,\vec{x}_h + 0.9\,\vec{x}_o = 0.1\begin{pmatrix} 0.2125 \\ 0.09 \end{pmatrix} + 0.9\begin{pmatrix} 0.225 \\ 0.05 \end{pmatrix} = \begin{pmatrix} 0.22375 \\ 0.054 \end{pmatrix}$

$f(\vec{x}_c) = 0.602595$

Fifth simplex: $\vec{x}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $\vec{x}_2 = \begin{pmatrix} 0.45 \\ 0.1 \end{pmatrix}$, $\vec{x}_3 = \vec{x}_c = \begin{pmatrix} 0.22375 \\ 0.054 \end{pmatrix}$

$f_1 = 0$, $f_2 = 0.3235$, $f_3 = 0.602595$

Current (optimal) solution:

$$\vec{x}_{opt} \simeq \frac{1}{3}\left(\vec{x}_1 + \vec{x}_2 + \vec{x}_3\right) = \begin{pmatrix} 0.22458 \\ 0.05133 \end{pmatrix}$$

$$f_{opt} \simeq f(\vec{x}_{opt}) = 0.601278$$

6.37 Minimize $f = (x_1 + 2x_2 - 7)^2 + (2x_1 + x_2 - 5)^2$

Assume $\alpha = 1.0$, $\beta = 0.5$, $\gamma = 2.0$ and $\epsilon = 0.2$

Iteration 1:

step 1: $\vec{x}_1 = \begin{Bmatrix} -2 \\ -2 \end{Bmatrix}$, $f_1 = (-2-4-7)^2 + (-4-2-5)^2 = 290$

$\vec{x}_2 = \begin{Bmatrix} -3 \\ 0 \end{Bmatrix}$, $f_2 = (-3-0-7)^2 + (-6+0-5)^2 = 221$

$\vec{x}_3 = \begin{Bmatrix} -1 \\ -1 \end{Bmatrix}$, $f_3 = (-1-2-7)^2 + (-2-1-5)^2 = 164$

$\vec{x}_h = \vec{x}_1 = \begin{Bmatrix} -2 \\ -2 \end{Bmatrix}$, $f_h = 290$

$\vec{x}_l = \vec{x}_3 = \begin{Bmatrix} -1 \\ -1 \end{Bmatrix}$, $f_l = 164$

step 2: centroid $= \vec{x}_0 = \frac{1}{2}(\vec{x}_2 + \vec{x}_3) = \begin{Bmatrix} -2 \\ -0.5 \end{Bmatrix}$, $f_0 = 190.25$

step 3: Reflection point, $\vec{x}_r = 2\vec{x}_0 - \vec{x}_h = \begin{Bmatrix} -4 \\ -1 \end{Bmatrix} - \begin{Bmatrix} -2 \\ -2 \end{Bmatrix} = \begin{Bmatrix} -2 \\ 1 \end{Bmatrix}$

$f_r = f(\vec{x}_r) = 113$

step 4: Since $f_r < f_l$, we find $\vec{x}_e$ by expansion as

$\vec{x}_e = 2\vec{x}_r - \vec{x}_0 = \begin{Bmatrix} -4 \\ 2 \end{Bmatrix} - \begin{Bmatrix} -2 \\ -0.5 \end{Bmatrix} = \begin{Bmatrix} -2 \\ 2.5 \end{Bmatrix}$

$f_e = f(\vec{x}_e) = 58.25$

step 5: Since $f_e < f_r$, we replace $\vec{x}_h$ by $\vec{x}_e$ and obtain the new simplex as:

$\vec{x}_1 = \begin{Bmatrix} -2 \\ 2.5 \end{Bmatrix}$, $\vec{x}_2 = \begin{Bmatrix} -3 \\ 0 \end{Bmatrix}$, $\vec{x}_3 = \begin{Bmatrix} -1 \\ -1 \end{Bmatrix}$

with $f_1 = 58.25$, $f_2 = 221.0$, $f_3 = 164.0$

step 6: Test for convergence:

$$Q = \left[\frac{(58.25 - 190.25)^2 + (221.0 - 190.25)^2 + (164.0 - 190.25)^2}{3} \right]^{\frac{1}{2}}$$

$= 79.7049 > \epsilon$

Hence we go to next iteration.

Iteration 2:

step 1: $\vec{x}_h = \vec{x}_2 = \begin{Bmatrix} -3 \\ 0 \end{Bmatrix}$, $\vec{x}_\ell = \vec{x}_1 = \begin{Bmatrix} -2 \\ 2.5 \end{Bmatrix}$

$f_h = 221.0$, $f_\ell = 58.25$

step 2: centroid: $\vec{x}_0 = \frac{1}{2}(\vec{x}_1 + \vec{x}_3) = \begin{Bmatrix} -1.5 \\ 0.75 \end{Bmatrix}$

$f_0 = f(\vec{x}_0) = 101.5625$

step 3: $\vec{x}_r = 2\vec{x}_0 - \vec{x}_h = \begin{Bmatrix} -3 \\ 1.5 \end{Bmatrix} - \begin{Bmatrix} -3 \\ 0 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 1.5 \end{Bmatrix}$

$f_r = f(\vec{x}_r) = 28.25$

step 4: Since $f_r < f_\ell$, we find $\vec{x}_e$ as

$\vec{x}_e = 2\vec{x}_r - \vec{x}_0 = \begin{Bmatrix} 0 \\ 3.0 \end{Bmatrix} - \begin{Bmatrix} -1.5 \\ 0.75 \end{Bmatrix} = \begin{Bmatrix} 1.5 \\ 2.25 \end{Bmatrix}$

$f_e = f(\vec{x}_e) = 1.0625$

step 5: As $f_e < f_r$, we replace $\vec{x}_h$ by $\vec{x}_e$ and obtain the new simplex as

$\vec{x}_1 = \begin{Bmatrix} -2 \\ 2.5 \end{Bmatrix}$, $\vec{x}_2 = \begin{Bmatrix} 1.5 \\ 2.25 \end{Bmatrix}$, $\vec{x}_3 = \begin{Bmatrix} -1 \\ -1 \end{Bmatrix}$

with $f_1 = 58.25$, $f_2 = 1.0625$, $f_3 = 164.0$

step 6: Test for convergence:

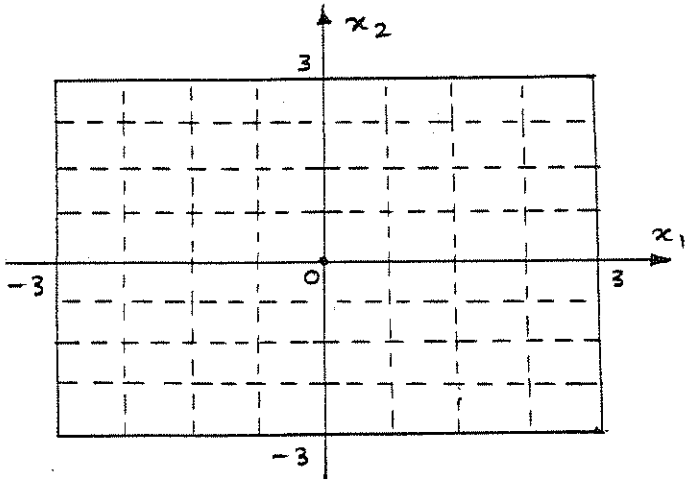
$$Q^2 = \frac{(58.25 - 101.5625)^2 + (1.0625 - 101.5625)^2 + (164.0 - 101.5625)^2}{3}$$

$$= 5291.5547$$

$$Q = 72.7431$$

since $Q > \epsilon$, we need to proceed to the next iteration.

6.38 Grid is set up in increments of 0.1 along x_1 and x_2 directions. The function value is evaluated at each of the grid points and the minimum of $f = 100\,(x_2 - x_1^2)^2 + (1 - x_1)^2$ is found as



$$f_{min} = 0.29064751 * 10^{-10} \text{ at } \vec{X}_{min} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \vec{X}^*.$$

The computer program and a partial output is given below.

```
      fmin=+100000.0
      print 31
      print 32
 32   format (/)
      x1=-3.1
      do 10 i=1,61
      x1=x1+0.1
      x2=-3.1
      do 20 j=1,61
      x2=x2+0.1
      f=100.0*(x2-x1**2)**2+(1.0-x1)**2
      if (f .lt. fmin) fmin=f
      if (i .eq. 11) go to 80
      if (i .eq. 21) go to 80
      if (i .eq. 31) go to 80
      if (i .eq. 41) go to 80
      if (i .eq. 51) go to 80
      if (i. eq. 61) go to 80
      go to 60
 80   continue
      if (j .eq. 11) go to 30
      if (j .eq. 21) go to 30
      if (j .eq. 31) go to 30
      if (j .eq. 41) go to 30
      if (j .eq. 51) go to 30
      if (j .eq. 61) go to 30
      go to 60
 31   format (3x,'i',4x,'j',9x,'x1',10x,'x2',12x,'f',8x)
 30   print 40,i,j,x1,x2,f
 40   format (2i5,2x,2f12.4,2x,e15.8)
 60   continue
 20   continue
 10          continue
```

```
      print 50, fmin
50    format (//,2x,' minimum value of f =',e15.8)
      stop
      end
```

i	j	x1	x2	f
11	11	-2.0000	-2.0000	0.36090056E+04
11	21	-2.0000	-1.0000	0.25090049E+04
11	31	-2.0000	0.0000	0.16090034E+04
11	41	-2.0000	1.0000	0.90900256E+03
11	51	-2.0000	2.0000	0.40900171E+03
11	61	-2.0000	3.0000	0.10900101E+03
21	11	-1.0000	-2.0000	0.90400140E+03
21	21	-1.0000	-1.0000	0.40400085E+03
21	31	-1.0000	0.0000	0.10400040E+03
21	41	-1.0000	1.0000	0.40000029E+01
21	51	-1.0000	2.0000	0.10399964E+03
21	61	-1.0000	3.0000	0.40399896E+03
31	11	0.0000	-2.0000	0.40100037E+03
31	21	0.0000	-1.0000	0.10100014E+03
31	31	0.0000	0.0000	0.10000012E+01
31	41	0.0000	1.0000	0.10099989E+03
31	51	0.0000	2.0000	0.40099985E+03
31	61	0.0000	3.0000	0.90099933E+03
41	11	1.0000	-2.0000	0.90000000E+03
41	21	1.0000	-1.0000	0.39999985E+03
41	31	1.0000	0.0000	0.99999908E+02
41	41	1.0000	1.0000	0.29064751E-10
41	51	1.0000	2.0000	0.10000014E+03
41	61	1.0000	3.0000	0.39999994E+03
51	11	2.0000	-2.0000	0.36009995E+04
51	21	2.0000	-1.0000	0.25009990E+04
51	31	2.0000	0.0000	0.16009994E+04
51	41	2.0000	1.0000	0.90099945E+03
51	51	2.0000	2.0000	0.40099957E+03
51	61	2.0000	3.0000	0.10099995E+03
61	11	3.0000	-2.0000	0.12103988E+05
61	21	3.0000	-1.0000	0.10003988E+05
61	31	3.0000	0.0000	0.81039902E+04
61	41	3.0000	1.0000	0.64039902E+04
61	51	3.0000	2.0000	0.49039912E+04
61	61	3.0000	3.0000	0.36039932E+04

```
 minimum value of f = 0.29064751E-10
```

6.39

Minimize $f = 6x_1^2 + 2x_2^2 - 6x_1x_2 - x_1 - 2x_2$

$$\nabla f = \begin{Bmatrix} 12x_1 - 6x_2 - 1 \\ 4x_2 - 6x_1 - 2 \end{Bmatrix}, \quad \vec{x}_1 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}, \quad f(\vec{x}_1) = 0$$

$$\nabla f(\vec{x}_1) = \begin{Bmatrix} -1 \\ -2 \end{Bmatrix}; \quad \vec{s}_1 = \begin{Bmatrix} 1 \\ 2 \end{Bmatrix}, \quad \vec{s}_2 = \begin{Bmatrix} 1 \\ 0 \end{Bmatrix}$$

First, we minimize along $\vec{s}_1$ and then along $\vec{s}_2$:

(1) $\vec{x} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} + \lambda_1 \begin{Bmatrix} 1 \\ 2 \end{Bmatrix} = \begin{Bmatrix} \lambda_1 \\ 2\lambda_1 \end{Bmatrix} = \vec{x}_1 + \lambda_1 \vec{s}_1$

$$f(\vec{x}) = f(\lambda) = 6\lambda_1^2 + 8\lambda_1^2 - 12\lambda_1^2 - \lambda_1 - 4\lambda_1 = 2\lambda_1^2 - 5\lambda_1$$

$$\frac{df}{d\lambda_1} = 4\lambda_1 - 5 = 0 \Rightarrow \lambda_1^* = \frac{5}{4}$$

$$\therefore \vec{x}_2 = \begin{Bmatrix} \lambda_1^* \\ 2\lambda_1^* \end{Bmatrix} = \begin{Bmatrix} 1.25 \\ 2.5 \end{Bmatrix}, \quad f(\vec{x}_2) = -3.125$$

(2) $\vec{x} = \vec{x}_2 + \lambda_2 \vec{s}_2 = \begin{Bmatrix} 1.25 + \lambda_2 \\ 2.5 \end{Bmatrix}$

$$f(\vec{x}) = f(\lambda_2) = 6(1.25 + \lambda_2)^2 + 2(2.5^2) - 6(1.25 + \lambda_2)(2.5) - (1.25 + \lambda_2) - 2(2.5) = 6\lambda_2^2 - \lambda_2 - 3.125$$

$$\frac{df}{d\lambda_2} = 12\lambda_2 - 1 = 0 \Rightarrow \lambda_2^* = \frac{1}{12} = 0.08333$$

$$\vec{x}_3 = \vec{x}_2 + \lambda_2^* \vec{s}_2 = \begin{Bmatrix} 1.3333 \\ 2.5 \end{Bmatrix}, \quad f(\vec{x}_3) = -3.1667$$

$\vec{x}_3$ can be verified to be the optimum solution of the problem.

$$\therefore \vec{x}^* = \vec{x}_3 = \begin{Bmatrix} 1.3333 \\ 2.5 \end{Bmatrix}, \quad f^* = -3.1667$$

Next, we minimize first along $\vec{s}_2 = \begin{Bmatrix} 1 \\ 0 \end{Bmatrix}$ and then along $\vec{s}_1 = \begin{Bmatrix} 1 \\ 2 \end{Bmatrix}$:

(1) $\vec{x} = \vec{x}_1 + \lambda_1 \vec{s}_2 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} + \lambda_1 \begin{Bmatrix} 1 \\ 0 \end{Bmatrix} = \begin{Bmatrix} \lambda_1 \\ 0 \end{Bmatrix}$

$$f(\vec{x}) = f(\lambda_1) = 6\lambda_1^2 - \lambda_1$$

$$\frac{df}{d\lambda_1} = 12\lambda_1 - 1 = 0 \Rightarrow \lambda_1^* = \frac{1}{12} = 0.08333$$

$$\vec{X}_2 = \vec{X}_1 + \lambda_1^* \vec{S}_2 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} + 0.08333 \begin{Bmatrix} 1 \\ 0 \end{Bmatrix} = \begin{Bmatrix} 0.08333 \\ 0.0 \end{Bmatrix}$$

$$f(\vec{X}_2) = -0.04167$$

(2) $$\vec{X} = \vec{X}_2 + \lambda_2 \vec{S}_1 = \begin{Bmatrix} 0.08333 \\ 0.0 \end{Bmatrix} + \lambda_2 \begin{Bmatrix} 1 \\ 2 \end{Bmatrix}$$

$$= \begin{Bmatrix} 0.08333 + \lambda_2 \\ 2\lambda_2 \end{Bmatrix}$$

$$f(\vec{X}) = f(\lambda_2) = 6(0.08333 + \lambda_2)^2 + 2(2\lambda_2)^2$$
$$- 6(0.08333 + \lambda_2)(2\lambda_2) - (0.08333 + \lambda_2) - 2(2\lambda_2)$$
$$= 2\lambda_2^2 - 5\lambda_2 - 0.0416667$$

$$\frac{df}{d\lambda_2} = 4\lambda_2 - 5 = 0 \Rightarrow \lambda_2^* = 1.25$$

$$\vec{X}_3 = \vec{X}_2 + \lambda_2^* \vec{S}_1 = \begin{Bmatrix} 1.3333 \\ 2.5 \end{Bmatrix}, \quad f(\vec{X}_3) = -3.1667$$

$\vec{X}_3$ can be verified to be the optimum point, $\vec{X}^*$.

This shows that the property of quadratic convergence is independent of the order in which the conjugate directions are used in the minimization process.

6.40 Consider the third search direction as a linear combination of $\vec{s}_1$, $\vec{s}_2$ and $-\nabla f_3$ is given by Eq. (6.88):

$$\vec{s}_3 = -\nabla f_3 + \beta_3 \vec{s}_2 + \delta_3 \vec{s}_1 \tag{1}$$

where the values of β_3 and δ_3 can be found by making $\vec{s}_3$ conjugate to $\vec{s}_1$ and $\vec{s}_2$. First we consider

$$\vec{s}_1^T [A] \vec{s}_3 = -\vec{s}_1^T [A] \nabla f_3 + \beta_3 \vec{s}_1^T [A] \vec{s}_2 + \delta_3 \vec{s}_1^T [A] \vec{s}_1 = 0 \tag{2}$$

If we assume that $\vec{s}_1$ and $\vec{s}_2$ are already made conjugate, $\vec{s}_1^T [A] \vec{s}_2 = 0$, and Eq. (2) gives

$$\delta_3 = \frac{\vec{s}_1^T [A] \nabla f_3}{\vec{s}_1^T [A] \vec{s}_1} \tag{3}$$

From Eq. (6.76), δ_3 can be expressed as

$$\delta_3 = \frac{(\vec{x}_2 - \vec{x}_1)^T}{\lambda_1^*} \frac{[A] \nabla f_3}{\vec{s}_1^T [A] \vec{s}_1} \tag{4}$$

By using Eq. (6.84), Eq. (4) can be rewritten as

$$\delta_3 = \frac{1}{\lambda_1^*} \frac{(\nabla f_2 - \nabla f_1)^T \nabla f_3}{\vec{s}_1^T [A] \vec{s}_1} \tag{5}$$

Since $\vec{s}_1 = -\nabla f_1$ from Eq. (6.74) and $\vec{s}_2 - \beta_2 \vec{s}_1 = -\nabla f_2$ from Eq. (6.80), we obtain

$$\nabla f_2 - \nabla f_1 = -\vec{s}_2 + \vec{s}_1 (1 + \beta_2) \tag{6}$$

and Eq. (5) gives

$$\delta_3 = \frac{1}{\lambda_1^*} \frac{\{-\vec{s}_2 + (1 + \beta_2) \vec{s}_1\}^T \nabla f_3}{\vec{s}_1^T [A] \vec{s}_1} \tag{7}$$

which can be seen to be equal to zero since $\vec{s}_1^T \nabla f_3 = 0$ and $\vec{s}_2^T \nabla f_3 = 0$. Therefore Eq. (1) reduces to

$$\vec{s}_3 = -\nabla f_3 + \beta_3 \vec{s}_2 \tag{8}$$

6.41 When $\vec{s}_3$ is made conjugate to $\vec{s}_2$, we obtain

$$\vec{s}_2^{\,T}\,[A]\,\vec{s}_3 = \vec{s}_2^{\,T}\,[A]\left(-\nabla f_3 + \beta_3\,\vec{s}_2\right) \tag{1}$$

But

$$\vec{s}_2 = \frac{\vec{x}_3 - \vec{x}_2}{\lambda_2^*} \tag{2}$$

Also

$$\begin{aligned}\nabla f_3 - \nabla f_2 &= \left([A]\,\vec{x}_3 + \vec{B}\right) - \left([A]\,\vec{x}_2 + \vec{B}\right)\\ &= [A]\left(\vec{x}_3 - \vec{x}_2\right)\end{aligned} \tag{3}$$

Using Eq. (2), Eq. (1) can be expressed as

$$\vec{s}_2^{\,T}\,[A]\,\vec{s}_3 = -\left(\frac{\vec{x}_3 - \vec{x}_2}{\lambda_2^*}\right)^T [A]\left(\nabla f_3 - \beta_3\,\vec{s}_2\right) = 0 \tag{4}$$

In view of Eq. (3), Eq. (4) becomes

$$\left(\nabla f_3 - \nabla f_2\right)^T\left(\nabla f_3 - \beta_3\,\vec{s}_2\right) = 0 \tag{5}$$

or

$$\nabla f_3^{\,T}\,\nabla f_3 - \nabla f_2^{\,T}\,\nabla f_3 - \beta_3\,\nabla f_3^{\,T}\,\vec{s}_2 + \beta_3\,\nabla f_2^{\,T}\,\vec{s}_2 = 0 \tag{6}$$

Since

$$\nabla f_2^{\,T}\,\nabla f_3 = -\,\vec{s}_2^{\,T}\,\nabla f_3 = 0 \tag{7}$$

from Theorem A given below, Eq. (6) yields

$$\beta_3 = -\,\frac{\nabla f_3^{\,T}\,\nabla f_3}{\nabla f_2^{\,T}\,\vec{s}_2} \tag{8}$$

However,

$$\vec{s}_2 = -\nabla f_2 + \lambda_1^*\,\vec{s}_1 \tag{9}$$

and

$$\nabla f_2^{\,T}\,\vec{s}_2 = -\nabla f_2^{\,T}\,\nabla f_2 - \lambda_1^*\,\nabla f_2^{\,T}\,\nabla f_1 \tag{10}$$

($\nabla f_2^{\,T}\,\nabla f_1 = 0$)

Equations (8) and (10) give the desired result:

$$\beta_3 = \frac{\nabla f_3^{\,T}\,\nabla f_3}{\nabla f_2^{\,T}\,\nabla f_2} \tag{11}$$

THEOREM A: *Suppose that the point* $\mathbf{X}_{i+1}$ *is reached after* i *steps while minimizing a quadratic function* $f(\mathbf{X}) = \frac{1}{2}\mathbf{X}^T \mathbf{A}\mathbf{X} + \mathbf{B}^T\mathbf{X} + C$. *If the search directions used in the minimization process,* $\mathbf{S}_1, \mathbf{S}_2, \ldots, \mathbf{S}_i$, *are mutually conjugate with respect to* $\mathbf{A}$, *then*

$$\mathbf{S}_k^T \nabla f_{i+1} = 0 \quad \text{for} \quad k = 1, 2, \ldots, i$$

Proof: The gradient of the function f, evaluated at the point $\mathbf{X}_{i+1}$, is given by

$$\nabla f_{i+1} = \mathbf{A}\mathbf{X}_{i+1} + \mathbf{B} \tag{12}$$

Since $\mathbf{X}_{i+1}$ is reached after i minimization steps, it can be written as

$$\begin{aligned}\mathbf{X}_{i+1} &= \mathbf{X}_1 + \lambda_1^* \mathbf{S}_1 + \lambda_2^* \mathbf{S}_2 + \ldots + \lambda_k^* \mathbf{S}_k + \lambda_{k+1}^* \mathbf{S}_{k+1} + \ldots + \lambda_i^* \mathbf{S}_i \\ &= \mathbf{X}_{k+1} + \sum_{j=k+1}^{i} \lambda_j^* \mathbf{S}_j \end{aligned} \tag{13}$$

where λ_j^* is the minimizing step length in the direction $\mathbf{S}_j$. In view of Eq. (12), Eq. (13) becomes

$$\begin{aligned}\nabla f_{i+1} &= \mathbf{A}\left(\mathbf{X}_{k+1} + \sum_{j=k+1}^{n} \lambda_j^* \mathbf{S}_j\right) + \mathbf{B} \\ &= \nabla f_{k+1} + \sum_{j=k+1}^{i} \lambda_j^* \mathbf{A}\mathbf{S}_j \end{aligned} \tag{14}$$

Premultiplying both sides of Eq. (14) by $\mathbf{S}_k^T$, we obtain

$$\mathbf{S}_k^T \nabla f_{i+1} = \mathbf{S}_k^T \nabla f_{k+1} + \sum_{j=k+1}^{i} \lambda_j^* \mathbf{S}_k^T \mathbf{A}\mathbf{S}_j \tag{15}$$

The first term on the right-hand side of Eq. (15) is zero since λ_k^* is the minimizing step length along the direction $\mathbf{S}_k$, and the second term is zero since the search directions $\mathbf{S}_1, \mathbf{S}_2, \ldots, \mathbf{S}_i$ are given to be A-conjugate. Therefore, we obtain the desired result

$$\mathbf{S}_k^T \nabla f_{i+1} = 0, \; k = 1, 2, \ldots, i \tag{16}$$

6.42

$f = 2x_1^2 + x_2^2$ (Minimize)

Iteration 1:

$\vec{X}_1 = \begin{Bmatrix} 1 \\ 2 \end{Bmatrix}$, $f_1 = f(\vec{X}_1) = 2*(1)^2 + (2)^2 = 6$

We use probe length $\epsilon = 0.01$ along $\vec{S}_1 = \begin{Bmatrix} 1 \\ 0 \end{Bmatrix}$:

$f^+ = f(\vec{X}_1 + \epsilon \vec{S}_1) = f(1.01, 2) = 6.0402 > f_1$

$f^- = f(\vec{X}_1 - \epsilon \vec{S}_1) = f(0.99, 2) = 5.9602 < f_1$

$\therefore -\vec{S}_1$ is the correct direction to minimize f from $\vec{X}_1$.

$f(\vec{X}_1 - \lambda_1 \vec{S}_1) = f(1-\lambda_1, 2) = 2(1-\lambda_1)^2 + (2)^2$

$= 2\lambda_1^2 - 4\lambda_1 + 6$

$\frac{df}{d\lambda_1} = 4\lambda_1 - 4 = 0 \Rightarrow \lambda_1^* = 1$

$\vec{X}_2 = \vec{X}_1 - \lambda_1^* \vec{S}_1 = \begin{Bmatrix} 1 \\ 2 \end{Bmatrix} - 1 \begin{Bmatrix} 1 \\ 0 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 2 \end{Bmatrix}$, $f_2 = 4$

Iteration 2:

We use probe length $\epsilon = 0.01$ along $\vec{S}_2 = \begin{Bmatrix} 0 \\ 1 \end{Bmatrix}$:

$f^+ = f(\vec{X}_2 + \epsilon \vec{S}_2) = f(0, 2.01) = 4.0401 > f_2$

$f^- = f(\vec{X}_2 - \epsilon \vec{S}_2) = f(0, 1.99) = 3.9601 < f_2$

$\therefore -\vec{S}_2$ is the correct direction to minimize f from $\vec{X}_2$.

$f(\vec{X}_2 - \lambda_2 \vec{S}_2) = f(0, 2-\lambda_2) = 0 + (2-\lambda_2)^2$

$= \lambda_2^2 - 4\lambda_2 + 4$

$\frac{df}{d\lambda_2} = 2\lambda_2 - 4 = 0 \Rightarrow \lambda_2^* = 2$

$\vec{X}_3 = \vec{X}_2 - \lambda_2^* \vec{S}_2 = \begin{Bmatrix} 0 \\ 2 \end{Bmatrix} - 2 \begin{Bmatrix} 0 \\ 1 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$

$f_3 = f(\vec{X}_3) = 0$

$\vec{X}_3$ can be verified to be the minimum of f.

6.43 Minimize $f = 2x_1^2 + x_2^2$

We use calculus method to find the one-dimensional step lengths.

$$\vec{x}_1 = \begin{Bmatrix} 1 \\ 2 \end{Bmatrix}, \quad f_1 = f(\vec{x}_1) = 6$$

$$\nabla f = \begin{Bmatrix} 4x_1 \\ 2x_2 \end{Bmatrix}, \quad \nabla f_1 = \nabla f(\vec{x}_1) = \begin{Bmatrix} 4 \\ 4 \end{Bmatrix}$$

Let $\vec{x} = \vec{x}_1 - \lambda_1 \nabla f_1 = \begin{Bmatrix} 1 \\ 2 \end{Bmatrix} - \lambda_1 \begin{Bmatrix} 4 \\ 4 \end{Bmatrix} = \begin{Bmatrix} 1-4\lambda \\ 2-4\lambda \end{Bmatrix}$

$$f(\lambda_1) = 2(1-4\lambda_1)^2 + (2-4\lambda_1)^2 = 48\lambda_1^2 - 32\lambda_1 + 6$$

$$\frac{df}{d\lambda_1} = 96\lambda_1 - 32 = 0 \Rightarrow \lambda_1^* = \frac{1}{3}$$

$$\vec{x}_2 = \vec{x}_1 - \lambda_1^* \nabla f_1 = \begin{Bmatrix} 1 \\ 2 \end{Bmatrix} - \frac{1}{3}\begin{Bmatrix} 4 \\ 4 \end{Bmatrix} = \begin{Bmatrix} -1/3 \\ 2/3 \end{Bmatrix}$$

$$f_2 = f(\vec{x}_2) = 8/9$$

$$\nabla f_2 = \nabla f(\vec{x}_2) = \begin{Bmatrix} -4/3 \\ 4/3 \end{Bmatrix}$$

Let $\vec{x} = \vec{x}_2 - \lambda_2 \nabla f_2 = \begin{Bmatrix} -\frac{1}{3} \\ \frac{2}{3} \end{Bmatrix} - \lambda_2 \begin{Bmatrix} -\frac{4}{3} \\ \frac{4}{3} \end{Bmatrix} = \begin{Bmatrix} -\frac{1}{3} + \frac{4\lambda_2}{3} \\ \frac{2}{3} - \frac{4\lambda_2}{3} \end{Bmatrix}$

$$f(\lambda_2) = 2\left(-\frac{1}{3} + \frac{4}{3}\lambda_2\right)^2 + \left(\frac{2}{3} - \frac{4}{3}\lambda_2\right)^2$$

$$= \frac{48}{9}\lambda_2^2 - \frac{32}{9}\lambda_2 + \frac{6}{9}$$

$$\frac{df}{d\lambda_2} = \frac{96}{9}\lambda_2 - \frac{32}{9} = 0 \Rightarrow \lambda_2^* = \frac{1}{3}$$

$$\vec{x}_3 = \vec{x}_2 - \lambda_2^* \nabla f_2 = \begin{Bmatrix} -\frac{1}{3} \\ \frac{2}{3} \end{Bmatrix} - \frac{1}{3}\begin{Bmatrix} -\frac{4}{3} \\ \frac{4}{3} \end{Bmatrix} = \begin{Bmatrix} \frac{1}{9} \\ \frac{2}{9} \end{Bmatrix}$$

$$f_3 = f(\vec{x}_3) = \frac{2}{27}$$

$$\nabla f_3 = \nabla f(\vec{x}_3) = \begin{Bmatrix} 4/9 \\ 4/9 \end{Bmatrix}$$

Since $\nabla f_3 \neq \vec{0}$, we need to go to next iteration.

6.44 Minimize $f = x_1^2 + 3x_2^2 + 6x_3^2$

with $\Delta x_1 = \Delta x_2 = \Delta x_3 = 0.5$. Let $\epsilon = 0.1$

Step 1: $\vec{x}_1 = \begin{Bmatrix} 2 \\ -1 \\ 1 \end{Bmatrix}$. step length $= 0.5$ along $\vec{u}_1$, $\vec{u}_2$ and $\vec{u}_3$.

set $k = 1$.

Step 2: $f_1 = f(\vec{x}_1) = (2)^2 + 3(-1)^2 + 6(1)^2 = 13$

$\vec{Y}_{10} = \vec{x}_1 = \begin{Bmatrix} 2 \\ -1 \\ 1 \end{Bmatrix}$

Step 3: To find the new temporary base point, we set $i = 1$ and evaluate

$f(\vec{Y}_{10}) = 13$

$f^+ = f(\vec{Y}_{10} + \Delta x_1 \vec{u}_1) = f(2.5, -1, 1) = 15.25$

$f^- = f(\vec{Y}_{10} - \Delta x_1 \vec{u}_1) = f(1.5, -1, 1) = 11.25$

Since $f^- < f$ and $< f^+$, we take

$\vec{Y}_{11} = \vec{Y}_{10} - \Delta x_1 \vec{u}_1 = \begin{Bmatrix} 1.5 \\ -1 \\ 1 \end{Bmatrix}$

Next, we evaluate

$f = f(\vec{Y}_{11}) = 11.25$

$f^+ = f(\vec{Y}_{11} + \Delta x_2 \vec{u}_2) = f(1.5, -0.5, 1) = 9.0 < f$

we set $\vec{Y}_{12} = \vec{Y}_{11} + \Delta x_2 \vec{u}_2 = \begin{Bmatrix} 1.5 \\ -0.5 \\ 1.0 \end{Bmatrix}$

Next, we evaluate

$f = f(\vec{Y}_{12}) = 9.0$

$f^+ = f(\vec{Y}_{12} + \Delta x_3 \vec{u}_3) = f(1.5, -0.5, 1.5) = 16.50$

$f^- = f(\vec{Y}_{12} - \Delta x_3 \vec{u}_3) = f(1.5, -0.5, 0.5) = 4.50$

Since $f^- < f$ and $< f^+$, we take

$\vec{Y}_{13} = \vec{Y}_{12} - \Delta x_3 \vec{u}_3 = \begin{Bmatrix} 1.5 \\ -0.5 \\ 0.5 \end{Bmatrix}$

Step 4: As $\vec{Y}_{13}$ is different from $\vec{X}_1$, the new base point is $\vec{X}_2 = \vec{Y}_{13} = \begin{Bmatrix} 1.5 \\ -0.5 \\ 0.5 \end{Bmatrix}$

Step 5: A pattern direction is established as

$$\vec{S} = \vec{X}_2 - \vec{X}_1 = \begin{Bmatrix} 1.5 \\ -0.5 \\ 0.5 \end{Bmatrix} - \begin{Bmatrix} 2 \\ -1 \\ 1 \end{Bmatrix} = \begin{Bmatrix} -0.5 \\ 0.5 \\ -0.5 \end{Bmatrix}$$

The optimal step length λ^* is found by minimizing

$$f(\vec{X}_2 + \lambda \vec{S}) = f(1.5 - 0.5\lambda, -0.5 + 0.5\lambda, 0.5 - 0.5\lambda)$$

$$= (1.5 - 0.5\lambda)^2 + 3(-0.5 + 0.5\lambda)^2 + 6(0.5 - 0.5\lambda)^2$$

$$\frac{df}{d\lambda} = 2(1.5 - 0.5\lambda)(-0.5) + 6(-0.5 + 0.5\lambda)(0.5)$$

$$+12(0.5 - 0.5\lambda)(-0.5) = 5\lambda - 6 = 0$$

$$\lambda^* = 6/5 = 1.2$$

$$\therefore \vec{Y}_{20} = \vec{X}_2 + \lambda^* \vec{S} = \begin{Bmatrix} 1.5 \\ -0.5 \\ 0.5 \end{Bmatrix} + 1.2 \begin{Bmatrix} -0.5 \\ 0.5 \\ -0.5 \end{Bmatrix} = \begin{Bmatrix} 0.9 \\ 0.1 \\ -0.1 \end{Bmatrix}$$

Step 6: Set $k = 2$, $f = f_2 = f(\vec{Y}_{20}) = 0.9$

and repeat step 3.

Taking $i = 1$, we evaluate

$$f^+ = f(\vec{Y}_{20} + \Delta x_1 \vec{u}_1) = f(1.4, 0.1, -0.1)$$

$$= 2.05$$

$$f^- = f(\vec{Y}_{20} - \Delta x_1 \vec{u}_1) = f(0.4, 0.1, -0.1)$$

$$= 0.25$$

Since $f^- < f < f^+$, we take

$$\vec{Y}_{21} = \vec{Y}_{20} - \Delta x_1 \vec{u}_1 = \begin{Bmatrix} 0.4 \\ 0.1 \\ -0.1 \end{Bmatrix}$$

Next, we set $i = 2$ and evaluate

$$f = f(\vec{Y}_{21}) = 0.25$$

$f^+ = f(\vec{Y}_{21} + \Delta x_2 \vec{u}_2) = f(0.4, 0.6, -0.1) = 1.30$

$f^- = f(\vec{Y}_{21} - \Delta x_2 \vec{u}_2) = f(0.4, -0.4, -0.1) = 0.70$

Since $f < \min(f^+, f^-)$, we take

$$\vec{Y}_{22} = \vec{Y}_{21} = \begin{Bmatrix} 0.4 \\ 0.1 \\ -0.1 \end{Bmatrix}$$

Next, we set $i = 3$ and evaluate

$$f = f(\vec{Y}_{22}) = 0.25$$

$f^+ = f(\vec{Y}_{22} + \Delta x_3 \vec{u}_3) = f(0.4, 0.1, 0.4) = 1.15$

$f^- = f(\vec{Y}_{22} - \Delta x_3 \vec{u}_3) = f(0.4, 0.1, -0.6) = 2.35$

Since $f < \min(f^+, f^-)$, we take

$$\vec{Y}_{23} = \vec{Y}_{22} = \begin{Bmatrix} 0.4 \\ 0.1 \\ -0.1 \end{Bmatrix}$$

Since $f(\vec{Y}_{23}) = 0.25 < f(\vec{X}_2) = 4.5$,

we take the new base point as

$$\vec{X}_3 = \vec{Y}_{23} = \begin{Bmatrix} 0.4 \\ 0.1 \\ -0.1 \end{Bmatrix}$$

and go to step 5.

6.45

$f = 4x_1^2 + 3x_2^2 - 5x_1x_2 - 8x_1$

$\vec{x}_1 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$, $f_1 = 0$

$\vec{s}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Probe along $+\vec{s}_2$: (take probe length, $\epsilon = 0.01$)

$\vec{x} = \vec{x}_1 + \epsilon \vec{s}_2 = \begin{pmatrix} 0 \\ 0.01 \end{pmatrix}$, $f(\vec{x}) = +0.0003 > f_1$

$+\vec{s}_2$ is not usable.

Probe along $-\vec{s}_2$ with $\epsilon = 0.01$:

$\vec{x} = \vec{x}_1 - \epsilon \vec{s}_2 = \begin{pmatrix} 0 \\ -0.01 \end{pmatrix}$, $f(\vec{x}) = 0.0003 > f_1$

$\vec{x}_1$ = optimum along $\vec{s}_2$

$\therefore \vec{x}_2 \equiv \vec{x}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $f(\vec{x}_2) = f_2 = 0.0$

$\vec{s}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

Probe along $\vec{s}_1$: (take probe length, $\epsilon = 0.01$)

$\vec{x} = \vec{x}_2 + \epsilon \vec{s}_1 = \begin{pmatrix} 0.01 \\ 0 \end{pmatrix}$, $f(\vec{x}) = -0.0796 < f_1$

$\therefore +\vec{s}_1$ is a usable direction.

$f(\lambda) = f(\vec{x}_1 + \lambda \vec{s}_1) = f\left(\begin{Bmatrix} \lambda \\ 0 \end{Bmatrix}\right) = 4\lambda^2 - 8\lambda$

$\frac{df}{d\lambda} = 8\lambda - 8 = 0 \Rightarrow \lambda^* = 1$

$\vec{x}_3 = \vec{x}_1 + \lambda^* \vec{s}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $f_3 = -4$

$\vec{s}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Probe along $\vec{s}_2$: (take probe length, $\epsilon = 0.01$)

$\vec{x} = \vec{x}_3 + \epsilon \vec{s}_2 = \begin{pmatrix} 1 \\ 0.01 \end{pmatrix}$, $f(\vec{x}) = -4.0497 < f_3$

$\therefore +\vec{s}_2$ is a usable direction.

$f(\lambda) = f(\vec{x}_3 + \lambda \vec{s}_2) = f\left(\begin{Bmatrix} 1 \\ \lambda \end{Bmatrix}\right) = 3\lambda^2 - 5\lambda - 4$

$\frac{df}{d\lambda} = 6\lambda - 5 = 0 \Rightarrow \lambda^* = \frac{5}{6} = 0.8333$

$$\vec{x}_4 = \vec{x}_3 + \lambda^* \vec{s}_2 = \begin{pmatrix} 1 \\ 0.8333 \end{pmatrix}, \quad f_4 = -6.0834$$

$$\vec{s}_p^{(1)} = \vec{x}_4 - \vec{x}_2 = \begin{pmatrix} 1 \\ 0.8333 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0.8333 \end{pmatrix}$$

We do not need to normalize $\vec{s}_p^{(1)}$ since one of its components is equal to 1.

Probe along $\vec{s}_p^{(1)}$ with $\epsilon = 0.01$:

$$\vec{x} = \vec{x}_4 + \epsilon \vec{s}_p^{(1)} = \begin{pmatrix} 1 \\ 0.8333 \end{pmatrix} + 0.01 \begin{pmatrix} 1 \\ 0.8333 \end{pmatrix} = \begin{pmatrix} 1.01 \\ 0.8416 \end{pmatrix}$$

$$f(\vec{x}) = -6.1248 < f_4$$

$\therefore \vec{s}_p^{(1)}$ is usable.

$$f(\lambda) = f\left(\vec{x}_4 + \lambda \vec{s}_p^{(1)}\right) = f\begin{pmatrix} 1+\lambda \\ 0.8333 + 0.8333\lambda \end{pmatrix}$$

$$= 1.9167\lambda^2 - 4.1666\lambda - 6.0834$$

$$\frac{df}{d\lambda} = 3.8334\lambda - 4.1666 = 0 \Rightarrow \lambda^* = 1.0869$$

$$\vec{x}_5 = \vec{x}_4 + \lambda^* \vec{s}_p^{(1)} = \begin{Bmatrix} 2.0869 \\ 1.7390 \end{Bmatrix}$$

$$f_5 = -8.3477$$

Note: Exact solution (Analytical method):

$$\frac{\partial f}{\partial x_1} = 8x_1 - 5x_2 - 8 = 0$$

$$\frac{\partial f}{\partial x_2} = 6x_2 - 5x_1 = 0$$

Solution of the algebraic linear equations gives:

$$\vec{x}^* = \begin{Bmatrix} 48/23 \\ 40/23 \end{Bmatrix}$$

with $f^* = -8.3483$.

6.46 Minimize $f = x_1^4 - 2x_1^2 x_2 + x_1^2 + x_2^2 - 2x_1 + 1$

Assume $\alpha = 1$, $\beta = 0.5$, $\gamma = 2$, $\epsilon = 0.2$

Iteration 1:

Step 1:

Let $\vec{X}_1 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$, $\vec{X}_2 = \begin{Bmatrix} 2 \\ 0 \end{Bmatrix}$, $\vec{X}_3 = \begin{Bmatrix} 0 \\ 2 \end{Bmatrix}$

with $f_1 = 1$, $f_2 = 17$, $f_3 = 5$

$\vec{X}_h = \vec{X}_2 = \begin{Bmatrix} 2 \\ 0 \end{Bmatrix}$, $f_h = 17$

$\vec{X}_l = \vec{X}_1 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$, $f_l = 1$

Step 2:

Centroid $= \vec{X}_0 = \frac{1}{2}(\vec{X}_1 + \vec{X}_3) = \begin{Bmatrix} 0 \\ 1 \end{Bmatrix}$, $f_0 = f(\vec{X}_0) = 2.0$

Step 3:

Reflection point:

$\vec{X}_r = 2\vec{X}_0 - \vec{X}_h = \begin{Bmatrix} 0 \\ 2 \end{Bmatrix} - \begin{Bmatrix} 2 \\ 0 \end{Bmatrix} = \begin{Bmatrix} -2 \\ 2 \end{Bmatrix}$, $f_r = f(\vec{X}_r) = 13$

Step 4:

since $f_r > f_l$, but $f_r < f_h$, we use contraction:

$\vec{X}_c = \beta \vec{X}_h + (1-\beta)\vec{X}_0 = 0.5 \begin{Bmatrix} 2 \\ 0 \end{Bmatrix} + 0.5 \begin{Bmatrix} 0 \\ 1 \end{Bmatrix} = \begin{Bmatrix} 1.0 \\ 0.5 \end{Bmatrix}$

$f_c = f(\vec{X}_c) = 0.25$

Step 5:

Since $f_c < f_r$ and also $< f_l$, we find the new simplex as

$\vec{X}_1 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$, $\vec{X}_2 = \begin{Bmatrix} 1 \\ 0.5 \end{Bmatrix}$, $\vec{X}_3 = \begin{Bmatrix} 0 \\ 2 \end{Bmatrix}$

with $f_1 = 1$, $f_2 = 0.25$, $f_3 = 5.0$

Step 6: Test for convergence:

$$Q = \left[\frac{(1-2)^2 + (0.25-2)^2 + (5-2)^2}{3}\right]^{\frac{1}{2}} = 2.0867$$

Since $Q > \epsilon$, we go to the next iteration.

Iteration 2:

Step 1: $\vec{x}_h = \vec{x}_3 = \begin{Bmatrix} 0 \\ 2 \end{Bmatrix}$, $f_h = 5$

$\vec{x}_l = \vec{x}_2 = \begin{Bmatrix} 1 \\ 0.5 \end{Bmatrix}$, $f_l = 0.25$

Step 2: Centroid $= \vec{x}_0 = \frac{1}{2}(\vec{x}_1 + \vec{x}_2) = \begin{Bmatrix} 0.5 \\ 0.25 \end{Bmatrix}$

$f_0 = f(\vec{x}_0) = 0.25$

Step 3:

$\vec{x}_r = 2\vec{x}_0 - \vec{x}_h = 2\begin{Bmatrix} 0.5 \\ 0.25 \end{Bmatrix} - \begin{Bmatrix} 0 \\ 2 \end{Bmatrix} = \begin{Bmatrix} 1.0 \\ -1.5 \end{Bmatrix}$

$f_r = f(\vec{x}_r) = 6.25$

Step 4: Since $f_r > f_h$, we use contraction.

$\vec{x}_c = \beta\vec{x}_h + (1-\beta)\vec{x}_0 = 0.5\begin{Bmatrix} 0 \\ 2 \end{Bmatrix} + 0.5\begin{Bmatrix} 0.5 \\ 0.25 \end{Bmatrix} = \begin{Bmatrix} 0.25 \\ 1.125 \end{Bmatrix}$

$f_c = f(\vec{x}_c) = 1.6914$

Step 5: Since $f_c < f_h$, we obtain the new simplex as

$\vec{x}_1 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$, $\vec{x}_2 = \begin{Bmatrix} 1 \\ 0.5 \end{Bmatrix}$, $\vec{x}_3 = \begin{Bmatrix} 0.25 \\ 1.125 \end{Bmatrix}$

with $f_1 = 1$, $f_2 = 0.25$, $f_3 = 1.6914$

Step 6: Test for convergence:

$$Q = \left[\frac{(1-0.25)^2 + (0.25-0.25)^2 + (1.6914-0.25)^2}{3}\right]^{\frac{1}{2}}$$

$= 0.9381$

Since $Q > \epsilon$, we go to next iteration.

6.47

$$f = (2x_1 - x_2 + x_3 + 1)^2 + (x_1 + 2x_2)^2 + (3x_1 + x_2 + 2x_3 - 3)^2$$

$$= 14x_1^2 + 6x_2^2 + 5x_3^2 + 6x_1x_2 + 2x_2x_3 + 16x_1x_3 - 14x_1 - 8x_2 - 10x_3 + 10$$

$$\frac{\partial f}{\partial x_1} = 28x_1 + 6x_2 + 16x_3 - 14$$

$$\frac{\partial f}{\partial x_2} = 6x_1 + 12x_2 + 2x_3 - 8$$

$$\frac{\partial f}{\partial x_3} = 16x_1 + 2x_2 + 10x_3 - 10$$

$$\frac{\partial^2 f}{\partial x_1^2} = 28, \quad \frac{\partial^2 f}{\partial x_2^2} = 12, \quad \frac{\partial^2 f}{\partial x_3^2} = 10$$

$$\frac{\partial^2 f}{\partial x_1 \partial x_2} = 6, \quad \frac{\partial^2 f}{\partial x_1 \partial x_3} = 16, \quad \frac{\partial^2 f}{\partial x_2 \partial x_3} = 2$$

$$\vec{X}_1 = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}, \quad \nabla f_1 = \begin{Bmatrix} -14 \\ -8 \\ -10 \end{Bmatrix}$$

$$[J_1] = \begin{bmatrix} 28 & 6 & 16 \\ 6 & 12 & 2 \\ 16 & 2 & 10 \end{bmatrix}, \quad [J_1]^{-1} = \frac{1}{50}\begin{bmatrix} 29 & -7 & -45 \\ -7 & 6 & 10 \\ -45 & 10 & 75 \end{bmatrix}$$

$$\vec{X}_2 = \vec{X}_1 - [J_1]^{-1} \nabla f_1$$

$$= \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix} - \frac{1}{50}\begin{bmatrix} 29 & -7 & -45 \\ -7 & 6 & 10 \\ -45 & 10 & 75 \end{bmatrix} = \begin{Bmatrix} -2 \\ 1 \\ 4 \end{Bmatrix}$$

6.48 Problem is:

Minimize $f = (x_1^2 + x_2^2 - 2)^2 + (10x_1^2 - 10x_2 - 5x_1 + 1)^2$

using DFP method.

$$\nabla f = \begin{Bmatrix} 4x_1(x_1^2 + x_2^2 - 2) + 2(20x_1 - 5)(10x_1^2 - 10x_2 - 5x_1 + 1) \\ 4x_2(x_1^2 + x_2^2 - 2) - 20(10x_1^2 - 10x_2 - 5x_1 + 1) \end{Bmatrix}$$

$$\vec{X}_1 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}, \quad f_1 = f(\vec{X}_1) = 5, \quad \nabla f_1 = \begin{Bmatrix} -10 \\ -20 \end{Bmatrix}$$

Let $[B_1] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$$\vec{S}_1 = -[B_1]\nabla f_1 = \begin{Bmatrix} 10 \\ 20 \end{Bmatrix}$$

$$\vec{X} = \vec{X}_1 + \lambda_1 \vec{S}_1 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} + \lambda_1 \begin{Bmatrix} 10 \\ 20 \end{Bmatrix} = \begin{Bmatrix} 10\lambda_1 \\ 20\lambda_1 \end{Bmatrix}$$

$$f(\lambda_1) = (500\lambda_1^2 - 2)^2 + (1000\lambda_1^2 - 250\lambda_1 + 1)^2$$

$$\lambda_1^* \simeq 0.0042$$

$$\vec{X}_2 = \vec{X}_1 + \lambda_1^* \vec{S}_1 = \begin{Bmatrix} 0.042 \\ 0.084 \end{Bmatrix}, \quad f_2 = 3.9658446$$

$$\nabla f_2 = \begin{Bmatrix} -0.065283388 \\ -0.021837294 \end{Bmatrix}$$

Since $|\nabla f_2| \neq 0$, we proceed to next iteration.

$$\vec{g}_1 = \nabla f_2 - \nabla f_1 = \begin{Bmatrix} -0.065283388 \\ -0.021837294 \end{Bmatrix} - \begin{Bmatrix} -10 \\ -20 \end{Bmatrix} = \begin{Bmatrix} 9.934716611 \\ 19.978162711 \end{Bmatrix}$$

$$\vec{S}_1^T \vec{g}_1 = 498.910\,4202$$

$$[B_1]\vec{g}_1 = \vec{g}_1 = \begin{Bmatrix} 9.934716611 \\ 19.978162711 \end{Bmatrix}$$

$$\vec{g}_1^T [B_1] \vec{g}_1 = 497.8255794$$

$$\vec{s}_1 \vec{s}_1^T = \begin{bmatrix} 100 & 200 \\ 200 & 400 \end{bmatrix}$$

$$[M_1] = \frac{\lambda_1^* \vec{s}_1 \vec{s}_1^T}{\vec{s}_1^T \vec{g}_1} = \frac{0.0042}{498.9104202} \begin{bmatrix} 100 & 200 \\ 200 & 400 \end{bmatrix}$$

$$= 10^{-6} \begin{bmatrix} 841.8 & 1683.6 \\ 1683.6 & 3367.2 \end{bmatrix}$$

$$[N_1] = -\frac{\left([B_1]\vec{g}_1\right)\left([B_1]\vec{g}_1\right)^T}{\vec{g}_1^T [B_1] \vec{g}_1} = -\frac{1}{497.8255794} \vec{g}_1 \vec{g}_1^T$$

$$= -10^{-3} \begin{bmatrix} 198.2594192 & 398.6886682 \\ 398.6886682 & 801.7407439 \end{bmatrix}$$

$$[B_2] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0.0008418 & 0.0016836 \\ 0.0016836 & 0.0033672 \end{bmatrix}$$

$$+ \begin{bmatrix} -0.19825942 & -0.39868867 \\ -0.39868867 & -0.80174074 \end{bmatrix}$$

$$= \begin{bmatrix} 0.80258238 & -0.39700507 \\ -0.39700507 & 0.20162646 \end{bmatrix}$$

$$\vec{s}_2 = -[B_2] \nabla f_2 = \begin{Bmatrix} 0.043\ 725\ 781 \\ -0.021\ 514\ 86 \end{Bmatrix}$$

$$\vec{x} = \vec{x}_2 + \lambda_2 \vec{s}_2 = \begin{Bmatrix} 0.042 + 0.043725781\ \lambda_2 \\ 0.084 - 0.02151486\ \lambda_2 \end{Bmatrix}$$

$$\lambda_2^* \simeq 2.6$$

$$\vec{x}_3 = \vec{x}_2 + \lambda_2^* \vec{s}_2 = \begin{Bmatrix} 0.15568702 \\ 0.028061368 \end{Bmatrix}$$

$$f_3 = 3.9341350$$

6.49 Problem is:

Minimize $f = (x_1^2 + x_2^2 - 2)^2 + (10x_1^2 - 10x_2 - 5x_1 + 1)^2$

using DFP method.

$$\nabla f = \begin{Bmatrix} 4x_1(x_1^2 + x_2^2 - 2) + 2(20x_1 - 5)(10x_1^2 - 10x_2 - 5x_1 + 1) \\ 4x_2(x_1^2 + x_2^2 - 2) - 20(10x_1^2 - 10x_2 - 5x_1 + 1) \end{Bmatrix}$$

$$\vec{x}_1 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}, \quad f_1 = f(\vec{x}_1) = 5, \quad \nabla f_1 = \begin{Bmatrix} -10 \\ -20 \end{Bmatrix}$$

Let $[B_1] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$$\vec{s}_1 = -[B_1]\nabla f_1 = \begin{Bmatrix} 10 \\ 20 \end{Bmatrix}$$

$$\vec{x} = \vec{x}_1 + \lambda_1 \vec{s}_1 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} + \lambda_1 \begin{Bmatrix} 10 \\ 20 \end{Bmatrix} = \begin{Bmatrix} 10\lambda_1 \\ 20\lambda_1 \end{Bmatrix}$$

$$f(\lambda_1) = (500\lambda_1^2 - 2)^2 + (1000\lambda_1^2 - 250\lambda_1 + 1)^2$$

$$\lambda_1^* \simeq 0.0042$$

$$\vec{x}_2 = \vec{x}_1 + \lambda_1^* \vec{s}_1 = \begin{Bmatrix} 0.042 \\ 0.084 \end{Bmatrix}, \quad f_2 = 3.96584446$$

$$\nabla f_2 = \begin{Bmatrix} -0.065283388 \\ -0.021837294 \end{Bmatrix}$$

Since $|\nabla f_2| \neq 0$, we proceed to next iteration.

$$\vec{g}_1 = \nabla f_2 - \nabla f_1 = \begin{Bmatrix} -0.065283388 \\ -0.021837294 \end{Bmatrix} - \begin{Bmatrix} -10 \\ -20 \end{Bmatrix} = \begin{Bmatrix} 9.93471661 \\ 19.978162 71 \end{Bmatrix}$$

$$\vec{d}_1 = \lambda_1^* \vec{s}_1 = 0.0042 \begin{Bmatrix} 10 \\ 20 \end{Bmatrix} = \begin{Bmatrix} 0.042 \\ 0.084 \end{Bmatrix}$$

$$\vec{d}_1 \vec{d}_1^T = \begin{bmatrix} 0.001764 & 0.003528 \\ 0.003528 & 0.007056 \end{bmatrix}$$

$$\vec{d}_1^T \vec{g}_1 = 2.09542377$$

$$\vec{d}_1 \vec{g}_1^T = \begin{bmatrix} 0.417258098 & 0.839082834 \\ 0.834516195 & 1.678165668 \end{bmatrix}$$

$$\vec{g}_1 \vec{d}_1^T = \begin{bmatrix} 0.417258098 & 0.834516195 \\ 0.839082834 & 1.678165668 \end{bmatrix}$$

$$\vec{g}_1^T [B] \vec{g}_1 = 497.8255794$$

$$\vec{d}_1 \vec{g}_1^T [B_1] = \vec{d}_1 \vec{g}_1^T$$

$$[B_1] \vec{g}_1 \vec{d}_1^T = \vec{g}_1 \vec{d}_1^T$$

$$[B_2] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \left(1 + \frac{497.8255794}{2.09542377}\right) \frac{1}{2.09542377} \vec{d}_1 \vec{d}_1^T$$

$$- \frac{1}{2.09542377} \left(\vec{d}_1 \vec{g}_1^T [B_1] + [B_1] \vec{g}_1 \vec{d}_1^T\right)$$

$$= \begin{bmatrix} 0.80258625 & -0.39700683 \\ -0.39700683 & 0.20162765 \end{bmatrix}$$

$$\vec{s}_2 = -[B_2] \nabla f_2 = \begin{Bmatrix} 0.04372599 \\ -0.02151495 \end{Bmatrix}$$

$$\vec{x} = \vec{x}_2 + \lambda_2 \vec{s}_2 = \begin{Bmatrix} 0.042 + 0.04372599\,\lambda_2 \\ 0.082 - 0.02151495\,\lambda_2 \end{Bmatrix}$$

$$\lambda^* \simeq 2.6$$

$$\vec{x}_3 = \vec{x}_2 + \lambda_2^* \vec{s}_2 = \begin{Bmatrix} 0.15568756 \\ 0.02806137 \end{Bmatrix}$$

$$f_3 = 3.93413345$$

6.50 Minimize $f(x_1, x_2) = (2x_1x_2 - 3)^2 + (x_1 - x_2 - 2)^2$

Iteration 1:

$\vec{X}_1 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$, $f_1 = f(\vec{X}_1) = 13$

Use a probe length $\epsilon = 0.01$ along $\vec{S}_1 = \begin{Bmatrix} 1 \\ 0 \end{Bmatrix}$:

$f^+ = f(\vec{X}_1 + \epsilon \vec{S}_1) = f(0.01, 0.0) = 12.9601 < f_1$

$\therefore +\vec{S}_1$ is the correct direction to minimize f from $\vec{X}_1$.

$f(\vec{X}_1 + \lambda_1 \vec{S}_1) = f(\lambda_1, 0.0) = 9 + (\lambda_1 - 2)^2 = \lambda_1^2 - 4\lambda_1 + 13$

$\frac{df}{d\lambda_1} = 2\lambda_1 - 4 = 0$ or $\lambda_1^* = 2$

$\vec{X}_2 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$, $f_2 = f(\vec{X}_2) = 9$

Iteration 2:

We use probe length $\epsilon = 0.01$ along $\vec{S}_2 = \begin{Bmatrix} 0 \\ 1 \end{Bmatrix}$:

$f^+ = f(\vec{X}_2 + \epsilon \vec{S}_2) = f(2, 0.01) = 8.7617 < f_2$

$\therefore +\vec{S}_2$ is the correct direction to minimize f from $\vec{X}_2$.

$f(\vec{X}_2 + \lambda_2 \vec{S}_2) = f(2, \lambda_2)$

$= (4\lambda_2 - 3)^2 + (-\lambda_2)^2 = 17\lambda_2^2 - 24\lambda_2 + 9$

$\frac{df}{d\lambda_2} = 34\lambda_2 - 24 = 0$ or $\lambda_2^* = 24/34 = 0.7059$

$\vec{X}_3 = \begin{Bmatrix} 2 \\ 0 \end{Bmatrix} + 0.7059 \begin{Bmatrix} 0 \\ 1 \end{Bmatrix} = \begin{Bmatrix} 2 \\ 0.7059 \end{Bmatrix}$

$f_3 = f(\vec{X}_3) = 0.52941$

6.51 Problem can be stated as:

Minimize $f = (7x_1^3 - 10x_1 - x_2 - 1)^2 + (8x_2^3 - 11x_2 + x_1 - 1)^2$

using Fletcher-Reeves method.

$$\vec{X}_1 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}, \quad f_1 = f(\vec{X}_1) = 2$$

$$\nabla f = \begin{Bmatrix} 2(21x_1^2 - 10)(7x_1^3 - 10x_1 - x_2 - 1) + 2(8x_2^3 - 11x_2 + x_1 - 1) \\ -2(7x_1^3 - 10x_1 - x_2 - 1) + 2(24x_2^2 - 11)(8x_2^3 - 11x_2 + x_1 - 1) \end{Bmatrix}$$

$$\nabla f_1 = \nabla f(\vec{X}_1) = \begin{Bmatrix} 18 \\ 24 \end{Bmatrix}$$

$$\vec{X} = \vec{X}_1 - \lambda_1 \nabla f_1 = \begin{Bmatrix} -18\lambda_1 \\ -24\lambda_1 \end{Bmatrix}$$

$$f(\vec{X}) = f(\lambda_1) = (-40824\lambda_1^3 + 204\lambda_1 - 1)^2 + (-110592\lambda_1^3 + 246\lambda_1 - 1)^2$$

$$\lambda^* \simeq 0.0045$$

$$\vec{X}_2 = \vec{X}_1 - \lambda_1^* \nabla f_1 = \begin{Bmatrix} -0.081 \\ -0.108 \end{Bmatrix}$$

$$f_2 = 0.016741879, \qquad \nabla f_2 = \begin{Bmatrix} 1.8846260 \\ -1.9065884 \end{Bmatrix}$$

$$|\nabla f_1| = \{(18)^2 + (24)^2\}^{\frac{1}{2}} = 30.0$$

$$|\nabla f_2| = \{1.884626^2 + (-1.9065884)^2\}^{\frac{1}{2}} = 2.680838397$$

$$\vec{S}_2 = -\nabla f_2 + \frac{|\nabla f_2|^2}{|\nabla f_1|^2} \vec{S}_1$$

$$= \begin{Bmatrix} -1.884626 \\ 1.9065884 \end{Bmatrix} + \left(\frac{7.186894487}{900.0}\right) \begin{Bmatrix} -18 \\ -24 \end{Bmatrix}$$

$$= \begin{Bmatrix} -2.02836389 \\ 1.71493788 \end{Bmatrix}$$

$$\vec{X} = \vec{X}_2 + \lambda_2 \vec{S}_2 = \begin{Bmatrix} -0.081 - 2.02836389\lambda_2 \\ -0.108 + 1.71493788\lambda_2 \end{Bmatrix}$$

$$\lambda_2^* \simeq 0.0045$$

$$\vec{x}_3 = \vec{x}_2 + \lambda_2^* \vec{s}_2 = \begin{Bmatrix} -0.090127639 \\ -0.100288278 \end{Bmatrix}$$

$$f_3 = 0.000036869191$$

6.52

$$f = (5x_1 + 3x_2 - 1)^2 + (4x_1 - 7x_2 - 76)^2$$

$$\frac{\partial f}{\partial x_1} = 82x_1 - 26x_2 - 618$$

$$\frac{\partial f}{\partial x_2} = -26x_1 + 116x_2 + 1058$$

$$\vec{x}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad [B_1] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Iteration 1:

$$\nabla f_1 = \begin{pmatrix} -618 \\ 1058 \end{pmatrix}, \quad \vec{s}_1 = -[B_1]\nabla f_1 = \begin{pmatrix} 618 \\ -1058 \end{pmatrix}$$

$$\vec{x} = \vec{x}_1 + \lambda \vec{s}_1 = \begin{pmatrix} 618\lambda \\ -1058\lambda \end{pmatrix}$$

$$f(\vec{x}) = f(\lambda) = (3090\lambda - 3174\lambda - 1)^2 + (2472\lambda + 7406\lambda - 76)^2$$

$$= (-84\lambda - 1)^2 + (9878\lambda - 76)^2$$

$$\frac{df}{d\lambda} = 2(-84)(-84\lambda - 1) + 2(9878)(9878\lambda - 76) = 0$$

i.e., $195\,163\,880\lambda = 1501288$

i.e., $\lambda^* = 0.00769248$

$$\vec{x}_2 = \vec{x}_1 + \lambda^* \vec{s}_1 = \begin{pmatrix} 4.753933 \\ -8.138610 \end{pmatrix}$$

$$\nabla f_2 = \begin{pmatrix} -16.573634 \\ -9.681018 \end{pmatrix}$$

Since $|\nabla f_2| \neq 0$, we proceed to the next iteration.

$$\vec{g}_1 = \nabla f_2 - \nabla f_1 = \begin{pmatrix} -16.573634 \\ -9.681018 \end{pmatrix} - \begin{pmatrix} -618 \\ 1058 \end{pmatrix} = \begin{pmatrix} 601.426366 \\ -1067.681018 \end{pmatrix}$$

$$\vec{d}_1 = \lambda_1^* \vec{s}_1 = \begin{pmatrix} 4.753934 \\ -8.138612 \end{pmatrix}$$

$$\vec{d}_1 \vec{d}_1^T = \begin{bmatrix} 22.599888 & -38.690424 \\ -38.690424 & 66.237005 \end{bmatrix}$$

$$\vec{d}_1^T \vec{g}_1 = 11548.5828$$

$$\vec{g}_1^T [B_1] \vec{g}_1 = 1501656.43$$

$$\vec{g}_1 \vec{d}_1^{\,T} = \begin{bmatrix} 2859.14125 & -4894.77584 \\ -5075.685093 & 8689.441545 \end{bmatrix}$$

$$\vec{d}_1 \vec{g}_1^{\,T} [B_1] = \begin{bmatrix} 2859.14125 & -5075.685093 \\ -4894.77584 & 8689.441545 \end{bmatrix}$$

$$[B_1] \vec{g}_1 \vec{d}_1^{\,T} = \begin{bmatrix} 2859.14125 & -4894.77584 \\ -5075.685093 & 8689.441545 \end{bmatrix}$$

$$1 + \frac{\vec{g}_1^{\,T} [B_1] \vec{g}_1}{\vec{d}_1^{\,T} \vec{g}_1} = 131.029$$

$$[B_2] = \begin{bmatrix} 0.761267 & 0.424370 \\ 0.424370 & 0.246671 \end{bmatrix}$$

Iteration 2:

$$\vec{s}_2 = -[B_2] \nabla f_2 = \begin{pmatrix} 16.725295 \\ 9.421379 \end{pmatrix}$$

$$\vec{x} = \vec{x}_2 + \lambda \vec{s}_2 = \begin{pmatrix} 4.753933 + \lambda (16.725295) \\ -8.138610 + \lambda (9.421379) \end{pmatrix}$$

$$f(\lambda) = (-1.646165 + 111.890612 \lambda)^2 + (-0.013998 + 0.951527 \lambda)^2$$

$$\frac{df}{d\lambda} = 0 \Rightarrow \lambda^* = 0.014712$$

$$\vec{x}_3 = \vec{x}_2 + \lambda^* \vec{s}_2 = \begin{pmatrix} 5.0 \\ -8.0 \end{pmatrix}$$

This point can be verified to be optimum since

$$\nabla f_3 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

6.53 $f = x_1^2 + x_2^2 - 2x_1 - 4x_2 + 5$ = Quadratic function with $n = 2$

(a) order of n

(b) 2

(c) 2

(d) 1

(e) order of (n^2)

(f) Indefinite

(g) 2

(h) Indefinite

6.54 $f = (x_2 - x_1^2)^2 + (1 - x_1)^2$ = Non-quadratic with $n = 2$

(a) Indefinite

(b) Indefinite ; fast

(c) Indefinite ; fast

(d) Indefinite ; very fast

(e) Indefinite; slow

(f) Indefinite

(g) Indefinite; fast

(h) Indefinite

6.55 $f = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$

$$[A] = \begin{bmatrix} \partial^2 f/\partial x_1^2 & \partial^2 f/\partial x_1 \partial x_2 \\ \partial^2 f/\partial x_1 \partial x_2 & \partial^2 f/\partial x_2^2 \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$$

(a) $\vec{S}_1^T [A] \vec{S}_2 = (-1 \quad 1) \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = -2 \neq 0$

$\vec{S}_1$ and $\vec{S}_2$ are not conjugate.

(b) $\vec{S}_1^T [A] \vec{S}_2 = (-1 \quad 1) \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 0$

$\vec{S}_1$ and $\vec{S}_2$ are conjugate.

6.56 Problem is:

Minimize $f = (x_1 + 2x_2 + 3x_3 - 14)^2 + (x_1 - x_2 + x_3 - 1)^2 + (3x_1 - 2x_2 + x_3 - 2)^2$

using Marquardt method.

Let $\alpha_1 = 10^4$, $c_1 = \frac{1}{4}$, $c_2 = 2$ and $\epsilon = 0.01$

$\vec{x}_1 = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}$, $f_1 = f(\vec{x}_1) = 201$

$$\nabla f = \begin{Bmatrix} \partial f/\partial x_1 \\ \partial f/\partial x_2 \\ \partial f/\partial x_3 \end{Bmatrix} = \begin{Bmatrix} 2(x_1 + 2x_2 + 3x_3 - 14) + 2(x_1 - x_2 + x_3 - 1) + 6(3x_1 - 2x_2 + x_3 - 2) \\ 4(x_1 + 2x_2 + 3x_3 - 14) - 2(x_1 - x_2 + x_3 - 1) - 4(3x_1 - 2x_2 + x_3 - 2) \\ 6(x_1 + 2x_2 + 3x_3 - 14) + 2(x_1 - x_2 + x_3 - 1) + 2(3x_1 - 2x_2 + x_3 - 2) \end{Bmatrix}$$

$$= \begin{Bmatrix} 22x_1 - 10x_2 + 14x_3 - 42 \\ -10x_1 + 18x_2 + 6x_3 - 46 \\ 14x_1 + 6x_2 + 22x_3 - 90 \end{Bmatrix}$$

$$\nabla f_1 = \begin{Bmatrix} -42 \\ -46 \\ -90 \end{Bmatrix}$$

Since $|\nabla f_1| = \{(-42)^2 + (-46)^2 + (-90)^2\}^{\frac{1}{2}} = 109.4532 > \epsilon$,

we compute

$$[J_1] = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_1 \partial x_3} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \frac{\partial^2 f}{\partial x_2 \partial x_3} \\ \frac{\partial^2 f}{\partial x_3 \partial x_1} & \frac{\partial^2 f}{\partial x_3 \partial x_2} & \frac{\partial^2 f}{\partial x_3^2} \end{bmatrix}_{\vec{x}_1} = \begin{bmatrix} 22 & -10 & 14 \\ -10 & 18 & 6 \\ 14 & 6 & 22 \end{bmatrix}$$

$$[A] = [J_1] + \alpha_1 [I] = \begin{bmatrix} 22 & -10 & 14 \\ -10 & 18 & 6 \\ 14 & 6 & 22 \end{bmatrix} + \begin{bmatrix} 10000 & 0 & 0 \\ 0 & 10000 & 0 \\ 0 & 0 & 10000 \end{bmatrix}$$

$$= \begin{bmatrix} 10022 & -10 & 14 \\ -10 & 10018 & 6 \\ 14 & 6 & 10022 \end{bmatrix}$$

$$[A]^{-1} = 10^{-6} \begin{bmatrix} 99.780780 & 0.099685018 & -0.13944612 \\ 0.099685018 & 99.820463 & -0.05990006 \\ -0.1394461 & -0.0599 0006 & 99.780715 \end{bmatrix}$$

$$\vec{x}_2 = \vec{x}_1 - [A]^{-1} \nabla f_1$$

$$= \begin{Bmatrix} 0.004182828 1 \\ 0.004590537 1 \\ 0.008971652 2 \end{Bmatrix}$$

$f_2 = 199.80756$

since $f_2 < f_1$, we set $\alpha_2 = c_1 \alpha_1 = 2500$ and proceed to next iteration.

$$\nabla f_2 = \begin{Bmatrix} -41.828281 \\ -45.905369 \\ -89.716522 \end{Bmatrix}$$

Since $|\nabla f_2| = 109.11445 > \epsilon$, we compute

$$[J_2] = \begin{bmatrix} 22 & -10 & 14 \\ -10 & 18 & 6 \\ 14 & 6 & 22 \end{bmatrix}$$

$$[A] = [J_2] + \alpha_2 [I] = \begin{bmatrix} 2522 & -10 & 14 \\ -10 & 2518 & 6 \\ 14 & 6 & 2522 \end{bmatrix}$$

$$[A]^{-1} = 10^{-5}\begin{bmatrix} 39.652921 & 0.15800325 & -0.22049521 \\ 0.15800324 & 39.714912 & -0.09536143 \\ -0.22049521 & -0.09536143 & 39.652523 \end{bmatrix}$$

$$\vec{x}_3 = \vec{x}_2 - [A]^{-1}\nabla f_2$$

$$= \begin{Bmatrix} 0.020643675 \\ 0.022802349 \\ 0.044410511 \end{Bmatrix}$$

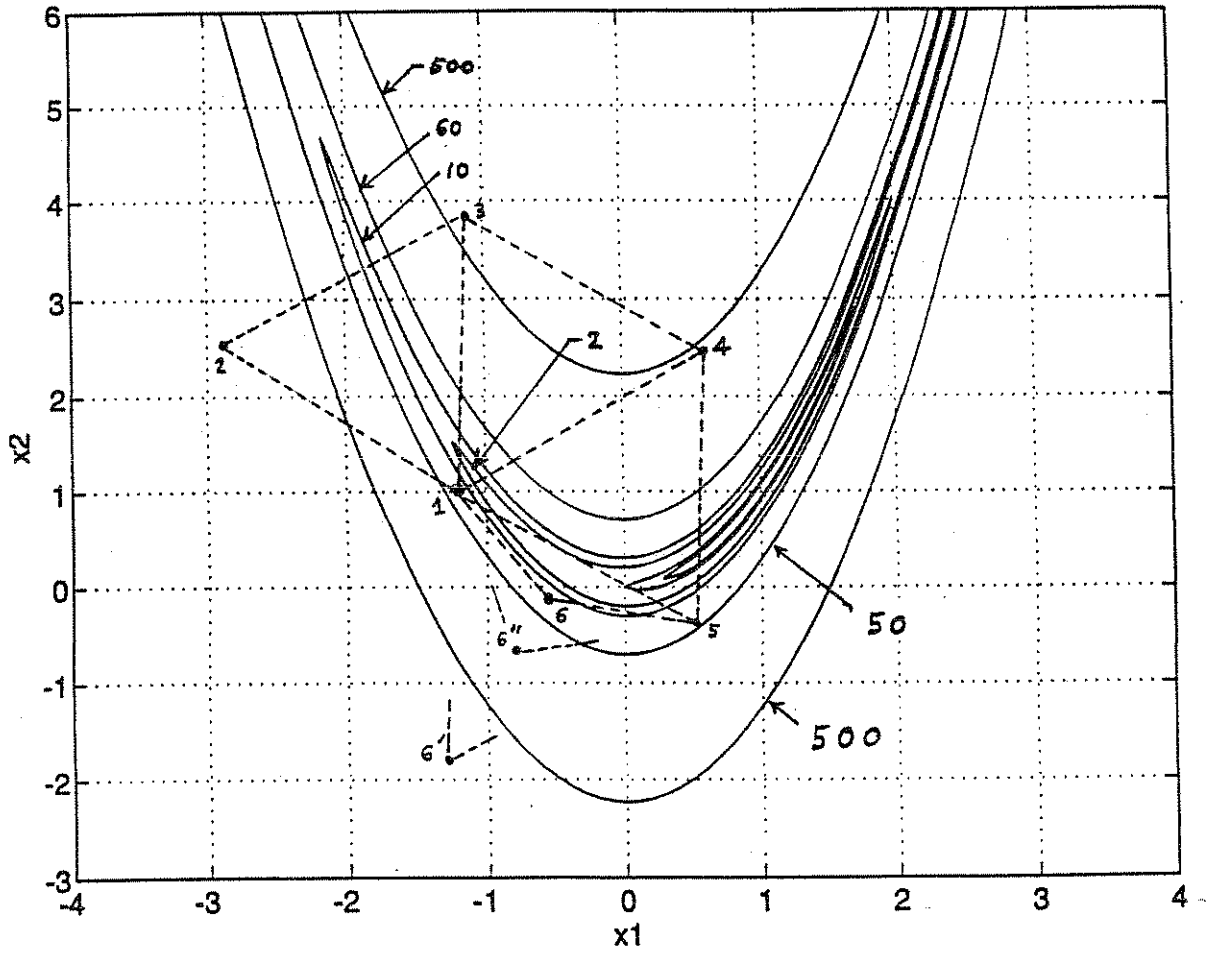
$$f_3 = 195.13237$$

$$\nabla f_3 = \begin{Bmatrix} -41.152115 \\ -45.529530 \\ -88.597145 \end{Bmatrix} \neq \vec{0}$$

Since $f_3 < f_2$, we set $\alpha_3 = c_1 \alpha_2 = 625$ and proceed to next iteration.

6.57 Minimize $f = 100\,(x_2 - x_1^2)^2 + (1 - x_1)^2$ using simplex method.

The procedure is shown graphically in the following figure. The simplex obtained after three steps is given by the vertices 1, 5 and 6 with $f_1 \simeq 24$, $f_5 \simeq 45$ and $f_6 \simeq 22$.



6.63

```
function p7
x0(1)=0;x0(2)=0;
fx=@(x)+(x(1)+2*x(2)-7)^2+(2*x(1)+x(2)-5)^2;
pro.objective=fx;
pro.x0=x0;
pro.solver='fminunc';
pro.options=optimset('Display','iter');
[xv,fval,exitflag,output]=fminunc(pro);

f0=fx(x0);
fprintf('Starting point: {%6.4f, %6.4f}\n', x0(1), x0(2));
fprintf('objective function value at starting point: %6.4f\n', f0);
fprintf('optimal x* ={%6.4f, %6.4f}\n', xv(1),xv(2));
fprintf('Optimized objective function value: %6.4f\n', fval)
```

Output

```
Starting point: {0.0000, 0.0000}

objective function value at starting point: 74.0000

optimal x* ={1.0000, 3.0000}

Optimized objective function value: 0.0000
```

6.64

```
function p8
x0(1)=3;x0(2)=-1;x0(3)=0;x0(4)=1;
fx=@(x)+(x(1)+10*x(2))^2+5*(x(3)-x(4))^2+(x(2)-2*x(3))^4+10*(x(1)-x(4))^4;
pro.objective=fx;
pro.x0=x0;
pro.solver='fminunc';
pro.options=optimset('Display','iter');
[xv,fval,exitflag,output]=fminunc(pro);

f0=fx(x0);
fprintf('Starting point: {%6.4f,%6.4f,%6.4f,%6.4f}\n', x0(1),x0(2),x0(3),x0(4));
fprintf('objective function value at starting point: %6.4f\n', f0);
fprintf('optimal x* ={%6.4f,%6.4f,%6.4f,%6.4f}\n', xv(1),xv(2),xv(3),xv(4));
fprintf('Optimized objective function value: %6.4f\n', fval);
```

Output

```
Starting point: {3.0000,-1.0000,0.0000,1.0000}

objective function value at starting point: 215.0000

optimal x* ={0.0015,-0.0002,-0.0031,-0.0031}        Optimized objective function value: 0.0000
```

6.65

```
function p9
x0(1)=-1;x0(2)=0;x0(3)=0;
options=optimset('Display','iter');
[xv,fval,exitflag,output]=fminunc(@objfun,x0,options);

f0=objfun(x0);
fprintf('Starting point: {%6.4f,%6.4f,%6.4f}\n', x0(1),x0(2),x0(3));
fprintf('objective function value at starting point: %6.4f\n', f0);
fprintf('optimal x* ={%6.4f,%6.4f,%6.4f}\n', xv(1),xv(2),xv(3));
fprintf('Optimized objective function value: %6.4f\n', fval)
end

function f=objfun(x)
f=100*((x(3)-10*theta(x))^2+(sqrt(x(1)^2+x(2)^2)-1)^2)+x(3)^2

function y=theta(x)
if x(1)>0
   y=(atan(x(2)/x(1)))/(2*pi);
else
   y=0.5+(atan(x(2)/x(1)))/(2*pi);
end

end

end
```

Output

```
Starting point: {-1.0000,0.0000,0.0000}

objective function value at starting point: 2500.0000

optimal x* ={1.0000,-0.0000,-0.0000}

Optimized objective function value: 0.0000
```

6.66

```
function p10
x0(1)=0;x0(2)=1;x0(3)=2;

options=optimset('Display','iter');
[xv,fval,exitflag,output]=fminunc(@objfun,x0,options);

f0=objfun(x0);
fprintf('Starting point: {%6.4f,%6.4f,%6.4f}\n', x0(1),x0(2),x0(3));
fprintf('objective function value at starting point: %6.4f\n', f0);
fprintf('optimal x* ={%6.4f,%6.4f,%6.4f}\n', xv(1),xv(2),xv(3));
fprintf('Optimized objective function value: %6.4f\n', fval);

end
function f=objfun(x)
a=x(1)-x(2);
b=(x(1)+x(3))/x(2);
ff=1/(1+a^2)+sin(pi*x(2)*x(3)/2)+exp(-(b-2)^2);
f=-ff;
end
```

Output

```
Starting point: {0.0000,1.0000,2.0000}

objective function value at starting point: -1.5000

optimal x* ={1.0000,1.0000,1.0000}

Optimized objective function value: -3.0000
```

6.67

```
function p11
clear;clc;
warning off;
x0(1)=-3;x0(2)=-1;x0(3)=-3;x0(4)=-1;


options=optimset('TolFun',10e-5,'Display','iter');
[xv,fval,exitflag,output]=fminunc(@objfun,x0,options)

f0=objfun(x0);
fprintf('Starting point: {%6.4f,%6.4f,%6.4f,%6.4f}\n', x0(1),x0(2),x0(3),x0(4));
fprintf('objective function value at starting point: %6.4f\n', f0);
fprintf('optimal x* ={%6.4f,%6.4f,%6.4f,%6.4f}\n', xv(1),xv(2),xv(3),xv(4));
fprintf('Optimized objective function value: %6.4f\n', fval);

end

function f=objfun(x)
a=x(2)-x(1)^2;
b=(x(4)-x(3)^2);
f=(10*a)^2+(1-x(1))^2+90*(b^2)+(1-x(3))^2+10*(x(2)+x(4)-2)^2+0.1*(x(2)-x(4))*(6.149);
end
```

Output

```
Starting point: {-3.0000,-1.0000,-3.0000,-1.0000}

objective function value at starting point: 19192.0000

optimal x* ={0.9799,0.9597,1.0227,1.0473}

Optimized objective function value: -0.0523
```

Chapter 7
Nonlinear Programming III:
Constrained Optimization Techniques

7.1 Minimize $f(\vec{x}) = x_1^2 + 2x_2^2 - 2x_1x_2 - 14x_1 - 14x_2 + 10$

subject to

$$g_1(\vec{x}) = 4x_1^2 + x_2^2 - 25 \leq 0$$

using a graphical procedure.

Plotting of constraint:

$$x_2 = \pm\sqrt{25 - 4x_1^2}$$

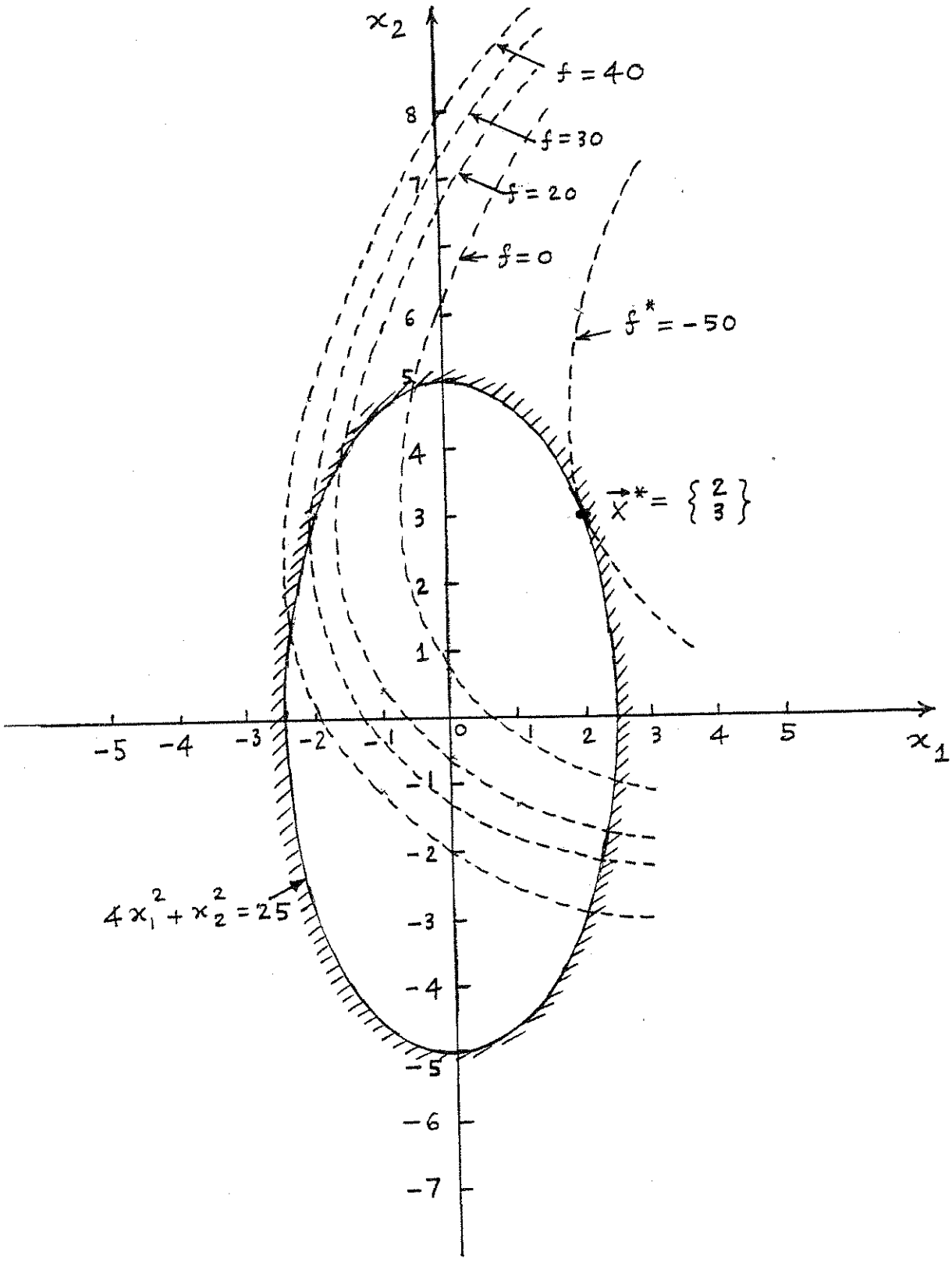
x_1	0	± 2.5	± 2.0	± 1.5	± 1.0	± 0.5
x_2	± 5	0	± 3.0	± 4.0	± 4.5826	± 4.8990

Plotting of objective function contours:

$$2x_2^2 + x_2(-2x_1 - 14) + (x_1^2 - 14x_1 + 10 - f) = 0$$

$$x_2 = \frac{(2x_1 + 14) \pm \sqrt{(2x_1 + 14)^2 - 8(x_1^2 - 14x_1 + 10 - f)}}{4}$$

$f = -50$:	x_1	2.0	3.0		
	x_2	6.0, 3.0	8.39, 1.61		
$f = -10$:	x_1	0.0	1.0	2.0	3.0
	x_2	5.0, 2.0	7.53, 0.464	9.22, −0.217	10.61, −0.612


x_2
$f = 40$
$f = 30$
$f = 20$
$f = 0$
$f^* = -50$
$\vec{x}^* = \begin{Bmatrix} 2 \\ 3 \end{Bmatrix}$
$4x_1^2 + x_2^2 = 25$
x_1
8
7
6
5
4
3
2
1
-1
-2
-3
-4
-5
-6
-7
-5
-4
-3
-2
-1
0
1
2
3
4
5

7.2 Welded beam problem:

```
feasible design vector:

        0.5750          2.5750          7.5250          0.5750

values of constraint functions:

   -0.1442E+04    -0.1452E+05     0.0000E+00    -0.1515E+01
   -0.4500E+00    -0.2410E+00    -0.6694E+05


feasible design vector:

        0.5750          2.5750         10.0000          0.5750

values of constraint functions:

   -0.4142E+04    -0.2123E+05     0.0000E+00    -0.3802E+00
   -0.4500E+00    -0.2462E+00    -0.8233E+05


feasible design vector:

        0.5750          5.0500          5.0500          1.0500

values of constraint functions:

   -0.3472E+04    -0.1118E+05    -0.4750E+00    -0.1057E+00
   -0.4500E+00    -0.2338E+00    -0.3185E+06


feasible design vector:

        0.5750          5.0500          7.5250          0.5750

values of constraint functions:

   -0.6498E+04    -0.1452E+05     0.0000E+00    -0.9998E+00
   -0.4500E+00    -0.2410E+00    -0.6694E+05


feasible design vector:

        0.5750          7.5250          7.5250          0.5750

values of constraint functions:

   -0.8325E+04    -0.1452E+05     0.0000E+00    -0.4846E+00
   -0.4500E+00    -0.2410E+00    -0.6694E+05


feasible design vector:

        1.0500          2.5750          5.0500          1.0500

values of constraint functions:

   -0.4812E+04    -0.1118E+05     0.0000E+00    -0.6562E+00
   -0.9250E+00    -0.2338E+00    -0.3185E+06


feasible design vector:

        1.0500          5.0500          5.0500          1.0500

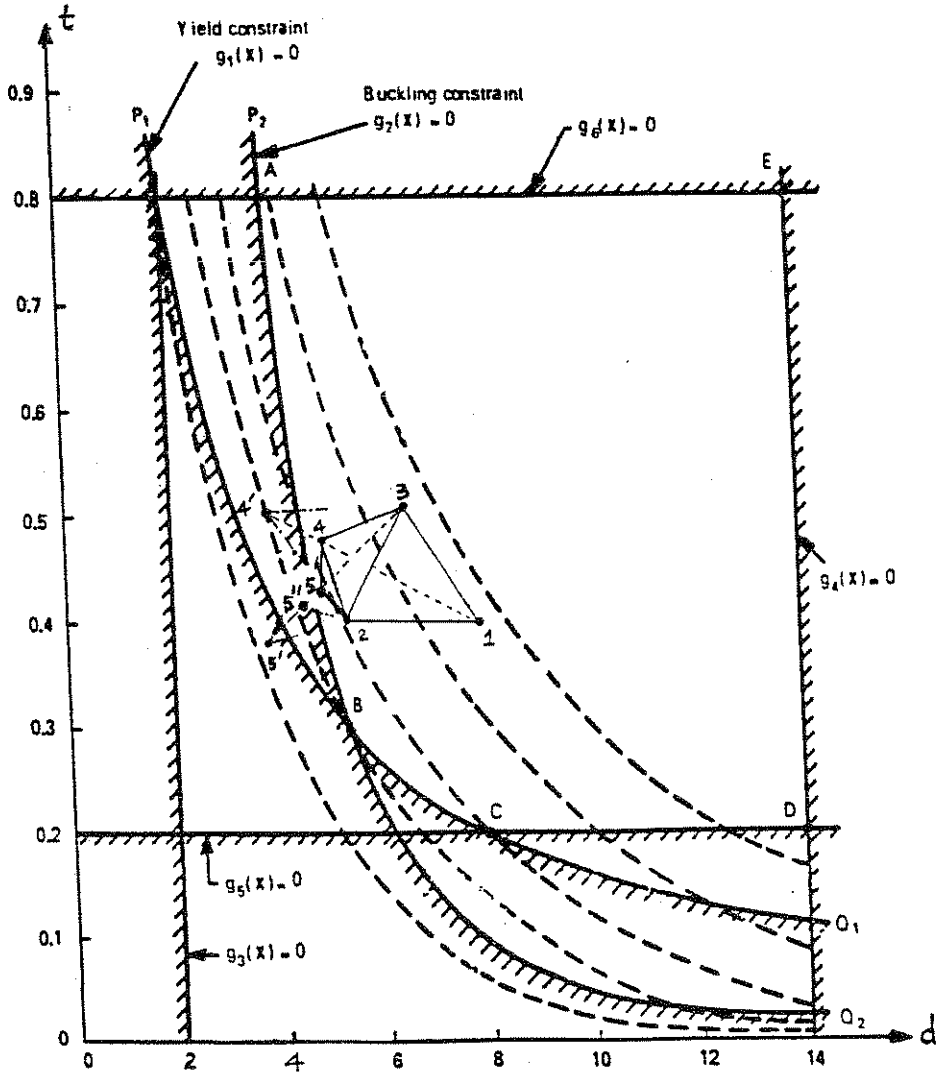
values of constraint functions:

   -0.8471E+04    -0.1118E+05     0.0000E+00    -0.2484E-01
   -0.9250E+00    -0.2338E+00    -0.3185E+06
```

7.3 Three-bar truss problem:

```
feasible design vector and constraint values:
    0.1325E+01    0.1325E+01       -0.9327E+01   -0.1375E+02   -0.1058E+02
feasible design vector and constraint values:
    0.1325E+01    0.2550E+01       -0.1042E+02   -0.1594E+02   -0.9481E+01
feasible design vector and constraint values:
    0.1325E+01    0.3775E+01       -0.1095E+02   -0.1700E+02   -0.8954E+01
feasible design vector and constraint values:
    0.1325E+01    0.5000E+01       -0.1126E+02   -0.1762E+02   -0.8644E+01
feasible design vector and constraint values:
    0.2550E+01    0.1325E+01       -0.1382E+02   -0.1548E+02   -0.1334E+02
feasible design vector and constraint values:
    0.2550E+01    0.2550E+01       -0.1445E+02   -0.1675E+02   -0.1270E+02
feasible design vector and constraint values:
    0.2550E+01    0.3775E+01       -0.1481E+02   -0.1746E+02   -0.1235E+02
feasible design vector and constraint values:
    0.2550E+01    0.5000E+01       -0.1504E+02   -0.1792E+02   -0.1212E+02
feasible design vector and constraint values:
    0.3775E+01    0.1325E+01       -0.1558E+02   -0.1646E+02   -0.1412E+02
feasible design vector and constraint values:
    0.3775E+01    0.2550E+01       -0.1600E+02   -0.1729E+02   -0.1371E+02
feasible design vector and constraint values:
    0.3775E+01    0.3775E+01       -0.1625E+02   -0.1781E+02   -0.1345E+02
feasible design vector and constraint values:
    0.3775E+01    0.5000E+01       -0.1643E+02   -0.1816E+02   -0.1327E+02
feasible design vector and constraint values:
    0.5000E+01    0.1325E+01       -0.1655E+02   -0.1709E+02   -0.1445E+02
feasible design vector and constraint values:
    0.5000E+01    0.2550E+01       -0.1684E+02   -0.1768E+02   -0.1416E+02
feasible design vector and constraint values:
    0.5000E+01    0.3775E+01       -0.1703E+02   -0.1807E+02   -0.1397E+02
feasible design vector and constraint values:
    0.5000E+01    0.5000E+01       -0.1717E+02   -0.1834E+02   -0.1383E+02
```

7.4 The procedure is shown graphically in the following figure. Starting from point 1 ($d = 8$, $t = 0.4$), the initial simplex is generated as 123. Since f_1 is the largest function value, the reflection process generates the point 4′. Since 4′ is in infeasible space, we use contraction and obtain 234 as the new simplex. Noting that f_3 is largest, the next reflection process yields the infeasible point 5′. The contraction process then gives the infeasible point 5″ first and then the feasible point 5. The next simplex is given by 245.



7.5 (a) $g = \underset{\sim}{f} = 3x_1^2 - 2x_1x_2 + x_2^2 - 1$

$$\frac{\partial \underset{\sim}{f}}{\partial x_1} = 6x_1 - 2x_2 \quad , \quad \frac{\partial \underset{\sim}{f}}{\partial x_2} = -2x_1 + 2x_2$$

$$\underset{\sim}{f}(\vec{x}) \simeq \underset{\sim}{f}(\vec{x}_0) + \left.\frac{\partial \underset{\sim}{f}}{\partial x_1}\right|_{\vec{x}_0} (x_1 - x_{10}) + \left.\frac{\partial \underset{\sim}{f}}{\partial x_2}\right|_{\vec{x}_0} (x_2 - x_{20})$$

At $\vec{x}_0 = \begin{Bmatrix} -2 \\ 2 \end{Bmatrix}$, $\underset{\sim}{f}_0 = 3(-2)^2 - 2(-2)(2) + 2^2 - 1 = 23$

$$\left.\frac{\partial \underset{\sim}{f}}{\partial x_1}\right|_{\vec{x}_0} = 6(-2) - 2(2) = -16 \; ; \quad \left.\frac{\partial \underset{\sim}{f}}{\partial x_2}\right|_{\vec{x}_0} = -2(-2) + 2(2) = 8$$

$$\underset{\sim}{f} \simeq 23 - 16(x_1 + 2) + 8(x_2 - 2) =$$

$\therefore$ LP problem is:

Minimize $f = x_1 - x_2$

subject to

$$\underset{\sim}{g} = -25 - 16x_1 + 8x_2 \leq 0$$

(b) $\underset{\sim}{g} = 0$ at $\left(x_1 = 0, \; x_2 = \frac{25}{8}\right)$, $\left(x_2 = 0, \; x_1 = -\frac{25}{16}\right)$

$f = x_1 - x_2$, $f = 0$ at $(0,0), (1,1), \cdots$

$f = 2$ at $(2,0), (0,-2), \cdots$

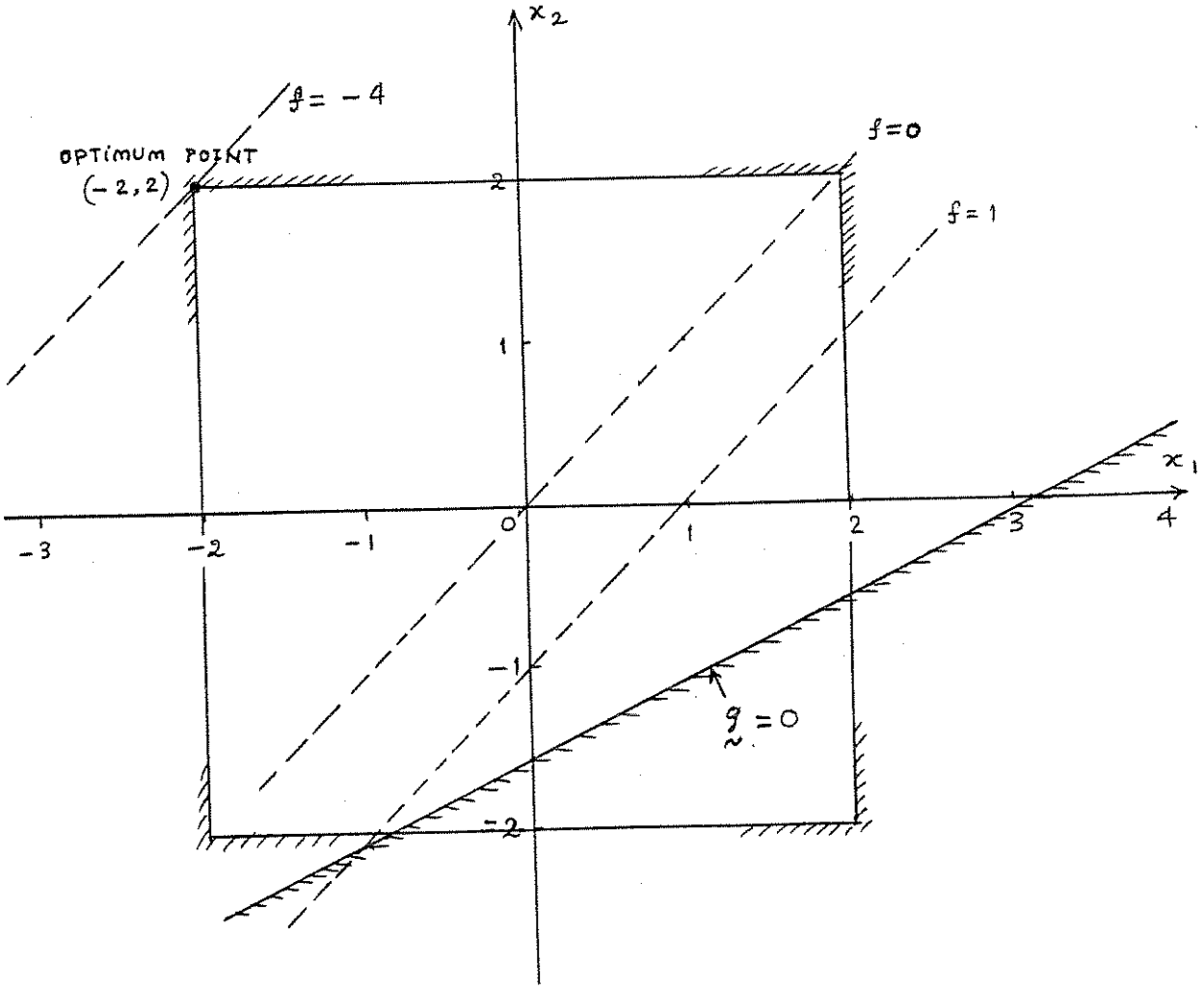
To avoid unbounded solution, we add the constraints,

$-2 \leq x_i \leq 2$, $i = 1, 2$.

Graphical solution is shown in the figure.

optimum solution: $(-2, 2)$

$g(0,0) = +23 \Rightarrow$ original constraint is violated.


x_2
$f = -4$
OPTIMUM POINT
$(-2, 2)$
$f = 0$
$f = 1$
x_1
$g = 0$
-3
-2
-1
0
1
2
3
4
2
1
-1
-2

7.6

$f = 2x_1^3 + 15x_2^2 - 8x_1x_2 - 4x_1 + 10 \quad ; \quad \vec{x}_0 = \begin{Bmatrix} 1 \\ -2 \end{Bmatrix}$

$\frac{\partial f}{\partial x_1} = 6x_1^2 - 8x_2 - 4 \quad , \quad \frac{\partial f}{\partial x_2} = 30x_2 - 8x_1$

$\frac{\partial^2 f}{\partial x_1^2} = 12x_1 \quad , \quad \frac{\partial^2 f}{\partial x_1 \partial x_2} = -8 \quad , \quad \frac{\partial^2 f}{\partial x_2^2} = 30$

Linear: $f(x_1, x_2) \simeq f|_0 + \frac{\partial f}{\partial x_1}\Big|_0 (x_1 - x_{10}) + \frac{\partial f}{\partial x_2}\Big|_0 (x_2 - x_{20})$

$\simeq 84 + 18(x_1 - 1) - 68(x_2 + 2)$

$\simeq 18x_1 - 68x_2 - 70$

Quadratic: $f(x_1, x_2) \simeq f|_0 + \frac{\partial f}{\partial x_1}\Big|_0 (x_1 - x_{10}) + \frac{\partial f}{\partial x_2}\Big|_0 (x_2 - x_{20})$

$+ \frac{1}{2!} \frac{\partial^2 f}{\partial x_1^2}\Big|_0 (x_1 - x_{10})^2 + \frac{1}{2!} \frac{\partial^2 f}{\partial x_2^2}\Big|_0 (x_2 - x_{20})^2$

$+ 2 \cdot \frac{1}{2!} \frac{\partial^2 f}{\partial x_1 \partial x_2}\Big|_0 (x_1 - x_{10})(x_2 - x_{20})$

i.e.,

$f(x_1, x_2) \simeq 84 + 18(x_1 - 1) - 68(x_2 + 2) + \frac{1}{2}(12)(x_1 - 1)^2$

$+ 2 \cdot \left(\frac{1}{2}\right)(-8)(x_1 - 1)(x_2 + 2) + \frac{1}{2}(30)(x_2 + 2)^2$

$\simeq 18x_1 - 68x_2 - 70 + 6(x_1^2 - 2x_1 + 1) - 8(x_1x_2$

$- x_2 + 2x_1 - 2) + 15(x_2^2 + 4x_2 + 4)$

$\simeq 12 - 10x_1 + 6x_1^2 - 8x_1x_2 + 15x_2^2$

$g_1 = x_1^2 + x_1x_2 + 1 = 0$

$\frac{\partial g_1}{\partial x_1}\Big|_0 = 2x_1 + x_2\Big|_0 = 0 \quad ; \quad \frac{\partial g_1}{\partial x_2}\Big|_0 = x_1\Big|_0 = 1$

$g_1|_0 = 0$

Linear: $g_1(x_1, x_2) \simeq g_1|_0 + \frac{\partial g_1}{\partial x_1}\Big|_0 (x_1 - x_{10}) + \frac{\partial g_1}{\partial x_2}\Big|_0 (x_2 - x_{20})$

$\simeq 0 + 0 + 1 \cdot (x_2 + 2)$

$\simeq x_2 + 2$

$$g_2 = 4x_1 - x_2^2 - 4 \le 0$$

$$g_2|_0 = 4(1) - (-2)^2 - 4 = -4$$

$$\left.\frac{\partial g_2}{\partial x_1}\right|_0 = 4 \;, \quad \left.\frac{\partial g_2}{\partial x_2}\right|_0 = -2x_2 = 4$$

Linear: $$g_2(x_1, x_2) \simeq g_2|_0 + \left.\frac{\partial g_2}{\partial x_1}\right|_0 (x_1 - x_{10}) + \left.\frac{\partial g_2}{\partial x_2}\right|_0 (x_2 - x_{20})$$

$$\simeq -4 + 4(x_1 - 1) + 4(x_2 + 2)$$

$$\simeq 4x_1 + 4x_2$$

Quadratic programming problem:

Minimize $$f(\vec{x}) = 12 - 10x_1 + 6x_1^2 - 8x_1x_2 + 15x_2^2$$

subject to $$x_2 + 2 = 0$$

$$x_1 + x_2 \le 0$$

Linear programming problem:

Minimize $$f(\vec{x}) = 18x_1 - 68x_2 - 70$$

subject to $$x_2 + 2 = 0$$

$$x_1 + x_2 \le 0$$

7.7

$$g_1 = \frac{20(x_2 + \sqrt{2}\,x_1)}{2x_1x_2 + \sqrt{2}\,x_1^2} - 20 \le 0 \quad ; \quad \vec{x}_0 = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$$

$$g_1|_{\vec{x}_0} = \frac{20(1+\sqrt{2})}{2+\sqrt{2}} - 20 = -5.8577$$

$$\frac{\partial g_1}{\partial x_1} = \frac{(2x_1x_2 + \sqrt{2}\,x_1^2)(20\sqrt{2}) - 20(x_2 + \sqrt{2}\,x_1)(2x_2 + 2\sqrt{2}\,x_1)}{(2x_1x_2 + \sqrt{2}\,x_1^2)^2}$$

$$\left.\frac{\partial g_1}{\partial x_1}\right|_{\vec{x}_0} = \frac{(2+\sqrt{2})(20\sqrt{2}) - 20(1+\sqrt{2})(2+2\sqrt{2})}{(2+\sqrt{2})^2} = -11.7157$$

$$\frac{\partial g_1}{\partial x_2} = \frac{(2x_1x_2 + \sqrt{2}\,x_1^2)(20) - 20(x_2 + \sqrt{2}\,x_1)(2x_1)}{(2x_1x_2 + \sqrt{2}\,x_1^2)^2}$$

$$\left.\frac{\partial g_1}{\partial x_2}\right|_{\vec{x}_0} = \frac{(2+\sqrt{2})(20) - 20(1+\sqrt{2})(2)}{(2+\sqrt{2})^2} = -2.4265$$

$$g_2 = \frac{20\,x_2}{2x_1x_2 + \sqrt{2}\,x_1^2} - 20 \leq 0$$

$$g_2\Big|_{\vec{x}_0} = \frac{20(1)}{2(1)(1) + \sqrt{2}\,(1)^2} - 20 = -14.1421$$

$$\frac{\partial g_2}{\partial x_1} = \frac{(2x_1x_2 + \sqrt{2}\,x_1^2)(0) - (20x_2)(2x_2 + 2\sqrt{2}\,x_1)}{(2x_1x_2 + \sqrt{2}\,x_1^2)^2}$$

$$\frac{\partial g_2}{\partial x_1}\Big|_{\vec{x}_0} = \frac{-20(2 + 2\sqrt{2})}{(2+\sqrt{2})^2} = -8.2843$$

$$\frac{\partial g_2}{\partial x_2} = \frac{(2x_1x_2 + \sqrt{2}\,x_1^2)(20) - 20x_2(2x_1)}{(2x_1x_2 + \sqrt{2}\,x_1^2)^2}$$

$$\frac{\partial g_2}{\partial x_2}\Big|_{\vec{x}_0} = \frac{(2+\sqrt{2})(20) - 20(2)}{(2+\sqrt{2})^2} = 2.4264$$

Linearized constraints become:

$g_1 \Rightarrow \quad -5.8577 - 11.7157\,(x_1 - 1) - 2.4265\,(x_2 - 1) \leq 0$

or $\quad -11.7157\,x_1 - 2.4265\,x_2 + 6.8580 \leq 0$

$g_2 \Rightarrow \quad -14.1421 - 8.2843\,(x_1 - 1) + 2.4264\,(x_2 - 1) \leq 0$

or $\quad -8.2843\,x_1 + 2.4264\,x_2 - 8.2842 \leq 0$

LP problem:

Minimize $\quad f = 282.8\,x_1 + 100.0\,x_2$

subject to

$$-11.7157\,x_1 - 2.4265\,x_2 + 6.8580 \leq 0$$

$$-8.2843\,x_1 + 2.4264\,x_2 - 8.2842 \leq 0$$

$$0 \leq x_i \leq 0.3 \quad ; \quad i = 1, 2$$

7.8 Minimize $f = x_1^2 + x_2^2 - 6x_1 - 8x_2 + 10$

subject to $g_1 = 4x_1^2 + x_2^2 - 16 \le 0$

$g_2 = 3x_1 + 5x_2 - 15 \le 0$

$x_i \ge 0, \quad i = 1, 2$

step 1: $\vec{x}_1 = \begin{Bmatrix} x_{11} \\ x_{21} \end{Bmatrix} = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$ = feasible. set $i = 1$.

step 2: Linearize f, g_1 and g_2.

$f(\vec{x}_1) = -2, \quad g_1(\vec{x}_1) = -11, \quad g_2(\vec{x}_1) = -7$

$\frac{\partial f}{\partial x_1} = 2x_1 - 6, \quad \frac{\partial f}{\partial x_2} = 2x_2 - 8$

$\frac{\partial g_1}{\partial x_1} = 8x_1, \quad \frac{\partial g_1}{\partial x_2} = 2x_2, \quad \frac{\partial g_2}{\partial x_1} = 3, \quad \frac{\partial g_2}{\partial x_2} = 5$

$$f(\vec{x}) \simeq f(\vec{x}_1) + \left.\frac{\partial f}{\partial x_1}\right|_{\vec{x}_1} (x_1 - x_{11}) + \left.\frac{\partial f}{\partial x_2}\right|_{\vec{x}_1} (x_2 - x_{21})$$

$$= -2 + (-4)(x_1 - 1) + (-6)(x_2 - 1) = -4x_1 - 6x_2 + 8$$

$$g_1(\vec{x}) \simeq g_1(\vec{x}_1) + \left.\frac{\partial g_1}{\partial x_1}\right|_{\vec{x}_1} (x_1 - x_{11}) + \left.\frac{\partial g_1}{\partial x_2}\right|_{\vec{x}_1} (x_2 - x_{21})$$

$$= -11 + 8(x_1 - 1) + 2(x_2 - 1) = 8x_1 + 2x_2 - 21$$

$$g_2(\vec{x}) \simeq g_2(\vec{x}_1) + \left.\frac{\partial g_2}{\partial x_1}\right|_{\vec{x}_1} (x_1 - x_{11}) + \left.\frac{\partial g_2}{\partial x_2}\right|_{\vec{x}_1} (x_2 - x_{21})$$

$$= -7 + 3(x_1 - 1) + 5(x_2 - 1) = 3x_1 + 5x_2 - 15$$

step 3: Approximate LP problem:

Minimize $\tilde{f} = -4x_1 - 6x_2 + 8$

subject to $\tilde{g}_1 = 8x_1 + 2x_2 - 21 \le 0$ (1)

$\tilde{g}_2 = 3x_1 + 5x_2 - 15 \le 0$

$x_i \ge 0, \quad i = 1, 2$

step 4: Solve the approximate LP problem.
Graphical method is used (see Figure).

$\tilde{g}_1 = 0$ at $A = (x_1 = 0, x_2 = 10.5)$, $B = (x_1 = 2.625, x_2 = 0)$

$\tilde{g}_2 = 0$ at $C = (x_1 = 0,\ x_2 = 3)$ and $D = (x_1 = 5,\ x_2 = 0)$

$\tilde{f}_A = -55$, $\tilde{f}_C = -10$, $\tilde{f}_D = -12$

$\tilde{f} = -10$ at $(x_1 = 1,\ x_2 = 2\frac{1}{3})$

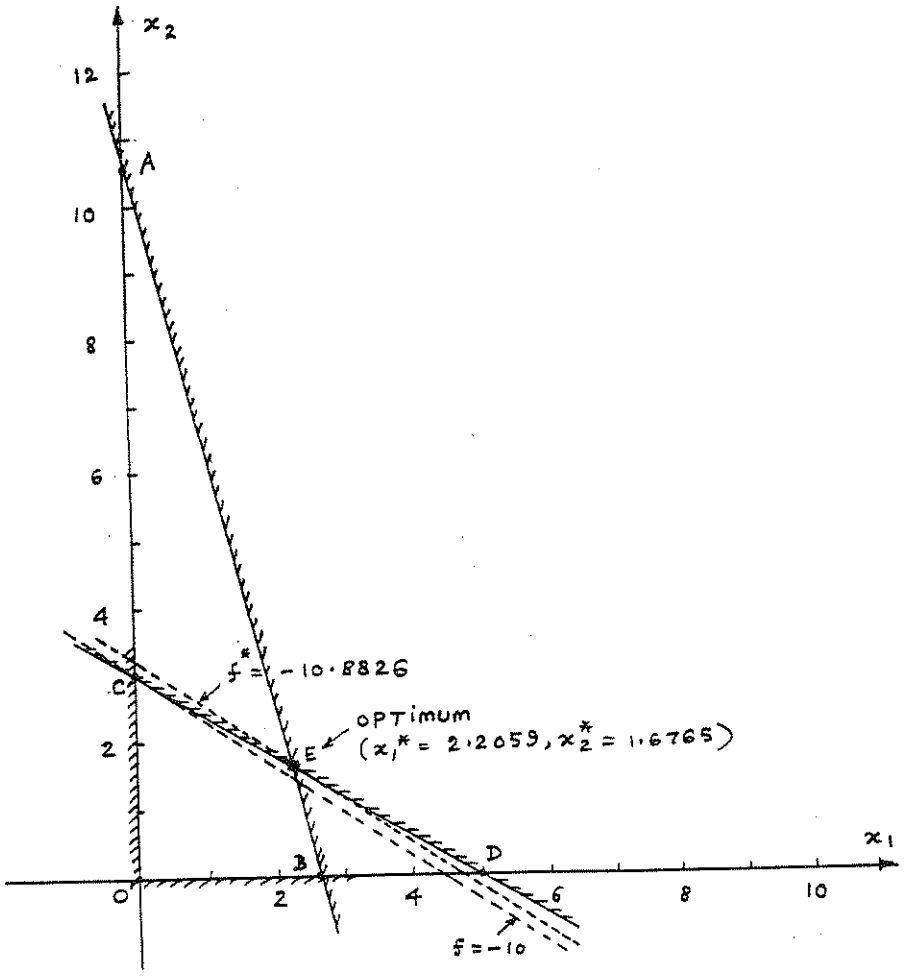
Point E = optimum = $(x_1 = 2.2059,\ x_2 = 1.6765)$

$\tilde{f}_E = \tilde{f}^* = -10.8826$

Step 5 : $\vec{x}_2 = \begin{Bmatrix} 2.2059 \\ 1.6765 \end{Bmatrix}$

$g_1(\vec{x}_2) = -8.3233 < 0$, $g_2(\vec{x}_2) = +0.0002 > 0$

since g_2 is violated, we relinearize g_2 about $\vec{x}_2$ and proceed to the next iteration.



7.9 Minimize $f = 9x_1^2 + 6x_2^2 + x_3^2 - 18x_1 - 12x_2 - 6x_3 - 8$

subject to $g_1 = x_1 + 2x_2 + x_3 - 4 \le 0$

$x_i \ge 0, \quad i = 1, 2, 3$

step 1: $\vec{X}_1 = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix} = \begin{Bmatrix} x_{11} \\ x_{21} \\ x_{31} \end{Bmatrix}$ = feasible. Set $i = 1$.

Step 2: Linearize f:

$$f(\vec{X}_1) = -8, \quad \left.\frac{\partial f}{\partial x_1}\right|_{\vec{X}_1} = 18x_1 - 18 = -18,$$

$$\left.\frac{\partial f}{\partial x_2}\right|_{\vec{X}_1} = 12x_2 - 12 = -12, \quad \left.\frac{\partial f}{\partial x_3}\right|_{\vec{X}_1} = 2x_3 - 6 = -6$$

$$\tilde{f}(\vec{X}) \simeq f(\vec{X}_1) + \sum_{i=1}^{3} \left.\frac{\partial f}{\partial x_i}\right|_{\vec{X}_1} (x_i - x_{i1})$$

$$= -8 + (-18)(x_1 - 0) - 12(x_2 - 0) - 6(x_3 - 0)$$

$$= -18x_1 - 12x_2 - 6x_3 - 8$$

$$\tilde{g}_1(\vec{X}) = g_1(\vec{X}) = x_1 + 2x_2 + x_3 - 4 \le 0 \; ; \; g_1(\vec{X}_1) = -4$$

Step 3: Approximate LP problem:

Minimize $\tilde{f} = -18x_1 - 12x_2 - 6x_3 - 8$

Subject to $\tilde{g}_1 = x_1 + 2x_2 + x_3 - 4 \le 0 \qquad (1)$

$x_i \ge 0, \quad i = 1, 2, 3$

Step 4: Solve the approximate LP problem of Eq. (1):

Basic variables	x_1	x_2	x_3	x_4	b_i
x_4	(1)	2	1	1	4
$-f$	−18 ↑	−12	−6	0	8
x_1	1	2	1	1	4
$-f$	0	24	12	18	80

Optimum solution of LP problem:

$x_1 = 4, \quad x_2 = x_3 = x_4 = 0, \quad \tilde{f}^* = -80$

Step 5: $\vec{x}_2 = \begin{Bmatrix} 4 \\ 0 \\ 0 \end{Bmatrix}$, $f(\vec{x}_2) = 64$, $g_1(\vec{x}_2) = 0$

Since the constraint is satisfied, the current solution is optimum to the original problem.

$\therefore \vec{x}^* = \vec{x}_2 = \begin{Bmatrix} 4 \\ 0 \\ 0 \end{Bmatrix}$, $f^* = 64$.

7.10 We consider the minimization of weight of the truss.

Minimize $f_1(\vec{x}) = 2\sqrt{2}\, x_1 + x_2$

subject to

$$20\left(\frac{x_2 + \sqrt{2}\, x_1}{\sqrt{2}\, x_1^2 + 2 x_1 x_2}\right) - 20 \le 0$$

or $g_1 = x_2 + \sqrt{2}\, x_1 - \sqrt{2}\, x_1^2 - 2 x_1 x_2 \le 0$

$$20\left(\frac{1}{x_1 + \sqrt{2}\, x_2}\right) - 20 \le 0$$

or $g_2 = 1 - x_1 - \sqrt{2}\, x_2 \le 0$

$$20\left(\frac{x_2}{\sqrt{2}\, x_1^2 + 2 x_1 x_2}\right) - 15 \le 0$$

or $g_3 = 4 x_2 - 3\sqrt{2}\, x_1^2 - 6 x_1 x_2 \le 0$

$g_4 = 0.1 - x_1 \le 0$

$g_5 = 0.1 - x_2 \le 0$

$g_6 = x_1 - 5.0 \le 0$

$g_7 = x_2 - 5.0 \le 0$

Step 1: $\vec{x}_1 = \begin{Bmatrix} x_{11} \\ x_{21} \end{Bmatrix} = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$ = feasible. Set $i = 1$

$g_1(\vec{x}_1) = -1$, $g_2(\vec{x}_1) = -\sqrt{2}$, $g_3(\vec{x}_1) = -3\sqrt{2} - 2$

Step 2: Since f is linear, we linearize g_1, g_2 and g_3:

$$\frac{\partial g_1}{\partial x_1} = \sqrt{2} - 2\sqrt{2}\, x_1 - 2 x_2 = \sqrt{2} - 2\sqrt{2} - 2 = -3.4142$$

$$\frac{\partial g_1}{\partial x_2} = 1 - 2 x_1 = -1$$

$$\frac{\partial g_2}{\partial x_1} = -1, \quad \frac{\partial g_2}{\partial x_2} = -\sqrt{2} = -1.4142$$

$$\frac{\partial g_3}{\partial x_1} = -6\sqrt{2}\, x_1 - 6x_2 = -14.4853, \quad \frac{\partial g_3}{\partial x_2} = 4 - 6x_1 = -2$$

$$\tilde{g}_1 = g_1(\vec{x}_1) + \left.\frac{\partial g_1}{\partial x_1}\right|_{\vec{x}_1} (x_1 - x_{11}) + \left.\frac{\partial g_1}{\partial x_2}\right|_{\vec{x}_1} (x_2 - x_{21})$$

$$= -1 - 3.4142\,(x_1 - 1) - 1\,(x_2 - 1) = -3.4142\, x_1 - x_2 + 3.4142 \le 0$$

$$\tilde{g}_2 = g_2(\vec{x}_1) + \left.\frac{\partial g_2}{\partial x_1}\right|_{\vec{x}_1} (x_1 - x_{11}) + \left.\frac{\partial g_2}{\partial x_2}\right|_{\vec{x}_1} (x_2 - x_{21})$$

$$= -1.4142 - 1\,(x_1 - 1) - 1.4142\,(x_2 - 1)$$

$$= -x_1 - 1.4142\, x_2 + 0.5858 \le 0$$

$$\tilde{g}_3 = g_3(\vec{x}_1) + \left.\frac{\partial g_3}{\partial x_1}\right|_{\vec{x}_1} (x_1 - x_{11}) + \left.\frac{\partial g_3}{\partial x_2}\right|_{\vec{x}_1} (x_2 - x_{21})$$

$$= -6.2426 - 14.4853\,(x_1 - 1) - 2\,(x_2 - 1)$$

$$= -14.4853\, x_1 - 2x_2 - 10.2427 \le 0$$

Step 3: Approximate LP problem:

Minimize $f = 2.8284\, x_1 + x_2$

subject to

$$\tilde{g}_1 = -3.4142\, x_1 - x_2 + 3.4142 \le 0$$

$$\tilde{g}_2 = -x_1 - 1.4142\, x_2 + 0.5858 \le 0$$

$$\tilde{g}_3 = -14.4853\, x_1 - 2x_2 - 10.2427 \le 0$$

$$g_4 = 0.1 - x_1 \le 0$$

$$g_5 = 0.1 - x_2 \le 0$$

$$g_6 = x_1 - 5.0 \le 0$$

$$g_7 = x_2 - 5.0 \le 0$$

Step 4: Solve the approximate LP problem graphically.

$\tilde{g}_1 = 0$ at $(x_1 = 0,\ x_2 = 3.4142)$, $(x_1 = 1.0,\ x_2 = 0)$

$\tilde{g}_2 = 0$ at $(x_1 = 0,\ x_2 = 0.4142)$, $(x_1 = 0.5858,\ x_2 = 0)$

$\tilde{g}_3 = 0$ at $(x_1 = 0,\ x_2 = -5.1213)$, $(x_1 = -0.7071,\ x_2 = 0)$

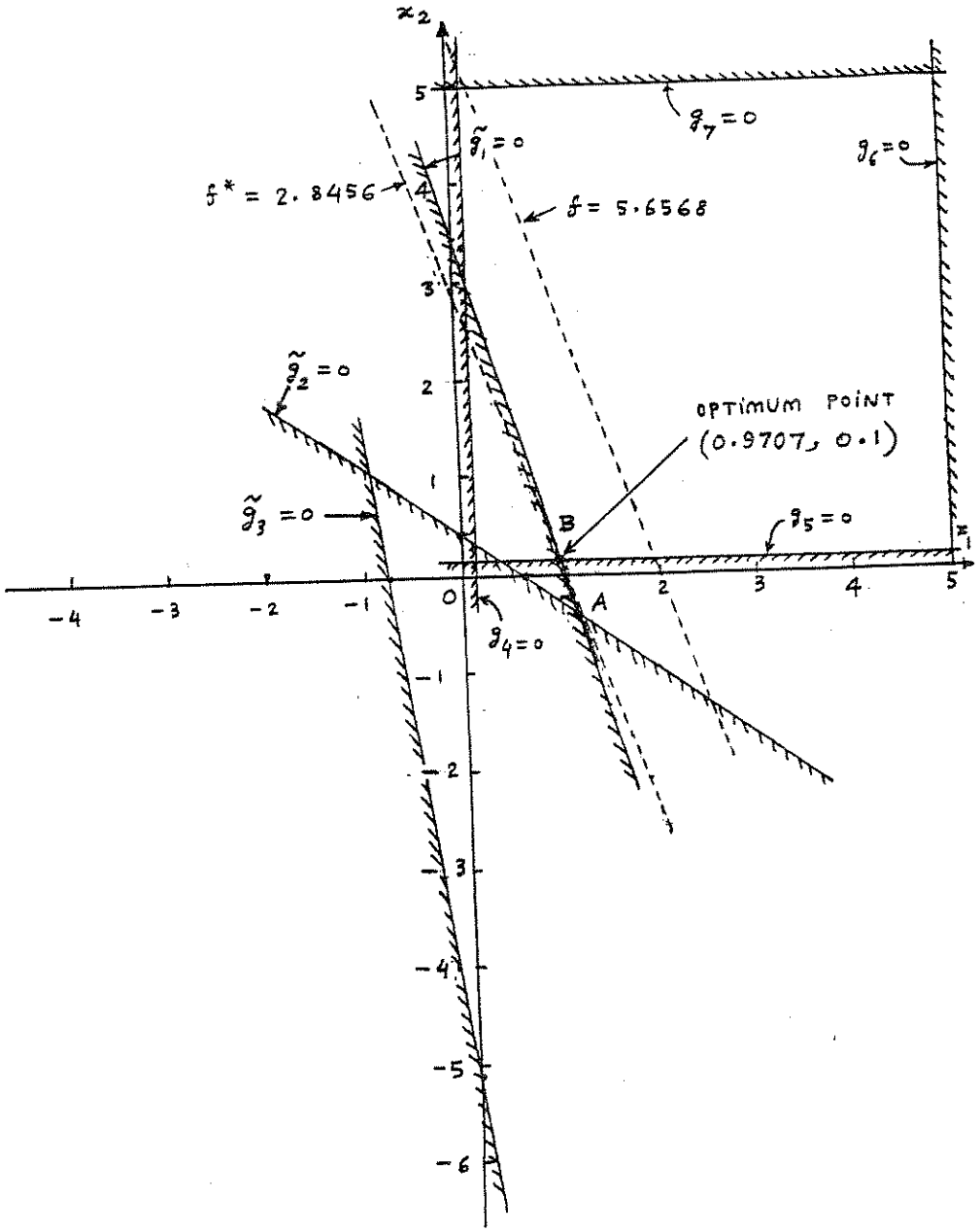
At (2,0), $f = 5.6568$, also at (0, 5.6568), $f = 5.6568$

B = intersection of $\tilde{g}_1 = 0$ and $g_5 = 0$:

$(x_1 = 0.9707,\ x_2 = 0.1)$ = optimum point

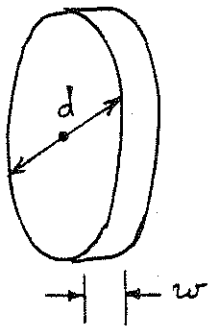
$f_B = 2.8456$

Step 5: $\vec{x}_2 = \begin{Bmatrix} 0.9707 \\ 0.1 \end{Bmatrix}$



7.11

$\vec{x} = \begin{Bmatrix} d \\ w \end{Bmatrix}$ r = radius



$I = \frac{1}{2} m r^2 = \frac{1}{2}\left(\rho \frac{\pi d^2}{4} w\right)\left(\frac{d}{2}\right)^2 = \frac{\pi}{32}\rho d^4 w$

$= \frac{\pi}{32}\left(\frac{0.283}{386.4}\right) d^4 w = 0.719035 \times 10^{-4}\, d^4 w$

$\omega = \frac{3000(2\pi)}{60} = 314.16$ rad/sec

$f = \frac{1}{2} I \omega^2 = \frac{1}{2}(0.719035 * 10^{-4})(314.16)^2 d^4 w$

or $f = 3.548312\, d^4 w$ (maximize f) (1)

$\sigma_r = \sigma_t = 0.283\left(\frac{3.3}{8 * 386.4}\right)(314.16)^2 d^2 = 29.817762\, d^2$

$\sigma_t^2 + \sigma_r^2 - \sigma_r \sigma_t = \sigma_t^2 = 889.098931\, d^4 \leq \sigma_{max}^2$

$\leq 400 * 10^6$

or $2.222747 \times 10^{-6} d^4 - 1 \leq 0$ (2)

weight $= mg = \frac{\gamma}{g}\left(\frac{\pi d^2}{4} w\right) g = \gamma \frac{\pi}{4} d^2 w$

$= 0.283\left(\frac{\pi}{4}\right) d^2 w = 0.2222268\, d^2 w$

$0.2222268\, d^2 w - 150 \leq 0$ (3)

$d \leq 25$ (4)

At $d = 15$ in and $w = 2$ in, the constraints, Eqs. (2), (3) and (4), are satisfied.

Constraint (2) can be rewritten as:

$d^4 \leq 1/(2.222747 \times 10^{-6}) = 44.98938 \times 10^4$

or $d \leq 25.8987$

This becomes redundant in view of (4).

∴ Problem becomes:

Maximize $f = 3.548312\, d^4 w$

subject to $g_1 = d^2 w - 674.8610 \leq 0$

$$g_2 = d - 25 \leq 0$$

Solution:

Step 1: $\vec{x}_1 = \begin{Bmatrix} d \\ w \end{Bmatrix} = \begin{Bmatrix} 15 \\ 2 \end{Bmatrix} \equiv \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix}$

$f(\vec{x}_1) = 359,266.59$

$g_1(\vec{x}_1) = -224.861, \quad g_2(\vec{x}_1) = -10$

Step 2: Linearize f and g_1:

$$\left.\frac{\partial f}{\partial x_1}\right|_{\vec{x}_1} = 14.193248\, d^3 w = 9.5804424 \times 10^4$$

$$\left.\frac{\partial f}{\partial x_2}\right|_{\vec{x}_1} = 3.548312\, d^4 = 17.96332 95 \times 10^4$$

$$\tilde{f} = f(\vec{x}_1) + \left.\frac{\partial f}{\partial x_1}\right|_{\vec{x}_1} \cdot (d-15) + \left.\frac{\partial f}{\partial x_2}\right|_{\vec{x}_1} \cdot (w-2)$$

$$= 35.9267 \times 10^4 + 9.5804 \times 10^4 \cdot (d-15) + 17.9633 \times 10^4 \cdot (w-2)$$

$$= 9.5804 \times 10^4\, d + 17.9633 \times 10^4\, w - 143.7059 \times 10^4$$

$$\left.\frac{\partial g_1}{\partial x_1}\right|_{\vec{x}_1} = 2dw = 60.0, \quad \left.\frac{\partial g_1}{\partial x_2}\right|_{\vec{x}_1} = d^2 = 225.0$$

$$\tilde{g}_1 = g_1(\vec{x}_1) + \left.\frac{\partial g_1}{\partial x_1}\right|_{\vec{x}_1} \cdot (d-15) + \left.\frac{\partial g_1}{\partial x_2}\right|_{\vec{x}_1} \cdot (w-2)$$

$$= -224.861 + 60(d-15) + 225(w-2)$$

$$= 60d + 225w - 1574.861$$

Step 3: Approximate LP problem:

Maximize $\tilde{f} = 10^4 (9.5804\, d + 17.9633\, w - 143.7059)$

subject to

$$\tilde{g}_1 = 60d + 225w - 1574.861 \leq 0$$

$$\tilde{g}_2 = g_2 = d - 25 \leq 0$$

step 4: Solve the approximate LP problem using a graphical procedure.

$\tilde{g}_1 = 0$ at $(d = 0, w = 6.9994)$, $(d = 26.2477, w = 0)$

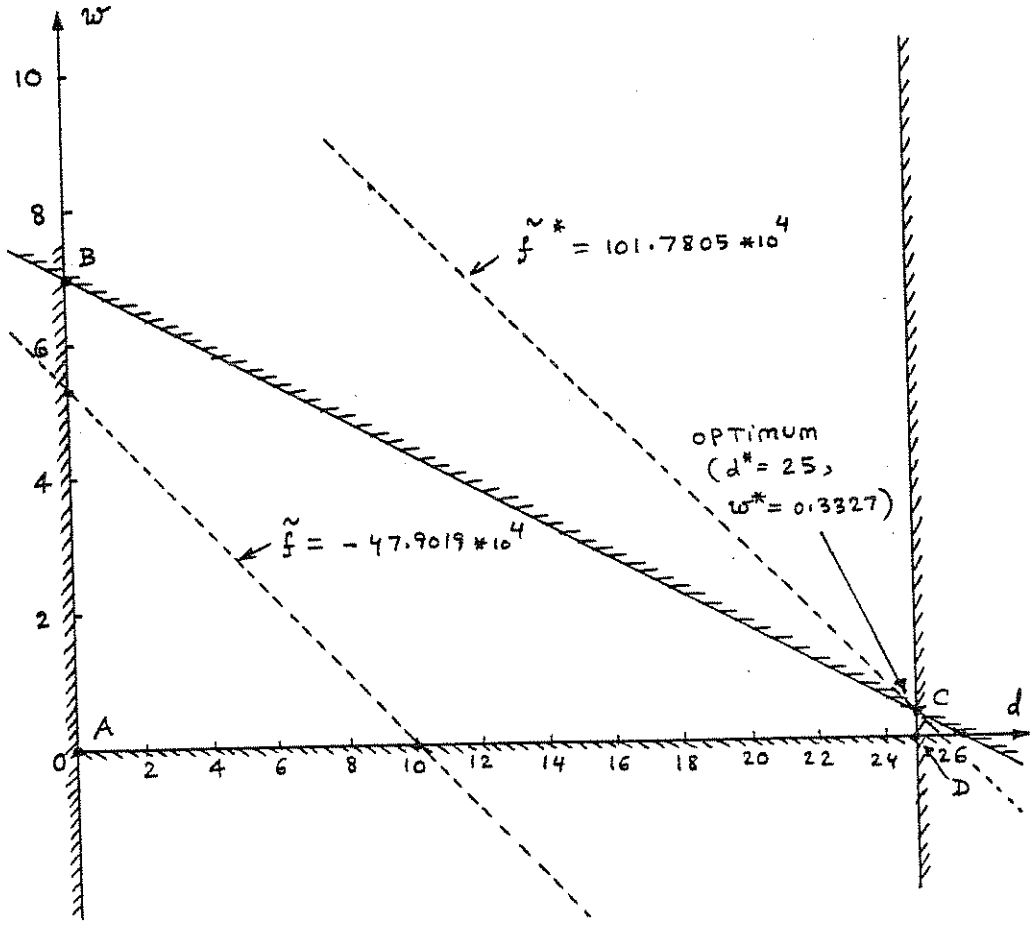
$\tilde{f} = -47.9019 \times 10^4$ at $(d = 10, w = 0)$, $(0, 5.3333)$

Point C : $(d = 25, w = 0.3327)$

$\tilde{f}_C = 101.7805 \times 10^4$

$\tilde{f} = 101.7805 \times 10^4$ at $(d = 10, w = 8.3327)$

Solution is shown in figure below.



Step 5 : $\vec{x}_2 = \begin{Bmatrix} 25 \\ 0.3327 \end{Bmatrix}$, $g_1(\vec{x}_2) = -466.9235 < 0$,

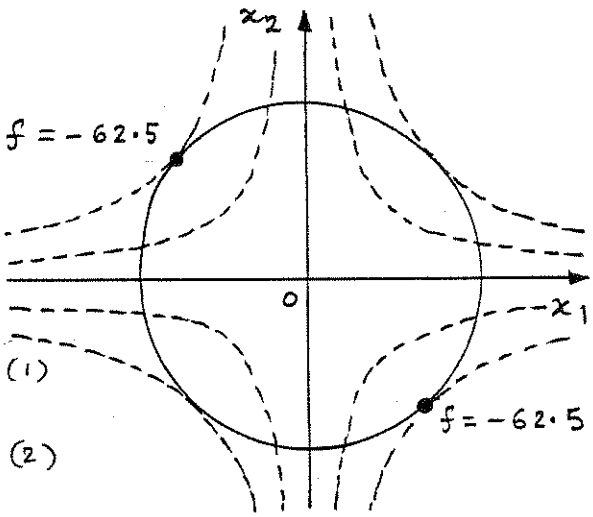
$g_2(\vec{x}_2) = 0$

Since $\vec{x}_2$ is feasible,

$$\vec{x}_{opt} = \vec{x}_2 = \begin{Bmatrix} 25.0 \text{ in} \\ 0.3327 \text{ in} \end{Bmatrix}$$

7.12 Minimize $f = 5x_1x_2$

subject to $g = 25 - x_1^2 - x_2^2 \geq 0$



Necessary conditions of optimality:

$$\frac{\partial f}{\partial x_1} + \lambda_1 \frac{\partial g}{\partial x_1} = 5x_2 - 2x_1\lambda_1 = 0 \quad (1)$$

$$\frac{\partial f}{\partial x_2} + \lambda_1 \frac{\partial g}{\partial x_2} = 5x_1 - 2x_2\lambda_1 = 0 \quad (2)$$

$$\lambda_1 g_1 = \lambda_1(25 - x_1^2 - x_2^2) = 0 \quad (3)$$

$$\lambda_1 \leq 0 \quad (4)$$

$$g \geq 0 \quad (5)$$

Eqs. (1) and (2): $\lambda = \frac{5}{2}\frac{x_2}{x_1} = \frac{5}{2}\frac{x}{x_2} \Rightarrow x_1^2 = x_2^2$

In order to have $\lambda_1 < 0$, x_1 and x_2 must be opposite in signs. Eq. (3) gives $\lambda_1(25 - 2x_1^2) = 0$.

$$x_1 = \pm (25/2)^{1/2} = \pm 3.5355$$

∴ optimum solution is: $x_1^* = -x_2^* = \pm 3.5355$;

$$f^* = -62.5$$

7.13 $\vec{X} = \begin{Bmatrix} 1 \\ 7 \end{Bmatrix}$;

$$g_1 = x_1 + 2x_2 - 15 = 1 + 2(7) - 15 = 0$$

$$g_2 = 1 - x_1 = 1 - 1 = 0$$

$$g_3 = x_1 - 10 = 1 - 10 = -9$$

$$g_4 = 1 - x_2 = 1 - 7 = -6$$

$$g_5 = x_2 - 10 = 7 - 10 = -3$$

∴ g_1 and g_2 are active.

$\vec{S}^T \nabla f < 0$ (usable)

$\vec{S}^T \nabla g_1 \leq 0$ (feasible)

$\vec{S}^T \nabla g_2 \leq 0$ (feasible)

$$\nabla f\big|_{\vec{x}} = \begin{Bmatrix} 2(x_1-5) \\ 2(x_2-5) \end{Bmatrix}_{\vec{x}} = \begin{Bmatrix} -8 \\ 4 \end{Bmatrix}$$

$$\nabla g_1\big|_{\vec{x}} = \begin{Bmatrix} 1 \\ 2 \end{Bmatrix}, \quad \nabla g_2\big|_{\vec{x}} = \begin{Bmatrix} -1 \\ 0 \end{Bmatrix}$$

$\therefore$ Conditions to be satisfied:

$$\begin{aligned} -8\delta_1 + 4\delta_2 &< 0 \\ \delta_1 + 2\delta_2 &\leq 0 \\ -\delta_1 &\leq 0 \end{aligned}$$

7.14 $\vec{x}_i = \begin{Bmatrix} -1 \\ 5 \end{Bmatrix}$, $\nabla f_i = \nabla f\big|_{\vec{x}_i} = \begin{Bmatrix} 2(x_1-1) \\ 2(x_2-5) \end{Bmatrix}_{\vec{x}_i} = \begin{Bmatrix} -4 \\ 0 \end{Bmatrix}$

$$g_1(\vec{x}_i) = -(-1)^2 + 5 - 4 = 0 \quad \text{(active)}$$

$$g_2(\vec{x}_i) = -(-1-2)^2 + 5 - 3 = -7 \quad \text{(not active)}$$

$$\nabla g_1(\vec{x}_i) = \begin{Bmatrix} -2x_1 \\ 1 \end{Bmatrix}_{\vec{x}_i} = \begin{Bmatrix} 2 \\ 1 \end{Bmatrix}$$

Direction finding problem (in Zoutendijk method):

Minimize $-\alpha$

subject to $\{\delta_1 \quad \delta_2\} \begin{Bmatrix} 2 \\ 1 \end{Bmatrix} + \theta_1 \alpha \leq 0$

and $\{\delta_1 \quad \delta_2\} \begin{Bmatrix} -4 \\ 0 \end{Bmatrix} + \alpha \leq 0$

i.e. Minimize $-\alpha$

subject to:

$$\begin{aligned} 2\delta_1 + \delta_2 + \theta_1 \alpha &\leq 0 \\ -4\delta_1 + \alpha &\leq 0 \end{aligned}$$

This can be restated in standard form as (with $\theta_1 = 1$):

Find $\begin{Bmatrix} t_1 \\ t_2 \\ \alpha \\ y_1 \\ y_2 \\ y_3 \\ y_4 \end{Bmatrix}$ to minimize $-\alpha$

subject to $2t_1 + t_2 + \alpha + y_1 = 3$

$-4t_1 + \alpha + y_2 = -4$

$t_1 + y_3 = 2$

$t_2 + y_4 = 2$

$t_1 \geq 0,\ t_2 \geq 0,\ \alpha \geq 0,\ y_i \geq 0,\ i = 1,2,3,4$

7.15 Minimize $f = (x_1-1)^2 + (x_2-5)^2$

subject to $g_1 = -x_1^2 + x_2 - 4 \leq 0$

$g_2 = -(x_1-2)^2 + x_2 - 3 \leq 0$

by Zoutendijk's method.

Step 1: $\vec{x}_1$ = initial feasible point = $\begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$

$f(\vec{x}_1) = 16,\quad g_1(\vec{x}_1) = -4,\quad g_2(\vec{x}_1) = -3$

Iteration number, $i = 1$.

Step 2: Since $g_j(\vec{x}_1) < 0$, $j = 1,2$, we take

$$\vec{s}_1 = -\nabla f(\vec{x}_1) = -\begin{Bmatrix} 2(x_1-1) \\ 2(x_2-5) \end{Bmatrix}_{\vec{x}_1} = \begin{Bmatrix} 0 \\ 8 \end{Bmatrix}$$

Normalized $\vec{s}_1 = \begin{Bmatrix} 0 \\ 1 \end{Bmatrix}$

Step 5: $\vec{x} = \vec{x}_1 + \lambda_1 \vec{s}_1 = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} + \lambda_1 \begin{Bmatrix} 0 \\ 1 \end{Bmatrix} = \begin{Bmatrix} 1 \\ 1+\lambda_1 \end{Bmatrix}$

$f(\vec{x}) = f(\lambda_1) = (1-1)^2 + (1+\lambda_1-5)^2 = (\lambda_1-4)^2$

$\frac{df}{d\lambda_1} = 2(\lambda_1 - 4) = 0$ or $\tilde{\lambda}_1^* = 4$

$\vec{x} = \vec{x}_1 + \tilde{\lambda}_1^* \vec{s}_1 = \begin{Bmatrix} 1 \\ 5 \end{Bmatrix}$

$g_1(\vec{x}) = 0,\quad g_2(\vec{x}) = 1 > 0$

Since g_2 is violated, we reduce λ arbitrarily, to $\tilde{\lambda}_1^* = 3$ so that

$\vec{x} = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} + 3 \begin{Bmatrix} 0 \\ 1 \end{Bmatrix} = \begin{Bmatrix} 1 \\ 4 \end{Bmatrix}$

$g_1(\vec{x}) = -1,\quad g_2(\vec{x}) = 0$

$\therefore \quad \lambda_1^* = \tilde{\lambda}_1^* = 3 \quad$ and $\quad \vec{x}_2 = \vec{x}_1 + \lambda_1^* \vec{s}_1 = \begin{Bmatrix} 1 \\ 4 \end{Bmatrix}$

Step 6: $f(\vec{x}_2) = 1$

Step 7: Test for convergence:

$$\left| \frac{f(\vec{x}_1) - f(\vec{x}_2)}{f(\vec{x}_1)} \right| = \left| \frac{16-1}{16} \right| = 0.9375$$

$$\| \vec{x}_1 - \vec{x}_2 \| = \{(1-1)^2 + (1-4)^2\}^{\frac{1}{2}} = 3.0$$

Assuming that convergence criteria are not satisfied, we go to step 8.

Step 8: Iteration number, $i = 2$.

step 2: Since $g_2(\vec{x}_2) = 0$, we go to step 3.

Step 3: We find a usable feasible direction, $\vec{s} = \begin{Bmatrix} s_1 \\ s_2 \end{Bmatrix}$, as follows:

$$\nabla g_2 \Big|_{\vec{x}_2} = \begin{Bmatrix} -2(x_1 - 2) \\ 1 \end{Bmatrix}_{\vec{x}_2} = \begin{Bmatrix} 2 \\ 1 \end{Bmatrix}$$

$$\nabla f \Big|_{\vec{x}_2} = \begin{Bmatrix} 2(x_1 - 1) \\ 2(x_2 - 5) \end{Bmatrix}_{\vec{x}_2} = \begin{Bmatrix} 0 \\ -2 \end{Bmatrix}$$

Problem:

Minimize $\quad -\alpha$

subject to

$$(s_1 \quad s_2) \begin{Bmatrix} 2 \\ 1 \end{Bmatrix} + \alpha \leq 0 \quad \text{or} \quad 2s_1 + s_2 + \alpha \leq 0$$

$$(s_1 \quad s_2) \begin{Bmatrix} 0 \\ -2 \end{Bmatrix} + \alpha \leq 0 \quad \text{or} \quad -2s_2 + \alpha \leq 0 \qquad (1)$$

$$-1 \leq s_i \leq 1, \quad i = 1, 2$$

The problem of Eq. (1) can be stated as a standard LP problem as follows:

Find $\{t_1, t_2, \alpha, y_1, y_2, y_3, y_4\}$

to minimize $\quad -\alpha$

subject to

$$
\begin{aligned}
2t_1 + t_2 + \alpha + y_1 &= 3 \\
-2t_2 + \alpha + y_2 &= -2 \\
t_1 + y_3 &= 2 \\
t_2 + y_4 &= 2
\end{aligned}
\qquad (2)
$$

$$t_i \geq 0 \quad (i = 1,2), \quad y_j \geq 0 \quad (j = 1, \ldots, 4), \quad \alpha \geq 0$$

Since the second constraint of Eq.(2) has a negative right hand side, we rewrite it as

$$2t_2 - \alpha - y_2 + y_5 = 2 \qquad (3)$$

where y_5 is an artificial variable. The infeasibility form (for Phase I of simplex method) is given by

$$w = y_5$$

which can be expressed as

$$-2t_2 + \alpha + y_2 + 2 = w \qquad (4)$$

The solution of the above LP problem can be found as (see next page for details):

$t_1^* = 0, \quad t_2^* = 1.667, \quad \alpha^* = 1.333, \quad y_1^* = 0, \quad y_2^* = 0,$

$y_3^* = 2.0, \quad y_4^* = 0.3333$

This gives $s_1 = t_1^* - 1 = -1, \quad s_2 = t_2^* - 1 = 0.667$

$$\therefore \vec{s} = \begin{Bmatrix} -1 \\ 0.667 \end{Bmatrix}$$

Step 4: Since $\alpha^* \neq 0$, we take the next usable feasible direction as

$$\vec{s_2} = \begin{Bmatrix} -1 \\ 0.667 \end{Bmatrix}$$

```
PHASE-I OF SIMPLEX METHOD
given system

  x 4:  0.2000E+01  0.1000E+01  0.1000E+01  0.1000E+01  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00  0.3000E+01
  x 8:  0.0000E+00  0.2000E+01 -0.1000E+01  0.0000E+00 -0.1000E+01  0.0000E+00
        0.0000E+00  0.1000E+01  0.2000E+01
  x 6:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01
        0.0000E+00  0.0000E+00  0.2000E+01
  x 7:  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.1000E+01  0.0000E+00  0.2000E+01
   -f:  0.0000E+00  0.0000E+00 -0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00  0.0000E+00
   -w:  0.0000E+00 -0.2000E+01  0.1000E+01  0.0000E+00  0.1000E+01  0.0000E+00
        0.0000E+00  0.0000E+00 -0.2000E+01

minimum value of ds:-0.20000000E+01  variable out of basis:     2
result of pivoting on element:     2     2


  x 4:  0.2000E+01  0.0000E+00  0.1500E+01  0.1000E+01  0.5000E+00  0.0000E+00
        0.0000E+00 -0.5000E+00  0.2000E+01
  x 2:  0.0000E+00  0.1000E+01 -0.5000E+00  0.0000E+00 -0.5000E+00  0.0000E+00
        0.0000E+00  0.5000E+00  0.1000E+01
  x 6:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01
        0.0000E+00  0.0000E+00  0.2000E+01
  x 7:  0.0000E+00  0.0000E+00  0.5000E+00  0.0000E+00  0.5000E+00  0.0000E+00
        0.1000E+01 -0.5000E+00  0.1000E+01
   -f:  0.0000E+00  0.0000E+00 -0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00  0.0000E+00
   -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.1000E+01  0.0000E+00

PHASE-II OF SIMPLEX METHOD
system at start of phase-II

  x 4:  0.2000E+01  0.0000E+00  0.1500E+01  0.1000E+01  0.5000E+00  0.0000E+00
        0.0000E+00  0.2000E+01
  x 2:  0.0000E+00  0.1000E+01 -0.5000E+00  0.0000E+00 -0.5000E+00  0.0000E+00
        0.0000E+00  0.1000E+01
  x 6:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01
        0.0000E+00  0.2000E+01
  x 7:  0.0000E+00  0.0000E+00  0.5000E+00  0.0000E+00  0.5000E+00  0.0000E+00
        0.1000E+01  0.1000E+01
   -f:  0.0000E+00  0.0000E+00 -0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00

minimum value of cs:-0.10000000E+01  variable out of basis:     3
result of pivoting on element:     1     3


  x 3:  0.1333E+01  0.0000E+00  0.1000E+01  0.6667E+00  0.3333E+00  0.0000E+00
        0.0000E+00  0.1333E+01
  x 2:  0.6667E+00  0.1000E+01  0.0000E+00  0.3333E+00 -0.3333E+00  0.0000E+00
        0.0000E+00  0.1667E+01
  x 6:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01
        0.0000E+00  0.2000E+01
  x 7: -0.6667E+00  0.0000E+00  0.0000E+00 -0.3333E+00  0.3333E+00  0.0000E+00
        0.1000E+01  0.3333E+00
   -f:  0.1333E+01  0.0000E+00  0.0000E+00  0.6667E+00  0.3333E+00  0.0000E+00
        0.0000E+00  0.1333E+01
```

Step 5: $\vec{x} = \vec{x}_2 + \lambda_2 \vec{s}_2 = \begin{Bmatrix} 1 - \lambda_2 \\ 4 + 0.667\lambda_2 \end{Bmatrix}$

$f(\vec{x}) = (1 - \lambda_2 - 1)^2 + (4 + 0.667\lambda_2 - 5)^2$

$= 1.4449\,\lambda_2^2 - 1.3334\,\lambda_2 + 1$

$\frac{df}{d\lambda_2} = 2.8898\,\lambda_2 - 1.3334 = 0 \quad , \quad \tilde{\lambda}_2^* = 0.4614$ (5)

$\vec{x} = \begin{Bmatrix} 1 - \tilde{\lambda}_2^* \\ 4 + 0.667\,\tilde{\lambda}_2^* \end{Bmatrix} = \begin{Bmatrix} 0.5386 \\ 4.3077 \end{Bmatrix}$

$g_1(\vec{x}) = +0.0176 > 0$

$g_2(\vec{x}) = -0.8280 < 0$

To make g_1 active at the new point $\vec{x}$, we try a value of $\tilde{\lambda}_2^* = 0.43833$ (95% of the value given by Eq. (5)).

This gives $\vec{x} = \begin{Bmatrix} 0.56167 \\ 4.29237 \end{Bmatrix}$

$g_1(\vec{x}) = -0.0231 \simeq 0$

$g_2(\vec{x}) = -0.77642 < 0$

Since $g_1(\vec{x}) \simeq 0$, we take $\lambda_2^* = \tilde{\lambda}_2^* = 0.43833$

and $\vec{x}_3 = \begin{Bmatrix} 0.56167 \\ 4.29237 \end{Bmatrix}$

This gives $f(\vec{x}_3) = 0.69287$.

7.16 Minimize $f = (x_1-1)^2 + (x_2-2)^2 - 4$

subject to $g_1 = x_1 + 2x_2 - 5 \le 0$

$g_2 = 4x_1 + 3x_2 - 10 \le 0$

$g_3 = 6x_1 + x_2 - 7 \le 0$

$x_i \ge 0, \; i = 1,2$

using Zoutendijk's method.

Step 1: $\vec{x}_1 = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$ = feasible. Set $i = 1$.

$f(\vec{x}_1) = -3, \quad g_1(\vec{x}_1) = -2 < 0, \quad g_2(\vec{x}) = -3 < 0, \quad g_3(\vec{x}) = 0$

Step 2: Since $g_3(\vec{x}_1) = 0$, we go to step 3.

Step 3: Find a usable feasible direction, $\vec{S} = \begin{Bmatrix} s_1 \\ s_2 \end{Bmatrix}$, by solving the following problem.

$$\nabla f(\vec{x}_1) = \begin{Bmatrix} 2(x_1-1) \\ 2(x_2-2) \end{Bmatrix} = \begin{Bmatrix} 0 \\ -2 \end{Bmatrix}$$

$$\nabla g_3 = \begin{Bmatrix} 6 \\ 1 \end{Bmatrix}$$

Problem: Minimize $-\alpha$

subject to

$$(s_1 \quad s_2)\begin{Bmatrix} 6 \\ 1 \end{Bmatrix} + \alpha \le 0 \quad \text{or} \quad 6s_1 + s_2 + \alpha \le 0$$

$$(s_1 \quad s_2)\begin{Bmatrix} 0 \\ -2 \end{Bmatrix} + \alpha \le 0 \quad \text{or} \quad -2s_2 + \alpha \le 0 \qquad (1)$$

$$-1 \le s_i \le 1, \quad i = 1,2$$

This problem can be stated as a standard LP problem as:

Find $\{t_1, \; t_2, \; \alpha, \; y_1, \; y_2, \; y_3, \; y_4\}$

to minimize $-\alpha$

subject to

$$6t_1 + t_2 + \alpha + y_1 = 7$$

$$-2t_2 + \alpha + y_2 = -2 \qquad (2)$$

$$t_1 + y_3 = 2$$

$$t_2 + y_4 = 2$$

$$t_i \geq 0 \quad (i = 1,2), \quad \alpha \geq 0, \quad y_i \geq 0 \quad (i = 1, \ldots, 4)$$

Since Eq. (2) has a negative right hand side, we rewrite it, using an artificial variable y_5, as

$$2t_2 - \alpha - y_2 + y_5 = 2$$

By defining the infeasibility form

$$w = y_5 = -2t_2 + \alpha + y_2 + 2$$

we state the final LP problem for the 2-phase simplex method as:

Find $\{t_1, t_2, \alpha, y_1, y_2, y_3, y_4, y_5\}$

to minimize $-\alpha$

subject to

$$6t_1 + t_2 + \alpha + y_1 = 7$$

$$2t_2 - \alpha - y_2 + y_5 = 2$$

$$t_1 + y_3 = 2$$

$$t_2 + y_4 = 2$$

with

$$w = -2t_2 + \alpha + y_2 + 2$$

The solution of this problem can be found as:

$t_1^* = 0$, $t_2^* = 2$, $\alpha^* = 2$, $y_1^* = 3$, $y_2^* = 0$, $y_3^* = 2$, $y_4^* = 0$. This gives $s_1 = t_1^* - 1 = -1$, $s_2 = t_2^* - 1 = 1$

Step 4: Since $\alpha^* \neq 0$, we can take the usable feasible direction as:

$$\vec{S_1} = \begin{Bmatrix} -1 \\ 1 \end{Bmatrix}$$

Step 5: $\vec{X} = \vec{X_1} + \lambda_1 \vec{S_1} = \begin{Bmatrix} 1 - \lambda_1 \\ 1 + \lambda_2 \end{Bmatrix}$

$$f = (1-\lambda_1-1)^2 + (1+\lambda_1-2)^2 - 4 = 2\lambda_1^2 - 2\lambda_1 - 3$$

$$\frac{df}{d\lambda_1} = 4\lambda_1 - 2 = 0 \;, \quad \tilde{\lambda}_1^* = 0.5$$

$$\vec{x} = \begin{Bmatrix} 0.5 \\ 1.5 \end{Bmatrix}$$

$$f(\vec{x}) = -3.5, \quad g_1(\vec{x}) = -1.5 < 0, \quad g_2(\vec{x}) = -3.5 < 0,$$

$$g_3(\vec{x}) = -2.5 < 0$$

Since all constraints are satisfied at $\tilde{\lambda}_1^* = 0.5$, we choose $\lambda_1^* = \tilde{\lambda}_1^* = 0.5$ and $\vec{x}_2 = \begin{Bmatrix} 0.5 \\ 1.5 \end{Bmatrix}$.

Step 6:

$$f(\vec{x}_2) = -3.5$$

Step 7: Test for convergence:

$$\left| \frac{f(\vec{x}_1) - f(\vec{x}_2)}{f(\vec{x}_1)} \right| = \left| \frac{-3.0 + 3.5}{-3.0} \right| = 0.1667$$

$$\| \vec{x}_1 - \vec{x}_2 \| = \left\| \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} - \begin{Bmatrix} 0.5 \\ 1.5 \end{Bmatrix} \right\| = \{0.25 + 0.25\}^{\frac{1}{2}} = 0.7071$$

Assuming that convergence criteria are not satisfied, we proceed to step 2 using $i = 2$.

Step 2: Since $\vec{x}_2$ is an interior feasible point, we select $\vec{s}_2$ as

$$\vec{s}_2 = -\nabla f(\vec{x}_2) = -\left\{ \begin{matrix} 2(x_1 - 1) \\ 2(x_2 - 2) \end{matrix} \right\}_{\vec{x}_2} = \begin{Bmatrix} 1.0 \\ 1.0 \end{Bmatrix}$$

Step 5:

$$\vec{x} = \vec{x}_2 + \lambda_2 \vec{s}_2 = \begin{Bmatrix} 0.5 \\ 1.5 \end{Bmatrix} + \lambda_2 \begin{Bmatrix} 1.0 \\ 1.0 \end{Bmatrix} = \begin{Bmatrix} 0.5 + \lambda_2 \\ 1.5 + \lambda_2 \end{Bmatrix}$$

$$f(\vec{x}) = (0.5 + \lambda_2 - 1)^2 + (1.5 + \lambda_2 - 2)^2 - 4$$

$$= 2\lambda_2^2 - 2\lambda_2 - 3.5$$

$$\frac{df}{d\lambda_2} = 4\lambda_2 - 2 = 0, \quad \tilde{\lambda}_2^* = 0.5$$

$$\vec{x} = \begin{Bmatrix} 1.0 \\ 2.0 \end{Bmatrix}, \quad f(\vec{x}) = -4.0$$

$g_1(\vec{x}) = 0$, $g_2(\vec{x}) = 0$, $g_3(\vec{x}) = 1 > 0$

since g_3 is violated at $\vec{x}$, we reduce the step length $\tilde{\lambda}_2^*$. Noting that g_3 is linear with values of -2.5 at $\tilde{\lambda}_2^* = 0$ and $+1.0$ at $\tilde{\lambda}_2^* = 0.5$, we choose

$\tilde{\lambda}_2^* = \left(\frac{2.5}{3.5}\right) 0.5 = 0.3571$ to make $g_3 = 0$.

At $\tilde{\lambda}_2^* = 0.3571$, we have

$$\vec{x} = \begin{Bmatrix} 0.8571 \\ 1.8571 \end{Bmatrix}$$

with $g_1(\vec{x}) = -0.4287 < 0$, $g_2(\vec{x}) = -1.0003 < 0$, $g_3(\vec{x}) = -0.0003 \simeq 0$. Thus we choose

$\lambda_2^* = \tilde{\lambda}_2^* = 0.3571$

and $\vec{x}_3 = \begin{Bmatrix} 0.8571 \\ 1.8571 \end{Bmatrix}$

Step 6: $f(\vec{x}_3) = -3.9592$

```
PHASE-I OF SIMPLEX METHOD
given system

  x 4:  0.6000E+01  0.1000E+01  0.1000E+01  0.1000E+01  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00  0.7000E+01
  x 8:  0.0000E+00  0.2000E+01 -0.1000E+01  0.0000E+00 -0.1000E+01  0.0000E+00
        0.0000E+00  0.1000E+01  0.2000E+01
  x 6:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01
        0.0000E+00  0.0000E+00  0.2000E+01
  x 7:  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.1000E+01  0.0000E+00  0.2000E+01
   -f:  0.0000E+00  0.0000E+00 -0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00  0.0000E+00
   -w:  0.0000E+00 -0.2000E+01  0.1000E+01  0.0000E+00  0.1000E+01  0.0000E+00
        0.0000E+00  0.0000E+00 -0.2000E+01

minimum value of ds:-0.20000000E+01  variable out of basis:     2
result of pivoting on element:     2     2


  x 4:  0.6000E+01  0.0000E+00  0.1500E+01  0.1000E+01  0.5000E+00  0.0000E+00
        0.0000E+00 -0.5000E+00  0.6000E+01
  x 2:  0.0000E+00  0.1000E+01 -0.5000E+00  0.0000E+00 -0.5000E+00  0.0000E+00
        0.0000E+00  0.5000E+00  0.1000E+01
  x 6:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01
        0.0000E+00  0.0000E+00  0.2000E+01
  x 7:  0.0000E+00  0.0000E+00  0.5000E+00  0.0000E+00  0.5000E+00  0.0000E+00
        0.1000E+01 -0.5000E+00  0.1000E+01
   -f:  0.0000E+00  0.0000E+00 -0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00  0.0000E+00
   -w:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.1000E+01  0.0000E+00

PHASE-II OF SIMPLEX METHOD
system at start of phase-II

  x 4:  0.6000E+01  0.0000E+00  0.1500E+01  0.1000E+01  0.5000E+00  0.0000E+00
        0.0000E+00  0.6000E+01
  x 2:  0.0000E+00  0.1000E+01 -0.5000E+00  0.0000E+00 -0.5000E+00  0.0000E+00
        0.0000E+00  0.1000E+01
  x 6:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01
        0.0000E+00  0.2000E+01
  x 7:  0.0000E+00  0.0000E+00  0.5000E+00  0.0000E+00  0.5000E+00  0.0000E+00
        0.1000E+01  0.1000E+01
   -f:  0.0000E+00  0.0000E+00 -0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00
        0.0000E+00  0.0000E+00

minimum value of cs:-0.10000000E+01  variable out of basis:     3
result of pivoting on element:     4     3


  x 4:  0.6000E+01  0.0000E+00  0.0000E+00  0.1000E+01 -0.1000E+01  0.0000E+00
       -0.3000E+01  0.3000E+01
  x 2:  0.0000E+00  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00
        0.1000E+01  0.2000E+01
  x 6:  0.1000E+01  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01
        0.0000E+00  0.2000E+01
  x 3:  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00  0.1000E+01  0.0000E+00
        0.2000E+01  0.2000E+01
   -f:  0.0000E+00  0.0000E+00  0.0000E+00  0.0000E+00  0.1000E+01  0.0000E+00
        0.2000E+01  0.2000E+01
```

7.17 Minimize $f = (x_1-1)^2 + (x_2-2)^2 - 4$

subject to $g_1 = x_1 + 2x_2 - 5 \le 0$

$g_2 = 4x_1 + 3x_2 - 10 \le 0$

$g_3 = 6x_1 + x_2 - 7 \le 0$

$g_4 = -x_1 \le 0$

$g_5 = -x_2 \le 0$

using Rosen's gradient projection method.

step 1: $\vec{x}_1 = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$ = feasible.

$f(\vec{x}_1) = -3.0$, $g_1(\vec{x}_1) = -2$, $g_2(\vec{x}_1) = -3$, $g_3(\vec{x}_1) = 0$,

$g_4(\vec{x}_1) = -1$, $g_5(\vec{x}_1) = -1$.

step 2: Iteration, $i = 1$.

step 3: Since $g_3(\vec{x}_1) = 0$, we go to step 4.

step 4: $\nabla g_3 = \begin{Bmatrix} 6 \\ 1 \end{Bmatrix}$, $\nabla f(\vec{x}_1) = \begin{Bmatrix} 2(x_1-1) \\ 2(x_2-2) \end{Bmatrix}_{\vec{x}_1} = \begin{Bmatrix} 0 \\ -2 \end{Bmatrix}$

$[N_1] = [\nabla g_3] = \begin{Bmatrix} 6 \\ 1 \end{Bmatrix}$, $[N_1]^T[N_1] = \{6 \quad 1\}\begin{Bmatrix} 6 \\ 1 \end{Bmatrix} = 37$

$[N_1]\left([N_1]^T[N_1]\right)^{-1}[N_1]^T = \frac{1}{37}[N_1][N_1]^T = \frac{1}{37}\begin{bmatrix} 36 & 6 \\ 6 & 1 \end{bmatrix}$

Projection matrix:

$$[P_1] = [I] - [N_1]\left([N_1]^T[N_1]\right)^{-1}[N_1]^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0.9730 & 0.1622 \\ 0.1622 & 0.0270 \end{bmatrix}$$

$$= \begin{bmatrix} 0.0270 & -0.1622 \\ -0.1622 & 0.9730 \end{bmatrix}$$

$$[P_1]\nabla f(\vec{x}_1) = \begin{bmatrix} 0.0270 & -0.1622 \\ -0.1622 & 0.9730 \end{bmatrix}\begin{Bmatrix} 0 \\ -2 \end{Bmatrix} = \begin{Bmatrix} 0.3244 \\ -1.9460 \end{Bmatrix}$$

$\| [P_1]\nabla f(\vec{x}_1) \| = 1.9728$

$$\vec{s}_1 = -\frac{[P_1]\nabla f(\vec{x}_1)}{\| [P_1]\nabla f(\vec{x}_1) \|} = \begin{Bmatrix} -0.1644 \\ 0.9864 \end{Bmatrix}$$

Step 5: Since $\vec{s}_1 \neq \vec{0}$, we go to step 6.

step 6: To find the step length λ_M, we set

$$\vec{x} = \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \vec{x}_1 + \lambda \vec{S}_1 = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} + \lambda \begin{Bmatrix} -0.1644 \\ 0.9864 \end{Bmatrix} = \begin{Bmatrix} 1 - 0.1644\,\lambda \\ 1 + 0.9864\,\lambda \end{Bmatrix}$$

For $j = 1$, $g_1(\vec{x}) = 1 - 0.1644\,\lambda + 2 + 1.9728\,\lambda - 5$

$= 1.8084\,\lambda - 2 = 0$ at $\lambda = \lambda_1 = 1.1059$

$g_2(\vec{x}) = 4 - 0.6576\,\lambda + 3 + 2.9592\,\lambda - 10$

$= 2.3016\,\lambda - 3 = 0$ at $\lambda = \lambda_2 = 1.3034$

$g_4(\vec{x}) = -1 + 0.1644\,\lambda = 0$ at $\lambda = \lambda_4 = 6.0827$

$g_5(\vec{x}) = -1 - 0.9864\,\lambda = 0$ at $\lambda = \lambda_5 = -1.0138$

$\therefore \lambda_M = \lambda_5 = 1.1059$

Also $f(\vec{x}) = f(\lambda) = (1 - 0.1644\,\lambda - 1)^2 + (1 + 0.9864\,\lambda - 2)^2 - 4$

$= \lambda^2 - 1.9728\,\lambda - 3$

$\frac{df}{d\lambda} = 2\lambda - 1.9728$, $\frac{df}{d\lambda}(\lambda_M) = 2\lambda_M - 1.9728 = +0.0548$

Since $\frac{df}{d\lambda}(\lambda_M) > 0$, we compute the minimizing step length λ_1^* by setting $\frac{df}{d\lambda} = 0$. This gives

$\lambda_1 = \lambda_1^* = 0.9864$

Step 7: The new point $\vec{x}_2$ is given by

$$\vec{x}_2 = \vec{x}_1 + \lambda_1 \vec{S}_1 = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} + 0.9864 \begin{Bmatrix} -0.1644 \\ 0.9864 \end{Bmatrix}$$

$$= \begin{Bmatrix} 0.8378 \\ 1.9730 \end{Bmatrix}$$

$f(\vec{x}_2) = -3.9730$

Since $\lambda_1 = \lambda_1^*$ and $\lambda_1^* < \lambda_M$, no new constraint has become active at $\vec{x}_2$ and hence the matrix $[N_1]$ remains unaltered. Next, we start iteration, $i = 2$.

7.18 Minimize $f = x_1^2 + x_2^2$

subject to $g_1 = x_1 x_2 - 9 = 0$

from $\vec{X}_1 = \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix}_1 = \begin{Bmatrix} 2.0 \\ 4.5 \end{Bmatrix}$ using GRG method.

We assume the bounds on x_i as $-9 \le x_i \le 9$; $i = 1, 2$

Step 1: We choose arbitrarily the independent and dependent variables as

$$\vec{Y} = \{y_1\} = \{x_1\}, \quad \vec{Z} = \{z_1\} = \{x_2\}$$

$$\vec{X}_1 = \begin{Bmatrix} 2.0 \\ 4.5 \end{Bmatrix}, \quad f(\vec{X}_1) = 24.25$$

Step 2: Compute the GRG at $\vec{X}_1$.

Since $\frac{\partial f}{\partial x_1} = 2x_1$, $\frac{\partial f}{\partial x_2} = 2x_2$, $\frac{\partial g_1}{\partial x_1} = x_2$, $\frac{\partial g_1}{\partial x_2} = x_1$,

we find, at $\vec{X}_1$,

$$\nabla_{\vec{Y}} f = \left\{ \frac{\partial f}{\partial x_1} \right\}_{\vec{X}_1} = \{4\}, \quad \nabla_{\vec{Z}} f = \left\{ \frac{\partial f}{\partial x_2} \right\}_{\vec{X}_1} = \{9\}$$

$$[C] = \left[\frac{\partial g_1}{\partial x_1} \right]_{\vec{X}_1} = [4.5], \quad [D] = \left[\frac{\partial g_1}{\partial x_2} \right]_{\vec{X}_1} = [2.0]$$

$$[D]^{-1} = [0.5], \quad [D]^{-1}[C] = [2.25]$$

$$[G_R] = \nabla_{\vec{Y}} f - \left([D]^{-1}[C] \right)^T \nabla_{\vec{Z}} f$$

$$= \{4\} - (2.25)\{9\} = \{-16.25\}$$

Step 3: Since the components of $[G_R]$ are not zero, the point $\vec{X}_1$ is not optimum, and we go to step 4.

Step 4: We use the steepest descent method and take the search direction as

$$\vec{S} = -[G_R] = \{16.25\}$$

Step 5: We find the optimal step length along $\vec{S}$.

(a) Considering the design variable (y_1), we find:

$$\lambda_1 = \frac{y_1^{(u)} - (y_1)_{old}}{s_1} = \frac{9-2}{16.25} = 0.4308$$

Considering the state variable, we get from Eq. (7.113),

$$[T] = -[D]^{-1}[C]\,\vec{S} = -(2.25)\{16.25\} = \{-36.5625\}$$

and Eq. (7.114) gives, for $z_1 = x_2$,

$$\lambda_2 = \frac{z_1^{(l)} - (z_1)_{old}}{t_1} = \frac{-9-4.5}{-36.5625} = 0.3692$$

(b) The upper bound on λ is given by the smaller of λ_1 and λ_2, which is equal to 0.3692. By expressing

$$\vec{X} = \begin{Bmatrix} \vec{Y} + \lambda \vec{S} \\ \vec{Z} + \lambda \vec{T} \end{Bmatrix}$$

we obtain

$$\vec{X} = \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} 2.0 \\ 4.5 \end{Bmatrix} + \lambda \begin{Bmatrix} 16.25 \\ -36.5625 \end{Bmatrix} = \begin{Bmatrix} 2.0 + 16.25\,\lambda \\ 4.5 - 36.5625\,\lambda \end{Bmatrix}$$

$$f(\vec{X}) = f(\lambda) = (2.0 + 16.25\,\lambda)^2 + (4.5 - 36.5625\,\lambda)^2$$

$$= 1600.8789\,\lambda^2 - 264.0625\,\lambda + 24.25$$

$$\frac{df}{d\lambda} = 3201.7578\,\lambda - 264.0625 = 0 \quad \text{at} \quad \lambda^* = 0.0825$$

Since λ^* is less than the upper bound value of 0.3692, we use λ^*.

(c) The new vector $\vec{X}_{new}$ is given by

$$\vec{X}_{new} = \begin{Bmatrix} \vec{Y}_{old} + d\vec{Y} \\ \vec{Z}_{old} + d\vec{Z} \end{Bmatrix} = \begin{Bmatrix} \vec{Y}_{old} + \lambda^* \vec{S} \\ \vec{Z}_{old} + \lambda^* \vec{T} \end{Bmatrix}$$

$$= \begin{Bmatrix} 2.0 + 0.0825\,(16.25) \\ 4.5 + 0.0825\,(-36.5625) \end{Bmatrix} = \begin{Bmatrix} 3.3406 \\ 1.4836 \end{Bmatrix}$$

where $d\vec{Y} = 1.3406$ and $d\vec{Z} = -3.0164$.

Now we need to check whether this vector is feasible. since

$g_1(\vec{x}_{new}) = -4.0439 \neq 0$

the vector $\vec{x}_{new}$ is infeasible. Hence we hold $\vec{Y}_{new}$ constant and modify $\vec{z}_{new}$ using Newton's method as:

$$d\vec{z} = [D]^{-1}\left(-\vec{g}_1(\vec{x}) - [C]\, d\vec{Y}\right)$$

where $[D] = \left[\frac{\partial g_1}{\partial z_1}\right] = [x_1] = [3.3406]$

$g_1(\vec{x}) = -4.0439$

$[C] = \left[\frac{\partial g_1}{\partial y_1}\right] = [x_2] = [1.4836]$

$$d\vec{z} = \frac{1}{3.3406}\{4.0439 - 1.4836 * 1.3406\} = \{0.6151\}$$

This gives $\vec{z}_{new} = \vec{z}_{old} + d\vec{z} = \{4.5 + 0.6151\} = \{5.1151\}$

and $\vec{x}_{new}$ becomes $\vec{x}_{new} = \begin{Bmatrix} \vec{Y}_{old} + d\vec{Y} \\ \vec{z}_{old} + d\vec{z} \end{Bmatrix} = \begin{Bmatrix} 3.3406 \\ 5.1151 \end{Bmatrix}$

The constraint becomes

$g_1 = (3.3406)(5.1151) - 9 = +8.0875 > 0$

Since this $\vec{x}_{new}$ is infeasible, we can use Newton's method. Instead, we try to satisfy the constraint as:

$x_2 = 9/x_1 = 9/3.3406 = 2.6941$

This gives $\vec{x}_{new} = \begin{Bmatrix} 3.3406 \\ 2.6941 \end{Bmatrix}$

$f(\vec{x}_{new}) = 18.4178$

Since $f(\vec{x}_{new}) < f(\vec{x}_1)$, we take $\vec{x}_2 = \vec{x}_{new} = \begin{Bmatrix} 3.3406 \\ 2.6941 \end{Bmatrix}$ and go to next iteration.

7.19 objective function is already in quadratic form.

$g_1: \; 4x_1^2 + x_2^2 - 16 \le 0 \quad ; \quad g_2: \; 3x_1^2 + 5x_2^2 - 15 \le 0$

Let $\vec{x}_1 = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$.

$g_1\big|_{\vec{x}_1} = -11, \quad \frac{\partial g_1}{\partial x_1}\Big|_{\vec{x}_1} = 8x_1\Big|_{\vec{x}_1} = 8, \quad \frac{\partial g_1}{\partial x_2}\Big|_{\vec{x}_1} = 2x_2\Big|_{\vec{x}_1} = 2$

$g_2\big|_{\vec{x}_1} = -7, \quad \frac{\partial g_2}{\partial x_1}\Big|_{\vec{x}_1} = 6x_1\Big|_{\vec{x}_1} = 6, \quad \frac{\partial g_2}{\partial x_2}\Big|_{\vec{x}_1} = 10x_2\Big|_{\vec{x}_1} = 10$

$g_1 \simeq -11 + 8(x_1 - 1) + 2(x_2 - 1) \le 1$

$g_2 \simeq -7 + 6(x_1 - 1) + 10(x_2 - 1) \le 0$

Quadratic programming problem:

Minimize $f = x_1^2 + x_2^2 - 6x_1 - 8x_2 + 15$

subject to

$8x_1 + 2x_2 - 21 \le 0$

$6x_1 + 10x_2 - 23 \le 0$

$x_i \ge 0, \; i = 1, 2$

7.20 Minimize $f = 0.1847\, x_1 + 0.1306\, x_2$

subject to $$g_1 = \frac{26.1546}{x_1} + \frac{30.1546}{x_2} - 1.0 \le 0$$

$$x_i \ge 0\,, \quad i = 1, 2$$

using SQP method.

Let $\vec{x}_1 = \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} 100.0 \\ 40.8348 \end{Bmatrix}$

with $g_1(\vec{x}_1) = 0$ and $f(\vec{x}_1) = 23.8030$

The gradients of f and g_1 at $\vec{x}_1$ are:

$$\nabla f = \begin{Bmatrix} 0.1847 \\ 0.1306 \end{Bmatrix}\,, \qquad \nabla g_1 = \begin{Bmatrix} -26.1546/x_1^2 \\ -30.1546/x_2^2 \end{Bmatrix} = \begin{Bmatrix} -0.00261546 \\ -0.01808 3925 \end{Bmatrix}$$

By assuming $[H_1] = [I]$, the objective function of Eq. (7.138) can be expressed as

$$Q(s_1, s_2) = 0.1847\, s_1 + 0.1306\, s_2 + 0.5\, s_1^2 + 0.5\, s_2^2 \qquad (1)$$

Eq. (7.139) gives $\beta_1 = 0$ since $g_1 = 0$ and hence the constraint of Eq. (7.138) can be expressed as

$$\tilde{g}_1 = -0.00261546\, s_1 - 0.01808 3925\, s_2 \le 0 \qquad (2)$$

We solve the quadratic programming problem given by Eqs. (1) and (2) directly with the use of Kuhn-Tucker conditions. The K-T conditions are:

$$\frac{\partial Q}{\partial s_1} + \lambda_1 \frac{\partial \tilde{g}_1}{\partial s_1} = 0 \quad \text{or} \quad 0.1847 + s_1 - 0.00261546\, \lambda_1 = 0 \qquad (3)$$

$$\frac{\partial Q}{\partial s_2} + \lambda_1 \frac{\partial \tilde{g}_1}{\partial s_2} = 0 \quad \text{or} \quad 0.1306 + s_2 - 0.01808 3925\, \lambda_1 = 0 \qquad (4)$$

$$\lambda_1 \tilde{g}_1 = 0 \qquad (5)$$

$$\tilde{g}_1 \le 0 \qquad (6)$$

$$\lambda_1 \ge 0 \qquad (7)$$

If $\tilde{g}_1 = 0$, Eq. (2) gives

$$s_1 = -6.91424\, s_2 \tag{8}$$

Eqs. (8) and (3) give

$$0.1847 - 6.91424\, s_2 = 0.00261546\, \lambda_1$$

or
$$s_2 = 0.02671 - 0.00037827\, \lambda_1 \tag{9}$$

Eq. (4) gives

$$s_2 = -0.1306 + 0.018083927\, \lambda_1 \tag{10}$$

Equating (9) and (10), we obtain

$$\lambda_1 = 8.5206 \tag{11}$$

Thus, Eqs. (10) and (8) yield

$$s_1 = -0.16239, \quad s_2 = 0.02349 \tag{12}$$

The design vector $\vec{x}$ can be expressed as

$$\vec{x} = \vec{x}_1 + \alpha \vec{s} = \begin{Bmatrix} 100.0 \\ 40.8348 \end{Bmatrix} + \alpha \begin{Bmatrix} -0.16239 \\ 0.02349 \end{Bmatrix}$$

$$= \begin{Bmatrix} 100.0 - 0.16239\,\alpha \\ 40.8348 + 0.02349\,\alpha \end{Bmatrix} \tag{13}$$

where α can be found by minimizing the function ϕ in Eq. (7.141):

$$\phi = f(\vec{x}) + \lambda_1 \cdot \max\left[0, g_1(\vec{x})\right]$$

$$= 0.1847\,(100.0 - 0.16239\alpha) + 0.1306\,(40.8348 + 0.02349\,\alpha)$$

$$+ 8.5206 * \max\left[0, \frac{26.1546}{100.0 - 0.16239\alpha} + \frac{30.1546}{40.8348 + 0.02349\,\alpha} - 1.0\right]$$

The minimizing step length α^* can be found as (without violating the constraint):

$$\alpha^* = 0.8 \quad \text{with} \quad g_1(\alpha^*) = 0$$

This gives the new design vector as:

$$\vec{x}_2 = \begin{Bmatrix} 100.0 - 0.16239\,\alpha^* \\ 40.8348 + 0.02349\,\alpha^* \end{Bmatrix} = \begin{Bmatrix} 99.870088 \\ 40.853592 \end{Bmatrix}$$

with $f(\vec{x}_2) = 23.781484$ and $g_1(\vec{x}_2) = 0.0$.

Next, we update the matrix [H] using Eq. (7.143) and proceed to the next iteration.

7.21

Let x_1, x_2 and x_3 denote the length, breadth and depth of the parcel respectively. The problem can be stated as follows:

Maximize $f(x_1, x_2, x_3) = x_1 x_2 x_3$ (E_1)

subject to $x_1 + 2x_2 + 2x_3 \leq 72$ (E_2)

$x_1 \leq 42$ (E_3)

$x_1 \geq 0$ (E_4)

$x_2 \geq 0$ (E_5)

$x_3 \geq 0$ (E_6)

By introducing the new variables y_1, y_2 and y_3 as

$$\left.\begin{aligned} y_1 &= x_1 \\ y_2 &= x_2, \\ y_3 &= x_1 + 2x_2 + 2x_3, \end{aligned}\right\} \text{ i.e. } \left.\begin{aligned} x_1 &= y_1 \\ x_2 &= y_2 \\ x_3 &= \tfrac{1}{2}(y_3 - y_1 - 2y_2) \end{aligned}\right\} \qquad (E_7)$$

the constraints (E_2) to (E_5) can be expressed as

$$\left.\begin{aligned} 0 \leq y_1 \leq 42 \\ 0 \leq y_2 \leq 36 \\ 0 \leq y_3 \leq 72 \end{aligned}\right\} \qquad (E_8)$$

where the upper bound on y_2 is obtained by setting $x_1 = x_3 = 0$ in (E_3). As x_1 and x_2 are constrained to be positive, intuitively we feel that a negative value of x_3 (negative volume) may not correspond to the maximum of f. Hence we do dot consider (E_6) in the subsequent computations.

The constraints (E_8) will be automatically satisfied if we define new variables z_1, z_2 and z_3 as

$$\left.\begin{aligned} y_1 &= 42 \sin^2 z_1 \\ y_2 &= 36 \sin^2 z_2 \\ y_3 &= 72 \sin^2 z_3 \end{aligned}\right\} \qquad (E_9)$$

If we use Eqs. (E_7) and (E_9), the original constrained optimization problem can be transformed into an equivalent unconstrained problem as:

Maximize $f(z_1, z_2, z_3)$

$$= \tfrac{1}{2}(42 \sin^2 z_1)\,(36 \sin^2 z_2)\,(72 \sin^2 z_3 - 42 \sin^2 z_1 - 72 \sin^2 z_2)$$

$$= 4536 \sin^2 z_1 \sin^2 z_2\,(12 \sin^2 z_3 - 7 \sin^2 z_1 - 12 \sin^2 z_2) \qquad (E_{10})$$

To find the maximum of f, we use the necessary conditions:

$$\frac{\partial f}{\partial z_1} = (4536)\,(4) \sin z_1 \cos z_1 \sin^2 z_2\,(6 \sin^2 z_3 - 7 \sin^2 z_1 - 6 \sin^2 z_2) = 0 \qquad (E_{11})$$

$$\frac{\partial f}{\partial z_2} = (4536)\,(2) \sin^2 z_1 \sin z_2 \cos z_2\,(12 \sin^2 z_3 - 7 \sin^2 z_1 - 24 \sin^2 z_2) = 0 \qquad (E_{12})$$

$$\frac{\partial f}{\partial z_3} = (4536)\,(24) \sin^2 z_1 \sin^2 z_2 \sin z_3 \cos z_3 = 0 \qquad (E_{13})$$

Equation (E_{13}) shows that $\sin z_1 = 0$ or $\sin z_2 = 0$ or $\sin z_3 = 0$ or $\cos z_3 = 0$. However if $\sin z_1 = 0$ or $\sin z_2 = 0$ is taken, we do not get any information about z_2 and z_3 from the other equations. Similarly, $\sin z_3 = 0$ will lead to the trivial solution $z_1 = z_2 = 0$ from Eqs. (E_{11}) and (E_{12}). Hence $\cos z_3$ must be zero to make Eq. (E_{13}) valid. This leads to $\sin^2 z_3 = 1$, $\sin^2 z_1 = 4/7$ and $\sin^2 z_2 = 1/3$. This solution can be verified to correspond to a relative maximum of f. Using Eqs. (E_9) and (E_7), the solution of the problem, in terms of the original variables, can be obtained as: $x_1^* =$ length $= 24$ cm, $x_2^* =$ breadth $= 12$ cm, $x_3^* =$ depth $= 12$ cm, and $f_{max} =$ maximum volume $= 3456$ cm^3.

7.22 Maximize $f = \{9 - (x_1 - 3)^2\} \dfrac{x_2^3}{27\sqrt{3}}$

subject to

$$0 \le x_1$$

$$0 \le x_2 \le \frac{x_1}{\sqrt{3}}$$

$$0 \le x_1 + \sqrt{3}\, x_2 \le 6$$

Define new variables y_1 and y_2 as:

$$x_1 = \tfrac{1}{2}(y_1 - \sqrt{3}\ y_2) \qquad (1)$$

$$x_2 = \tfrac{1}{2}\left|\frac{y_1}{\sqrt{3}} + y_2\right| \qquad (2)$$

with $\quad 0 \le y_1 \le 6 \quad$ and $\quad -2\sqrt{3} \le y_2 \le 0 \qquad (3)$

By using $\quad y_1 = 6 \sin^2 z_1 \qquad (4)$

$$y_2 = -2\sqrt{3} + 2\sqrt{3} \sin^2 z_2 \qquad (5)$$

Eqs. (3) are satisfied. Eqs. (1), (2), (4) and (5) give:

$$x_1 = \tfrac{1}{2}\left(6 \sin^2 z_1 + \sqrt{3}(2\sqrt{3}) - \sqrt{3}(2\sqrt{3}) \sin^2 z_2\right)$$

$$= \tfrac{1}{2}\left(6 \sin^2 z_1 + 6 - 6 \sin^2 z_2\right) \qquad (6)$$

$$x_2 = \tfrac{1}{2}\left|\frac{6}{\sqrt{3}} \sin^2 z_1 - 2\sqrt{3} + 2\sqrt{3} \sin^2 z_2\right|$$

$$= \tfrac{1}{2}\left|2\sqrt{3} \sin^2 z_1 - 2\sqrt{3} + 2\sqrt{3} \sin^2 z_2\right| \qquad (7)$$

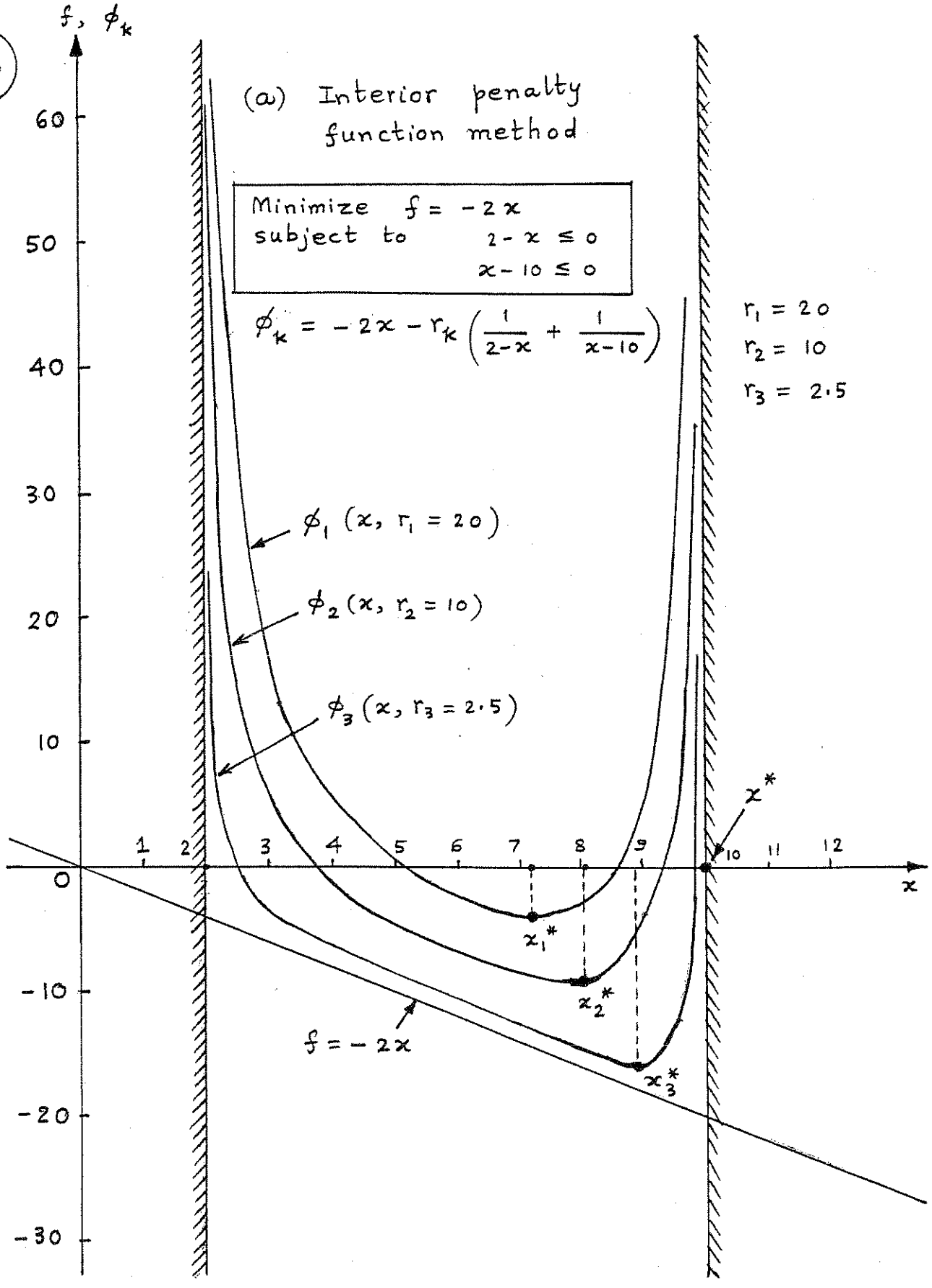
Using Eqs. (6) and (7), the objective function becomes:

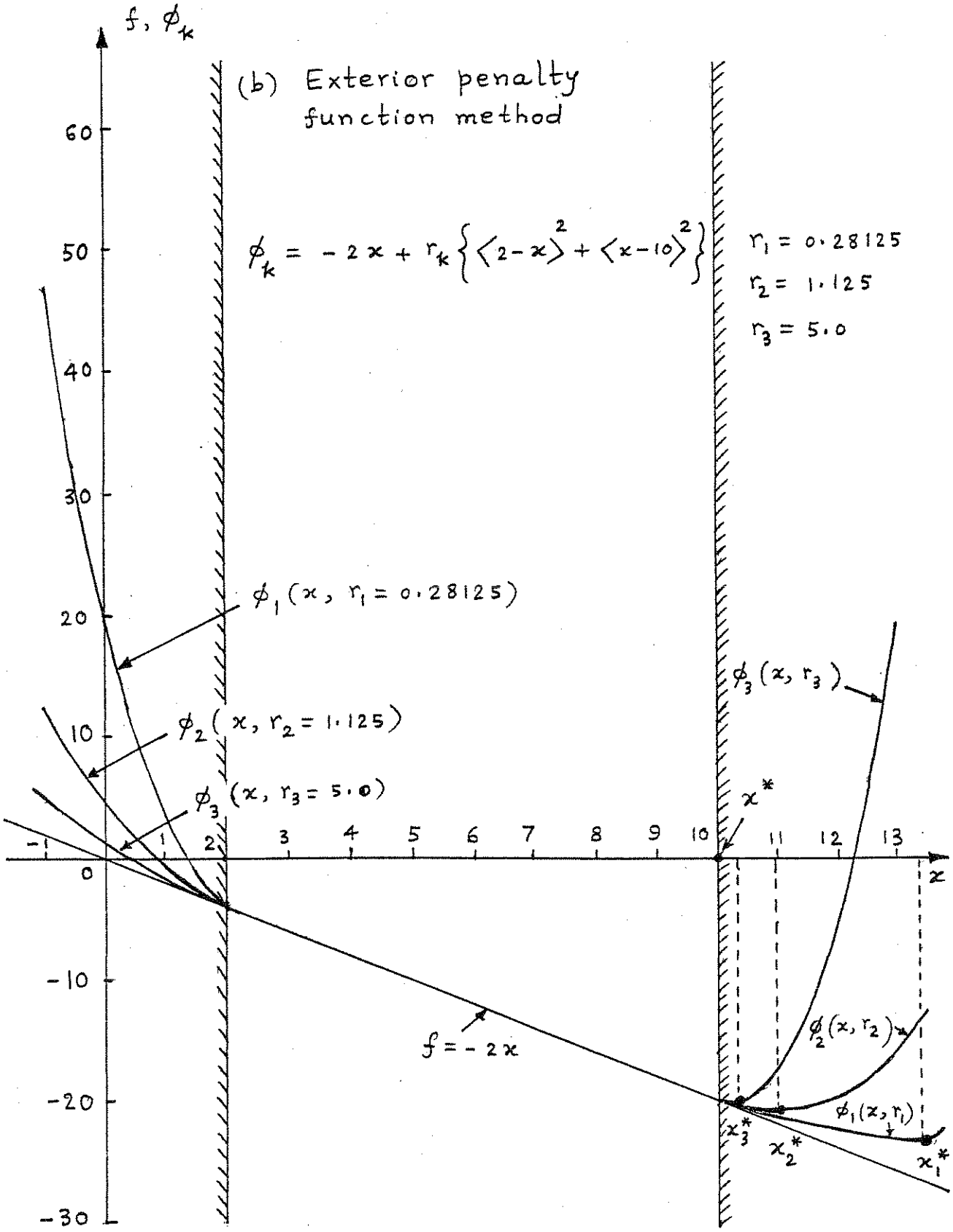
$$f = \left[9 - \{3 \sin^2 z_1 + 3 - 3 \sin^2 z_2 - 3\}^2\right] * \frac{(2\sqrt{3})^3}{(2)^3\, 27\sqrt{3}} \left|\sin^2 z_1 - 1 + \sin^2 z_2\right|^3$$

$$= \{9 - 9(\sin^2 z_1 - \sin^2 z_2)^2\} \cdot \frac{1}{9} \left|\sin^2 z_1 - 1 + \sin^2 z_2\right|^3$$

$$= \{1 - (\sin^2 y_1 - \sin^2 y_2)^2\} \cdot \left|\sin^2 z_1 + \sin^2 z_2 - 1\right|^3$$

7.23


f, ϕ_k
(a) Interior penalty function method
Minimize $f = -2x$
subject to $2 - x \le 0$
$x - 10 \le 0$
$\phi_k = -2x - r_k\left(\frac{1}{2-x} + \frac{1}{x-10}\right)$
$r_1 = 20$
$r_2 = 10$
$r_3 = 2.5$
$\phi_1(x, r_1 = 20)$
$\phi_2(x, r_2 = 10)$
$\phi_3(x, r_3 = 2.5)$
x^*
x_1^*
x_2^*
x_3^*
$f = -2x$
x
60
50
40
30
20
10
0
-10
-20
-30
1
2
3
4
5
6
7
8
9
10
11
12


f, ϕ_k
(b) Exterior penalty function method
$\phi_k = -2x + r_k\left\{\langle 2-x\rangle^2 + \langle x-10\rangle^2\right\}$
$r_1 = 0.28125$
$r_2 = 1.125$
$r_3 = 5.0$
$\phi_1(x, r_1 = 0.28125)$
$\phi_2(x, r_2 = 1.125)$
$\phi_3(x, r_3 = 5.0)$
$\phi_3(x, r_3)$
$\phi_2(x, r_2)$
$\phi_1(x, r_1)$
$f = -2x$
x^*
x_3^*
x_2^*
x_1^*
x
60
50
40
30
20
10
0
-10
-20
-30
-1
1
2
3
4
5
6
7
8
9
10
11
12
13

7.24 Minimize $f = (x-1)^2$

subject to $g_1 = 2 - x \le 0$

$g_2 = x - 4 \le 0$

$$\phi_k = (x-1)^2 + r_k \left\{ [\max(0, 2-x)]^2 + [\max(0, x-4)]^2 \right\}$$

$$= \begin{cases} (x-1)^2 + r_k (2-x)^2 & \text{if } x \le 2 \\ (x-1)^2 & \text{if } 2 < x < 4 \\ (x-1)^2 + r_k (x-4)^2 & \text{if } x \ge 4 \end{cases}$$

$$\frac{d\phi_k}{dx} = \begin{cases} 2(x-1) + 2 r_k (2-x)(-1) & \text{if } x \le 2 \\ 2(x-1) & \text{if } 2 < x < 4 \\ 2(x-1) + 2 r_k (x-4) & \text{if } x \ge 4 \end{cases}$$

$= 0$ at optimum point

This gives:

$2x - 2 - 4 r_k + 2x r_k = 0$ if $x \le 2$ (1)

$2(x-1) = 0$ if $2 < x < 4$ (2)

$2x - 2 + 2x r_k - 8 r_k = 0$ if $x \ge 4$ (3)

Equations (2) and (3) are not valid in this case.

Let $r_1 = 100$. Eq. (1) gives:

$$x^*(2 + 2 r_k) = 2 + 4 r_k \quad \text{or} \quad x^* = \frac{2 + 4 r_k}{2 + 2 r_k} = 1.99$$

if $x \le 2$

More generally, Eq. (1) gives:

$$x^* = \lim_{r_k \to \infty} \frac{\frac{2}{r_k} + 4}{\frac{2}{r_k} + 2} = 2 \quad \text{if } x \le 2 \text{ (for min.)}$$

$\therefore$ Minimum of $f = (2-1)^2 = 1$ at $x^* = 2$

7.25 Minimize $f = (x-1)^2$

subject to $g_1 = 2 - x \le 0$ and $g_2 = x - 4 \le 0$

$$\phi_k = (x-1)^2 + r_k \, \bar{g}_1(x) + r_k \, \bar{g}_2(x)$$

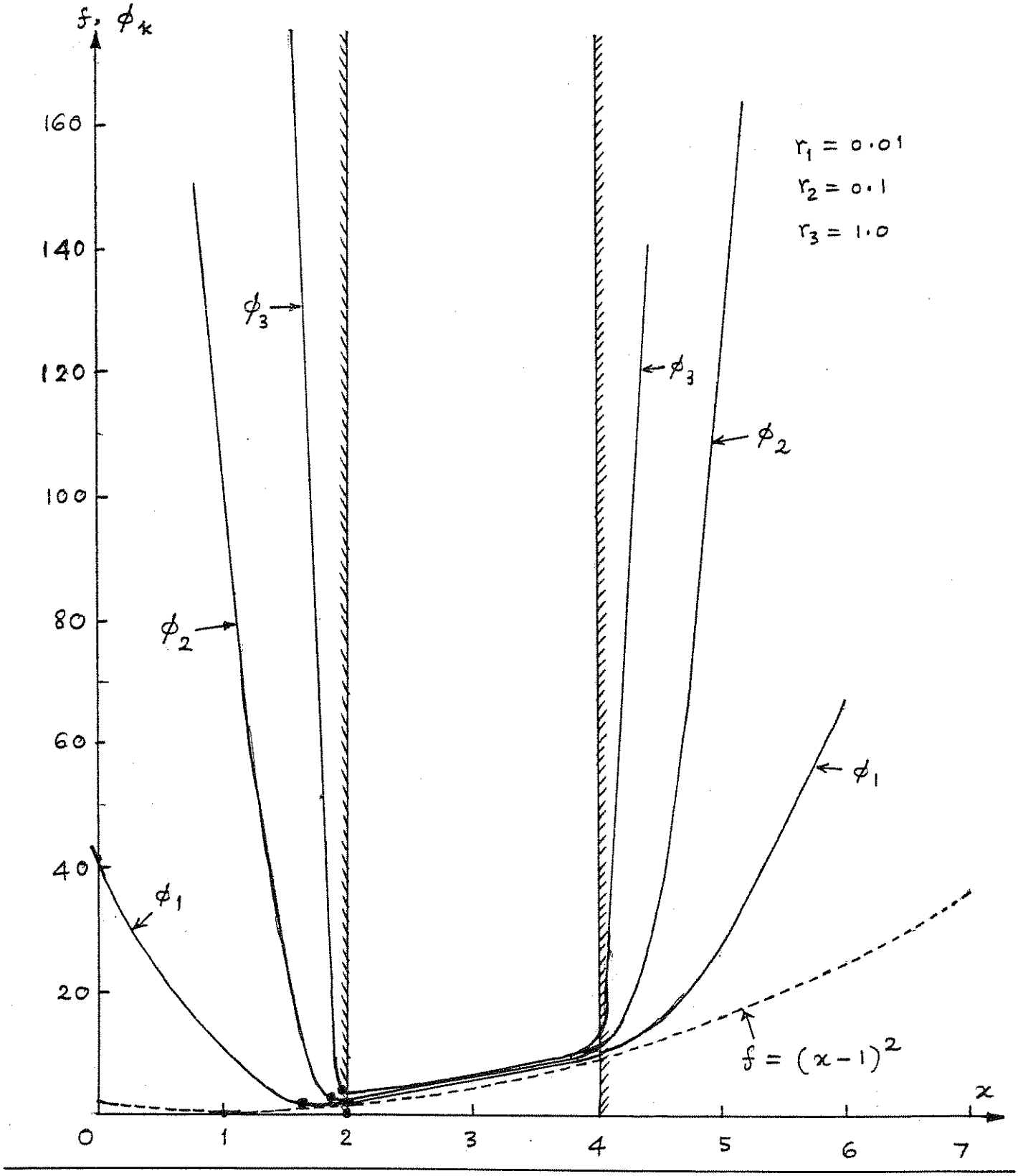
where

$$\bar{g}_1(x) = \begin{cases} -\dfrac{1}{2-x} & \text{if } 2 - x \le \epsilon \\ -\dfrac{1}{\epsilon}\left(\dfrac{2-x}{\epsilon}\right)^2 - 3\left(\dfrac{2-x}{\epsilon}\right) + 3 & \text{if } 2 - x > \epsilon \end{cases}$$

$$\bar{g}_2(x) = \begin{cases} -\dfrac{1}{x-4} & \text{if } x - 4 \le \epsilon \\ -\dfrac{1}{\epsilon}\left(\dfrac{x-4}{\epsilon}\right)^2 - 3\left(\dfrac{x-4}{\epsilon}\right) + 3 & \text{if } x - 4 > \epsilon \end{cases}$$

Let $\epsilon = -0.1$

Using $r_1 = 0.01$, $r_2 = 0.1$ and $r_3 = 1.0$, the contours of ϕ_k obtained are shown in the figure.


f, φ_k
160
140
120
100
80
60
40
20
0
1
2
3
4
5
6
7
x
r_1 = 0.01
r_2 = 0.1
r_3 = 1.0
φ_3
φ_2
φ_1
f = (x-1)^2

7.26 Minimize $f = x^2 - 10x - 1$

subject to $g_1 = 1 - x \le 0$ and $g_2 = x - 10 \le 0$

$$\phi_k = (x^2 - 10x - 1) + r_k \bar{g}_1(x) + r_k \bar{g}_2(x)$$

Where

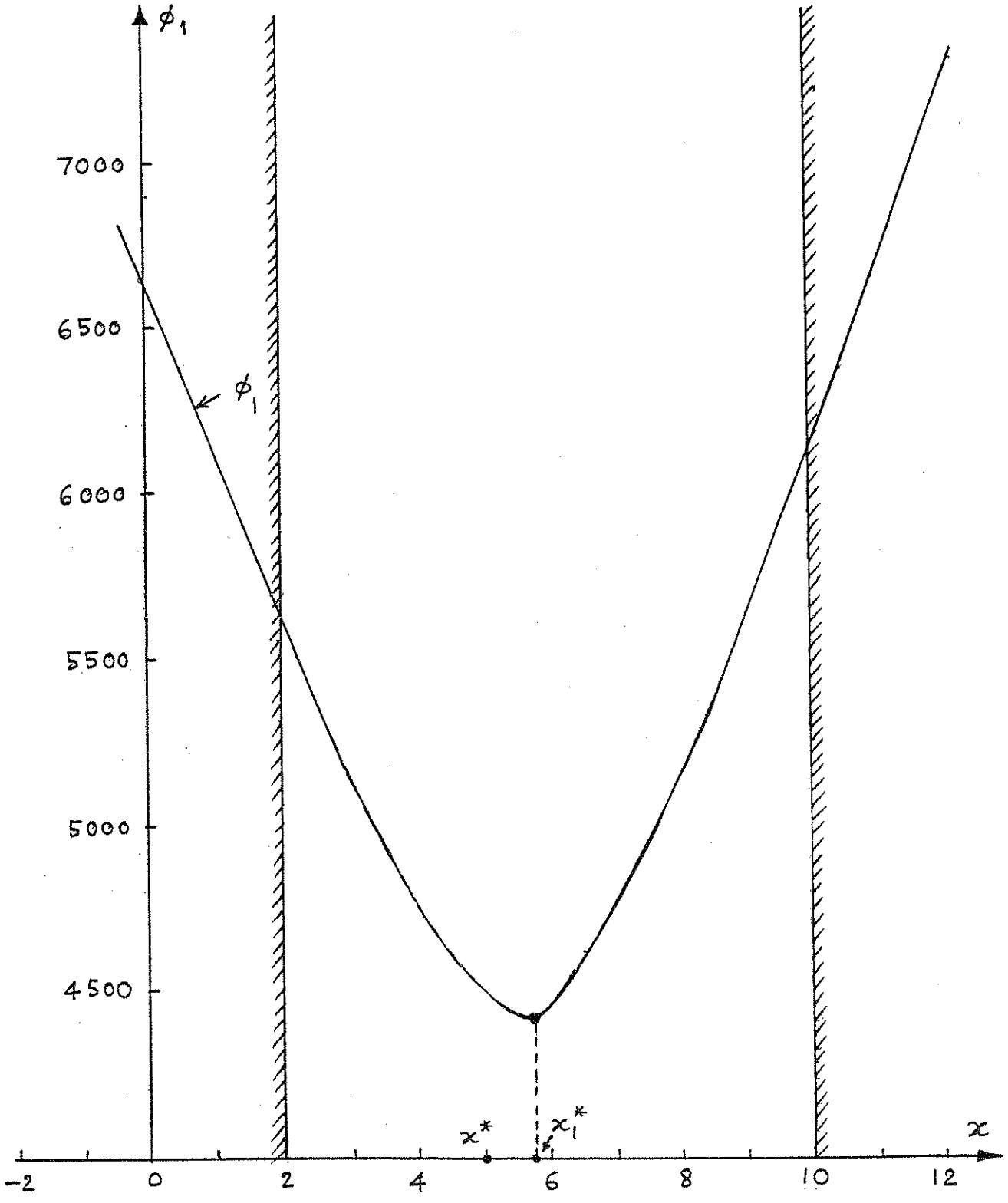
$$\bar{g}_1(x) = \begin{cases} -\dfrac{1}{1-x} & \text{if } 1 - x \le \epsilon \\ -\dfrac{2\epsilon - (1-x)}{\epsilon^2} & \text{if } 1 - x > \epsilon \end{cases}$$

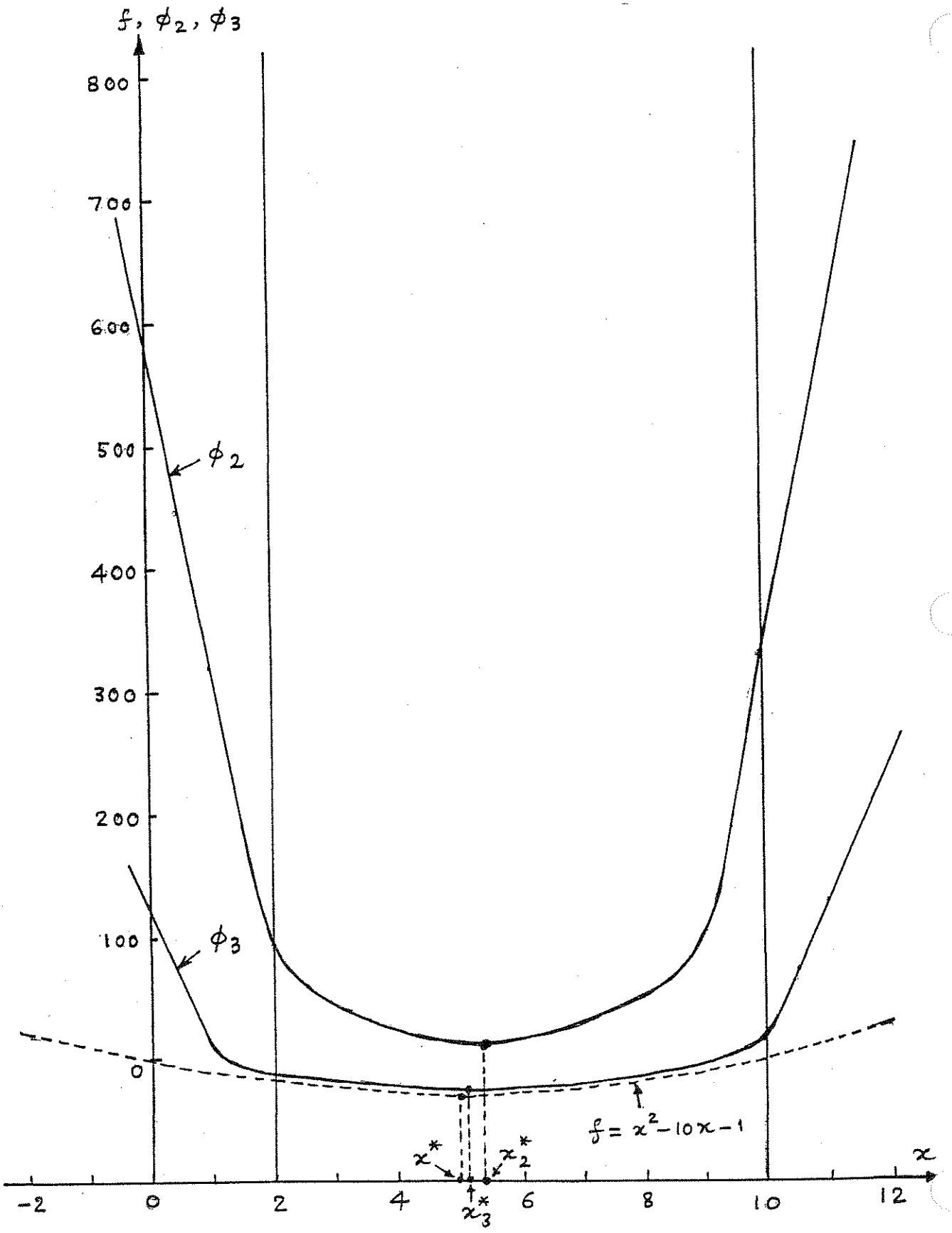
$$\bar{g}_2(x) = \begin{cases} -\dfrac{1}{x-10} & \text{if } x - 10 \le \epsilon \\ -\dfrac{2\epsilon - (x-10)}{\epsilon^2} & \text{if } x - 10 > \epsilon \end{cases}$$

$\epsilon = -c\, r_k^a$ with $c = 0.1$ and $a = 0.4$

$r_1 = 10{,}000.0$, $r_2 = 100.0$, $r_3 = 1.0$

Contours of ϕ_1, ϕ_2 and ϕ_3 are shown in figures.


ϕ_1
7000
6500
6000
5500
5000
4500
ϕ_1
x^*
x_1^*
x
−2
0
2
4
6
8
10
12


f, φ2, φ3
800
700
600
500
400
300
200
100
0
φ2
φ3
f = x² − 10x − 1
x*
x2*
x3*
x
−2
0
2
4
6
8
10
12

7.27 Minimize $f = (x_1-1)^2 + (x_2-2)^2$

subject to $g_1 = 2x_1 - x_2 = 0$ and $g_2 = x_1 - 5 \le 0$

$$\phi_k = (x_1-1)^2 + (x_2-2)^2 - \frac{r_k}{x_1-5} + \frac{1}{\sqrt{r_k}}(2x_1 - x_2)^2$$

Let $r_1 = 1.0$

$$\phi_1 = (x_1-1)^2 + (x_2-2)^2 - \frac{1}{x_1-5} + (2x_1 - x_2)^2 \qquad (1)$$

$$\frac{\partial \phi_1}{\partial x_1} = 2(x_1-1) + \frac{1}{(x_1-5)^2} + 4(2x_1 - x_2) = 0 \qquad (2)$$

$$\frac{\partial \phi_1}{\partial x_2} = 2(x_2-2) - 2(2x_1 - x_2) = 0 \qquad (3)$$

Eqs. (2) and (3) can be simplified as:

$$2(x_1-1)(x_1-5)^2 + 1 + 4(2x_1 - x_2)(x_1-5)^2 = 0 \qquad (4)$$

$$4x_2 - 4x_1 = 4 \qquad (5)$$

Eq. (5) $\Rightarrow$ $x_2 = x_1 + 1$ (6)

With the help of Eq. (6), Eq. (4) can be simplified as

$$(2x_1-2)(x_1^2 + 25 - 10x_1) + 1 + 4(2x_1 - x_1 - 1)(x_1-5)^2 = 0$$

or $$6x_1^3 - 66x_1^2 + 210x_1 - 149 = 0$$

or $$x_1^3 - 11x_1^2 + 35x_1 - 24.8333 = 0 \qquad (7)$$

Roots of Eq. (7) are:

$$x_1 = 0.989637,$$
$$5.005\,18 \pm i\,0.203801$$

$\therefore$ Solution that minimizes $\phi_1(x_1, x_2)$ of Eq. (1) is:

$$x_1^* = 0.989637, \quad x_2^* = 1.989637$$

Solution of the original problem:

$$x_1^* = 1.0, \quad x_2^* = 2.0, \quad f^* = 0.0$$

7.28 Minimize $f = x^2 - 2x - 1$

subject to $g: -1 + x \le 0$

$$\phi = x^2 - 2x - 1 - r\left(\frac{1}{x-1}\right)$$

$$\frac{\partial \phi}{\partial x} = 2x - 2 + \frac{r}{(x-1)^2} = 0 \Rightarrow (x-1)^3 = -\frac{r}{2}$$

$$\Rightarrow x - 1 = \left(-\frac{r}{2}\right)^{\frac{1}{3}}$$

$$x = 1 + \left(-\frac{r}{2}\right)^{1/3}$$

As $r \to 0$, $x \to 1$ $\therefore x^* = 1$, $f^* = -2$

7.29 Minimize $f = x_1^2 + x_2^2 - 6x_1 - 8x_2 + 15$

subject to $g_1: 4x_1^2 + x_2^2 - 16 \le 0$

$g_2: 3x_1 + 5x_2 - 15 \le 0$

$\underset{\sim}{g}_1: \frac{1}{4}x_1^2 + \frac{1}{16}x_2^2 - 1 \le 0$, $\underset{\sim}{g}_2 = \frac{1}{5}x_1 + \frac{1}{3}x_2 - 1 \le 0$

$\vec{x}_1 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$

$f(\vec{x}_1) = 15$, $\left(\frac{1}{\underset{\sim}{g}_1} + \frac{1}{\underset{\sim}{g}_2}\right)_{\vec{x}_1} = \left(\frac{1}{-1} + \frac{1}{-1}\right) = -2$

Let $\phi_1 = f(\vec{x}_1) - r_1\left(\frac{1}{\underset{\sim}{g}_1} + \frac{1}{\underset{\sim}{g}_2}\right) \simeq 1.2\, f(\vec{x}_1)$

i.e., $15 - r_1(-2) = 1.2(15)$

i.e., $r_1 = 1.5$

7.30 $f = -4x_1 + x_1^2 - 2x_1x_2 + 2x_2^2$

$$\frac{\partial f}{\partial x_1} = -4 + 2x_1 - 2x_2, \quad \frac{\partial f}{\partial x_2} = -2x_1 + 4x_2$$

$$\frac{\partial^2 f}{\partial x_1^2} = 2, \quad \frac{\partial^2 f}{\partial x_1 \partial x_2} = -2, \quad \frac{\partial^2 f}{\partial x_2^2} = 4$$

$$[J]_f = \begin{bmatrix} 2 & -2 \\ -2 & 4 \end{bmatrix}; \quad J_1 = +2, \quad J_2 = 8 - 4 = 4$$

$\therefore$ f is convex

$g_1: \; 2x_1 + x_2 - 6 \le 0$

$g_2: \; x_1 - 4x_2 \le 0$

$g_3: \; x_1 \ge 0$

$g_4: \; x_2 \ge 0$

} All constraints are linear, hence they are convex.

$\therefore$ optimization problem is convex.

7.31

$$\phi = (2x_1 - x_2)^2 + (x_2+1)^2 + r(x_1 + x_2 - 10)^2$$

$$\frac{\partial \phi}{\partial x_1} = 4(2x_1 - x_2) + 2r(x_1 + x_2 - 10) = 0$$

i.e., $$x_1(8 + 2r) + x_2(-4 + 2r) = 20r \qquad (1)$$

$$\frac{\partial \phi}{\partial x_2} = -2(2x_1 - x_2) + 2(x_2+1) + 2r(x_1 + x_2 - 10) = 0$$

i.e., $$x_1(-4 + 2r) + x_2(4 + 2r) = 20r - 2 \qquad (2)$$

Eqs. (1) and (2): $$(4+r)x_1 + (-2+r)x_2 = 10r \qquad (3)$$

$$(-2+r)x_1 + (2+r)x_2 = 10r - 1 \qquad (4)$$

Multiply Eq. (3) by $(2+r)$ and Eq. (4) by $(r-2)$, and subtract to obtain:

$$x_1 = \frac{41r - 2}{10r + 4} = \frac{41 - \frac{2}{r}}{10 + \frac{4}{r}} = \frac{41}{10} \quad \text{as} \quad r \to \infty$$

and

$$x_2 = \frac{59r^2 - 122r + 8}{10r^2 - 16r - 8} = \frac{59 - \frac{122}{r} + \frac{8}{r^2}}{10 - \frac{16}{r} - \frac{8}{r^2}} = \frac{59}{10} \quad \text{as } r \to \infty$$

$\therefore \quad x_1^* = 4.1, \quad x_2^* = 5.9$

$$f_{min} = (2 * 4.1 - 5.9)^2 + (5.9 + 1)^2$$

$$= 53.53$$

7.32

$$\phi = 3x_1^2 + 4x_2^2 + r(x_1 + 2x_2 - 8)^2$$

$$\frac{\partial \phi}{\partial x_1} = 6x_1 + 2r(x_1 + 2x_2 - 8) = 0$$

or $$x_1(6 + 2r) + x_2(4r) = 16r \qquad (1)$$

$$\frac{\partial \phi}{\partial x_2} = 8x_2 + 4r(x_1 + 2x_2 - 8) = 0$$

or $$x_1(4r) + x_2(8 + 8r) = 32r \qquad (2)$$

Multiply Eq. (1) by $4r$ and Eq. (2) by $(6+2r)$ and subtract the resulting equations one from the other:

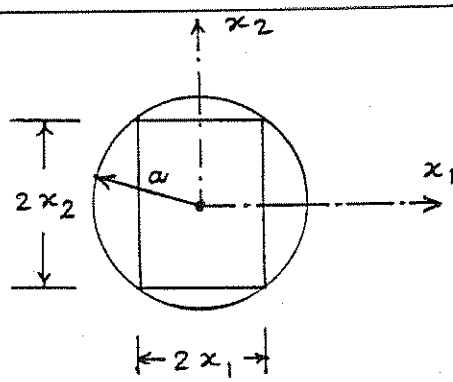
$$x_2 = \frac{12r}{3 + 4r} = \frac{12}{\frac{3}{r} + 4} \quad \text{and} \quad x_1 = \frac{8r}{3 + 4r} = \frac{8}{\frac{3}{r} + 4}$$

As $r \to \infty$, $x_1^* = \frac{8}{4} = 2$ and $x_2^* = \frac{12}{4} = 3$

$$f^* = 3(2)^2 + 4(3)^2 = 48$$

7.33

Let σ = stress, M = bending moment, I = moment of inertia of cross-section.



$$\frac{\sigma}{x_2} = \frac{M}{I} \quad \text{where} \quad I = \frac{1}{12}(2x_1)(2x_2)^3 = \frac{4}{3}x_1 x_2^3$$

$$\sigma = \frac{c}{x_1 x_2^2} \quad \text{where} \quad c = \frac{3M}{4}$$

Problem: Minimize $f = c\, x_1^{-1} x_2^{-2}$ (1)

subject to $g: x_1^2 + x_2^2 - a^2 = 0$ (2)

Exterior penalty function approach:

$$\phi = f(\vec{x}) + r\, g^2(\vec{x}) = c\, x_1^{-1} x_2^{-2} + r(x_1^2 + x_2^2 - a^2)^2$$

$$\frac{\partial \phi}{\partial x_1} = -c\, x_1^{-2} x_2^{-2} + 2r(x_1^2 + x_2^2 - a^2)(2x_1) = 0$$

or $$\frac{*}{x_1^2 x_2^2} = 4r x_1 (x_1^2 + x_2^2 - a^2) \qquad (3)$$

$$\frac{\partial \phi}{\partial x_2} = -2c\, x_1^{-1} x_2^{-3} + 2r\,(x_1^2 + x_2^2 - a^2)(2x_2) = 0$$

or $$\frac{2c}{x_1 x_2^3} = 4r\, x_2 (x_1^2 + x_2^2 - a^2) \qquad (4)$$

Divide Eq. (3) by (4):

$$\frac{x_2}{2x_1} = \frac{x_1}{x_2} \Rightarrow x_2^2 = 2x_1^2 \Rightarrow x_2 = \sqrt{2}\, x_1 \qquad (5)$$

Eqs. (3) and (5):

$$\frac{c}{x_1^2 (2x_1^2)} = 4r\, x_1 (x_1^2 + 2x_1^2 - a^2)$$

i.e. $$(3x_1^2 - a^2) = \frac{c}{8r\, x_1^5} \qquad (6)$$

As $r \to \infty$, Eq. (6) gives $$x_1^* = \frac{a}{\sqrt{3}} \qquad (7)$$

Eqs. (5) and (7): $$x_2^* = \sqrt{\frac{2}{3}}\, a \qquad (8)$$

7.34

Assume $$\vec{x}(r) = \vec{A}_0 + \frac{1}{r}\vec{A}_1 \qquad (1)$$

$$\vec{x}_1^* = \vec{A}_0 + \frac{1}{r_1}\vec{A}_1 \Rightarrow \vec{A}_0 = \vec{x}_1^* - \frac{1}{r_1}\vec{A}_1 \qquad (2)$$

$$\vec{x}_2 = \vec{A}_0 + \frac{1}{r_2}\vec{A}_1 \Rightarrow \vec{x}_2^* = \left(\vec{x}_1^* - \frac{1}{r_1}\vec{A}_1\right) + \frac{1}{r_2}\vec{A}_1 \qquad (3)$$

or $$\vec{x}_2^* - \vec{x}_1^* = \left(\frac{1}{r_2} - \frac{1}{r_1}\right)\vec{A}_1 \qquad (4)$$

$$\vec{A}_1 = \frac{1}{\left(\frac{1}{r_2} - \frac{1}{r_1}\right)}\left(\vec{x}_2^* - \vec{x}_1^*\right) \qquad (5)$$

$$\vec{A}_0 = \vec{x}_1^* - \frac{1}{r_1}\cdot\frac{1}{\left(\frac{1}{r_2} - \frac{1}{r_1}\right)}\left(\vec{x}_2^* - \vec{x}_1^*\right) \qquad (6)$$

As $r \to \infty$, $\vec{x}^* = \vec{A}_0$ from Eq. (1).

Using the given data:

$$\vec{x}_2^* = \begin{Bmatrix} 0.8322 \\ -0.05 \end{Bmatrix}, \quad \vec{x}_1^* = \begin{Bmatrix} 0.2361 \\ -0.5 \end{Bmatrix}, \quad r_2 = 10, \quad r_1 = 1,$$

we obtain

$$\vec{x}^{\,*} = \begin{Bmatrix} 0.2361 \\ -0.5 \end{Bmatrix} - \frac{1}{\frac{1}{1}\left(\frac{1}{10} - \frac{1}{1}\right)} \left(\begin{Bmatrix} 0.8322 \\ -0.05 \end{Bmatrix} - \begin{Bmatrix} 0.2361 \\ -0.5 \end{Bmatrix} \right)$$

$$= \begin{Bmatrix} 0.8984 \\ 0 \end{Bmatrix}$$

Similarly, using the relation $f(r) = a_0 + \frac{1}{r} a_1$, we obtain

$$f^* = f_1^* - \frac{1}{\frac{1}{r_1}\left(\frac{1}{r_2} - \frac{1}{r_1}\right)} \left(f_2^* - f_1^* \right)$$

$$= 0.1295 - \frac{1}{\frac{1}{1}\cdot\left(\frac{1}{10} - \frac{1}{1}\right)} \left(2.0001 - 0.1295 \right) = 2.2079$$

7.35 Assume $\vec{x}(r) = \vec{A}_0 + \frac{1}{r}\vec{A}_1$ and $f(r) = a_0 + \frac{1}{r} a_1$

Using the relations derived in the solution of Problem 7.34, we obtain

$$f^* = f_1^* - \frac{1}{\frac{1}{r_1}\cdot\left(\frac{1}{r_2} - \frac{1}{r_1}\right)} \left(f_2^* - f_1^* \right)$$

$$= -49.9977 - \frac{1}{\frac{1}{0.01}\left(\frac{1}{1} - \frac{1}{0.01}\right)} \left(0.1295 + 49.9977 \right) = 0.6358$$

$$\vec{x}^{\,*} = \vec{x}_1^{\,*} - \frac{1}{\frac{1}{r_1}\cdot\left(\frac{1}{r_2} - \frac{1}{r_1}\right)} \left(\vec{x}_2^{\,*} - \vec{x}_1^{\,*} \right)$$

$$= \begin{Bmatrix} -0.80975 \\ -50.0 \end{Bmatrix} - \frac{1}{\frac{1}{0.01}\left(\frac{1}{1} - \frac{1}{0.01}\right)} \left(\begin{Bmatrix} 0.23607 \\ -0.5 \end{Bmatrix} - \begin{Bmatrix} -0.80975 \\ -50.0 \end{Bmatrix} \right)$$

$$= \begin{Bmatrix} 0.246634 \\ -0.010101 \end{Bmatrix}$$

7.36

Assume $\vec{x}(r) = \vec{A}_0 + \frac{1}{r}\vec{A}_1$ (1)

At $r_1 = 10^{-10}$; $\begin{Bmatrix} 0.66 \\ 28.6 \end{Bmatrix} = \vec{A}_0 + \frac{1}{10^{-10}}\vec{A}_1 = \vec{A}_0 + 10^{10}\vec{A}_1$ (2)

At $r_2 = 10^{-9}$; $\begin{Bmatrix} 1.57 \\ 18.7 \end{Bmatrix} = \vec{A}_0 + \frac{1}{10^{-9}}\vec{A}_1 = \vec{A}_0 + 10^{9}\vec{A}_1$ (3)

subtracting Eq. (3) from (2), we get

$$\begin{Bmatrix} -0.91 \\ 9.9 \end{Bmatrix} = (10^{10} - 10^{9})\vec{A}_1 = 9(10^{9})\vec{A}_1$$

$$\vec{A}_1 = \begin{Bmatrix} -0.1011 \\ 1.1 \end{Bmatrix} 10^{-9}$$

$$\vec{A}_0 = \begin{Bmatrix} 0.66 \\ 28.6 \end{Bmatrix} - 10^{10}\vec{A}_1 = \begin{Bmatrix} 0.66 \\ 28.6 \end{Bmatrix} - 10\begin{Bmatrix} -0.1011 \\ 1.1 \end{Bmatrix} = \begin{Bmatrix} 1.671 \\ 17.6 \end{Bmatrix}$$

$$\vec{x}^* = \vec{x}(r \to \infty) = \vec{A}_0 = \begin{Bmatrix} 1.671 \\ 17.6 \end{Bmatrix}$$

7.37

Assume $\vec{x}(r) = \vec{A}_0 + \frac{1}{r}\vec{A}_1 + \frac{1}{r^2}\vec{A}_2$ (1)

From given data, we have

$$\vec{x}_i^* = \vec{A}_0 + \frac{1}{r_i}\vec{A}_1 + \frac{1}{r_i^2}\vec{A}_2 \quad ; \; i = 1, 2, 3 \qquad (2)$$

subtract $\vec{x}_2^*$ from $\vec{x}_3^*$ given by Eq. (2):

$$\vec{x}_3^* - \vec{x}_2^* = \left(\frac{1}{r_3} - \frac{1}{r_2}\right)\vec{A}_1 + \left(\frac{1}{r_3^2} - \frac{1}{r_2^2}\right)\vec{A}_2 \qquad (3)$$

Subtract $\vec{x}_1^*$ from $\vec{x}_3$ given by Eq. (2):

$$\vec{x}_3^* - \vec{x}_1^* = \left(\frac{1}{r_3} - \frac{1}{r_1}\right)\vec{A}_1 + \left(\frac{1}{r_3^2} - \frac{1}{r_1^2}\right)\vec{A}_2 \qquad (4)$$

Eqs. (3) and (4):

$$\vec{A}_1 + \left(\frac{1}{r_3} + \frac{1}{r_2}\right)\vec{A}_2 = \frac{\vec{x}_3^* - \vec{x}_2^*}{\left(\frac{1}{r_3} - \frac{1}{r_2}\right)} \qquad (5)$$

$$\vec{A}_1 + \left(\frac{1}{r_3} + \frac{1}{r_1}\right)\vec{A}_2 = \frac{\vec{x}_3^* - \vec{x}_1^*}{\left(\frac{1}{r_3} - \frac{1}{r_1}\right)} \qquad (6)$$

Solution of Eqs. (5) and (6) gives:

$$\vec{A}_2 = \frac{\vec{c}_1 - \vec{c}_2}{c_3 - c_4} \quad , \quad \vec{A}_1 = \vec{c}_1 - c_3 \left(\frac{\vec{c}_1 - \vec{c}_2}{c_3 - c_4} \right) , \tag{7}$$

$$\vec{A}_0 = \vec{x}_3^{\,*} - \frac{1}{r_3} \vec{A}_1 - \frac{1}{r_3^2} \vec{A}_2 \tag{8}$$

where

$$\vec{c}_1 = \frac{\vec{x}_3^{\,*} - \vec{x}_2^{\,*}}{\left(\frac{1}{r_3} - \frac{1}{r_2} \right)} \quad , \quad \vec{c}_2 = \frac{\vec{x}_3^{\,*} - \vec{x}_1^{\,*}}{\left(\frac{1}{r_3} - \frac{1}{r_1} \right)} , \tag{9}$$

$$c_3 = \frac{1}{r_3} + \frac{1}{r_2} \quad , \quad c_4 = \frac{1}{r_3} + \frac{1}{r_1} \tag{10}$$

For the given data, we find

$$c_3 = 0.11 * 10^{9} \quad , \quad c_4 = 0.1011 * 10^{10}$$

$$\vec{c}_1 = \begin{Bmatrix} -0.3222 * 10^{-8} \\ -0.1111 * 10^{-8} \end{Bmatrix} \quad , \quad \vec{c}_2 = \begin{Bmatrix} -0.1212 * 10^{-8} \\ 0.9899 * 10^{-8} \end{Bmatrix}$$

$$\vec{A}_0 = \begin{Bmatrix} 1.8945 \\ 18.8233 \end{Bmatrix} \quad , \quad \vec{A}_1 = \begin{Bmatrix} -0.3468 * 10^{-8} \\ -0.2457 * 10^{-8} \end{Bmatrix} ,$$

$$\vec{A}_2 = \begin{Bmatrix} 0.2233 * 10^{-17} \\ 0.1223 * 10^{-16} \end{Bmatrix}$$

$$\vec{x}^{\,*} = \vec{x}(r \to \infty) = \vec{A}_0 = \begin{Bmatrix} 1.8945 \\ 18.8233 \end{Bmatrix}$$

7.38

$$\vec{x} = \begin{Bmatrix} d \\ t \end{Bmatrix} \qquad (1)$$

Minimize

$$f = \text{volume} = 2\pi\, d\, t\, \sqrt{b^2 + h^2}$$

$$= 671.046054\, d\, t \qquad (2)$$

$$P_1 = 25{,}000 \cos 2\theta \left(\frac{\sin\theta}{0.936329} + \frac{\cos\theta}{0.351123} \right) \qquad (3)$$

$$P_2 = 25{,}000 \cos 2\theta \left(\frac{\sin\theta}{0.936329} - \frac{\cos\theta}{0.351123} \right) \qquad (4)$$

or

$$P_1 = 26700.01677 \cos 2\theta \cdot \sin\theta + 71200.12076 \cos 2\theta \cdot \cos\theta \qquad (5)$$

$$P_2 = 26700.01677 \cos 2\theta \cdot \sin\theta - 71200.12076 \cos 2\theta \cdot \cos\theta \qquad (6)$$

Assuming pin-ends,

$$P_{cri} = \frac{\pi^2 E I}{\ell^2} = \frac{\pi^2 E}{\ell^2} \left\{ \frac{\pi}{8}\, d\, t\, (d^2 + t^2) \right\}$$

with $E = 30 \times 10^6$ and $\ell = 106.80047$

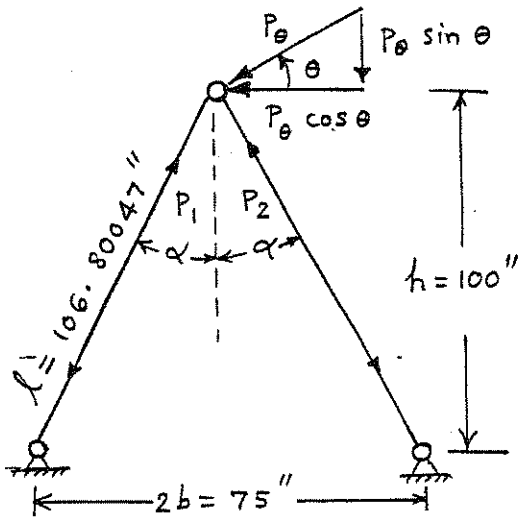
$$\therefore P_{cri} = 10193.835\, d\, t\, (d^2 + t^2) \qquad (7)$$

Buckling constraints:

$$P_1 \le P_{cri} \qquad (8)$$

$$P_2 \le P_{cri} \qquad (9)$$

Eqs. (8) and (9) can be rewritten, using Eqs. (5) to (7), as:



Member reactions:

$$P_\theta \sin\theta = P_1 \cos\alpha + P_2 \cos\alpha$$

$$P_\theta \cos\theta = P_1 \sin\alpha - P_2 \sin\alpha$$

or

$$P_1 + P_2 = \frac{P_\theta \sin\theta}{\cos\alpha}$$

$$P_1 - P_2 = \frac{P_\theta \cos\theta}{\sin\alpha}$$

$$P_1 = \frac{P_\theta}{2} \left(\frac{\sin\theta}{\cos\alpha} + \frac{\cos\theta}{\sin\alpha} \right)$$

$$P_2 = \frac{P_\theta}{2} \left(\frac{\sin\theta}{\cos\alpha} - \frac{\cos\theta}{\sin\alpha} \right)$$

$$P_\theta = P_0 \cos 2\theta = 50{,}000 \cos 2\theta \ \text{lb}$$

$$\cos\alpha = h / \sqrt{h^2 + b^2} = 100/106.800047 = 0.936329$$

$$\sin\alpha = b / \sqrt{h^2 + b^2} = 37.5/106.800047 = 0.351123$$

$$26700.01677 \cos 2\theta \cdot \sin\theta + 71200.12076 \cos 2\theta \cdot \cos\theta - 10193.835\, dt\,(d^2+t^2) \le 0 \quad (10)$$

$$26700.01677 \cos 2\theta \cdot \sin\theta - 71200.12076 \cos 2\theta \cdot \cos\theta - 10193.835\, dt\,(d^2+t^2) \le 0 \quad (11)$$

Yielding constraints:

$$\sigma_1 \le \sigma_{yield} \quad (12)$$

$$\sigma_2 \le \sigma_{yield} \quad (13)$$

where $\sigma_{yield} = 30{,}000$ psi

and $$\sigma_i = \frac{P_i}{\pi dt}, \quad i = 1, 2 \quad (14)$$

Eqs. (12) and (13) can be rewritten as

$$26700.01677 \cos 2\theta \cdot \sin\theta + 71200.12076 \cos 2\theta \cdot \cos\theta - 30000\,\pi dt \le 0 \quad (15)$$

$$26700.01677 \cos 2\theta \cdot \sin\theta - 71200.12076 \cos 2\theta \cdot \cos\theta - 30000\,\pi dt \le 0 \quad (16)$$

The NLP problem is defined by Eqs. (1), (2), (10), (11), (15) and (16).

Graphical solution:

1. Divide the range of the parameter (angle) θ into several convenient discrete values, θ_i. For example, use $\theta = 0°, 45°, 90°, 135°, 180°, 225°, 270°, 315°$ where $\theta_1 = 0°$, $\theta_2 = 45°, \ldots, \theta_8 = 315°$.
2. Write the constraints, Eqs. (10), (11), (15) and (16), at each angle θ_i so that the total number of constraints would be 32.
3. Solve the problem with d and t as design variables to minimize f (given by Eq. (2)) subject to the discretized constraints using a graphical procedure.

7.39 Minimize $f = (x_1-1)^2 + (x_2-2)^2$ (1)

subject to $x_1 + 2x_2 - 2 = 0 = h(\vec{x})$ (2)

Augmented Lagrangian function is:

$$A = (x_1-1)^2 + (x_2-2)^2 + \lambda (x_1 + 2x_2 - 2) + (x_1 + 2x_2 - 2)^2 \quad (3)$$

Necessary conditions for the stationary point of A:

$$\frac{\partial A}{\partial x_1} = 2(x_1-1) + \lambda + 2(x_1 + 2x_2 - 2) = 0 \quad (4)$$

$$\frac{\partial A}{\partial x_2} = 2(x_2-2) + 2\lambda + 4(x_1 + 2x_2 - 2) = 0 \quad (5)$$

$$\frac{\partial A}{\partial \lambda} = (x_1 + 2x_2 - 2) = 0 \quad (6)$$

Eqs. (4) and (5): $4x_1 + 4x_2 = -\lambda + 6$ (7)

$$4x_1 + 10x_2 = -2\lambda + 12 \quad (8)$$

Solution of Eqs. (7) and (8):

$$x_1 = \frac{1}{2} - \frac{\lambda}{12}, \quad x_2 = 1 - \frac{\lambda}{6} \quad (9)$$

Let $\lambda^{(1)} = 0$. Eq. (9) gives $x_1^* = 0.5$, $x_2^* = 1.0$

$\lambda^{(2)} = \lambda^{(1)} + 2 r_p h(\vec{x}_1^*)$ where $\vec{x}_1^* = \begin{Bmatrix} 0.5 \\ 1.0 \end{Bmatrix}$ & $r_p = 1$.

$= 0 + 2(1)(0.5 + 2 * 1.0 - 2) = 1.0$

$\lambda^{(2)} = 1.0$. Eq. (9) gives $x_1^* = \frac{1}{2} - \frac{1}{12} = 0.4167$,

$x_2^* = 1 - \frac{1}{6} = 0.8333$

$\lambda^{(3)} = \lambda^{(2)} + 2 r_p h(\vec{x}_2^*)$ where $\vec{x}_2^* = \begin{Bmatrix} 0.4167 \\ 0.8333 \end{Bmatrix}$.

$= 1.0 + 2(1)(0.4167 + 2 * 0.8333 - 2) = 1.1667$

$\lambda^{(3)} = 1.1667$. Eq. (9) gives $x_1^* = \frac{1}{2} - \frac{1.1667}{12} = 0.4028$

$x_2^* = 1 - \frac{1.1667}{6} = 0.8056$

$\therefore \quad \vec{x}_3^* = \begin{Bmatrix} 0.4028 \\ 0.8056 \end{Bmatrix}$

7.40

$$A = (x_1-1)^2 + (x_2-2)^2 + \lambda(-x_1+2x_2-2) + r_p(-x_1+2x_2-2)^2$$

$$\frac{\partial A}{\partial x_1} = 2(x_1-1) - \lambda - 2r_p(-x_1+2x_2-2) = 0$$

or $$x_1(2+2r_p) + x_2(-4r_p) = \lambda - 4r_p + 2 \qquad (1)$$

$$\frac{\partial A}{\partial x_2} = 2(x_2-2) + 2\lambda + 4r_p(-x_1+2x_2-2) = 0$$

or $$x_1(-4r_p) + x_2(2+8r_p) = -2\lambda + 8r_p + 4 \qquad (2)$$

Solution of Eqs. (1) and (2) gives:

$$x_1 = \frac{\lambda r_p + 12 r_p^2 + 14 r_p + \lambda + 2}{2(1+r_p)(5r_p+1)} \qquad (3)$$

$$x_2 = \frac{8r_p - \lambda + 2}{5r_p + 1} \qquad (4)$$

$r_p = 1$ (fixed), $\lambda^{(1)} = 0$

Eqs. (3) and (4) give

$$x_1^{*(1)} = \frac{0+12+14+0+2}{2(2)(6)} = \frac{7}{6} = 1.166667$$

$$x_2^{*(1)} = \frac{8-0+2}{5+1} = \frac{5}{3} = 1.666667$$

$$h = -x_1 + 2x_2 - 2 = \frac{1}{6}$$

$$\lambda^{(2)} = \lambda^{(1)} + 2r_p \cdot h(\vec{x}^{*(1)}) = 0 + 2(1)\left(\frac{1}{6}\right) = \frac{1}{3}$$

Using $r_p = 1$ and $\lambda^{(2)} = \lambda = \frac{1}{3}$, Eqs. (3) and (4) give:

$$x_2^{*(2)} = \frac{32 - \frac{4}{3} + 8}{20+4} = 1.611125$$

$$x_1^{*(2)} = \frac{\frac{1}{3}+12+14+\frac{1}{3}+2}{2(2)(6)} = 1.194444$$

$$h = 0.02778$$

$$\lambda^{(3)} = \lambda^{(2)} + 2r_p \cdot h(\vec{x}^{*(2)}) = \frac{1}{3} + 2(0.02778) = 0.388893$$

Using $r_p = 1$ and $\lambda = 0.388893$, Eqs. (3) and (4) give:

$$x_2^{*(3)} = \frac{32 - 4(0.388893) + 8}{20 + 4} = 1.601851$$

$$x_1^{*(3)} = \frac{0.388893 + 12 + 14 + 0.388893 + 2}{2(2)(6)} = 1.199074$$

7.41 (a)

$$A = (x_1 - 1)^2 + (x_2 - 5)^2 + \lambda(x_1 + x_2 - 5) + r_p (x_1 + x_2 - 5)^2$$

$$\frac{\partial A}{\partial x_1} = 2(x_1 - 1) + \lambda + 2 r_p (x_1 + x_2 - 5) = 0$$

or $$x_1(2 + 2r_p) + x_2(2 r_p) = 10 r_p - \lambda + 2 \qquad (1)$$

$$\frac{\partial A}{\partial x_2} = 2(x_2 - 5) + \lambda + 2 r_p (x_1 + x_2 - 5) = 0$$

or $$x_1(2 r_p) + x_2(2 + 2 r_p) = 10 r_p - \lambda + 10 \qquad (2)$$

Solution of Eqs. (1) and (2):

$$x_2 = \frac{18 r_p - \lambda + 10}{4 r_p + 2} \qquad (3)$$

$$x_1 = \frac{4 r_p^2 - 2\lambda r_p + 8 r_p - 2\lambda + 4}{4(2 r_p + 1)(r_p + 1)} \qquad (4)$$

(b) $\lambda_1^{(1)} = 0$, $r_p = 1$

Eqs. (3) and (4) give:

$$x_1^{*(1)} = \frac{4 - 0 + 8 - 0 + 4}{4(3)(2)} = \frac{2}{3} = 0.666667$$

$$x_2^{*(1)} = \frac{18 - 0 + 10}{4 + 2} = \frac{14}{3} = 4.666667$$

$$h(\vec{x}^{*(1)}) = x_1 + x_2 - 5 = \frac{2}{3} + \frac{14}{3} - 5 = \frac{1}{3}$$

$$\lambda^{(2)} = \lambda^{(1)} + 2 r_p h(\vec{x}^{*(1)}) = 0 + 2(1)\left(\frac{1}{3}\right) = 0.666667$$

Using $r_p = 1$ and $\lambda = 0.666667$, Eqs. (3) and (4) give:

$$x_1^{*(2)} = \frac{4 - 1.333334 + 8 - 1.333334 + 4}{4(3)(2)} = 0.555555$$

$$x_2^{*(2)} = \frac{18 - 0.666667 + 10}{4 + 2} = 4.555555$$

(c) $h(\vec{x}^{*(2)}) = 0.555555 + 4.555555 - 5 = 0.111110$

$$\lambda^{(3)} = \lambda^{(2)} + 2 r_p h(\vec{x}^{*(2)}) = 2(1)(0.111110) = 0.222220$$

7.42 Minimize $f = x_1^3 - 6x_1^2 + 11x_1 + x_3$

subject to $g_1 = x_1^2 + x_2^2 - x_3^2 \le 0$

$g_2 = 4 - x_1^2 - x_2^2 - x_3^2 \le 0$

$g_3 = x_3 - 5 \le 0$

$g_4 = -x_1 \le 0$, $g_5 = -x_2 \le 0$, $g_6 = -x_3 \le 0$

At $\vec{X} = \begin{Bmatrix} 0 \\ \sqrt{2} \\ \sqrt{2} \end{Bmatrix}$, g_1, g_2 and g_4 are active.

$\frac{\partial f}{\partial x_1} = 3x_1^2 - 12x_1 + 11 = 11$, $\frac{\partial f}{\partial x_2} = 0$, $\frac{\partial f}{\partial x_3} = 1$

$\vec{F} = \begin{Bmatrix} -11 \\ 0 \\ -1 \end{Bmatrix}$

$$[G] = \begin{bmatrix} \frac{\partial g_1}{\partial x_1} & \frac{\partial g_2}{\partial x_1} & \frac{\partial g_4}{\partial x_1} \\ \frac{\partial g_1}{\partial x_2} & \frac{\partial g_2}{\partial x_2} & \frac{\partial g_4}{\partial x_2} \\ \frac{\partial g_1}{\partial x_3} & \frac{\partial g_2}{\partial x_3} & \frac{\partial g_4}{\partial x_3} \end{bmatrix} = \begin{bmatrix} 2x_1 & -2x_1 & -1 \\ 2x_2 & -2x_2 & 0 \\ -2x_3 & -2x_3 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & -1 \\ 2\sqrt{2} & -2\sqrt{2} & 0 \\ -2\sqrt{2} & -2\sqrt{2} & 0 \end{bmatrix}$$

$$[G]^T[G] = \begin{bmatrix} 16 & 0 & 0 \\ 0 & 16 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \left([G]^T[G]\right)^{-1} = \frac{1}{256}\begin{bmatrix} 16 & 0 & 0 \\ 0 & 16 & 0 \\ 0 & 0 & 256 \end{bmatrix}$$

$$[G]^T\vec{F} = \begin{Bmatrix} 2\sqrt{2} \\ 2\sqrt{2} \\ 11 \end{Bmatrix}$$

$$\vec{\lambda} = \left([G]^T[G]\right)^{-1}\vec{G}^T\vec{F} = \begin{Bmatrix} 1/4\sqrt{2} \\ 1/4\sqrt{2} \\ 11 \end{Bmatrix} > \vec{0}$$

$\therefore$ Solution is optimal.

Note: In this example, the number of active constraints is same as the number of design variables. Hence the equations $[G]\vec{\lambda} = \vec{F}$ can be directly solved.

7.43 Minimize $f = x_1^3 - 6x_1^2 + 11x_1 + x_3$

subject to $g_1 = x_1^2 + x_2^2 - x_3^2 \leq 0$

$g_2 = 4 - x_1^2 - x_2^2 - x_3^2 \leq 0$

$g_3 = x_3 - 5 \leq 0$

$g_4 = -x_1 \leq 0$

$g_5 = -x_2 \leq 0$

$g_6 = -x_3 \leq 0$

$$\vec{x}^* = \begin{Bmatrix} 0.0 \\ 1.41421356 \\ 1.41421356 \end{Bmatrix}, \qquad f^* = 1.41421356$$

$\Delta x_i = 0.001, \quad i = 1, 2, 3$

Each design variable is perturbed (reduced and increased) and the objective function and constraints are evaluated. The results are given below:

design vector: 0.100000E-02 0.141421E+01 0.141421E+01 ($\vec{x}_1^{\,+}$)
objective function: 0.142521E+01
constraints:
0.107288E-05 -0.715256E-06 -0.358579E+01 -0.100000E-02 -0.141421E+01 -0.141421E+01

design vector: 0.000000E+00 0.141521E+01 0.141421E+01 ($\vec{x}_2^{\,+}$)
objective function: 0.141421E+01
constraints:
0.282967E-02 -0.282943E-02 -0.358579E+01 0.000000E+00 -0.141521E+01 -0.141421E+01

design vector: 0.000000E+00 0.141421E+01 0.141521E+01 ($\vec{x}_3^{\,+}$)
objective function: 0.141521E+01
constraints:
-0.282967E-02 -0.282955E-02 -0.358479E+01 0.000000E+00 -0.141421E+01 -0.141521E+01

design vector: -0.100000E-02 0.141421E+01 0.141421E+01 ($\vec{x}_1^{\,-}$)
objective function: 0.140321E+01
constraints:
0.107288E-05 -0.715256E-06 -0.358579E+01 0.100000E-02 -0.141421E+01 -0.141421E+01

design vector: 0.000000E+00 0.141321E+01 0.141421E+01 ($\vec{x}_2^{\,-}$)
objective function: 0.141421E+01
constraints:
-0.282753E-02 0.282776E-02 -0.358579E+01 0.000000E+00 -0.141321E+01 -0.141421E+01

design vector: 0.000000E+00 0.141421E+01 0.141321E+01 ($\vec{x}_3^{\,-}$)
objective function: 0.141321E+01
constraints:
0.282753E-02 0.282764E-02 -0.358679E+01 0.000000E+00 -0.141421E+01 -0.141321E+01

Since either $f > f^*$ or some $g_j > 0$ at each of the perturbed design vectors $\vec{x}_i^{\,+}$ and $\vec{x}_i^{\,-}$ ($i = 1, 2, 3$), the given design vector $\vec{x}^* = \begin{Bmatrix} 0 \\ \sqrt{2} \\ \sqrt{2} \end{Bmatrix}$ is optimum.

7.44

(a) $\vec{X}(r) = \vec{A}_0 + r\,\vec{A}_1$

At $r = 0.0001$,
$$\begin{Bmatrix} 1.3478 \\ 0.7720 \\ 0.4293 \end{Bmatrix} = \vec{A}_0 + 0.0001\,\vec{A}_1 \qquad (1)$$

At $r = 0.01$,
$$\begin{Bmatrix} 1.3313 \\ 0.7539 \\ 0.3710 \end{Bmatrix} = \vec{A}_0 + 0.01\,\vec{A}_1 \qquad (2)$$

subtracting Eq.(2) from (1), we find:
$$\vec{A}_1 = \begin{Bmatrix} -1.6667 \\ -1.8283 \\ -5.8889 \end{Bmatrix} \qquad (3)$$

Eqs. (2) and (3):
$$\vec{A}_0 = \begin{Bmatrix} 1.3480 \\ 0.7722 \\ 0.4299 \end{Bmatrix} \qquad (4)$$

As $r \to 0$,
$$\vec{X}^* = \vec{A}_0 = \begin{Bmatrix} 1.3480 \\ 0.7722 \\ 0.4299 \end{Bmatrix} \qquad (5)$$

For f, assume $f(r) = a_0 + r\,a_1$ (6)

At $r = 0.0001$, $0.1158 = a_0 + 0.0001\,a_1$ (7)

At $r = 0.01$, $0.1564 = a_0 + 0.01\,a_1$ (8)

Eqs. (7) and (8) give: $a_0 = 0.1154$, $a_1 = 4.1010$ (9)

As $r \to 0$, $f^* = a_0 = 0.1154$ (10)

(b) $\vec{X}(r) = \vec{A}_0 + \sqrt{r}\,\vec{A}_1$ (11)

At $r = 0.0001$,
$$\begin{Bmatrix} 1.3478 \\ 0.7720 \\ 0.4293 \end{Bmatrix} = \vec{A}_0 + \sqrt{0.0001}\,\vec{A}_1 \qquad (12)$$

At $r = 0.01$,
$$\begin{Bmatrix} 1.3313 \\ 0.7539 \\ 0.3710 \end{Bmatrix} = \vec{A}_0 + \sqrt{0.01}\,\vec{A}_1 \qquad (13)$$

Multiply Eq. (12) by 0.1 and Eq. (13) by 0.01 and subtract to find:
$$\vec{A}_0 = \vec{X}^* = \vec{X}(r \to 0) = \begin{Bmatrix} 1.3496 \\ 0.7740 \\ 0.4358 \end{Bmatrix} \qquad (14)$$

Similarly, $f(r) = a_0 + \sqrt{r}\, a_1$

Using $0.1158 = a_0 + \sqrt{0.0001}\, a_1$

and $0.1564 = a_0 + \sqrt{0.01}\, a_1$

we obtain $a_0 = 0.1113$ and $a_1 = 0.4510$

$\therefore\ f^* = f(r \to 0) = a_0 = 0.1113$

7.45

$$\vec{X}(r) = \vec{A}_0 + r\,\vec{A}_1 + r^2\,\vec{A}_2 \tag{1}$$

$$\vec{X}_k^* = \vec{A}_0 + r_k\,\vec{A}_1 + r_k^2\,\vec{A}_2 \tag{2}$$

$$\vec{X}_{k-1}^* = \vec{A}_0 + r_{k-1}\,\vec{A}_1 + r_{k-1}^2\,\vec{A}_2 \tag{3}$$

$$\vec{X}_{k-2}^* = \vec{A}_0 + r_{k-2}\,\vec{A}_1 + r_{k-2}^2\,\vec{A}_2 \tag{4}$$

Eq. (2) − Eq. (3): $$\vec{X}_k^* - \vec{X}_{k-1}^* = (r_k - r_{k-1})\vec{A}_1 + (r_k^2 - r_{k-1}^2)\vec{A}_2 \tag{5}$$

Eq. (2) − Eq. (4): $$\vec{X}_k^* - \vec{X}_{k-2}^* = (r_k - r_{k-2})\vec{A}_1 + (r_k^2 - r_{k-2}^2)\vec{A}_2 \tag{6}$$

Solution of Eqs. (5) and (6):

$$\vec{A}_1 = \frac{c_2\,\vec{c}_3 - c_4\,\vec{c}_1}{c_2 - c_4}, \qquad \vec{A}_2 = \frac{\vec{c}_1 - \vec{c}_3}{c_2 - c_4} \tag{7}$$

Eqs. (2) and (7):

$$\vec{A}_0 = \vec{X}_k^* - r_k\left(\frac{c_2\,\vec{c}_3 - c_4\,\vec{c}_1}{c_2 - c_4}\right) - r_k^2\left(\frac{\vec{c}_1 - \vec{c}_3}{c_2 - c_4}\right) \tag{8}$$

Here $r_k = 0.0001$, $r_{k-1} = 0.01$, $r_{k-2} = 1$,

$c_2 = r_k + r_{k-1} = 0.0101$,

$c_4 = r_k + r_{k-2} = 1.0001$,

$$\vec{c}_1 = \frac{\vec{X}_k^* - \vec{X}_{k-1}^*}{r_k - r_{k-1}} = \begin{Bmatrix} -1.6667 \\ -1.8283 \\ -5.8889 \end{Bmatrix}$$

$$\vec{c}_3 = \frac{\vec{X}_k^* - \vec{X}_{k-2}}{r_k - r_{k-2}} = \begin{Bmatrix} -0.4594 \\ -0.0532 \\ 0.2967 \end{Bmatrix}$$

$$\vec{X}^* = \vec{A}_0 = \vec{X}(r=0) = \begin{Bmatrix} 1.3480 \\ 0.7722 \\ 0.4299 \end{Bmatrix} \quad \text{from Eq. (8).}$$

$f(r) = a_0 + r\,a_1 + r^2 a_2$

Using equations similar to those of $\vec{x}(r)$, we obtain

$$a_0 = f_k^* - r_k \begin{pmatrix} c_2 c_3 - c_4 c_1 \\ c_2 - c_4 \end{pmatrix} - r_k^2 \begin{pmatrix} c_1 - c_3 \\ c_2 - c_4 \end{pmatrix}$$

with $c_2 = r_k + r_{k-1} = 0.0001 + 0.01 = 0.0101$

$c_4 = r_k + r_{k-2} = 0.0001 + 1.0 = 1.0001$

$$c_1 = \frac{f_k^* - f_{k-1}^*}{r_k - r_{k-1}} = \frac{0.1158 - 0.1564}{0.0001 - 0.01} = 4.1010$$

$$c_3 = \frac{f_k^* - f_{k-2}^*}{r_k - r_{k-2}} = \frac{0.1158 - 0.7072}{0.0001 - 1.0} = 0.5915$$

$$a_0 = f_k^* - r_k \left(\frac{c_2 c_3 - c_4 c_1}{c_2 - c_4}\right) - r_k^2 \left(\frac{c_1 - c_3}{c_2 - c_4}\right) = 0.1154$$

$\therefore f^* = f(r=0) = a_0 = 0.1154$

7.46

THEOREM 7.3 *If*

$$\phi(\mathbf{X}, r_k) = f(\mathbf{X}) + r_k\, G[\mathbf{g}(\mathbf{X})] = f(\mathbf{X}) + r_k \sum_{j=1}^{m} \langle g_j(\mathbf{X}) \rangle^q$$

the following relations will be valid for any $0 < r_k < r_{k+1}$:

(i) $\phi(\mathbf{X}_k^*, r_k) \leqslant \phi(\mathbf{X}_{k+1}^*, r_{k+1})$,

(ii) $f(\mathbf{X}_k^*) \leqslant f(\mathbf{X}_{k+1}^*)$,

(iii) $G[\mathbf{g}(\mathbf{X}_k^*)] \geqslant G[\mathbf{g}(\mathbf{X}_{k+1}^*)]$.

Proof:

(i) Since $r_k < r_{k+1}$, and $G[\mathbf{g}(\mathbf{X})] \geqslant 0$,

$$r_k\, G[\mathbf{g}(\mathbf{X}_{k+1}^*)] \leqslant r_{k+1}\, G[\mathbf{g}(\mathbf{X}_{k+1}^*)]$$

Hence

$$\begin{aligned} f(\mathbf{X}_{k+1}^*) + r_k\, G[\mathbf{g}(\mathbf{X}_{k+1}^*)] &\leqslant f(\mathbf{X}_{k+1}^*) + r_{k+1}\, G[\mathbf{g}(\mathbf{X}_{k+1}^*)] \\ &= \phi(\mathbf{X}_{k+1}^*, r_{k+1}) \end{aligned} \tag{1}$$

Furthermore, $\mathbf{X}_k^*$ minimizes $\phi(\mathbf{X}, r_k)$, so that

$$\begin{aligned} \phi(\mathbf{X}_k^*, r_k) = f(\mathbf{X}_k^*) + r_k\, G[\mathbf{g}(\mathbf{X}_k^*)] &\leqslant f(\mathbf{X}_{k+1}^*) + r_k\, G[\mathbf{g}(\mathbf{X}_{k+1}^*)] \\ &\leqslant \phi(\mathbf{X}_{k+1}^*, r_{k+1}) \end{aligned} \tag{2}$$

(ii) Because $\mathbf{X}_{k+1}^*$ minimizes $\phi(\mathbf{X}, r_{k+1})$,

$$f(\mathbf{X}_k^*) + r_{k+1}\, G[\mathbf{g}(\mathbf{X}_k^*)] \geqslant f(\mathbf{X}_{k+1}^*) + r_{k+1}\, G[\mathbf{g}(\mathbf{X}_{k+1}^*)] \tag{3}$$

Similarly,

$$f(\mathbf{X}_{k+1}^*) + r_k\, G[\mathbf{g}(\mathbf{X}_{k+1}^*)] \geqslant f(\mathbf{X}_k^*) + r_k\, G[\mathbf{g}(\mathbf{X}_k^*)] \tag{4}$$

By adding and rearranging Eqs. (3) and (4), we obtain

$$r_k \{G[\mathbf{g}(\mathbf{X}_{k+1}^*)] - G[\mathbf{g}(\mathbf{X}_k^*)]\} \geqslant r_{k+1} \{G[\mathbf{g}(\mathbf{X}_{k+1}^*)] - G[\mathbf{g}(\mathbf{X}_k^*)]\} \tag{5}$$

But $r_k < r_{k+1}$. Hence the inequality (5) gives

$$G[\mathbf{g}(\mathbf{X}_{k+1}^*)] - G[\mathbf{g}(\mathbf{X}_k^*)] \leqslant 0$$

or

$$G[\mathbf{g}(\mathbf{X}_k^*)] \geqslant G[\mathbf{g}(\mathbf{X}_{k+1}^*)] \tag{6}$$

(iii) Finally from inequality (1), we have

$$f(\mathbf{X}_k^*) - f(\mathbf{X}_{k+1}^*) \leqslant r_k \{G[\mathbf{g}(\mathbf{X}_{k+1}^*)] - G[\mathbf{g}(\mathbf{X}_k^*)]\} \tag{7}$$

Since $G[\mathbf{g}(\mathbf{X}_{k+1}^*)] \leqslant G[\mathbf{g}(\mathbf{X}_k^*)]$, and $r_k > 0$, inequality (7) gives

$$f(\mathbf{X}_k^*) \leqslant f(\mathbf{X}_{k+1}^*) \tag{8}$$

The above results can be used to prove the following theorem (using a procedure similar to that of Theorem 7.1).

THEOREM 7.4 *If the function* $\phi(\mathbf{X}, r_k)$ *given by Eq.* (7.199) *is minimized for an increasing sequence of values of* r_k, *the unconstrained minima* $\mathbf{X}_k^*$ *converge to the optimum solution* $(\mathbf{X}^*)$ *of the constrained problem as* $r_k \to \infty$.

7.52

problem752.m

```
function problem752
clear;clc;

lb=[-100 -100 -100 -100];ub=[100 100 100 100];

x0=[0 0 0 0];
f0=obj(x0);g0=gcons(x0);

[xv,fval]=fmincon(@obj,x0,[],[],[],[],lb,ub,@cons)
gx=gcons(xv);

fprintf('**********Results for Problem 752**********\n');
fprintf('*************************************************\n');
fprintf('  x0(1)     x0(2)     x0(3)     x0(4)\n');
fprintf(' %8.4f   %8.4f  %8.4f   %8.4f\n', x0(1),x0(2),x0(3),x0(4));
fprintf('*************************************************\n');
fprintf('   f(x0)\n');
fprintf('  %8.4f\n',f0);
fprintf('*************************************************\n');
fprintf('   g1(x0)          g2(x0)          g3(x0)\n');
fprintf('   %8.4f       %8.4f       %8.4f\n',g0(1),g0(2),g0(3));
fprintf('*************************************************\n');
fprintf('  x*(1)     x*(2)     x*(3)     x*(4)\n');
fprintf(' %8.4f   %8.4f  %8.4f   %8.4f\n', xv(1),xv(2),xv(3),xv(4));
fprintf('*************************************************\n');
fprintf('   f(x*)\n');
fprintf(' %8.4f\n',fval);
fprintf('*************************************************\n');
fprintf('   g1(x*)          g2(x*)          g3(x*)\n');
fprintf('   %8.4f       %8.4f       %8.4f\n',gx(1),gx(2),gx(3));
end
%%%
```

```
%%%
```

7.53

```
function problem753

lb=[0.00 0.00];
x0=[0.5 1.0];
f0=obj(x0);g0=gcons(x0);

[xv,fval]=fmincon(@obj,x0,[],[],[],[],lb,[],@cons)
gx=gcons(xv);

fprintf('***************Results Problem 5***************\n');
fprintf('***************************************************\n');
fprintf('  x0(1)                x0(2)\n');
fprintf(' %8.4f              %8.4f\n', x0(1),x0(2));
fprintf('***************************************************\n');
fprintf('  f(x0)\n');
fprintf('  %8.4f\n',f0);
fprintf('***************************************************\n');
fprintf('  g1(x0)         g2(x0)         \n');
fprintf('  %8.4f      %8.4f       %8.4f\n',g0(1),g0(2));
fprintf('***************************************************\n');
fprintf('  x*(1)              x*(2)\n');
fprintf(' %8.4f           %8.4f\n', xv(1),xv(2));
fprintf('***************************************************\n');
fprintf('  f(x*)\n');
fprintf(' %8.4f\n',fval);
fprintf('***************************************************\n');
fprintf('  g1(x*)               g2(x*)          \n');
fprintf(' %8.4f  %8.4f  %8.4f\n',gx(1),gx(2));
end
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
function f=obj(x)
f=x(1)^2+x(2)^2-4*x(1)-6*x(2);
end
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
function [c,ceq]=cons(x)
c(1)=x(1)+x(2)-2;
c(2)=2*x(1)+3*x(2)-12;
ceq=0;
end
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
function [g,geq]=gcons(xv);
g(1)=xv(1)+xv(2)-2;
g(2)=2*xv(1)+3*xv(2)-12;
geq=0;
end
```

**********Results Problem 7.53**********

****

x0(1) x0(2)

0.5000 1.0000

****

f(x0)

-6.7500

****

g1(x0) g2(x0)

-0.5000 -8.0000

****

x*(1) x*(2)

0.5000 1.5000

****

f(x*)

-8.5000

****

g1(x*) g2(x*)

-0.0000 -6.5000

7.54

```
function problem754
clear;clc;

x0=[0.5 1.0 1.0];
f0=obj(x0);[g0,geq0]=gcons(x0);

[xv,fval]=fmincon(@obj,x0,[],[],[],[],[],[],@cons)
gxeq=gcons(xv);

fprintf('**********Results for Problem 754**********\n');
fprintf('************************************************\n');
fprintf('  x0(1)      x0(2)      x0(3)\n');
fprintf(' %8.4f   %8.4f   %8.4f\n', x0(1),x0(2),x0(3));
fprintf('************************************************\n');
fprintf('   f(x0)\n');
fprintf('  %8.4f\n',f0);
fprintf('************************************************\n');
fprintf('   geq(x0)\n');
fprintf('   %8.4f\n',geq0);
fprintf('************************************************\n');
fprintf('   x*(1)      x*(2)      x*(3)\n');
fprintf(' %8.4f   %8.4f   %8.4f\n', xv(1),xv(2),xv(3));
fprintf('************************************************\n');
fprintf('   f(x*)\n');
fprintf(' %8.4f\n',fval);
fprintf('************************************************\n');
fprintf('   geq(x*)\n');
fprintf('   %8.4f\n',gxeq);
end
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
function f=obj(x)
f=x(1)^2+3*x(2)^2+x(3);
end
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
function [c,ceq]=cons(x)
c=0;
ceq=x(1)^2+x(2)^2+x(3)^2-16;
end
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
function [g,geq]=gcons(xv);
g=0;
geq=xv(1)^2+xv(2)^2+xv(3)^2-16;
end
```

**********Results for Problem 754**********

****

x0(1) x0(2) x0(3)

0.5000 1.0000 1.0000

****

f(x0)

4.2500

****

geq(x0)

-13.7500

****

x*(1) x*(2) x*(3)

0.0000 -0.0000 4.0000

****

f(x*)

4.0000

****

geq(x*)

0.0000

7.55

```
function problem 7.55%%%GA

x0=[1 1];
f0=obj(x0);g0=gcons(x0);

[xv,fval]=fmincon(@obj,x0,[],[],[],[],[],[],@cons);
gx=gcons(xv);

fprintf('****** Results Problem 7.55 ********\n');
fprintf('*********************************\n')
fprintf(' x0(1)       x0(2)\n');
fprintf(' %6.4f       %6.4f\n', x0(1),x0(2));
fprintf('*********************************\n')
fprintf(' f(x0)\n');
fprintf('%8.4f\n',f0);
fprintf('*********************************\n')
fprintf(' g1(x0)       g2(x0)      g3(x0)\n');
fprintf(' %6.4f       %6.4f\n',g0(1),g0(2),g0(3));
fprintf('*********************************\n')
fprintf('  x*(1)       x*(2)\n');
fprintf('  %6.4f       %6.4f\n', xv(1),xv(2));
fprintf('*********************************\n')
fprintf('  f(x*)\n');
fprintf(' %8.4f\n',fval);
fprintf('*********************************\n')
fprintf('  g1(x*)       g2(x*)      g3(x*)\n');
fprintf('  %6.4f       %6.4f\n',gx(1),gx(2),gx(3));
end

%%%%%%%%%%%%%%%%%%%%%
function f=obj(x);
f=x(1)^2+x(2)^2;
end
%%%%%%%%%%%%%%%%%%%%%
function [c,ceq]=cons(x);
c(1)=4-x(1)-x(2)^2;
c(2)=3*x(2)-x(1);
c(3)=-3*x(2)-x(1);
ceq=0;
end
%%%%%%%%%%%%%%%%%%%%%
function g=gcons(v);
g(1)=4-v(1)-v(2)^2;
g(2)=3*v(2)-v(1);
g(3)=-3*v(2)-v(1);
end
```

****** Results Problem 7.55 ********

****

x0(1) x0(2)

1.0000 1.0000

****

f(x0)

2.0000

****

g1(x0) g2(x0) g3(x0)

2.0000 2.0000 -4.0000

****

x*(1) x*(2)

3.0000 1.0000

****

f(x*)

10.0000

****

g1(x*) g2(x*) g3(x*)

0.0000 0.0000 -6.0000

Chapter 8
Geometric Programming

8.1 Arithmetic - geometric inequality:

$$\Delta_1 u_1 + \Delta_2 u_2 + \cdots + \Delta_N u_N \geq u_1^{\Delta_1} u_2^{\Delta_2} \cdots u_N^{\Delta_N} \quad \text{with} \quad \sum_{i=1}^{N} \Delta_i = 1 \qquad (1)$$

$$f = \frac{1}{3} x^{-2} + \frac{2}{3} x^{-3} + \frac{4}{3} x^{3/2}$$

$$= \frac{3}{16}\left(\frac{16}{9} x^{-2}\right) + \frac{3}{16}\left(\frac{32}{9} x^{-3}\right) + \frac{10}{16}\left(\frac{64}{30} x^{3/2}\right)$$

$$\geq \left(\frac{16}{9} x^{-2}\right)^{\frac{3}{16}} \left(\frac{32}{9} x^{-3}\right)^{\frac{3}{16}} \left(\frac{64}{30} x^{3/2}\right)^{\frac{10}{16}}$$

$$\geq \left(\frac{512}{81}\right)^{3/16} x^{-\frac{15}{16}} \left(\frac{64}{30}\right)^{\frac{10}{16}} x^{\frac{15}{16}}$$

$$\geq \left(\frac{512}{81}\right)^{3/16} \left(\frac{64}{30}\right)^{10/16} = (1.413016)(1.605690)$$

$$\geq 2.268866$$

8.2

$$f = 1 + x + \frac{1}{x} + \frac{1}{x^2} = \frac{1}{6}(6) + \frac{3}{6}(2x) + \frac{1}{6}\left(\frac{6}{x}\right) + \frac{1}{6}\left(\frac{6}{x^2}\right)$$

$$\geq (6)^{\frac{1}{6}} (2x)^{3/6} \left(\frac{6}{x}\right)^{1/6} \left(\frac{6}{x^2}\right)^{1/6}$$

$$\geq 6^{1/6}\ 2^{3/6}\ 6^{1/6}\ 6^{1/6} = (216)^{1/6}\ (2)^{3/6} = (216 * 8)^{1/6}$$

$$\geq 3.464102$$

8.3

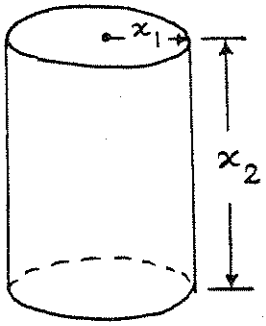
$$f = \frac{1}{2} x^{-3} + x^2 + 2x = \frac{1}{3}\left(\frac{3}{2} x^{-3}\right) + \frac{1}{3}\left(3x^2\right) + \frac{1}{3}(6x)$$

$$\geq \left(\frac{3}{2} x^{-3}\right)^{\frac{1}{3}} \left(3x^2\right)^{\frac{1}{3}} (6x)^{\frac{1}{3}}$$

$$\geq \left(\frac{3}{2}\right)^{\frac{1}{3}} (3)^{\frac{1}{3}} (6)^{\frac{1}{3}} = (27)^{\frac{1}{3}}$$

$$\geq 3.0$$

8.4 Design variables: x_1 = radius (m), x_2 = height (m)



f = cost = cost of cylinder + cost of transportation

$$= 80\pi x_1^2 + 20\pi x_1 x_2 + \frac{80}{\pi x_1^2 x_2}$$

$c_1 = 80\pi$, $c_2 = 20\pi$, $c_3 = 80/\pi$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} = \begin{bmatrix} 2 & 1 & -2 \\ 0 & 1 & -1 \end{bmatrix}$$

Orthogonality and normality conditions:

$$2\Delta_1 + \Delta_2 - 2\Delta_3 = 0 \qquad (1)$$

$$\Delta_2 - \Delta_3 = 0 \qquad (2)$$

$$\Delta_1 + \Delta_2 + \Delta_3 = 1 \qquad (3)$$

Eq. (2): $\Delta_2 = \Delta_3$, Eq. (1): $2\Delta_1 = 2\Delta_3 - \Delta_2 = \Delta_3$

Eq. (3): $\Delta_3/2 + \Delta_3 + \Delta_3 = 1 \Rightarrow \Delta_3^* = \frac{2}{5}$

$\Delta_1^* = \frac{1}{5}$, $\Delta_2^* = \frac{2}{5}$, $\Delta_3^* = \frac{2}{5}$

$$f^* = \left(\frac{c_1}{\Delta_1^*}\right)^{\Delta_1^*}\left(\frac{c_2}{\Delta_2^*}\right)^{\Delta_2^*}\left(\frac{c_3}{\Delta_3^*}\right)^{\Delta_3^*} = \left(\frac{80\pi}{1/5}\right)^{1/5}\left(\frac{20\pi}{2/5}\right)^{2/5}\left(\frac{80}{2\pi/5}\right)^{2/5}$$

$= \$\ 165.8983$

$U_i^* = \Delta_i^* f^*$; $i = 1, 2, 3$

i.e., $80\pi x_1^{*2} = \frac{1}{5}(165.8983) \Rightarrow x_1^* = 0.3633$ m

$20\pi x_1^* x_2^* = \frac{2}{5}(165.8983) \Rightarrow x_2^* = 2.9070$ m

---

8.5 Here $c_2 = 40\pi$

$\Delta_1^* = \frac{1}{5}$, $\Delta_2^* = \frac{2}{5}$, $\Delta_3^* = \frac{2}{5}$ (same as in Problem 8.4)

$$f^* = \left(\frac{80\pi}{1/5}\right)^{1/5}\left(\frac{40\pi}{2/5}\right)^{2/5}\left(\frac{80}{2\pi/5}\right)^{2/5} = \$\ 218.904$$

$U_1^* = 80\pi x_1^{*2} = \frac{1}{5}(218.904) \Rightarrow x_1^* = 0.4174$ m

$U_2^* = 40\pi x_1^* x_2^* = \frac{2}{5}(218.904) \Rightarrow x_2^* = 1.6695$ m

---

8.6

Min. $f = 80\pi r^2 + 20\pi r h + \dfrac{80}{\pi r^2 h}$

subject to: $\dfrac{80}{\pi r^2 h} \le 10$ or $\dfrac{8}{\pi} r^{-2} h^{-1} \le 1$

$x_1 = r$, $x_2 = h$; $n = 2$, $N = 4$, $N_0 = 3$, $N_1 = 1$, $m = 1$, DOD = 1

$\vec{\lambda} = \{ \lambda_{01}, \lambda_{02}, \lambda_{03}, \lambda_{11} \}^T$

$$v = \left[\frac{c_{01}}{\lambda_{01}}(\lambda_{01}+\lambda_{02}+\lambda_{03})\right]^{\lambda_{01}} \left[\frac{c_{02}}{\lambda_{02}}(\lambda_{01}+\lambda_{02}+\lambda_{03})\right]^{\lambda_{02}} \cdot \left[\frac{c_{03}}{\lambda_{03}}(\lambda_{01}+\lambda_{02}+\lambda_{03})\right]^{\lambda_{03}} \left[\frac{c_{11}}{\lambda_{11}}(\lambda_{11})\right]^{\lambda_{11}} \quad (1)$$

Here $c_{01} = 80\pi$, $c_{02} = 20\pi$, $c_{03} = \dfrac{80}{\pi}$, $c_{11} = \dfrac{8}{\pi}$

$a_{011} = 2$ $a_{012} = 1$ $a_{013} = -2$ $a_{111} = -2$

$a_{021} = 0$ $a_{022} = 1$ $a_{023} = -1$ $a_{121} = -1$

$$\lambda_{01} + \lambda_{02} + \lambda_{03} = 1 \quad (2)$$

$$a_{011}\lambda_{01} + a_{012}\lambda_{02} + a_{013}\lambda_{03} + a_{111}\lambda_{11} = 0 \quad (3)$$

$$a_{021}\lambda_{01} + a_{022}\lambda_{02} + a_{023}\lambda_{03} + a_{121}\lambda_{11} = 0 \quad (4)$$

i.e.,

$$\lambda_{01} + \lambda_{02} + \lambda_{03} = 1 \quad (5)$$

$$2\lambda_{01} + \lambda_{02} - 2\lambda_{03} - 2\lambda_{11} = 0 \quad (6)$$

$$\lambda_{02} - \lambda_{03} - \lambda_{11} = 0 \quad (7)$$

Eqs. (5) to (7) can be solved as follows:

$$\lambda_{03} = 1 - \lambda_{01} - \lambda_{02} \quad (8)$$

$$\lambda_{03} = \lambda_{01} + \tfrac{1}{2}\lambda_{02} - \lambda_{11} \quad (9)$$

$$\lambda_{03} = \lambda_{02} - \lambda_{11} \quad (10)$$

Eqs. (9) and (10) give: $\lambda_{02} = 2\lambda_{01}$ (11)

Eqs. (8) and (9): $\lambda_{02} = \tfrac{1}{2} - \tfrac{1}{2}\lambda_{01} + \tfrac{1}{2}\lambda_{11}$ (12)

Eqs. (11) and (12): $\lambda_{11} = 5\lambda_{01} - 1$ (13)

Eqs. (10), (11) and (13): $\lambda_{03} = 1 - 3\lambda_{01}$ (14)

$$v = \left(\frac{80\pi}{\lambda_{01}}\right)^{\lambda_{01}} \left(\frac{20\pi}{2\lambda_{01}}\right)^{2\lambda_{01}} \left(\frac{80}{\pi\{1-3\lambda_{01}\}}\right)^{1-3\lambda_{01}} \left(\frac{8}{\pi}\right)^{5\lambda_{01}-1}$$

v is maximized with respect to λ_{01}.

By varying λ_{01} from 0.00 in steps of 0.01, $\lambda_{01}^* = 0.2$ is found to maximize v.

$$\lambda_{01}^* = 0.2, \quad \lambda_{02}^* = 0.4, \quad \lambda_{03}^* = 0.4, \quad \lambda_{11}^* = 0$$

$$v^* = \left(\frac{80\pi}{0.2}\right)^{0.2} \left(\frac{20\pi}{0.4}\right)^{0.4} \left(\frac{80}{\pi(0.4)}\right)^{0.4} \left(\frac{8}{\pi}\right)^{0}$$

$$= (4.1672)(7.5587)(5.2669)(1) = 165.898743$$

$$\lambda_{01}^* = 0.2 = \frac{80\pi r^{*2}}{v^*} \Rightarrow r^* = 0.363342 \text{ m}$$

$$\lambda_{02}^* = 0.4 = \frac{20\pi r^* h^*}{v^*} \Rightarrow h^* = 2.906739 \text{ m}$$

$\lambda_{11}^* = 0 \Rightarrow$ indicates that the constraint is inactive.

8.7

$$\text{Min.} \quad f = x_1^{-1.5} x_2^{-1} \qquad (1)$$

subject to:

$$\frac{1}{4} x_1 x_2^{-1} \leq 1 \qquad (2)$$

$$\frac{x_1}{2.5 * 10^6} + \frac{x_2}{2.5 * 10^6} \leq 1 \qquad (3)$$

Here $m = 2$, $N_0 = 1$, $N_1 = 1$, $N_2 = 2$, $N = 4$, $n = 2$, $DOD = 1$

$$c_{01} = 1, \quad c_{11} = 0.25, \quad c_{21} = \frac{1}{2.5 * 10^6}, \quad c_{22} = \frac{1}{2.5 * 10^6}$$

Normalization and orthogonalization conditions:

$$\lambda_{01} = 1 \qquad (4)$$

$$-1.5\lambda_{01} + \lambda_{11} + \lambda_{21} = 0 \qquad (5)$$

$$-\lambda_{01} - \lambda_{11} + \lambda_{22} = 0 \qquad (6)$$

Eqs. (4) to (6): $\lambda_{01} = 1, \quad \lambda_{11} = \lambda_{22} - 1, \quad \lambda_{21} = 2.5 - \lambda_{22}$ (7)

$$\text{Max.} \quad v = \left(\frac{c_{01}}{\lambda_{01}} \cdot \lambda_{01}\right)^{\lambda_{01}} \left(\frac{c_{11}}{\lambda_{11}} \cdot \lambda_{11}\right)^{\lambda_{11}} \left(\frac{c_{21}}{\lambda_{21}} \{\lambda_{21} + \lambda_{22}\}\right)^{\lambda_{21}} \cdot$$

$$\left(\frac{c_{22}}{\lambda_{22}} \{\lambda_{21} + \lambda_{22}\}\right)^{\lambda_{22}}$$

$$= (1)\left(\frac{1}{4}\right)^{\lambda_{11}} \left(\frac{2.5}{2.5 * 10^6 \lambda_{21}}\right)^{\lambda_{21}} \left(\frac{2.5}{2.5 * 10^6 \lambda_{22}}\right)^{\lambda_{22}} \qquad (8)$$

Where $\lambda_{11} = \lambda_{22} - 1$ and $\lambda_{21} = 2.5 - \lambda_{22}$

Since all $\lambda_{ij} \geq 0$, we have the solution:

$$\lambda_{01} = 1, \quad \lambda_{11} = 0, \quad \lambda_{21} = \frac{3}{2}, \quad \lambda_{22} = 1$$

$$v_{max} = \left(\frac{1}{4}\right)^0 \left(\frac{2}{3*10^6}\right)^{3/2} \left(\frac{1}{10^6}\right)^1 = 0.544331 * 10^{-15}$$

$$\lambda_{01}^* = 1 = \frac{x_1^{-1.5}\, x_2^{-1}}{0.544331 * 10^{-15}} \Rightarrow x_1^{-1.5}\, x_2^{-1} = 0.544331 * 10^{-15}$$

$\Delta_{11}^* = 0 \Rightarrow$ constraint g_1 is inactive.

$$\Delta_{21}^* = \frac{3/2}{5/2} = \frac{x_1}{2.5 * 10^6} \Rightarrow x_1^* = \left(\frac{3}{5}\right)\left(2.5 * 10^6\right) = \$\ 1.5 * 10^6$$

$$\Delta_{22}^* = \frac{1}{5/2} = \frac{x_2}{2.5 * 10^6} \Rightarrow x_2^* = \left(\frac{2}{5}\right)\left(2.5 * 10^6\right) = \$\ 1.0 * 10^6$$

8.8

Min. $f = \alpha_2 D^0 L^1 d^0 N^1 + \alpha_3 D^{2.5} L^1 d^0 N^0 + \alpha_4 D^1 L^1 d^0 N^0$

$+ \alpha_5 \alpha_{10} D^0 L^1 d^{-5} N^{-2} + \alpha_6 \alpha_{10} D^0 L^1 d^1 N^1$

subject to:

g_1: $\frac{1}{\alpha_1} D^0 L^1 d^0 N^1 \geq 1$

g_2: $\frac{\alpha_7}{\alpha_9} D^0 L^{-1.4} d^{-1} N^{-1.2} + \frac{\alpha_8}{\alpha_9} D^0 L^{-1} d^{-0.2} N^0 \geq 1$

g_3: $\frac{\pi}{4} D^2 L^0 d^{-2} N^{-1} \geq 1$

$D > 0, \quad L > 0, \quad d > 0, \quad N > 0$

Here $N_0 = 5, \; N_1 = 1, \; N_2 = 2, \; N_3 = 1, \; N = 9, \; n = 4,$

$DOD = N - n - 1 = 4, \; m = 3; \; x_1 = D, \; x_2 = L, \; x_3 = d, \; x_4 = N$

$\sigma_0 = 1, \; \sigma_1 = -1, \; \sigma_2 = -1, \; \sigma_3 = -1$

$c_{01} = \alpha_2, \; c_{02} = \alpha_3, \; c_{03} = \alpha_4, \; c_{04} = \alpha_5 \alpha_{10}, \; c_{05} = \alpha_6 \alpha_{10}$

$c_{11} = \frac{1}{\alpha_1}, \; c_{21} = \frac{\alpha_7}{\alpha_9}, \; c_{22} = \frac{\alpha_8}{\alpha_9}, \; c_{31} = \frac{\pi}{4}$

$a_{011} = 0$ $a_{012} = 2.5$ $a_{013} = 1$ $a_{014} = 0$ $a_{015} = 0$

$a_{021} = 1$ $a_{022} = 1$ $a_{023} = 1$ $a_{024} = 1$ $a_{025} = 1$

$a_{031} = 0$ $a_{032} = 0$ $a_{033} = 0$ $a_{034} = -5$ $a_{035} = 1$

$a_{041} = 1$ $a_{042} = 0$ $a_{043} = 0$ $a_{044} = -2$ $a_{045} = 1$

$a_{111} = 0$ $a_{211} = 0$ $a_{212} = 0$ $a_{311} = 2$

$a_{121} = 1$ $a_{221} = -1.4$ $a_{222} = -1$ $a_{321} = 0$

$a_{131} = 0$ $a_{231} = -1$ $a_{232} = -0.2$ $a_{331} = -2$

$a_{141} = 1$ $a_{241} = -1.2$ $a_{242} = 0$ $a_{341} = -1$

Dual problem:

$$\text{Max.} \quad v(\vec{\lambda}) = \prod_{k=0}^{m} \prod_{j=1}^{N_k} \left\{ \frac{c_{kj}}{\lambda_{kj}} \sum_{l=1}^{N_k} \lambda_{kl} \right\}^{\sigma_k \lambda_{kj}} \tag{1}$$

Subject to

$$\sum_{j=1}^{N_0} \lambda_{0j} = 1 \tag{2}$$

$$\sum_{k=0}^{m} \sum_{j=1}^{N_k} \sigma_k \, a_{kij} \, \lambda_{kj} = 0 \; ; \; i = 1, 2, \ldots, m \tag{3}$$

Equations (2) to (3) can be rewritten as:

$$\lambda_{01} + \lambda_{02} + \lambda_{03} + \lambda_{04} + \lambda_{05} = 1 \tag{4}$$

For $i=1$:

$$\sum_{j=1}^{N_0} \sigma_0 \, a_{01j} \, \lambda_{0j} + \sum_{j=1}^{N_1} \sigma_1 \, a_{11j} \, \lambda_{1j} + \sum_{j=1}^{N_2} \sigma_2 \, a_{21j} \, \lambda_{2j} + \sum_{j=1}^{N_3} \sigma_3 \, a_{31j} \, \lambda_{3j} = 0 \tag{5}$$

For $i=2$:

$$\sum_{j=1}^{N_0} \sigma_0 \, a_{02j} \, \lambda_{0j} + \sum_{j=1}^{N_1} \sigma_1 \, a_{12j} \, \lambda_{1j} + \sum_{j=1}^{N_2} \sigma_2 \, a_{22j} \, \lambda_{2j} + \sum_{j=1}^{N_3} \sigma_3 \, a_{32j} \, \lambda_{3j} = 0 \tag{6}$$

For $i=3$:

$$\sum_{j=1}^{N_0} \sigma_0\, a_{03j}\, \lambda_{0j} + \sum_{j=1}^{N_1} \sigma_1\, a_{13j}\, \lambda_{1j} + \sum_{j=1}^{N_2} \sigma_2\, a_{23j}\, \lambda_{2j} + \sum_{j=1}^{N_3} \sigma_3\, a_{33j}\, \lambda_{3j} = 0 \qquad (7)$$

For $i=4$:

$$\sum_{j=1}^{N_0} \sigma_0\, a_{04j}\, \lambda_{0j} + \sum_{j=1}^{N_1} \sigma_1\, a_{14j}\, \lambda_{1j} + \sum_{j=1}^{N_2} \sigma_2\, a_{24j}\, \lambda_{2j} + \sum_{j=1}^{N_3} \sigma_3\, a_{34j}\, \lambda_{3j} = 0 \qquad (8)$$

with $\vec{\lambda} = \{\lambda_{01}, \lambda_{02}, \cdots, \lambda_{05};\ \lambda_{11};\ \lambda_{21}, \lambda_{22};\ \lambda_{31}\}^T > \vec{0}$.

8.9 Min. $f = 1 \cdot x_1^{1} x_2^{1} x_3^{-2} + 2 \cdot x_1^{-1} x_2^{-1} x_3^{1} + 5 \cdot x_1^{0} x_2^{1} x_3^{0} + 3 \cdot x_1^{1} x_2^{-2} x_3^{0}$

Here $n=3$, $N=4$, $DOD = N-n-1 = 0$

$c_1 = 1$, $c_2 = 2$, $c_3 = 5$, $c_4 = 3$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 & 1 \\ 1 & -1 & 1 & -2 \\ -2 & 1 & 0 & 0 \end{bmatrix}$$

orthogonality and normality conditions:

$$\Delta_1 - \Delta_2 + \Delta_4 = 0 \qquad (1) \Rightarrow \Delta_4 = -\Delta_1 + \Delta_2 \qquad (5)$$

$$\Delta_1 - \Delta_2 + \Delta_3 - 2\Delta_4 = 0 \qquad (2) \Rightarrow \Delta_3 = -\Delta_1 + \Delta_2 + 2\Delta_4 \qquad (6)$$

$$-2\Delta_1 + \Delta_2 = 0 \qquad (3) \Rightarrow \Delta_2 = 2\Delta_1 \qquad (7)$$

$$\Delta_1 + \Delta_2 + \Delta_3 + \Delta_4 = 1 \qquad (4)$$

Eqs. (5) to (7): $\Delta_2 = 2\Delta_1$, $\Delta_3 = 3\Delta_1$, $\Delta_4 = \Delta_1$ (8)

Eqs. (8) and (4): $\Delta_1^* = \frac{1}{7}$, $\Delta_2^* = \frac{2}{7}$, $\Delta_3^* = \frac{3}{7}$, $\Delta_4^* = \frac{1}{7}$

$$f^* = \prod_{i=1}^{4} \left(\frac{c_i}{\Delta_i^*}\right)^{\Delta_i^*} = (7)^{\frac{1}{7}} (7)^{\frac{2}{7}} \left(\frac{5}{3} * 7\right)^{\frac{3}{7}} (3*7)^{\frac{1}{7}} = 10.193803$$

$$U_1^* = \Delta_1^* f^* = \frac{1}{7}(10.193803) = 1.456258 = x_1^* x_2^* x_3^{*-2} \quad (9)$$

$$U_2^* = \Delta_2^* f^* = \frac{2}{7}(10.193803) = 2.912515 = 2.0\, x_1^{*-1} x_2^{*-1} x_3^* \quad (10)$$

$$U_3^* = \Delta_3^* f^* = \frac{3}{7}(10.193803) = 4.368773 = 5\, x_2^* \quad (11)$$

$$U_4^* = \Delta_4^* f^* = \frac{1}{7}(10.193803) = 1.456258 = 3\, x_1^* x_2^{*-2} \quad (12)$$

Eq. (11): $x_2^* = 0.873755$ (13)

Eqs. (12) and (13): $x_1^* = 5.722431$ (14)

Eqs. (10), (13) and (14): $x_3^* = 7.281288$ (15)

8.10

Min. $f = \frac{1}{2} x_1^2 x_2^0 + 1 \cdot x_1^0 x_2^1 + \frac{3}{2} x_1^{-1} x_2^{-1}$

$c_1 = \frac{1}{2}$, $c_2 = 1$, $c_3 = \frac{3}{2}$; $N = 3$, $n = 2$, $DOD = 0$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 1 & -1 \end{bmatrix}$$

orthogonality and normality conditions:

$2\Delta_1 - \Delta_3 = 0 \Rightarrow \Delta_3 = 2\Delta_1$

$\Delta_2 - \Delta_3 = 0 \Rightarrow \Delta_3 = \Delta_2 = 2\Delta_1$

$\Delta_1 + \Delta_2 + \Delta_3 = 1 \Rightarrow \Delta_1 + (2\Delta_1) + (2\Delta_1) = 1$

$\Delta_1^* = \frac{1}{5}$, $\Delta_2^* = \frac{2}{5}$, $\Delta_3^* = \frac{2}{5}$

$$f^* = \left(\frac{1}{2} * 5\right)^{\frac{1}{5}} \left(\frac{5}{2}\right)^{\frac{2}{5}} \left(\frac{3}{2} * \frac{5}{2}\right)^{\frac{2}{5}} = \left(\frac{28125}{128}\right)^{\frac{1}{5}} = 2.940197$$

$$\Delta_2^* f^* = x_2^* = \frac{2}{5}(2.940197) = 1.1760788$$

$$\Delta_1^* f^* = \frac{1}{2} x_1^{*2} = \frac{1}{5}(2.940197) = 0.5880394$$

$$\therefore x_1^* = 1.0844717$$

8.11

Min. $f = 20\, x_1^0\, x_2^1\, x_3^1\, x_4^4 + 20\, x_1^2\, x_2^0\, x_3^{-1}\, x_4^0 + 5\, x_1^0\, x_2^1\, x_3^2\, x_4^0$

subject to:

$$g_1: \quad 5\, x_1^0\, x_2^{-5}\, x_3^{-1}\, x_4^0 \leq 1$$

$$g_2: \quad 10\, x_1^{-1}\, x_2^3\, x_3^0\, x_4^{-1} \leq 1$$

Here $n = 4$, $m = 2$, $N_0 = 3$, $N_1 = N_2 = 1$, $N = 5$, $DOD = 1$

$c_{01} = 20$ $\quad a_{011} = 0$ $\quad a_{012} = 2$ $\quad a_{013} = 0$ $\quad a_{111} = 0$ $\quad a_{211} = -1$

$c_{02} = 20$ $\quad a_{021} = 1$ $\quad a_{022} = 0$ $\quad a_{023} = 1$ $\quad a_{121} = -5$ $\quad a_{221} = 3$

$c_{03} = 5$ $\quad a_{031} = 1$ $\quad a_{032} = -1$ $\quad a_{033} = 2$ $\quad a_{131} = -1$ $\quad a_{231} = 0$

$c_{11} = 5$ $\quad a_{041} = 4$ $\quad a_{042} = 0$ $\quad a_{043} = 0$ $\quad a_{141} = 0$ $\quad a_{241} = -1$

$c_{21} = 10$

$$v(\vec{\lambda}) = \left\{ \frac{c_{01}}{\lambda_{01}} (\lambda_{01} + \lambda_{02} + \lambda_{03}) \right\}^{\lambda_{01}} \left\{ \frac{c_{02}}{\lambda_{02}} (\lambda_{01} + \lambda_{02} + \lambda_{03}) \right\}^{\lambda_{02}} \cdot$$

$$\left\{ \frac{c_{03}}{\lambda_{03}} (\lambda_{01} + \lambda_{02} + \lambda_{03}) \right\}^{\lambda_{03}} \left\{ \frac{c_{11}}{\lambda_{11}} \cdot \lambda_{11} \right\}^{\lambda_{11}} \left\{ \frac{c_{21}}{\lambda_{21}} \cdot \lambda_{21} \right\}^{\lambda_{21}}$$

$$= \left(\frac{20}{\lambda_{01}}\right)^{\lambda_{01}} \left(\frac{20}{\lambda_{02}}\right)^{\lambda_{02}} \left(\frac{5}{\lambda_{03}}\right)^{\lambda_{03}} (5)^{\lambda_{11}} (10)^{\lambda_{21}}$$

$$\lambda_{01} + \lambda_{02} + \lambda_{03} = 1$$

$$a_{011}\lambda_{01} + a_{012}\lambda_{02} + a_{013}\lambda_{03} + a_{111}\lambda_{11} + a_{211}\lambda_{21} = 0$$

$$a_{021}\lambda_{01} + a_{022}\lambda_{02} + a_{023}\lambda_{03} + a_{121}\lambda_{11} + a_{221}\lambda_{21} = 0$$

$$a_{031}\lambda_{01} + a_{032}\lambda_{02} + a_{033}\lambda_{03} + a_{131}\lambda_{11} + a_{231}\lambda_{21} = 0$$

$$a_{041}\lambda_{01} + a_{042}\lambda_{02} + a_{043}\lambda_{03} + a_{141}\lambda_{11} + a_{241}\lambda_{21} = 0$$

$$\lambda_{01} + \lambda_{02} + \lambda_{03} = 1 \qquad (1)$$

$$2\lambda_{02} - \lambda_{21} = 0 \qquad (2)$$

$$\lambda_{01} + \lambda_{03} - 5\lambda_{11} + 3\lambda_{21} = 0 \qquad (3)$$

$$\lambda_{01} - \lambda_{02} + 2\lambda_{03} - \lambda_{11} = 0 \qquad (4)$$

$$4\lambda_{01} - \lambda_{21} = 0 \qquad (5)$$

Eq. (2) $\Rightarrow$ $\lambda_{21} = 2\lambda_{02}$ (6)

Eq. (5) $\Rightarrow$ $\lambda_{21} = 4\lambda_{01}$ (7)

Eqs. (6) and (7): $\lambda_{02} = 2\lambda_{01}$ (8)

Eq. (3) $\Rightarrow$ $5\lambda_{11} = \lambda_{01} + \lambda_{03} + 3\lambda_{21} = \lambda_{03} + 13\lambda_{01}$ (9)

Eq. (4) $\Rightarrow$ $\lambda_{11} = \lambda_{01} - \lambda_{02} + 2\lambda_{03} = 2\lambda_{03} - \lambda_{01}$ (10)

Eqs. (9) and (10) $\Rightarrow$ $\lambda_{03} = 2\lambda_{01}$ (11)

Eqs. (1), (8) and (11) $\Rightarrow$ $\lambda_{01} + 2\lambda_{01} + 2\lambda_{01} = 1$

$$\lambda_{01} = \frac{1}{5}\ ,\quad \lambda_{02} = \frac{2}{5}\ ,\quad \lambda_{03} = \frac{2}{5}$$

Eq. (2) $\Rightarrow$ $\lambda_{21} = \frac{4}{5}$

Eq. (10) $\Rightarrow$ $\lambda_{11} = \frac{3}{5}$

v^* can be expressed as:

$$v^* = (20*5)^{\frac{1}{5}} (10*5)^{\frac{2}{5}} \left(\frac{5*5}{2}\right)^{\frac{2}{5}} (5)^{\frac{3}{5}} (10)^{\frac{4}{5}} = 546.6810$$

$$\lambda_{02}^* = \frac{20\, x_1^2\, x_3^{-1}}{x_0^*} \Rightarrow \frac{2}{5} = \frac{20}{546.6810}\, x_1^2\, x_3^{-1}$$

or $x_3 = 0.09146\, x_1^2$ (12)

$$\lambda_{03}^* = \frac{2}{5} = \frac{5\, x_2\, x_3^2}{546.6810} \Rightarrow x_2 x_3^2 = 43.7344$$

or $x_2 = \frac{43.7344}{x_3^2} = \frac{5228.3130}{x_1^4}$ (13)

$$\lambda_{01}^* = \frac{1}{5} = \frac{20\, x_2 x_3\, x_4^4}{546.6810} \Rightarrow x_4^4 = \frac{5.46681}{x_2 x_3} = 0.0114325\, x_1^2$$

or $x_4 = 0.32699\sqrt{x_1}$ (14)

$$1 = 5\, x_2^{-5}\, x_3^{-1} \quad \Rightarrow \quad x_3\, x_2^5 = 5$$

$$\text{or} \quad \left(0.09146\, x_1^2\right)\left(\frac{5228.3130}{x_1^4}\right)^5 = 5$$

$$\text{or} \quad x_1^{18} = 0.071461 * 10^{18}$$

$$\text{or} \quad x_1^* = 8.63649 \qquad (15)$$

$$\text{Eq. (13)} \Rightarrow \quad x_2^* = 0.93975 \qquad (16)$$

$$\text{Eq. (12)} \Rightarrow \quad x_3^* = 6.82191 \qquad (17)$$

$$\text{Eq. (14)} \Rightarrow \quad x_4^* = 0.96095 \qquad (18)$$

8.12

Min. $f = x_1^{-2} + \frac{1}{4}\, x_2^2\, x_3$

subject to: $g: \quad \frac{3}{4}\, x_1^2\, x_2^{-2} + \frac{3}{8}\, x_2\, x_3^{-2} \le 1$

Here $n = 3, \; N_0 = 2, \; N_1 = 2, \; N = 4, \; m = 1, \; DOD = 0$

$c_{01} = 1$	$a_{011} = -2$	$a_{012} = 0$	$a_{111} = 2$	$a_{112} = 0$
$c_{02} = \frac{1}{4}$	$a_{021} = 0$	$a_{022} = 2$	$a_{121} = -2$	$a_{122} = 1$
$c_{11} = \frac{3}{4}$	$a_{031} = 0$	$a_{032} = 1$	$a_{131} = 0$	$a_{132} = -2$
$c_{12} = \frac{3}{8}$				

$$\lambda_{01} + \lambda_{02} = 1 \qquad (1)$$

$$-2\lambda_{01} + 2\lambda_{11} = 0 \quad \Rightarrow \quad \lambda_{11} = \lambda_{01} \qquad (2)$$

$$2\lambda_{02} - 2\lambda_{11} + \lambda_{12} = 0 \qquad (3)$$

$$\lambda_{02} - 2\lambda_{12} = 0 \quad \Rightarrow \quad \lambda_{12} = \frac{1}{2}\lambda_{01} \qquad (4)$$

$$\text{Eq. (3):} \quad 2\lambda_{02} = 2\lambda_{11} - \lambda_{12} = 2\lambda_{01} - \frac{1}{2}\lambda_{01} = \frac{3}{2}\lambda_{01}$$

$$\text{or} \quad \lambda_{02} = \frac{3}{4}\lambda_{01}$$

$$\text{Eq. (1):} \quad \lambda_{01} + \lambda_{02} = \frac{7}{4}\lambda_{01} = 1$$

$$\lambda_{01} = \frac{4}{7}, \quad \lambda_{02} = \frac{3}{7}, \quad \lambda_{11} = \frac{4}{7}, \quad \lambda_{12} = \frac{2}{7}$$

$$v^* = \left(\frac{1}{\lambda_{01}}\right)^{\lambda_{01}} \left(\frac{1}{4\,\lambda_{02}}\right)^{\lambda_{02}} \left(\frac{3}{4}\cdot\frac{1}{\lambda_{11}}\right)^{\lambda_{11}} \left(\frac{3}{8}\cdot\frac{1}{\lambda_{12}}\right)^{\lambda_{12}}$$

$$= \left(\frac{7}{4}\right)^{4/7} \left(\frac{7}{12}\right)^{3/7} \left(\frac{3*7}{16}\right)^{4/7} \left(\frac{3}{8}*\frac{7}{2}\right)^{2/7} = 1.379703$$

$$\lambda_{01}^* = \frac{x_1^{-2}}{x_0^*} \Rightarrow \frac{4}{7} = \frac{x_1^{-2}}{1.379703} \Rightarrow x_1^* = 1.126228$$

$$\lambda_{02}^* = \frac{1}{4}\,\frac{x_2^2\,x_3}{x_0^*} \Rightarrow x_2^2\,x_3 = 4\,x_0^*\left(\frac{3}{7}\right) = 2.365205$$

$$\frac{\lambda_{11}^*}{\lambda_{11}^* + \lambda_{12}^*} = \frac{4/7}{6/7} = \frac{2}{3} = \frac{3}{4}\,x_1^2\,x_2^{-2} \Rightarrow x_2^2 = \frac{9}{8}\,x_1^2$$

$$\Rightarrow x_2^* = 1.194545$$

$$x_3^* = \frac{2.365205}{x_2^2} = 1.657538$$

8.13 Min. $f = x_1^{-3}\,x_2 + x_1^{3/2}\,x_3^{-1}$

subject to: $g: \quad x_1^2\,x_2^{-1} + \frac{1}{2}\,x_1^{-2}\,x_3^3 \le 1$

Here $c_{01} = 1, \; c_{02} = 1, \; c_{11} = 1, \; c_{12} = \frac{1}{2}$

$a_{011} = -3$	$a_{012} = \frac{3}{2}$	$a_{111} = 2$	$a_{112} = -2$
$a_{021} = 1$	$a_{022} = 0$	$a_{121} = -1$	$a_{122} = 0$
$a_{031} = 0$	$a_{032} = -1$	$a_{131} = 0$	$a_{132} = 3$

$$\lambda_{01} + \lambda_{02} = 1 \qquad (1)$$

$$-3\,\lambda_{01} + \tfrac{3}{2}\,\lambda_{02} + 2\,\lambda_{11} - 2\,\lambda_{12} = 0 \qquad (2)$$

$$\lambda_{01} - \lambda_{11} = 0 \qquad (3)$$

$$-\lambda_{02} + 3\,\lambda_{12} = 0 \qquad (4)$$

Eq. (3): $\lambda_{11} = \lambda_{01}$ (5)

Eq. (4): $\lambda_{12} = \frac{1}{3}\,\lambda_{02}$ (6)

Eqs. (2), (5), and (6): $3\lambda_{01} = \frac{3}{2}\lambda_{02} + 2\lambda_{11} - 2\lambda_{12}$

$= \frac{3}{2}\lambda_{02} + 2\lambda_{01} - \frac{2}{3}\lambda_{02}$

i.e., $\lambda_{01} = \frac{5}{6}\lambda_{02}$ (7)

Eqs. (1) and (7): $\frac{5}{6}\lambda_{02} + \lambda_{02} = 1 \Rightarrow \lambda_{02}^* = \frac{6}{11}$

Eqs. (7), (5) and (6): $\lambda_{01}^* = \frac{5}{11}$, $\lambda_{11}^* = \frac{5}{11}$, $\lambda_{12}^* = \frac{2}{11}$

$v^* = \left(\frac{11}{5}\right)^{\frac{5}{11}} \left(\frac{11}{6}\right)^{\frac{6}{11}} \left(\frac{5}{7}\right)^{\frac{5}{11}} \left(\frac{2}{7}\right)^{\frac{2}{11}} = x_0^* = 1.361099$

$\lambda_{01}^* = \frac{5}{11} = \frac{x_1^{-3} x_2}{x_0^*} \Rightarrow \frac{x_2}{x_1^3} = \frac{5}{11}(x_0^*) = 0.618681$

$\lambda_{02}^* = \frac{6}{11} = \frac{x_1^{3/2} x_3^{-1}}{x_0^*} \Rightarrow \frac{x_1^{3/2}}{x_3} = 0.742418$

$\frac{\lambda_{11}^*}{\lambda_{11}^* + \lambda_{12}^*} = x_1^2 x_2^{-1} \Rightarrow \frac{5}{7} = x_1^2 x_2^{-1}$ or $x_1^2 = \frac{5}{7} x_2$

$= \frac{5}{7}(0.618681\, x_1^3)$

$\therefore x_1^* = 2.262879$, $x_2^* = 7.168867$,

$x_3^* = \frac{x_1^{3/2}}{0.742418} = 4.585044$

8.14 Min. $f = x_1^{-1} x_2^{-2} x_3^{-2}$

subject to $x_1^2 + x_2^2 + x_3 \leq 1$

Here $N_0 = 1$, $N_1 = 3$, $N = 4$, $n = 3$, $DOD = 0$, $m = 1$

$c_{01} = 1$ $a_{011} = -1$ $a_{111} = 2$ $a_{112} = 0$ $a_{113} = 0$

$c_{11} = 1$ $a_{021} = -2$ $a_{121} = 0$ $a_{122} = 2$ $a_{123} = 0$

$c_{12} = 1$ $a_{031} = -2$ $a_{131} = 0$ $a_{132} = 0$ $a_{133} = 1$

$c_{13} = 1$

$\lambda_{01} = 1$

$-\lambda_{01} + 2\lambda_{11} = 0 \quad \Rightarrow \lambda_{11}^* = 0.5$

$-2\lambda_{01} + 2\lambda_{12} = 0 \quad \Rightarrow \lambda_{12}^* = 1$

$-2\lambda_{01} + \lambda_{13} = 0 \quad \Rightarrow \lambda_{13}^* = 2$

$$v^* = \left(\frac{1}{1}\right)^1 \left(\frac{1}{0.5} * 3.5\right)^{0.5} \left(\frac{1}{1} * 3.5\right)^1 \left(\frac{1}{2} * 3.5\right)^2$$

$$= 28.359143 = x_0^*$$

$$\frac{\lambda_{11}^*}{\lambda_{11}^* + \lambda_{12}^* + \lambda_{13}^*} = \frac{0.5}{3.5} = x_1^2 \quad \Rightarrow \quad x_1^* = 0.377964$$

$$\frac{\lambda_{12}^*}{\lambda_{11}^* + \lambda_{12}^* + \lambda_{13}^*} = \frac{1}{3.5} = x_2^2 \quad \Rightarrow \quad x_2^* = 0.534522$$

$$\frac{\lambda_{13}^*}{\lambda_{11}^* + \lambda_{12}^* + \lambda_{13}^*} = \frac{2}{3.5} = x_3 \quad \Rightarrow \quad x_3^* = 0.571429$$

8.15 $y = \sum_{i=1}^{n} c_i e^{a_i x_i}$; $c_i \geq 0$

consider a typical term, $y_i = c_i e^{a_i x_i}$

$$\frac{\partial y_i}{\partial x_i} = c_i a_i e^{a_i x_i}, \quad \frac{\partial^2 y_i}{\partial x_i^2} = c_i a_i^2 e^{a_i x_i} \geq 0$$

Hence y_i is convex.

Since a positive sum of convex functions is convex, y is convex (see Theorem 7.2).

8.16 $f = \ln x$

$$\frac{df}{dx} = \frac{1}{x}, \quad \frac{d^2 f}{dx^2} = -\frac{1}{x^2} = \text{negative for } x > 0$$

$\therefore$ f is concave.

8.17 Minimize $$f = \frac{\pi^2 (7.7 * 10^4)(20 * 10^{10})(20)}{14680 * 0.3} d^6 + \frac{\pi^2 (7.7 * 10^4)(2)}{4} D d^2$$

subject to

$$\frac{14.5 * 0.3}{(15 * 10^7) d^{2.885} D^{0.115}} \leq 1$$

i.e., Minimize $f = 6.902480 * 10^{14} d^6 + 37.998155 * 10^4 D d^2$

subject to: $0.29 * 10^{-7} D^{-0.115} d^{-2.885} \leq 1$

Here $N_0 = 2$, $N_1 = 1$, $N = 3$, $m = 1$, $n = 2$, DOD = 0

$x_1 = d$, $x_2 = D$

$c_{01} = 6.902480 * 10^{14}$, $c_{02} = 37.998155 * 10^4$, $c_{11} = 0.29 * 10^{-7}$

$a_{011} = 6$, $a_{021} = 0$, $a_{012} = 2$, $a_{022} = 1$, $a_{111} = -2.885$,

$a_{121} = -0.115$

$$v = \left\{\frac{6.902480 * 10^{14}}{\lambda_{01}} (\lambda_{01} + \lambda_{02})\right\}^{\lambda_{01}} \left\{\frac{37.998155 * 10^4}{\lambda_{02}} (\lambda_{01} + \lambda_{02})\right\}^{\lambda_{02}} \cdot \left\{\frac{2.9 * 10^{-8}}{\lambda_{11}} (\lambda_{11})\right\}^{\lambda_{11}} \quad (1)$$

subject to:

$$\lambda_{01} + \lambda_{02} = 1 \quad (2)$$

$$6\lambda_{01} + 2\lambda_{02} - 2.885\lambda_{11} = 0 \quad (3)$$

$$\lambda_{02} - 0.115\lambda_{11} = 0 \Rightarrow \lambda_{02} = 0.115\lambda_{11} \quad (4)$$

$$\lambda_{01} \geq 0, \quad \lambda_{02} \geq 0, \quad \lambda_{11} \geq 0 \quad (5)$$

Eqs. (2) and (4): $\lambda_{01} = 1 - \lambda_{02} = 1 - 0.115\lambda_{11}$ (6)

Eqs. (3), (4) and (6): $6(1 - 0.115\lambda_{11}) + 2(0.115\lambda_{11}) - 2.885\lambda_{11} = 0$

or $\lambda_{11}^* = 1.793722$

and $\lambda_{01}^* = 0.793722$, $\lambda_{02}^* = 0.206278$

$$v^* = x_o^* = \left(\frac{6.902480 * 10^{14}}{0.793722}\right)^{0.793722} \left(\frac{37.998155 * 10^{4}}{0.206278}\right)^{0.206278} \cdot$$

$$\left(2.9 * 10^{-8}\right)^{1.793722} = 0.426299 \quad N$$

$$\lambda_{01}^* = 0.793722 = \frac{6.902480 * 10^{14}}{0.426299} \; d^6$$

$$\Rightarrow \; d^* = 0.002808 \quad m$$

$$\lambda_{02}^* = 0.206278 = \frac{37.998155 * 10^{4}}{0.426299} \; D\, d^2$$

$$\Rightarrow D^* = \frac{0.206278 * 0.426299}{(37.998155 * 10^4)(0.002808)^2} = 0.02935 \quad m$$

8.18 Minimize $f = c_{01} V^{-1} F^{-1} + c_{02} V^{\left(\frac{t}{c} - 1\right)} F^{\left(\frac{b}{c} - 1\right)}$

subject to $c_{11} F \leq 1$

with $c_{01} = \frac{K_m \pi D L}{12} = \frac{0.1 (\pi)(6.0)(8.0)}{12} = 1.25664$

$$c_{02} = \frac{\pi D L (K_m t_c + K_t)}{12 \; a^{1/c}} = \frac{\pi (6)(8)(0.1 * 0.5 + 0.5)}{12 * (140)^{(1/0.25)}}$$

$$= 178.98073 * 10^{-10}$$

$$c_{11} = \frac{1}{F_{max}} = \frac{1}{0.005} = 200.0$$

Maximize

$$v = \left\{\frac{c_{01}}{\lambda_{01}} (\lambda_{01} + \lambda_{02})\right\}^{\lambda_{01}} \left\{\frac{c_{02}}{\lambda_{02}} (\lambda_{01} + \lambda_{02})\right\}^{\lambda_{02}} \left\{\frac{c_{11}}{\lambda_{11}} (\lambda_{11})\right\}^{\lambda_{11}} \quad (1)$$

subject to: $\lambda_{01} + \lambda_{02} = 1 \quad (2)$

$$-\lambda_{01} + 3\,\lambda_{02} = 0 \quad \Rightarrow \quad \lambda_{01} = 3\,\lambda_{02} \quad (3)$$

$$-\lambda_{01} + 0.16\,\lambda_{02} + \lambda_{11} = 0 \quad (4)$$

Eqs. (2) and (3): $\lambda_{02}^* = 0.25$, $\lambda_{01}^* = 0.75 \quad (5)$

Eqs. (4) and (5): $\lambda_{11}^* = \lambda_{01}^* - 0.16\,\lambda_{02}^* = 0.71$ (6)

Note that $N_0 = 2$, $N_1 = 1$, $N = 3$, $n = 2$, $m = 1$, $DOD = 0$

$a_{011} = -1$, $a_{021} = -1$, $a_{012} = \frac{1}{c} - 1 = 3$,

$a_{022} = \frac{b}{c} - 1 = \frac{0.29}{0.25} - 1 = 0.16$, $a_{111} = 0$, $a_{121} = 1$

in this problem.

$$v^* = x_0^* = \left(\frac{1.25664}{0.75}\right)^{0.75} \left(\frac{178.98073 * 10^{-10}}{0.25}\right)^{0.25} \left(200.0\right)^{0.71}$$

$= 1.036447$ $ per piece

$$\lambda_{01}^* = 0.75 = \frac{1.25664\ V^{-1} F^{-1}}{1.036447} \Rightarrow V^{-1} F^{-1} = 0.618582 \quad (7)$$

$$\lambda_{02}^* = 0.25 = \frac{178.98073 * 10^{-10}\ V^3 F^{0.16}}{1.036447}$$

$$\Rightarrow V^3 F^{0.16} = 1447.708 * 10^4 \quad (8)$$

$$200 F^* = 1 \Rightarrow F^* = 0.005 \text{ in/rev} \quad (9)$$

Eqs. (7) and (9): $V^{-1} = \frac{0.618582}{F^{-1}} = (0.618582)(0.005)$

or $V^* = 323.3201$ ft/min (10)

8.19 Minimize $f = c_{01} V^{-1} F^{-1} + c_{02} V^{3} F^{0.16}$

subject to

$$c_{11} V^{0} F^{1} \leq 1$$

$$c_{21} V^{0.91} F^{0.78} \leq 1$$

Here $n = 2$, $N_0 = 2$, $N_1 = 1$, $N_2 = 1$, $N = 4$, $DOD = N - n - 1 = 1$.

$x_1 = V$, $x_2 = F$.

$c_{01} = 1.25664$, $c_{02} = 178.98073 \times 10^{-10}$, $c_{11} = 200.0$

$$c_{21} = \frac{a_1}{P_{max}} = \frac{3.58}{1} = 3.58$$

$a_{111} = -1$, $a_{021} = -1$, $a_{012} = 3$, $a_{022} = 0.16$, $a_{111} = 0$,

$a_{121} = 1$, $a_{211} = 0.91$, $a_{221} = 0.78$

$$\vec{\lambda} = \{ \lambda_{01} \quad \lambda_{02} \quad \lambda_{11} \quad \lambda_{21} \}^T$$

$$v = \left(\frac{c_{01}}{\lambda_{01}}\right)^{\lambda_{01}} \left(\frac{c_{02}}{\lambda_{02}}\right)^{\lambda_{02}} \left(\frac{c_{11}}{\lambda_{11}}\ \lambda_{11}\right)^{\lambda_{11}} \left(\frac{c_{21}}{\lambda_{21}}\ \lambda_{21}\right)^{\lambda_{21}} \quad (1)$$

subject to

$$\lambda_{01} + \lambda_{02} = 1 \quad (2)$$

$$-\lambda_{01} + 3\lambda_{02} + 0.91\lambda_{21} = 0 \quad (3)$$

$$-\lambda_{01} + 0.16\lambda_{02} + \lambda_{11} + 0.78\lambda_{21} = 0 \quad (4)$$

Equations (2) to (4) give

$\lambda_{02} = 1 - \lambda_{01}$, $\lambda_{11} = 2.411426 - 2.268568\lambda_{01}$,

$\lambda_{21} = 4.3956\lambda_{01} - 3.2967$

Maximization of v with respect to λ_{01} leads to the solution:

$\lambda_{01}^* = 0.82$, $\lambda_{02}^* = 0.18$, $\lambda_{11}^* = 0.55$, $\lambda_{21}^* = 0.31$,

$v^* = \$1.05$ per piece, $V^* = 290.0$ ft/min, $F^* = 0.005$ in/rev.

8.20 $\vec{x} = \begin{Bmatrix} d \\ f \\ p \\ s \\ t \end{Bmatrix}$, $n = 5$

$$f = \frac{\pi}{4} p d^2 \geq F \quad \Rightarrow \quad \frac{\pi}{4F} p d^2 \geq 1 \qquad (1)$$

$$s = \frac{p d}{2t} \leq S \quad \Rightarrow \quad \frac{p \cdot d}{2 S t} \leq 1 \qquad (2)$$

$$d + 2t \leq D \quad \Rightarrow \quad \frac{d}{D} + \frac{2t}{D} \leq 1 \qquad (3)$$

$$p \leq P \quad \Rightarrow \quad \frac{p}{P} \leq 1 \qquad (4)$$

$$t \geq T \quad \Rightarrow \quad \frac{t}{T} \geq 1 \qquad (5)$$

$$\text{objective function} = \pi t d + \pi t^2 \qquad (6)$$

$N_0 = 2,\ N_1 = 1,\ N_2 = 1,\ N_3 = 2,\ N_4 = 1,\ N_5 = 1,\ N = 8$

$DOD = N - n - 1 = 2$

8.21 $\vec{x} = \begin{Bmatrix} t \\ n \\ Z \end{Bmatrix}$, $n = 3$

Minimize $f = c_{01} t^1 n^0 Z^0 + c_{02} t^0 n^1 Z^{2/3}$ (1)

subject to: $c_{11} t^{-3} n^{-4} Z^0 + c_{12} t^0 n^{-1} Z^{-4/3} \leq 1$ (2)

with

$$c_{01} = k_f \gamma A B = (0.06)(7850)(10)(50) = 235{,}500.0$$

$$c_{02} = k_b \gamma A k_1 = (0.105)(7850)(10)(0.78) = 3{,}061.5$$

$$c_{11} = \frac{56.25\, W B^4}{E A} = \frac{(56.25)(1000)(50^4)}{(2.1 * 10^{11})(10)} = 0.167411$$

$$c_{12} = \frac{4.69\, W B A^3}{E k_2} = \frac{(4.69)(1000)(50)(10^3)}{(2.1 * 10^{11})(1.95)} = 5.726496 * 10^{-4}$$

$\vec{\lambda} = \begin{Bmatrix} \lambda_{01} \\ \lambda_{02} \\ \lambda_{11} \\ \lambda_{12} \end{Bmatrix}$ = Dual variables

Dual problem (constraints):

$$\lambda_{01} + \lambda_{02} = 1 \tag{3}$$

$$a_{011}\lambda_{01} + a_{012}\lambda_{02} + a_{111}\lambda_{11} + a_{112}\lambda_{12} = 0 \tag{4}$$

$$a_{021}\lambda_{01} + a_{022}\lambda_{02} + a_{121}\lambda_{11} + a_{122}\lambda_{12} = 0 \tag{5}$$

$$a_{031}\lambda_{01} + a_{032}\lambda_{02} + a_{131}\lambda_{11} + a_{132}\lambda_{12} = 0 \tag{6}$$

Here $N_0 = 2$, $N_1 = 2$, $N = 4$, $m = 1$, $n = 3$, DOD = 0

$a_{011} = 1$ $\quad a_{012} = 0$ $\quad a_{111} = -3$ $\quad a_{112} = 0$

$a_{021} = 0$ $\quad a_{022} = 1$ $\quad a_{121} = -4$ $\quad a_{122} = -1$

$a_{031} = 0$ $\quad a_{032} = \frac{2}{3}$ $\quad a_{131} = 0$ $\quad a_{132} = -\frac{4}{3}$

Eqs. (3) to (6):

$$\lambda_{01} + \lambda_{02} = 1 \tag{7}$$

$$\lambda_{01} - 3\lambda_{11} = 0 \Rightarrow \lambda_{11} = \tfrac{1}{3}\lambda_{01} \quad \text{or} \quad \lambda_{01} = 3\lambda_{11} \tag{8}$$

$$\lambda_{02} - 4\lambda_{11} - \lambda_{12} = 0 \Rightarrow \lambda_{02} = 4\lambda_{11} + \lambda_{12} \tag{9}$$

$$\tfrac{2}{3}\lambda_{02} - \tfrac{4}{3}\lambda_{12} = 0 \Rightarrow \lambda_{12} = \tfrac{1}{2}\lambda_{02} \tag{10}$$

Eqs. (9) and (10): $\lambda_{02} = 2\lambda_{12} = 4\lambda_{11} + \lambda_{12}$

or $\lambda_{12} = 4\lambda_{11} = \tfrac{1}{2}\lambda_{02}$

or $\lambda_{11} = \tfrac{1}{8}\lambda_{02}$ or $\lambda_{02} = 8\lambda_{11}$ (11)

Eqs. (7), (8) and (11): $3\lambda_{11} + 8\lambda_{11} = 1$ or $\lambda_{11}^* = \frac{1}{11}$

$$\therefore \quad \lambda_{01}^* = \tfrac{3}{11}, \quad \lambda_{02}^* = \tfrac{8}{11}, \quad \lambda_{11}^* = \tfrac{1}{11}, \quad \lambda_{12}^* = \tfrac{4}{11} \tag{12}$$

Dual problem (objective for maximization):

$$v^* = v_{max} = \left\{\frac{c_{01}}{\lambda_{01}}(\lambda_{01} + \lambda_{02})\right\}^{\lambda_{01}} \left\{\frac{c_{02}}{\lambda_{02}}(\lambda_{01} + \lambda_{02})\right\}^{\lambda_{02}} \cdot$$

$$\left\{\frac{c_{11}}{\lambda_{11}}(\lambda_{11} + \lambda_{12})\right\}^{\lambda_{11}} \left\{\frac{c_{12}}{\lambda_{12}}(\lambda_{11} + \lambda_{12})\right\}^{\lambda_{12}}$$

$$= \left(\frac{2.355 * 10^5}{3/11}\right)^{3/11} \left(\frac{3061.5}{8/11}\right)^{8/11} \left(\frac{0.167411}{1/11} * \frac{5}{11}\right)^{1/11} \cdot$$

$$\left(\frac{5.726496 * 10^{-4}}{4/11} * \frac{5}{11}\right)^{4/11} = \$\ 3,619.11813$$

$$\lambda_{01}^* = \frac{U_{01}^*}{v^*}, \quad \lambda_{02}^* = \frac{U_{02}^*}{v^*}, \quad \frac{\lambda_{11}^*}{\lambda_{11}^* + \lambda_{12}^*} = U_{11}^*,$$

$$\frac{\lambda_{12}^*}{\lambda_{11}^* + \lambda_{12}^*} = U_{12}^*$$

From these equations, the solution can be found as:

$t^* = 0.0042$ m, $n^* = 103.1$, $Z^* = 764.0 * 10^{-6}$ m^3

8.22 Minimize $f = 0.45\, R^3 L^{-2} + 0.025\, R^{-1} L^0$

subject to $8.62\, R^{-1} L^3 \le 1$

Here $x_1 = R$, $x_2 = L$, $n = 2$, $N_0 = 2$, $N_1 = 1$, $N = 3$, DOD $= 0$, $m = 1$

$c_{01} = 0.45$, $c_{02} = 0.025$, $c_{11} = 8.62$

$a_{011} = 3$, $a_{012} = -1$, $a_{021} = -2$, $a_{022} = 0$, $a_{111} = -1$, $a_{121} = 3$

Dual problem:

$$\vec{\lambda} = \begin{Bmatrix} \lambda_{01} \\ \lambda_{02} \\ \lambda_{11} \end{Bmatrix}$$

$$v = \left\{\frac{c_{01}}{\lambda_{01}}(\lambda_{01} + \lambda_{02})\right\}^{\lambda_{01}} \left\{\frac{c_{02}}{\lambda_{02}}(\lambda_{01} + \lambda_{02})\right\}^{\lambda_{02}} \left\{\frac{c_{11}}{\lambda_{11}}(\lambda_{11})\right\}^{\lambda_{11}}$$

$$= \left(\frac{0.45}{\lambda_{01}}\right)^{\lambda_{01}} \left(\frac{0.025}{\lambda_{02}}\right)^{\lambda_{02}} (8.62)^{\lambda_{11}} \qquad (1)$$

$$\lambda_{01} + \lambda_{02} = 1 \quad \Rightarrow \quad \lambda_{02} = 1 - \lambda_{01} \qquad (2)$$

$$a_{011}\lambda_{01} + a_{012}\lambda_{02} + a_{111}\lambda_{11} = 0 \Rightarrow 3\lambda_{01} - \lambda_{02} - \lambda_{11} = 0 \qquad (3)$$

$$a_{021}\lambda_{01} + a_{022}\lambda_{02} + a_{121}\lambda_{11} = 0 \Rightarrow -2\lambda_{01} + 3\lambda_{11} = 0 \qquad (4)$$

Eq. (4): $\lambda_{11} = \frac{2}{3}\lambda_{01}$ (5)

Eqs. (2), (3) and (5): $3\lambda_{01} - (1-\lambda_{01}) - \left(\frac{2}{3}\lambda_{01}\right) = 0$

$\therefore \quad \lambda_{01}^* = \frac{3}{10}, \quad \lambda_{02}^* = \frac{7}{10}, \quad \lambda_{11}^* = \frac{1}{5}$

$$v^* = x_o^* = \left(\frac{0.45}{3/10}\right)^{3/10}\left(\frac{0.025}{7/10}\right)^{7/10}(8.62)^{1/5} = 0.168623$$

$$\lambda_{01}^* = \frac{0.45\,R^3 L^{-2}}{x_o^*} \Rightarrow R^3 L^{-2} = \frac{(0.3)(0.168623)}{0.45} = 0.112415$$

$$\lambda_{02}^* = \frac{0.025\,R^{-1} L^{0}}{x_o^*} \Rightarrow R^{-1} = \frac{(0.7)(0.168623)}{0.025} = 4.721444$$

or $R^* = 0.2118$

$$L^{*2} = \frac{R^{*3}}{0.112415} = 0.084518 \quad \text{or} \quad L^* = 0.290720$$

8.23 Minimize $f = 0.44\,R^3 L^{-2} + 10\,R^{-1}L^0 + 0.592\,R^1 L^{-3}$

subject to: $8.62\,R^{-1}L^3 \le 1$

Here $x_1 = R, \quad x_2 = L, \quad n = 2, \quad N_0 = 3, N_1 = 1, N = 4,$

$DOD = 1$

$c_{01} = 0.44, \quad c_{02} = 10.0, \quad c_{03} = 0.592, \quad c_{11} = 8.62$

$a_{011} = 3 \quad a_{012} = -1 \quad a_{013} = 1 \quad a_{111} = -1$

$a_{021} = -2 \quad a_{022} = 0 \quad a_{023} = -3 \quad a_{121} = 3$

Dual problem:

$$\vec{\lambda} = \begin{Bmatrix} \lambda_{01} \\ \lambda_{02} \\ \lambda_{03} \\ \lambda_{11} \end{Bmatrix}$$

$$\text{Max.} \quad v = \left\{\frac{c_{01}}{\lambda_{01}}(\lambda_{01}+\lambda_{02}+\lambda_{03})\right\}^{\lambda_{01}}\left\{\frac{c_{02}}{\lambda_{02}}(\lambda_{01}+\lambda_{02}+\lambda_{03})\right\}^{\lambda_{02}} \cdot$$

$$\left\{\frac{c_{03}}{\lambda_{03}}(\lambda_{01}+\lambda_{02}+\lambda_{03})\right\}^{\lambda_{03}}\left\{\frac{c_{11}}{\lambda_{11}}(\lambda_{11})\right\}^{\lambda_{11}}$$

$$= \left(\frac{0.44}{\lambda_{01}}\right)^{\lambda_{01}} \left(\frac{10.0}{\lambda_{02}}\right)^{\lambda_{02}} \left(\frac{0.592}{\lambda_{03}}\right)^{\lambda_{03}} (8.62)^{\lambda_{11}} \qquad (1)$$

subject to:

$$\lambda_{01} + \lambda_{02} + \lambda_{03} = 1 \qquad (2)$$

$$a_{011}\lambda_{01} + a_{012}\lambda_{02} + a_{013}\lambda_{03} + a_{111}\lambda_{11} = 0$$

$$\text{or} \quad 3\lambda_{01} - \lambda_{02} + \lambda_{03} - \lambda_{11} = 0 \qquad (3)$$

$$a_{021}\lambda_{01} + a_{022}\lambda_{02} + a_{023}\lambda_{03} + a_{121}\lambda_{11} = 0$$

$$\text{or} \quad -2\lambda_{01} - 3\lambda_{03} + 3\lambda_{11} = 0 \qquad (4)$$

Eq.(2): $$\lambda_{01} = 1 - \lambda_{02} - \lambda_{03} \qquad (5)$$

Eq.(3): $$\lambda_{02} = 3\lambda_{01} + \lambda_{03} - \lambda_{11} = 3 - 3\lambda_{02} - 2\lambda_{03} - \lambda_{11}$$

$$\text{or} \quad \lambda_{02} = \frac{3}{4} - \frac{1}{2}\lambda_{03} - \frac{1}{4}\lambda_{11} \qquad (6)$$

Eq.(4): $$3\lambda_{03} = -2\lambda_{01} + 3\lambda_{11} = -2 + 2\lambda_{02} + 2\lambda_{03} + 3\lambda_{11}$$

$$\text{or} \quad \lambda_{03} = -2 + 2\lambda_{02} + 3\lambda_{11} \qquad (7)$$

Eqs.(6) and (7) give:

$$\lambda_{02} = \frac{3}{4} - \frac{1}{2}(-2 + 2\lambda_{02} + 3\lambda_{11}) - \frac{1}{4}\lambda_{11}$$

$$\text{or} \quad \lambda_{02} = \frac{7}{8} - \frac{7}{8}\lambda_{11} \qquad (8)$$

Eqs.(7) and (8): $$\lambda_{03} = -\frac{1}{4} + \frac{5}{4}\lambda_{11} \qquad (9)$$

Eqs.(5) and (9): $$\lambda_{01} = \frac{3}{8} - \frac{3}{8}\lambda_{11} \qquad (10)$$

v of Eq.(1) is maximized with respect to λ_{11}. For this, λ_{11} is varied from 0.01 in increments of 0.01 and v_{max} is found to be at $\lambda_{11}^* = 0.45$. This gives: $v^* = 16.205618$,

$\lambda_{01}^* = 0.2063$, $\lambda_{02}^* = 0.4813$, $\lambda_{03}^* = 0.3125$, $\lambda_{11}^* = 0.45$

$$\Delta_{01}^* = \lambda_{01}^* = \frac{0.44\ R^3\ L^{-2}}{16.205618} = 0.2063$$

$$\text{or} \quad R^3 L^{-2} = 7.598319 \qquad (11)$$

$$\Delta_{02}^* = \lambda_{02}^* = \frac{10\ R^{-1}\ L^\circ}{16.205818} = 0.4813$$

or $R^* = 1.282074$ (12)

Eqs.(11) and (12): $L^* = 0.526636$

∴ Solution is: $R^* = 1.282074$, $L^* = 0.526636$, $f^* = 16.205618$

8.24 Minimize $f = \sum_{i=1}^{n} l_i x_i$ (1)

subject to $\frac{F_i^{(k)}}{x_i\ \sigma_i^*} \le 1 \; ; \quad i = 1, \cdots, n; \quad k = 1, \cdots, \ell$ (2)

$$\sum_{i=1}^{n} \frac{F_i^{(k)}\ l_i}{x_i\ E\ \Delta_j^*}\ \delta_{ij} \le 1 \; ; \quad j = 1, \cdots, m; \quad k = 1, \cdots, \ell \quad (3)$$

Let $x_i = \frac{1}{\ln z_i}$ (4)

$$f = \sum_{i=1}^{n} \frac{l_i}{\ln z_i} \quad (5)$$

$$\frac{F_i^{(k)}}{\sigma_i^*}\ \frac{1}{x_i} = \frac{a_i^*}{x_i} \le 1 \quad \text{where} \quad a_i^* = \frac{F_i^{(k)}}{\sigma_i^*}$$

i.e., $a_i^* \ln z_i \le 1$ or $a_i^* \le \frac{1}{\ln z_i} = \ln_{z_i}(e)$

Taking antilogarithms, we get

$$\frac{z_i^{\,a_i^{(k)}}}{e} \le 1 \; ; \quad i = 1, \cdots, n; \quad k = 1, \cdots, \ell \quad (6)$$

Let $b_{ij}^{(k)} = \frac{F_i^{(k)}\ l_i\ \delta_{ij}}{E\ \Delta_j^*}$ (7)

Eq.(3): $\sum_{i=1}^{n} \frac{b_{ij}^{(k)}}{x_i} = \sum_{i=1}^{n} b_{ij}^{(k)} \ln z_i \le 1$

Taking antilogarithms, we get

$$\prod_{i=1}^{n} \frac{z_i^{\,b_{ij}^{(k)}}}{e} \le 1 \; ; \quad i = 1, \cdots, n; \quad k = 1, \cdots, \ell \quad (8)$$

Equation (5) is not in posynomial form. For this, we introduce new variables y_i through the following inequalities:

$$\left.\begin{aligned} \ln z_i &\geq \frac{2 y_i}{z_i + 1} \\ y_i &\leq z_i - 1 \end{aligned}\right\}, \quad i = 1, \cdots, n \tag{9}$$

Hence the geometric programming problem can be stated as follows:

Design variables: $y_1, \cdots, y_n, z_1, \cdots, z_n$ (10)

$$f = \sum_{i=1}^{n} l_i \left(\frac{z_i + 1}{2 y_i} \right) \rightarrow \text{minimize} \tag{11}$$

subject to

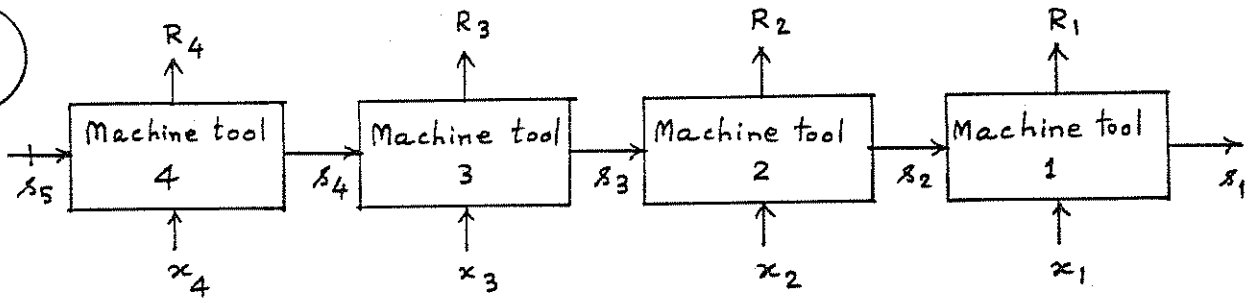
$$\frac{y_i}{z_i} + \frac{1}{z_i} \leq 1 \quad ; \quad i = 1, \cdots, n \tag{12}$$

$$\frac{z_i^{a_i^{(k)}}}{e} \leq 1 \quad ; \quad i = 1, \cdots, n \; ; \quad k = 1, \cdots, \ell \tag{13}$$

$$\prod_{i=1}^{n} \frac{z_i^{b_{ij}^{(k)}}}{e} \leq 1 \quad ; \quad i = 1, \cdots, n \; ; \quad k = 1, \cdots, \ell \tag{14}$$

Chapter 9
Dynamic Programming

9.1



Suboptimization of stage 1 (Machine tool 1):

$$f_1^*(s_2) = \min_{x_1} \left[R_1(x_1, s_2) \right]$$

s_2 ($)	x_1	R_1 (number of speeds)	s_1
10,000	0	0	10,000
	1	9	6,500
	2 *	18	3,000
9,500	0	0	9,500
	1	9	6,000
	2 *	18	2,500
9,000	0	0	9,000
	1	9	5,500
	2 *	18	2,000
8,500	0	0	8,500
	1	9	5,000
	2 *	18	1,500

8,000	0	0	8,000
	1	9	4,500
	2 *	18	1,000
7,500	0	0	7,500
	1	9	4,000
	2 *	18	500
7,000	0	0	7,000
	1	9	3,500
	2 *	18	0
6,500	0	0	6,500
	1 *	9	3,000
6,000	0	0	6,000
	1 *	9	2,500
5,500	0	0	5,500
	1 *	9	2,000
5,000	0	0	5,000
	1 *	9	1,500
4,500	0	0	4,500
	1 *	9	1,000
4,000	0	0	4,000
	1 *	9	500
3,500	0	0	3,500
	1 *	9	0
3,000	0	0	3,000
2,500	0	0	2,500
2,000	0	0	2,000
1,500	0	0	1,500
1,000	0	0	1,000
500	0	0	500
0	0	0	0

Suboptimization of stages 2 and 1 (Machine tools 1 and 2):

$$f_2^*(s_3) = \min_{x_2} \left[R_2(x_2, s_3) + f_1^*(s_2) \right]$$

s_3 ($)	x_2	R_2 (number of speeds)	s_2	f_1^*	$R_2 + f_1^*$
10,000	0	0	10,000	18	18
	1 *	4	7,500	18	22
	2	8	5,000	9	17
	3	12	2,500	0	12
	4	16	0	0	16
9,500	0	0	9,500	18	18
	1 *	4	7,000	18	22
	2	8	4,500	9	17
	3	12	2,000	0	12
9,000	0 *	0	9,000	18	18
	1	4	6,500	9	13
	2	8	4,000	9	17
	3	12	1,500	0	12
8,500	0 *	0	8,500	18	18
	1	4	6,000	9	13
	2	8	3,500	9	17
	3	12	1,000	0	12
8,000	0 *	0	8,000	18	18
	1	4	5,500	9	13
	2	8	3,000	0	8
	3	12	500	0	12
7,500	0 *	0	7,500	18	18
	1	4	5,000	9	13
	2	8	2,500	0	8
	3	12	0	0	12
7,000	0 *	0	7,000	18	18
	1	4	4,500	9	13
	2	8	2,000	0	8
6,500	0	0	6,500	9	9
	1 *	4	4,000	9	13
	2	8	1,500	0	8

6,000	0	0	6,000	9	9
	1 *	4	3,500	9	13
	2	8	1,000	0	8
5,500	0	0	5,500	9	9
	1 *	4	3,000	9	13
	2	8	500	0	8
5,000	0	0	5,000	9	9
	1 *	4	2,500	9	13
	2	8	0	0	8
4,500	0 *	0	4,500	9	9
	1	4	2,000	0	4
4,000	0 *	0	4,000	9	9
	1	4	1,500	0	4
3,500	0 *	0	3,500	9	9
	1	4	1,000	0	4
3,000	0	0	3,000	0	0
	1 *	4	500	0	4
2,500	0	0	2,500	0	0
	1 *	4	0	0	4
2,000	0	0	2,000	0	0
1,500	0	0	1,500	0	0
1,000	0	0	1,000	0	0
500	0	0	500	0	0
0	0	0	0	0	0

Suboptimization of stages 3, 2 and 1 (Machine tools 1, 2 and 3):

$$f_3^*(s_4) = \min_{x_3} \left[R_3(x_3, s_4) + f_2^*(s_3) \right]$$

s_4 ($)	x_3	R_3 (number of speeds)	s_3	f_2^*	$R_3 + f_2^*$
10,000	0 *	0	10,000	22	22
	1	3	8,000	13	16
	2	6	6,000	13	19
	3	9	4,000	4	13
	4	12	2,000	0	12
	5	15	0	0	15

9,500	0 *	0	9,500	22	22
	1	3	7,500	13	16
	2	6	5,500	13	19
	3	9	3,500	4	13
	4	12	1,500	0	12
9,000	0	0	9,000	18	18
	1 *	3	7,000	18	21
	2	6	5,000	13	19
	3	9	3,000	4	13
	4	12	1,000	0	12
8,500	0 *	0	8,500	18	18
	1	3	6,500	13	16
	2	6	4,500	9	15
	3	9	2,500	4	13
	4	12	500	0	12
8,000	0 *	0	8,000	18	18
	1	3	6,000	13	16
	2	6	4,000	9	15
	3	9	2,000	0	9
	4	12	0	0	12
7,500	0 *	0	7,500	18	18
	1	3	5,500	13	16
	2	6	3,500	9	15
	3	9	1,500	0	9
7,000	0 *	0	7,000	18	18
	1	3	5,000	13	16
	2	6	3,000	4	10
	3	9	1,000	0	9
6,500	0 *	0	6,500	13	13
	1	3	4,500	9	12
	2	6	2,500	4	10
	3	9	500	0	9
6,000	0 *	0	6,000	13	13
	1	3	4,000	9	12
	2	6	2,000	0	6
	3	9	0	0	9

5,500	0 *	0	5,500	13	13
	1	3	3,500	9	12
	2	6	1,500	0	6
5,000	0 *	0	5,000	13	13
	1	3	3,000	4	7
	2	6	1,000	0	6
4,500	0 *	0	4,500	9	9
	1	3	2,500	4	7
	2	6	500	0	6
4,000	0 *	0	4,000	9	9
	1	3	2,000	0	3
	2	6	0	0	6
3,500	0 *	0	3,500	9	9
	1	3	1,500	0	3
3,000	0 *	0	3,000	4	4
	1	3	1,000	0	3
2,500	0 *	0	2,500	4	4
	1	3	500	0	3
2,000	0	0	2,000	0	0
	1 *	3	0	0	3
1,500	0	0	1,500	0	0
1,000	0	0	1,000	0	0
500	0	0	500	0	0
0	0	0	0	0	0

Suboptimization of stages 4, 3, 2 and 1 (Machine tools 1, 2, 3 and 4):

$$f_4^*(s_5) = \min_{x_4} \left[R_4(x_4, s_5) + f_3^*(s_4) \right]$$

s_5 ($)	x_4	R_4 (number of speeds)	s_4 ($)	f_3^*	$R_4 + f_3^*$
10,000	0	0	10,000	22	22
	1	2	9,000	21	23
	2	4	8,000	18	22

3 *	6	7,000	18	24
4	8	6,000	13	21
5	10	5,000	13	23
6	12	4,000	9	21
7	14	3,000	4	18
8	16	2,000	3	19
9	18	1,000	0	18
10	20	0	0	20

Retrace the steps to find:

$s_5 = \$\,10{,}000$, $x_4 = 3$, $R_4 = 6$

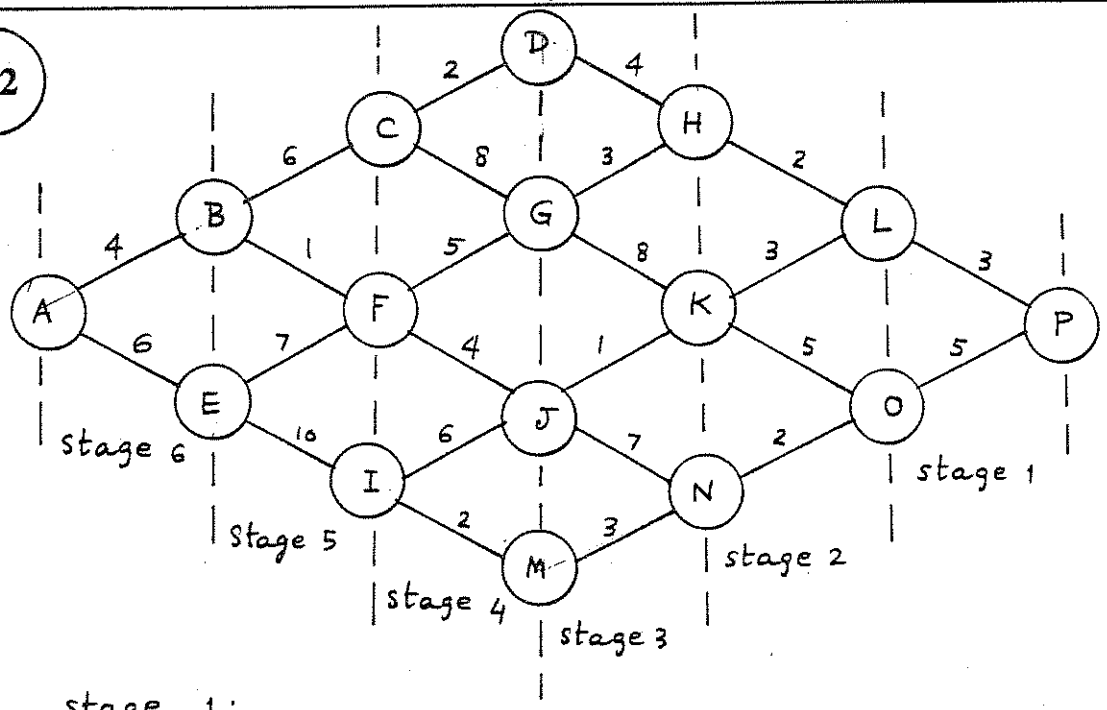
$s_4 = \$\,7{,}000$, $x_3 = 0$, $R_3 = 0$

$s_3 = \$\,7{,}000$, $x_2 = 0$, $R_2 = 0$

$s_2 = \$\,7{,}000$, $x_1 = 2$, $R_1 = 18$

$f_{opt} = f_4^*(s_5 = 10{,}000) = 24$.

9.2



stage 1:

	From	To	cost
*	L	P	3
*	O	P	5

Stages 2 and 1:

From	To	cost	Minimum downward cost	Total cost
* H	L	2	3	5
* K	L	3	3	6
	O	5	5	10
* N	O	2	5	7

Stages 3, 2 and 1:

* D	H	4	5	9
* G	H	3	5	8
	K	8	6	14
* J	K	1	6	7
	N	7	7	14
* M	N	3	7	10

Stages 4, 3, 2 and 1:

* C	D	2	9	11
	G	8	8	16
F	G	5	8	13
*	J	4	7	11
I	J	6	7	13
*	M	2	10	12

Stages 5, 4, 3, 2 and 1:

B	C	6	11	17
*	F	1	11	12
* E	F	7	11	18
	I	10	12	22

Stages 6, 5, 4, 3, 2 and 1:

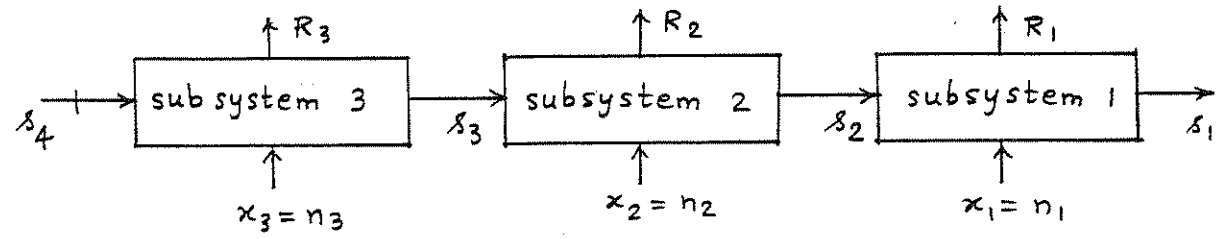
* A	B	4	12	16
	E	6	18	24

Retrace steps to find the optimum route:

AB (cost 4), BF (cost 1), FJ (cost 4), JK (cost 1), KL (cost 3), LP (cost 3).

Total minimum cost = 16.

9.3



s_i = weight available for allocation to subsystem i

Suboptimization of stage 1 (subsystem 1):

$$f_1^*(s_2) = \max_{x_1} \left[R_1(x_1, s_2)\right]$$

subject to $s_2 \le 20$

$$= \max_{x_1} \left[R_1 = 1 - (1-r_1)^{x_1}\right]$$

subject to $s_2 \le 20$

R_1 will be maximum when x_1 is maximum. Since $w_1 = 4$ lb,

max $x_1 = 5$,

$$(R_1)_{max} = 1 - (1-0.96)^{x_1} = 1 - 0.04^{x_1} = 1 - 0.04^5 = 0.999\,999\,898$$

s_2	x_1^*	f_1^*	s_1
20	5	0.999 999 898	0
19	4	0.999 997 440	3
18	4	"	2
17	4	"	1
16	4	"	0
15	3	0.999 936	3

14	3	"	2
13	3	"	1
12	3	"	0
11	2	0.9984	3
10	2	"	2
9	2	"	1
8	2	"	0
7	1	0.96	3
6	1	"	2
5	1	"	1
4	1	"	0
3	0	0.0	3
2	0	"	2
1	0	"	1
0	0	"	0

Suboptimization of stages 2 and 1 (subsystems 1 and 2):

$$f_2^*(s_3) = \max_{x_2} \left[R_2(x_2, s_3) + f_1^*(s_2) \right]$$

subject to $s_3 \le 20$

$$= \max_{x_2} \left[1 - (1 - r_2)^{x_2} + f_1^*(s_2) \right]$$

subject to $s_3 \le 20$

s_3	x_2	s_2	R_2	x_1^*	f_1^*	$R_2 f_1^*$ = reliability of 2 and 1
20	0	20	0	5	0.999 999 878	0
	1	18	0.92	4	0.999 997440	0.919 997 645
	2	16	0.9936	4	"	0.993 597 456
	3	14	0.999 488	3	0.999 936	0.999 424 033
	4 *	12	0.999 959 04	3	"	0.999 895 043
	5	10	0.999 996 723	2	0.9984	0.998 396 728
	6	8	0.999999738	2	"	0.998399 738
	7	6	0.999 999 979	1	0.96	0.959 999 980
	8	4	0.999 999 998	1	"	0.959 999 998
	9	2	1	0	0	0
	10	0	1	0	0	0

19	0	19	0	4	0.999 987440	0
	1	17	0.92	4	"	0.919 988 445
	2	15	0.9936	3	0.999 936	0.993 536 410
	3 *	13	0.999 488	3	"	0.999 424 033
	4	11	0.999 959 04	2	0.9984	0.998 359 106
	5	9	0.999 996 723	2	"	0.998 396 728
	6	7	0.999 999 738	1	0.96	0.959 999 748
	7	5	0.999 999 979	1	"	0.959 999 998
	8	3	0.999 999 998	0	0	0
	9	1	1	0	"	0

18	0	18	0	4	0.999 997 440	0
	1	16	0.92	4	0.999 987 440	0.919 988 445
	2	14	0.9936	3	0.999 936	0.993 536 410
	3 *	12	0.999 488	3	"	0.999 424 033
	4	10	0.999 959 04	2	0.9984	0.998 359 106
	5	8	0.999 996 723	2	"	0.998 396 728
	6	6	0.999 999 738	1	0.96	0.959 999 748
	7	4	0.999 999 979	1	"	0.959 999 998
	8	2	0.999 999 998	0	0	0
	9	0	1	0	"	0

17	0	17	0	4	0.999 997 440	0
	1	15	0.92	3	0.999 936	0.919 941 120
	2	13	0.9936	3	"	0.993 536 410
	3	11	0.999 488	2	0.9984	0.997 888 819
	4 *	9	0.999 959 04	2	"	0.998 359 106
	5	7	0.999 996 723	1	0.96	0.959 996 854
	6	5	0.999 999 738	1	"	0.959 996 748
	7	3	0.999 999 979	0	0	0
	8	1	0.999 999 998	0	0	0

16	0	16	0	4	0.999 997 440	0
	1	14	0.92	3	0.999 936	0.919 941 120
	2	12	0.9936	3	"	0.993 536 410
	3	10	0.999 488	2	0.9984	0.997 888 819
	4 *	8	0.999 959 04	2	"	0.998 359 106

	5	6	0.999 996 723	1	0.96	0.959 996 854
	6	4	0.999 999 738	1	"	0.959 996 748
	7	2	0.999 999 979	0	0	0
	8	0	0.999 999 998	0	"	0
15	0	15	0	3	0.999 936	0
	1	13	0.92	3	"	0.919 941 120
	2	11	0.9936	2	0.9984	0.992 010 240
	3 *	9	0.999 488	2	"	0.997 888 819
	4	7	0.999 959 04	1	0.96	0.959 960 678
	5	5	0.999 996 723	1	"	0.959 996 854
	6	3	0.999 999 738	0	0	0
	7	1	0.999 999 979	0	0	0
14	0	14	0	3	0.999 936	0
	1	12	0.92	3	"	0.919 941 120
	2	10	0.9936	2	0.9984	0.992 010 240
	3 *	8	0.999 488	2	"	0.997 888 819
	4	6	0.999 959 04	1	0.96	0.959 960 678
	5	4	0.999 996 723	1	"	0.959 996 854
	6	2	0.999 999 738	0	0	0
	7	0	0.999 999 979	0	"	0
13	0	13	0	3	0.999 936	0
	1	11	0.92	2	0.9984	0.918 528
	2 *	9	0.9936	2	"	0.992 010 24
	3	7	0.999 488	1	0.96	0.959 508 48
	4	5	0.999 959 04	1	"	0.959 960 678
	5	3	0.999 996 723	0	0	0
	6	1	0.999 999 738	0	0	0
12	0	12	0	3	0.999 936	0
	1	10	0.92	2	0.9984	0.918 528
	2 *	8	0.9936	2	"	0.992 010 24
	3	6	0.999 488	1	0.96	0.959 508 48
	4	4	0.999 959 04	1	"	0.959 960 678
	5	2	0.999 996 723	0	0	0
	6	0	0.999 999 738	0	"	0

11	0	11	0	2	0.9984	0
	1	9	0.92	2	"	0.918 528
	2	7	0.9936	1	0.96	0.953 856
	3 *	5	0.999 488	1	"	0.959 508 48
	4	3	0.999 959 04	0	0	0
	5	1	0.999 996 723	0	"	0
10	0	10	0	2	0.9984	0
	1	8	0.92	2	"	0.918 528
	2	6	0.9936	1	0.96	0.953 856
	3 *	4	0.999 488	1	"	0.959 508 48
	4	2	0.999 959 04	0	0	0
	5	0	0.999 996 723	0	"	0
9	0	9	0	2	0.9984	0
	1	7	0.92	1	0.96	0.8832
	2 *	5	0.9936	1	0.96	0.953 856
	3	3	0.999 488	0	0	0
	4	1	0.999 959 04	0	0	0
8	0	8	0	2	0.9984	0
	1	6	0.92	1	0.96	0.883 2
	2 *	4	0.9936	1	0.96	0.953 856
	3	2	0.999 488	0	0	0
	4	0	0.999 959 04	0	0	0
7	0	7	0	1	0.96	0
	1 *	5	0.92	1	0.96	0.8832
	2	3	0.9936	0	0	0
	3	1	0.999 488	0	0	0
6	0	6	0	1	0.96	0
	1 *	4	0.92	1	0.96	0.8832
	2	2	0.9936	0	0	0
	3	0	0.999 488	0	0	0
5	0	5	0	1	0.96	0
	1	3	0.92	0	0	0
	2	1	0.9936	0	0	0

4	0	4	0	1	0.96	0
	1	2	0.92	0	0	0
	2	0	0.9936	0	0	0
3	0	2	0	0	0	0
	1	0	0.92	0	0	0
2	0	2	0	0	0	0
	1	0	0.92	0	0	0
1	0	1	0	0	0	0

Suboptimization of stages 3, 2 and 1 (subsystems 1, 2 and 3):

$$f_3^*(s_4) = \max_{x_3} \left[R_3(x_3, s_4) + f_2^*(s_3) \right]$$

subject to $s_4 \le 20$

$$= \max_{x_3} \left[1 - (1 - r_3)^{x_3} + f_2^*(s_3) \right]$$

subject to $s_4 \le 20$

s_4	x_3	s_3	R_3	$x_2^* + x_1^*$	f_2^*	$R_3 f_2^*$ = reliability of 3, 2 and 1
20	0	20	0	4+3	0.999 895 043	0
	1 *	14	0.98	3+2	0.997 888 819	0.977 931 043
	2	8	0.9996	2+1	0.953 856	0.953 474 458
	3	2	0.999 992	1+0	0	0

Retrace the steps to find:

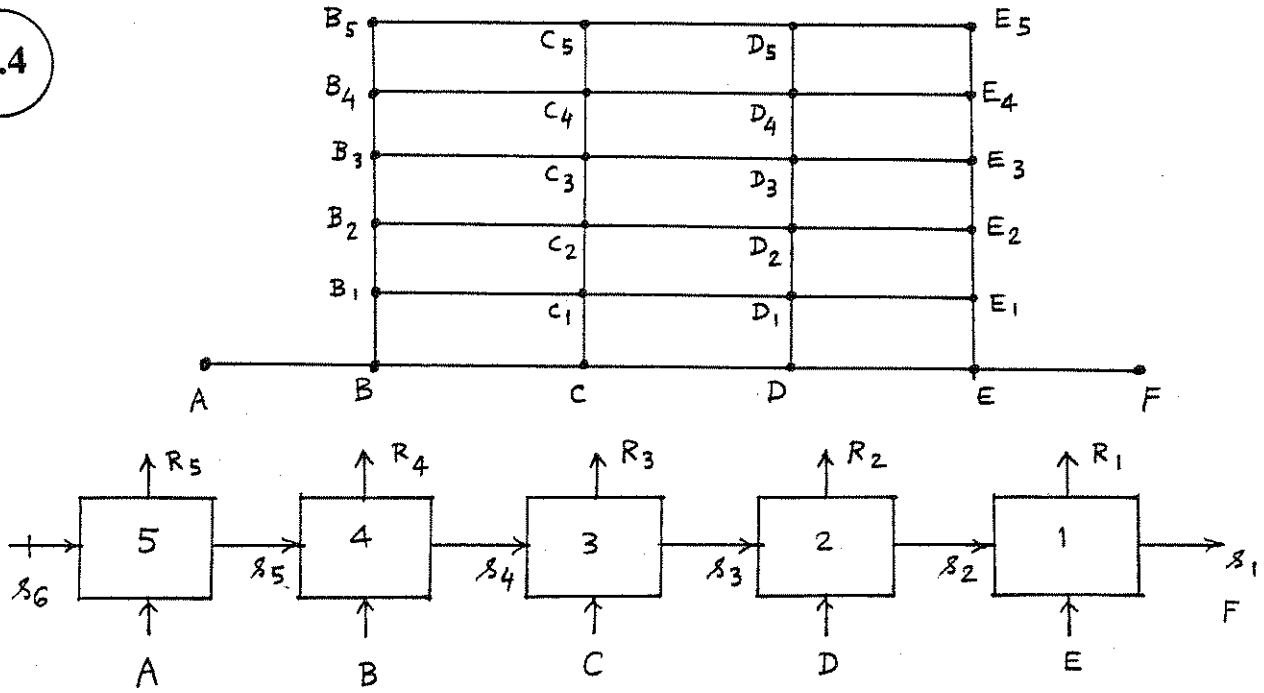
$s_4 = 20$ lb, $x_3^* = 1$, $R_3 = 0.98$

$s_3 = 14$ lb, $x_2^* = 3$, $R_2 = 0.999\,488$

$s_2 = 8$ lb, $x_1^* = 2$, $R_1 = 0.9984$

$\therefore \quad R_0 = R_1 R_2 R_3 = 0.977\,931\,043$

9.4



stage 1 (E):

s_2	$R_1 = f_1^*$
E_1	800
E_2	320
E_3	0
E_4	0
E_5	0

stages 2 and 1:

s_3	x_2	R_2	x_1^*	f_1^*	$R_2 + f_1^*$
D_1	E_1*	1600	F	800	2400
	E_2	2680	F	320	3000
	E_3	4000	F	0	4000
	E_4	4720	F	0	4720
	E_5	6080	F	0	6080
D_2	E_1	480	F	800	1280
	E_2*	800	F	320	1120
	E_3	2240	F	0	2240
	E_4	3120	F	0	3120
	E_5	4640	F	0	4640

D_3	E_1	160	F	800	960
	E_2	320	F	320	640
	E_3 *	560	F	0	560
	E_4	1600	F	0	1600
	E_5	3040	F	0	3040
D_4	E_1	0	F	800	800
	E_2	80	F	320	400
	E_3 *	240	F	0	240
	E_4	480	F	0	480
	E_5	1600	F	0	1600
D_5	E_1	0	F	800	800
	E_2	0	F	320	320
	E_3 *	0	F	0	0
	E_4	160	F	0	160
	E_5	240	F	0	240

stages 3, 2 and 1:

δ_4	x_3	R_3	x_2^*	f_2^*	$R_3 + f_2^*$
C_1	D_1	1600	E_1	2400	4000
	D_2 *	2680	E_2	1120	3800
	D_3	4000	E_3	560	4560
	D_4	4720	E_3	240	4960
	D_5	6080	E_3	0	6080
C_2	D_1	480	E_1	2400	2880
	D_2 *	800	E_2	1120	1920
	D_3	2240	E_3	560	2800
	D_4	3120	E_3	240	3360
	D_5	4640	E_3	0	4640
C_3	D_1	160	E_1	2400	2560
	D_2	320	E_2	1120	1440
	D_3 *	560	E_3	560	1120
	D_4	1600	E_3	240	1840
	D_5	3040	E_3	0	3040

C_4	D_1	0	E_1	2400	2400
	D_2	80	E_2	1120	1200
	D_3	240	E_3	560	800
	D_4 *	480	E_3	240	720
	D_5	1600	E_3	0	1600
C_5	D_1	0	E_1	2400	2400
	D_2	0	E_2	1120	1120
	D_3	0	E_3	560	560
	D4	160	E_3	240	400
	D_5 *	240	E_3	0	240

Stages 4, 3, 2 and 1:

s_5	x_4	R_4	x_3^*	f_3^*	$R_4 + f_3^*$
B_1	C_1	1600	D_2	3800	5400
	C_2 *	2680	D_2	1920	4600
	C_3	4000	D_3	1120	5120
	C_4	4720	D_4	720	5440
	C_5	6080	D_5	240	6320
B_2	C_1	480	D_2	3800	4280
	C_2 *	800	D_2	1920	2720
	C_3	2240	D_3	1120	3360
	C_4	3120	D_4	720	3840
	C_5	4640	D_5	240	4880
B_3	C_1	160	D_2	3800	3960
	C_2	320	D_2	1920	2240
	C_3 *	560	D_3	1120	1680
	C_4	1600	D_4	720	2320
	C_5	3040	D_5	240	3280
B_4	C_1	0	D_2	3800	3800
	C_2	80	D_2	1920	2000
	C_3	240	D_3	1120	1360
	C_4 *	480	D_4	720	1200
	C_5	1600	D_5	240	1840

B_5	C_1	0	D_2	3800	3800
	C_2	0	D_2	1920	1920
	C_3	0	D_3	1120	1120
	C_4	160	D_4	720	880
	C_5 *	240	D_5	240	480

stages 5, 4, 3, 2 and 1:

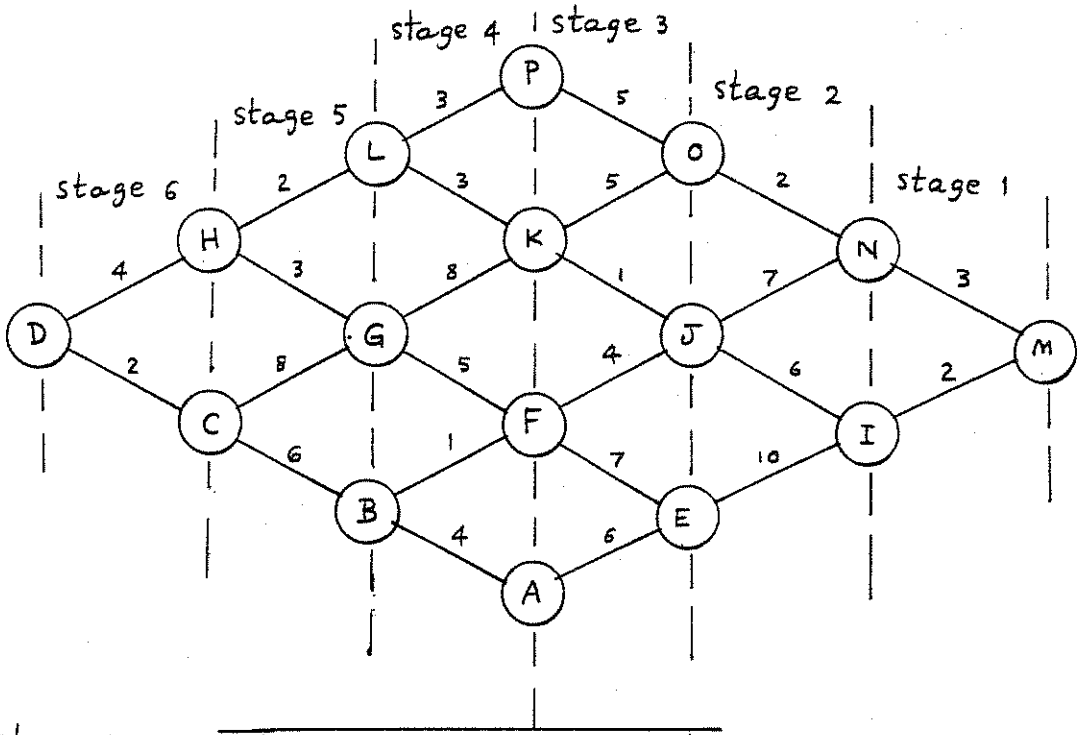
s_6	x_5	R_5	x_4^*	f_4^*	$R_5 + f_4^*$
A	B_1	4000	C_2	4600	8600
	B_2	4800	C_2	2720	7520
	B_3 *	5520	C_3	1680	7200
	B_4	6160	C_4	1200	7360
	B_5 *	6720	C_5	480	7200

Optimal solution(s):

AB_3 (cost 5520)
B_3C_3 (cost 560)
C_3D_3 (cost 560)
D_3E_3 (cost 560)
E_3F (cost 0)
Minimum total cost: 7200

AB_5 (cost 6720)
B_5C_5 (cost 240)
C_5D_5 (cost 240)
D_5E_3 (cost 0)
E_3F (cost 0)
Minimum total cost: 7200

9.5



stage 1:

	From	To	Cost
*	N	M	3
*	I	M	2

Stages 2 and 1:

	From	To	cost	Minimum downward cost	Total cost
*	O	N	2	3	5
	J	N	7	3	10
*		I	6	2	8
*	E	I	10	2	12

Stages 3, 2 and 1:

	From	To	cost	Minimum downward cost	Total cost
*	P	O	5	5	10
	K	O	5	5	10
*		J	1	8	9
*	F	J	4	8	12
		E	7	12	19
*	A	E	6	12	18

Stages 4, 3, 2 and 1:

	From	To	cost	Minimum downward cost	Total cost
	L	P	3	10	13
*		K	3	9	12
	G	K	8	9	17
*		F	5	12	17
*	B	F	1	12	13
		A	4	18	22

Stages 5, 4, 3, 2 and 1:

	From	To	cost	Minimum downward cost	Total cost
*	H	L	2	12	14
		G	3	17	20
	C	G	8	17	25
*		B	6	13	19

Stages 6, 5, 4, 3, 2 and 1:

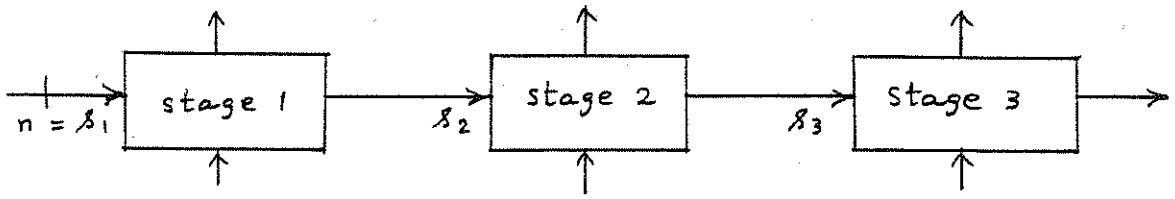
	From	To	cost	Minimum downward cost	Total cost
*	D	H	4	14	18
		C	2	19	21

Retrace the steps to find the optimum route:

D H (cost 4)
H L (cost 2)
L K (cost 3)
K J (cost 1)
J I (cost 6)
I M (cost 2)

Total minimum cost = 18

9.6



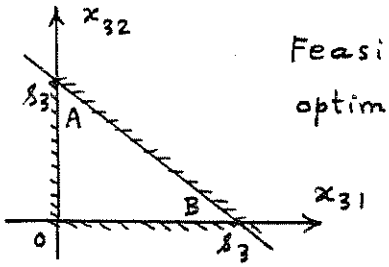
Let x_{ij} be the number of lathes used to produce part j in stage i $(j = 1, 2\ ;\ i = 1, 2, 3)$

s_i = number of lathes available for use in stage i

stage 3:

Number of lathes allocated = $\begin{cases} x_{31} & \text{for part 1} \\ x_{32} & \text{for part 2} \end{cases}$

$$R_3(s_3) = (3x_{31} + 2.5x_{32})$$
$$\text{with} \quad x_{31} + x_{32} \le s_3\ ; \quad x_{31} \ge 0,\ x_{32} \ge 0 \qquad (1)$$



Feasible space of Eq.(1) is shown in figure. optimal solution is B $(s_3, 0)$.

$$\therefore\ f_3^*(s_3) = 3s_3 = \max R_3(s_3) \qquad (2)$$

stages 2 and 3:

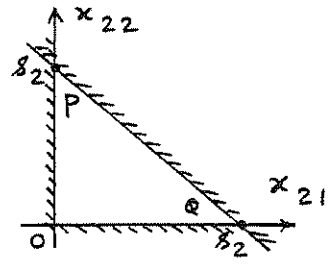
Number of lathes available at beginning = s_2

$$R_2(s_2) = 3x_{21} + 2.5x_{22} + f_3^*\left(\tfrac{1}{3}x_{21} + \tfrac{2}{3}x_{22}\right)$$

But $f_3^*\left(\frac{1}{3}x_{21} + \frac{2}{3}x_{22}\right) = 3\left(\frac{1}{3}x_{21} + \frac{2}{3}x_{22}\right)$

$= x_{21} + 2x_{22}$ from Eq. (2)

$$R_2(s_2) = 3x_{21} + 2.5x_{22} + (x_{21} + 2x_{22}) = 4x_{21} + 4.5x_{22}$$
$$\text{with} \quad x_{21} + x_{22} \le s_2\ ; \quad x_{21} \ge 0\ ,\ x_{22} \ge 0 \qquad (3)$$



Feasible space of Eq.(3) is shown in figure. optimal solution is P $(0, s_2)$.

$$\therefore\ f_2^*(s_2) = 4.5s_2 = \max R_2(s_2) \qquad (4)$$

Stages 1, 2 and 3:

Number of lathes available at beginning = $s_1 = n$

$$R_1(s_1) = 3x_{11} + 2.5x_{12} + f_2^*\left(\tfrac{1}{3}x_{11} + \tfrac{2}{3}x_{12}\right)$$

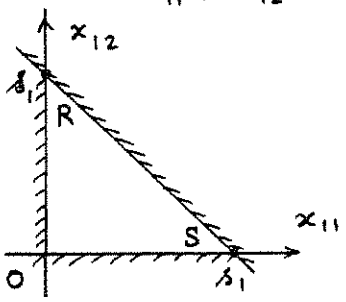
But $$f_2^*\left(\tfrac{1}{3}x_{11} + \tfrac{2}{3}x_{12}\right) = 4.5\left(\tfrac{1}{3}x_{11} + \tfrac{2}{3}x_{12}\right)$$

$$= 1.5x_{11} + 3x_{12} \quad \text{from Eq. (4)}$$

$$R_1(s_1) = 3x_{11} + 2.5x_{12} + (1.5x_{11} + 3x_{12}) = 4.5x_{11} + 5.5x_{12} \quad (5)$$

with

$$x_{11} + x_{12} \le s_1 \; ; \; x_{11} \ge 0, \quad x_{12} \ge 0$$



Feasible space of Eq. (5) is shown in figure. Optimal solution is R $(0, s_1)$.

$$\therefore f_1^*(s_1) = 5.5\, s_1 = \max R_1(s_1) \quad (6)$$

Final solution:

since $s_1 = n$, we have

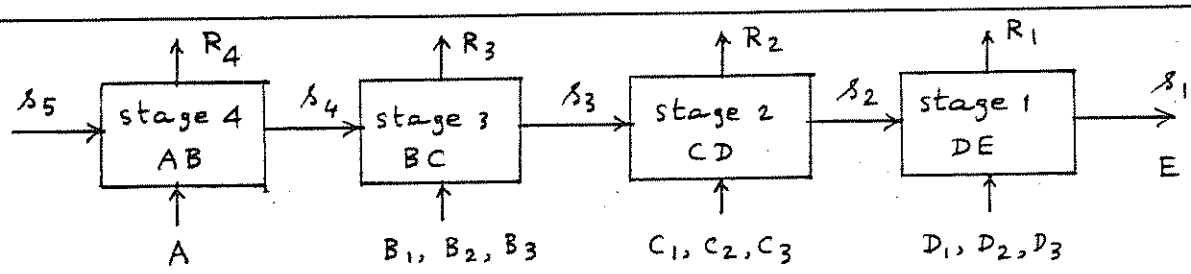
$$x_{11} = 0, \quad x_{12} = n, \quad f_1^*(n) = 5.5\, n$$

Also, we find

$$x_{21} = 0, \quad x_{22} = s_2 = \tfrac{2}{3}n$$

$$x_{31} = s_3 = \tfrac{2}{3}\left(\tfrac{2}{3}n\right) = \frac{4n}{9}, \quad x_{32} = 0$$

9.7



Stage 1:

s_2	x_1	$R_1 = f_1^*$
D_1	E	9
D_2	E	6
D_3	E	12

Stages 2 and 1 :

s_3	x_2	R_2	x_1	f_1^*	$R_2 + f_1^*$
C_1	D_1 *	8	E	9	17
	D_2	12	E	6	18
	D_3	19	E	12	31
C_2	D_1	9	E	9	18
	D_2 *	11	E	6	17
	D_3	13	E	12	25
C_3	D_1 *	7	E	9	16
	D_2	15	E	6	21
	D_3	14	E	12	26

Stages 3, 2 and 1:

s_4	x_3	R_3	x_2^*	f_2^*	$R_3 + f_2^*$
B_1	C_1 *	8	D_1	17	25
	C_2	12	D_2	17	29
	C_3	19	D_1	16	35
B_2	C_1 *	9	D_1	17	26
	C_2	11	D_2	17	28
	C_3	13	D_1	16	29
B_3	C_1 *	7	D_1	17	24
	C_2	15	D_2	17	32
	C_3	14	D_1	16	30

Stages 4, 3, 2 and 1 :

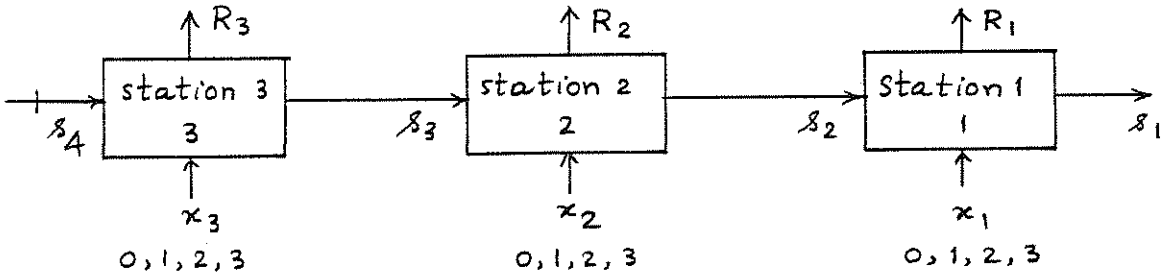
s_5	x_4	R_4	x_3^*	f_3^*	$R_4 + f_3^*$
A	B_1 *	10	C_1	25	35
	B_2	15	C_1	26	41
	B_3	12	C_1	24	36

Optimum solution:

AB_1 (cost: 10), B_1C_1 (cost : 8), C_1D_1 (cost: 8), D_1E (cost: 9)

Minimum cost = 35.

9.9



Stage 1:

s_2	x_1	s_1	$R_1 = f_1^*$
3	0	3	0
	1	2	2
	2	1	4
	3 *	0	6
2	0	2	0
	1	1	2
	2 *	0	4
1	0	1	0
	1 *	0	2
0	0	0	0

Stages 2 and 1:

s_3	x_2	s_2	R_2	f_1^*	$R_2 + f_1^*$
3	0	3	0	6	6
	1	2	1	4	5
	2 *	1	5	2	7
	3	0	6	0	6
2	0	2	0	4	4
	1	1	1	2	3
	2 *	0	5	0	5
1	0 *	1	0	2	2
	1	0	1	0	1
0	0	0	0	0	0

Stages 3, 2 and 1:

s_4	x_3	s_3	R_3	f_2^*	$R_3 + f_2^*$
3	0	3	0	7	7
	1*	2	3	5	8
	2	1	5	2	7
	3	0	6	0	6

Optimum solution: Thermal station 3: units invested (x_3^*) = 1
Return: 3 units
Thermal station 2: units invested (x_2^*) = 2
Return: 5 units
Thermal station 1: units invested (x_1^*) = 0
Return: 0 units

Total profit = 8 units.

9.10 $n = 2$, $m = 3$; Two-stage dynamic programming problem.

First stage problem:

$$\max \quad f_1(\beta_1, \beta_2, \beta_3, x_1) = \max_{0 \le x_1 \le \overline{\beta}} (10\, x_1)$$

where $\beta_1, \beta_2, \beta_3$ = resources available for allocation at stage 1

$\beta_1 = 25 - x_2$, $\beta_2 = 45 - 2x_2$, β_3 = unrestricted (no limit)

x_1 should satisfy:

$$2x_1 \le 25 - x_2 = \beta_1 \ , \quad 3x_1 \le 45 - 2x_2 = \beta_2$$

$$\beta_1^* = x_1^* = \min \left\{ \frac{25 - x_2}{2} , \frac{45 - 2x_2}{3} \right\} \qquad (1)$$

$$f_1^* = 10\, x_1^* = 10 \ \min \left\{ \frac{25 - x_2}{2} , \frac{45 - 2x_2}{3} \right\} \qquad (2)$$

Second stage problem:

$$\max \quad f_2(\beta_1, \beta_2, \beta_3) = \max_{0 \le x_2 \le \overline{\beta}} \left(8x_2 + f_1^* \right) \qquad (3)$$

where $\beta_1, \beta_2, \beta_3$ = resources available for allocation at stage 2

= 25, 45, 10 (respectively)

$\bar{\beta}$ = maximum value of x_2 without violating any constraint

$= \min\left(\frac{25}{1}, \frac{45}{2}, \frac{10}{1}\right) = 10$

Equation (3) becomes:

$$\max f_2(25, 45, 10) = \max_{0 \le x_2 \le 10} \left\{ 8x_2 + 10 \cdot \min\left(\frac{25 - x_2}{2}, \frac{45 - 2x_2}{3}\right)\right\}$$

$$= \max_{0 \le x_2 \le 10} \left\{ 8x_2 + 10\left(\frac{25 - x_2}{2}\right)\right\}$$

$$= \max_{0 \le x_2 \le 10} \left(3x_2 + 125\right) = 155 \text{ at } x_2^* = 10$$

$$x_1^* = \min\left(\frac{25 - x_2^*}{2}\right) = 7.5$$

$\therefore$ Optimum solution: $x_1^* = 7.5$, $x_2^* = 10.0$, $f^* = f_2^* = 155$

Graphical solution:

$g_1 = 2x_1 + x_2 - 25 \le 0$

$g_1 = 0$ at $(0, 25)$, $(12\frac{1}{2}, 0)$

$g_2 = 3x_1 + 2x_2 - 45 \le 0$

$g_2 = 0$ at $(0, 22\frac{1}{2})$, $(15, 0)$

$g_3 = x_2 - 10 \le 0$

$g_3 = 0$ at $x_2 = 10$

$f = 10x_1 + 8x_2$

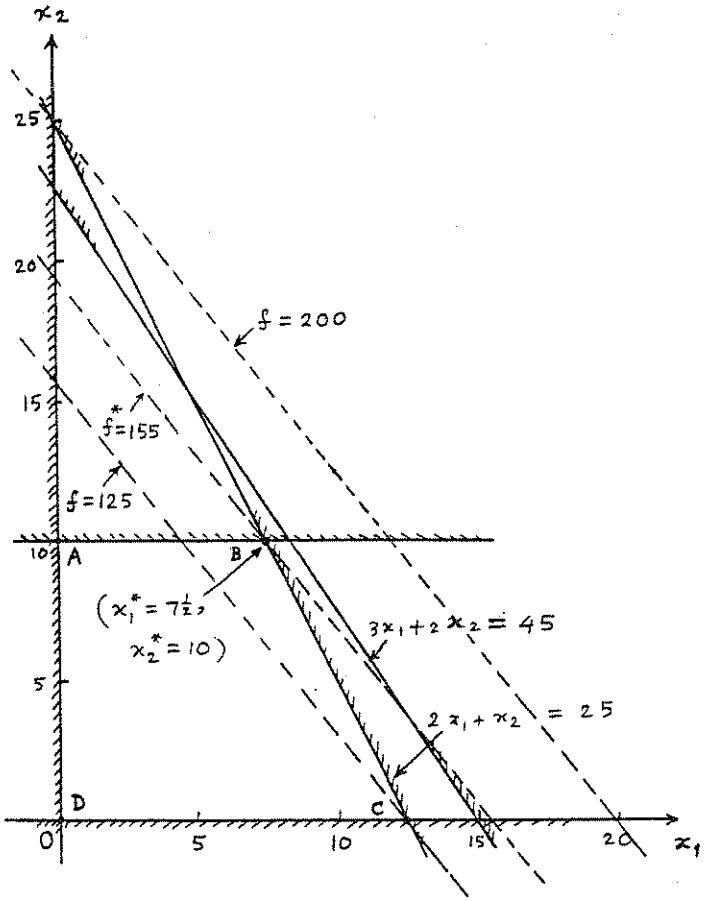
$f = 200$ at $(0, 25)$, $(10, 12\frac{1}{2})$

$f = 125$ at $(12\frac{1}{2}, 0)$, $(0, 15\frac{5}{8})$

$f = 180$ at $(0, 22\frac{1}{2})$, $(10, 10)$

Feasible space = ABCDA

Optimum point = B.



9.11 Minimize $f = (4500\,x_1 + 20\,x_1^2) + (4500\,x_2 + 20\,x_2^2) + (4500\,x_3 + 20\,x_3^2) + 400\,(x_1 - 50) + 400\,(x_1 + x_2 - 120)$

subject to

$$x_1 \geq 50$$

$$x_1 + x_2 \geq 120$$

$$x_1 + x_2 + x_3 = 210$$

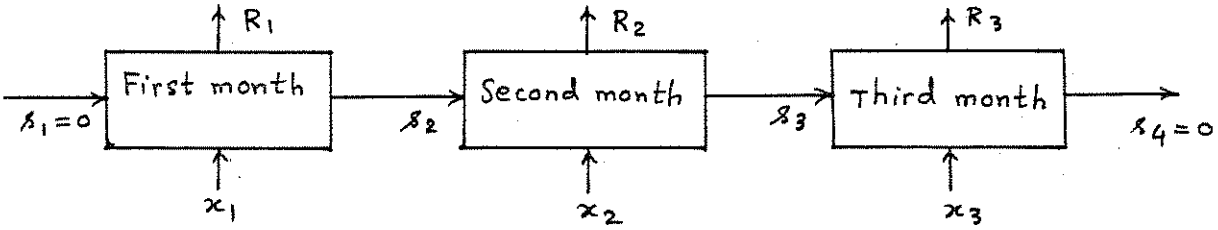
$$x_1 \geq 0,\quad x_2 \geq 0,\quad x_3 \geq 0$$

where x_i = number of tons of fertilizer produced in month i $(i = 1, 2, 3)$

Assume: No initial inventory at beginning of 1st month $(s_1 = 0)$

No final inventory at end of 3rd month $(s_4 = 0)$

Problem is solved as initial value problem.



$$R_1(x_1) = 4500\,x_1 + 20\,x_1^2 \qquad (1)$$

Let s_2 = inventory at beginning of second month

$$x_1^* = 50 + s_2 \qquad (2)$$

$$f_1^*(x_1) = R_1(x_1^*) = 4500\,(50 + s_2) + 20\,(50 + s_2)^2$$

$$= 275{,}000 + 6{,}500\,s_2 + 20\,s_2^2 \qquad (3)$$

Cost in second month $= 400\,s_2 + 4500\,x_2 + 20\,x_2^2 = R_2(x_2)$

$$R_2(x_2) + f_1^*(x_1) = 275{,}000 + 6{,}500\,s_2 + 20\,s_2^2 + 400\,s_2 + 4{,}500\,x_2 + 20\,x_2^2$$

$$= 275{,}000 + 6{,}900\,s_2 + 20\,s_2^2 + 4{,}500\,x_2 + 20\,x_2^2 \qquad (4)$$

$s_3 = s_2 + x_2 - 70$ = inventory at beginning of 3rd month

$$s_2 = s_3 + 70 - x_2 \qquad (5)$$

Substitute Eq. (5) into (4):

$$R_2(x_2) + f_1^*(x_1) = f_2(s_2, x_2, x_1^*)$$
$$= 275000 + 6900(s_3 + 70 - x_2) + 20(s_3 + 70 - x_2)^2 + 4500 x_2 + 20 x_2^2$$
$$= 856,000 + 9,700 s_3 - 5,200 x_2 - 40 s_3 x_2 + 20 s_3^2 + 40 x_2^2 \quad (6)$$

$$\frac{\partial f_2}{\partial x_2} = 0 \Rightarrow -5200 - 40 s_3 + 80 x_2 = 0$$
$$\text{or} \quad x_2^* = 65 + \tfrac{1}{2} s_3 \quad (7)$$

$$f_2^* = f_2(x_2^*) = 856000 + 9700 s_3 - 5200(65 + \tfrac{1}{2} s_3) - 40 s_3 (65 + \tfrac{1}{2} s_3) + 20 s_3^2 + 40(65 + \tfrac{1}{2} s_3)^2$$
$$= 687,000 + 7,100 s_3 + 10 s_3^2 \quad (8)$$

Cost in third month $= 400 s_3 + 4500 x_3 + 20 x_3^2 = R_3(s_3)$

$$f_3(x_3) = R_3(x_3) + f_2^*(x_2) = 400 s_3 + 4500 x_3 + 20 x_3^2 + 687000 + 7100 s_3 + 10 s_3^2$$
$$= 687,000 + 7,500 s_3 + 10 s_3^2 + 4,500 x_3 + 20 x_3^2 \quad (9)$$

Since $s_3 = 90 - x_3$, (10)

substitution of Eq. (10) into (9) gives the total cost for three months as:

$$f_3(x_3) = 687000 + 7500(90 - x_3) + 10(90 - x_3)^2 + 4500 x_3 + 20 x_3^2$$
$$= 1,443,000 - 4,800 x_3 + 30 x_3^2 \quad (11)$$

$$\frac{df_3}{dx_3} = -4800 + 60 x_3 = 0 \quad (12)$$

Eq. (12): $x_3^* = 80$

Eq. (10): $s_3 = 90 - x_3^* = 10$

Eq. (7): $x_2^* = 65 + 5 = 70$

Eq. (5): $s_2 = s_3 + 70 - x_2^* = 10$

Eq. (2): $x_1^* = 50 + s_2 = 60$

Total optimum cost $= f_3(x_3^*) = 1,251,000$ from Eq. (11).

9.12 We start at third month.

Optimal amount of fertilizer produced in 3rd month

$$= x_3^* = 90 - s_3 \qquad (1)$$

where s_3 = inventory at beginning of 3rd month.

R_3 = cost in 3rd month $= 400\, s_3 + 4500\, x_3 + 20\, x_3^2$ (2)

substitute Eq. (1) in (2):

$$R_3(s_3) = 400\, s_3 + 4500\,(90 - s_3) + 20\,(90 - s_3)^2$$

$$= 567\,000 - 7700\, s_3 + 20\, s_3^2 \qquad (3)$$

R_2 = cost in 2nd month $= 400\, s_2 + 4500\, x_2 + 20\, x_2^2$

where s_2 = inventory at beginning of 2nd month

Total cost in 2nd and 3rd months

$$= f(s_2, x_2, s_3, x_3^*) = R_2 + R_3(s_3)$$

$$= 400\, s_2 + 4500\, x_2 + 20\, x_2^2 + 567\,000 - 7700\, s_3 + 20\, s_3^2 \qquad (4)$$

But $s_3 = s_2 + x_2 - 70$ (5)

Eqs. (4) and (5) yield:

$$f = 400\, s_2 + 4500\, x_2 + 20\, x_2^2 + 567\,000 - 7700\,(s_2 + x_2 - 70) + 20\,(s_2 + x_2 - 70)^2$$

$$= 1{,}204{,}000 - 10{,}100\, s_2 - 6{,}000\, x_2 + 40\, x_2^2 + 20\, s_2^2 + 40\, s_2 x_2 \qquad (6)$$

$$\frac{\partial f}{\partial x_2} = -6000 + 80\, x_2 + 40\, s_2 = 0$$

or $$x_2^* = 75 - \tfrac{1}{2}\, s_2 \qquad (7)$$

Substitute Eq. (7) into (6) to find:

$$f(x_2^*) = 1{,}204{,}000 - 10{,}100\, s_2 - 6{,}000\,(75 - \tfrac{1}{2}\, s_2) + 40\,(75 - \tfrac{1}{2}\, s_2)^2 + 20\, s_2^2 + 40\, s_2\,(75 - \tfrac{1}{2}\, s_2)$$

$$= 979{,}000 - 7{,}100\, s_2 + 10\, s_2^2 \qquad (8)$$

R_1 = cost in 1st month $= 4500\, x_1 + 20\, x_1^2$

Total cost of 3 months $= f = R_1 + f(x_2^*)$

$$= 4500\, x_1 + 20\, x_1^2 + 979\,000 - 7100\, s_2 + 10\, s_2^2 \qquad (9)$$

Since $s_2 = x_1 - 50$ (10)

Eq. (9) becomes:

$$f = 4500\,x_1 + 20\,x_1^2 + 979\,000 - 7100\,(x_1 - 50) + 10\,(x_1 - 50)^2$$

$$= 1{,}359{,}000 - 3{,}600\,x_1 + 30\,x_1^2 \quad (11)$$

$$\frac{\partial f}{\partial x_1} = -3600 + 60\,x_1 = 0 \quad (12)$$

Eq. (12): $x_1^* = 60$

Eq. (10): $s_2 = 10$

Eq. (7): $x_2^* = 70$

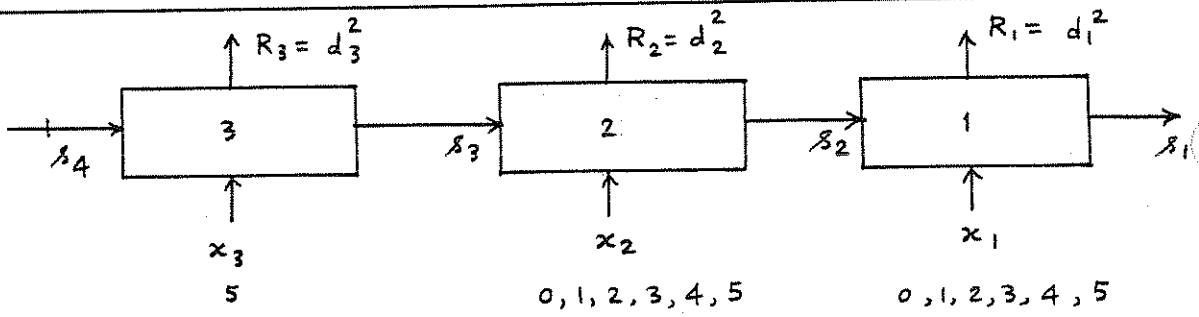
Eq. (5): $s_3 = 10$

Eq. (1): $x_3^* = 80$

} optimum solution

Optimum total cost = $f(x_1^*)$ in Eq. (11) = 1,251,000

9.13



Stage 1:

$s_2 = x_2$	x_1	$d_1 = x_2 - x_1$	$R_1^* = d_1^2 = f_1^*$
0	0	0	0
	1	−1	1
	2	−2	4
	3	−3	9
	4	−4	16
	5 *	−5	25
1	0	1	1
	1	0	0
	2	−1	1
	3	−2	4
	4	−3	9
	5 *	−4	16

2	0	2	4
	1	1	1
	2	0	0
	3	−1	1
	4	−2	4
	5 *	−3	9
3	0 *	3	9
	1	2	4
	2	1	1
	3	0	0
	4	−1	1
	5	−2	4
4	0 *	4	16
	1	3	9
	2	2	4
	3	1	1
	4	0	0
	5	−1	1
5	0 *	5	25
	1	4	16
	2	3	9
	3	2	4
	4	1	1
	5	0	0

Stages 2 and 1:

$s_3 = x_3$	$x_2 = s_2$	$d_2 = x_3 - x_2$	$R_2 = d_2^2$	x_1^*	f_1^*	$f_2^* = R_2 + f_1^*$
5	0 *	5	25	5	25	50
	1	4	16	5	16	32
	2	3	9	5	9	18
	3	2	4	0	9	13
	4	1	1	0	16	17
	5	0	0	0	25	25

Stages 3, 2 and 1:

$s_4 = x_4$	x_3	$d_3 = x_4 - x_3$	$R_3 = d_3^2$	x_2^*	f_2^*	$f_3^* = R_3 + f_2^*$
0	5	-5	25	0	50	75

Optimal solution:

$$x_4^* = 0$$
$$x_3^* = 5$$
$$x_2^* = 0$$
$$x_1^* = 5$$
$$f_{max} = f_3^* = 75$$

Chapter 10
Integer Programming

10.1 Minimize $f = 4x_1 + 5x_2$

subject to $g_1: 3x_1 + x_2 \geq 2$

$g_2: x_1 + 4x_2 \geq 5$

$g_3: 3x_1 + 2x_2 \geq 7$

$x_1 \geq 0,\ x_2 \geq 0$, both integers

Plotting of $g_1 = 0 \Rightarrow 3x_1 + x_2 = 2 \Rightarrow (x_1 = 0,\ x_2 = 2),\ (\frac{2}{3}, 0)$

Plotting of $g_2 = 0 \Rightarrow x_1 + 4x_2 = 5 \Rightarrow (0, \frac{5}{4}),\ (5, 0)$

Plotting of $g_3 = 0 \Rightarrow 3x_1 + 2x_2 = 7 \Rightarrow (0, \frac{7}{2}),\ (\frac{7}{3}, 0)$

Plotting of contours of $f = 4x_1 + 5x_2$:

$f = 10 \Rightarrow (0, 2),\ (2\frac{1}{2}, 0)$

$f = 6\frac{1}{4} \Rightarrow (0, 5/4),\ (25/16, 0)$

$f = 17\frac{1}{2} \Rightarrow (0, 7/2),\ (35/8, 0)$

Feasible space is shown in Figure.

Point C = continuous variable optimum

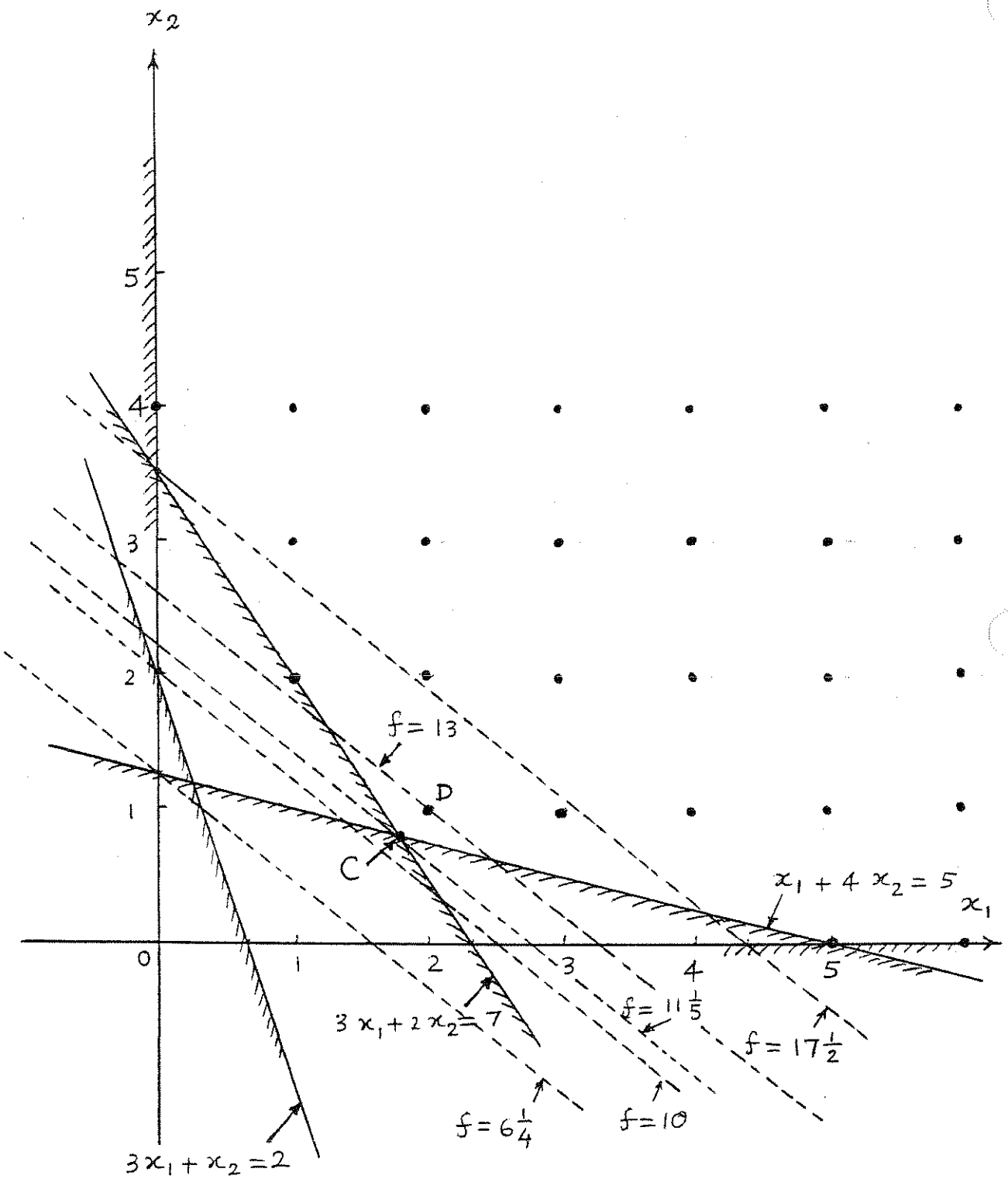
= point of intersection of $g_2 = 0$ and $g_3 = 0$

$= (x_1 = \frac{9}{5},\ x_2 = \frac{4}{5})$

$f_C = 11\frac{1}{5}$

Point D = integer variable optimum $= \vec{X}^* = \begin{Bmatrix} 2 \\ 1 \end{Bmatrix}$

$f_D = f^* = 13$


x_2
5
4
3
2
1
0
1
2
3
4
5
x_1
$f = 13$
D
C
$x_1 + 4x_2 = 5$
$3x_1 + 2x_2 = 7$
$f = 11\frac{1}{5}$
$f = 17\frac{1}{2}$
$f = 6\frac{1}{4}$
$f = 10$
$3x_1 + x_2 = 2$

10.2 Maximize $f = 4x_1 + 8x_2$

g_1: $4x_1 + 5x_2 \le 40$

g_2: $x_1 + 2x_2 \le 12$

$x_1 \ge 0$, $x_2 \ge 0$, both integers

$g_1 = 0$ at $(0, 8)$, $(10, 0)$

$g_2 = 0$ at $(0, 6)$, $(12, 0)$

contours of $f = 4x_1 + 8x_2$

$f = 64$ at $(0, 8)$, $(16, 0)$

$f = 40$ at $(10, 0)$, $(5, 0)$

$f = 48$ at $(0, 6)$, $(12, 0)$

Graphical optimization shown in the figure.

Point C = intersection point of $g_1 = 0$ and $g_2 = 0$

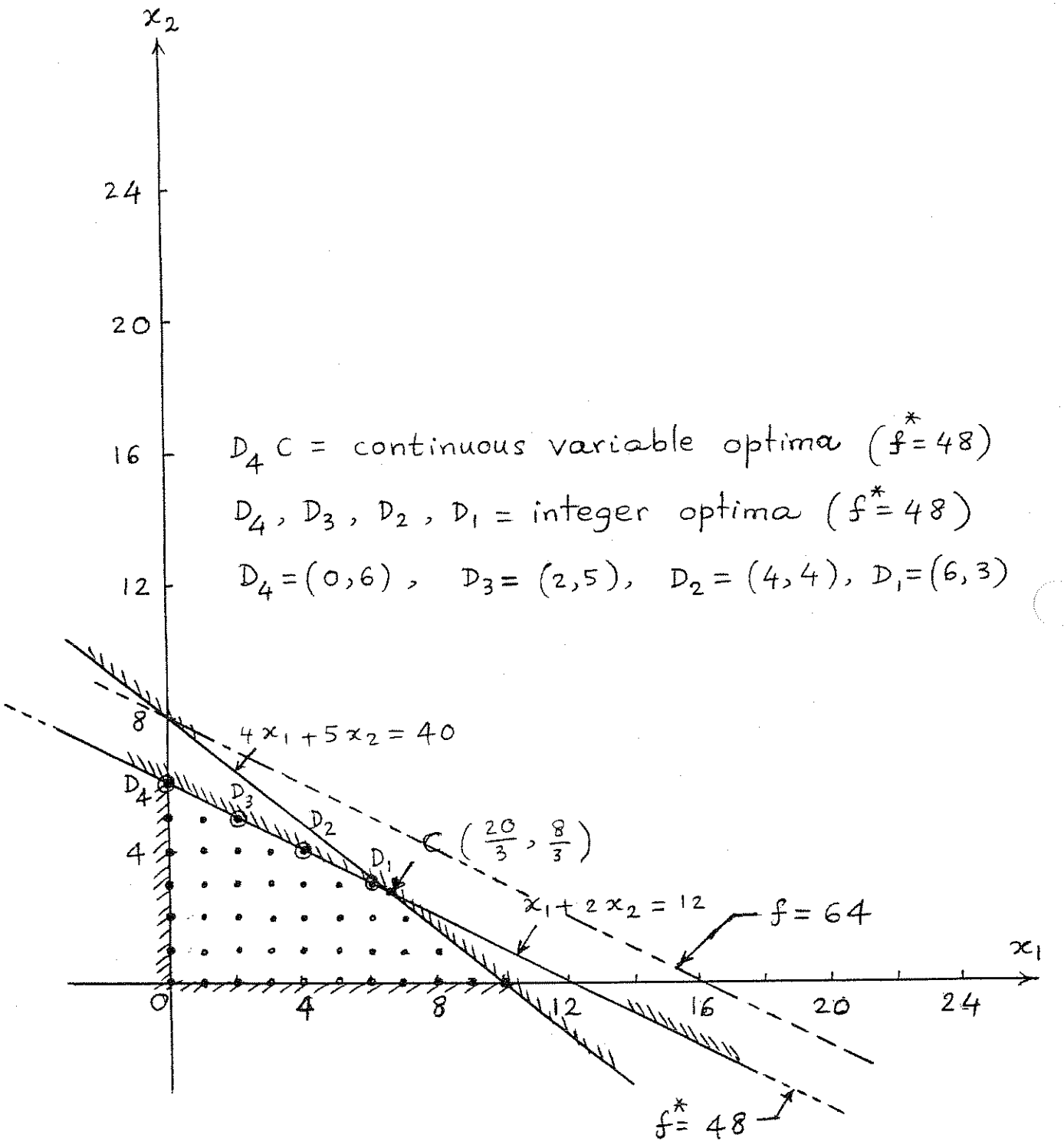
$= \left(\frac{20}{3}, \frac{8}{3}\right)$

Continuous optimum solution = line D_4C with $f = 48$.

Integer optimum solution = D_1 or D_2 or D_3 or D_4

$\therefore \vec{X}^* = \begin{Bmatrix} 6 \\ 3 \end{Bmatrix}, \begin{Bmatrix} 4 \\ 4 \end{Bmatrix}, \begin{Bmatrix} 2 \\ 5 \end{Bmatrix}, \begin{Bmatrix} 0 \\ 6 \end{Bmatrix}$

$f^* = 48$


x_2
24
20
16
12
8
4
D_4 C = continuous variable optima ($f^* = 48$)
D_4, D_3, D_2, D_1 = integer optima ($f^* = 48$)
$D_4 = (0,6)$, $D_3 = (2,5)$, $D_2 = (4,4)$, $D_1 = (6,3)$
$4x_1 + 5x_2 = 40$
D_4
D_3
D_2
D_1
C $\left(\frac{20}{3}, \frac{8}{3}\right)$
$x_1 + 2x_2 = 12$
$f = 64$
$f^* = 48$
x_1
O
4
8
12
16
20
24

10.3 Maximize $f = 4x_1 + 3x_2$

subject to $g_1: \; 3x_1 + 2x_2 \leq 18$

$x_1, x_2 \geq 0$, integers

$g_1 = 0$ at $(0, 9)$, $(6, 0)$

Contours of $f = 4x_1 + 3x_2$:

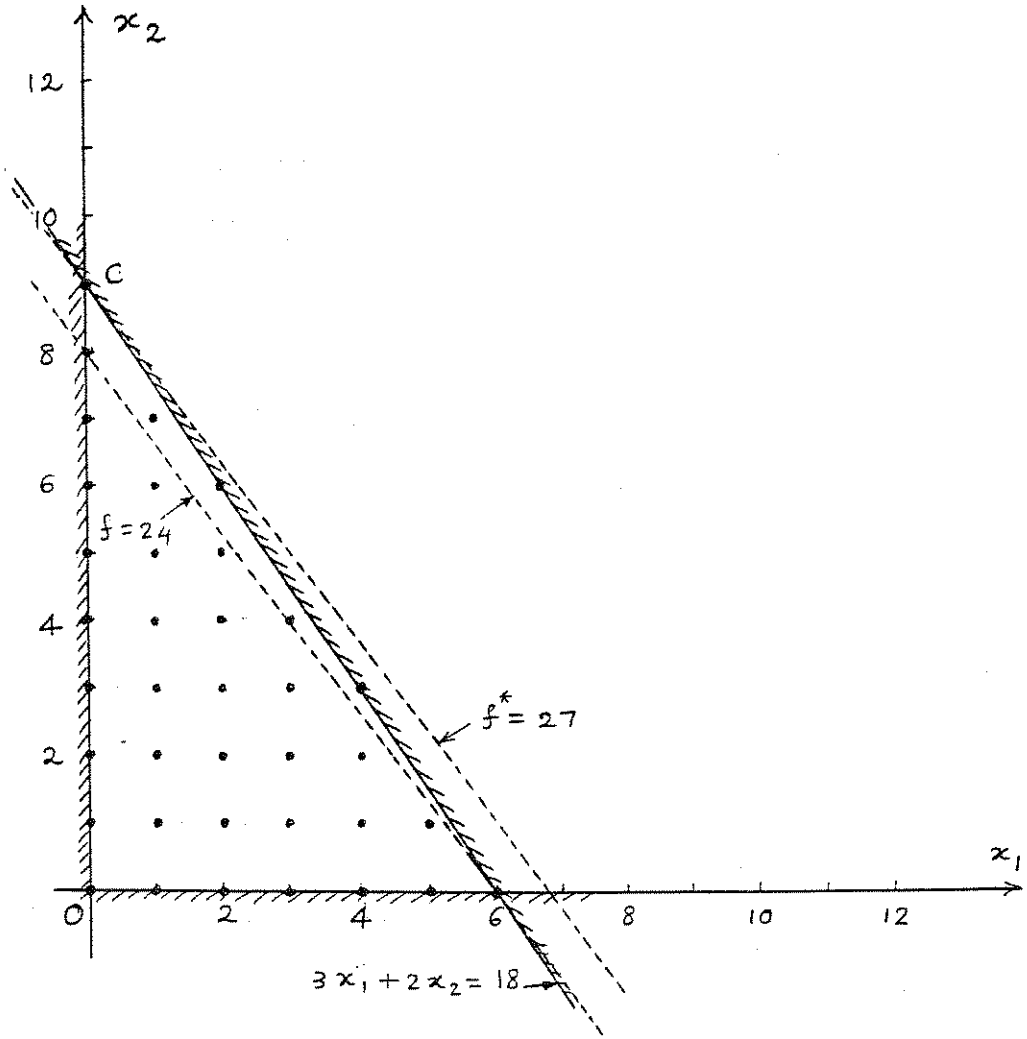
$f = 27$ at $(0, 9)$, $(27/4, 0)$

$f = 24$ at $(6, 0)$, $(0, 8)$

Point C = continuous optimum point = $(0, 9)$

= also *discrete* optimum point.

$\therefore \vec{X}^* = \begin{Bmatrix} 0 \\ 9 \end{Bmatrix}$, $f^* = 27$



10.4 Maximize $f = 3x_1 - x_2$

subject to $g_1: \; 3x_1 - 2x_2 \le 3$

$g_2: \; -5x_1 - 4x_2 \le -10$

$x_1, x_2 \ge 0$, integers

$g_1 = 0$ at $(x_1 = 0, \; x_2 = -\frac{3}{2})$, $(1, 0)$

$g_2 = 0$ at $(x_1 = 0, \; x_2 = 2\frac{1}{2})$, $(2, 0)$

Contours of $f = 3x_1 - x_2$:

$f = \frac{3}{2}$ at $(0, -\frac{3}{2})$, $(\frac{1}{2}, 0)$

$f = 3$ at $(1, 0)$, $(0, -3)$

$f = -\frac{5}{2}$ at $(0, \frac{5}{2})$, $(-\frac{5}{6}, 0)$

$f = 6$ at $(2, 0)$, $(0, -6)$

Graphical optimization is shown in the figure.

Point C = intersection point of $g_1 = 0$ and $g_2 = 0$

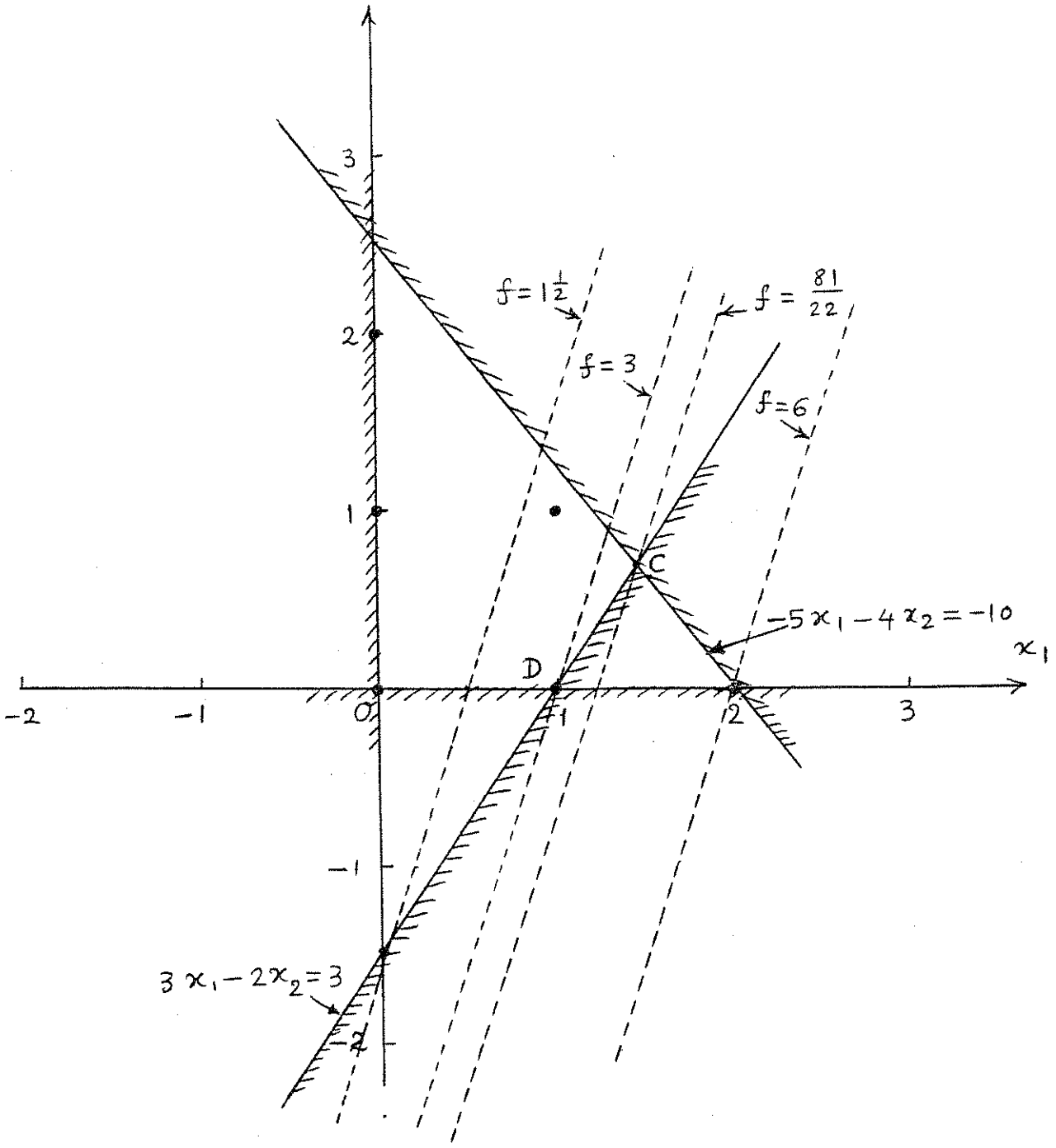
$= (\frac{16}{11}, \frac{15}{22})$

= continuous optimum point

$f_C = \frac{81}{22}$

Point D = integer optimum point $= (1, 0)$

$\vec{X}^* = \begin{Bmatrix} 1 \\ 0 \end{Bmatrix}$, $\quad f^* = 3$


3
2
1
−1
−2
−2
−1
O
1
2
3
x_1
$f = 1\frac{1}{2}$
$f = 3$
$f = \frac{81}{22}$
$f = 6$
C
D
$-5x_1 - 4x_2 = -10$
$3x_1 - 2x_2 = 3$

10.5 Maximize $f = 2x_1 + x_2$

subject to $g_1: \quad 8x_1 + 5x_2 - 15 \leq 0$

$x_1, x_2 \geq 0$, integers

$g_1 = 0$ at $(0,3)$, $\left(\frac{15}{8}, 0\right)$

Contours of $f = 2x_1 + x_2$:

$f = 3$ at $(0,3)$, $(3/2, 0)$

$f = 15/4$ at $(15/8, 0)$, $(0, 15/4)$

$f = 0$ at $(0,0)$

$f = 1$ at $(1,-1)$, $\left(\frac{1}{2}, 0\right)$

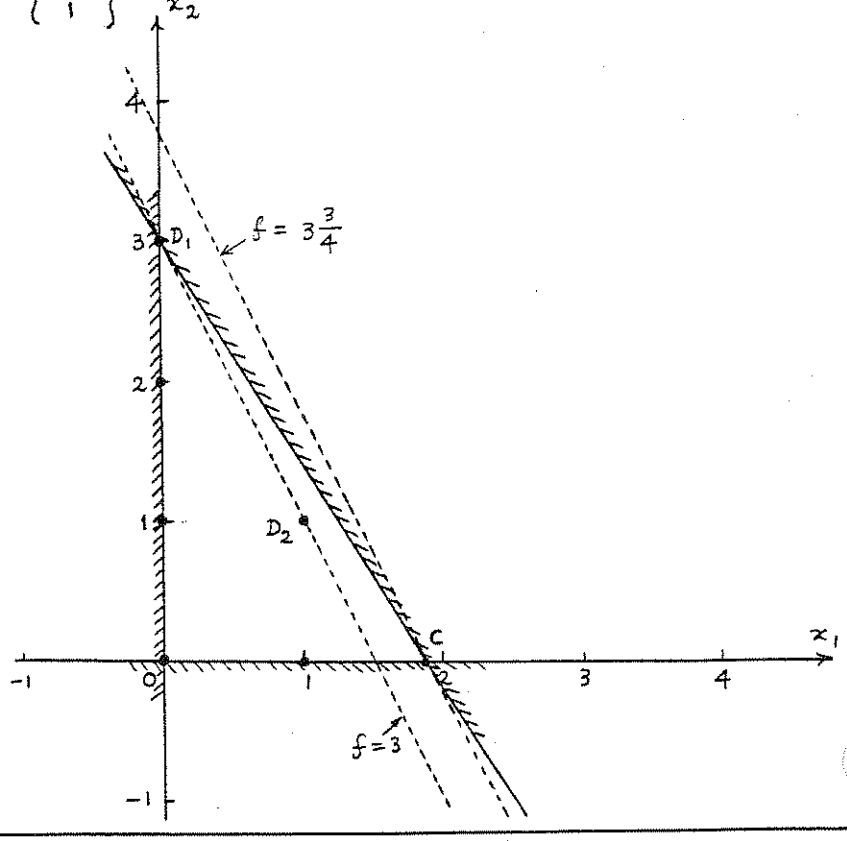
Point C = continuous optimum point = $\left(\frac{15}{8}, 0\right)$

$f_C = 3\frac{3}{4}$

Integer optimum point = D_1 or D_2

$\vec{X}^* = \begin{Bmatrix} 0 \\ 3 \end{Bmatrix}, \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$

$f^* = 3$



10.6 Maximize $f = 6x_1 + 7x_2$

subject to $g_1: \quad 7x_1 + 6x_2 - 42 \le 0$

$g_2: \quad 5x_1 + 9x_2 - 45 \le 0$

$g_3: \quad x_1 - x_2 - 4 \le 0$

$x_1, x_2 \ge 0$, integers

$g_1 = 0$ at $(0, 7)$, $(6, 0)$

$g_2 = 0$ at $(0, 5)$, $(9, 0)$

$g_3 = 0$ at $(0, -4)$, $(4, 0)$

Contours of $f = 6x_1 + 7x_2$:

$f = 49$ at $(0, 7)$, $(8\frac{1}{6}, 0)$

$f = 35$ at $(0, 5)$, $(5\frac{5}{6}, 0)$

$f = -28$ at $(0, -4)$, $(-4\frac{2}{3}, 0)$

$f = 24$ at $(4, 0)$, $(0, 3\frac{3}{7})$

Graphical optimization is shown in the figure.

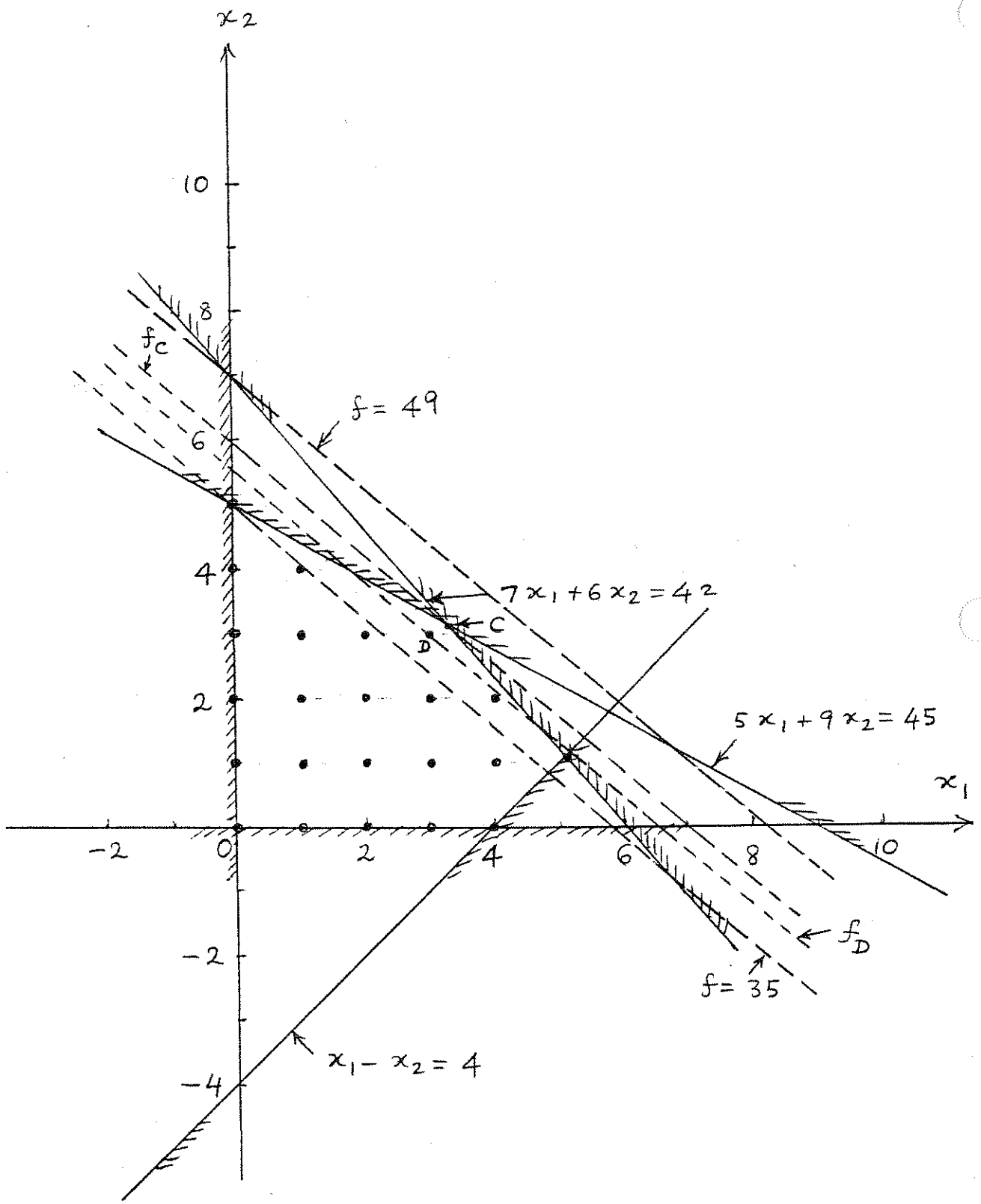
Point C = intersection of $g_1 = 0$ and $g_2 = 0$

$= \left(3\frac{3}{11}, 3\frac{2}{11}\right)$ = continuous optimum solution

$f_C = 41\frac{10}{11}$

Point D = integer optimum solution = $(3, 3)$

$\therefore \vec{x}^* = \begin{Bmatrix} 3 \\ 3 \end{Bmatrix}$, $f^* = 39$


x_2
10
8
6
4
2
−2
−4
f_C
$f = 49$
$7x_1 + 6x_2 = 42$
C
D
$5x_1 + 9x_2 = 45$
x_1
−2
0
2
4
6
8
10
f_D
$f = 35$
$x_1 - x_2 = 4$

10.7 Maximize $f = x_1 + 2x_2$

subject to $g_1: \quad x_1 + x_2 - 7 \le 0$

$g_2: \quad 2x_1 - 11 \le 0$

$g_3: \quad 2x_2 - 7 \le 0$

$x_1, x_2 \ge 0$, integers

Graphical optimization (shown in figure):

$g_1 = 0$ at $(0,7)$, $(7,0)$

$g_2 = 0$ at $(5\frac{1}{2}, 0)$

$g_3 = 0$ at $(0, 3\frac{1}{2})$

Contours of $f = x_1 + 2x_2$:

$f = 14$ at $(0,7)$, $(14,0)$

$f = 7$ at $(7,0)$, $(0, 3\frac{1}{2})$

Point C = intersection point of $g_1 = 0$ and $g_3 = 0$

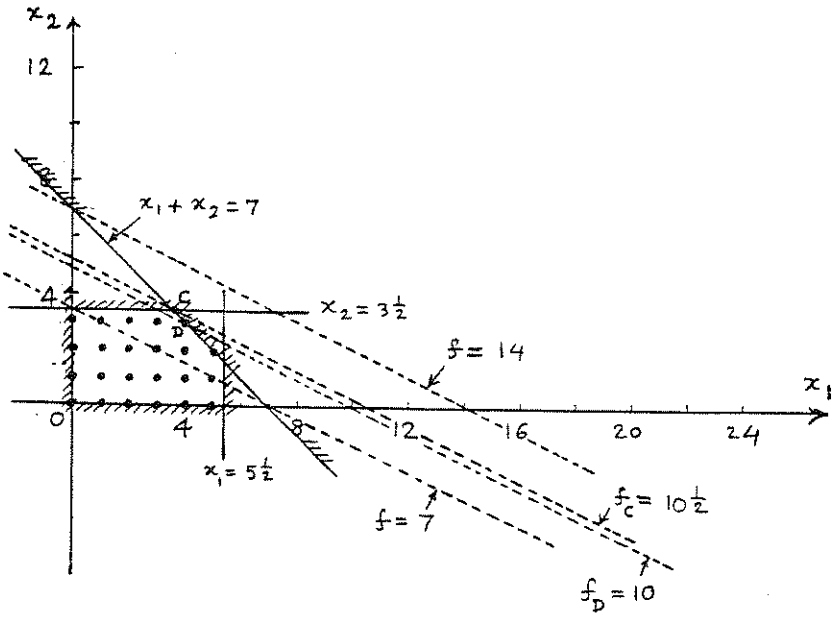
$= (3\frac{1}{2}, 3\frac{1}{2})$ = continuous optimum point

$f_C = 10\frac{1}{2}$

Point D = integer optimum point = (4,3)

$\therefore \vec{X}^* = \begin{Bmatrix} 4 \\ 3 \end{Bmatrix}$

$f^* = 10$



Gomory's cutting plane method:

By adding slack variables $x_3 \geq 0$, $x_4 \geq 0$ and $x_5 \geq 0$, the problem can be stated as:

$$\text{max.} \quad f = x_1 + 2x_2$$

subject to

$$x_1 \quad x_2 + x_3 = 7$$

$$2x_1 \quad + x_4 = 11$$

$$2x_2 \quad + x_5 = 7$$

$$x_i \geq 0, \quad i = 1, \ldots, 5$$

Initial basic feasible solution is:

$$\vec{x}_B = \begin{Bmatrix} x_3 \\ x_4 \\ x_5 \end{Bmatrix} = \begin{Bmatrix} 7 \\ 11 \\ 7 \end{Bmatrix}$$

Initial simplex tableau:

Basic variables	coefficients of variables							
	x_1	x_2	x_3	x_4	x_5	$-f$	$\bar{b}_i$	$\frac{\bar{b}_i}{\bar{a}_{is}}$ for $\bar{a}_{is} > 0$
x_3	1	1	1	0	0	0	7	7 ←
x_4	2	0	0	1	0	0	11	
x_5	0	2	0	0	1	0	7	7/2
$-f$	−1	−2	0	0	0	1	0	

↑ most negative

Simplex method gives the following solution:

x_1	1	0	0	0	$-\frac{1}{2}$	0	7/2
x_4	0	0	−2	1	1	0	4
x_2	0	1	0	0	$\frac{1}{2}$	0	7/2
$-f$	0	0	1	0	$\frac{1}{2}$	1	21/2

Continuous optimum solution:

$$\vec{x}_B = \begin{Bmatrix} x_1 \\ x_4 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} 7/2 \\ 4 \\ 7/2 \end{Bmatrix}$$

We arbitrarily select x_2 as the variable having the largest fractional value. Last row can be written as

$$\frac{7}{2} = 3 + \frac{1}{2} = x_2 + \frac{1}{2} x_5$$

Gomory constraint is $\frac{1}{2} x_5 \geq \frac{1}{2}$

or $s_1 = -\frac{1}{2} + \frac{1}{2} x_5$ (1)

Introducing Eq. (1), the tableau becomes:

	x_1	x_2	x_3	x_4	x_5	s_1	$\bar{b}_i$
x_1	1	0	1	0	$-\frac{1}{2}$	0	$\frac{7}{2}$
x_4	0	0	-2	1	1	0	4
x_2	0	1	0	0	$\frac{1}{2}$	0	$\frac{7}{2}$
s_1	0	0	0	0	$-\frac{1}{2}$	1	$-\frac{1}{2}$
$-f$	0	0	1	0	$\frac{1}{2}$	0	$\frac{21}{2}$

Since all $\bar{c}_j \geq 0$, and $s_1 < 0$, the solution is optimum, but infeasible. Hence we apply the dual simplex method to find the optimum feasible solution. We introduce x_5 and drop s_1 from basis to obtain the following result:

x_1	1	0	1	0	0	-1	4
x_4	0	0	-2	1	0	2	3
x_2	0	1	0	0	0	1	3
x_5	0	0	0	0	1	-2	1
$-f$	0	0	1	0	0	1	10

This gives the desired integer optimum solution as:

$x_1^* = 4$, $x_2^* = 3$, $f^* = 10$.

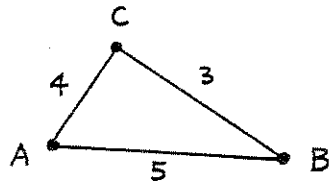
10.8 The recursive relations of Eq. (10.35) can be computed in tabular form as shown below:

k	$b_k=(b_{k-1}-q_{k-1})/2$	q_k
0	187	1
1	93	1
2	46	0
3	23	1
4	11	1
5	5	1
6	2	0
7	1	1

Hence $187=(1)\,2^7+(0)\,2^6+(1)\,2^5+(1)\,2^4+(1)\,2^3+(0)\,2^2+(1)\,2^1+(1)\,2^0$.

$\therefore\ 187 = (1\ 0\ 1\ 1\ 1\ 0\ 1\ 1)$

10.9 Let the capacities of pipes used between (A, B), (B, C) and (C, A) be x_1, x_2 and x_3, respectively.



Problem:

Minimize $f = k\,(x_1 + x_2 + x_3)$ where k = constant $= 1$ (for simplicity)

constraints:

$x_1 + x_3 \geq 3$ or $g_1: \quad x_1 + x_3 - 3 \geq 0$

$x_1 + x_2 = 2$ or 3 or $g_2: \quad x_1 + x_2 = 2$ or 3

$x_2 + x_3 = 2$ or 3 or $g_3: \quad x_2 + x_3 = 2$ or 3

$x_i \geq 0, \quad i = 1, 2, 3$

$x_i \leq 2, \quad i = 1, 2, 3$

$x_i = 0, 1, 2 \ ; \ i = 1, 2, 3$

All possible solutions are enumerated and the optimum solution found as:

$$\vec{X}^* = \begin{Bmatrix} 1 \\ 2 \\ 0 \end{Bmatrix}, \quad f^* = 3$$

x_1	x_2	x_3	g_1	g_2	g_3	f	
0.00	0.00	0.00	-3.00	0.00	0.00	0.00	
0.00	0.00	1.00	-3.00	0.00	1.00	1.00	
0.00	0.00	2.00	-3.00	0.00	2.00	2.00	
0.00	1.00	0.00	-2.00	1.00	1.00	1.00	
0.00	1.00	1.00	-2.00	1.00	2.00	2.00	
0.00	1.00	2.00	-2.00	1.00	3.00	3.00	
0.00	2.00	0.00	-1.00	2.00	2.00	2.00	
0.00	2.00	1.00	-1.00	2.00	3.00	3.00	
0.00	2.00	2.00	-1.00	2.00	4.00	4.00	
1.00	0.00	0.00	-2.00	1.00	0.00	1.00	
1.00	0.00	1.00	-2.00	1.00	1.00	2.00	
1.00	0.00	2.00	-2.00	1.00	2.00	3.00	
1.00	1.00	0.00	-1.00	2.00	1.00	2.00	
1.00	1.00	1.00	-1.00	2.00	2.00	3.00	
1.00	1.00	2.00	-1.00	2.00	3.00	4.00	
1.00	2.00	0.00	0.00	3.00	2.00	3.00	←
1.00	2.00	1.00	0.00	3.00	3.00	4.00	
1.00	2.00	2.00	0.00	3.00	4.00	5.00	
2.00	0.00	0.00	-1.00	2.00	0.00	2.00	
2.00	0.00	1.00	-1.00	2.00	1.00	3.00	
2.00	0.00	2.00	-1.00	2.00	2.00	4.00	
2.00	1.00	0.00	0.00	3.00	1.00	3.00	
2.00	1.00	1.00	0.00	3.00	2.00	4.00	
2.00	1.00	2.00	0.00	3.00	3.00	5.00	
2.00	2.00	0.00	1.00	4.00	2.00	4.00	
2.00	2.00	1.00	1.00	4.00	3.00	5.00	
2.00	2.00	2.00	1.00	4.00	4.00	6.00	

10.10 Minimize $f = 2x_1^2 + 3x_2^2 + 4x_1x_2 - 6x_1 - 3x_2$ (1)

subject to $x_1 + x_2 \le 1$ (2)

$2x_1 + 3x_2 \le 4$ (3)

$x_1, x_2 \ge 0$, integers (4)

Bounds on x_1 and x_2:

Eq.(2) $\Rightarrow$ $x_1 \le 1$, $x_2 \le 1$

Eq.(3) $\Rightarrow$ $x_1 \le 2$, $x_2 \le \frac{4}{3}$

Upper bounds: $x_1 \le 1$, $x_2 \le 1$

$\therefore$ Already the problem is in 0-1 form.

Suppress exponents to obtain:

Minimize $f = 2x_1 + 3x_2 + 4x_1x_2 - 6x_1 - 3x_2$ (5)

subject to $x_1 + x_2 \le 1$ (6)

$2x_1 + 3x_2 \le 4$ (7)

$x_1, x_2 = 0$ or 1 (8)

Define $x_3 = x_1 x_2$ (9)

with $x_1 + x_2 - x_3 \le 1$ (10)

and $-x_1 - x_2 + 2x_3 \le 0$ (11)

Final 0-1 linear problem is:

Minimize $f = 6x_1 + 3x_2 + 4x_3 - 6x_1 - 3x_2 = -4x_1 + 4x_3$

subject to

$$x_1 + x_2 \le 1$$

$$2x_1 + 3x_2 \le 4$$

$$x_1 + x_2 - x_3 \le 1$$

$$-x_1 - x_2 + 2x_3 \le 0$$

$$x_1, x_2, x_3 = 0 \text{ or } 1$$

10.11 Minimize $f = 6x_1 - x_2$ (1)

subject to

$3x_1 - x_2 \ge 4$ (2)

$2x_1 + x_2 \ge 3$ (3)

$-x_1 - x_2 \ge -3$ or $x_1 + x_2 \le 3$ (4)

x_1, x_2 = nonnegative integers (5)

Bounds on x_1 and x_2:

Eq. (4) $\Rightarrow$ $x_1 \le 3$, $x_2 \le 3$

Hence we represent x_1 and x_2 using binary variables as:

$$x_1 = y_{01} + 2y_{11}, \quad x_2 = z_{01} + 2z_{11} \quad (6)$$

where $y_{01}, y_{11}, z_{01}, z_{11} = 0$ or 1 (7)

Eqs. (1) to (6) lead to the final 0-1 LP problem:

Minimize $f = 6y_{01} + 12y_{11} - z_{01} - 2z_{11}$ (8)

subject to

$3y_{01} + 6y_{11} - z_{01} - z_{11} \ge 4$ (9)

$2y_{01} + 4y_{11} + z_{01} + 2z_{11} \ge 3$ (10)

$y_{01} + 2y_{11} + z_{01} + 2z_{11} \le 3$ (11)

$y_{01}, y_{11}, z_{01}, z_{11} = 0$ or 1 (12)

10.12 Maximize $f = -10x_1 - 5x_2 - 3x_3$

subject to $g_1 = x_1 + 2x_2 + x_3 \geq 4$

$g_2 = 2x_1 + x_2 + x_3 \leq 6$

$x_i = 0$ or 1 , $i = 1, 2, 3$

Solution by enumeration:

x_1	x_2	x_3	g_1	g_2	f	
0	0	0	0	0	0	
1	0	0	1	2	−10	
0	1	0	2	1	− 5	
0	0	1	1	1	− 3	
1	1	0	3	3	−15	
0	1	1	3	2	− 8	
1	0	1	2	3	−13	
1	1	1	4	4	−18	← only feasible point

$\therefore \vec{X}^* = \begin{Bmatrix} 1 \\ 1 \\ 1 \end{Bmatrix}$, $f^* = -18$

10.13 Minimize $f = -5x_1 + 7x_2 + 10x_3 - 3x_4 + x_5$

subject to $x_1 + 3x_2 - 5x_3 + x_4 + 4x_5 \leq 0$

$2x_1 + 6x_2 - 3x_3 + 2x_4 + 2x_5 \geq 4$

$x_2 - 2x_3 - x_4 + x_5 \leq -2$

$x_i = 0$ or 1 , $i = 1, \ldots, 5$

Solution by enumeration:

$$\vec{X}^* = \begin{Bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 0 \end{Bmatrix}$$

$g_1 \leq 0 \quad (g_1 = 0)$

$g_2 \leq 0 \quad (g_2 = -3)$

$g_3 \leq 0 \quad (g_3 = 0)$

$f^* = 9.0$

x_1	x_2	x_3	x_4	x_5	g_1	g_2	g_3	f
0.0	0.0	0.0	0.0	0.0	0.00	4.00	2.00	0.000
0.0	0.0	0.0	0.0	1.0	4.00	2.00	3.00	1.000
0.0	0.0	0.0	1.0	0.0	1.00	2.00	1.00	-3.000
0.0	0.0	0.0	1.0	1.0	5.00	0.00	2.00	-2.000
0.0	0.0	1.0	0.0	0.0	-5.00	7.00	0.00	10.000
0.0	0.0	1.0	0.0	1.0	-1.00	5.00	1.00	11.000
0.0	0.0	1.0	1.0	0.0	-4.00	5.00	-1.00	7.000
0.0	0.0	1.0	1.0	1.0	0.00	3.00	0.00	8.000
0.0	1.0	0.0	0.0	0.0	3.00	-2.00	3.00	7.000
0.0	1.0	0.0	0.0	1.0	7.00	-4.00	4.00	8.000
0.0	1.0	0.0	1.0	0.0	4.00	-4.00	2.00	4.000
0.0	1.0	0.0	1.0	1.0	8.00	-6.00	3.00	5.000
0.0	1.0	1.0	0.0	0.0	-2.00	1.00	1.00	17.000
0.0	1.0	1.0	0.0	1.0	2.00	-1.00	2.00	18.000
0.0	1.0	1.0	1.0	0.0	-1.00	-1.00	0.00	14.000
0.0	1.0	1.0	1.0	1.0	3.00	-3.00	1.00	15.000
1.0	0.0	0.0	0.0	0.0	1.00	2.00	2.00	-5.000
1.0	0.0	0.0	0.0	1.0	5.00	0.00	3.00	-4.000
1.0	0.0	0.0	1.0	0.0	2.00	0.00	1.00	-8.000
1.0	0.0	0.0	1.0	1.0	6.00	-2.00	2.00	-7.000
1.0	0.0	1.0	0.0	0.0	-4.00	5.00	0.00	5.000
1.0	0.0	1.0	0.0	1.0	0.00	3.00	1.00	6.000
1.0	0.0	1.0	1.0	0.0	-3.00	3.00	-1.00	2.000
1.0	0.0	1.0	1.0	1.0	1.00	1.00	0.00	3.000
1.0	1.0	0.0	0.0	0.0	4.00	-4.00	3.00	2.000
1.0	1.0	0.0	0.0	1.0	8.00	-6.00	4.00	3.000
1.0	1.0	0.0	1.0	0.0	5.00	-6.00	2.00	-1.000
1.0	1.0	0.0	1.0	1.0	9.00	-8.00	3.00	0.000
1.0	1.0	1.0	0.0	0.0	-1.00	-1.00	1.00	12.000
1.0	1.0	1.0	0.0	1.0	3.00	-3.00	2.00	13.000
1.0	1.0	1.0	1.0	0.0	0.00	-3.00	0.00	9.000
1.0	1.0	1.0	1.0	1.0	4.00	-5.00	1.00	10.000

10.16 Minimize $f = 4x_1 + 5x_2$

subject to $g_1 = 10x_1 + x_2 - 10 \geq 0$

$g_2 = 5x_1 + 4x_2 - 20 \geq 0$

$g_3 = 3x_1 + 7x_2 - 21 \geq 0$

$g_4 = x_1 + 12x_2 - 12 \geq 0$

$x_1 \geq 0$ and integer

$x_2 \geq 0$ (real)

Graphical solution:

$g_1 = 0$ at $(0, 10)$, $(1, 0)$

$g_2 = 0$ at $(0, 5)$, $(4, 0)$

$g_3 = 0$ at $(7, 0)$, $(0, 3)$

$g_4 = 0$ at $(0, 1)$, $(12, 0)$

Contours of $f = 4x_1 + 5x_2$:

$f = 50$ at $(0, 10)$, $(12\frac{1}{2}, 0)$

$f = 25$ at $(0, 5)$, $(6\frac{1}{4}, 0)$

$f = 15$ at $(0, 3)$, $(3\frac{3}{4}, 0)$

$f = 5$ at $(0, 1)$, $(1\frac{1}{4}, 0)$

Point B = intersection of $g_2 = 0$ and $g_3 = 0$

$= \left(\frac{168}{69}, \frac{45}{23}\right) = (2.43, 1.96)$

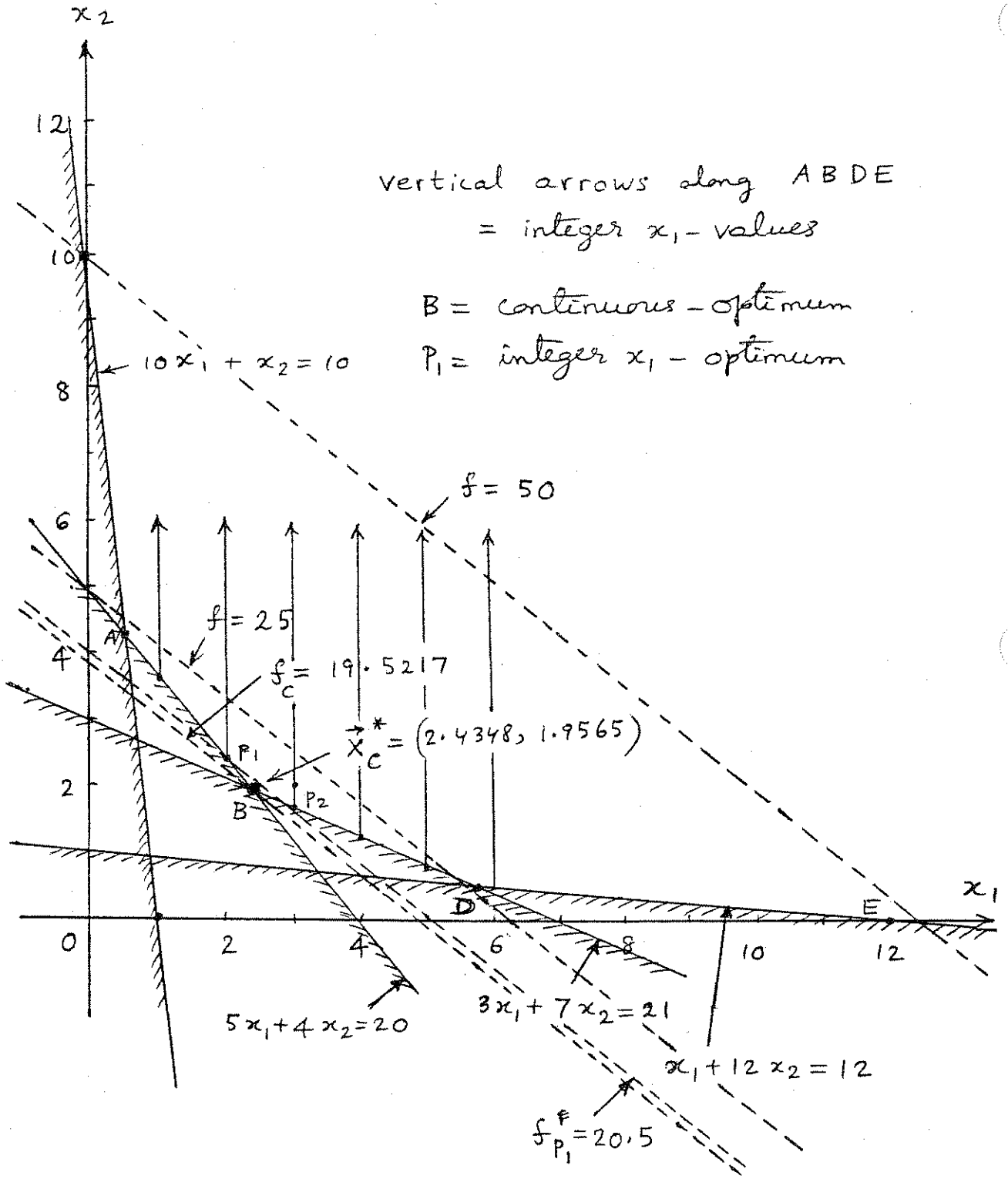
= continuous optimum point

$$f_B = 4\left(\frac{168}{69}\right) + 5\left(\frac{45}{23}\right) = 19.5217$$

$P_1 = \begin{Bmatrix} 2 \\ 2\frac{1}{2} \end{Bmatrix}$, $f_{P_1} = 20\frac{1}{2}$; $P_2 = \begin{Bmatrix} 3 \\ 12/7 \end{Bmatrix}$, $f_{P_2} = 20.5714$

Optimum point with x_1 integer = point P_1

$\therefore \quad \vec{X}^* = \begin{Bmatrix} 2 \\ 2\frac{1}{2} \end{Bmatrix}$, $f^* = 20\frac{1}{2}$


x_2
12
10
8
6
4
2
0
2
4
6
8
10
12
x_1
vertical arrows along A B D E
= integer x_1 - values
B = continuous - optimum
P_1 = integer x_1 - optimum
$10x_1 + x_2 = 10$
$f = 50$
$f = 25$
$f_c = 19.5217$
$\vec{x}_c^* = (2.4348, 1.9565)$
A
P_1
P_2
B
D
E
$5x_1 + 4x_2 = 20$
$3x_1 + 7x_2 = 21$
$x_1 + 12x_2 = 12$
$f_{P_1}^F = 20.5$

10.18 Maximize $f = x_1 x_2$ (1)

subject to $x_1^2 + x_2^2 \le 25$ (2)

$x_i \ge 0$, integer ; $i = 1, 2$ (3)

Bounds on x_i :

From Eq. (2): $x_i^2 \le 25$ or $x_i \le 5$, $i = 1, 2$ (4)

Since the smallest integer k satisfying the relation

$2^{k+1} \ge (5+1)$ is $k = 2$, we get (5)

$x_1 = y_{10} + 2\,y_{11} + 4\,y_{12}$, $x_2 = z_{10} + 2\,z_{11} + 4\,z_{12}$ (6)

where y_{ij} and z_{ij} can take values of 0 or 1.

Substitute Eqs. (5) and (6) into (1) and (2):

$$f = x_1 x_2 = (y_{10} + 2y_{11} + 4y_{12})(z_{10} + 2z_{11} + 4z_{12})$$

$$= y_{10}z_{10} + 2y_{10}z_{11} + 4y_{10}z_{12} + 2y_{11}z_{10} + 4y_{11}z_{11} + 8y_{11}z_{12} + 4y_{12}z_{10} + 8y_{12}z_{11} + 16y_{12}z_{12} \quad (7)$$

subject to

$$(y_{10} + 2y_{11} + 4y_{12})^2 + (z_{10} + 2z_{11} + 4z_{12})^2 \le 25$$

or

$$y_{10}^2 + 4y_{11}^2 + 16y_{12}^2 + 4y_{10}y_{11} + 16y_{11}y_{12} + 8y_{10}y_{12} + z_{10}^2 + 4z_{11}^2 + 16z_{12}^2 + 4z_{10}z_{11} + 16z_{11}z_{12} + 8z_{10}z_{12} \le 25 \quad (8)$$

Ignoring the exponents in Eq. (8), we get

$$y_{10} + 4y_{11} + 16y_{12} + 4y_{10}y_{11} + 16y_{11}y_{12} + 8y_{10}y_{12} + z_{10} + 4z_{11} + 16z_{12} + 4z_{10}z_{11} + 16z_{11}z_{12} + 8z_{10}z_{12} \le 25 \quad (9)$$

all y_{ij} and $z_{ij} = 0$ or 1

Define new variables for product terms as follows:

$$w_1 = y_{10}z_{10} \quad \text{with} \quad y_{10} + z_{10} - w_1 \le 1, \quad -y_{10} - z_{10} + 2w_1 \le 0 \quad (10)$$

$w_2 = y_{10}\, z_{11}$ with $y_{10} + z_{11} - w_2 \le 1$

$-y_{10} - z_{11} + 2w_2 \le 0$ (11)

$w_3 = y_{10}\, z_{12}$ with $y_{10} + z_{12} - w_3 \le 1$

$-y_{10} - z_{12} + 2w_3 \le 0$ (12)

$w_4 = y_{11}\, z_{10}$ with $y_{11} + z_{10} - w_4 \le 1$

$-y_{11} - z_{10} + 2w_4 \le 0$ (13)

$w_5 = y_{11}\, z_{11}$ with $y_{11} + z_{11} - w_5 \le 1$

$-y_{11} - z_{11} + 2w_5 \le 0$ (14)

$w_6 = y_{11}\, z_{12}$ with $y_{11} + z_{12} - w_6 \le 1$

$-y_{11} - z_{12} + 2w_6 \le 0$ (15)

$w_7 = y_{12}\, z_{10}$ with $y_{12} + z_{10} - w_7 \le 1$

$-y_{12} - z_{10} + 2w_7 \le 0$ (16)

$w_8 = y_{12}\, z_{11}$ with $y_{12} + z_{11} - w_8 \le 1$

$-y_{12} - z_{11} + 2w_8 \le 0$ (17)

$w_9 = y_{12}\, z_{12}$ with $y_{12} + z_{12} - w_9 \le 1$

$-y_{12} - z_{12} + 2w_9 \le 0$ (18)

$w_{10} = y_{10}\, y_{11}$ with $y_{10} + y_{11} - w_{10} \le 1$

$-y_{10} - y_{11} + 2w_{10} \le 0$ (19)

$w_{11} = y_{11}\, y_{12}$ with $y_{11} + y_{12} - w_{11} \le 1$

$-y_{11} - y_{12} + 2w_{11} \le 0$ (20)

$w_{12} = y_{10}\, y_{12}$ with $y_{10} + y_{12} - w_{12} \le 1$

$-y_{10} - y_{12} + 2w_{12} \le 0$ (21)

$w_{13} = z_{10}\, z_{11}$ with $z_{10} + z_{11} - w_{13} \le 1$

$-z_{10} - z_{11} + 2w_{13} \le 0$ (22)

$$w_{14} = z_{11} z_{12} \quad \text{with} \quad z_{11} + z_{12} - w_{14} \le 1$$
$$-z_{11} - z_{12} + 2 w_{14} \le 0 \qquad (23)$$

$$w_{15} = z_{10} z_{12} \quad \text{with} \quad z_{10} + z_{12} - w_{15} \le 1$$
$$-z_{10} - z_{12} + 2 w_{15} \le 0 \qquad (24)$$

$$w_{16} = y_{10}, \; w_{17} = y_{11}, \; w_{18} = y_{12}, \; w_{19} = z_{10},$$
$$w_{20} = z_{11}, \; w_{21} = z_{12} \qquad (25)$$

Using Eqs. (10) to (25), the final 0-1 LP problem [Eqs. (7) and (8)] can be stated as follows:

Maximize
$$f = w_1 + 2 w_2 + 4 w_3 + 2 w_4 + 4 w_5 + 8 w_6 + 4 w_7 + 8 w_8 + 16 w_9 \qquad (26)$$

subject to

$$w_{16} + 4 w_{17} + 16 w_{18} + 4 w_{10} + 16 w_{11} + 8 w_{12} + w_{19} + 4 w_{20} + 16 w_{21} + 4 w_{13} + 16 w_{14} + 8 w_{15} \le 25 \qquad (27)$$

and the inequalities of Eqs. (10) to (24) with all y_{ij}, z_{ij} and w_i are (0, 1)-variables.

10.19 Maximize $f = x_1 x_2$

subject to $g = x_1^2 + x_2^2 - 4 \le 0$

$x_1 \in \{0.1, 0.5, 1.1, 1.6, 2.0\}$

$x_2 \in \{0.4, 0.8, 1.5, 2.0\}$

$$\vec{X}^{(0)} = \begin{Bmatrix} 1.1 \\ 0.8 \end{Bmatrix}$$

$$f(\vec{X}^{(0)}) = 1.1\,(0.8) = 0.88$$

$$\nabla f(\vec{X}^{(0)}) = \begin{Bmatrix} x_2 \\ x_1 \end{Bmatrix}_{\vec{X}^{(0)}} = \begin{Bmatrix} 0.8 \\ 1.1 \end{Bmatrix}$$

$$g(\vec{X}^{(0)}) = (1.1)^2 + (0.8)^2 - 4 = -2.15$$

$$\nabla g(\vec{X}^{(0)}) = \begin{Bmatrix} 2x_1 \\ 2x_2 \end{Bmatrix}_{\vec{X}^{(0)}} = \begin{Bmatrix} 2.2 \\ 1.6 \end{Bmatrix}$$

Using only the current and adjacent higher and lower values for approximation, we can represent x_1 and x_2 as:

$$x_1 = y_{11}(0.5) + y_{12}(1.1) + y_{13}(1.6)$$

$$x_2 = y_{21}(0.4) + y_{22}(0.8) + y_{23}(1.5)$$

$$\delta x_1 = y_{11}(0.5 - 1.1) + y_{12}(1.1 - 1.1) + y_{13}(1.6 - 1.1) = -0.6\,y_{11} + 0.5\,y_{13}$$

$$\delta x_2 = y_{21}(0.4 - 0.8) + y_{22}(0.8 - 0.8) + y_{23}(1.5 - 0.8) = -0.4\,y_{21} + 0.7\,y_{23}$$

Approximate f and g can be expressed as:

Maximize
$$f \simeq 0.88 + \{0.8 \quad 1.1\} \begin{Bmatrix} -0.6\,y_{11} + 0.5\,y_{13} \\ -0.4\,y_{21} + 0.7\,y_{23} \end{Bmatrix}$$

$$\simeq 0.88 - 0.48\,y_{11} + 0.4\,y_{13} - 0.44\,y_{21} + 0.77\,y_{23}$$

subject to

$$g \simeq -2.15 + (2.2 \quad 1.6) \begin{Bmatrix} -0.6\,y_{11} + 0.5\,y_{13} \\ -0.4\,y_{21} + 0.7\,y_{23} \end{Bmatrix}$$

$$\simeq -2.15 - 1.32\,y_{11} + 1.1\,y_{13} - 0.64\,y_{21} + 1.12\,y_{23} \leq 0$$

and

$$y_{11} + y_{12} + y_{13} = 1$$

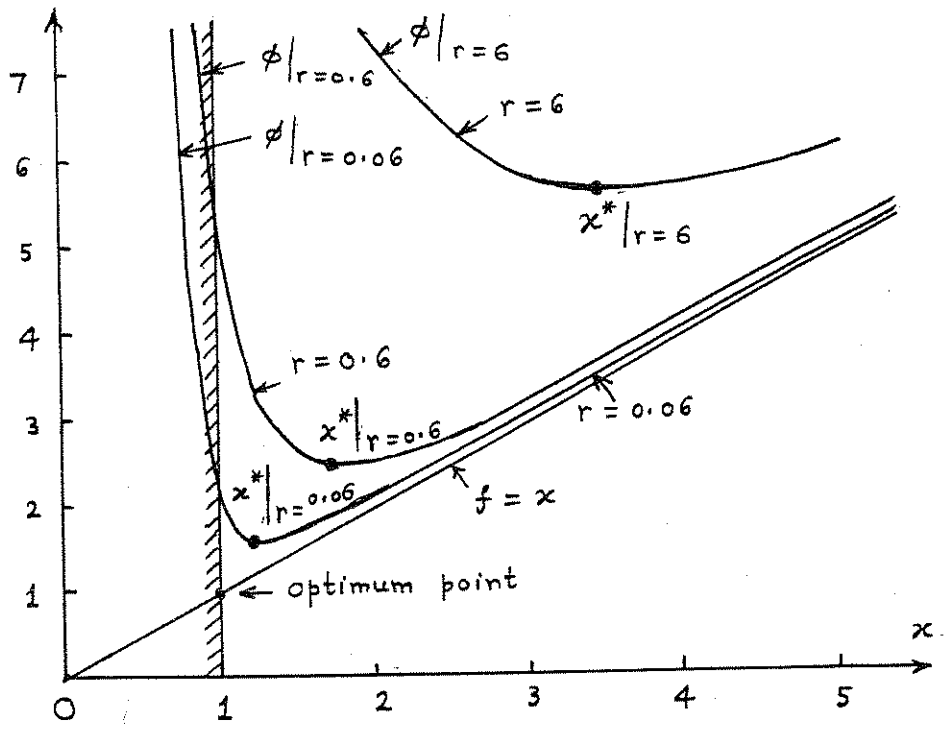
$$y_{21} + y_{22} + y_{23} = 1$$

$$y_{ij} = 0 \text{ or } 1 \quad ; \quad i = 1, 2 \; ; \; j = 1, 2, 3$$

10.20 Minimize $f = x$

subject to $x - 1 \geq 0$ with $x = \{1, 2, 3, \cdots\}$

(a) Continuous optimization:



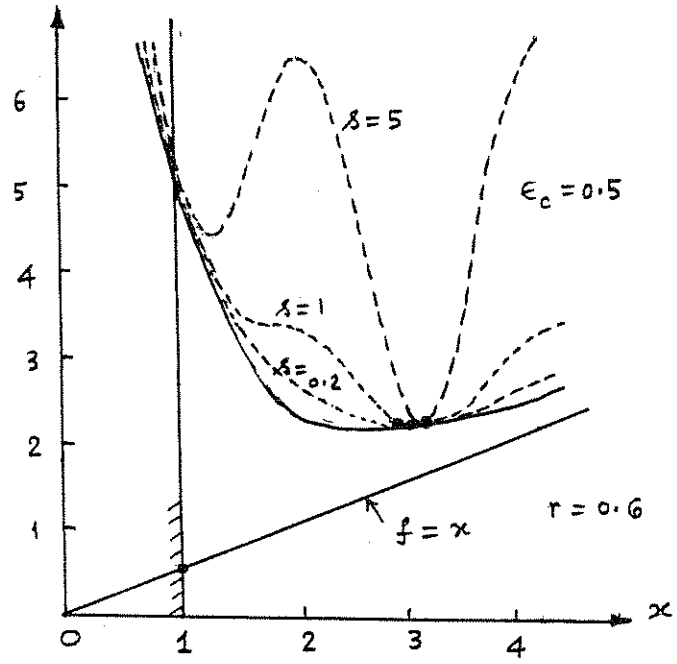
(b) Discrete optimization:

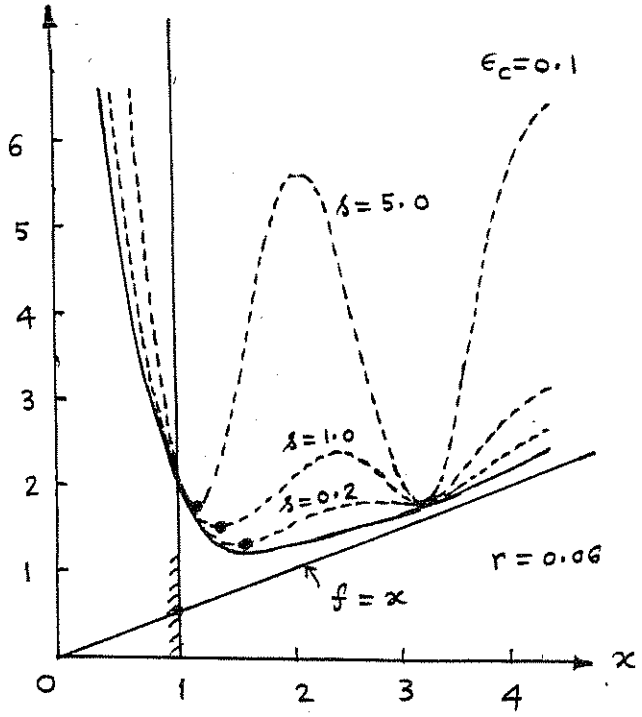
$$\phi - \text{function} = f(x) + r \sum_j p(g_j) + s \sum_i \Phi_d^i(x)$$

where $p(g_j) = \begin{cases} \dfrac{1}{g_j} & \text{for interior method} \\ \{\max(-g_j, 0)\}^2 & \text{for exterior method} \end{cases}$

$$\Phi_d^i(x) = \frac{1}{2} \sin\left(\frac{2\pi \left\{ x_i - \frac{1}{4}\left(d_{ij+1} + 3\, d_{ij} \right) \right\}}{d_{ij+1} - d_{ij}} + 1 \right)$$

with $d_{ij} < x_i < d_{ij+1}$.


6
5
4
3
2
1
0
1
2
3
4
x
δ = 5
ϵ_c = 0.5
δ = 1
δ = 0.2
f = x
r = 0.6


ϵ_c = 0.1
6
5
4
3
2
1
0
1
2
3
4
x
δ = 5.0
δ = 1.0
δ = 0.2
f = x
r = 0.06

10.21

M-file: Binary02.m

```
clear all; clc; close all;
f = [-1 -2 0 0 -2 -1 1 -3 2]';
A = [-1 1 0 0 0 0 0 0 0;...
   0 -1 1 0 0 0 0 0 0;...
   0 0 0 -1 1 0 0 0 0;...
   0 0 0 0 -1 1 0 0 0;...
   0 0 0 0 0 0 -1 1 0;...
   0 0 0 0 0 0 0 -1 1];
b = [0 0 0 0 0 0]';
Aeq = [1 1 1 1 1 1 1 1 1];
beq = 5;
[x,fval] = bintprog(f,A,b,Aeq,beq)
```

Output (Results):

```
Optimization terminated.

x =

     1
     1
     0
     1
     1
     1
     0
     0
     0

fval =

    -6
```

10.22

M-file: Binary03.m

```
clear all; clc; close all;
f = [-2 -3 -1 -4 -3 -2 -2 -1 -3]';
A = [0 -3 0 -1 -1 0 0 0 0;...
   1 1 0 0 0 0 0 0 0;...
   0 1 0 1 -1 -1 0 0 0;...
   0 -1 0 0 0 -2 -3 -1 -2;...
   0 0 -1 0 2 1 2 -2 1];

b = [-3 1 -1 -4 5]';
[x,fval] = bintprog(f,A,b,[])
```

Output (Results):

```
Optimization terminated.

x =

     0
     1
     1
     0
     1
     1
     1
     1
     1

fval =

   -15
```

Chapter 11
Stochastic Programming

11.1 Events: N = tractor functioning for a year with no breakdown

B = tractor not functioning due to breakdown in a year

X = number of tractors functioning at end of year

$P(N) = p = 0.8$, $P(B) = q = 0.2$

Probability mass function:

$$P(x=0) = p_x(0) = \binom{4}{0} p^0 q^{4-0} = \frac{4!}{4!\,0!}(0.8)^0(0.2)^4 = 0.0016$$

$$P(x=1) = p_x(1) = \binom{4}{1} p^1 q^{4-1} = \frac{4!}{3!\,1!}(0.8)^1(0.2)^3 = 0.0256$$

$$P(x=2) = p_x(2) = \binom{4}{2} p^2 q^{4-2} = \frac{4!}{2!\,2!}(0.8)^2(0.2)^2 = 0.1536$$

$$P(x=3) = p_x(3) = \binom{4}{3} p^3 q^{4-3} = \frac{4!}{1!\,3!}(0.8)^3(0.2)^1 = 0.4096$$

$$P(x=4) = p_x(4) = \binom{4}{4} p^4 q^{4-4} = \frac{4!}{0!\,4!}(0.8)^4(0.2)^0 = 0.4096$$

Probability distribution function:

$F_x(x) = P(X \le x)$

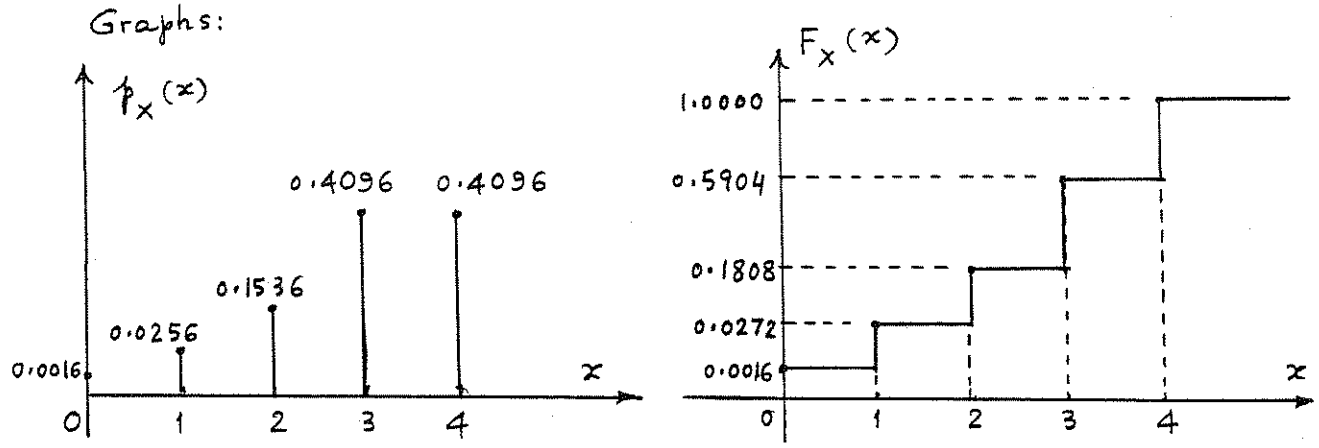
$F_x(x=0) = 0.0016$, $F_x(x=1) = 0.0016 + 0.0256 = 0.0272$

$F_x(x=2) = 0.0272 + 0.1536 = 0.1808$

$F_X(x=3) = 0.1808 + 0.4096 = 0.5904$

$F_X(x=4) = 0.5904 + 0.4096 = 1.0000$

Graphs:



11.2

$$f_V(v) = \frac{4h^3}{\sqrt{\pi}} v^2 e^{-h^2 v^2} \quad \text{with} \quad h^2 = \frac{m}{2kT} \quad \text{and} \quad v \geq 0$$

$$\overline{V} = \int_0^\infty v \, f_V(v) \, dv = c \int_0^\infty v^3 e^{-kv^2} \, dv = \frac{c}{2h^4} \qquad (1)$$

where $c = \frac{4h^3}{\sqrt{\pi}}$ and $k = h^2$.

$$\sigma_V^2 = \int_0^\infty (v - \overline{V})^2 f_V(v) \, dv$$

$$= c \int_0^\infty v^4 e^{-h^2 v^2} \, dv - 2c\overline{V} \int_0^\infty v^3 e^{-h^2 v^2} \, dv + c\overline{V}^2 \int_0^\infty v^2 e^{-h^2 v^2} \, dv$$

$$= c \left(\frac{1 * 3}{2^3 h^4} \sqrt{\frac{\pi}{h^2}} \right) - 2c\overline{V} \left(\frac{1}{2h^4} \right) + c\overline{V}^2 \left(\frac{\sqrt{\pi}}{4h^3} \right)$$

$$= c \cdot \frac{\sqrt{\pi}}{h^3} \left(\frac{3}{8h^2} - \frac{\overline{V}}{\sqrt{\pi} h} + \frac{\overline{V}^2}{4} \right) \qquad (2)$$

11.3

$$\bar{X} = \sum_{x_i=0}^{4} x_i \, p_X(x_i) = 0(0.0016) + 1(0.0256) + 2(0.1536) + 3(0.4096) + 4(0.4096) = 3.2$$

$$\sigma_X^2 = E\left[(X-\bar{X})^2\right] = \sum_{x_i=0}^{4} (x_i - \bar{X})^2 \, p_X(x_i)$$

$$= (0-3.2)^2(0.0016) + (1-3.2)^2(0.0256) + (2-3.2)^2(0.1536) + (3-3.2)^2(0.4096) + (4-3.2)^2(0.4096) = 0.64$$

$$\therefore \sigma_X = 0.8$$

11.4

Normalization:

$$\int_{-\infty}^{\infty} f_X(x)\,dx = \int_{0.9}^{1.1} a\,(x-0.9)(1.1-x)\,dx = 1$$

i.e., $$a \int_{0.9}^{1.1} (2x - 0.99 - x^2)\,dx = 1$$

i.e., $$a\left(x^2 - 0.99x - \frac{x^3}{3}\right)_{0.9}^{1.1} = 1$$

i.e., $$a\left[1.21 - 1.089 - 0.4437 - 0.81 + 0.891 + 0.243\right] = 1$$

i.e., $$a = 769.2308$$

Mean value:

$$\mu_X = \int_{-\infty}^{\infty} x \cdot f_X(x) \cdot dx = a \int_{0.9}^{1.1} (2x^2 - 0.99x - x^3)\,dx$$

$$= a\left\{\frac{2}{3}x^3 - \frac{0.99}{2}x^2 - \frac{1}{4}x^4\right\}_{0.9}^{1.1} = 0.0013\,a = 1$$

Standard deviation:

$$\sigma_X^2 = \int_{-\infty}^{\infty} (x-1)^2 f_X(x)\, dx = a \int_{0.9}^{1.1} (x^2 - 2x + 1)(2x - 0.99 - x^2)\, dx$$

$$= a \int_{0.9}^{1.1} (-x^4 + 4x^3 - 5.99x^2 + 3.98x - 0.99)\, dx$$

$$= a \left\{ -\frac{1}{5}x^5 + \frac{4}{4}x^4 - \frac{5.99}{3}x^3 + \frac{3.98}{2}x^2 - 0.99x \right\}_{0.9}^{1.1}$$

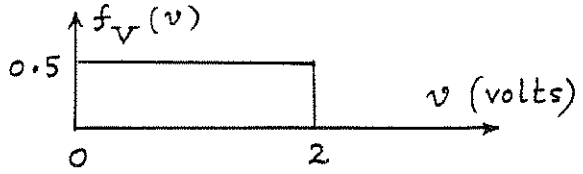
$$= a\,(3.0 * 10^{-6}) = 0.00230769$$

$$\therefore \sigma_X = 0.048038$$

11.5 (a) Y = power = $\frac{V^2}{R}$

where R = constant

and V = random (uniformly distributed)



$$F_Y(y) = P\left[\frac{V^2}{R} \le y\right] = \int_{-\sqrt{yR}}^{+\sqrt{yR}} f_V(v)\, dv$$

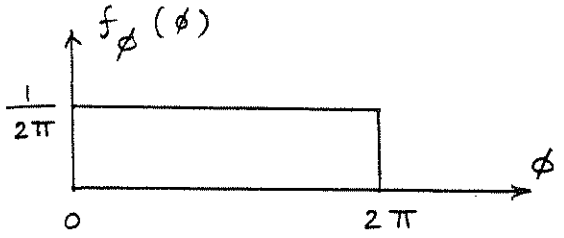
$$= \int_{-\sqrt{yR}}^{+\sqrt{yR}} (0.5)\, dv = \sqrt{Ry}$$

$$f_Y(y) = \frac{\partial}{\partial y}\left[F_Y(y)\right] = \sqrt{R}\ \frac{1}{2}\ \frac{1}{\sqrt{y}} = \frac{1}{2}\sqrt{\frac{R}{y}}\ ; \quad 0 < y \le \frac{4}{R}$$

(b) $V = A\cos(\omega t + \phi)$

A, ω, t = constants

ϕ = random (uniformly distributed)



$$\cos(\omega t + \phi) = \frac{V}{A}$$

$$\omega t + \phi = \cos^{-1}\left(\frac{V}{A}\right) \quad \text{or} \quad \phi = \cos^{-1}\left(\frac{V}{A}\right) - \omega t$$

$$F_\phi(\tilde{\phi}) = \frac{\tilde{\phi}}{2\pi}\ ; \quad 0 \le \tilde{\phi} \le 2\pi$$

$$F_V(v) = P\{A\cos(\omega t + \phi) < v\}$$

$$= P\left\{\phi < \left[\cos^{-1}\left(\frac{v}{A}\right) - \omega t\right]\right\}$$

$$= F_\phi\left\{\cos^{-1}\left(\frac{v}{A}\right) - \omega t\right\}$$

$$= \frac{1}{2\pi}\left\{\cos^{-1}\left(\frac{v}{A}\right) - \omega t\right\}$$

$$f_V(v) = \frac{d}{dv}\{F_V(v)\} = \frac{1}{2\pi}\left\{-\frac{1}{\sqrt{1-\left(\frac{v}{A}\right)^2}}\cdot\frac{1}{A}\right\}\ ;$$

$$0 \le \cos^{-1}\left(\frac{v}{A}\right) \le \pi$$

11.6 $H = cV^2$ where $c = \dfrac{fL}{2gD}$

$$f_V(v) = \begin{cases} \frac{1}{V_0}e^{-kv} & \text{for } v \ge 0 \quad \left(k = \frac{1}{V_0}\right) \\ 0 & \text{for } v < 0 \end{cases}$$

$$V = \pm\sqrt{\frac{H}{c}}\ , \quad \left|\frac{dV}{dH}\right| = \frac{1}{2\sqrt{Hc}}$$

$$f_H(h) = \left\{f_V\left(\sqrt{\frac{h}{c}}\right) + f_V\left(-\sqrt{\frac{h}{c}}\right)\right\}\frac{1}{2\sqrt{hc}}$$

But $f_V(v) = \frac{1}{V_0}e^{-kv}$

$$f_H(h) = \left\{ \frac{1}{V_0} e^{-k\sqrt{\frac{h}{c}}} + \frac{1}{V_0} e^{+k\sqrt{\frac{h}{c}}} \right\} \cdot \frac{1}{2\sqrt{hc}}$$

But $$k\sqrt{\frac{h}{c}} = \frac{1}{V_0}\sqrt{\frac{h}{c}} = \frac{\sqrt{2gD}\sqrt{h}}{\sqrt{fL}\ V_0}$$

$$\sqrt{hc} = \sqrt{h}\ \frac{\sqrt{fL}}{\sqrt{2gD}} = \frac{\sqrt{hfL}}{\sqrt{2gD}}$$

$$f_H(h) = \frac{1}{V_0}\sqrt{\frac{gD}{2hfL}} \left\{ e^{-\frac{1}{V_0}\sqrt{\frac{2gDh}{fL}}} + e^{+\frac{1}{V_0}\sqrt{\frac{2gDh}{fL}}} \right\};$$

for $h \geq 0$

11.7 $$f_{X,Y}(x,y) = \begin{cases} 3x^2y + 3xy^2 & \text{for } 0 \leq x \leq 1,\ 0 \leq y \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

Normalization (verification):

$$\int_0^1 \int_0^1 f_{X,Y}(x,y)\,dx\,dy = \int_{x=0}^{1} 3x^2\,dx \left(\frac{y^2}{2}\right)_0^1 + \int_{x=0}^{1} 3x\,dx \left(\frac{y^3}{3}\right)_0^1 = \frac{3}{2}\int_0^1 x^2\,dx + \int_0^1 x\,dx$$

$$= \frac{3}{2}\left(\frac{x^3}{3}\right)_0^1 + \left(\frac{x^2}{2}\right)_0^1 = 1$$

Marginal density function of X:

$$\int_{y=0}^{1} (3x^2y + 3xy^2)\,dy = 3x^2\left(\frac{y^2}{2}\right)_0^1 + 3x\left(\frac{y^3}{3}\right)_0^1$$

$$= 3x^2\left(\frac{1}{2}\right) + x = x + \frac{3}{2}x^2$$

Marginal density function of Y:

$$\int_{x=0}^{1} (3x^2y + 3xy^2)\,dy = 3y\left(\frac{x^3}{3}\right)_0^1 + 3y^2\left(\frac{x^2}{2}\right)_0^1$$

$$= y + \frac{3}{2}y^2$$

11.8 (a) X = diameter of rods

$P(X \le 2.99) = 0.05$ (undersize)

$$P\left(\frac{X-\bar{X}}{\sigma_X} \le \frac{2.99-3.00}{\sigma_X}\right) = 0.05$$

$P(Z \le z_1) = 0.05 \Rightarrow z_1 \simeq -1.645$

i.e., $\dfrac{2.99-3.00}{\sigma_X} \simeq -1.645$

$\therefore \sigma_X = 0.006079$ cm

(b) Undersize rejects:

$$P\left(Z \le \frac{2.985-3.00}{0.006079}\right) = P(Z \le z_1) \text{ with } z_1 = -2.4675$$

Rejects $= 1 - 0.9934 = 0.0066 = 0.66\%$

Similarly, oversize rejects $= 0.66\%$

$\therefore$ Total rejects $= 1.32\%$

11.9 $$f_{X,Y}(x,y) = \begin{cases} 4xy & \text{if } 0 \le x \le 1,\ 0 \le y \le 1 \\ 0 & \text{elsewhere} \end{cases}$$

Normalization (verification):

$$\int_0^1 \int_0^1 f_{X,Y}(x,y)\,dx\,dy = \int_0^1 4x\,dx \left(\frac{y^2}{2}\right)_0^1 = \int_0^1 2x\,dx = 1$$

Marginal density functions:

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)\,dy = \int_{y=0}^{1} 4xy\,dy = 4x\left(\frac{y^2}{2}\right)_0^1 = 2x$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\,dy = \int_{x=0}^{1} 4y\,x\,dx = 4y\left(\frac{x^2}{2}\right)_0^1 = 2y$$

Since $f_{X,Y}(x,y) = f_X(x) \cdot f_Y(y)$,

X and Y are independent.

11.10 $f_{X,Y}(x,y) = \begin{cases} \frac{1}{4\pi^2}\{1-\sin(x+y)\} ; & -\pi \le x, y \le \pi \\ 0 & \text{elsewhere} \end{cases}$

$$f_X(x) = \int_{y=-\pi}^{\pi} \frac{1}{4\pi^2}\{1-\sin(x+y)\}\,dy$$

$$= \frac{1}{4\pi^2}\{y + \cos(x+y)\}_{y=-\pi}^{\pi} = \frac{1}{4\pi^2}(2\pi) = \frac{1}{2\pi}$$

$$f_Y(y) = \int_{x=-\pi}^{\pi} \frac{1}{4\pi^2}\{1-\sin(x+y)\}\,dx$$

$$= \frac{1}{4\pi^2}\{x + \cos(x+y)\}_{x=-\pi}^{\pi} = \frac{1}{4\pi^2}(2\pi) = \frac{1}{2\pi}$$

Since $f_{X,Y}(x,y) \neq f_X(x) \cdot f_Y(y)$,

X and Y are dependent.

11.11 X = yield stress, $\mu_X = \bar{X} = 4000$ kg$_f$/cm^2, $\sigma_X = 300$ kg$_f$/cm^2

(a) $P(3000 \le X \le 5000) = F_X(5000) - F_X(3000)$

$= F_Z(z_2) - F_Z(z_1)$

where Z = standard normal variate

$z_1 = \frac{3000-4000}{300} = -3.33$, $z_2 = \frac{5000-4000}{300} = +3.33$

$F_Z(z_1) = 0.00048$, $F_Z(z_2) = 0.99952$

$\therefore P(3000 \le X \le 5000) = 0.99904$

(b) $P(X \ge 4500) = P(Z \ge z_3)$ where $z_3 = \frac{4500-4000}{300} = 1.67$

$\therefore P(X \ge 4500) = 0.0475$

(c) Value of X at which $F_X(x) = 0.1$:

$F_Z(z) = 0.1$ when $z = -1.28 = (X-4000)/300$

$\therefore X = 4000 - 300(1.28) = 3,616$ kg$_f$/cm^2.

11.12

$$P(x=d) = p_x(d) = \frac{e^{-2}\,2^d}{d!} \quad ; \; d = 0, 1, 2, \ldots$$

$$p_x(0) = \frac{e^{-2}\,2^0}{0!} = e^{-2}$$

$$p_x(1) = e^{-2}\,2^1/(1!) = 2e^{-2}$$

$$p_x(2) = e^{-2}\,2^2/(2!) = 2e^{-2}$$

$$P(x \le 2) = p_x(0) + p_x(1) + p_x(2) = e^{-2}(1+2+2) = \frac{5}{e^2}$$

$$= 0.6767$$

11.13

$$R = \frac{V_0^2}{g}\sin 2\phi$$

Random variables: V_0, ϕ

$\bar{V}_0 = 100$ ft/sec, $\sigma_{V_0} = 10$ ft/sec

$\bar{\phi} = 30^\circ$, $\sigma_\phi = 3^\circ$

$$\frac{\partial R}{\partial V_0} = \frac{2V_0}{g}\sin 2\phi$$

$$\frac{\partial R}{\partial \phi} = \frac{2V_0^2}{g}\cos 2\phi$$

$$\left.\frac{\partial R}{\partial V_0}\right|_{mean} = \frac{2(100)\sin 60^\circ}{32.2} = 5.3790$$

$$\left.\frac{\partial R}{\partial \phi}\right|_{mean} = \frac{2(100)^2}{32.2}\cos 60^\circ = 310.5590$$

Taylor's series expansion:

$$R \simeq R(\bar{V}_0, \bar{\phi}) + \left.\frac{\partial R}{\partial V_0}\right|_{mean}(V_0 - \bar{V}_0) + \left.\frac{\partial R}{\partial \phi}\right|_{mean}(\phi - \bar{\phi})$$

$$+ \frac{1}{2}\left.\frac{\partial^2 R}{\partial V_0^2}\right|_{mean}(V_0 - \bar{V}_0)^2 + \frac{1}{2}\left.\frac{\partial^2 R}{\partial \phi^2}\right|_{mean}(\phi - \bar{\phi})^2$$

First order mean:

$$\bar{R} = \frac{\bar{V}_0^2}{g}\sin 2\bar{\phi} = \frac{(100)^2\sin 60^\circ}{32.2} = 268.9520 \text{ ft}$$

First order standard deviation:

$$\sigma_R^2 = \left(\left.\frac{\partial R}{\partial \bar{V}_0}\right|_{mean}\right)^2 \sigma_{V_0}^2 + \left(\left.\frac{\partial R}{\partial \phi}\right|_{mean}\right)^2 \sigma_\phi^2$$

$$= (5.379)^2(10)^2 + (310.559)^2\left(3/57.2956\right)^2 = 3157.7804$$

$$\sigma_R = 56.1941 \text{ ft}$$

Second order mean:

$$\bar{R} = \frac{\bar{V}_o^2}{g} \sin 2\bar{\phi} + \frac{1}{2} \left.\frac{\partial^2 R}{\partial V_o^2}\right|_{mean} \sigma_{V_o}^2 + \frac{1}{2} \left.\frac{\partial^2 R}{\partial \phi^2}\right|_{mean} \sigma_{\phi}^2$$

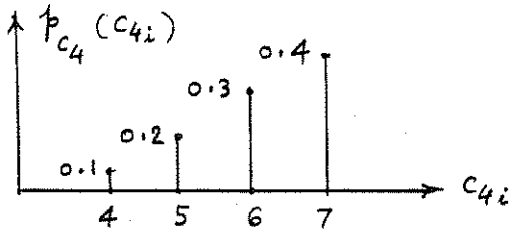
Here $$\left.\frac{\partial^2 R}{\partial V_o^2}\right|_{mean} = \frac{2}{g} \sin 2\bar{\phi} = \frac{2}{32.2} \sin 60^\circ = 0.0538$$

$$\left.\frac{\partial^2 R}{\partial \phi^2}\right|_{mean} = -\frac{4 V_o^2}{g} \sin 2\bar{\phi} = -\frac{4(100)^2 \sin 60^\circ}{32.2} = -1075.808$$

$$\therefore \bar{R} = 268.9520 + \frac{1}{2}(0.0538)(10^2) + \frac{1}{2}(-1075.808)\left(\frac{3}{57.2956}\right)^2$$

$$= 270.1673 \text{ ft}$$

11.14 C_4 = discrete random variable



$$\bar{C}_4 = \sum_i c_{4i}\, p_{C_4}(c_{4i})$$

$$= 4(0.1) + 5(0.2) + 6(0.3) + 7(0.4)$$

$$= 6.0$$

Expected value of $f = 4x_1 + 2x_2 + 3x_3 + 6x_4$

LP problem becomes:

Maximize $f = 4x_1 + 2x_2 + 3x_3 + 6x_4$

subject to $x_1 + x_3 + x_4 \le 24$

$$3x_1 + x_2 + 2x_3 + 4x_4 \le 48$$

$$2x_1 + 2x_2 + 3x_3 + 2x_4 \le 36$$

$$x_i \ge 0, \quad i = 1, 2, 3, 4$$

Solution is: $$\vec{X}^* = \begin{Bmatrix} x_1^* \\ x_2^* \\ x_3^* \\ x_4^* \end{Bmatrix} = \begin{Bmatrix} 0 \\ 8 \\ 0 \\ 10 \end{Bmatrix}, \quad f^* = 76.0$$

11.15 Variance of $c_4 = \sigma_{c_4}^2 = \sum_i (c_{4i} - \bar{c}_4)^2 \, p_{c_4}(c_{4i})$

$= (4-6)^2 (0.1) + (5-6)^2 (0.2) + (6-6)^2 (0.3) + (7-6)^2 (0.4)$

$= 1.0$

Variance of $f = 1.0\, x_4^2 \Rightarrow$ Maximize $\sqrt{\text{Var}(f)} = x_4$

LP problem becomes:

Maximize $f = x_4$

subject to $x_1 + x_3 + x_4 \leq 24$

$3x_1 + x_2 + 2x_3 + 4x_4 \leq 48$

$2x_1 + 2x_2 + 3x_3 + 2x_4 \leq 36$

$x_i \geq 0, \; i = 1, 2, 3, 4$

Solution is:

$$\vec{x}^* = \begin{Bmatrix} x_1^* \\ x_2^* \\ x_3^* \\ x_4^* \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 12 \end{Bmatrix}, \quad f^* = 12$$

11.16 Let x = number of units produced in a month

y = demand (number of units) in a month

f = profit = y * unit revenue $-$ x * unit cost of production

$= 1000\, y - 400\, x$

Let $x = 1$. E(profit) $= 0.2\,(1000 - 400) = \$120$

where 0.2 = probability that demand equals to 1

Let $x = 2$. E(profit) $= 0.2\,(2000 - 800) = \$240$

where 0.2 = probability that demand equals to 2

Let $x = 3$. E(profit) $= 0.6\,(3000 - 1200) = \$1080$

where 0.6 = probability that demand equals to 3

$\therefore$ Output to maximize the expected total profit = 3 units

---

11.17 Let x_1, x_2 and x_3 be the amounts (number of units) of products A, B and C produced per day, respectively.

Maximize: profit $= f = c_1 x_1 + c_2 x_2 + c_3 x_3$ (1)

subject to

$$\left.\begin{aligned} a_{11} x_1 + a_{12} x_2 + a_{13} x_3 &\le b_1 \\ a_{21} x_1 + a_{22} x_2 + a_{23} x_3 &\le b_2 \\ a_{31} x_1 + a_{32} x_2 + a_{33} x_3 &\le b_3 \end{aligned}\right\} \quad (2)$$

where c_i = profit per unit of product i ($i = 1 = A$, $i = 2 = B$, $i = 3 = C$), and a_{ij} = time required at stage i to produce 1 unit of product j.

Here all c_j, a_{ij} and b_i are random variables.

objective function:

$$\bar{f} = \text{mean} = \sum_{j=1}^{3} \bar{c}_j x_j = 6 x_1 + 4 x_2 + 10 x_3 \quad (3)$$

$$\sigma_f = \text{standard deviation} = \sqrt{\vec{X}^T [V] \vec{X}}$$

$$= \left(x_1^2 \sigma_{c_1}^2 + x_2^2 \sigma_{c_2}^2 + x_3^2 \sigma_{c_3}^2 \right)^{\frac{1}{2}} = \sqrt{4 x_1^2 + x_2^2 + 9 x_3^2} \quad (4)$$

Constraints:

$$h_i = \sum_{j=1}^{3} a_{ij} x_j - b_i \quad (5)$$

$$\left.\begin{aligned} \bar{h}_1 &= 4 x_1 + 8 x_2 + 4 x_3 - 1720 \\ \bar{h}_2 &= 12 x_1 + 0 \cdot x_2 + 8 x_3 - 1840 \\ \bar{h}_3 &= 4 x_1 + 16 x_2 + 0 \cdot x_3 - 1680 \end{aligned}\right\} \quad (6)$$

$$\sigma_{h_i}^2 = \sum_{j=1}^{3} x_j^2 \sigma_{a_{ij}}^2 + \sigma_{b_i}^2 \quad (7)$$

$$\left.\begin{aligned} \sigma_{h_1}^2 &= x_1^2 + 9 x_2^2 + 16 x_3^2 + (172)^2 \\ \sigma_{h_2}^2 &= 4 x_1^2 + 0 \cdot x_2^2 + 4 x_3^2 + (276)^2 \\ \sigma_{h_3}^2 &= 4 x_1^2 + 16 x_2^2 + 0 \cdot x_3^2 + (336)^2 \end{aligned}\right\} \quad (8)$$

Probabilistic constraints:

$$\bar{h}_i + e_i \sigma_{h_i} \leq 0 \quad \text{with} \quad e_i = 1.645 \qquad (9)$$

PROBLEMS:

(a) Maximize the expected profit:

Find $\vec{x} = \{x_1, x_2, x_3\}^T$

to maximize $\bar{f} = 6x_1 + 4x_2 + 10x_3$

subject to $\bar{h}_i + 1.645\, \sigma_{h_i} \leq 0 \quad ; \quad i = 1,2,3$

and $\vec{x} \geq \vec{0}$

where $\bar{h}_i$ and σ_{h_i} are given by Eqs. (6) and (8).

Solution of this NLP problem:

$$\vec{x}_{start} = \begin{Bmatrix} 1.0 \\ 1.0 \\ 1.0 \end{Bmatrix}, \quad \vec{x}^* = \begin{Bmatrix} 0.0 \\ 36.9262 \\ 174.4011 \end{Bmatrix},$$

$$\vec{g}(\vec{x}^*) = \begin{Bmatrix} 0 \\ -0.0001831 \\ -722.1475 \end{Bmatrix}, \quad \bar{f}^* = \$\ 1,891.7162$$

(b) Maximize the standard deviation of profit:

Find $\vec{x} = \{x_1, x_2, x_3\}^T$

to maximize $\sigma_f = \sqrt{4x_1^2 + x_2^2 + 9x_3^2}$

subject to $\bar{h}_i + 1.645\, \sigma_{h_i} \leq 0, \quad i = 1,2,3$

and $\vec{x} \geq \vec{0}$

Where $\bar{h}_i$ and σ_{h_i} are given by Eqs. (6) and (8).

Solution of the NLP problem:

$$\vec{x}_{start} = \begin{Bmatrix} 1.0 \\ 1.0 \\ 1.0 \end{Bmatrix}, \quad \vec{x}^* = \begin{Bmatrix} 0.0 \\ 36.9263 \\ 174.4011 \end{Bmatrix}, \quad \vec{g}(\vec{x}^*) = \begin{Bmatrix} 0.0006714 \\ -0.0000610 \\ -722.1463 \end{Bmatrix},$$

$$\sigma_f^* = \$\ 524.5049$$

(c) Maximize the sum of expected value and standard deviation of profit:

Find $\vec{x} = \{ x_1, \; x_2, \; x_3 \}^T$

to maximize $\quad \bar{f} + \sigma_f = 6x_1 + 4x_2 + 10x_3 + \sqrt{4x_1^2 + x_2^2 + 9x_3^2}$

subject to $\quad \bar{h}_i + 1.645 \, \sigma_{h_i} \leq 0, \quad i = 1, 2, 3$

and $\quad \vec{x} \geq \vec{0}$

where $\bar{h}_i$ and σ_{h_i} are given by Eqs. (6) and (8).

Solution of this NLP problem:

$$\vec{x}_{start} = \begin{Bmatrix} 1.0 \\ 1.0 \\ 1.0 \end{Bmatrix}, \qquad \vec{x}^* = \begin{Bmatrix} 0.0 \\ 36.9262 \\ 174.4011 \end{Bmatrix},$$

$$\vec{g}(\vec{x}^*) = \begin{Bmatrix} -0.0000610 \\ -0.0000610 \\ -722.1478 \end{Bmatrix}, \qquad \left(\bar{f} + \sigma_f\right)^* = \$ \, 2,416.2212$$

11.18 Angle of contact on shorter pulley = $165^\circ = 2.8798$ radians

v = linear speed of belt = (1700, 51) m/min

ρ = density of belt = (1, 0.05) g/cm^3

σ_{per} = permissible stress in belt = (25, 2.5) kg_f/cm^2

μ = coefficient of friction = (0.25, 0.05)

γ = coefficient of variation of belt dimensions = 0.02

probability with which constraints are to be satisfied = 0.95 ($z = 1.645$)

$$P = \text{horse power} = (T_1 - T_2)\frac{v}{75} = T_1\left\{1 - \frac{1}{\left(\frac{T_1}{T_2}\right)}\right\}\frac{v}{75}$$

$$= T_1\left\{1 - e^{-\mu\theta}\right\}\frac{v}{75}$$

$$= \left(T_{max} - \frac{w v^2}{g}\right)\left(1 - e^{-\mu\theta}\right)\frac{v}{75}$$

where $T_{max} = \sigma_{per}\ b\ t$

b = width of belt, t = thickness of belt

$w = \rho g\ bt\ (1) = \rho\ g\ bt$

$$\therefore P = \left(\sigma_{per}\ b\ t - \rho\ bt\ v^2\right)\left(1 - e^{-2.8798\,\mu}\right)\frac{v}{75}$$

$$\left.\begin{array}{l} P\ (b \le 10) \ge 0.95 \\ P\ (t \le 0.5) \ge 0.95 \end{array}\right\} \Rightarrow \begin{array}{l} \bar{b} - 10 - 1.645\,\sigma_{\bar{b}} \le 0 \\ \bar{t} - 0.5 - 1.645\,\sigma_{\bar{t}} \le 0 \end{array}$$

where $\sigma_{\bar{b}} = 0.02\,\bar{b}$,

and $\sigma_{\bar{t}} = 0.02\,\bar{t}$

Final problem:

Find $\vec{x} = \left\{\begin{array}{l} \bar{b} \\ \bar{t} \end{array}\right\}$

to maximize $f = \bar{P} = \left(\overline{\sigma_{per}}\ \bar{b}\ \bar{t} - \bar{\rho}\ \bar{b}\ \bar{t}\ \bar{v}^2\right)\left(1 - e^{-2.8798\,\bar{\mu}}\right)\frac{\bar{v}}{75}$

subject to

$$0.9671\ \bar{b} - 10.0 \le 0$$

$$0.9671\ \bar{t} - 0.5 \le 0$$

consistent units to be used:

$\overline{\sigma_{per}} = 25.0 * 10^3$ grams$_f$/cm^2

$\bar{\rho} = \frac{1.0}{981.0}$ grams/cm^3

$\bar{v} = 1700\left(\frac{100}{60}\right) = 2{,}833.33$ cm/sec

$\bar{\mu} = 0.25$

Solution of this NLP problem:

$\vec{x}_{start} = \left\{\begin{array}{l} \bar{b} \\ \bar{t} \end{array}\right\} = \left\{\begin{array}{l} 5.0 \\ 0.25 \end{array}\right\}$ cm, $f\left(\vec{x}_{start}\right) = 40.756297 \times 10^4$

$\vec{x}_{opt} = \left\{\begin{array}{l} \bar{b}^* \\ \bar{t}^* \end{array}\right\} = \left\{\begin{array}{l} 10.340193 \\ 0.517010 \end{array}\right\}$ cm, $f_{opt} = 174.30582 \times 10^4$

11.20

Design vector: $\vec{x} = \begin{Bmatrix} \bar{D} \\ \bar{d} \end{Bmatrix}$ (1)

objective: Minimize volume (f):

$$f = \frac{\pi^2}{4} d^2 D n = 2\pi^2 D d^2 \quad (n = 8 = \text{deterministic})$$

$$\bar{f} = 2\pi^2 \bar{D} \bar{d}^2 \tag{2}$$

$$\sigma_f^2 = (2\pi^2 \bar{d}^2)^2 \sigma_D^2 + (4\pi^2 \bar{D} \bar{d})^2 \sigma_d^2$$

$$= 4\pi^4 \bar{d}^4 k^2 \bar{D}^2 + 16 \pi^4 \bar{D}^2 \bar{d}^2 k^2 \bar{d}^2$$

$$\sigma_f^2 = \bar{f}^2 k^2 + 4 \bar{f}^2 k^2 = 5 \bar{f}^2 k^2$$

$$\sigma_f = \sqrt{5}\, \bar{f} k = \sqrt{5}\,(0.05)\, \bar{f} = 0.1118\, \bar{f} \tag{3}$$

New objective for minimization:

$$F(\vec{x}) = \bar{f} + \sigma_f = 1.1118\, \bar{f} = 21.946155\, \bar{D} \bar{d}^2 \tag{4}$$

Probabilistic constraints:

$$P[g_j(\vec{x}) \geq 0] \geq p_j \Rightarrow -\bar{g}_j + z_j \sigma_{g_j} \leq 0 \tag{5}$$

(i) Deflection constraint:

$$g_1 = \delta_{max} - \delta \geq 0 \quad \text{with} \quad \delta = \frac{8 P D^3 n}{G d^4} \tag{6}$$

$$\bar{g}_1 = \bar{\delta}_{max} - \bar{\delta} = \bar{\delta}_{max} - \frac{64 \bar{P} \bar{D}^3}{\bar{G} \bar{d}^4} \tag{7}$$

$$\sigma_{g_1} = \sqrt{\sigma_{\delta_{max}}^2 + \sigma_\delta^2} \tag{8}$$

where

$$\sigma_\delta^2 = \left(\frac{\partial \delta}{\partial P}\Big|_m\right)^2 \sigma_P^2 + \left(\frac{\partial \delta}{\partial D}\Big|_m\right)^2 \sigma_D^2 + \left(\frac{\partial \delta}{\partial G}\Big|_m\right)^2 \sigma_G^2 + \left(\frac{\partial \delta}{\partial d}\Big|_m\right)^2 \sigma_d^2$$

$$\sigma_D = k \bar{D}, \quad \sigma_d = k \bar{d}$$

$$\frac{\partial \delta}{\partial P} = \frac{8 D^3 n}{G d^4}, \quad \frac{\partial \delta}{\partial D} = \frac{24 P D^2 n}{G d^4}, \quad \frac{\partial \delta}{\partial G} = \frac{-8 P D^3 n}{G^2 d^2}, \quad \frac{\partial \delta}{\partial d} = \frac{-32 P D^3 n}{G d^5}$$

(ii) stress constraint:

$$g_2 = \sigma_{max} - \sigma \geq 0 \quad \text{with} \quad \sigma = \frac{8PDK}{\pi d^3}$$

$$\text{and} \quad K = \frac{4D - d}{4(D-d)} + \frac{0.615\, d}{D}$$

$$\bar{g}_2 = \bar{\sigma}_{max} - \bar{\sigma} = \bar{\sigma}_{max} - \frac{8\bar{P}\bar{D}\bar{K}}{\pi \bar{d}^3} \tag{9}$$

$$\text{with} \quad \bar{K} = \frac{4\bar{D} - \bar{d}}{4(\bar{D} - \bar{d})} + \frac{0.615\, \bar{d}}{\bar{D}} \tag{10}$$

$$\sigma_{g_2}^2 = \sigma_{\sigma_{max}}^2 + \sigma_{\sigma}^2 \tag{11}$$

$$\sigma_{\sigma}^2 = \left(\frac{\partial \sigma}{\partial P}\Big|_m\right)^2 \sigma_P^2 + \left(\frac{\partial \sigma}{\partial D}\Big|_m\right)^2 \sigma_D^2 + \left(\frac{\partial \sigma}{\partial d}\Big|_m\right)^2 \sigma_d^2 + \left(\frac{\partial \sigma}{\partial K}\Big|_m\right)^2 \sigma_K^2$$

$$\sigma_K^2 = \left(\frac{\partial K}{\partial D}\Big|_m\right)^2 \sigma_D^2 + \left(\frac{\partial K}{\partial d}\Big|_m\right)^2 \sigma_d^2$$

$$\sigma_D = k\bar{D}, \quad \sigma_d = k\bar{d}$$

$$\frac{\partial \sigma}{\partial P} = \frac{8DK}{\pi d^3}, \quad \frac{\partial \sigma}{\partial D} = \frac{8PK}{\pi d^3}, \quad \frac{\partial \sigma}{\partial K} = \frac{8PD}{\pi d^3}, \quad \frac{\partial \sigma}{\partial d} = -\frac{24PDK}{\pi d^4}$$

$$\frac{\partial K}{\partial D} = \frac{16(D-d) - 4(4D-d)}{\{4(D-d)\}^2} - \frac{0.615\, d}{D^2}$$

$$\frac{\partial K}{\partial d} = \frac{-4(D-d) + 4(4D-d)}{16(D-d)^2} + \frac{0.615\, D}{D^2}$$

(iii) solid height constraint:

$$g_3 = n d - h \geq 0 \tag{12}$$

$$\bar{g}_3 = 8\bar{d} - \bar{h}, \quad \sigma_{g_3}^2 = \sigma_{(nd)}^2 + \sigma_h^2 \tag{13}$$

$$\sigma_{(nd)}^2 = \left(\frac{\partial (nd)}{\partial d}\Big|_m\right)^2 \sigma_d^2 = n^2 k^2 \bar{d}^2 = 64 k^2 \bar{d}^2$$

Final problem:

Find $\vec{x} = \begin{Bmatrix} \overline{D} \\ \overline{d} \end{Bmatrix}$

to minimize $f = 21.946155\ \overline{D}\ \overline{d}^{2}$

subject to $\overline{g}_j - z_j\ \sigma_{\overline{g}_j} \geq 0 \quad ; \quad j = 1, 2, 3$

with $z_j = 2.33 \quad (j = 1, 2, 3)$

Data: $k = 0.05,\ \overline{G} = 84 * 10^{4},\ \sigma_{\overline{G}} = 84 * 10^{3},$

$\overline{\delta}_{max} = 2,\ \sigma_{\overline{\delta}_{max}} = 0.1,\ \overline{\sigma}_{max} = 3000,\ \sigma_{\overline{\sigma}_{max}} = 150,$

$\overline{P} = 12,\ \sigma_{\overline{P}} = 3,\ \overline{h} = 2,\ \sigma_{\overline{h}} = 0.4$

Chapter 12

Optimal Control and Optimality Criteria Methods

12.2 For simplicity, consider the points in a plane. The length of arc is

$$L = \int_{x_1}^{x_2} \sqrt{1+(y')^2}\, dx \quad ; \quad x_1 < x_2 \; ; \; y' = \frac{\partial y}{\partial x}$$

Assuming $F(x, y, y') = \sqrt{1+(y')^2}$

$$\frac{\partial F}{\partial y} = 0, \quad \frac{\partial F}{\partial y'} = \frac{y'}{\sqrt{1+(y')^2}}$$

Euler's equation gives

$$\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) - \frac{\partial F}{\partial y} = 0 \Rightarrow \frac{d}{dx}\left(\frac{y'}{\sqrt{1+(y')^2}}\right) = 0 \qquad (1)$$

Integration of Eq. (1) with respect to x gives

$$\frac{y'}{\sqrt{1+(y')^2}} = c = \text{constant} \qquad (2)$$

Solution of Eq. (2) gives $y' = \text{constant} = a$

$$\text{or} \quad y = ax + b \qquad (3)$$

where a and b are constants. Eq. (3) shows that the curve is a straight line.

To show that the distance is a minimum but not a maximum, we need to use the second variation of L. We can also use the following argument for this purpose. If the curve is denoted parametrically as $x = x(t)$ and $y = y(t)$ with $t_1 \le t \le t_2$. Then, the length of the curve is given by $L = \int_{t_1}^{t_2} \sqrt{\dot{x}^2 + \dot{y}^2}\, dt$, $\dot{x} = \frac{dx}{dt}$, $\dot{y} = \frac{dy}{dt}$. Since length of the curve is invariant under rigid body rotation and translation of axes, we can choose the two points as $(0,0)$ and $(c,0)$ with $c > 0$. This leads to $\sqrt{\dot{x}^2 + \dot{y}^2} \ge \dot{x}$; $\int_{t_1}^{t_2} \sqrt{\dot{x}^2+\dot{y}^2}\, dt > \int_{t_1}^{t_2} \dot{x}\, dt = c$ or $L \ge c$. The equality holds only when $\dot{y} = 0$ or the curve is a straight line.

12.3

$$A = \int_0^T \left[x^2 + 2xt + \left(\frac{dx}{dt}\right)^2 \right] dt$$

Euler's equation:

$$\frac{\partial F}{\partial u} - \frac{d}{dx}\frac{\partial F}{\partial u'} = 0 \quad \text{where} \quad A = \int F(x, u, u')\, dx$$

Here $x = t$, $u = x$, $u' = x' = \frac{dx}{dt}$

$$F = x^2 + 2xt + (x')^2$$

$$\frac{\partial F}{\partial x} = 2x + 2t, \quad \frac{\partial F}{\partial x'} = 2x', \quad \frac{d}{dt}\left(\frac{\partial F}{\partial x'}\right) = 2x''$$

Euler's equation becomes:

$$2(x + t) - 2x'' = 0$$

or

$$\frac{d^2x}{dt^2} - x = t$$

Solution of homogeneous equation:

$$x_h(t) = c_1 e^t + c_2 e^{-t} \quad \text{where} \quad c_1, c_2 = \text{constants}$$

Solution of particular integral:

$$x_p(t) = c_3 + c_4 t$$

$$\frac{dx_p}{dt} = c_4, \quad \frac{d^2x_p}{dt^2} = 0 \Rightarrow c_3 = 0, \; c_4 = -1$$

$$x_p(t) = -t$$

Complete solution:

$$x(t) = c_1 e^t + c_2 e^{-t} - t$$

$$x(0) = c_1 e^0 + c_2 e^{-0} - 0 = 2$$

or $c_1 + c_2 = 2$

or $c_2 = 2 - c_1$

$$\therefore \; x(t) = c_1 e^t + (2 - c_1) e^{-t} - t$$

Where c_1 = constant.

12.4 Let the curve be denoted in parametric form as

$$x = x(t) \;,\quad y = y(t) \;;\quad t_1 \le t \le t_2 \tag{1}$$

where t is the parameter and

$$x(t_1) = x(t_2) \;,\quad y(t_1) = y(t_2) \tag{2}$$

If A indicates the area enclosed by the curve, of length L, we have

$$A = \frac{1}{2}\int_{t_1}^{t_2} (x\,\dot{y} - \dot{x}\,y)\,dt \tag{3}$$

and

$$L = \int_{t_1}^{t_2} \sqrt{\dot{x}^2 + \dot{y}^2}\;dt \tag{4}$$

Let

$$I = \int_{t_1}^{t_2} \left\{ \frac{1}{2}(x\dot{y} - \dot{x}\,y) + \lambda\sqrt{\dot{x}^2 + \dot{y}^2} \right\} dt \tag{5}$$

where λ = Lagrange multiplier. Then

$$F(x, \dot{x}, y, \dot{y}) = \frac{1}{2}(x\dot{y} - \dot{x}\,y) + \lambda\sqrt{\dot{x}^2 + \dot{y}^2} \tag{6}$$

To use the Euler equation, Eq. (12.47), we find

$$\frac{\partial F}{\partial x} = \frac{\dot{y}}{2} \;,\quad \frac{\partial F}{\partial \dot{x}} = -\frac{y}{2} + \frac{\lambda\,\dot{x}}{\sqrt{\dot{x}^2 + \dot{y}^2}} \tag{7}$$

$$\frac{\partial F}{\partial y} = -\frac{\dot{x}}{2} \;,\quad \frac{\partial F}{\partial \dot{y}} = \frac{x}{2} + \frac{\lambda\,\dot{y}}{\sqrt{\dot{x}^2 + \dot{y}^2}} \tag{8}$$

Eq. (12.47) $\Rightarrow$

$$\frac{\partial F}{\partial x} - \frac{d}{dt}\left(\frac{\partial F}{\partial \dot{x}}\right) = \frac{\dot{y}}{2} - \frac{d}{dt}\left\{-\frac{y}{2} + \frac{\lambda\,\dot{x}}{\sqrt{\dot{x}^2 + \dot{y}^2}}\right\} = 0 \tag{9}$$

$$\frac{\partial F}{\partial y} - \frac{d}{dt}\left(\frac{\partial F}{\partial \dot{y}}\right) = -\frac{\dot{x}}{2} - \frac{d}{dt}\left\{\frac{x}{2} + \frac{\lambda\,\dot{y}}{\sqrt{\dot{x}^2 + \dot{y}^2}}\right\} = 0 \tag{10}$$

The parameter t can be selected such that the arc length $\sqrt{\dot{x}^2 + \dot{y}^2} = 1$. Then Eqs. (9) and (10) yield:

$$\dot{y} - \lambda\,\ddot{x} = 0 \;,\qquad \dot{x} + \lambda\,\ddot{y} = 0 \tag{11}$$

Eqs. (11) give, after integration,

$$y = \lambda \dot{x} + a_1 \quad , \quad x = -\lambda \dot{y} + a_2 \tag{12}$$

where a_1 and a_2 are integration constants. Differentiating the first equation of (12) and substituting the result into the second equation of (12) gives

$$\lambda^2 \ddot{x} + x = a_2 \tag{13}$$

The solution of Eq. (13) is:

$$x(t) = c_1 \sin \frac{t}{\lambda} + c_2 \cos \frac{t}{\lambda} + a_2 \tag{14}$$

Eqs. (12) and (14):

$$y(t) = c_1 \cos \frac{t}{\lambda} - c_2 \sin \frac{t}{\lambda} + a_1 \tag{15}$$

where c_1 and c_2 are integration constants. Equations (14) and (15) can be rewritten as

$$x(t) = c_3 \sin\left(\frac{t}{\lambda} + c_4\right) + a_2 \; ; \quad y(t) = c_3 \cos\left(\frac{t}{\lambda} + c_4\right) + a_1 \tag{16}$$

where c_3 and c_4 are new constants. It can be seen that Eqs. (16) denote a circle of radius c_3 and center at (a_2, a_1), with $c_3 = L/2\pi$.

12.5 $\delta \Pi_0 = 0$ gives

$$\pi D \int_{r_1}^{r_2} \left\{ 2r\, \omega_{rr}\, \delta\omega_{rr} + \frac{2}{r}\, \omega_r\, \delta\omega_r + \nu \frac{d}{dr}\left(2\, \omega_r\, \delta\omega_r\right) \right\} dr$$

$$- 2\pi \int_{r_1}^{r_2} \{ q\, r\, \delta w\, dr + 2\pi \left(r M\, \delta\omega_r - r Q\, \delta w \right)\Big|_{r=r_2} = 0 \quad (1)$$

where $\omega_r = \frac{dw}{dr}$, $\omega_{rr} = \frac{d^2 w}{dr^2}$, and $\frac{d}{dr}(\omega_r^2) = 2\omega_r \omega_{rr}$.

Using integration by parts, Eq. (1) can be expressed as

$$\left(r\, \omega_{rr} + \nu\, \omega_r\right) \delta\omega_r + \left(\nu\, \omega_{rr} + \frac{1}{r}\, \omega_r\right) \delta w \Big|_{r_1}^{r_2}$$

$$- \int_{r_1}^{r_2} \left\{ \frac{d}{dr}\left(r\, \omega_{rr} + \nu\, \omega_r\right) \cdot \delta\omega_r + \frac{d}{dr}\left(\nu\, \omega_{rr} + \frac{1}{r}\, \omega_r\right) \delta w \right\} dr$$

$$-\int_{r_1}^{r_2} \frac{q\, r\, \delta w}{D}\, dr + \left(\frac{r M}{D}\, \delta w_r - \frac{r Q}{D}\, \delta w\right)\Big|_{r=r_2} = 0 \qquad (2)$$

where

$$\int_{r_1}^{r_2} \frac{d}{dr}\left(r\, w_{rr} + \nu\, w_r\right)\, \delta w_r\, dr = \int_{r_1}^{r_2} \frac{d^2}{dr^2}\left(r\, w_{rr} + \nu\, w_r\right) \delta w\, dr$$

$$- \frac{d}{dr}\left(r\, w_{rr} + \nu\, w_r\right) \cdot \delta w \Big|_{r_1}^{r_2} \qquad (3)$$

Equations (2) and (3) can be combined and the governing differential equation can be identified as:

$$\frac{d^2}{dr^2}\left(r\, w_{rr} + \nu\, w_r\right) - \frac{d}{dr}\left(\nu\, w_{rr} + \frac{1}{r}\, w_r\right) - \frac{q\, r}{D} = 0$$

or $$r\frac{d^4 w}{dr^4} + 2\frac{d^3 w}{dr^3} - \frac{1}{r}\cdot\frac{d^2 w}{dr^2} + \frac{1}{r^2}\frac{dw}{dr} - \frac{q\, r}{D} = 0 \qquad (4)$$

The boundary conditions can be identified as:

$$\left\{\nu\frac{d^2 w}{dr^2} + \frac{1}{r}\frac{dw}{dr} - \frac{d}{dr}\left(r\frac{d^2 w}{dr^2} + \nu\frac{dw}{dr}\right) - \frac{r Q}{D}\right\} \delta w \Big|_{r=r_2} = 0 \qquad (5)$$

$$\left(\frac{r M}{D} + r\frac{d^2 w}{dr^2} + \nu\frac{dw}{dr}\right) \delta w_r \Big|_{r=r_2} = 0 \qquad (6)$$

$$\left\{\nu\frac{d^2 w}{dr^2} + \frac{1}{r}\frac{dw}{dr} - \frac{d}{dr}\left(r\frac{d^2 w}{dr^2} + \nu\frac{dw}{dr}\right)\right\} \delta w \Big|_{r=r_1} = 0 \qquad (7)$$

$$\left(r\frac{d^2 w}{dr^2} + \nu\frac{dw}{dr}\right) \delta w_r \Big|_{r=r_1} = 0 \qquad (8)$$

Eq. (5) $\Rightarrow$ Either $\delta w = 0$ (w is prescribed) or the quantity in { } is zero (shear force Q is prescribed).

Eq. (6) $\Rightarrow$ Either $\delta w_r = 0$ (slope is fixed) or the quantity in ( ) is zero (bending moment fixed).

Eq. (7) $\Rightarrow$ Either $\delta w = 0$ or $Q = 0$.

Eq. (8) $\Rightarrow$ Either $\delta w_r = 0$ or $M = 0$.

12.6

Minimize $f = 60\sqrt{2}\,(x_1 + x_2)$

$$g = 10^{-3}\left(\frac{1}{x_1} + \frac{1}{x_2}\right) - 10^{-2} \le 0$$

$$0.1 \le x_i \le 1.0 \quad ; \quad i = 1, 2$$

$$\frac{\partial f}{\partial x_1} = \frac{\partial f}{\partial x_2} = 60\sqrt{2}$$

$$x_i = \frac{1}{z_i} \quad ; \quad \frac{\partial g}{\partial z_i} = \frac{\partial g}{\partial x_i}\,\frac{\partial x_i}{\partial z_i} = -x_i^2\,\frac{\partial g}{\partial x_i} \quad ; \quad i = 1, 2$$

$$\frac{\partial g}{\partial x_1} = -\frac{10^{-3}}{x_1^2} \quad , \quad \frac{\partial g}{\partial x_2} = -\frac{10^{-3}}{x_2^2}$$

At any design $\vec{X}_i$, Eq. (12.103) gives

$$g_0 = g(\vec{X}_i) + \left.\frac{\partial g}{\partial x_1}\right|_{\vec{X}_i} x_{i1} + \left.\frac{\partial g}{\partial x_2}\right|_{\vec{X}_i} x_{i2}$$

$$= 10^{-3}\left(\frac{1}{x_{i1}} + \frac{1}{x_{i2}}\right) - 10^{-2} - \frac{10^{-3}}{x_{i1}} - \frac{10^{-3}}{x_{i2}}$$

The values of λ and (x_1, x_2) can be determined iteratively, using Eqs. (12.104) and (12.101):

$$\lambda = \left\{\frac{1}{g_0}\sum_{i=1}^{2}\left(\frac{\partial f}{\partial x_i}\,\frac{\partial g}{\partial z_i}\right)^{1/2}\right\}^2$$

$$x_i = \left\{\lambda\,\frac{\partial g/\partial z_i}{\partial f/\partial x_i}\right\}^{1/2} \quad ; \quad i = 1, 2$$

Let $(x_1, x_2) = (1.0,\ 1.0)$ in^2 initially

Solution: $\vec{X}^* = \begin{Bmatrix} 0.2 \\ 0.2 \end{Bmatrix}$ in^2

The computer program used and the output are attached.

```
      dimension x(2)
      data x/0.5,0.5/
      x1=x(1)
      x2=x(2)
      i=0
      print 20
20    format (4x,'z1',11x,'z2',13x,'g0',11x,'xlam',
     2 11x,'x1',11x,'x2',11x)
      print 99
1     continue
      i=i+1
      z1=1.0/x1
      z2=1.0/x2
      g0=-0.01
      dfdx1=60.*sqrt(2.)
      dfdx2=dfdx1
      dgdz1=0.001
      dgdz2=0.001
      c1=(dfdx1*dgdz1)**0.5
      c2=(dfdx2*dgdz2)**0.5
      xlam=((c1+c2)/g0)**2
      x1=(xlam*dgdz1/dfdx1)**0.5
      x2=(xlam*dgdz2/dfdx2)**0.5
99    format (/)
      print 10,z1,z2,g0,xlam,x1,x2
10    format (2x,2e13.5,2x,2e13.5,2x,2e13.5)
      if(x1 .lt. 0.1)x1=0.1
      if(x2 .lt. 0.1)x2=0.1
      if(i .lt. 5) go to 1
      stop
      end
```

z1	z2	g0	xlam	x1	x2
0.20000E+01	0.20000E+01	-0.10000E-01	0.33941E+04	0.20000E+00	0.20000E+00
0.50000E+01	0.50000E+01	-0.10000E-01	0.33941E+04	0.20000E+00	0.20000E+00
0.50000E+01	0.50000E+01	-0.10000E-01	0.33941E+04	0.20000E+00	0.20000E+00
0.50000E+01	0.50000E+01	-0.10000E-01	0.33941E+04	0.20000E+00	0.20000E+00
0.50000E+01	0.50000E+01	-0.10000E-01	0.33941E+04	0.20000E+00	0.20000E+00

12.7 Minimize $f = 80.0445\, x_1 + 28.3\, x_2$

subject to $g = \left\{ \frac{1}{x_1^2} + \frac{1}{(x_1 + \sqrt{2}\, x_2)^2} \right\}^{\frac{1}{2}} \leq 1$

or $\frac{1}{x_1^2} + \frac{1}{(x_1 + \sqrt{2}\, x_2)^2} - 1.0 \leq 0$

$\frac{\partial f}{\partial x_1} = 80.0445, \quad \frac{\partial f}{\partial x_2} = 28.3$

$z_i = \frac{1}{x_i} \quad ; \quad \frac{\partial g}{\partial z_i} = \frac{\partial g}{\partial x_i} \frac{\partial x_i}{\partial z_i} = - x_i^2 \frac{\partial g}{\partial x_i} \quad ; \quad i = 1, 2$

$\frac{\partial g}{\partial x_1} = - \frac{2}{x_1^3} - \frac{2}{(x_1 + \sqrt{2}\, x_2)^3} \quad ; \quad \frac{\partial g}{\partial x_2} = - \frac{2\sqrt{2}}{(x_1 + \sqrt{2}\, x_2)^3}$

Eq. (12.103) gives, at any design $\vec{X}_i$,

$$g_0 = g(\vec{X}_i) + \left.\frac{\partial g}{\partial x_1}\right|_{\vec{X}_i} x_{i1} + \left.\frac{\partial g}{\partial x_2}\right|_{\vec{X}_i} x_{i2}$$

$$= \frac{1}{x_{i1}^2} + \frac{1}{(x_{i1} + \sqrt{2}\, x_{i2})^2} - 1 - 2 x_{i1} \left(\frac{1}{x_{i1}^3} + \frac{1}{(x_{i1} + \sqrt{2}\, x_{i2})^3} \right) - \frac{2\sqrt{2}\, x_{i2}}{(x_{i1} + \sqrt{2}\, x_{i2})^3}$$

λ and (x_1, x_2) can be determined iteratively, using Eqs. (12.104) and (12.101).

Let $(x_1, x_2) = (1, 1)$ initially.

Solution is $\vec{X}^* = \begin{Bmatrix} 1.2169 \\ 0.38053 \end{Bmatrix}$ in^2

Computer program and output are given below.

```
      dimension x(2)
      data x/1.0,1.0/
      x1=x(1)
      x2=x(2)
      i=0
      print 20
20    format (4x,'z1',11x,'z2',13x,'g0',11x,'xlam',
     2 11x,'x1',11x,'x2',11x)
      print 99
1     continue
      i=i+1
      z1=1.0/x1
      z2=1.0/x2
      con=(x1+sqrt(2.)*x2)
      g0=1./(x1**2)+1./(con**2)-1.0-2.*x1*(1./(x1**3)+1./
     2 (con**3))-2.*sqrt(2.)*x2/(con**3)
      dfdx1=80.0445
      dfdx2=28.3
      dgdx1=-2./(x1**3)-2./(con**3)
      dgdz1=-(x1**2)*dgdx1
      dgdx2=-2.*sqrt(2.)/(con**3)
      dgdz2=-(x2**2)*dgdx2
      c1=(dfdx1*dgdz1)**0.5
      c2=(dfdx2*dgdz2)**0.5
      xlam=((c1+c2)/g0)**2
      x1=(xlam*dgdz1/dfdx1)**0.5
      x2=(xlam*dgdz2/dfdx2)**0.5
99    format (/)
      print 10,z1,z2,g0,xlam,x1,x2
10    format (2x,2e13.5,2x,2e13.5,2x,2e13.5)
      if(x1 .lt. 0.05)x1=0.05
      if(x2 .lt. 0.05)x2=0.05
      if(i .lt. 30) go to 1
      stop
      end
```

z1	z2	g0	xlam	x1	x2
0.10000E+01	0.10000E+01	-0.21716E+01	0.50812E+02	0.11661E+01	0.60076E+00
0.85755E+00	0.16646E+01	-0.19815E+01	0.54879E+02	0.11847E+01	0.49163E+00
0.84409E+00	0.20341E+01	-0.19954E+01	0.54302E+02	0.11966E+01	0.44432E+00
0.83570E+00	0.22506E+01	-0.19986E+01	0.54156E+02	0.12040E+01	0.41929E+00
0.83057E+00	0.23850E+01	-0.19995E+01	0.54112E+02	0.12086E+01	0.40479E+00
0.82740E+00	0.24704E+01	-0.19998E+01	0.54096E+02	0.12115E+01	0.39598E+00
0.82540E+00	0.25254E+01	-0.19999E+01	0.54090E+02	0.12134E+01	0.39047E+00
0.82413E+00	0.25610E+01	-0.20000E+01	0.54088E+02	0.12146E+01	0.38697E+00
0.82331E+00	0.25842E+01	-0.20000E+01	0.54087E+02	0.12154E+01	0.38472E+00
0.82278E+00	0.25993E+01	-0.20000E+01	0.54086E+02	0.12159E+01	0.38326E+00
0.82244E+00	0.26092E+01	-0.20000E+01	0.54086E+02	0.12162E+01	0.38231E+00
0.82221E+00	0.26156E+01	-0.20000E+01	0.54086E+02	0.12164E+01	0.38170E+00
0.82206E+00	0.26199E+01	-0.20000E+01	0.54086E+02	0.12166E+01	0.38130E+00
0.82197E+00	0.26226E+01	-0.20000E+01	0.54086E+02	0.12167E+01	0.38103E+00
0.82191E+00	0.26245E+01	-0.20000E+01	0.54086E+02	0.12167E+01	0.38086E+00
0.82186E+00	0.26256E+01	-0.20000E+01	0.54086E+02	0.12168E+01	0.38075E+00
0.82184E+00	0.26264E+01	-0.20000E+01	0.54086E+02	0.12168E+01	0.38067E+00
0.82182E+00	0.26269E+01	-0.20000E+01	0.54086E+02	0.12168E+01	0.38063E+00
0.82181E+00	0.26273E+01	-0.20000E+01	0.54086E+02	0.12168E+01	0.38059E+00
0.82180E+00	0.26275E+01	-0.20000E+01	0.54086E+02	0.12168E+01	0.38057E+00
0.82180E+00	0.26276E+01	-0.20000E+01	0.54086E+02	0.12169E+01	0.38056E+00
0.82179E+00	0.26277E+01	-0.20000E+01	0.54086E+02	0.12169E+01	0.38055E+00
0.82179E+00	0.26278E+01	-0.20000E+01	0.54086E+02	0.12169E+01	0.38055E+00
0.82179E+00	0.26278E+01	-0.20000E+01	0.54086E+02	0.12169E+01	0.38054E+00
0.82179E+00	0.26278E+01	-0.20000E+01	0.54086E+02	0.12169E+01	0.38054E+00
0.82179E+00	0.26279E+01	-0.20000E+01	0.54086E+02	0.12169E+01	0.38054E+00
0.82179E+00	0.26279E+01	-0.20000E+01	0.54086E+02	0.12169E+01	0.38054E+00
0.82179E+00	0.26279E+01	-0.20000E+01	0.54086E+02	0.12169E+01	0.38054E+00
0.82179E+00	0.26279E+01	-0.20000E+01	0.54086E+02	0.12169E+01	0.38053E+00
0.82179E+00	0.26279E+01	-0.20000E+01	0.54086E+02	0.12169E+01	0.38053E+00

12.8

Minimize $f = 0.1\,x_1 + 0.05773\,x_2$

$$g = \frac{0.6}{x_1} + \frac{0.3464}{x_2} - 0.1 \le 0$$

$$x_i \ge 4 \quad ; \quad i = 1, 2$$

$$\frac{\partial f}{\partial x_1} = 0.1, \quad \frac{\partial f}{\partial x_2} = 0.05773$$

$$z_i = \frac{1}{x_i} \quad ; \quad \frac{\partial g}{\partial z_i} = \frac{\partial g}{\partial x_i}\frac{\partial x_i}{\partial z_i} = -x_i^2\,\frac{\partial g}{\partial x_i} \quad ; \quad i = 1, 2$$

$$\frac{\partial g}{\partial x_1} = -\frac{0.6}{x_1^2}, \quad \frac{\partial g}{\partial x_2} = -\frac{0.3464}{x_2^2}$$

Eq. (12.103) gives, at any design $\vec{X}_i$,

$$g_o = g(\vec{X}_i) + \left.\frac{\partial g}{\partial x_1}\right|_{\vec{X}_i} x_{i1} + \left.\frac{\partial g}{\partial x_2}\right|_{\vec{X}_i} x_{i2}$$

$$= \frac{0.6}{x_{i1}} + \frac{0.3464}{x_{i2}} - 0.1 - \frac{0.6}{x_{i1}} - \frac{0.3464}{x_{i2}}$$

λ and (x_1, x_2) can be determined, iteratively, using Eqs. (12.104) and (12.101).

Using $\begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} 6.0 \\ 6.0 \end{Bmatrix}$ initially, the following solution is obtained: $\vec{X}^* = \begin{Bmatrix} 9.4639 \\ 9.4642 \end{Bmatrix}$ in^2.

Computer program and output are given below.

```
      dimension x(2)
      data x/2.0,2.0/
      x1=x(1)
      x2=x(2)
      i=0
      print 20
20    format (4x,'z1',11x,'z2',11x,2x,'g0',11x,'xlam',
     2 11x,'x1',11x,'x2',11x)
      print 99
1     continue
      i=i+1
      z1=1.0/x1
      z2=1.0/x2
      g0=-0.1
       dfdx1=0.1
       dfdx2=0.05773
       dgdx1=-0.6/(x1**2)
       dgdx2=-0.3464/(x2**2)
       dgdz1=-(x1**2)*dgdx1
       dgdz2=-(x2**2)*dgdx2
      c1=(dfdx1*dgdz1)**0.5
      c2=(dfdx2*dgdz2)**0.5
      xlam=((c1+c2)/g0)**2
      x1=(xlam*dgdz1/dfdx1)**0.5
      x2=(xlam*dgdz2/dfdx2)**0.5
99    format (/)
      print 10,z1,z2,g0,xlam,x1,x2
10    format (2x,2e13.5,2x,2e13.5,2x,2e13.5)
      if(x1 .lt. 4.0)x1=4.0
      if(x2 .lt. 4.0)x2=4.0
      if(i .lt. 5) go to 1
      stop
      end
```

z1	z2	g0	xlam	x1	x2
0.50000E+00	0.50000E+00	-0.10000E+00	0.14928E+02	0.94639E+01	0.94642E+01
0.10566E+00	0.10566E+00	-0.10000E+00	0.14928E+02	0.94639E+01	0.94642E+01
0.10566E+00	0.10566E+00	-0.10000E+00	0.14928E+02	0.94639E+01	0.94642E+01
0.10566E+00	0.10566E+00	-0.10000E+00	0.14928E+02	0.94639E+01	0.94642E+01
0.10566E+00	0.10566E+00	-0.10000E+00	0.14928E+02	0.94639E+01	0.94642E+01

Chapter 13

Modern Methods of Optimization

13.1

$\vec{x}_1 = \{ 1 \; 0 \mid 0 \; 0 \; 1 \mid 0 \; 1 \; 1 \; 0 \; 1 \}$

$\vec{x}_2 = \{ 0 \; 1 \mid 1 \; 1 \; 1 \mid 1 \; 0 \; 1 \; 1 \; 0 \}$

crossover location

$\vec{x}_3 = \{ 1 \; 0 \mid 1 \; 1 \; 1 \mid 1 \; 0 \; 1 \; 1 \; 0 \}$ Result of crossover

$\vec{x}_4 = \{ 0 \; 1 \mid 0 \; 0 \; 1 \mid 0 \; 1 \; 1 \; 0 \; 1 \}$

Decimal equivalents:

$$\vec{x}_1 = \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix}_1 = \begin{Bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 \end{Bmatrix}$$

$$b_4 \quad b_3 \quad b_2 \quad b_1 \quad b_0$$

$$= \begin{Bmatrix} 2^4(1) + 2^3(0) + 2^2(0) + 2^1(0) + 2^0(1) \\ 2^4(0) + 2^3(1) + 2^2(1) + 2^1(0) + 2^0(1) \end{Bmatrix}$$

$$= \begin{Bmatrix} 16 + 0 + 0 + 0 + 1 \\ 0 + 8 + 4 + 0 + 1 \end{Bmatrix} = \begin{Bmatrix} 17 \\ 13 \end{Bmatrix}$$

$$\vec{x}_2 = \begin{Bmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 \end{Bmatrix}$$

$$= \begin{Bmatrix} 2^4(0) + 2^3(1) + 2^2(1) + 2^1(1) + 2^0(1) \\ 2^4(1) + 2^3(0) + 2^2(1) + 2^1(1) + 2^0(0) \end{Bmatrix}$$

$$= \begin{Bmatrix} 0 + 8 + 4 + 2 + 1 \\ 16 + 0 + 4 + 2 + 0 \end{Bmatrix} = \begin{Bmatrix} 15 \\ 22 \end{Bmatrix}$$

$$\vec{x}_3 = \begin{Bmatrix} 1 & 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 \end{Bmatrix} = \begin{Bmatrix} 16+0+4+2+1 \\ 16+0+4+2+0 \end{Bmatrix} = \begin{Bmatrix} 23 \\ 22 \end{Bmatrix}$$

$$\vec{x}_4 = \begin{Bmatrix} 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 \end{Bmatrix} = \begin{Bmatrix} 0+8+0+0+1 \\ 0+8+4+0+1 \end{Bmatrix} = \begin{Bmatrix} 9 \\ 13 \end{Bmatrix}$$

13.2 $\mu_{A \cup B} = \max\left(\mu_A, \mu_B\right)$

$\therefore A \cup B = \{ (60, 0.1), (62, 0.5), (64, 0.7), (66, 0.9), (68, 1.0), (70, 1.0) \}$

$\mu_{A \cap B} = \min\left(\mu_A, \mu_B\right)$

$\therefore A \cap B = \{ (60, 0.0), (62, 0.2), (64, 0.4), (66, 0.8), (68, 0.9), (70, 0.8) \}$

13.3 Formula to determine the size of binary string (q) to achieve an accuracy of Δx:

$$2^q \geq \frac{x^{(u)} - x^{(l)}}{\Delta x} + 1$$

(a) When $x^{(u)} = 5$, $x^{(l)} = 0$, and $\Delta x = 0.01$:

$$2^q \geq \frac{5-0}{0.01} + 1 = 501 \Rightarrow q = 9$$

(b) When $x^{(u)} = 10$, $x^{(l)} = 0$, and $\Delta x = 0.01$:

$$2^q \geq \frac{10-0}{0.01} + 1 = 1001 \Rightarrow q = 10$$

(c) When $x^{(u)} = 20$, $x^{(l)} = 0$, and $\Delta x = 0.01$:

$$2^q \geq \frac{20-0}{0.01} + 1 = 2001 \Rightarrow q = 11$$

13.4 $2^q \geq \dfrac{x^{(u)} - x^{(l)}}{\Delta x} + 1$

For $x^{(u)} = 13$, $x^{(l)} = 2$, and $\Delta x = 0.02$:

$$2^q \geq \frac{13-2}{0.02} + 1 = 551$$

$$q \ln 2 \geq \ln 551 \Rightarrow q \geq 9.103288 \text{ or } q = 10$$

13.5 Minimum of $f(x) = x^5 - 5x^3 - 20x + 5$

Differential calculus approach:

$$\frac{df}{dx} = 5x^4 - 15x^2 - 20 = 0 \Rightarrow \text{quadratic eqn. in } x^2$$

Solution: $x^2 = \dfrac{15 \pm \sqrt{225 + 400}}{10} = -1, 4$

$\therefore x^* = 2$

Ant colony approach:

We use the range as $(0, 3)$, number of ants as $4 = N$,

$x_{11} = 0.0$, $x_{12} = 0.5$, $x_{13} = 1.0$, $x_{14} = 1.5$, $x_{15} = 2.0$,

$x_{16} = 2.5$, $x_{17} = 3.0$

Iteration $l = 1$:

Assume equal pheromone for all edges or rays so that

$\tau_{1j} = 1$, $j = 1, 2, \cdots, 7$

Probability of choosing x_{1j} is given by $p_{1j} = \frac{1}{7}$

Possible discrete paths are:

$x_{11} = (0, \frac{1}{7}) = (0, 0.1428)$; $x_{12} = (0.1428, 0.2857)$;

$x_{13} = (0.2856, 0.4286)$; $x_{14} = (0.4286, 0.5714)$;

$x_{15} = (0.5714, 0.7143)$; $x_{16} = (0.7143, 0.8571)$;

$x_{17} = (\frac{6}{7}, 1) = (0.8571, 1.0)$

Random numbers (using TI-83 Plus) are generated as:

$r_1 = 0.6432$, $r_2 = 0.1467$, $r_3 = 0.5147$, $r_4 = 0.4058$

Paths (arcs) selected by four ants:

Ant 1: $x_{15} = 2.0$, Ant 2: $x_{12} = 0.5$; Ant 3: $x_{14} = 1.5$;

Ant 4: $x_{13} = 1.0$

Corresponding objective function values:

$f_1 = f(x_{15}) = f(2.0) = -43 \Rightarrow f_{best}$

$f_2 = f(x_{12}) = f(0.5) = -5.59 \Rightarrow f_{worst}$

$f_3 = f(x_{14}) = f(1.5) = -34.28$

$f_4 = f(x_{13}) = f(1.0) = -19$

Assuming that all ants return home and start again in search of food, we set:

Iteration $l = 2$:

The pheromone array is updated as

$$\tau_{1j}^{(2)} = \tau_{1j}^{(old)} + \sum_k \Delta\tau^{(k)}$$

where $\sum_k \Delta\tau^{(k)} = \Delta\tau = \frac{\zeta f_{best}}{f_{worst}} = \frac{2(-43)}{-5.59} = 15.3846$

where the scaling parameter is assumed as $\zeta = 2$.

Using $\rho = 0.5$, $\tau_{1j}^{(old)}$ can be computed as

$\tau_{1j}^{(old)} = (1 - 0.5)\,\tau_{1j}^{(1)} = 0.5(1.0) = 0.5$; $j = 1, 2, 3, 4, 6, 7$

$\tau_{1j}^{(2)} = 1.0 + 3.0 = 4.0$ for $j = 5$

$\tau_{1j}^{(2)} = 0.5$ for $j = 1, 2, 3, 4, 6, 7$

Probability of selecting path x_{1j}:

$p_{1j} = \frac{0.5}{7.0} = 0.0714$, $j = 1, 2, 3, 4, 6, 7$

$p_{15} = \frac{4.0}{7.0} = 0.5716$

Thus, possible discrete paths are:

$x_{11} = (0, 0.0714)$; $x_{12} = (0.0714, 0.1429)$;

$x_{13} = (0.1429, 0.2142)$; $x_{14} = (0.2142, 0.2856)$;

$x_{15} = (0.2856, 0.8572)$; $x_{16} = (0.8572, 0.9226)$;

$x_{17} = (0.9226, 1.0)$

Random numbers (using TI-83 Plus) generated are:

$r_1 = 0.3394$, $r_2 = 0.2003$, $r_3 = 0.7981$, $r_4 = 0.9519$

Paths (arcs) selected by the ants:

Ant 1: $x_{15} = 2.0$, Ant 2: $x_{13} = 1.0$, Ant 3: $x_{15} = 2.0$, Ant 4: $x_{17} = 3.0$

Corresponding f-values:

$f_1 = f(2.0) = -43$ ← f_{best}

$f_2 = f(1.0) = -19$

$f_3 = f(2.0) = -43$ ← f_{best}

$f_4 = f(3.0) = +53$ ← f_{worst}

Since not all ants have selected the same best path, we neet to go to the next iteration by setting $l = 3$.

13.6 Pheromone along the various arcs at node i:

$\tau_{ij} = 1, 2, 4, 3, 5, 2$; $\sum_{j=1}^{6} \tau_{ij} = 17$

Probability of selecting different arcs x_{ij} $(j = 1, 2, \cdots, 6)$ are:

$$p_{i1} = \frac{\tau_{i1}}{\sum_{j=1 \text{ to } 6} \tau_{ij}} = \frac{1}{17} = 0.05882$$

$$p_{i2} = \tau_{i,2} \Big/ \sum_{j=1}^{6} \tau_{ij} = \frac{2}{17} = 0.1176$$

$p_{i3} = \frac{4}{17} = 0.2353$, $p_{i4} = \frac{3}{17} = 0.1765$,

$p_{i5} = \frac{5}{17} = 0.2941$, $p_{i6} = \frac{2}{17} = 0.1176$

Possible discrete paths or arcs are:

$x_{i1} = (0, p_{i1}) = (0, 0.05882)$

$x_{i2} = (p_{i1}, p_{i1} + p_{i2}) = (0.05882, 0.1765)$

$x_{i3} = (p_{i1} + p_{i2}, p_{i1} + p_{i2} + p_{i3}) = (0.1765, 0.4118)$

Similarly, $x_{i4} = (0.4118, 0.5882)$,

$x_{i5} = (0.5882, 0.8824)$, $x_{i6} = (0.8824, 1.0)$

Roulette-wheel selection process:

Using the random number, $r = 0.4921$,

Arc chosen by the ant: $x_{i4} = (i\ 4) = \text{arc}(i, j)$

13.7 Minimize $f(x) = x^5 - 5x^3 - 20x + 5$; $0 \le x \le 3$

Number of ants = $N = 4$.

Permissible discrete values of $x_1 = x$ assumed as:

$x_{11} = 0.0$, $x_{12} = 0.5$, $x_{13} = 1.0$, $x_{14} = 1.5$, $x_{15} = 2.0$, $x_{16} = 2.5$, $x_{17} = 3.0$

Iteration $l = 1$:

Assume equal pheromone for all edges so that $\tau_{1j} = 1$ for $j = 1, 2, \ldots, 7$.

Probability of choosing x_{1j} is given by $p_{1j} = \frac{1}{7}$ for all j.

Probability ranges for paths:

$x_{11} = (0, \frac{1}{7}) = (0, 0.1428)$; $x_{12} = (\frac{1}{7}, \frac{2}{7}) = (0.1428, 0.2857)$;

$x_{13} = (\frac{2}{7}, \frac{3}{7}) = (0.2857, 0.4286)$; $x_{14} = (\frac{3}{7}, \frac{4}{7}) = (0.4286, 0.5714)$;

$x_{15} = (\frac{4}{7}, \frac{5}{7}) = (0.5714, 0.7143)$; $x_{16} = (\frac{5}{7}, \frac{6}{7}) = (0.7143, 0.8571)$;

$x_{17} = (\frac{6}{7}, 1) = (0.8571, 1)$

Generate four random numbers, one for each ant, as:

$r_1 = 0.3122$, $r_2 = 0.8705$, $r_3 = 0.4132$, $r_4 = 0.6170$

Arcs selected by ants:

Ant 1: $x_{13} = 1.0$; Ant 2: $x_{17} = 3.0$; Ant 3: $x_{14} = 1.5$; Ant 4: $x_{15} = 2.0$

Corresponding objective function values:

$f_1 = f(1.0) = -19$; $f_2 = f(3.0) = +53$; $f_3 = f(1.5) = -34.28$

$f_4 = f(2.0) = -43$

$x_{best} = x_{15} = 2.0, \quad f_{best} = -43$

$x_{worst} = x_{17} = 3.0, \quad f_{worst} = +53$

Iteration $l = 2$:

$$\sum_k \Delta\tau^{(k)} = \Delta\tau = \frac{\zeta f_{best}}{f_{worst}} = \frac{(-2)(-43)}{53} = 1.62$$

where the scaling parameter is assumed as $\zeta = -2$.

Because pheromone evaporation is neglected,

$\tau_{1j}^{(2)} = 1 + 1.62 = 2.62$ for $j = 5$

$\tau_{1j}^{(2)} = 1$ for $j = 1, 2, 3, 4, 6, 7$

Probability of selecting path $(1j)$:

$$p_{1j} = \tau_{1j} \Big/ \sum_{k=1}^{7} \tau_{1j} \quad ; \quad j = 1, 2, \ldots, 7$$

This gives: $p_{1j} = \frac{1}{8.62} = 0.1160$ for $j = 1, 2, 3, 4, 6, 7$

and $p_{15} = \frac{2.62}{8.62} = 0.3039$.

Probability ranges associated with different paths:

$x_{11} = (0, 0.1160)$; $x_{12} = (0.1160, 0.2320)$;

$x_{13} = (0.2320, 0.3480)$; $x_{14} = (0.3480, 0.4640)$;

$x_{15} = (0.4640, 0.7679)$; $x_{16} = (0.7679, 0.8839)$;

$x_{17} = (0.8839, 1.0000)$

Roulette-wheel selection:

Generate 4 random numbers in the range $(0, 1)$:

$r_1 = 0.3688, \quad r_2 = 0.8577, \quad r_3 = 0.0771, \quad r_4 = 0.5790$

Paths selected by ants: Ant 1: $x_{14} = 1.5$; Ant 2: $x_{16} = 2.5$;

Ant 3: $x_{11} = 0.0$; Ant 4: $x_{15} = 2.0$

Corresponding objective function values:

$f_1 = f(1.5) = -34.2812$; $f_2 = f(2.5) = -25.4688$;

$f_3 = f(0.0) = 5$; $f_4 = f(2.0) = -43$

$\therefore x_{best} = 2.0$, $f_{best} = -43$

$x_{worst} = 0.0$, $f_{worst} = 5$

since not all ants have selected the same best path, we need to go to the next iteration, $l = 3$.

13.8 PSO method to maximize $f(x) = -x^5 + 5x^3 + 20x - 5$; $-4 \le x \le 4$

1. $N = 4$:
2. Initial positions: $x_1 = -2$, $x_2 = 0$, $x_3 = 1$, $x_4 = 3$

$f(x_1) = -53$, $f(x_2) = -5$, $f(x_3) = 19$, $f(x_4) = -53$

3. set initial velocities of particles as zero:

$V_1(0) = V_2(0) = V_3(0) = V_4(0) = 0$

4. (a) $P_{best,1} = -2$, $P_{best,2} = 0$, $P_{best,3} = 1$, $P_{best,4} = 3$,

$G_{best} = 1$

(b) Find velocities of particles using the random numbers $r_1 = 0.3294$ and $r_2 = 0.9542$:

$V_1(1) = 0 + 0.3294(-2+2) + 0.9542(1+2) = 2.8626$

$V_2(1) = 0 + 0.3294(0-0) + 0.9542(1-0) = 0.9542$

$V_3(1) = 0 + 0.3294(1-1) + 0.9542(1-1) = 0$

$V_4(1) = 0 + 0.3294(3-3) + 0.9542(1-3) = -1.9084$

(c) Find the new values of $x_j(1)$:

$x_1(1) = 0.8626$, $x_2(1) = 0.9542$, $x_3(1) = 1.0$,

$x_4(1) = 1.0916$

5. Evaluate objective function values:

$f(x_1(1)) = 14.9836$, $f(x_2(1)) = 17.6370$,

$f(x_3(1)) = 19.0$, $f(x_4(1)) = 21.7857$

4. (a) $P_{best,1} = 0.8626$, $P_{best,2} = 0.9542$, $P_{best,3} = 1.0$,

$P_{best,4} = 1.0916$, $G_{best} = 1.0916$

(b) New velocities using the random numbers $r_1 = 0.1482$ and $r_2 = 0.4867$:

$$V_1(2) = 2.8626 + 0.1482(0.8626 - 0.8626) + 0.4867(1.0916 - 0.8626) = 2.9741$$

$$V_2(2) = 0.9542 + 0.1482(0.9542 - 0.9542) + 0.4867(1.0916 - 0.9542) = 1.0211$$

$$V_3(2) = 0 + 0.1482(1 - 1) + 0.4867(1.0916 - 1) = 0.04458$$

$$V_4(2) = -1.9084 + 0.1482(1.0916 - 1.0916) + 0.4867(1.0916 - 1.0916) = -1.9084$$

(c) New values of $x_j(i)$:

$$x_1(2) = 0.8626 + 2.9741 = 3.8367$$

$$x_2(2) = 0.9542 + 1.0211 = 1.9753$$

$$x_3(2) = 1 + 0.04458 = 1.04458$$

$$x_4(2) = 1.0916 + (-1.9084) = -0.8168$$

5. objective function values:

$$f(x_1(2)) = -477.2399$$

$$f(x_2(2)) = 42.9699$$

$$f(x_3(2)) = 20.3469$$

$$f(x_4(2)) = -23.6971$$

Since the values of $f(x_j(i))$ did not converge, we need to go to Step 4 for another iteration.

13.9 Maximize $f(x) = -x^2 + 2x + 11 \; ; \; -2 \le x \le 2$

Use: $0.4 \le \theta \le 0.9$

$$\theta(i) = \theta_{max} - \left(\frac{\theta_{max} - \theta_{min}}{i_{max}}\right) i = 0.9 - \frac{0.5}{i_{max}} i$$

so that

$$V_j(i) = \theta(i) \cdot V_j(i-1) + c_1 r_1 \left(\vec{P}_{best,j} - \vec{x}_j(i-1)\right) + c_2 r_2 \left(\vec{G}_{best} - \vec{x}_j(i-1)\right)$$

$N = 4$

$x_1(0) = -1.5$, $x_2(0) = 0.0$, $x_3(0) = 0.5$, $x_4(0) = 1.25$

$f_1 = f(x_1(0)) = 5.75$, $f_2 = f(x_2(0)) = 11.0$, $f_3 = f(x_3(0)) = 11.75$, $f_4 = f(x_4(0)) = 11.9375$

$V_1(0) = V_2(0) = V_3(0) = V_4(0) = 0$

Step 4: Iteration $i = 1$:

(a) $P_{best,1} = -1.5$, $P_{best,2} = 0.0$, $P_{best,3} = 0.5$, $P_{best,4} = 1.25$,

$G_{best} = 1.25$

(b) $c_1 = c_2 = 1$, $r_1 = 0.3294$, $r_2 = 0.9542$; $i = 1$, $i_{max} = 1$, $\theta(1) = 0.4$:

$V_1(1) = 0.4(0) + 0.3294(-1.5 + 1.5) + 0.9542(1.25 + 1.5) = 2.6241$

$V_2(1) = 0.4(0) + 0.3294(0.0 - 0.0) + 0.9542(1.25 - 0.0) = 1.1927$

$V_3(1) = 0.4(0) + 0.3294(0.0 - 0.0) + 0.9542(1.25 - 0.5) = 0.7156$

$V_4(1) = 0.0$

(c) $x_1(1) = -1.5 + 2.6241 = 1.1241$

$x_2(1) = 0.0 + 1.1927 = 1.1927$

$x_3(1) = 0.5 + 0.7156 = 1.2156$

$x_4(1) = 1.25 + 0.0 = 1.25$

$f(x_1(1)) = 11.9846$, $f(x_2(1)) = 11.9629$,

$f(x_3(1)) = 11.9535$, $f(x_4(1)) = 11.9375$

Convergence not achieved. Go to next iteration.

$i = 2$: Step 4:

(a) $P_{best,1} = 1.1241$, $P_{best,2} = 1.1927$, $P_{best,3} = 1.2156$,

$P_{best,4} = 1.25$, $G_{best} = 1.1241$

$\theta(2) = 0.9 - \frac{0.5}{2} = 0.65$, $r_1 = 0.1482$, $r_2 = 0.4867$

(b) $V_1(2) = 0.65(2.6241) + 0.1482(1.1241 - 1.1241)$

$+ 0.4867(1.1241 - 1.1241) = 1.705665$

$V_2(2) = 0.65(1.1927) + 0.1482(1.1927 - 1.1927)$

$+ 0.4867(1.1241 - 1.1927) = 0.741855$

$V_3(2) = 0.42064$

$V_4(2) = -0.0613$

(c) $x_1(2) = x_1(1) + V_1(2) = 1.1241 + 1.705665 = 2.829765$

$x_2(2) = x_2(1) + V_2(2) = 1.1927 + 0.741855 = 1.934555$

$x_3(2) = x_3(1) + V_3(2) = 1.2156 + 0.42064 = 1.63624$

$x_4(2) = x_4(1) + V_4(2) = 1.25 - 0.0613 = 1.1887$

$f(x_1(2)) = 8.651959$

$f(x_2(2)) = 11.126607$

$f(x_3(2)) = 11.595199$

$f(x_4(2)) = 11.964393$

Convergence not achieved.

Go to next iteration ($i = 3$).

13.12

String number, i	Fitness value	Probability	Cumulative probability	Range of cumulative prob. of string i
1	8	0.123	0.123	0 – 0.123
2	12	0.185	0.308	0.123 – 0.308
3	6	0.0923	0.4003	0.308 – 0.4003
4	2	0.0308	0.4311	0.4003 – 0.4311
5	18	0.277	0.7081	0.4311 – 0.7081
6	9	0.138	0.8461	0.7081 – 0.8461
7	10	0.154	1.0	0.8461 – 1.0

Following random numbers are used:

Random no. (cumulative prob. of the string to be copied)	0.41	0.65	0.42	0.80	0.67	0.39	0.63	0.53
string no. to be copied to mating pool.	4	5	4	6	5	3	5	5

0.86	0.88	0.75	0.55
7	7	6	5

This shows that the mating pool consists of:

1 copy of string 3,

2 copies of string 4,

5 copies of string 5,

2 copies of string 6, and

2 copies of string 7

13.13 Fitness function to be used in GA:

$$\phi(\vec{x}) = f(\vec{x}) + \sum_{i=1}^{m} r_i \langle g_i(\vec{x}) \rangle^2$$

$$F(\vec{x}) = \frac{1}{1+\phi(\vec{x})} \quad = \text{Final fitness function}$$

where

$$\phi(\vec{x}) = x_1^3 - 6x_1^2 + 11x_1 + x_3 + r\{\max(0, x_1^2 + x_2^2 - x_3^2)\}^2$$
$$+ r[\max(0, 4 - x_1^2 - x_2^2 - x_3^2)]^2$$
$$+ r[\max(0, x_3 - 5)]^2 + r[\max(0, -x_1)]^2$$
$$+ r[\max(0, -x_2)]^2 + r[\max(0, -x_3)]^2$$

13.14 Minimum binary string length of design vector:

$-10 \le x_1 \le 10$ with accuracy of $\Delta x_1 = 0.01$

$$2^q \ge \frac{10-(-10)}{0.01} + 1 = 2001 \Rightarrow q = 11$$

$0 \le x_2 \le 8$ with accuracy of $\Delta x_2 = 0.01$

$$2^q \ge \frac{8-0}{0.01} + 1 = 801 \Rightarrow q = 10$$

$150 \le x_3 \le 750$ with accuracy of $\Delta x_3 = 0.01$

$$2^q \ge \frac{750-150}{0.01} + 1 = 60001 \Rightarrow q = 16$$

String length of design vector is $11+10+16 = 37$ binary digits.

Chapter 14

Practical Aspects of Optimization

14.1

$$\vec{X} = \begin{Bmatrix} A_1 \\ A_2 \\ A_3 \\ A_4 \end{Bmatrix} = c_1 \begin{Bmatrix} 1 \\ 1 \\ 0.5 \\ 0.5 \end{Bmatrix} + c_2 \begin{Bmatrix} 0.5 \\ 0.5 \\ 1 \\ 1 \end{Bmatrix} = \begin{Bmatrix} c_1 + 0.5\,c_2 \\ c_1 + 0.5\,c_2 \\ 0.5\,c_1 + c_2 \\ 0.5\,c_1 + c_2 \end{Bmatrix} = \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix}$$

$$f(\vec{X}) = \text{volume} = A_1 l_1 + A_2 l_2 + A_3 l_3 + A_4 l_4$$

$$= 55.9017(x_1 + x_2) + 167.7051\,x_3 + 141.4214\,x_4$$

$$= 55.9017\,(c_1 + 0.5\,c_2 + c_1 + 0.5\,c_2) + 167.7051\,(0.5\,c_1 + c_2) + 141.4214\,(0.5\,c_1 + c_2)$$

$$= 266.3667\,c_1 + 365.0282\,c_2$$

Contours of f:

$f = 302.9734$ at $(c_1 = 0,\ c_2 = 0.83)$, $(c_1 = 1.1374,\ c_2 = 0)$

$f = 346.2767$ at $(c_1 = 1.3,\ c_2 = 0)$, $(c_1 = 0,\ c_2 = 0.9486)$

$f = 372.9134$ at $(c_1 = 1.4,\ c_2 = 0)$, $(c_1 = 0,\ c_2 = 1.0216)$

$f = 306.3275$ at $(c_1 = 0.04,\ c_2 = 0.81)$

Constraint:

Vertical deflection of node 4 = 0.1 inch

To find the displacement of node 4, equilibrium equations are solved:

$$\left[K(c_1, c_2)\right] \vec{Y} = \vec{P} \qquad (1)$$

where $\vec{Y} = \begin{Bmatrix} y_5 \\ y_6 \\ y_7 \\ y_8 \end{Bmatrix}$ and $\vec{P} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ -1000 \end{Bmatrix}$

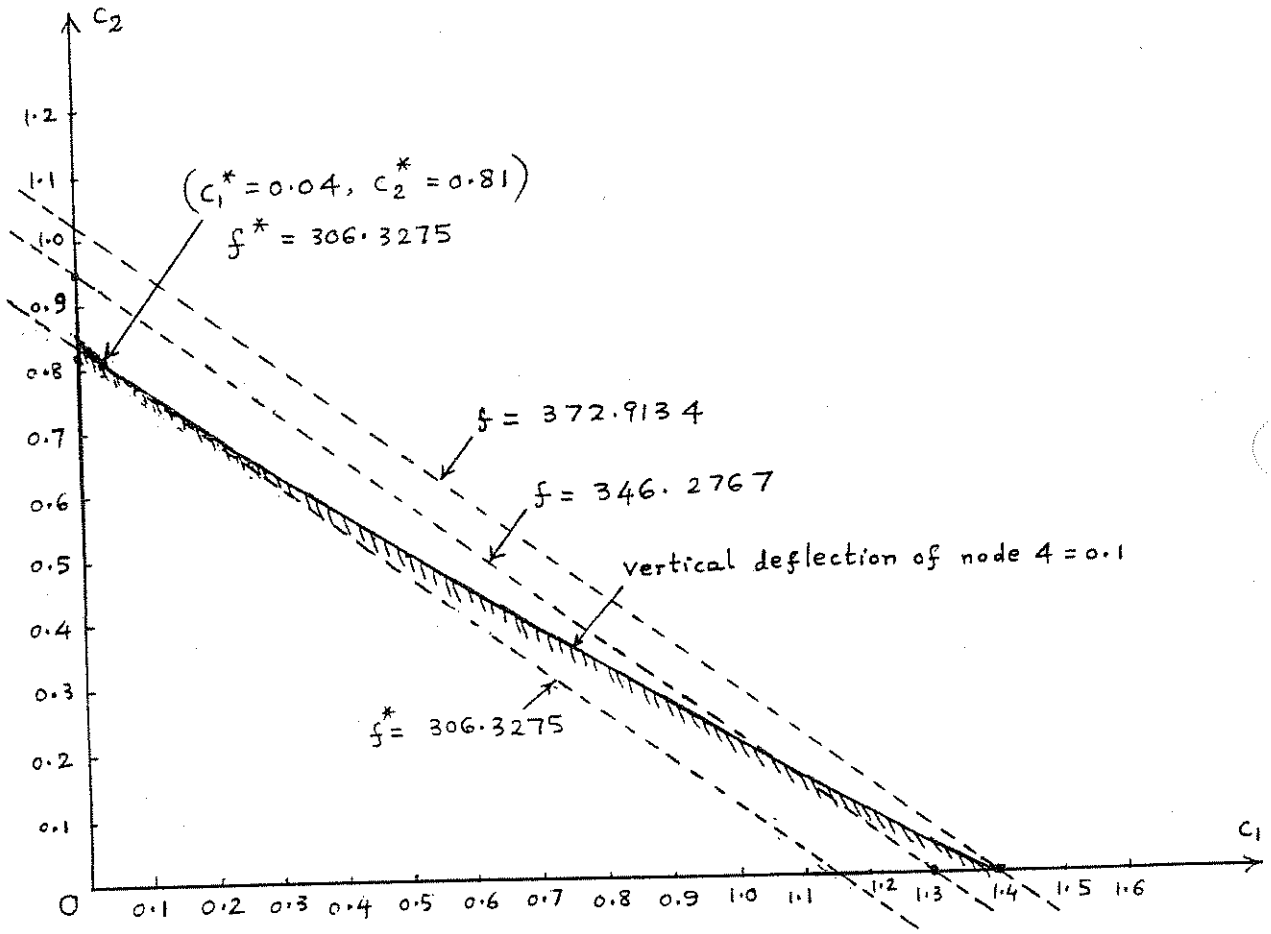
Program used to find the solution of Eqs. (1) for varying values of c_1 and c_2:

```
      DIMENSION C(2),A(4,4),B(4),LA(4),LB(4,2),S(4)
      DIMENSION X(4)
       C(1)=-0.01
       DO 41 I1=1,10
       C(1)=C(1)+0.01
       C(2)=0.75
       DO 42 I2=1,10
       C(2)=C(2)+0.01
        B(1)=0.
        B(2)=0.
        B(3)=0.
        B(4)=-1000.
       X(1)=C(1)+0.5*C(2)
       X(2)=C(1)+0.5*C(2)
       X(3)=0.5*C(1)+C(2)
        X(4)=0.5*C(1)+C(2)
      A(1,1)=0.8*X(1)/55.9017 + 0.8*X(2)/55.9017 + 0.8*X(3)/167.7051
      A(1,2)=0.4*X(1)/55.9017 - 0.4*X(2)/55.9017 + 0.4*X(3)/167.7051
      A(1,3)=-0.8*X(3)/167.7051
      A(1,4)=-0.4*X(3)/167.7051
      A(2,1)=A(1,2)
      A(2,2)=0.2*X(1)/55.9017 + 0.2*X(2)/55.9017 + 0.2*X(3)/167.7051
      A(2,3)=-0.4*X(3)/167.7051
      A(2,4)=-0.2*X(3)/167.7051
      A(3,1)=A(1,3)
      A(3,2)=A(2,3)
      A(3,3)=0.8*X(3)/167.7051 + 0.5*X(4)/141.4214
      A(3,4)=0.4*X(3)/167.7051 + 0.5*X(4)/141.4214
      A(4,1)=A(1,4)
      A(4,2)=A(2,4)
      A(4,3)=A(3,4)
      A(4,4)=0.2*X(3)/167.7051 + 0.5*X(4)/141.4214
      DO 55 I =1,4
      DO 55 J =1,4
 55   A(I,J)=A(I,J)*30.0E+06
      N=4
      IND=1
C END OF PROBLEM-DEPENDENT DATA
```

$A(i,j)$ = stiffness matrix

```
      CALL SIMUL (A,B,N,IND,LA,LB,S)
       OBJ=266.3667*C(1)+365.0282*C(2)
       CON=ABS(B(4))-0.1
        PRINT 500,C(1),C(2),OBJ,CON
500     FORMAT (2f12.4,2x,2f12.4)
42      CONTINUE
41      CONTINUE
      STOP
      END
```

Graphical optimization : shown in figure.



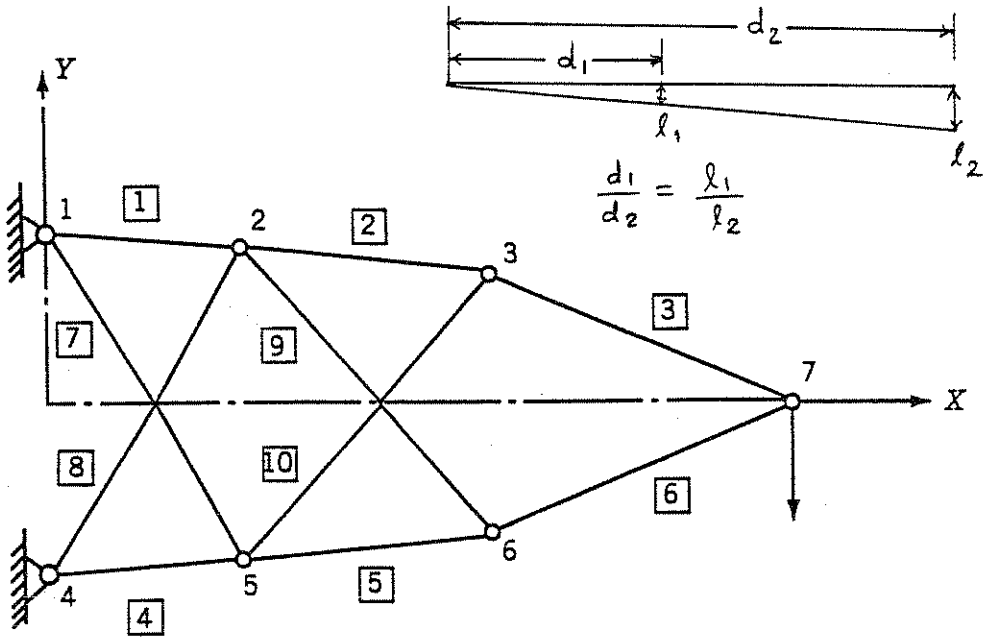
$$c_1^* = 0.04\ ,\ c_2^* = 0.81$$

$$\therefore \quad \vec{X}^* = \begin{Bmatrix} c_1^* + 0.5\ c_2^* \\ c_1^* + 0.5\ c_2^* \\ 0.5\ c_1^* + c_2^* \\ 0.5\ c_1^* + c_2^* \end{Bmatrix} = \begin{Bmatrix} 0.445 \\ 0.445 \\ 0.83 \\ 0.83 \end{Bmatrix} \text{ in}^2$$

14.2 Variables:

$X_1 = 0$, Y_1, X_2, Y_2, X_3, Y_3, $X_4 = 0$, $Y_4 = -Y_1$, $X_5 = X_2$,

$Y_5 = -Y_2$, $X_6 = X_3$, $Y_6 = -Y_3$, X_7, $Y_7 = 0$



Also, $\dfrac{X_3 - X_2}{X_3 - X_1} = \dfrac{Y_2 - Y_3}{Y_1 - Y_3}$ or $X_3 - X_2 = X_3\left(\dfrac{Y_2 - Y_3}{Y_1 - Y_3}\right)$

or $X_3 = X_2\left(\dfrac{Y_1 - Y_3}{Y_1 - Y_2}\right)$

∴ Independent variables: Y_1, X_2, Y_2, Y_3, X_7

Dependent variables:

$X_3 = X_2\left(\dfrac{Y_1 - Y_3}{Y_1 - Y_2}\right)$, $Y_4 = -Y_1$, $X_5 = X_2$, $Y_5 = -Y_2$,

$X_6 = X_3$, $Y_6 = -Y_3$, $X_1 = 0$, $X_4 = 0$, $Y_7 = 0$

14.3 SOLUTION OF SIMULTANEOUS LINEAR EQUATIONS

ORIGINAL COEFFICIENT MATRIX

```
 0.1574E+07   0.3578E+06  -0.2862E+06  -0.1431E+06
 0.3578E+06   0.3935E+06  -0.1431E+06  -0.7155E+05
-0.2862E+06  -0.1431E+06   0.3923E+06   0.2492E+06
-0.1431E+06  -0.7155E+05   0.2492E+06   0.1776E+06
```

RIGHT HAND SIDE VECTOR

```
 0.0000E+00   0.0000E+00   0.0000E+00  -0.1000E+04
```

INVERSE OF COEFFICIENT MATRIX

```
 0.8735E-06  -0.5823E-06   0.1165E-05  -0.1165E-05
-0.5823E-06   0.3494E-05   0.2329E-05  -0.2329E-05
 0.1165E-05   0.2329E-05   0.2806E-04  -0.3749E-04
-0.1165E-05  -0.2329E-05  -0.3749E-04   0.5635E-04
```

(a) SOLUTION VECTOR $(\vec{Y}_o)$:

```
 0.116462E-02   0.232924E-02   0.374901E-01  -0.563463E-01
```

NEW STIFFNESS MATRIX, [K0]+[DeltaK]

```
 0.186041E+07 0.329149E+06-0.228973E+06-0.114487E+06
 0.329149E+06 0.465102E+06-0.114487E+06-0.572433E+05
-0.228973E+06-0.114487E+06 0.292613E+06 0.178126E+06
-0.114487E+06-0.572433E+05 0.178126E+06 0.120883E+06
```

CHANGE IN STIFFNESS MATRIX, [DeltaK]

```
 0.286217E+06-0.286216E+05 0.572433E+05 0.286217E+05
-0.286216E+05 0.715542E+05 0.286217E+05 0.143108E+05
 0.572433E+05 0.286217E+05-0.996697E+05-0.710481E+05
 0.286217E+05 0.143108E+05-0.710481E+05-0.567372E+05
```

(b) EXACT SOLUTION AT NEW DESIGN VECTOR, $(\vec{Y}_0 + \Delta\vec{Y})$:

```
  0.970514E-03   0.194103E-02   0.527783E-01  -0.842052E-01
```

NEW SOLUTION VECTOR IN ITERATION, I= 1

```
 0.931694E-03 0.186339E-02 0.468959E-01-0.732945E-01
```

CORRECTION VECTOR, DeltaY, IN ITERATION I= 1

```
-0.232925E-03-0.465851E-03 0.940582E-02-0.169483E-01
```

NEW SOLUTION VECTOR IN ITERATION, I= 2

```
 0.978278E-03 0.195656E-02 0.506582E-01-0.800738E-01
```

CORRECTION VECTOR, DeltaY, IN ITERATION I= 2

```
 0.465846E-04 0.931691E-04 0.376232E-02-0.677929E-02
```

NEW SOLUTION VECTOR IN ITERATION, I= 3

```
 0.968961E-03 0.193792E-02 0.519395E-01-0.825619E-01
```

CORRECTION VECTOR, DeltaY, IN ITERATION I= 3

```
-0.931710E-05-0.186342E-04 0.128132E-02-0.248811E-02
```

NEW SOLUTION VECTOR IN ITERATION, I= 4

```
 0.970824E-03 0.194165E-02 0.524521E-01-0.835572E-01
```

CORRECTION VECTOR, DeltaY, IN ITERATION I= 4

```
 0.186335E-05 0.372669E-05 0.512526E-03-0.995240E-03
```

(c) NEW SOLUTION VECTOR IN ITERATION, I= 5, $(\vec{Y}_0 + \Delta\vec{Y})$:

```
 0.970452E-03 0.194090E-02 0.526481E-01-0.839463E-01
```

CORRECTION VECTOR, DeltaY, IN ITERATION I= 5

```
-0.372700E-06-0.745396E-06 0.196066E-03-0.389151E-03
```

NEW SOLUTION VECTOR IN ITERATION, I= 6

```
 0.970526E-03 0.194105E-02 0.527266E-01-0.841020E-01
```

CORRECTION VECTOR, DeltaY, IN ITERATION I= 6

```
 0.745304E-07 0.149057E-06 0.784261E-04-0.155660E-03
```

SOLUTION OF SIMULTANEOUS LINEAR EQUATIONS

14.4 ORIGINAL COEFFICIENT MATRIX:

```
 0.60000E+01  0.00000E+00 -0.40000E+01  0.00000E+00  0.00000E+00
 0.00000E+00 -0.10000E+01  0.17321E+01  0.00000E+00  0.00000E+00
 0.00000E+00  0.60000E+01  0.00000E+00  0.00000E+00  0.00000E+00
 0.00000E+00  0.17321E+01 -0.30000E+01  0.00000E+00  0.00000E+00
-0.40000E+01  0.00000E+00  0.10000E+02  0.00000E+00 -0.40000E+01
 0.00000E+00 -0.10000E+01 -0.17321E+01 -0.10000E+01  0.17321E+01
 0.00000E+00  0.00000E+00  0.00000E+00  0.60000E+01  0.00000E+00
 0.00000E+00 -0.17321E+01 -0.30000E+01  0.17321E+01 -0.30000E+01
 0.00000E+00  0.00000E+00 -0.40000E+01  0.00000E+00  0.60000E+01
 0.00000E+00  0.00000E+00  0.00000E+00 -0.10000E+01 -0.17321E+01
 0.00000E+00  0.00000E+00  0.00000E+00  0.00000E+00  0.00000E+00
 0.60000E+01  0.00000E+00  0.00000E+00 -0.17321E+01 -0.30000E+01
-0.10000E+01  0.17321E+01 -0.10000E+01 -0.17321E+01  0.00000E+00
 0.00000E+00  0.10000E+02  0.00000E+00 -0.40000E+01  0.00000E+00
 0.17321E+01 -0.30000E+01 -0.17321E+01 -0.30000E+01  0.00000E+00
 0.00000E+00  0.00000E+00  0.60000E+01  0.00000E+00  0.00000E+00
 0.00000E+00  0.00000E+00 -0.10000E+01  0.17321E+01 -0.10000E+01
-0.17321E+01 -0.40000E+01  0.00000E+00  0.10000E+02  0.00000E+00
 0.00000E+00  0.00000E+00  0.17321E+01 -0.30000E+01 -0.17321E+01
-0.30000E+01  0.00000E+00  0.00000E+00  0.00000E+00  0.60000E+01
```

RIGHT HAND SIDE VECTOR

```
 0.00000E+00 -0.66667E-02  0.00000E+00  0.00000E+00 -0.66667E-02
 0.00000E+00  0.00000E+00  0.00000E+00  0.00000E+00 -0.13333E-01
```

INVERSE OF COEFFICIENT MATRIX:

```
 0.36111E+00 -0.11226E+00  0.19444E+00 -0.14434E+00  0.11111E+00
-0.32075E-01  0.83333E-01 -0.17641E+00  0.83333E-01 -0.11226E+00
-0.11226E+00  0.34259E+00 -0.16038E-01  0.25000E+00  0.32075E-01
 0.74074E-01 -0.48113E-01  0.32407E+00 -0.48113E-01  0.17593E+00
 0.19444E+00 -0.16038E-01  0.27778E+00  0.10762E-07  0.19444E+00
 0.16038E-01  0.83333E-01  0.16038E-01  0.83333E-01 -0.16038E-01
-0.14434E+00  0.25000E+00  0.53810E-08  0.72222E+00  0.14434E+00
 0.25000E+00  0.48113E-01  0.52778E+00 -0.48113E-01  0.52778E+00
 0.11111E+00  0.32075E-01  0.19444E+00  0.14434E+00  0.36111E+00
 0.11226E+00  0.83333E-01  0.11226E+00  0.83333E-01  0.17641E+00
-0.32075E-01  0.74074E-01  0.16038E-01  0.25000E+00  0.11226E+00
 0.34259E+00  0.48113E-01  0.17593E+00  0.48113E-01  0.32407E+00
 0.83333E-01 -0.48113E-01  0.83333E-01  0.48113E-01  0.83333E-01
 0.48113E-01  0.16667E+00  0.37253E-08  0.83333E-01  0.48113E-01
-0.17641E+00  0.32407E+00  0.16038E-01  0.52778E+00  0.11226E+00
 0.17593E+00  0.18626E-07  0.64815E+00 -0.48113E-01  0.37963E+00
 0.83333E-01 -0.48113E-01  0.83333E-01 -0.48113E-01  0.83333E-01
 0.48113E-01  0.83333E-01 -0.48113E-01  0.16667E+00  0.74506E-08
-0.11226E+00  0.17593E+00 -0.16038E-01  0.52778E+00  0.17641E+00
 0.32407E+00  0.48113E-01  0.37963E+00 -0.22352E-07  0.64815E+00
```

(a) SOLUTION VECTOR, $\vec{U}_o$:

```
 0.15045E-02 -0.48435E-02 -0.97555E-03 -0.96660E-02 -0.49734E-02
-0.55632E-02 -0.87631E-03 -0.79706E-02 -0.23481E-03 -0.10991E-01
```

NEW STIFFNESS MATRIX, [K0]+[DeltaK]:

```
 0.62000E+01  0.00000E+00 -0.44000E+01  0.00000E+00  0.00000E+00
 0.00000E+00 -0.90000E+00  0.15588E+01  0.00000E+00  0.00000E+00
```

```
 0.00000E+00  0.54000E+01  0.00000E+00  0.00000E+00  0.00000E+00
 0.00000E+00  0.15588E+01 -0.27000E+01  0.00000E+00  0.00000E+00
-0.44000E+01  0.00000E+00  0.10600E+02  0.00000E+00 -0.44000E+01
 0.00000E+00 -0.90000E+00 -0.15588E+01 -0.90000E+00  0.15588E+01
 0.00000E+00  0.00000E+00  0.00000E+00  0.54000E+01  0.00000E+00
 0.00000E+00 -0.15588E+01 -0.27000E+01  0.15588E+01 -0.27000E+01
 0.00000E+00  0.00000E+00 -0.44000E+01  0.00000E+00  0.62000E+01
 0.00000E+00  0.00000E+00  0.00000E+00 -0.90000E+00 -0.15588E+01
 0.00000E+00  0.00000E+00  0.00000E+00  0.00000E+00  0.00000E+00
 0.54000E+01  0.00000E+00  0.00000E+00 -0.15588E+01 -0.27000E+01
-0.90000E+00  0.15588E+01 -0.90000E+00 -0.15588E+01  0.00000E+00
 0.00000E+00  0.10600E+02  0.00000E+00 -0.44000E+01  0.00000E+00
 0.15588E+01 -0.27000E+01 -0.15588E+01 -0.27000E+01  0.00000E+00
 0.00000E+00  0.00000E+00  0.54000E+01  0.00000E+00  0.00000E+00
 0.00000E+00  0.00000E+00 -0.90000E+00  0.15588E+01 -0.90000E+00
-0.15588E+01 -0.44000E+01  0.00000E+00  0.10600E+02  0.00000E+00
 0.00000E+00  0.00000E+00  0.15588E+01 -0.27000E+01 -0.15588E+01
-0.27000E+01  0.00000E+00  0.00000E+00  0.00000E+00  0.54000E+01
```

CHANGE IN STIFFNESS MATRIX, [DeltaK]:

```
 0.20000E+00  0.00000E+00 -0.40000E+00  0.00000E+00  0.00000E+00
 0.00000E+00  0.10000E+00 -0.17321E+00  0.00000E+00  0.00000E+00
 0.00000E+00 -0.60000E+00  0.00000E+00  0.00000E+00  0.00000E+00
 0.00000E+00 -0.17321E+00  0.30000E+00  0.00000E+00  0.00000E+00
-0.40000E+00  0.00000E+00  0.60000E+00  0.00000E+00 -0.40000E+00
 0.00000E+00  0.10000E+00  0.17321E+00  0.10000E+00 -0.17321E+00
 0.00000E+00  0.00000E+00  0.00000E+00 -0.60000E+00  0.00000E+00
 0.00000E+00  0.17321E+00  0.30000E+00 -0.17321E+00  0.30000E+00
 0.00000E+00  0.00000E+00 -0.40000E+00  0.00000E+00  0.20000E+00
 0.00000E+00  0.00000E+00  0.00000E+00  0.10000E+00  0.17321E+00
 0.00000E+00  0.00000E+00  0.00000E+00  0.00000E+00  0.00000E+00
-0.60000E+00  0.00000E+00  0.00000E+00  0.17321E+00  0.30000E+00
 0.10000E+00 -0.17321E+00  0.10000E+00  0.17321E+00  0.00000E+00
 0.00000E+00  0.60000E+00  0.00000E+00 -0.40000E+00  0.00000E+00
-0.17321E+00  0.30000E+00  0.17321E+00  0.30000E+00  0.00000E+00
 0.00000E+00  0.00000E+00 -0.60000E+00  0.00000E+00  0.00000E+00
 0.00000E+00  0.00000E+00  0.10000E+00 -0.17321E+00  0.10000E+00
 0.17321E+00 -0.40000E+00  0.00000E+00  0.60000E+00  0.00000E+00
 0.00000E+00  0.00000E+00 -0.17321E+00  0.30000E+00  0.17321E+00
 0.30000E+00  0.00000E+00  0.00000E+00  0.00000E+00 -0.60000E+00
```

(b) EXACT SOLUTION AT NEW DESIGN VECTOR: $\left(\vec{U}_0 + \Delta\vec{U}\right)$:

```
 0.11433E-02 -0.50765E-02 -0.11113E-02 -0.99096E-02 -0.47457E-02
-0.57309E-02 -0.79664E-03 -0.81439E-02 -0.21346E-03 -0.11339E-01
```

NEW SOLUTION VECTOR IN ITERATION, I= 1

```
 0.11318E-02 -0.50258E-02 -0.11002E-02 -0.98105E-02 -0.46983E-02
-0.56736E-02 -0.78868E-03 -0.80625E-02 -0.21132E-03 -0.11225E-01
```

CORRECTION VECTOR, Delta Y, in ITERATION, I= 1

```
-0.37267E-03 -0.18232E-03 -0.12467E-03 -0.14452E-03  0.27512E-03
-0.11034E-03  0.87631E-04 -0.91825E-04  0.23480E-04 -0.23424E-03
```

NEW SOLUTION VECTOR IN ITERATION, I= 2

```
 0.11469E-02 -0.50742E-02 -0.11100E-02 -0.99071E-02 -0.47480E-02
-0.57292E-02 -0.79744E-03 -0.81422E-02 -0.21367E-03 -0.11335E-01
```

```
CORRECTION VECTOR, Delta Y, in ITERATION, I=  2

      0.15045E-04 -0.48435E-04 -0.97555E-05 -0.96660E-04 -0.49734E-04
     -0.55632E-04 -0.87631E-05 -0.79707E-04 -0.23480E-05 -0.10991E-03

NEW SOLUTION VECTOR IN ITERATION, I=  3

     0.11432E-02 -0.50760E-02 -0.11112E-02 -0.99086E-02 -0.47453E-02
    -0.57303E-02 -0.79656E-03 -0.81431E-02 -0.21344E-03 -0.11337E-01

CORRECTION VECTOR, Delta Y, in ITERATION, I=  3

      -0.37267E-05 -0.18232E-05 -0.12467E-05 -0.14452E-05  0.27512E-05
      -0.11034E-05  0.87630E-06 -0.91825E-06  0.23480E-06 -0.23424E-05

NEW SOLUTION VECTOR IN ITERATION, I=  4

     0.11433E-02 -0.50765E-02 -0.11113E-02 -0.99095E-02 -0.47458E-02
    -0.57309E-02 -0.79665E-03 -0.81439E-02 -0.21346E-03 -0.11338E-01

CORRECTION VECTOR, Delta Y, in ITERATION, I=  4

      0.15045E-06 -0.48435E-06 -0.97555E-07 -0.96660E-06 -0.49734E-06
     -0.55633E-06 -0.87630E-07 -0.79707E-06 -0.23480E-07 -0.10991E-05

NEW SOLUTION VECTOR IN ITERATION, I=  5

     0.11433E-02 -0.50765E-02 -0.11113E-02 -0.99096E-02 -0.47457E-02
    -0.57309E-02 -0.79664E-03 -0.81439E-02 -0.21346E-03 -0.11339E-01

CORRECTION VECTOR, Delta Y, in ITERATION, I=  5

      -0.37268E-07 -0.18232E-07 -0.12467E-07 -0.14452E-07  0.27512E-07
      -0.11034E-07  0.87630E-08 -0.91826E-08  0.23480E-08 -0.23424E-07
```

14.5 – 14.8

$$\frac{\partial \vec{Y}}{\partial x_i} = [K]^{-1}\left(\frac{\partial \vec{P}}{\partial x_i} - \frac{\partial [K]}{\partial x_i}\vec{Y}\right) \qquad (13.35)$$

Here $\frac{\partial \vec{P}}{\partial x_i} = 0$ and hence Eq. (13.35) can be written as

$$\frac{\partial \vec{Y}}{\partial x_i} \simeq -[K]^{-1}\frac{\Delta [K]}{\Delta x_i}\vec{Y} = -[K]_b^{-1}\frac{[K]_n - [K]_b}{\Delta x_i}\vec{Y}_b$$

where subscripts "b" and "n" denote "base" and "new" quantities.

BASE STIFFNESS MATRIX: $[K]_b$:

```
 0.186041E+07 0.715542E+05-0.143108E+06-0.715542E+05
 0.715542E+05 0.465102E+06-0.715542E+05-0.357771E+05
-0.143108E+06-0.715542E+05 0.249174E+06 0.177620E+06
-0.715542E+05-0.357771E+05 0.177620E+06 0.141843E+06
```

INVERSE OF BASE STIFFNESS MATRIX

```
 0.5823E-06  -0.7461E-13   0.1165E-05  -0.1165E-05
-0.8527E-13   0.2329E-05   0.2329E-05  -0.2329E-05
 0.1165E-05   0.2329E-05   0.4204E-04  -0.5147E-04
-0.1165E-05  -0.2329E-05  -0.5147E-04   0.7032E-04
```

BASE SOLUTION VECTOR: $\vec{Y}_b$:

```
 0.1165E-02   0.2329E-02   0.5147E-01  -0.7032E-01
```

14.5

```
SENSITIVITY OF SOLUTION W.R.T. VARIABLE:     1
AREA VECTOR:      2.0100      2.0000      1.0000      1.0000
```

NEW STIFFNESS MATRIX (AT PERTURBED VECTOR), $[K]_n$:

```
 0.186470E+07 0.737008E+05-0.143108E+06-0.715542E+05
 0.737008E+05 0.466175E+06-0.715542E+05-0.357771E+05
-0.143108E+06-0.715542E+05 0.249174E+06 0.177620E+06
-0.715542E+05-0.357771E+05 0.177620E+06 0.141843E+06
```

CHANGE IN STIFFNESS MATRIX, [Delta K]:

```
 0.429338E+04 0.214663E+04 0.000000E+00 0.000000E+00
 0.214663E+04 0.107334E+04 0.000000E+00 0.000000E+00
 0.000000E+00 0.000000E+00 0.000000E+00 0.000000E+00
 0.000000E+00 0.000000E+00 0.000000E+00 0.000000E+00
```

```
DYI/DXJ  -0.582318E-03  -0.116464E-02  -0.232927E-02   0.232927E-02
```

$\Rightarrow \frac{\partial \vec{Y}}{\partial x_1}$

14.6

SENSITIVITY OF SOLUTION W.R.T. VARIABLE: 2
AREA VECTOR: 2.0000 2.0100 1.0000 1.0000

NEW STIFFNESS MATRIX (AT PERTURBED VECTOR), $[K]_n$:

```
 0.186470E+07 0.694075E+05-0.143108E+06-0.715542E+05
 0.694075E+05 0.466175E+06-0.715542E+05-0.357771E+05
-0.143108E+06-0.715542E+05 0.249174E+06 0.177620E+06
-0.715542E+05-0.357771E+05 0.177620E+06 0.141843E+06
```

CHANGE IN STIFFNESS MATRIX, [Delta K]:

```
 0.429338E+04-0.214663E+04 0.000000E+00 0.000000E+00
-0.214663E+04 0.107334E+04 0.000000E+00 0.000000E+00
 0.000000E+00 0.000000E+00 0.000000E+00 0.000000E+00
 0.000000E+00 0.000000E+00 0.000000E+00 0.000000E+00
```

DYI/DXJ -0.746923E-08 -0.147163E-07 -0.296548E-07 0.296547E-07 $\Rightarrow \frac{\partial \vec{Y}}{\partial x_2}$

14.7

SENSITIVITY OF SOLUTION W.R.T. VARIABLE: 3
AREA VECTOR: 2.0000 2.0000 1.0100 1.0000

NEW STIFFNESS MATRIX (AT PERTURBED VECTOR), $[K]_n$:

```
 0.186184E+07 0.722697E+05-0.144539E+06-0.722697E+05
 0.722697E+05 0.465460E+06-0.722697E+05-0.361349E+05
-0.144539E+06-0.722697E+05 0.250605E+06 0.178336E+06
-0.722697E+05-0.361349E+05 0.178336E+06 0.142201E+06
```

CHANGE IN STIFFNESS MATRIX, [Delta K]:

```
 0.143112E+04 0.715539E+03-0.143108E+04-0.715539E+03
 0.715539E+03 0.357781E+03-0.715539E+03-0.357770E+03
-0.143108E+04-0.715539E+03 0.143109E+04 0.715547E+03
-0.715539E+03-0.357770E+03 0.715547E+03 0.357766E+03
```

DYI/DXJ 0.469299E-07 0.947693E-07 -0.279483E-01 0.279472E-01 $\Rightarrow \frac{\partial \vec{Y}}{\partial x_3}$

14.8

SENSITIVITY OF SOLUTION W.R.T. VARIABLE: 4
AREA VECTOR: 2.0000 2.0000 1.0000 1.0100

NEW STIFFNESS MATRIX (AT PERTURBED VECTOR), $[K]_n$:

```
 0.186041E+07 0.715542E+05-0.143108E+06-0.715542E+05
 0.715542E+05 0.465102E+06-0.715542E+05-0.357771E+05
-0.143108E+06-0.715542E+05 0.250235E+06 0.178681E+06
-0.715542E+05-0.357771E+05 0.178681E+06 0.142904E+06
```

CHANGE IN STIFFNESS MATRIX, [Delta K]:

```
 0.000000E+00 0.000000E+00 0.000000E+00 0.000000E+00
 0.000000E+00 0.000000E+00 0.000000E+00 0.000000E+00
 0.000000E+00 0.000000E+00 0.106064E+04 0.106066E+04
 0.000000E+00 0.000000E+00 0.106066E+04 0.106064E+04
```

DYI/DXJ 0.222826E-06 0.445289E-06 -0.188470E-01 0.377002E-01 $\Rightarrow \frac{\partial \vec{Y}}{\partial x_4}$

14.9 (a) For given data,

$$k_{11} = \frac{A_1 E_1}{\ell_1} + \frac{A_2 E_2}{\ell_2} = \frac{(2)(30\times10^6)}{25} + \frac{(1)(30\times10^6)}{50} = 0.3\times10^7$$

$$k_{12} = k_{21} = -\frac{A_2 E_2}{\ell_2} = -\frac{(1)(30\times10^6)}{50} = -0.6\times10^6$$

$$k_{22} = \frac{A_2 E_2}{\ell_2} = \frac{(1)(30\times10^6)}{50} = 0.6\times10^6$$

$$[K] = \begin{bmatrix} 3 & -0.6 \\ -0.6 & 0.6 \end{bmatrix} \times 10^6$$

$$[K]^{-1} = [a] = \begin{bmatrix} 0.416667 & 0.416667 \\ 0.416667 & 2.08333 \end{bmatrix} \times 10^{-6}$$

$$\vec{Y} = [K]^{-1}\vec{P} = \begin{Bmatrix} 0.000125 \\ 0.000458333 \end{Bmatrix} \text{ in.}$$

(b) $\vec{P} = \begin{Bmatrix} 100 \\ 200 \end{Bmatrix}$

$$\frac{\partial \vec{P}}{\partial A_1} = \frac{\partial \vec{P}}{\partial A_2} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

Exact derivatives:

$$\frac{\partial [K]}{\partial A_1} = \begin{bmatrix} E_1/\ell_1 & 0 \\ 0 & 0 \end{bmatrix}, \quad \frac{\partial [K]}{\partial A_2} = \begin{bmatrix} E_2/\ell_2 & -E_2/\ell_2 \\ -E_2/\ell_2 & E_2/\ell_2 \end{bmatrix}$$

Since $\frac{E_1}{\ell_1} = \frac{30\times10^6}{25} = 1.2\times10^6$, $\frac{E_2}{\ell_2} = \frac{30\times10^6}{50} = 0.6\times10^6$,

$$\frac{\partial [K]}{\partial A_1} = 1.2\times10^6 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad \frac{\partial [K]}{\partial A_2} = 0.6\times10^6 \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$\frac{\partial \vec{Y}}{\partial A_1} = [K]^{-1}\left(\frac{\partial \vec{P}}{\partial A_1} - \frac{\partial [K]}{\partial A_1}\vec{Y}\right) = \begin{Bmatrix} -0.000229167 \\ -0.000229167 \end{Bmatrix}$$

$$\frac{\partial \vec{Y}}{\partial A_2} = [K]^{-1}\left(\frac{\partial \vec{P}}{\partial A_2} - \frac{\partial [K]}{\partial A_2}\vec{Y}\right) = \begin{Bmatrix} 0.0 \\ 0.000333333 \end{Bmatrix}$$

(c) $\epsilon_1 = \dfrac{y_1}{\ell_1}$, $\sigma_1 = E_1 \epsilon_1 = \dfrac{E_1 y_1}{\ell_1}$

$\epsilon_2 = \dfrac{y_2 - y_1}{\ell_2}$, $\sigma_2 = E_2 \epsilon_2 = \dfrac{E_2}{\ell_2}(y_2 - y_1)$

$$\vec{\sigma} = [R]\vec{Y} \equiv \begin{bmatrix} E_1/\ell_1 & 0 \\ -E_2/\ell_2 & E_2/\ell_2 \end{bmatrix} \begin{Bmatrix} y_1 \\ y_2 \end{Bmatrix}$$

$$\frac{\partial \vec{\sigma}}{\partial A_i} = [R] \frac{\partial \vec{Y}}{\partial A_i} \quad , \quad i = 1,2$$

Here $$[R] = \begin{bmatrix} 1.2 \times 10^6 & 0 \\ -0.6 \times 10^6 & 0.6 \times 10^6 \end{bmatrix}$$

$$\frac{\partial \vec{\sigma}}{\partial A_1} = [R] \frac{\partial \vec{Y}}{\partial A_1} = 0.6 \times 10^6 \begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} \partial y_1/\partial A_1 \\ \partial y_2/\partial A_1 \end{Bmatrix}$$

Since $\dfrac{\partial y_1}{\partial A_1} = \dfrac{\partial y_2}{\partial A_2} = -0.000229167$, we obtain

$$\frac{\partial \vec{\sigma}}{\partial A_1} = \begin{Bmatrix} -275.0 \\ 0.0 \end{Bmatrix}$$

$$\frac{\partial \vec{\sigma}}{\partial A_2} = [R] \frac{\partial \vec{Y}}{\partial A_2} = 0.6 \times 10^6 \begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} \partial y_1/\partial A_2 \\ \partial y_2/\partial A_2 \end{Bmatrix}$$

Since $\dfrac{\partial y_1}{\partial A_2} = 0.0$ and $\dfrac{\partial y_2}{\partial A_2} = 0.000333333$, we obtain

$$\frac{\partial \vec{\sigma}}{\partial A_2} = \begin{Bmatrix} 0.0 \\ 200.0 \end{Bmatrix}$$

14.10 (a) Let $[K] = \begin{bmatrix} a_1 & -a_2 \\ -a_2 & a_2 \end{bmatrix}$, $[M] = \begin{bmatrix} b_1 & b_2 \\ b_2 & b_2 \end{bmatrix}$

$$\left| [K] - \lambda [M] \right| = 0$$

$$\begin{vmatrix} a_1 - \lambda b_1 & -a_2 - \lambda b_2 \\ -a_2 - \lambda b_2 & a_2 - \lambda b_2 \end{vmatrix} = 0$$

or $\lambda^2 (b_1 b_2 - b_2^2) + \lambda (-b_1 a_2 - b_2 a_1 - 2 a_2 b_2) + (a_1 a_2 - a_2^2) = 0$ (1)

For given data,

$a_1 = 0.3 \times 10^7$, $a_2 = 0.6 \times 10^6$,

$b_1 = 2\left(\frac{0.283}{386.4}\right)(2)(25) + \left(\frac{0.283}{386.4}\right)(1)(50)$

$= 0.07324017 + 0.03662008 = 0.10986025$

$b_2 = 0.03662008$

Eq. (1) becomes $0.241385 \times 10^{10}\, \lambda^2 - 0.171382 \times 10^{12} + 0.144 \times 10^{13} = 0$

Roots: $\lambda_1 = 9.73788$, $\lambda_2 = 61.2614$

$\vec{Y}^{(1)} = \begin{Bmatrix} 0.474272 \\ 1.0 \end{Bmatrix}$, $\vec{Y}^{(2)} = \begin{Bmatrix} -0.383363 \\ 1.0 \end{Bmatrix}$

[M]-orthogonal eigenvectors:

$\vec{Y}^{(1)} = \begin{Bmatrix} 0.00176180 \\ 0.00371475 \end{Bmatrix}$, $\vec{Y}^{(2)} = \begin{Bmatrix} -0.00244919 \\ 0.00638870 \end{Bmatrix}$

(b) $\frac{\partial [K]}{\partial A_1} = \frac{E_1}{\ell_1} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = 1.2 \times 10^6 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

$\frac{\partial [K]}{\partial A_2} = \frac{E_2}{\ell_2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = 0.6 \times 10^6 \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$

$\frac{\partial [M]}{\partial A_1} = \begin{bmatrix} 2\rho_1 \ell_1 & 0 \\ 0 & 0 \end{bmatrix} = 0.03662 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

$\frac{\partial [M]}{\partial A_2} = \rho_2 \ell_2 \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = 0.03662 \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

$\lambda_1 = 9.73788$, $\lambda_2 = 61.2614$

$$\vec{Y}_1 \equiv \vec{Y}^{(1)} = \begin{Bmatrix} 0.00176180 \\ 0.00371475 \end{Bmatrix}, \quad \vec{Y}_2 \equiv \vec{Y}^{(2)} = \begin{Bmatrix} -0.00244919 \\ 0.00638870 \end{Bmatrix}$$

$$\frac{\partial \lambda_1}{\partial A_1} = \vec{Y}_1^T \left[\frac{\partial [K]}{\partial A_1} - \lambda_1 \frac{\partial [M]}{\partial A_1} \right] \vec{Y}_1 = 3.72473$$

$$\frac{\partial \lambda_2}{\partial A_1} = \vec{Y}_2^T \left[\frac{\partial [K]}{\partial A_1} - \lambda_2 \frac{\partial [M]}{\partial A_1} \right] \vec{Y}_2 = 7.19822$$

$$[P_i] = [K] - \lambda_i [M] \quad ; \; i = 1, 2$$

(c) Eq. (13.51) gives:

$$\frac{\partial \vec{Y}_1}{\partial A_1} = \begin{Bmatrix} 0.8809 \times 10^{-3} \\ 0.8809 \times 10^{-3} \end{Bmatrix}$$

$$\frac{\partial \vec{Y}_2}{\partial A_1} = \begin{Bmatrix} -0.122460 \times 10^{-2} \\ -0.122460 \times 10^{-2} \end{Bmatrix}$$

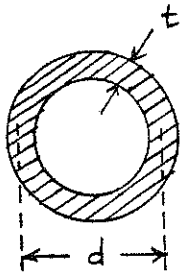
14.11 (a) $$\frac{\partial \lambda_1}{\partial A_2} = 2.28840 = \vec{Y}_1^T \left[\frac{\partial [K]}{\partial A_2} - \lambda_1 \frac{\partial [M]}{\partial A_2} \right] \vec{Y}_1$$

$$\frac{\partial \lambda_2}{\partial A_2} = 46.8649 = \vec{Y}_2^T \left[\frac{\partial [K]}{\partial A_2} - \lambda_2 \frac{\partial [M]}{\partial A_2} \right] \vec{Y}_2$$

(b) $$\frac{\partial \vec{Y}_1}{\partial A_2} = \begin{Bmatrix} -0.162981 \times 10^{-8} \\ 0.195295 \times 10^{-2} \end{Bmatrix}$$

$$\frac{\partial \vec{Y}_2}{\partial A_2} = \begin{Bmatrix} 0.698492 \times 10^{-8} \\ 0.883790 \times 10^{-2} \end{Bmatrix}$$

14.12 $I = \frac{\pi}{8} d t \sqrt{d^2 + t^2} = \frac{\pi}{64}(d_o^4 - d_i^4)$



$$6 Y_1 - 3 l Y_2 = \frac{P_1 l^3}{2 E I} \qquad (1)$$

$$-3 l Y_1 + 2 l^2 Y_2 = \frac{P_2 l^2}{2 E I} \qquad (2)$$

Solution of Eqs. (1) and (2) gives:

$$Y_1 = \frac{P_1 l^3}{3 E I}, \quad Y_2 = \frac{P_1 l^2}{2 E I} \qquad (3)$$

or

$$Y_1 = \frac{8 P_1 l^3}{3 \pi E d t (d^2 + t^2)}, \quad Y_2 = \frac{8 P_1 l^2}{2 \pi E d t (d^2 + t^2)} \qquad (4)$$

since $I = \frac{1}{8} \pi d t (d^2 + t^2)$.

Using $P_1 = 100$, $l = 20$, $E = 30 \times 10^6$, $d = 2$, $t = 0.1$, we obtain

$$Y_1 = 0.028224 \text{ in}, \quad Y_2 = 0.002117 \text{ rad}$$

$$\frac{\partial Y_1}{\partial d} = \frac{3\pi E d t (d^2 + t^2)(0) - 8 P_1 l^3 (3 \pi E t)(3 d^2 + t^2)}{\{3 \pi E d t (d^2 + t^2)\}^2}$$

$$= -\frac{8}{3} \frac{P_1 l^3 (3 d^2 + t^2)}{\pi E d^2 t (d^2 + t^2)^2}$$

$$\frac{\partial Y_2}{\partial d} = \frac{2\pi E d t (d^2 + t^2)(0) - 8 P_1 l^2 (2 \pi E t)(3 d^2 + t^2)}{2 \pi E d t (d^2 + t^2)^2}$$

$$= -4 \frac{P_1 l^2 (3 d^2 + t^2)}{\pi E t d^2 (d^2 + t^2)^2}$$

$$\frac{\partial Y_1}{\partial t} = -\frac{8}{3} \frac{P_1 l^3 (3 t^2 + d^2)}{\pi E t^2 d (d^2 + t^2)^2}$$

$$\frac{\partial Y_2}{\partial t} = -4 \frac{P_1 l^2 (3 t^2 + d^2)}{\pi E d t^2 (d^2 + t^2)^2}$$

For given data, we obtain $\frac{\partial Y_1}{\partial d} = -0.042265$, $\frac{\partial Y_1}{\partial t} = -0.283643$, $\frac{\partial Y_2}{\partial d} = -0.003194$, and $\frac{\partial Y_2}{\partial t} = -0.021273$.

14.13 (a)

$$\left| \frac{2EI}{\ell^3}\begin{bmatrix} 6 & -3\ell \\ -3\ell & 2\ell^2 \end{bmatrix} - \frac{\lambda \rho A \ell}{420}\begin{bmatrix} 156 & -22\ell \\ -22\ell & 4\ell^2 \end{bmatrix} \right| = 0$$

or

$$\begin{vmatrix} 6-156\alpha & -3\ell+22\ell\alpha \\ -3\ell+22\ell\alpha & 2\ell^2-4\ell^2\alpha \end{vmatrix} = 0$$

or $$2\ell^2(6-156\alpha)(1-2\alpha) - \ell^2(-3+22\alpha)^2 = 0$$

or $$140\,\alpha^2 - 204\,\alpha + 3 = 0 \qquad (1)$$

where $$\alpha = \frac{\lambda \rho A \ell^4}{840\,EI} \qquad (2)$$

Roots of Eq. (1): $\alpha_1 = 0.0148575$, $\alpha_2 = 1.442285$ (3)

For given data,

$$\rho = \frac{0.283}{386.4}, \quad \ell = 20, \quad E = 30 \times 10^6, \quad A = \pi\, dt = \pi(2)(0.1) = 0.62832$$

$$I = \frac{\pi}{8}\, dt\,(d^2+t^2) = \frac{\pi}{8}(2)(0.1)\left(2^2+0.1^2\right) = 0.3149454$$

$$\frac{\rho A \ell^4}{EI} = 7.7928 \times 10^{-6}$$

From Eqs. (2) and (3),

$$\lambda_1 = \alpha_1(840)\frac{EI}{\rho A \ell^4} = 1.601516785 \times 10^6$$

$$\lambda_2 = \alpha_2(840)\frac{EI}{\rho A \ell^4} = 155.4665075 \times 10^6$$

Eigenvectors:

$\alpha_1 = 0.0148575$, $\ell = 20$:

$$\begin{bmatrix} 6-156\alpha_1 & -3\ell+22\ell\alpha_1 \\ -3\ell+22\ell\alpha_1 & 2\ell^2-4\ell^2\alpha_1 \end{bmatrix}\begin{Bmatrix} Y_1 \\ Y_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

$$\begin{bmatrix} 3.68223 & -53.4627 \\ -53.4627 & 776.228 \end{bmatrix}\begin{Bmatrix} Y_1 \\ Y_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \qquad (4)$$

Let $Y_2 = 1$

Eq. (4) gives $3.68223\ Y_1 = 53.4627\ Y_2$ or $Y_1 = 14.519109$

$$\vec{Y}^{(1)} = \begin{Bmatrix} 14.519109 \\ 1.0 \end{Bmatrix}$$

$\alpha_2 = 1.442285$, $\ell = 20$:

$$\begin{bmatrix} 6 - 156\,\alpha_2 & -3\ell + 22\ell\alpha_2 \\ -3\ell + 22\ell\alpha_2 & 2\ell^2 - 4\ell^2\alpha_2 \end{bmatrix} \begin{Bmatrix} Y_1 \\ Y_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

$$\begin{bmatrix} -218.99646 & 574.6054 \\ 574.6054 & -1507.656 \end{bmatrix} \begin{Bmatrix} Y_1 \\ Y_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \qquad (5)$$

Let $Y_2 = 1$

Eq. (5) gives $218.99646\ Y_1 = 574.6054\ Y_2$

or $Y_1 = 2.623811$

$$\vec{Y}^{(2)} = \begin{Bmatrix} 2.623811 \\ 1.0 \end{Bmatrix}$$

(b) $$\lambda_1 = 840\,\alpha_1 \frac{EI}{\rho A \ell^4} = 12.4803\,(E)\,\frac{\pi}{8}\frac{dt\,(d^2+t^2)}{\rho\,\pi\,dt\,(\ell^4)}$$

$$= 1.5600375\ E\,(d^2+t^2)/(\rho\ell^4)$$

$$\frac{\partial\lambda_1}{\partial d} = \frac{3.120075\ E\ d}{\rho\ell^4} = 1.5975225 \times 10^6$$

$$\frac{\partial\lambda_1}{\partial t} = \frac{3.120075\ E\ t}{\rho\ell^4} = 7.9876125 \times 10^4$$

$$\lambda_2 = 840\,\alpha_2 \frac{EI}{\rho A \ell^4} = 1211.5194\,\frac{EI}{\rho A \ell^4}$$

$$= 151.439925\ E\,(d^2+t^2)/(\rho\ell^4)$$

$$\frac{\partial\lambda_2}{\partial d} = 151.439925\,(2)\,Ed/(\rho\ell^4) = 155.07876422 \times 10^6$$

$$\frac{\partial\lambda_2}{\partial t} = 151.439925\,(2)\,Et/(\rho\ell^4) = 7.7539382 \times 10^6$$

14.15 $\lambda_1 = \omega_1^2 = \dfrac{k}{m} = \dfrac{d^4 G}{8 D^3 n m}$

$$\frac{\partial (\omega_1^2)}{\partial D} = 2\omega_1 \frac{\partial \omega_1}{\partial D} = \frac{d^4 G}{8 n m}\left(-\frac{3}{D^4}\right)$$

$$\frac{\partial \omega_1}{\partial D} = \frac{1}{2\omega_1}\left(-\frac{3 d^4 G}{8 n m D^4}\right) = -\frac{3\sqrt{2}}{8} d^2 \sqrt{\frac{G}{n m D^5}}$$

$$\lambda_2 = \omega_2^2 = \frac{3 d^4 G}{8 D^3 n m}$$

$$\frac{\partial(\omega_2^2)}{\partial D} = 2\omega_2 \frac{\partial \omega_2}{\partial D} = \frac{3 d^4 G}{8 n m}\left(-\frac{3}{D^4}\right)$$

$$\frac{\partial \omega_2}{\partial D} = \frac{1}{2\omega_2}\left(-\frac{9 d^4 G}{8 n m D^4}\right) = -\frac{3\sqrt{6}}{8} d^2 \sqrt{\frac{G}{n m D^5}}$$

For given data,

$$\sqrt{\frac{G}{n m D^5}} = \sqrt{\frac{11.5 * 10^6}{10\,(32.2)\,(0.4)^5}} = 1867.5447$$

$$\therefore \quad \frac{\partial \omega_1}{\partial D} = -1.584664$$

$$\frac{\partial \omega_2}{\partial D} = -2.744719$$

14.16 The stresses induced in the bars can be expressed as

$$\sigma_1 = \frac{P\sqrt{y^2+81}}{10\,y\,A_1} \le \sigma_0 \quad , \quad \sigma_2 = \frac{9P\sqrt{y^2+1}}{10\,y\,A_2} \le \sigma_0 \qquad (1)$$

Thus the constraints are

$$\frac{P\sqrt{y^2+81}}{10\,y\,z_1\,\sigma_0} - 1 \le 0 \quad , \quad \frac{9P\sqrt{y^2+1}}{10\,y\,z_2\,\sigma_0} - 1 \le 0 \qquad (2)$$

where $z_1 = A_1$ and $z_2 = A_2$.

Problem: Find $\vec{x} = \{y, z_1, z_2\}$

to minimize $f(\vec{x}) = z_1\sqrt{y^2+81} + z_2\sqrt{y^2+1}$ (3)

subject to

constraints of Eq. (2) and

$1 \le y \le 4$, $0 \le z_i \le 0.2$, $i = 1, 2$

Subproblem 1: Find z_1 which minimizes

$$f^{(1)}(y^*, z_1) = z_1\sqrt{(y^*)^2+81}$$

subject to (4)

$$g_1 = \frac{1000\sqrt{(y^*)^2+81}}{10\,y\,z_1\,(10^5)} - 1 \le 0$$

and $0 \le z_1 \le 0.2$

Subproblem 2: Find z_2 which minimizes

$$f^{(2)}(y^*, z_2) = z_2\sqrt{(y^*)^2+1}$$

subject to (5)

$$g_2 = \frac{9000\sqrt{(y^*)^2+1}}{10\,y\,z_1\,(10^5)} - 1 \le 0$$

and $0 \le z_2 \le 0.2$

To minimize $f^{(1)}$, we need to make z_1 as small as possible without violating the constraints in Eqs. (4) and (5). This gives the solution of subproblem 1, z_1^*, (which makes g_1 active) as:

$$z_1^* = \frac{10^{-3}\sqrt{(y^*)^2 + 81}}{y^*} \tag{6}$$

Similarly, the solution of subproblem 2, z_2^*, (which makes g_2 active) is:

$$z_2^* = \frac{9 \times 10^{-3}\sqrt{(y^*)^2 + 1}}{y^*} \tag{7}$$

Now we state the second level problem as:

$$\left.\begin{array}{ll} \text{Find } y \text{ to minimize} & f = f^{(1)}(y, z_1^*) + f^{(2)}(y, z_2^*) \\ \text{subject to} & 1 \le y \le 4 \end{array}\right\} \tag{8}$$

Using Eqs. (6) and (7), this problem can be restated as (using y for y^*):

Find y which minimizes

$$f = z_1^*\sqrt{y^2 + 81} + z_2^*\sqrt{y^2 + 1}$$

$$= 10^{-3}\left(\frac{y^2 + 81}{y}\right) + 9 \times 10^{-3}\left(\frac{y^2 + 1}{y}\right) \tag{9}$$

subject to $1 \le y \le 4$

and f must be defined.

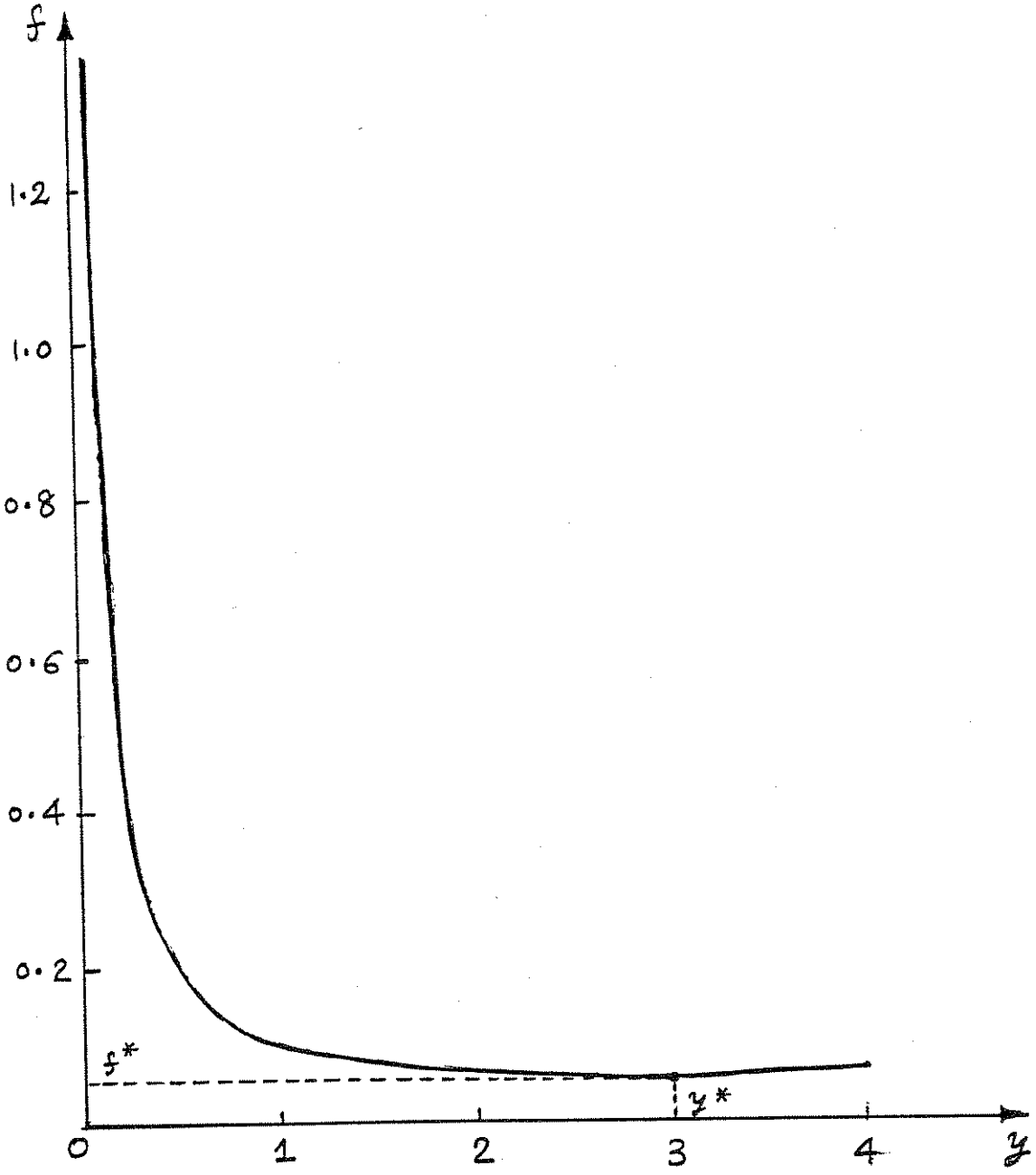
By varying y from 1 to 4, we minimize f of Eq. (9) and obtain the solution:

$f^* = 0.06 \ m^3$

$y^* = h^* = 3.0 \ m$

$z_1^* = A_1^* = 0.00316228 \ m^2$

$z_2^* = A_2^* = 0.00948683 \ m^2$


f
1.2
1.0
0.8
0.6
0.4
0.2
f*
y*
0
1
2
3
4
y

14.18

Subproblem 1:

Find $z_1 = A_1$ which minimizes

$$f^{(1)}(x^*, z_1) = 28.3\, z_1 \sqrt{1 + (x^*)^2} \tag{1}$$

subject to

$$g_1 = \frac{0.1768\,(1 + x^*)\sqrt{1 + (x^*)^2}}{z_1\, x^*} - 1 \le 0 \tag{2}$$

$$1.0 \le z_1 \le 2.5 \tag{3}$$

Subproblem 2:

Find $z_2 = A_2$ which minimizes

$$f^{(2)}(x^*, z_2) = 14.15\, z_2 \sqrt{1 + (x^*)^2} \tag{4}$$

subject to

$$g_2 = \frac{0.1768\,|x^* - 1|\sqrt{1 + (x^*)^2}}{z_2\, x^*} - 1 \le 0 \tag{5}$$

$$1.0 \le z_2 \le 2.5 \tag{6}$$

To minimize $f^{(1)}$, we need to make z_1 as small as possible without violating the constraints, Eqs. (2) and (3). This gives z_1^* as:

$$z_1^* = \frac{0.1768\,(1 + x^*)\sqrt{1 + (x^*)^2}}{x^*} \tag{7}$$

Similarly, the solution of subproblem 2, z_2^*, (which makes z_2 active) is:

$$z_2^* = \frac{0.1768\,|x^* - 1|\sqrt{1 + (x^*)^2}}{x^*} \tag{8}$$

Now we state the second level problem as:

$$\left.\begin{aligned} &\text{Find } x \text{ to minimize} \\ &f = f^{(1)}(x, z_1^*) + f^{(2)}(x, z_2^*) \\ &\text{subject to} \quad 0.1 \le x \le 2.5 \end{aligned}\right\} \tag{9}$$

Using Eqs. (7) and (8), this problem can be restated as (using x for x^*):

Find x which minimizes

$$f = z_1^* \left(28.3\sqrt{1+x^2}\right) + z_2^* \left(14.15\sqrt{1+x^2}\right)$$

$$= 28.3\sqrt{1+x^2}\left(\frac{0.1768\,(x+1)\sqrt{1+x^2}}{x}\right)$$

$$+ 14.15\sqrt{1+x^2}\left(\frac{0.1768\,|x-1|\sqrt{1+x^2}}{x}\right) \qquad (10)$$

subject to

$0.1 \le x \le 2.5$

and f must be defined.

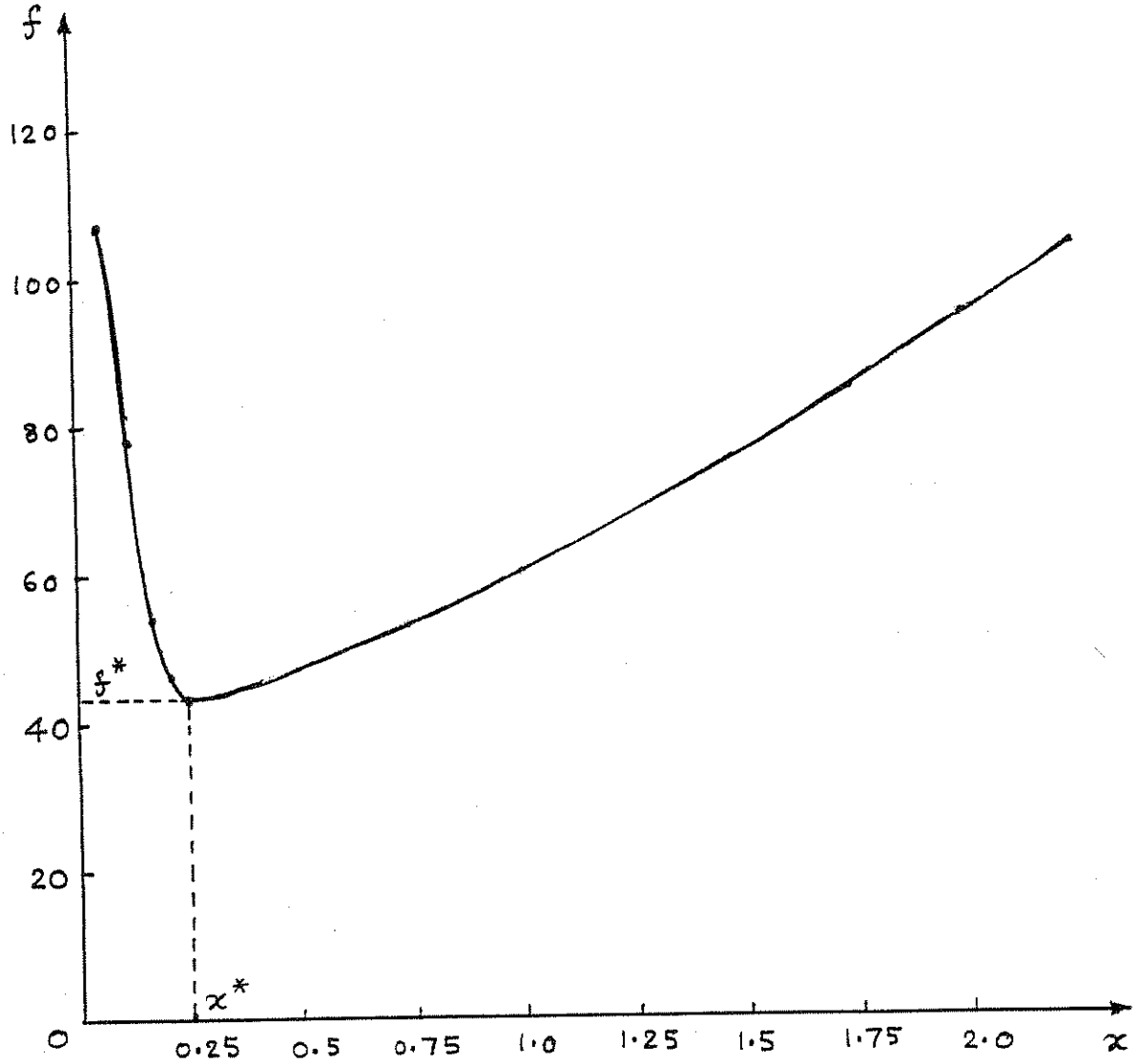
By varying x from 0.1 to 2.5, we find the solution of problem, Eq. (10), as:

$f^* = 43.7565$

$x^* = 0.25$

$z_1^* = A_1^* = 1.0$

$z_2^* = A_2^* = 1.0$


f
120
100
80
60
40
20
0
f*
x*
0.25
0.5
0.75
1.0
1.25
1.5
1.75
2.0
x

14.19

$$f_1(\vec{x}) = 2\rho h\, x_2 \sqrt{1+x_1^2} = 2(0.283)(100)\, x_2 \sqrt{1+x_1^2}$$

$$= 56.6\, x_2 \sqrt{1+x_1^2}$$

$$f_2(\vec{x}) = \frac{P h \left(1+x_1^2\right)^{1.5} \sqrt{1+x_1^4}}{2\sqrt{2}\, E\, x_1^2 x_2} = \frac{(10000)(100)\left(1+x_1^2\right)^{1.5} \sqrt{1+x_1^4}}{2\sqrt{2}\,(30\times 10^6)\, x_1^2 x_2}$$

$$= 0.0117851\, \frac{\left(1+x_1^2\right)^{1.5} \sqrt{1+x_1^4}}{x_1^2 x_2}$$

$$g_1(\vec{x}) = \frac{10{,}000\,(1+x_1)\sqrt{1+x_1^2}}{2\sqrt{2}\, x_1 x_2} - 20{,}000 \le 0$$

or

$$\frac{(1+x_1)\sqrt{1+x_1^2}}{x_1 x_2} - 5.656854 \le 0$$

$$g_2(\vec{x}) = \frac{10{,}000\,(x_1-1)\sqrt{1+x_1^2}}{2\sqrt{2}\, x_1 x_2} - 20{,}000 \le 0$$

or

$$\frac{(x_1-1)\sqrt{1+x_1^2}}{x_1 x_2} - 5.656854 \le 0$$

$x_1 \ge 0.1$

$x_2 \ge 1.0$

Graphical optimization gives the following results:

$$\vec{x}_1^{\,*} = \begin{Bmatrix} 0.6743 \\ 0.5295 \end{Bmatrix}, \quad f_1^{\,*} = 36.1493 \text{ lb}$$

Value of f_2 at $\vec{x}_1^{\,*} = 0.0943$ inch

$$\vec{x}_2^{\,*} = \begin{Bmatrix} 0.8612 \\ 2.5 \end{Bmatrix}, \quad f_2^{\,*} = 0.0182 \text{ inch}$$

Value of f_1 at $\vec{x}_2^{\,*} = 186.7361$ lb.

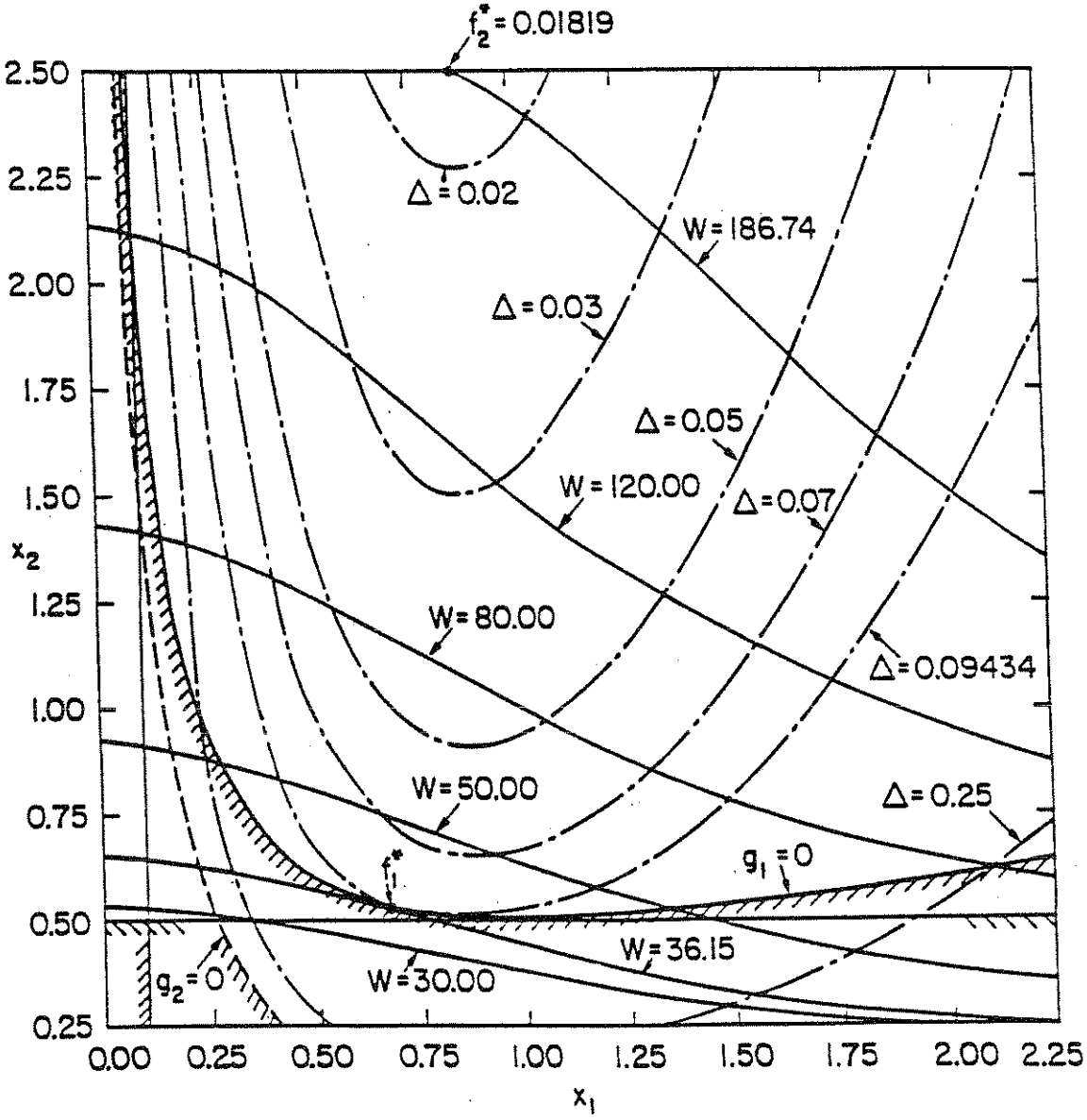


Fig. Contours of the objective functions.

14.20 New objective function is: $f = c_1 f_1 + c_2 f_2$

$$= 0.5\, w_1 \left(2 \rho h\, x_2 \sqrt{1 + x_1^2}\right) + 0.5\, w_2 \left(\frac{P h \left(1 + x_1^2\right)^{1.5} \left(1 + x_1^4\right)^{0.5}}{2\sqrt{2}\, E\, x_1^2\, x_2}\right)$$

where $w_1 = 1.25$ and $w_2 = 2122.0$

(to make $f_1 \simeq f_2$)

Graphical optimization gives:

$$\vec{X}^* = \begin{Bmatrix} 0.7635 \\ 1.0540 \end{Bmatrix}, \quad f^* = \left(f_1 + f_2\right)^* = 1.25 * 75.0595 + 2122.0 * 0.0442 = 187.5670$$

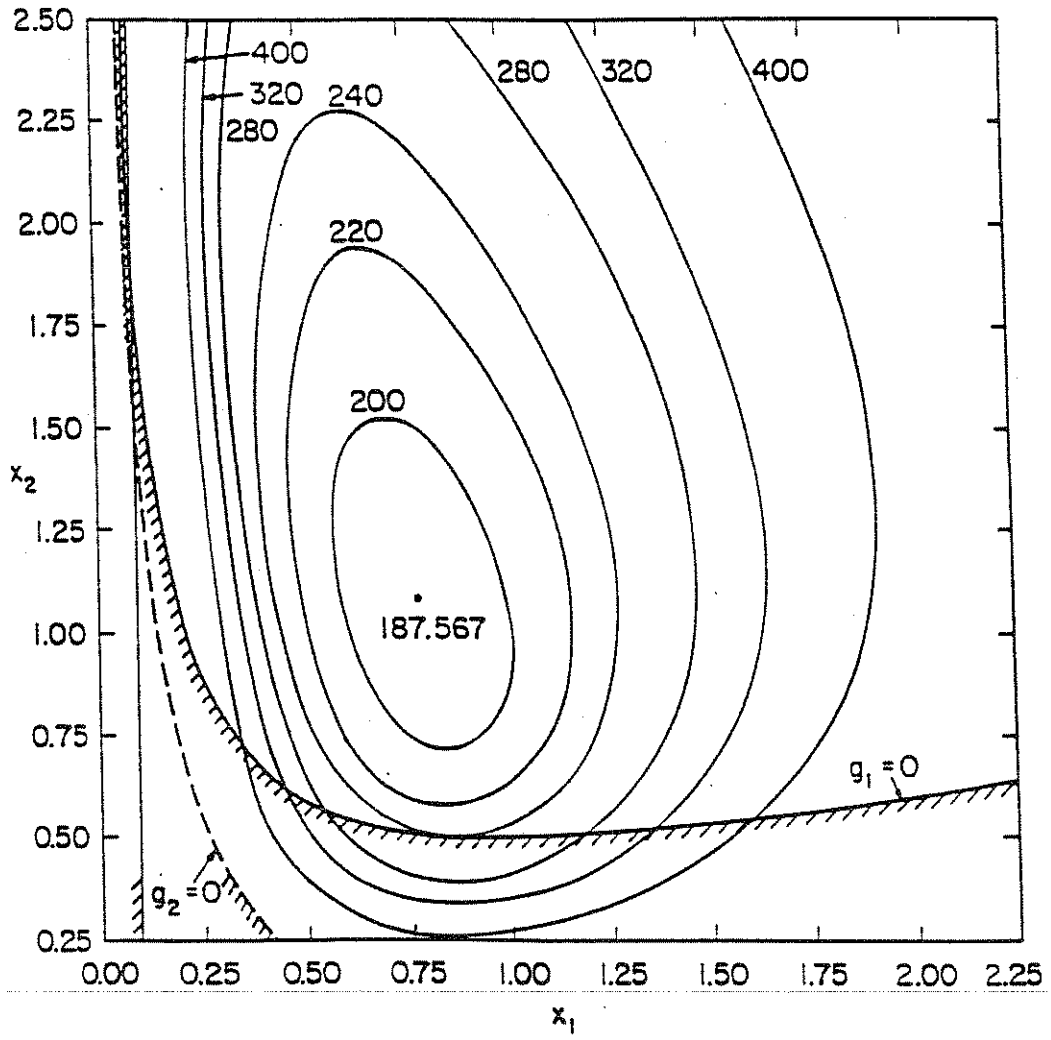


Fig. Contours of the objective function in the weighting method.

14.21

Minimize $$F(\vec{x}) = \left\{\frac{f_1(\vec{x}_1^*) - f_1(\vec{x})}{f_1(\vec{x}_1^*)}\right\}^2 + \left\{\frac{f_2(\vec{x}_2^*) - f_2(\vec{x})}{f_2(\vec{x}_2^*)}\right\}^2$$

subject to $$g_j(\vec{x}) \le 0 \;, \quad j = 1, 2, \ldots, m$$

Here $f_1(\vec{x}_1^*) = 36.1493$, $f_2(\vec{x}_2^*) = 0.0182$

(from solution of Problem 12.12)

$$f_1(\vec{x}) = 2\rho h x_2\sqrt{1+x_1^2}\;, \quad f_2(\vec{x}) = \frac{P h (1+x_1^2)^{1.5}(1+x_1^4)^{0.5}}{2\sqrt{2}\, E\, x_1^2 x_2}$$

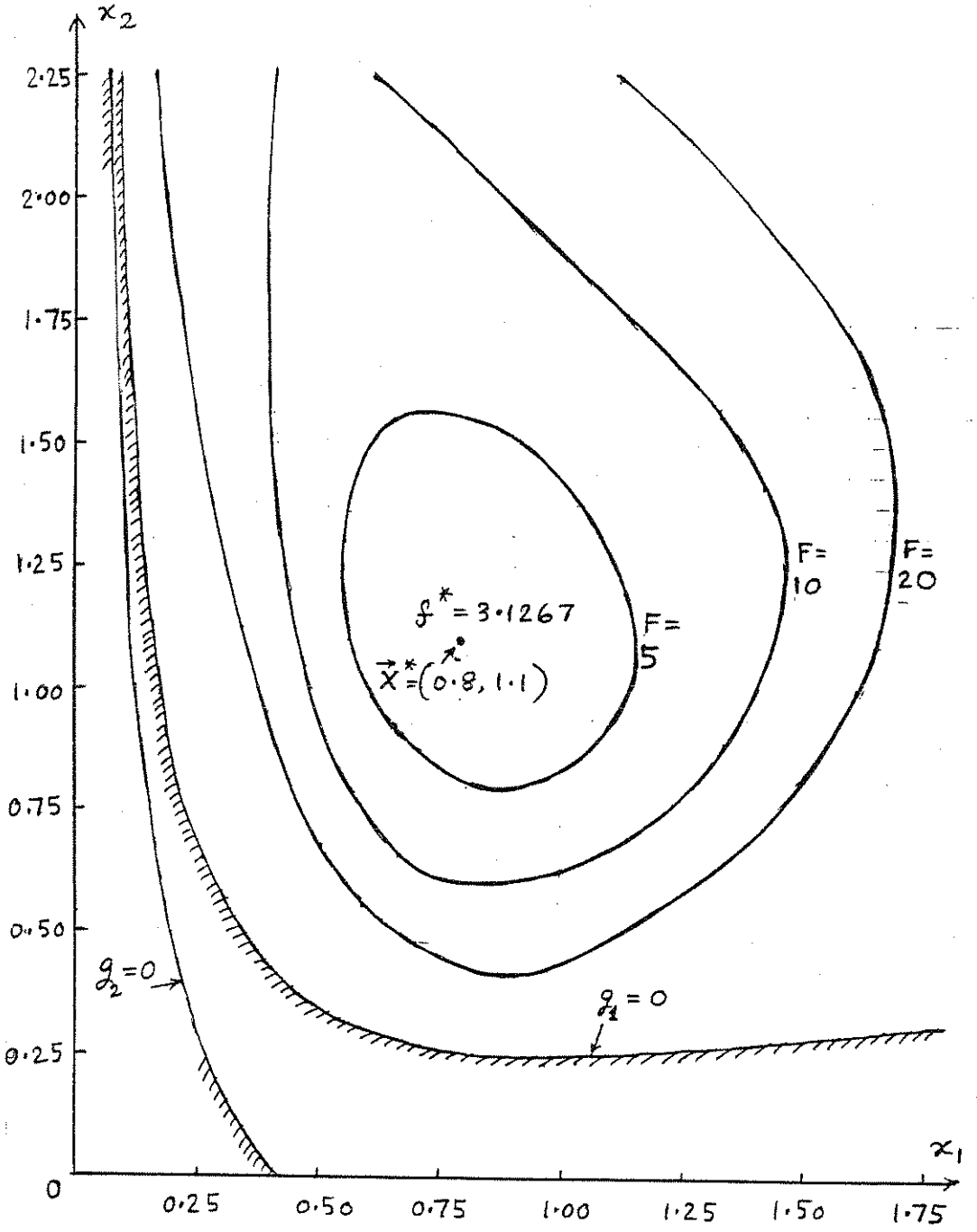
with $\rho = 0.283$, $h = 100$, $P = 10000$, $E = 30 \times 10^6$

Range of $x_1 = 0$ to 2.25, Range of $x_2 = 0$ to 2.5

Graphical optimization gives:

$$\vec{x}^* = \begin{Bmatrix} 0.8 \\ 1.1 \end{Bmatrix}$$

$F^* \simeq 3.1267$, $f_1 = 79.7317$, $f_2 = 0.0417$


x_2
2·25
2·00
1·75
1·50
1·25
1·00
0·75
0·50
0·25
0
$f^* = 3·1267$
$\vec{X}^* = (0·8, 1·1)$
F= 5
F= 10
F= 20
$g_2 = 0$
$g_1 = 0$
x_1
0·25
0·50
0·75
1·00
1·25
1·50
1·75

14.22 Goals are: $b_1 = 30.0$ lb, $b_2 = 0.015$ inch

Let $p = 2$

$f_1(\vec{x}) - b_1 = d_1^- - d_1^+$, $f_2(\vec{x}) - b_2 = d_2^- - d_2^+$

Minimize $F(\vec{x}) = (d_1^+ + d_1^-)^2 + (d_2^+ + d_2^-)^2$

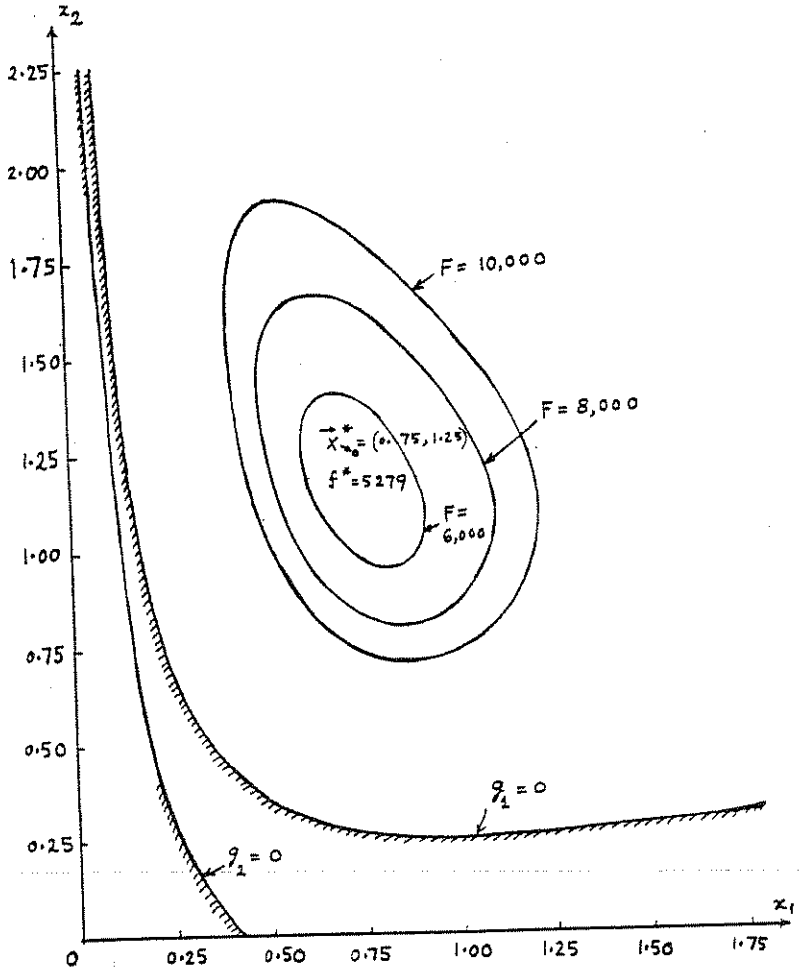
subject to $g_j(\vec{x}) \leq 0$; $j = 1, 2, \ldots, m$

Range of x_1: 0 to 2.25, Range of x_2: 0 to 2.5

Graphical optimization gives:

$$\vec{x}^* = \begin{Bmatrix} 0.75 \\ 1.25 \end{Bmatrix}$$

$F^* = 5279.0$, $f_1 = 79.5938$, $f_2 = 0.0417$



14.23 p1weight.m

```
function p1weight
clear,clc;
%%%%%%%%initial
x0=[2 2 4 1 4 5];
lb=[0 0 1 0 1 0];
ub=[10 10 5 6 5 10];
a=0;b=0;

f01=obj1(x0); %%%%%% function1 value at initial point
f02=obj2(x0); %%%%%% function2 value at initial point
[cons0,ceq0]=cons(x0);  %%%%%% constraints value at initial point
%%%%%%%%%% weight method
[xd1,fd1]=fmincon(@obj1,x0,[],[],[],[],lb,ub,@cons);
[xd2,fd2]=fmincon(@obj2,x0,[],[],[],[],lb,ub,@cons);


[xw,fw]=fmincon(@objweight,x0,[],[],[],[],lb,ub,@cons)

%%%%%%%%%%
fv1=obj1(xw,a,b);
fv2=obj2(xw,a,b);
[consv,ceqv]=cons(xw);

fprintf('********************Results Problem 1(weight)***************\n');
fprintf('                  initial value result                \n');
fprintf(' x0(1)    x0(2)   x0(3)   x0(4)   x0(5)    x0(6) \n');
fprintf(' %4.4f  %4.4f  %4.4f  %4.4f   %4.4f   %4.4f\n', x0(1),x0(2),x0(3),x0(4),x0(5),x0(6));
fprintf('************************************************************\n');
fprintf('  f1(x0)      f2(x0)\n');
fprintf('  %8.4f    %8.4f\n', f01, f02);
fprintf('************************************************************\n');
fprintf('  g1(x0)    g2(x0)    g3(x0)   g4(x0)   g5(x0)    g6(x0)\n');
fprintf('  %4.4f  %4.4f  %4.4f  %4.4f   %4.4f
%4.4f\n',cons0(1),cons0(2),cons0(3),cons0(4),cons0(5),cons0(6));
fprintf('************************************************************\n');
fprintf('                                                \n');
fprintf('          Single objective optimization result          \n');
fprintf('********************  For objective 1  *********************\n');
fprintf(' x1*(1)    x1*(2)   x1*(3)   x1*(4)   x1*(5)   x1*(6) \n');
fprintf(' %4.4f  %4.4f  %4.4f  %4.4f   %4.4f   %4.4f\n', xd1(1),xd1(2),xd1(3),xd1(4),xd1(5),xd1(6));
fprintf('************************************************************\n');
fprintf('  f1(x*)\n');
fprintf('  %8.4f\n', fd1);
fprintf('********************  For objective 2  *********************\n');
fprintf(' x2*(1)    x2*(2)   x2*(3)   x2*(4)   x2*(5)   x2*(6) \n');
```

```
fprintf(' %4.4f  %4.4f  %4.4f  %4.4f   %4.4f   %4.4f\n', xd2(1),xd2(2),xd2(3),xd2(4),xd2(5),xd2(6));
fprintf('*************************************************************\n');
fprintf('  f2(x*)\n');
fprintf('  %8.4f\n', fd2);
fprintf('*************************************************************\n');
fprintf('                                                \n');
fprintf('         Multi-objective  optimization result          \n');
fprintf('*************************************************************\n');
fprintf(' x*(1)    x*(2)   x*(3)   x*(4)   x*(5)   x*(6) \n');
fprintf(' %4.4f  %4.4f  %4.4f  %4.4f   %4.4f   %4.4f\n', xw(1),xw(2),xw(3),xw(4),xw(5),xw(6));
fprintf('*************************************************************\n');
fprintf('  f1(x*)      f2(x*)\n');
fprintf('  %8.4f    %8.4f\n', fv1, fv2);
fprintf('*************************************************************\n');
fprintf('  g1(x*)    g2(x*)    g3(x*)   g4(x*)   g5(x*)    g6(x*)\n');
fprintf('  %4.4f  %4.4f  %4.4f  %4.4f   %4.4f
%4.4f\n',consv(1),consv(2),consv(3),consv(4),consv(5),consv(6));
end



%%%%%%%%%%%%%%
%%%%%%%% f01*, f02*
function f1=obj1(x,a,b);
f1=-25*(x(1)-2)^2-(x(2)-2)^2-(x(3)-1)^2-(x(4)-4)^2-(x(5)-1)^2;
end
%%%%%%%%%%%%%%%%%%%%  f(2)
function f2=obj2(x,a,b);
f2=x(1)^2+x(2)^2+x(3)^2+x(4)^2+x(5)^2+x(6)^2;
end


%%%%%%%%%%%%%%%%%%%%%%%%
function fw=objweight(x,a,b)
fw=obj1(x)+obj2(x);
end
%%%%%%%%%%%%
function [c,ceq]=cons(x)
c(1)=-x(1)-x(2)+2;
c(2)=x(1)+x(2)-6;
c(3)=-x(1)+x(2)-2;
c(4)=x(1)-3*x(3)-2;
c(5)=(x(3)-3)^2+x(4)^2-4;
c(6)=-(x(5)-3)^2-x(6)+4;
ceq=[];
end
```

initial value result

x0(1) x0(2) x0(3) x0(4) x0(5) x0(6)

2.0000 2.0000 4.0000 1.0000 4.0000 5.0000

****

f1(x0) f2(x0)

-27.0000 66.0000

****

g1(x0) g2(x0) g3(x0) g4(x0) g5(x0) g6(x0)

-2.0000 -2.0000 -2.0000 -12.0000 -2.0000 -2.0000

****

Single objective optimization result

********** For objective 1 **********

x1*(1) x1*(2) x1*(3) x1*(4) x1*(5) x1*(6)

2.0000 2.0000 5.0000 0.0000 5.0000 5.0000

****

f1(x*)

-48.0000

********** For objective 2 **********

x2*(1) x2*(2) x2*(3) x2*(4) x2*(5) x2*(6)

1.0000 1.0000 1.0000 0.0000 4.6010 1.4369

****

f2(x*)

26.2336

****

14.24

Listing of program as p1weight.m given in solution of Problem 14.23

```
        Multi-objective  optimization result

************************************************************

x*(1)   x*(2)  x*(3)  x*(4)  x*(5)  x*(6)

-0.0000  2.0000  1.0000  0.0000  4.9343  0.2585

************************************************************

 f1(x*)     f2(x*)

 -131.4788     29.4142

************************************************************

 g1(x*)   g2(x*)   g3(x*)   g4(x*)   g5(x*)   g6(x*)

 0.0000  -4.0000  0.0000  -5.0000  0.0000  -0.0000
```

14.24 (1)

14.25

p1fglobal.m

```
function p1fglobal
clear;clc;
%%%%%%%%initial
x0=[2 2 4 1 4 5];
lb=[0 0 1 0 1 0];
ub=[10 10 5 6 5 10];
a=0;b=0;
%%%%%%%% f01*, f02*
options=optimset('Gradobj','on','LargeScale','off');
options = optimset('TolX',1e-200);
options=optimset('MaxFunEvals',1e1000);
options = optimset(options,'TolCon',1e-28000 );
options=optimset('MaxIter',2000);

f001=obj1(x0);
f002=obj2(x0);

[xd1,f01]=fmincon(@obj1,x0,[],[],[],[],lb,ub,@cons,options,a,b);
[xd2,f02]=fmincon(@obj2,x0,[],[],[],[],lb,ub,@cons,options,a,b);
a=f01;  %%%%%%% a feedback value of single fval of f(1);
b=f02;  %%%%%%% b is feedback value of single of f(2);
[cons0,ceq0]=cons(x0);
```

```
%%%%%%%%% weight method
% options=optimset('Gradobj','off','LargeScale','on');
[xg,fg,exitflag]=fmincon(@objg,x0,[],[],[],[],lb,ub,@cons,options,a,b);

%%%%%%%%%%
fv1=obj1(xg,a,b);
fv2=obj2(xg,a,b);
[consv,ceqv]=cons(xg);

fprintf('          Results Problem 1(global criterion)       \n');
fprintf('*************************************************************\n');
fprintf('              value at initial point                \n');
fprintf(' x0(1)   x0(2)   x0(3)   x0(4)   x0(5)   x0(6) \n');
fprintf(' %4.4f  %4.4f  %4.4f  %4.4f  %4.4f  %4.4f\n', x0(1),x0(2),x0(3),x0(4),x0(5),x0(6));
fprintf('*************************************************************\n');
fprintf('  f1(x0)     f2(x0)\n');
fprintf('  %8.4f    %8.4f\n', f001, f002);
fprintf('*************************************************************\n');
fprintf('  g1(x0)   g2(x0)   g3(x0)   g4(x0)   g5(x0)   g6(x0)\n');
fprintf('  %4.4f  %4.4f  %4.4f  %4.4f  %4.4f
%4.4f\n',cons0(1),cons0(2),cons0(3),cons0(4),cons0(5),cons0(6));
fprintf('*************************************************************\n');
fprintf('                                                \n');
fprintf('          Single objective optimization result          \n');
fprintf('********************  For objective 1  *********************\n');
fprintf(' x1*(1)    x1*(2) \n');
fprintf(' %4.4f  %4.4f  %4.4f  %4.4f  %4.4f  %4.4f\n', xd1(1),xd1(2),xd1(3),xd1(4),xd1(5),xd1(6));
fprintf('*************************************************************\n');
fprintf('  f1(x1*)\n');
fprintf('  %8.4f\n', f01);
fprintf('********************  For objective 2  *********************\n');
fprintf(' x2*(1)   x2*(2)  x2*(3)  x2*(4)  x2*(5)  x2*(6) \n');
fprintf(' %4.4f  %4.4f  %4.4f  %4.4f  %4.4f  %4.4f\n', xd2(1),xd2(2),xd2(3),xd2(4),xd2(5),xd2(6));
fprintf('*************************************************************\n');
fprintf('  f2(x2*)\n');
fprintf('  %8.4f\n', f02);
fprintf('*************************************************************\n');
fprintf('                                                \n');
fprintf('          multi-objectives optimization result          \n');
fprintf('*************************************************************\n');
fprintf(' x*(1)    x*(2)   x*(3)   x*(4)   x*(5)   x*(6) \n');
fprintf(' %4.4f  %4.4f  %4.4f  %4.4f  %4.4f  %4.4f\n', xg(1),xg(2),xg(3),xg(4),xg(5),xg(6));
fprintf('*************************************************************\n');
fprintf('  f1(x*)     f2(x*)\n');
fprintf('  %8.4f    %8.4f\n', fv1, fv2);
fprintf('*************************************************************\n');
fprintf('  g1(x*)   g2(x*)   g3(x*)   g4(x*)   g5(x*)   g6(x*)\n');
fprintf('  %4.4f  %4.4f  %4.4f  %4.4f  %4.4f
%4.4f\n',consv(1),consv(2),consv(3),consv(4),consv(5),consv(6));
end
```

```
%%%%%%%%%%%%%%%
function f=objg(x,a,b); %%%%%
f1=obj1(x);f2=obj2(x);
f=(((a-obj1(x))/a)^2)+((b-obj2(x))/b)^2;
end
%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%% f(1)
function f1=obj1(x,a,b);
f1=-25*((x(1)-2)^2)-((x(2)-2)^2)-((x(3)-1)^2)-((x(4)-4)^2)-((x(5)-1)^2);
end
%%%%%%%%%%%%%%%%%%%%  f(2)
function f2=obj2(x,a,b);
f2=x(1)^2+x(2)^2+x(3)^2+x(4)^2+x(5)^2+x(6)^2;
end
%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%% cons
function [c,ceq]=cons(x,a,b);
c(1)=-x(1)-x(2)+2;
c(2)=x(1)+x(2)-6;
c(3)=-x(1)+x(2)-2;
c(4)=x(1)-3*x(3)-2;
c(5)=(x(3)-3)^2+(x(4)^2)-4;
c(6)=-(x(5)-3)^2-x(6)+4;
ceq=[];
end
```

Results Problem 1(global criterion)

```
************************************************************

          value at initial point

 x0(1)   x0(2)   x0(3)   x0(4)   x0(5)   x0(6)

 2.0000  2.0000  4.0000  1.0000  4.0000  5.0000

************************************************************

  f1(x0)     f2(x0)

  -27.0000    66.0000

************************************************************

  g1(x0)   g2(x0)   g3(x0)   g4(x0)   g5(x0)   g6(x0)

  -2.0000  -2.0000  -2.0000  -12.0000  -2.0000  -2.0000

************************************************************
```

Single objective optimization result

********** For objective 1 **********

x1*(1) x1*(2)

2.0000 2.0000 5.0000 0.0000 5.0000 5.0000

****

f1(x1*)

-48.0000

********** For objective 2 **********

x2*(1) x2*(2) x2*(3) x2*(4) x2*(5) x2*(6)

1.0000 1.0000 1.0000 0.0000 4.6010 1.4369

****

f2(x2*)

26.2336

****

multi-objectives optimization result

****

x*(1) x*(2) x*(3) x*(4) x*(5) x*(6)

1.1531 0.8469 1.0001 0.0222 4.5937 1.4602

****

f1(x*) f2(x*)

-47.9998 26.2818

****

g1(x*) g2(x*) g3(x*) g4(x*) g5(x*) g6(x*)

-0.0000 -4.0000 -2.3061 -3.8473 0.0000 -0.0001

14.26

p1fbounded.m

```
function p1fbounded
clear;clc;
%%%%%%%%initial
x0=[2 2 4 1 4 5];
lb=[0 0 1 0 1 0];
ub=[10 10 5 6 5 10];
a=0;b=0;

f001=obj1(x0);
f002=obj2(x0);
[cons0,ceq0]=cons1(x0);
%%%%%%%% f01*, f02*
options=optimset('Gradobj','off','LargeScale','off');
% [x1,f01]=fmincon(@obj1,x0,[],[],[],[],lb,ub,@cons,options,a,b);
[x2,f02]=fmincon(@obj2,x0,[],[],[],[],lb,ub,@cons1,options,a,b);
% a=f01;  %%%%%%% a feedback value of single fval of f(1);
b=f02;  %%%%%%% b is feedback value of single of f(2);

%%%%%%%%% bounded method
[xd1,fd1]=fmincon(@obj1,x0,[],[],[],[],lb,ub,@cons1);
[xd2,fd2]=fmincon(@obj2,x0,[],[],[],[],lb,ub,@cons1);

options=optimset('Gradobj','off','LargeScale','on');
[xb,fb]=fmincon(@obj1,x0,[],[],[],[],lb,ub,@cons2,options,a,b);
fv1=obj1(xb,a,b);
fv2=obj2(xb,a,b);
[consv,ceqv]=cons1(xb);


fprintf('*******************Results Problem 1(Bounded)*************\n');
fprintf('                                                \n');
fprintf('                value at initial point                \n');
fprintf('*************************************************************\n');
fprintf(' x0(1)    x0(2)   x0(3)   x0(4)   x0(5)    x0(6) \n');
fprintf(' %4.4f  %4.4f  %4.4f  %4.4f   %4.4f   %4.4f\n', x0(1),x0(2),x0(3),x0(4),x0(5),x0(6));
fprintf('*************************************************************\n');
fprintf('  f1(x0)      f2(x0)\n');
fprintf('  %8.4f    %8.4f\n', f001, f002);
fprintf('*************************************************************\n');
fprintf('  g1(x0)    g2(x0)    g3(x0)   g4(x0)   g5(x0)   g6(x0)\n');
fprintf('  %4.4f  %4.4f  %4.4f  %4.4f   %4.4f
%4.4f\n',cons0(1),cons0(2),cons0(3),cons0(4),cons0(5),cons0(6));
fprintf('*************************************************************\n');
fprintf('                                                \n');
fprintf('          single objective optimization result          \n');
```

```
fprintf('******************  For objective 1  ********************\n');
fprintf(' x1*(1)   x1*(2)  x1*(3)  x1*(4)  x1*(5)  x1*(6) \n');
fprintf(' %4.4f  %4.4f  %4.4f  %4.4f  %4.4f  %4.4f\n', xd1(1),xd1(2),xd1(3),xd1(4),xd1(5),xd1(6));
fprintf('************************************************************\n');
fprintf('  f1(x*)\n');
fprintf('  %8.4f\n', fd1);
fprintf('******************  For objective 2  ********************\n');
fprintf(' x2*(1)   x2*(2)  x2*(3)  x2*(4)  x2*(5)  x2*(6) \n');
fprintf(' %4.4f  %4.4f  %4.4f  %4.4f  %4.4f  %4.4f\n', xd2(1),xd2(2),xd2(3),xd2(4),xd2(5),xd2(6));
fprintf('************************************************************\n');
fprintf('  f2(x*)\n');
fprintf('  %8.4f\n', fd2);
fprintf('************************************************************\n');
fprintf('                                          \n');
fprintf('        multi-objectives optimization result          \n');
fprintf('************************************************************\n');
fprintf(' x*(1)   x*(2)  x*(3)  x*(4)  x*(5)  x*(6) \n');
fprintf(' %4.4f  %4.4f  %4.4f  %4.4f  %4.4f  %4.4f\n', xb(1),xb(2),xb(3),xb(4),xb(5),xb(6));
fprintf('************************************************************\n');
fprintf('  f1(x*)     f2(x*)\n');
fprintf('  %8.4f   %8.4f\n', fv1, fv2);
fprintf('************************************************************\n');
fprintf('  g1(x*)   g2(x*)   g3(x*)   g4(x*)   g5(x*)   g6(x*)\n');
fprintf('  %4.4f  %4.4f  %4.4f  %4.4f  %4.4f
%4.4f\n',consv(1),consv(2),consv(3),consv(4),consv(5),consv(6));

end

%%%%%%%%%%%%%%%%
% function f=objbound(x,a,b); %%%%%%
% f1=obj1(x);f2=obj2(x);
% f=((a-obj1(x))/a)^2+((b-obj2(x))/b)^2
% end
%%%%%%%%%%%%%%%%

%%%%%%%%%%%%%%%%%%%% f(1)
function f1=obj1(x,a,b);
f1=-25*(x(1)-2)^2-(x(2)-2)^2-(x(3)-1)^2-(x(4)-4)^2-(x(5)-1)^2;
end
%%%%%%%%%%%%%%%%%%%%  f(2)
function f2=obj2(x,a,b);
f2=x(1)^2+x(2)^2+x(3)^2+x(4)^2+x(5)^2+x(6)^2;
end
%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%% cons1
```

```
function [c,ceq]=cons1(x,a,b);
c(1)=-x(1)-x(2)+2;
c(2)=x(1)+x(2)-6;
c(3)=-x(1)+x(2)-2;
c(4)=x(1)-3*x(3)-2;
c(5)=(x(3)-3)^2+x(4)^2-4;
c(6)=-(x(5)-3)^2-x(6)+4;
ceq=[];
end
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
function [c,ceq]=cons2(x,a,b);

c(1)=-x(1)-x(2)+2;
c(2)=x(1)+x(2)-6;
c(3)=-x(1)+x(2)-2;
c(4)=x(1)-3*x(3)-2;
c(5)=(x(3)-3)^2+x(4)^2-4;
c(6)=-(x(5)-3)^2-x(6)+4;
c(7)=0.8*b-obj2(x);
c(8)=obj2(x)-1.2*b;
ceq=[];
end
```

**********Results Problem 1(Bounded)**********

value at initial point

```
************************************************************

 x0(1)   x0(2)   x0(3)   x0(4)   x0(5)   x0(6)

 2.0000  2.0000  4.0000  1.0000  4.0000  5.0000

************************************************************

  f1(x0)     f2(x0)

  -27.0000     66.0000

************************************************************

   g1(x0)   g2(x0)   g3(x0)   g4(x0)   g5(x0)   g6(x0)

   -2.0000  -2.0000  -2.0000  -12.0000  -2.0000  -2.0000

************************************************************
```

single objective optimization result

********** For objective 1 **********

x1*(1) x1*(2) x1*(3) x1*(4) x1*(5) x1*(6)

2.0000 2.0000 5.0000 0.0000 5.0000 5.0000

****

f1(x*)

-48.0000

********** For objective 2 **********

x2*(1) x2*(2) x2*(3) x2*(4) x2*(5) x2*(6)

1.0000 1.0000 1.0000 0.0000 4.6010 1.4369

****

f2(x*)

26.2336

****

multi-objectives optimization result

****

x*(1) x*(2) x*(3) x*(4) x*(5) x*(6)

-0.0000 2.0000 1.5749 0.0000 5.0000 0.0000

****

f1(x*) f2(x*)

-132.3305 31.4803

****

g1(x*) g2(x*) g3(x*) g4(x*) g5(x*) g6(x*)

0.0000 -4.0000 0.0000 -6.7247 -1.9691 -0.0000

14.27

p1fgoalattain.m

```
function p1fgoalattain
clear;clc;
format short;

x0=[2 2 4 1 4 5];  %%%%%%%% ***remind add x(7)
lb=[0 0 1 0 1 0];
ub=[10 10 5 6 5];
% a=0;b=0;
f001=obj1(x0);f002=obj2(x0);  %%%%%%%% f(x0)
% f002=obj2(x0);
[cons0,ceq0]=cons2(x0); %%%%%%% constraints value @ x0

%%%%%%%%% f01*, f02*
A=[-1 -1 0 0 0 0; 1 1 0 0 0 0; -1 1 0 0 0 0; 1 0 -3 0 0 0];
b=[-2 6 2 2]';
options=optimset('Gradobj','off','LargeScale','on');
aa=0,bb=0;
[xd1,f01]=fmincon(@obj1,x0,A,b,[],[],lb,ub,@cons2,options,aa,bb);
[xd2,f02]=fmincon(@obj2,x0,A,b,[],[],lb,ub,@cons2,options,aa,bb);
aa=f01;  %%%%%%% a is feedback value of single fval of f(1);
bb=f02;  %%%%%%% b is feedback value of single fval of f(2);

%%%%%%%%%% goalattain method
options=optimset('Gradobj','off','LargeScale','on');
options=optimset('MaxFunEvals',1000);

goal=[0.8*aa 0.8*bb];
weight=[0.5 0.5];

options=optimset('GoalsExactAchieve','6');
ProdPlanObj=inline('[-25*(x(1)-2)^2-(x(2)-2)^2-(x(3)-1)^2-(x(4)-4)^2-(x(5)-1)^2,
x(1)^2+x(2)^2+x(3)^2+x(4)^2+x(5)^2+x(6)^2]','x');
[xga,fval]= fgoalattain(ProdPlanObj,x0,goal,weight,A,b,[],[],lb,ub,[],options);

fv1=obj1(xga);
fv2=obj2(xga);
[consv,ceqv]=cons2(xga);

fprintf('          Results Problem 1(goal attainment)          \n');
fprintf('                                                      \n');
fprintf('              value at initial point                  \n');
fprintf(' x0(1)    x0(2)    x0(3)    x0(4)    x0(5)    x0(6) \n');
fprintf(' %4.4f  %4.4f  %4.4f  %4.4f  %4.4f  %4.4f\n', x0(1),x0(2),x0(3),x0(4),x0(5),x0(6));
fprintf('*********************************************************\n');
fprintf('  f1(x0)      f2(x0)\n');
```

```
fprintf('  %8.4f    %8.4f\n', f001, f002);
fprintf('**********************************************************\n');
fprintf('  g1(x0)    g2(x0)    g3(x0)    g4(x0)    g5(x0)    g6(x0)\n');
fprintf('  %4.4f  %4.4f  %4.4f  %4.4f   %4.4f
%4.4f\n',cons0(1),cons0(2),cons0(3),cons0(4),cons0(5),cons0(6));
fprintf('**********************************************************\n');
fprintf('                                          \n');
fprintf('          Single objective optimization result          \n');
fprintf('********************  For objective 1  ********************\n');
fprintf(' x1*(1)    x1*(2)    x1*(3)    x1*(4)    x1*(5)    x1*(6) \n');
fprintf(' %4.4f  %4.4f  %4.4f  %4.4f   %4.4f   %4.4f\n', xd1(1),xd1(2),xd1(3),xd1(4),xd1(5),xd1(6));
fprintf('**********************************************************\n');
fprintf('  f1(x1*)\n');
fprintf('  %8.4f\n', f01);
fprintf('********************  For objective 2  ********************\n');
fprintf(' x2*(1)    x2*(2)    x2*(3)    x2*(4)    x2*(5)    x2*(6) \n');
fprintf(' %4.4f  %4.4f  %4.4f  %4.4f   %4.4f   %4.4f\n', xd2(1),xd2(2),xd2(3),xd2(4),xd2(5),xd2(6));
fprintf('**********************************************************\n');
fprintf('  f2(x2*)\n');
fprintf('  %8.4f\n', f02);
fprintf('**********************************************************\n');
fprintf('                                          \n');
fprintf('          multi-objectives optimization result          \n');
fprintf('**********************************************************\n');
fprintf(' x*(1)    x*(2)    x*(3)    x*(4)    x*(5)    x*(6) \n');
fprintf(' %4.4f  %4.4f  %4.4f  %4.4f   %4.4f   %4.4f\n', xga(1),xga(2),xga(3),xga(4),xga(5),xga(6));
fprintf('**********************************************************\n');
fprintf('  f1(x*)      f2(x*)\n');
fprintf('  %8.4f    %8.4f\n', fv1, fv2);
fprintf('**********************************************************\n');
fprintf('  g1(x*)    g2(x*)    g3(x*)    g4(x*)    g5(x*)    g6(x*)\n');
fprintf('  %4.4f  %4.4f  %4.4f  %4.4f   %4.4f
%4.4f\n',consv(1),consv(2),consv(3),consv(4),consv(5),consv(6));

end

%%%
```

```
function [c,ceq]=cons1(x);
c(1)=(x(3)-3)^2+x(4)^2-4;
c(2)=-(x(5)-3)^2-x(6)+4;
ceq=[];
end
% %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
function [c,ceq]=cons2(x,aa,bb);
c(1)=-x(1)-x(2)+2;
c(2)=x(1)+x(2)-6;
c(3)=-x(1)+x(2)-2;
c(4)=x(1)-3*x(3)-2;
c(5)=(x(3)-3)^2+x(4)^2-4;
c(6)=-(x(5)-3)^2-x(6)+4;
ceq=[];
end
```

Results Problem 1(goal attainment)

value at initial point

```
x0(1)   x0(2)   x0(3)   x0(4)   x0(5)   x0(6)

2.0000  2.0000  4.0000  1.0000  4.0000  5.0000

************************************************************

 f1(x0)     f2(x0)

 -27.0000    66.0000

************************************************************

 g1(x0)   g2(x0)   g3(x0)   g4(x0)   g5(x0)   g6(x0)

 -2.0000  -2.0000  -2.0000  -12.0000  -2.0000  -2.0000

************************************************************
```

Single objective optimization result

```
******************** For objective 1 **********************

 x1*(1)   x1*(2)  x1*(3)  x1*(4)  x1*(5)  x1*(6)

 2.0000  2.0000  5.0000  0.0000  5.0000  5.0000

************************************************************
```

f1(x1*)

-48.0000

********** For objective 2 **********

x2*(1) x2*(2) x2*(3) x2*(4) x2*(5) x2*(6)

1.0000 1.0000 1.0000 0.0000 4.6010 1.4369

****

f2(x2*)

26.2336

****

multi-objectives optimization result

****

x*(1) x*(2) x*(3) x*(4) x*(5) x*(6)

1.2014 0.8833 2.8984 0.2579 2.8984 1.3766

****

f1(x*) f2(x*)

-38.4000 20.9869

****

g1(x*) g2(x*) g3(x*) g4(x*) g5(x*) g6(x*)

-0.0848 -3.9152 -2.3181 -9.4938 -3.9231 2.6131

14.28

p2weight.m

```
function p2weight
clear,clc;

%%%%%%%%%%% main%%%%%%%%%%%
%%%%%%%%%initial %%%%%%%%%%%
x0=[3 1.5];
lb=[-10 -10];
ub=[10 10];
% a=0;b=0;

f01=obj1(x0);
f02=obj2(x0);
f03=obj3(x0);
[cons0,ceq0]=cons(x0);

%%%%%%%%% weight method

[xw,fw]=fmincon(@objweight,x0,[],[],[],[],lb,ub,@cons)

%%%%%%%%% feedback value from optimal results
fv1=obj1(xw);
fv2=obj2(xw);
fv3=obj3(xw);
[consv,ceqv]=cons(xw);

fprintf('*******************Results Problem 2(weight)***************\n');
fprintf('*************************************************************\n');
fprintf('x0(1)          x0(2) \n');
fprintf('%4.4f          %4.4f\n', x0(1),x0(2));
fprintf('*************************************************************\n');
fprintf(' f1(x0)       f2(x0)       f3(x0)\n');
fprintf('%8.4f     %8.4f     %8.4f\n', f01, f02, f03);
fprintf('*************************************************************\n');
fprintf('g1(x0)          g2(x0)\n');
fprintf('%4.4f          %4.4f\n', cons0(1),cons0(2));
fprintf('*************************************************************\n');
fprintf('*************************************************************\n');
fprintf('x*(1)          x*(2) \n');
fprintf('%4.4f          %4.4f\n', xw(1),xw(2));
fprintf('*************************************************************\n');
fprintf(' f1(x*)       f2(x*)       f3(x*)\n');
fprintf('%8.4f     %8.4f     %8.4f\n', fv1, fv2,fv3);
fprintf('*************************************************************\n');
fprintf('g1(x*)          g2(x*)\n');
fprintf('%4.4f          %4.4f\n',consv(1),consv(2));
```

```
end
%%%
```

Single objective optimization result

********** For objective 1 **********

x1*(1) x1*(2)

2.6617 -1.3679

****

f1(x1*)

-4.8024

********** For objective 2 **********

x2*(1) x2*(2)

3.4550 0.0278

****

f2(x2*)

-1.2023

********** For objective 3 **********

x3*(1) x3*(2)

1.8835 -1.8387

****

f3(x3*)

-10.9666

14.29

Listing of program as p2weight.m given in solution of Problem 14.28

```
        Multi-objective  optimization result

************************************************************

x*(1)         x*(2)

2.1179        -1.7362

************************************************************

 f1(x*)      f2(x*)      f3(x*)

 -4.2948      6.5159     -10.5760

************************************************************

 g1(x*)         g2(x*)

-0.4857         0.0000
```

14.30

p2fgoalattain.m

```
function p2fgoalattain
clear;clc;
format short;

x0=[3 1.5];   %%%%%%%%
lb=[-10 -10];
ub=[10 10];
aa=0;bb=0;cc=0; %%%%%%% prepare for single optimization for each f1 f2 f3

f001=obj1(x0);f002=obj2(x0);f003=obj3(x0); %%%%%% feedback optimal value of f1 f2 f3
[cons0,ceq0]=cons1(x0);   %%%%%% feedback optimal value of constraints

%%%%%%%%% f01*, f02* f03*
options=optimset('Gradobj','off','LargeScale','on');
[x01,f01]=fmincon(@obj1,x0,[],[],[],[],lb,ub,@cons1,options,aa,bb,cc); %%%%%%use aa in case to
different with A
[x02,f02]=fmincon(@obj2,x0,[],[],[],[],lb,ub,@cons1,options,aa,bb,cc); %%%%%%use bb in case to
different with vector b
[x03,f03]=fmincon(@obj3,x0,[],[],[],[],lb,ub,@cons1,options,aa,bb,cc); %%%%%%use cc in case to
different with vector c
aa=f01;  %%%%%%% a is feedback value of single fval of f(1);
bb=f02;  %%%%%%% b is feedback value of single of f(2);
cc=f03;  %%%%%%% c is feedback value of single of f(3);
```

```
% a
% b
% c
%%%%%%%%% goalattain method
goal=[0.8*aa 0.8*bb 0.8*cc];  %%%%%%%% corresponding to new setup constraints with f1-x3-f*1
f2-x2-f*2 f3-x3-f*3
weight=[1/3 1/3 1/3];

options=optimset('GoalsExactAchieve','3');
ProdPlanObj=inline('[1.5-x(1)*(1-x(2)),
2.25-x(1)*(1-x(2)^2),2.625-x(1)*(1-(x(2)^3))]','x');%%%%%corresponding to new multiobjective
[xga,fval]= fgoalattain(ProdPlanObj,x0,goal,weight,[],[],[],[],lb,ub,[],options);%%%%%% no linear
parts

fv1=obj1(xga,aa,bb,cc); %%%%%%%% feedback of optimal value for f1 f2 f3 and constraints
fv2=obj2(xga,aa,bb,cc);
fv3=obj3(xga,aa,bb,cc);
[consv,ceqv]=cons1(xga);

fprintf('*****************Results Problem 2(goal attain)*********\n');
fprintf('************************************************************\n');
fprintf('x0(1)          x0(2) \n');
fprintf('%4.4f          %4.4f\n', x0(1),x0(2));
fprintf('************************************************************\n');
fprintf(' f1(x0)       f2(x0)       f3(x0)\n');
fprintf('%8.4f     %8.4f     %8.4f\n', f001, f002, f003);
fprintf('************************************************************\n');
fprintf('g1(x0)          g2(x0)\n');
fprintf('%4.4f          %4.4f\n', cons0(1),cons0(2));
fprintf('************************************************************\n');
fprintf('************************************************************\n');
fprintf('x*(1)          x*(2) \n');
fprintf('%4.4f          %4.4f\n', xga(1),xga(2));
fprintf('************************************************************\n');
fprintf(' f1(x*)       f2(x*)       f3(x*)\n');
fprintf('%8.4f     %8.4f     %8.4f\n', fv1, fv2,fv3);
fprintf('************************************************************\n');
fprintf('g1(x*)          g2(x*)\n');
fprintf('%4.4f          %4.4f\n',consv(1),consv(2));
end

%%%%%%%%%% construct goal attainment objective fun

%%%%%%%%%%%%%%%%%%%% f(1)
function f1=obj1(x,aa,bb,cc)
f1=1.5-x(1)*(1-x(2));
end
%%%%%%%%%%%%%%%%%%%%  f(2)
```

```
function f2=obj2(x,aa,bb,cc)
f2=2.25-x(1)*(1-x(2)^2);
end
%%%%%%%%%%%%%%%%% f(3)
function f3=obj3(x,aa,bb,cc)
f3=2.625-x(1)*(1-(x(2)^3));
end
%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%% cons1
function [c,ceq]=cons1(x,aa,bb,cc)
c(1)=-x(1)^2-(x(2)-0.5)^2+9;
c(2)=(x(1)-1)^2+(x(2)-0.5)^2-6.25;
ceq=[];
end

%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% function [c,ceq]=cons2(x,a,b,cc)
% c(1)=-x(1)^2-(x(2)-0.5)^2+9;
% c(2)=(x(1)-1)^2+(x(2)-0.5)^2-6.25;
% c(3)=obj1(x,a,b,cc)-(x(3)/3)-0.8*a;
% c(4)=obj2(x,a,b,cc)-(x(3)/3)-0.8*b;
% c(5)=obj3(x,a,b,cc)-(x(3)/3)-0.8*cc;%%%%% be care unknown error when use 'c' as f03
% ceq=[];
% end
```

Results Problem 2(goal attain)

Initial value result

```
************************************************************

x0(1)        x0(2)

3.0000        1.5000

************************************************************

 f1(x0)      f2(x0)      f3(x0)

  3.0000     6.0000      9.7500

************************************************************

g1(x0)        g2(x0)

-1.0000       -1.2500

************************************************************
```